2025
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NEUROLOGY

NEU-316

SOHAG UNIVERSITY
FACULITY OF
MEDICINE

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Neuroscience

Anatomy & embryology development

NEU-316

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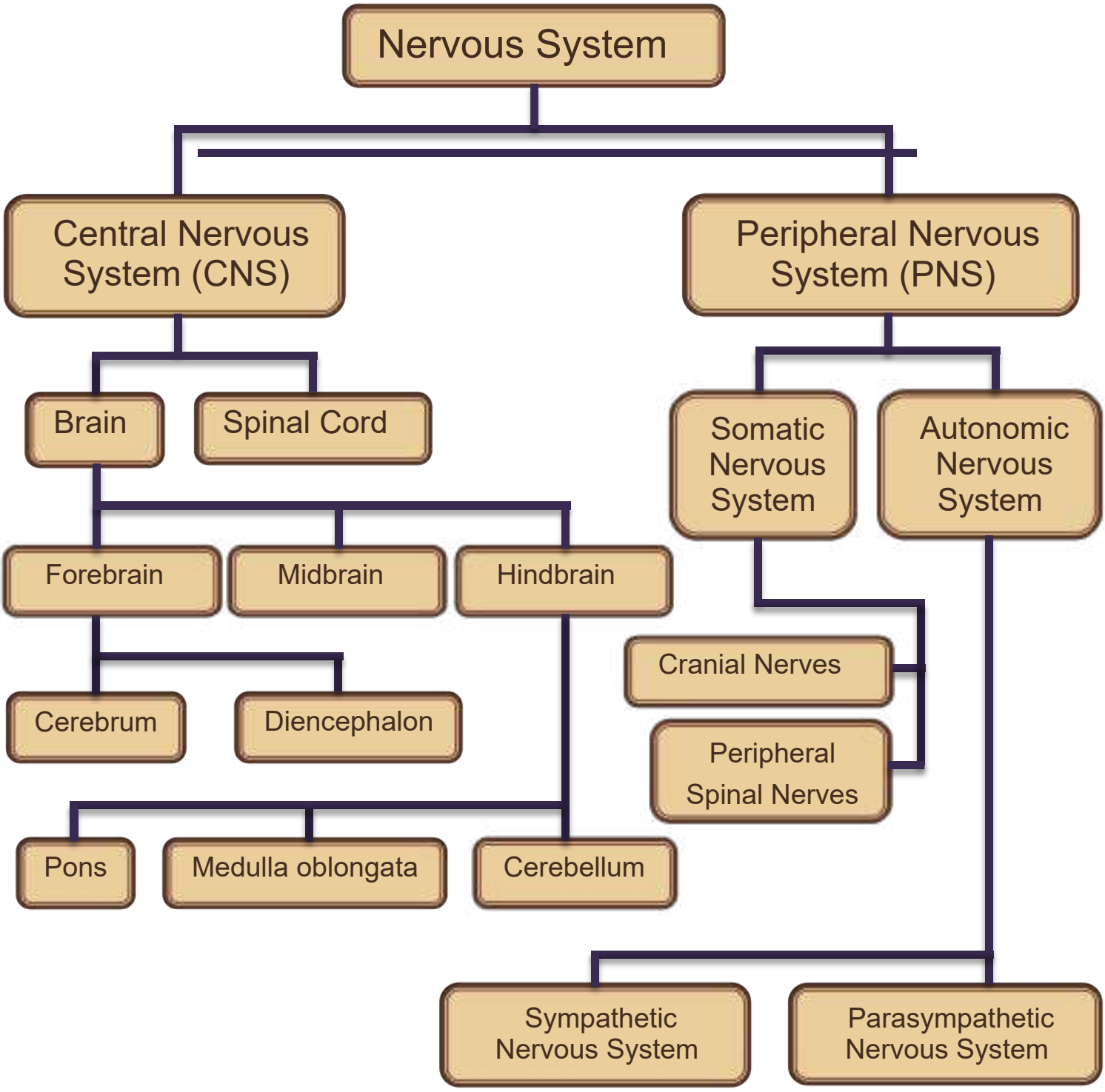
Nervous System



1) Meninges

The brain in the skull is surrounded by three protective membranes, or meninges: **the dura mater, the arachnoid mater, and the pia mater.**

2) Dura mater

The dura mater is conventionally described as two layers:

a- The endosteal layer: covering the inner surface of the skull bones. It becomes continuous with the periosteum on the outside of the skull bones.

b- The meningeal layer: It is adense, strong, fibrous membrane covering the brain and is continuous through the foramen magnum with the dura mater of the spinal cord. It provides tubular sheaths for the cranial nerves as the latter pass through the foramina in the skull.

- The meningeal layer sends inward four septa that divide the cranial cavity into freely communicating spaces lodging the subdivisions of the brain. This septa is called **Dural folds**.

- **Dural fold or septa:** The function of these septa is to restrict the rotatory displacement of the brain.

1-The falx cerebri

shape: is a sickle-shapedfold of dura mater **Site:** lies in the midlinebetween the two cerebral hemispheres **attachment:** in frontisattached to the internal frontal crest and the crista galli. The posteriorly part blendsinthe midline with the upper surface of the tentorium cerebelli. The superior sagittalsinusruns in its upper margin. The inferior sagittal sinus runs in its lower concave freemargin.

2-the tentorium cerebelli

Shape: Thetentorium cerebelli is a crescent-shaped fold

Site: Itcoversthe upper surface of the cerebellum and supports the

Attachment:

a- fixedborder: The fixed border is attached to the posterior clinoid processes, the superiorborders of the petrous bones, and the margins of the grooves for the transverse sinuses on the occipital bone. The free border runs forward at its two ends, crosses the attached border, and is affixed to the anterior clinoid process on
b- each side. At the point where the two borders cross, the third and fourth cranial nerves .

3-The falx cerebelli

shape: is a small, sickle-shaped fold of dura **site:** present between the two cerebellar hemispheres. **attachment:** superiorly (base)attached to tentorium cerbelli inferiorly (apex) attached to
margin of foramen magnum posteriorly attached to internal occipital crest it enclose the occipital sinus anterior border free

4-The diaphragma sellae

isa small circular foldof dura mater that forms the roof for the sella turcica . A small opening in its center allows passage of the stalk of the pituitary gland.

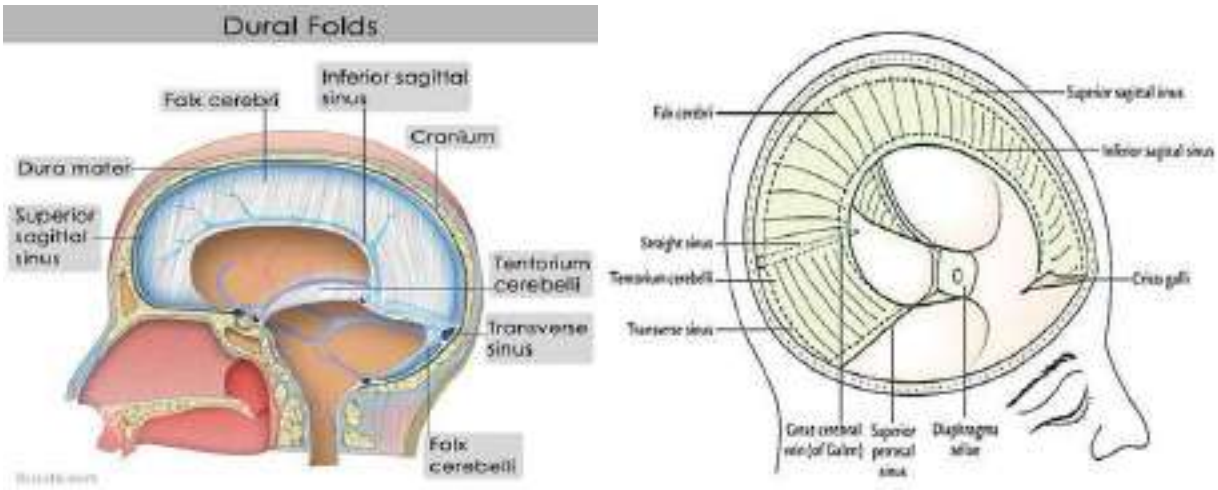


Figure (49): Dural fold

Dural Nerve Supply

Branches of the trigeminal, vagus, and first three cervical nerves and branches from the sympathetic system pass to the dura.

Dural Arterial Supply

Numerous arteries supply the dura mater from the internal carotid, maxillary(through middle accessory meningeal artery), ascending pharyngeal, occipital, and vertebral arteries.

Dural Venous Drainage

The meningeal veins lie in the endosteal layer of dura. The middle meningeal vein follows the branches of the middle meningeal artery and drains into the pterygoid venous plexus or the sphenoparietal sinus.

Applied anatomy

Skull trauma lead to:

Extradural hemorrhage (present between skull and dura matter): results from injuries to the meningeal arteries. The most common artery to be damaged is the anterior division of the middle meningeal artery.

Subdural hemorrhage: results from tearing of the superior cerebral veins at their point of entrance into the superior sagittal sinus.

Dural venous sinuses

Its channel between two layers of dura

Single sinus	Paired sinus
• Superior Sagittal sinuses	• Sphenoparietal Sinus
• Inferior Sagittal sinuses	• Cavernous sinus
• straight sinus	• Petrosal sinus
• occipital sinus	• Transverse sinus
	• Sigmoid sinus

1- The superior sagittal sinus: lies in the upper fixed border of the falx cerebri. It runs backward and becomes continuous with the right transverse sinus. The sinus communicates on each side with the venous lacunae.Tributaries: Numerous arachnoid villi and granulations project into the lacunae ,superior cerebral veins

2-The inferior sagittal sinus: lies in the free lower margin of the falx cerebri. It runs backward and joins the great cerebral vein to form the straight sinus. It receives cerebral veins from the medial surface of the cerebral hemisphere.

3-The straight sinus: lies at the junction of the falx cerebri with the tentorium cerebelli. Formed by the union of the inferior sagittal sinus with the great cerebral vein, it drains into the left transverse sinus.

4-The occipital sinus: lies in the attached margin of the falx cerebelli it run from foremen magnum to occipital protuberance . It communicates with the vertebral veins through the foramen magnum and the transverse sinuses.

5-The transverse sinus: right transverse sinus begins as a continuation of the superior sagittal sinus while the left transverse sinus is usually a continuation of the straight sinus each sinus lies in the lateral attached margin of the tentorium cerebelli, and they end on each side by becoming the sigmoid sinus.

6-The sigmoid sinuses: are a direct continuation of the transverse sinuses. Each sinus turns downward behind the mastoid antrumof the temporal bone and then leaves the skull through the jugular foramen to becomethe internal jugular vein.

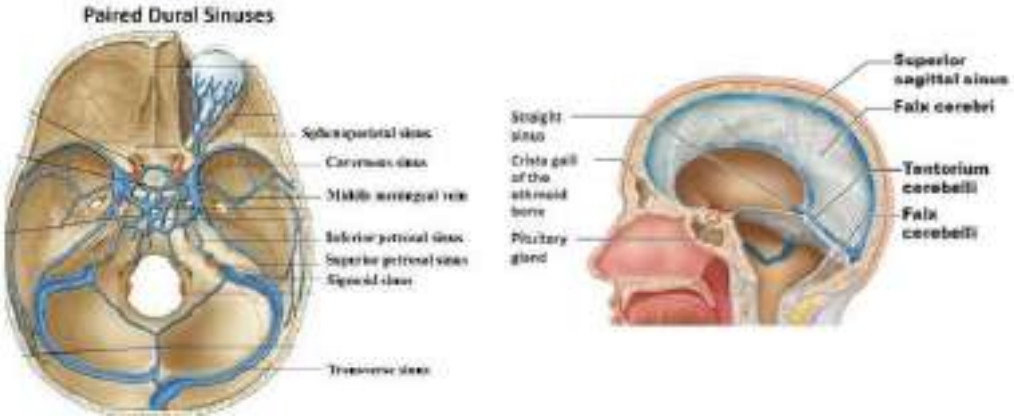


Figure (50): Dural venous sinus

7-Cavernous sinus:

Site: liesonthelateral side of the body of the sphenoid bone

Extension: anteriorend reach medial end of superior orbital fissure. posterior end reach apexofpetrousbone.

Structure at lateralwallofsinus:
1- oculomotor nerve
2- trochlear nerve
3- ophthalmic nerve
4- maxillary nerve

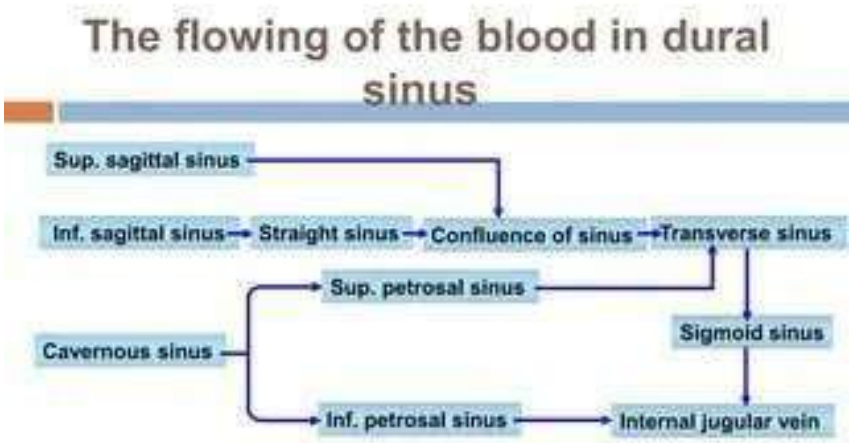
Structure pass throughsinus:internalcarotid artery, abducent nerve

These structures separatefromthebloodby endothelial lining.

Tributaries from the orbit:
1- superior inferior ophthalmic veins
2- superficial middle cerebral vein
3- inferior cerebral vein
4- Central retinal vein

Tributaries from meninges:
1- sphenoparitalsinus
2- middle meningeal vein

Drainage:
1- superior petrosal sinus
2- inferior petrosal sinus
Communication: facial vein through superior ophthalmic vein



N.B.

confluence of sinuses: is connecting point of superior sagittal sinus straight sinus occipital sinus and two transverse sinuses.

3) *Arachnoid mater*

The arachnoid mater is a delicate, impermeable membrane covering the brain and lying between the pia mater internally and the dura mater externally. It is separated from the dura by a potential space, the subdural space, and from the pia by the subarachnoid space, which is filled with cerebrospinal fluid. All the cerebral arteries and veins lie in this space

The arachnoid villi: it is arachnoid projection into the venous sinuses are most numerous along the superior sagittal sinus. Aggregations of arachnoid villi are referred to as arachnoid granulations. it serve as sites where the cerebrospinal fluid diffuses into the bloodstream.

Applied anatomy

Subarachnoid hemorrhage results from leakage or rupture of a congenital aneurysm on the circle of Willis

4) *Pia mater*

The pia mater is a vascular membrane that closely invests the brain, covering the gyri and descending into the deepest sulci. It extends over the cranial nerves and fuses with their epineurium. Pia matter + blood vessels + ependyma form choroid plexuses which project into ventricle and responsible for production of CSF.

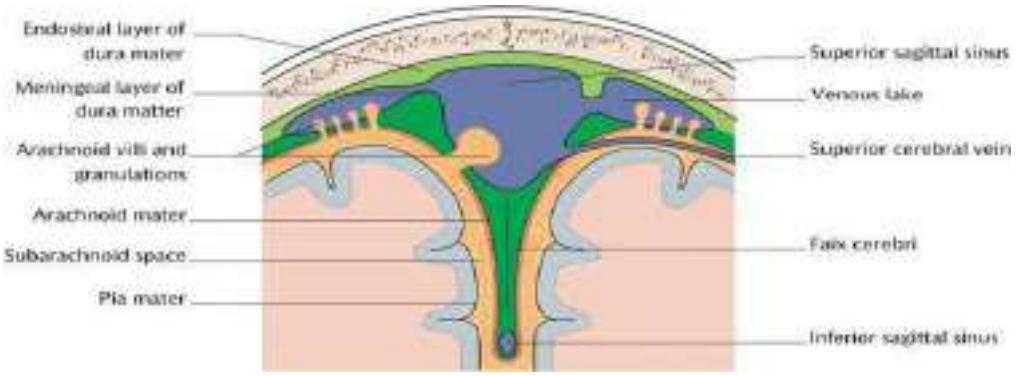


Figure (51): Dural, arachnoid, pia matter

5) *Cerebrum*

The cerebrum is the largest part of the brain and consists of two cerebral hemispheres connected by a mass of white matter called the corpus callosum**.** The hemispheres are separated by a deep cleft, the longitudinal fissure, into which projects the falx cerebri. The surface layer of each hemisphere is called the cortex and is composed of gray matter that through into groove called sulci separated by fold called gyri.

External features

Poles: frontal, temporal, occipital.

Surfaces: superior lateral, medial, inferior.

Lobes: frontal, parietal, temporal, occipital and insular.

Sucli and gyri of cerebral hemisphere

Sulci: main sulcus:

1-The central sulcusor the fissure of Rolando separating the parietal lobe from the frontal lobe 2- The lateral sulcus(also called Sylvian fissure) separates the frontal and parietal lobes from the temporal lobee, it divides into the anterior horizontal ramus and the anterior ascending ramus andcontinues as the posterior ramus.

Sucli and gyri of superior lateral surface

Frontal lobe:

sulci:
1- Theprecentral sulcus lies parallel to, and in front of, the central sulcus.
2- superior and inferior frontal sulcus

Gyri:
1- precentral gyrus :between central, precentral sulcus
2- superior frontal gyrus , the middle and inferior frontal gyrus: lie above and in between ,below superior and inferior frontal sulcus

Parietal lobe:

sulci:
1- postcentral sulcus
2- intrapariatal sulcus

Gyri:
1- postcentral gyrus: between central, postcentral sulcus
2- superior and inferior parietal lobules: above and below interparietal sulcus

Temporal lobe:

sulci: superior and inferior temporal sulcus

Gyri:superior ,middle and inferior temporal gyrus : above and in between ,below superiorand inferior temporal sulcus

occipital lobe:

sulci: 1- Lunatesulcus : semilunar sulcus at superiorlateral surface of occipital lobe 2- lateraloccipital:at lateral surface

Gyri: superior and inferior occipital gyrus above and below lateral occipital sulcus

N.B.

An area of cortex called the insula lies at the bottom of the deep lateral sulcus and cannot be seen from the surface unless the lips of the sulcus are separated

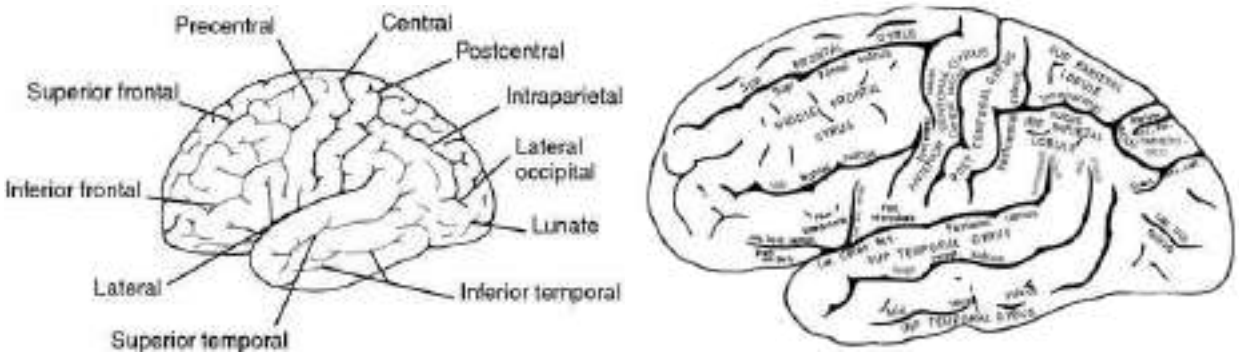


Figure (52): superior lateral surface

Medial Surfaces of the Hemisphere sulci and gyri

1- Thecorpus callosum: surface. which is the largest commissure of the brain, forms on this

2- The cingulate gyrus: begins beneath the anterior end of the corpus callosum and continues above the corpus callosum until it reaches its posterior end.

3- the callosal sulcus: separated the cingulate gyrus from the corpus callosum

4- cingulate sulcus: Y shape sulcus above corpus callosum

5- cingulate gyrus: between corpus callosum and cingulate sulcus

6- The paracentral lobule: between two limp of cingulate sulcus The anterior part of this lobule is a continuation of the precentral gyrus on the superior lateral surface, and the posterior part of the lobule is a continuation of the postcentral gyrus.

7- medial frontal gyrus: above stem of cingulate sulcus

8- parietooccipital sulcus: at occipital lobe

9- calcarine sulcus:atoccipital lobe

10- precunus: is anareaof cortex bounded anteriorly by posterior end of the cingulate sulcus and posteriorlybythe parieto-occipital sulcus.

11- The cuneus: is atriangular area of cortex bounded above by the parieto-occipital sulcus, inferiorly by thecalcarine sulcus

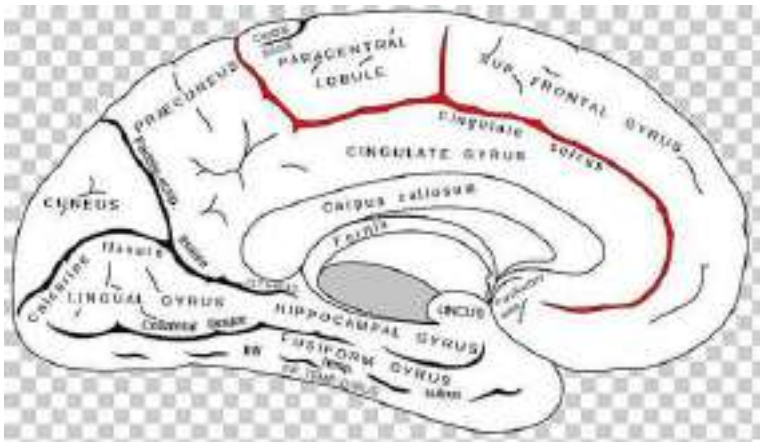


Figure (53): medial surface

Inferior Surfaces of the Hemisphere sulci and gyri

Tentorial part:

1- The collateral sulcus is situated on the inferior surface of the hemisphere.
2- the parahippocampal gyrus medial to The collateral sulcus
3- the occipitotemporal sulcus.
4- The medial and lateral occipitotemporal gyrus on each side of the occipitotemporal sulcus
5- rhinal sulci: above the collateral sulcus
6- uncus area medial to rhinal sulci

Orbital part :

On the inferior surface of the frontal lobe,
1- the olfactory sulcus
2- the olfactory bulb and tract Medial to the olfactory sulcus
3- orbital sulcus: H shape sulcus
4- anterior, lateral, medial, posterior orbital gyrus

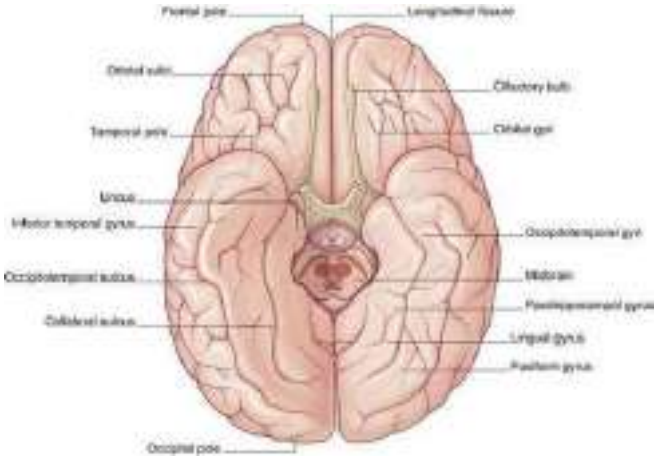


Figure (54): superior lateral surface

Cortical area

Frontal Lobe:

a-The precentral area: is situated in the precentral gyrus and the posterior parts of the superior, middle, and inferior frontal gyri. The precentral area may be divided into:

1- primary motor area 4, occupies the precentral gyrus extending over the superior border into the paracentral lobule. body represented in inverted manner

2- The premotor area 6, it e anterior to motor area in sup, middle, inferior frontal gyrus

3- The supplementary motor area is situated in the medial frontal gyrus on the medial surface of the hemisphere and anterior to the paracentral lobule.

4- The frontal eye field area 8 in the superior and the middle frontal gyrus

5- The motor speech area44 and 45 (Broca s area) : is located in the inferior frontal gyrus between the anterior and ascending rami and the ascending and posterior rami of the lateral fissure). It present only in dominant hemisphere.

6- The prefrontal cortex is an extensive area that lies anterior to the precentral area. It includes the greater parts of the superior ,middle, and inferior frontal gyri; the orbital gyri; most of the medial frontal gyrus; and the anterior half of the cingulate gyrus (Brodmann areas 9, 10, 11, and 12). Large numbers of afferent and efferent pathways connect the prefrontal area with other areas of the cerebral cortex, the thalamus, the hypothalamus, and the corpus striatum. The frontopontine fibers also connect this area to the cerebellum through the pontine nuclei. The commissural fibers of the forceps minor and genu of the corpus callosum unite these areas in both cerebral hemispheres.

b-The prefrontal area: present in superior, middle, inferior frontal gyrus orbital gyrus, medial frontal gyrus ,it is concerned with the makeup of the individual's personality.

Parietal lobe:

1- The primary somatic sensory area (Brodmann areas 3, 1, and 2) occupies the postcentral gyrus (on the lateral surface of the hemisphere and the posterior part of the paracentral lobule on the medial surface. body represented in inverted manner

2- secondary somatic sensory cortex S2 is in the superior lip of the posterior limb of the lateral fissure. The face area lies most anterior, and the leg area is posterior.

3- The parietal association area occupies the superior parietal lobule extending onto the medial surface of the hemisphere (areas 5 and 7). This area has many connections with other sensory areas of the cortex. It is believed that its main function is to receive and integrate different sensory modalities. For example, it enables one to recognize objects placed in the hand without the help of vision.

4- taste area (43): in superior lip of post.ramus of lateral sulcus

Occipital Lobe

1- The primary visual area (area 17) is situated in the walls of the posterior part of the calcarine sulcus and occasionally extends around the occipital pole onto the lateral surface of the hemisphere

2- The secondary visual area (areas 18 and 19) surrounds the primary visual area on the medial and lateral surfaces of the hemisphere

Temporal Lobe

1- The primary auditory area (areas 41 and 42) is situated in the inferior wall of the lateral sulcus

2- The secondary auditory area (auditory association cortex) is situated posterior to the primary auditory area in the lateral sulcus and in the superior temporal gyrus (area 22).

3- The sensory speech area of Wernicke (is localized in the left dominant hemisphere, mainly in the superior temporal gyrus, with extensions around the posterior end of the lateral sulcus into the parietal region.

4- The vestibular area is believed to be situated near the part of the postcentral gyrus concerned with sensations of
the face. Its location lies close to the auditory area in the superior temporal gyrus.

5- olfactory areas :1ry olfactory :uncus 2ry olfactory areas :para hippocampal gyrus.

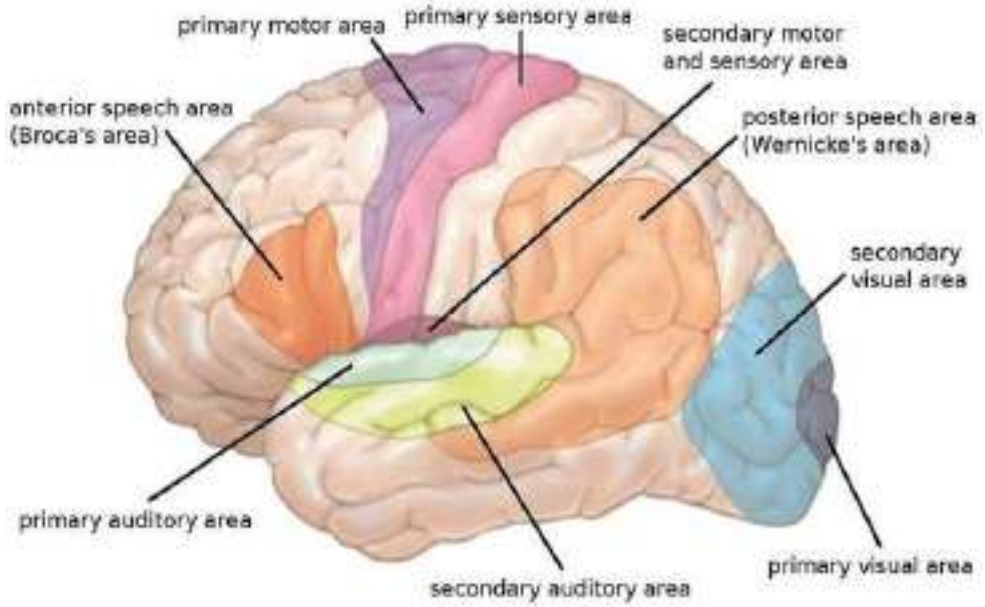


Figure (55): area in lateral surface

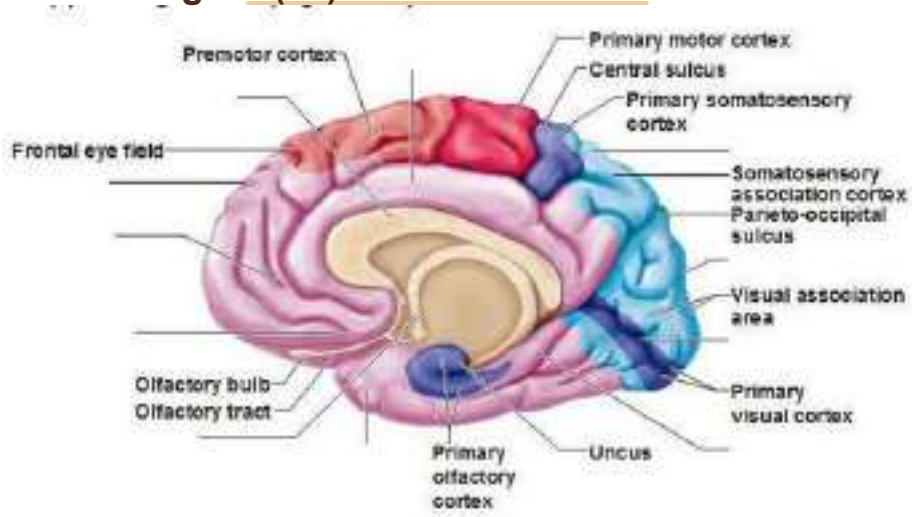


Figure (56): area in medial surface

Components of cerebral hemisphere

1- Outer layer

2- Inner white

3- Lateral of gray matter matter bentricel

4- Basal ganglia: collections of gray matter embedded in white matter

White matter fibers: are classified into three categories:

1- Projection fibers

2- Association fibers

3- Commissural fibers

Projections fibers

descending tact extend vertically between higher and lower brain areas and spinal cord centers, and carry information between the cerebrum to the rest of the body. ascending tracts carry signals upward to the cerebral cortex. This tract pass in collection called internal capsule

Association fibers

connect different regions within the same hemisphere of the brain. Long association fibers connect different lobes of a hemisphere to each other, whereas short association fibers connect different gyri within a single lobe.

Commissural fibers

cross from one cerebral hemisphere to the other. The great majority of commissural tracts pass through the large commissure called corpus callosum.

Corpus callosum

Is the largest commissure which divided into four parts: rostrum, genu, body/trunk and splenium **a- The rostrum:** is continuous with the lamina terminalis and connects the orbital surfaces of the frontal lobes. **b- The genu:** is the anterior part of corpus callosum and the forceps minor is a tract that projects fibres from the genu to connect the medial and lateral surfaces of the frontal lobes. **c- The body:** forms the long central section and its fibres pass through the corona

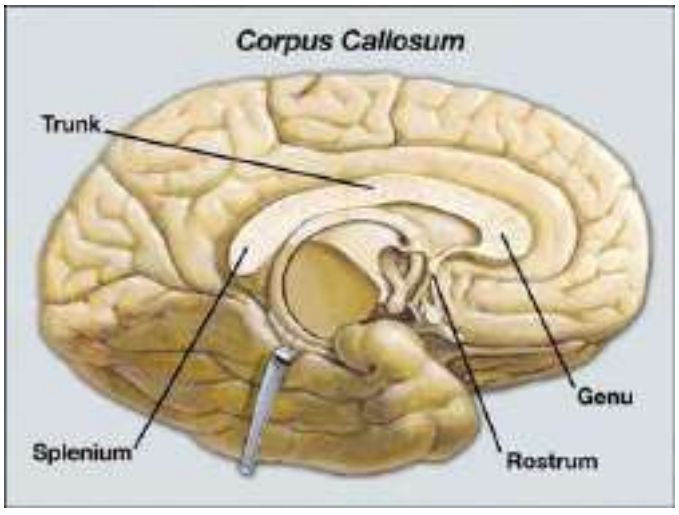


Figure (57): corpus callosum

radiata to reach the surface of the hemispheres.

d- The splenium: is the posterior part from it ,the forceps major projecting fibres from the splenium to connect the two occipital lobes.

Blood supply: callosal branch from anterior and posterior cerebral artery

Internal capsule

lamina of white fiber made up from projecting fiber.

Parts:

1- anterior limb: between head of caudate and lentiform nucleus

2- posterior limp: between thalamus and lentiform nucleus

3- genu: betweenanterior and posterior limp

4- retrolentiform part: behind lentiform

5- sub lentiform part: below lentiform

Blood supply: anterior limp supply by anterior ,middle cerebral artery Posterior limb, genu supply by middle cerebral artery Retro and sub lentiform supply by anterior choroidal branch of internal carotid artery

Applied anatomy

Injury in the blood supplyof internal capsule lead to contralateral hemiplegia due to affection of tract pass through it

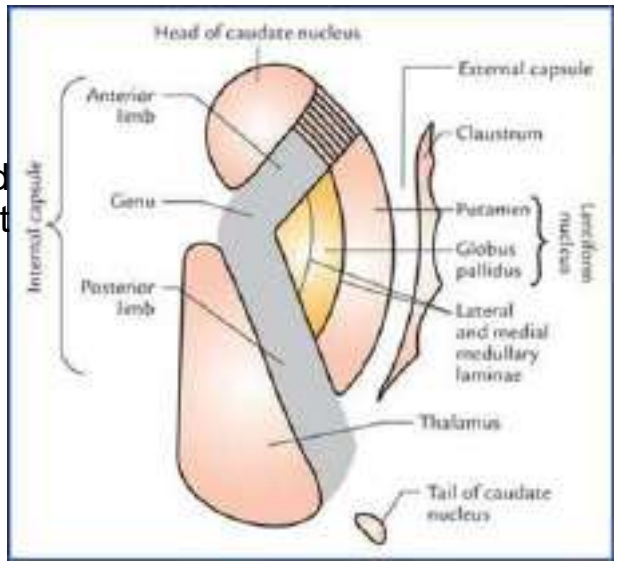


Figure (58): internal capsule

Basal ganglia

mass of gray matter present in cerebral hemisphere **Blood supply:** Central branches of

Anterior and middle cerebral artery ,vein drain into
thalamostriate vein

Function: help motor area in learning motor skill (writing alphabet)also prepare body posture to suit movement

Parts:

1- corpus Striatum:

a- the caudatenucleus

b- the lentiformnucleus.

2- amygdaloid nuclei

3- claustrum.

Caudate nucleus

The caudate nucleus is a large C-shaped mass of gray matter that is closely related to the lateral ventricle and lies lateral to the thalamus

Relation: The lateral surface: is related to the internal capsule, which separates it from the lentiform nucleus .

Parts:

1- head:
The head of the nucleus contributes to the lateral wall of the lateral ventricle.

2- body:
the caudate nucleus is long and narrow and is continuous with the head in the region of the interventricular foramen. The body of the caudate nucleus forms part of the floor of the body of the lateral ventricle.

3- tail: of the caudate nucleus is long and slender and is continuous with the body in the region of the posterior end of the thalamus. It follows the contour of the lateral ventricle and continues forward in the roof of the inferior horn of the lateral ventricle. It terminates anteriorly in the amygdaloid nucleus.

Lentiform nucleus

Itis looked by convex lens formed of two part:

Putamen: dark lateral part
Globus pallidus: paler medial part
Itpresents between two limps of internal capsule

Amygdaloid nucleus

The amygdaloid nucleus is situated in the temporal lobe close to the uncus at the end of the tail of the caudate.it also considered to be part of the limbic system

Claustrum

The claustrum is a thin sheet of gray matter that is separated from the lateral surface of the lentiform nucleus by the external capsule

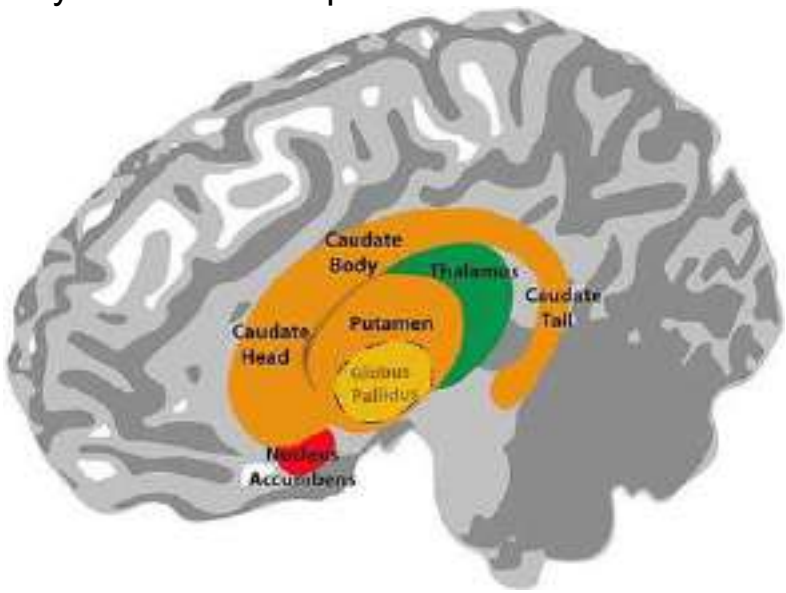
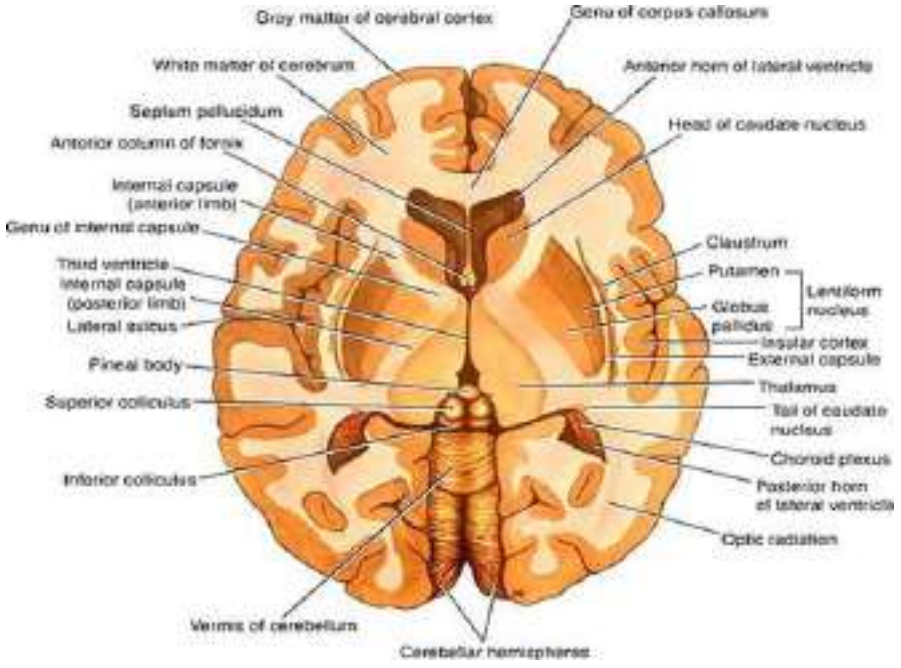


Figure (61): anatomy of basal ganglia

Applied anatomy

lesion in corpus straitum

Caudate nucleus: chorea the patient exhibits involuntary, quick, jerky, irregular movements that are nonrepetitive.
Putamen: athetosis (slow sinuous writing movement in distal part of limp



6) *Diencephalon*

Part between cerebral hemisphere and brain stem

1- Thalamus

2- Hypothalamus

3- Epithalamus

4- Subthalamus

5- Metathalamus

Thalamus

Thethalamus is a large, egg-shaped mass of gray matter that forms the major part of thediencephalon. There are two thalami one is situated on each side of the third ventricle **ends:**

1-Theanterior end of the thalamus is narrow and rounded and forms the posterior boundaryof the interventricular foramen.

2- The posterior end is expanded to form the pulvinar

Surfaces:

1- superior surface: form floor of the body of lateral ventricle 2- The inferior surface separate from subthalamus ,hypothalamus by hypothalamic sulcus

3- The medial surface of the thalamus forms part of the lateral wall of the third ventricle and is usually connected to the opposite thalamus by a band of gray matter called the inter thalamic adhesion.

4- lateral surface separate from lentiform nucleus by internal capsule

Function of thalamus:

1- integration and correlation between sensory activity . 2- perception of affective sensation such as crude touch ,pain, temperature 3- Anatomically and functionally, the thalamus and the cerebral cortex are closely linked. 4- The thalamus important for function of hypothalamus, limbic system, reticular formation.

Blood supply: posterior cerebral artery

Venous drainage: thalamostriate vein

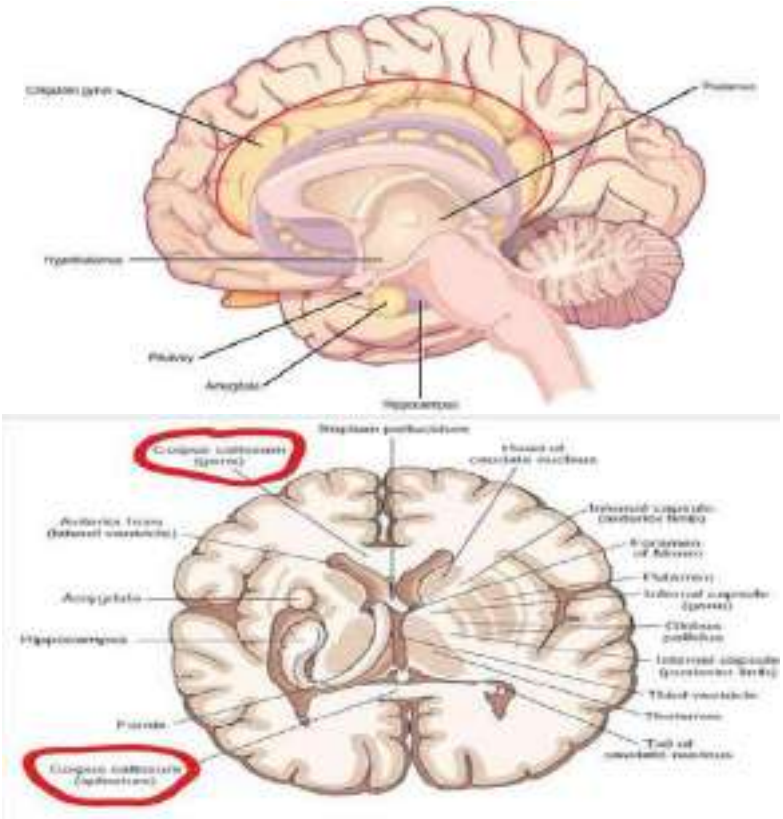


Figure (59): surfaces of thalamus

Applied anatomy

Lesions of the Thalamus These lesions usually result from thrombosis or hemorrhage of one of the arteries supplying the thalamus. will result in the loss of all forms of sensation, including light touch, tactile localization and discrimination, and muscle joint sense from the opposite side of the body.

Hypothalamus

Is the partofthediencephalon that lies below the thalamus**.**

Relations:

1- Posteriorly,thehypothalamus merges with the tegmentum of the midbrain. 2- Anteriorly,itextends up to the lamina terminalis, 3- Inferiorly,thehypothalamus is related to structures in the floor of the third ventricle.

Parts: thetubercinereum, the infundibulum, and the mammillary bodies, optic chiasma

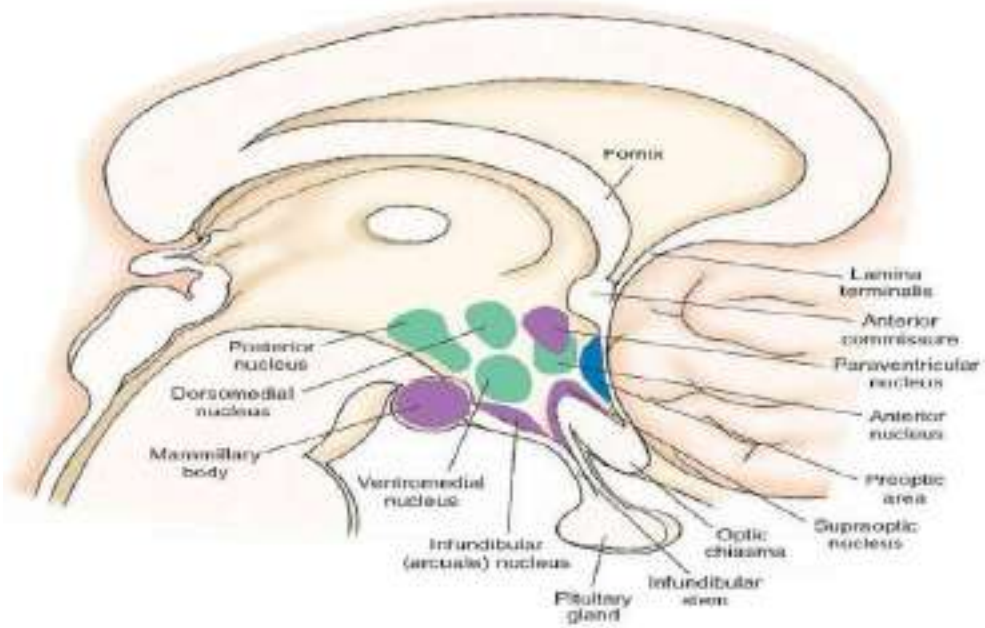


Figure (60): hypothalamus

Functions:

1- regulation of eating and drinking behaviour, 2- regulation of sexual activity and reproduction, and the control of autonomic activity. 3- regulation of emotional behaviour, response to stress, temperature regulation, as well as the control of circadian rhythms.

4- endocrine activities through relasing of pituitary stimulating and releasing hormones

Applied anatomy

Clinical Disorders Associated With Hypothalamic Lesions:

- Obesity and Wasting
- Sexual Disorders
- Hyperthermia and Hypothermia
- Disturbances of Sleep
- Diabetes Insipidus the patient passes large volumes of urine of low specific gravity
- Emotional Disorders

Metathalamus

Formed of: medial geniculate body: receive auditory impulse from inferior colliculus of mid brain through inferior brachium

Lateral geniculate body: receive visual impulse from optic tract

Connect to suprior colliculus of mid brain through superior brachium.

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7) *Limbic system*

also known as Papez'sCircuit, it group of curved structure seen in the medial aspect of cerebralhemisphereitisconcernedwith emotional andbehavioral expression.it include: **cortical components, subcortical components, white matter bundle.**

Cortical components

A- limbic lobe: it includes:

- Cingulate gyrus
- Parahippocampal gyrus (provides path for communication between cortical association areas and hippocampus)
- uncus-subcallosal area (paraolfactory area)

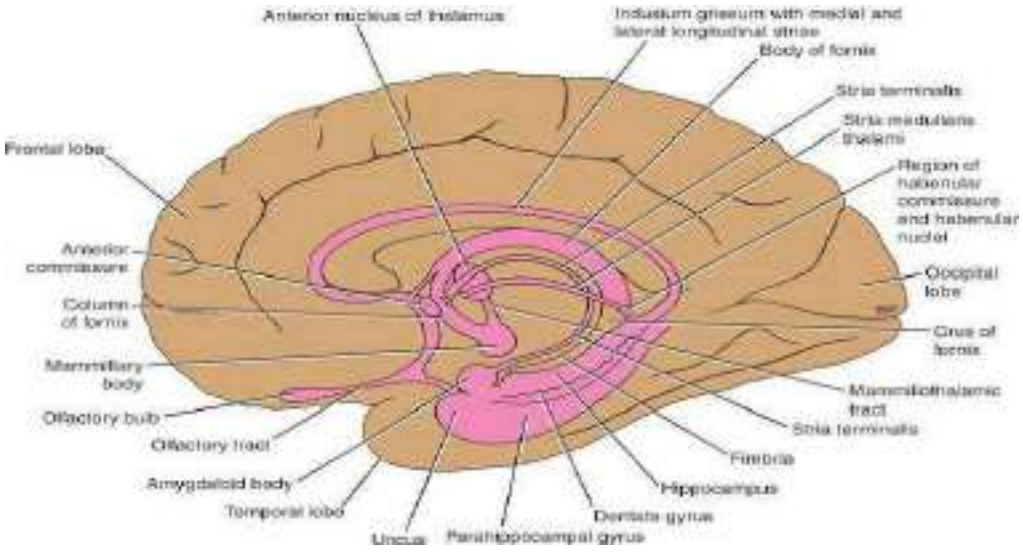
B-pre frontal cortex
C-Hippocampus
D-septal area: It'srelatively small gyrus that is found anterior to the lamina terminalis

Subcortical components

A- Amygdala
B- Olfactory bulb
C- Hypothalamus
D- Anterior and dorsomedial nuclei of the thalamus

Hippocampus

A- fornix
B- mammillothalamic tract
C- striasemicircularis



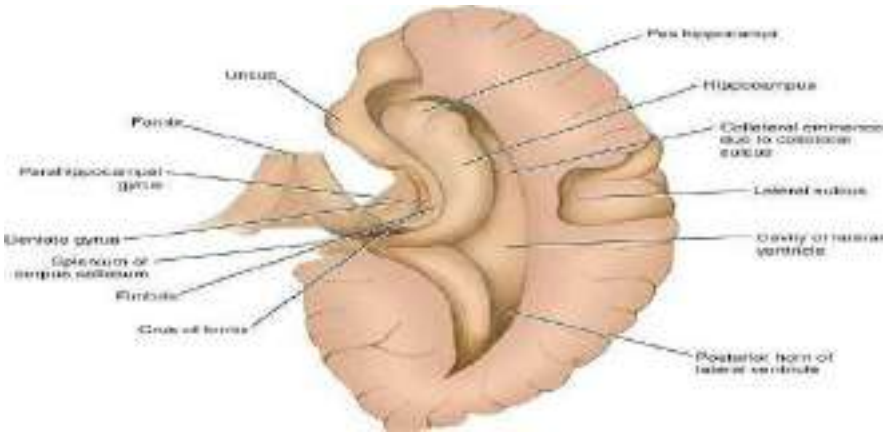


Figure (62): hippocampus and limbic system

Fornix

The fornix is a prominent bundle of white matter fibers seen on the medial aspects of the cerebral hemispheres**.** Parts:

1- crus(posterior column): It begins in the hippocampus as the alveus, which is a collection of myelinated fibers found medially to the floor of the temporal horn of the lateral ventricle**.** The fibers of the alveus travel posteromedially and aggregate to form the fimbria. The main bulk of fimbriae continues on each side of the cerebral hemisphere to form the crus of the fornix.

2- body: Both crura continue anteriorly and merge in the midline to form the body of the fornix(two crura connected with each other by the partially decussating fibers called the hippocampal commissure).The body of the fornix arches over the thalamus and under the septum pellucidum, which connects it to the corpus callosum.

3-column: The body of the fornix extends anteriorly and bifurcates into the right and left halves called the columns of the fornix. Each column turns downwards just in front of the interventricular foramen of Monro and passes through the hypothalamus to reach the mammillary body.

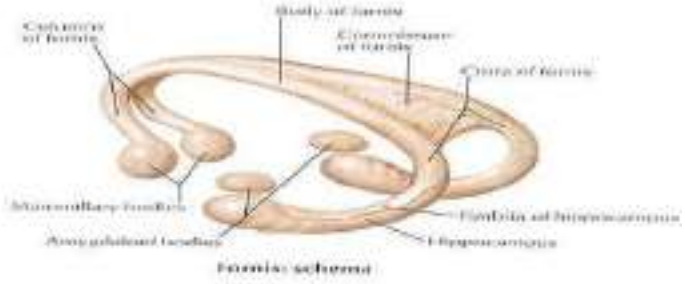


Figure (63): fornix

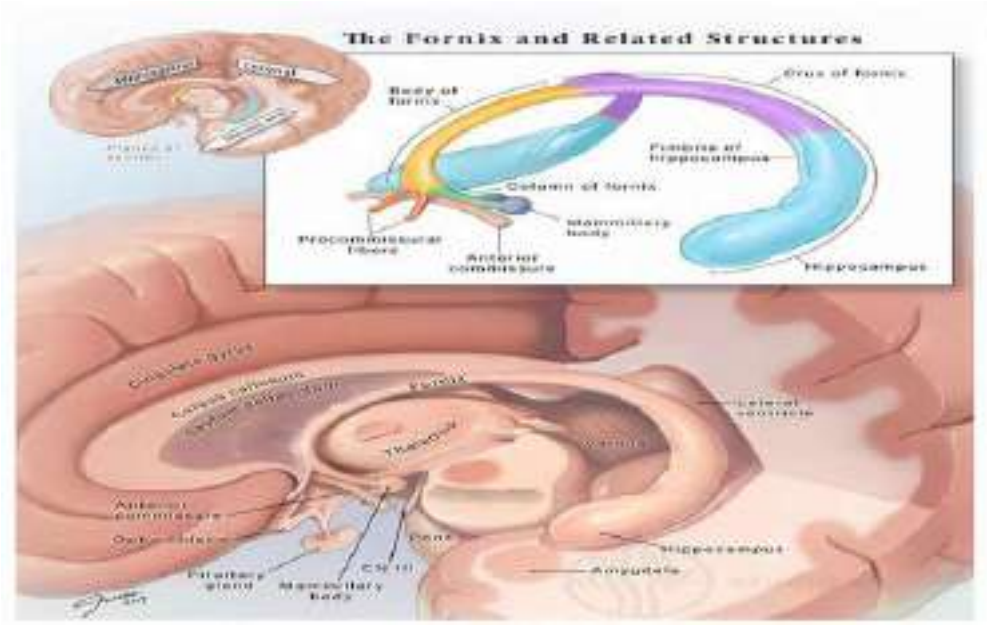


Figure (64): hippocampus and fornix

8) *Brain stem*

The brainstem (brain stem) is the distal part of the brain that is made up of the midbrain, pons, and medulla oblongata.

9) *Mid brain*

The midbrain is the narrow part of the brain that passes through the tentorial notch and connects the forebrain to the hindbrain

External features

anterior surface: The cerebral peduncles two lateral halves separate by interpeduncular fossa each peduncles is divided into an anterior part(the crus cerebri) and a posterior part (the Tegmentum) separated by a pigmented band of gray matter called the substantia nigra. The narrow cavity in the midbrain is called the cerebral aqueduct, which connects the third and fourth ventricles. Divide mid brain into anterior and posterior part

Posterior surface:

The tectum is the part of the midbrain posterior to the cerebral aqueduct; it has four small surface swellings called two superior and two inferior colliculi.

a-The superior colliculi are centers for visual reflexes

b-The inferior colliculi are lower auditory centers. In the midline below the inferior colliculi.

lateral surface: On the lateral aspect of the midbrain, the superior brachium passes from the superior colliculus to the lateral geniculate body and the optic tract. The inferior brachium connects the inferior colliculus to the medialgeniculate body.

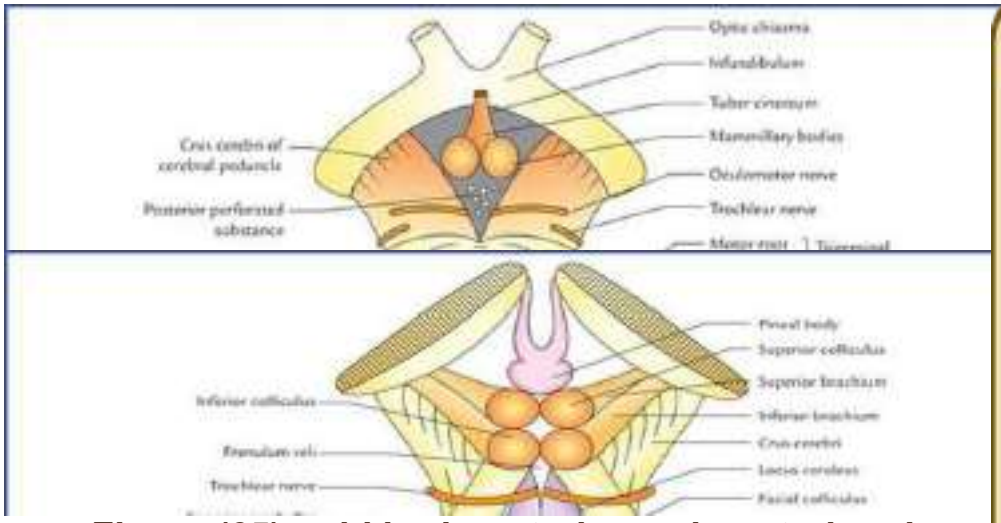


Figure (65): mid brain anterior and posterior view

N.B.

oculomotor nerve merge from groove in the medial aspect of cru s cerebri. Trochlear nerve leaves the midbrain just below inferior colliculi and then pass laterally and posteriorly around to cerebral peduncle ventral appear in aspect.

Blood supply: Basilar artery ,posterior communicating artery.

10) *Pons*

The pons is the middlepart of brainstem and situated between the midbrain and the medulla.

External features

Ventral (anterior) surface :

Ventral surface is convex in both the directions, It is transversely striated due to underlying ponto-cerebellar fibres.

In the median plane, it presents a vertical groove, the basilar groove, which lodges the basilar artery.

The trigeminal nerve is attached to this surface by two roots: a small motor and a large sensory root near upper border of pons

the abducent (VI) nerve appear in the pontomedullary junction facial (VII), nervus intermedius, and vestibulocochlear (VIII) nerves emerge at pontocerebellar junction

Dorsal (posterior) surface :

The dorsal surface of the pons is covered by the cerebellum, and separated from it by the cavity of the fourth ventricle. The dorsal surface of the pons is triangular in shape and forms the upper part of the floor of the fourth ventricle contain (medial longitoudinal bundle ,medial eminence produce by abducent nucleus . fascial coliculus. medullary stria separate pons from medulla

Superior border

Thecerebralpeduncles are attached to this border.

Inferior border

Theupperendof the medulla is continuouswith this border.

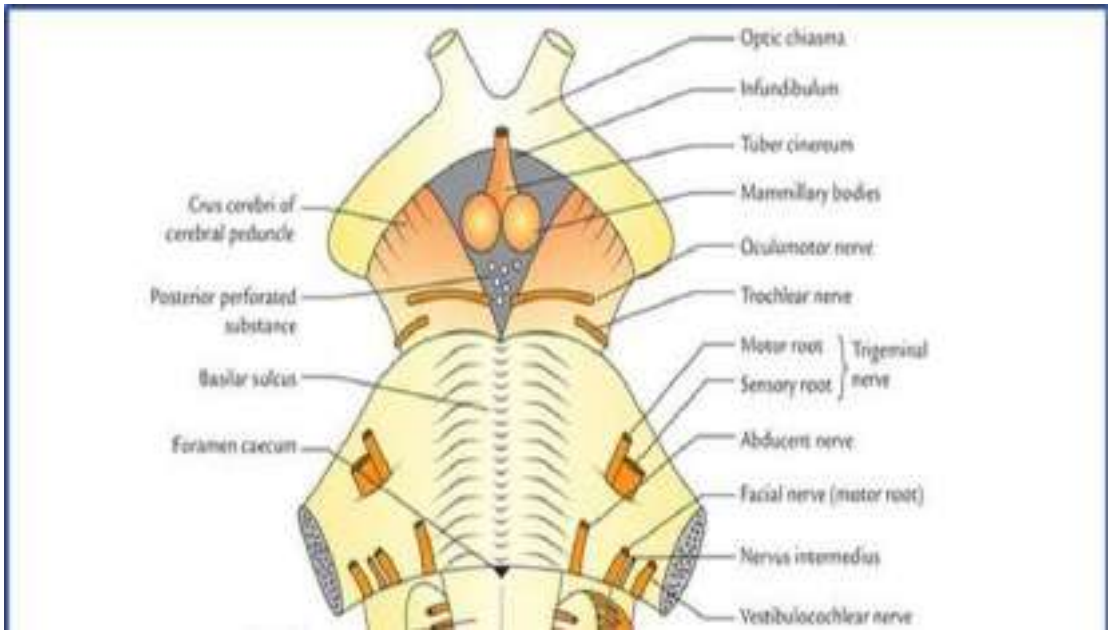


Figure (67): anterior view to the pons

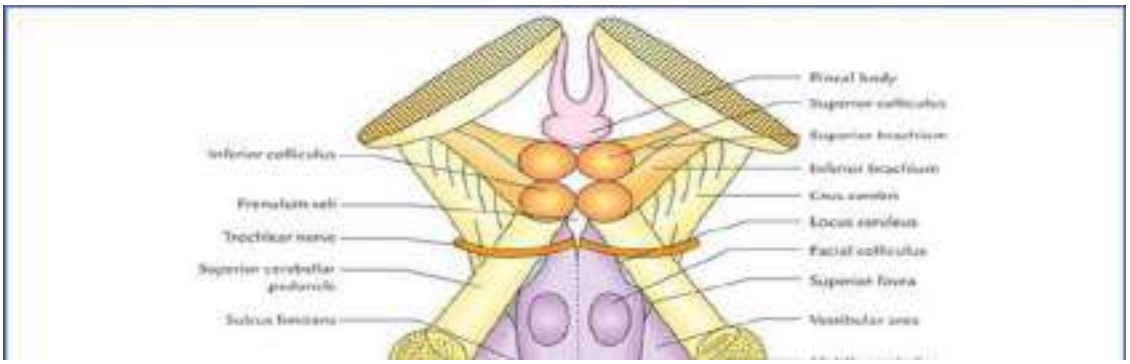


Figure (68): posterior view to the pons

Blood supply: Numerous (pontine) branches from the basilar artery. Anterior inferior cerebellar artery.

11) *Medulla oblongata*

The medulla oblongata is the direct upward continuation of the spinal cord, extending from foramen magnum to the lower border of the pons. It forms the lowest part of the brainstem

Parts:
1- open medulla :form floor of 4th ventricle
2- closed medulla contain central canal continuous with 4th ventricle

External features

1-anterior lateral surface:

- The anterior median fissure is continuous below with the corresponding fissure on the spinal cord.
- The anterolateral sulcus extends along the lateral border of the pyramid and along it emerges the rootlets of the hypoglossal (XII cranial) nerve.
- The posterolateral sulcus lies between the olive and the inferior cerebellar peduncle and along it emerges the rootlets of glossopharyngeal (IX), vagus (X) and accessory (XI) cranial nerves.
- Pyramids These are two elongated elevations, one on either side of anterior median fissure are produced by the corticospinal (pyramidal) fibres.
- Olives These are oval elevations, posterolateral to the pyramids and are produced by an underlying mass of grey matter called inferior olivary nucleus.
- Rootlets of the hypoglossal nerve . These emerge from the anterolateral sulcus between the pyramid and the olive.

- Inferior cerebellar peduncles. These are thick bundles of fibres lying posterolateral to the olive, and attach the medulla with the cerebellum.
- Lateral to olive are rootles glossopharyngeal, vagus, accessory nerve

2-posterior surface:

Featureof closed part: The posterior median sulcus continues below with the corresponding sulcus of the cord and is present only in the lower half of the medulla. on each side of posterior median sulcus the gracile and cuneate tubercles inferior cerebellar peduncle

Features of the open part: The open part of the medulla forms the lower part of the floor of fourth ventricle, which presents number of features like, median sulcus, hypoglossal and vagal triangles, vestibular areas, , stria medullaris,. For details see floor of fourth ventricle .

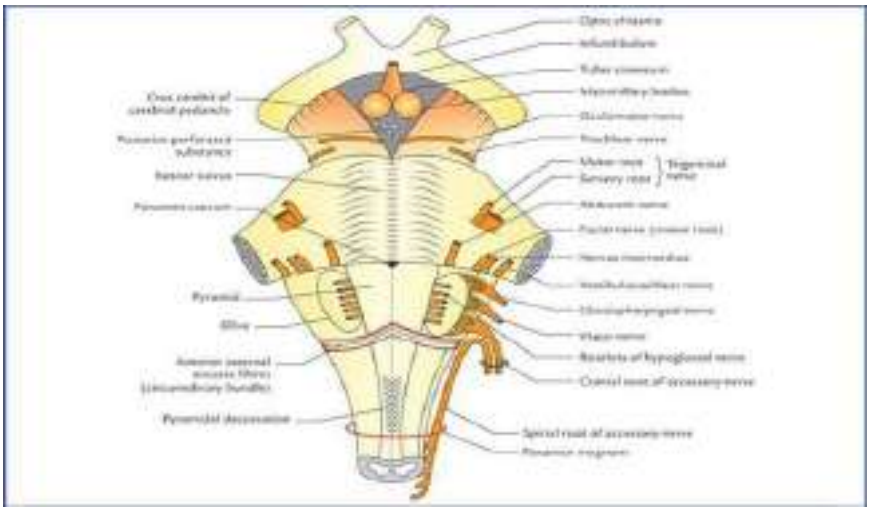


Figure (69): anterior view to brain stem(medulla oblongata)

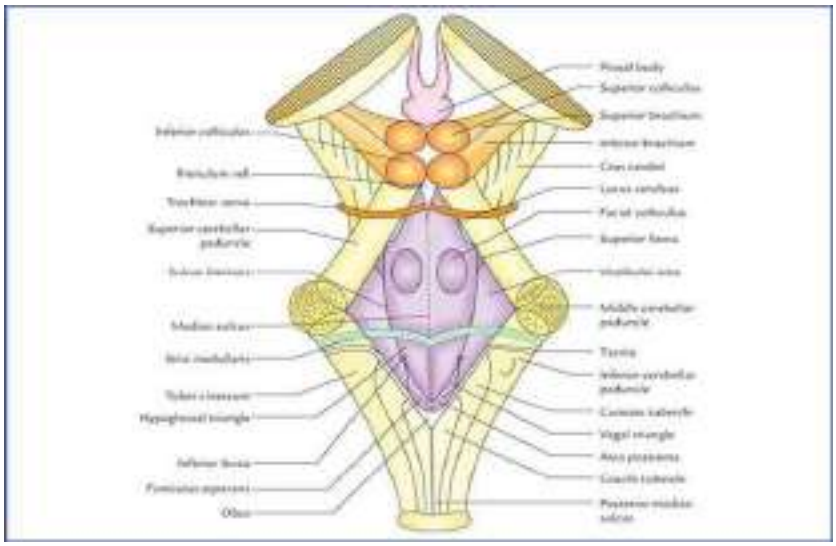


Figure (68): posterior view to the brain stem(medulla oblongata)

In the groove between the pons and medulla oblongata merge :from medial to lateral abducent ,facial, vestibulocochlear nerves.

Blood Supply of the Medulla

- Two vertebral arteries.
- Anterior and posterior spinal arteries.
- Anterior and posterior inferior cerebellar arteries.
- Basilar artery.

Ventricular system

The ventricles are four fluid-filled cavities located within the brain; these are the two lateral ventricles, the third ventricle, and the fourth ventricle.

Lateral ventricles

- There are two lateral ventricles one in each cerebral hemisphere. Each lateral ventricle is a roughly C-shaped cavity situated within each cerebral hemisphere.
- The lateral ventricle wraps itself around the thalamus, the lentiform nucleus, and the caudate nucleus.
- It is lined with ependyma and filled with cerebrospinal fluid.
- The main parts of two ventricles are separated from each other by a septum extending between corpus callosum and fornix called **septum pellucidum**.
- Each lateral ventricle communicates with the third ventricle through the interventricular foramen (of Monro). Most of the CSF in the CNS is produced by the choroid plexuses of two lateral ventricles.

Parts of Lateral Ventricle: anterior horn ,body, posterior horn ,inferior horn

Anterior (frontal) horn

Extend: infrontallobe

Boundaries:

The roof: isformed by the undersurfaceof the anterior part of the body of corpus callosum.

Floor: isformed by the upper surface oftherostrum of corpus callosum.

The anterior wall: is formed by the genuofthe corpus callosum.

The medial wall: is formed by the septumpellucidum.

The lateral wall: is formed by the bulginghead of the caudate nucleus.

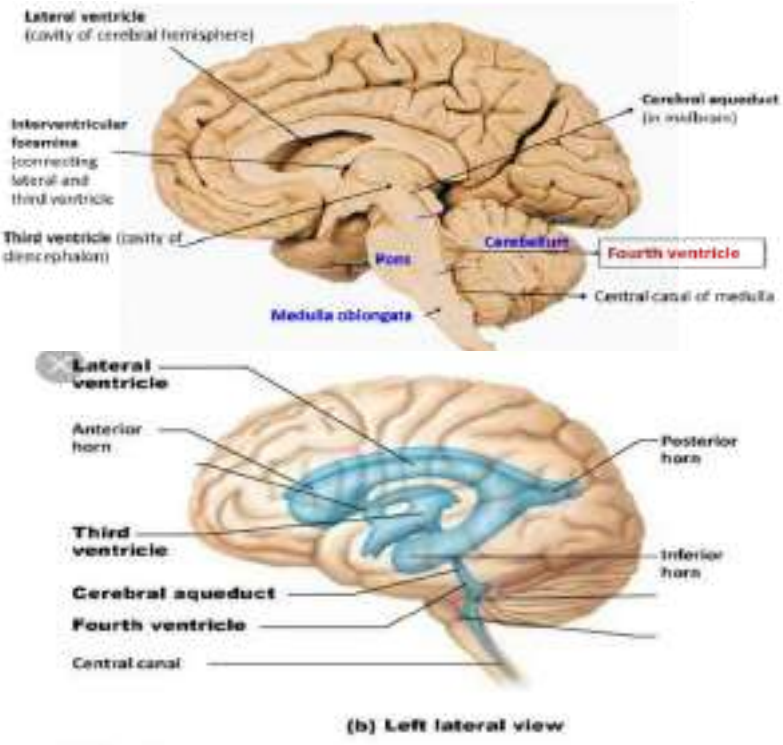


Figure (72): lateral ventricle

Central part of the lateral ventricle

Extend in parietal lobe. It appeared triangle in the cross cection.

The roof is formed by the under surface of the body (trunk) of the corpus callosum.

The floor is formed in that order by:

- Body of caudate nucleus.
- Stria terminalis and thalamostriate vein.
- Lateral part of the upper surface of the thalamus.
- Choroid plexus covering the medial part of the upper surface of the thalamus,Upper surface of the body of fornix.

The medial wall is formed by the septum pellucidum

Posterior part of the lateral ventricle

The roof, lateral wall, and floor are formed by a sheet of fibers (tapetum) from the splenium of the corpus callosum.

The medial wall is invaginated by, the bulb of posterior horn is formed by the fibres of forceps major and the lower one the calcar Avis is produced by the anterior part of the calcarine sulcus.

Inferior part of the lateral ventricle

Itrun in the temporal lobe.

The roof bounded by the tail of caudate nucleus and stria terminalis.

The floor presents following features from lateral to medial side:

- Collateral eminence, an elongated swelling in the lateral part of the floor, produced by the collateral sulcus
- Hippocampus, an another longitudinal elevation lying parallel and medial to the collateral eminence.
- fimbria converging fibres of alveus

The lateral part is formed by tapetum of corpus callosum

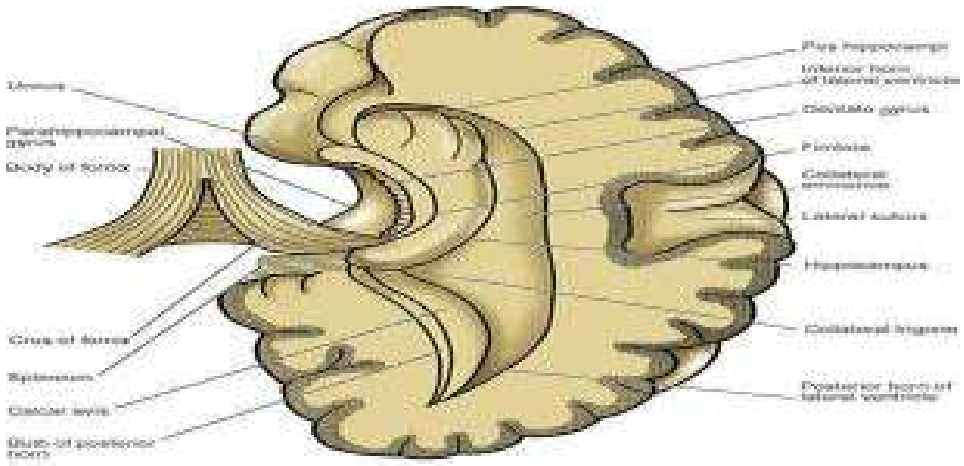


Figure (73): posterior and inferior horn of lateral ventricle

Chroid plexus

collection of blood vessels covered by pia matter and ependyma The choroid plexus of lateral ventricle is derived from anterior choroidal artery, a branch of internal carotid artery and the posterior choroidal artery a branch of posterior cerebral artery.

Third ventricle

The third ventricle is the cavity of diencephalon. It is a midline slit-like cavity situated between the two thalami and the part of hypothalamus.

Boundaries:

Anterior wall is formed from above downwards by: anteriorcolumn of fornix, anterior commissure, and lamina terminalis. pineal gland, posterior

Posterior wall is formed from above downwards by:

commissure, and commencement of cerebral aqueduct.

Roof: isformedbytheependymathatstretches acrosstheupper limits of two thalami.

the roofextendsfrominterventricularforamento thehabenular commissure.

Floor is formed frombefore backwards by:

optic chiasma,tubercinereumandinfundibulum (pituitarystalk), mammillary bodies

Communications:

1- communicates with the lateral ventricle through interventricular foramen (ofMonro),
2- with the fourth ventricle through cerebral aqueduct (of Sylvius)

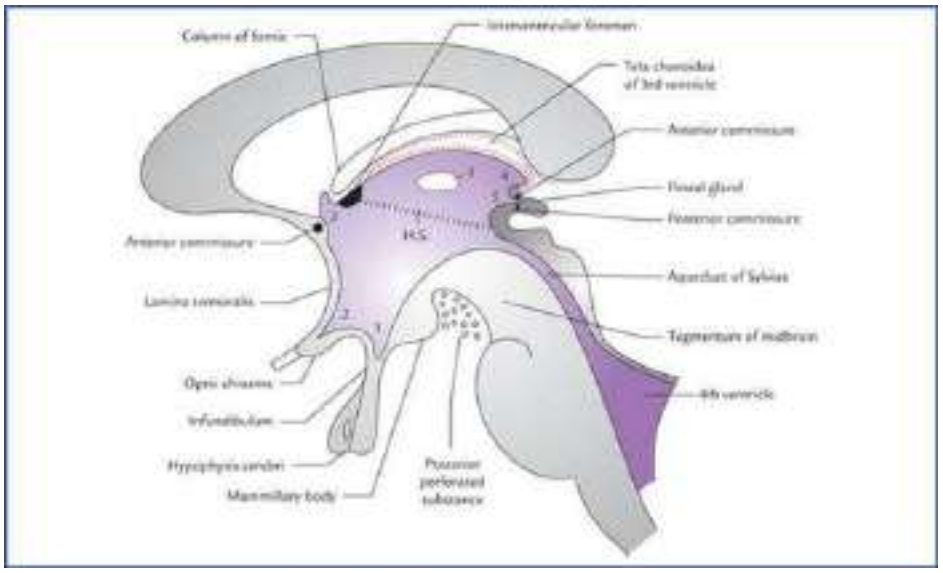


Figure (74): third ventricle

Choroid Plexus and Tela Choroidea of the Third Ventricle:

The tela choroidea in the roof of the third ventricle is triangular in shape. The choroid plexus of the third ventricle hangs downwards from the tela as two longitudinal anteroposterior vascular fringes. The blood vessels contributing to its formation are derived from the anterior and posterior choroidal arteries.

Fourth ventricle

The fourth ventricle is a tent-like cavity of hindbrain lined with ependyma and filled with cerebrospinal fluid. It is situated in the posterior cranial fossa in front of the cerebellum and behind the pons and the upper part of medulla oblongata.

Boundaries:

Angles of the fourth ventricle:

- The superior angle is continuous above with the cerebral aqueduct of midbrain.
- The inferior angle is continuous below with the central canal of the closed part of the medulla oblongata.
- The two lateral angles, one on each side is carried outwards across the dorsal surface of the inferior cerebellar peduncle as tubular pouches called lateral recess each lateral recess open in subarachnoid space by lateral aperture called (foramen of Luschka for).

Lateral boundaries (lateral walls): On each side, the fourth ventricle is bounded, inferolaterally by inferior cerebellar peduncle, superolaterally, by superior cerebellar peduncle.

Roof (posterior wall): the upper part of the roof is formed by the convergence of two superior cerebellar peduncles and a thin sheet of white matter(the superior medullary velum) that bridges the triangular gap between the two superior cerebellar peduncles.
The lower part of the roof is formed by a thin sheet of non-nervous tissue, the inferior medullary velum which is formed conjointly by the ventricular ependyma and the pia mater (of tela choroidea) that form choroid plexuses of 4th ventricle.
The lower part of the inferior medullary velum has a large aperture in the median plane, the foramen of Magendie through which the cavity of fourth ventricle communicates with the subarachnoid space

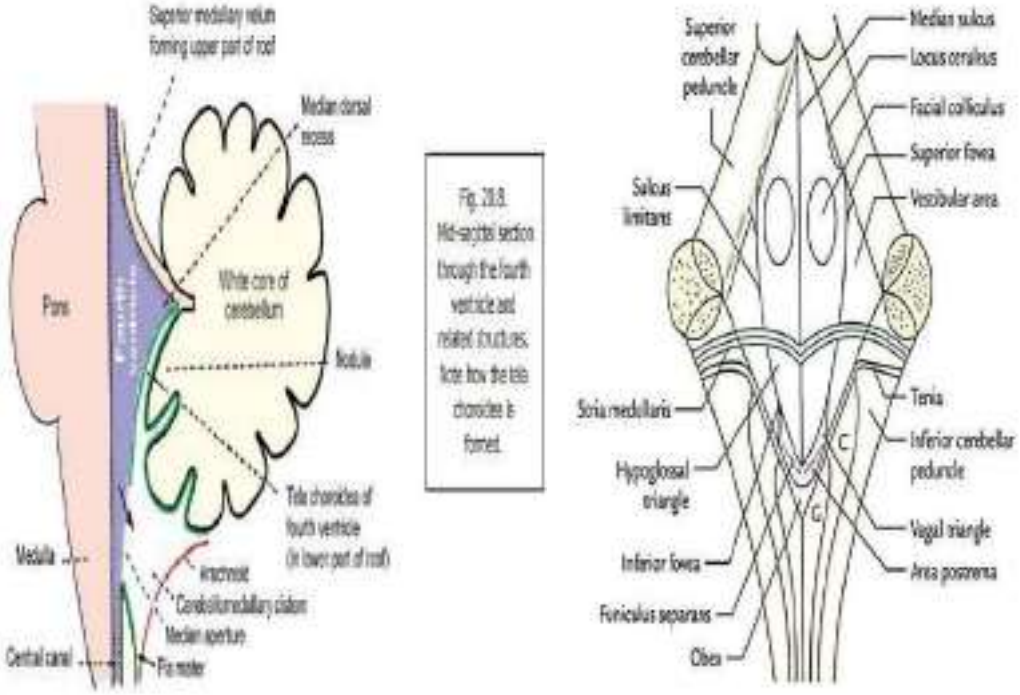


Figure (75): 4th ventricle

Floor (rhomboid fossa):

The floor of the fourth ventricle is formed by posterior surfaces of the pons and the upper part of the medulla. It is rhomboid in shape (diamond-shaped) It is divided into two parts separated by delicate bundles of transversely arranged fibres called the striae medullaris.

- The upper triangular part is formed by the posterior surface of the pons.
- median sulcus**.** The entire floor is divided into right and left symmetrical halves by a
- The facial colliculus On either side, the medial eminence shows an oval swelling is produced by the fibres from the motor nucleus of facial nerve hooking around the abducent nucleus
- superior fovea a triangular depression in the upper end of sulcus limitans
- locus ceruleus a bluish grey area Above the superior fovea the sulcus limitans. contain large quantities of norepinephrine (noradrenaline).
- vestibular area The region, lateral to sulcus limitans overlies the vestib-ular nuclei. This area lies partly in the pons and partly in the medulla oblongata.
- The lower triangular part is formed by the upper part of the posterior surface of the medulla.
- on each side of median sulcus a small depression called inferior fovea.
- From inferior fovea divided into three triangles, the hypoglossal triangle above and the vagal triangle below ,vestibular triangle.

1- The hypoglossal triangle overlies the nucleus of hypoglossal nerve

2- The vagal triangle overlies the nuclei of vagus, glossopha-ryngeal and cranial accessory nerves.

Choroid plexuses of the fourth ventricle

The capillary plexus of blood vessels between the two layers of tela choroidea (blood vessels + pia matter) form the rich vascular fringe that projects through the lower part of the roof of the fourth ventricle to form the choroid plexus. It is derived from the branches of posterior inferior cerebellar arteries.

Cerebrospinal circulation

Cerebrospinal Fluid

- Cerebrospinal fluid is clear, colourless and slightly alkaline fluid with a specific gravity of 1005-1008.It is present in the ventricular system within the CNS and in the subarachnoid space surrounding the CNS. bout 80 to 90% of the CSF is produced by the choroid plexuses within the lateral ventricles, with remaining being produced by the choroid plexuses in the third and fourth ventricles.

Circulation and Absorption

- The CSF is produced mainly in the lateral ventricles, from where it passes through the interventricular foramina (of Monro) into the third ventricle,
- then via cerebral aqueduct into the fourth ventricle.
- Here the fluid escapes via the median aperture (foramen of Magendie) and lateral apertures (foramina of Luschka) in the roof of 4thventricle, into the subarachnoid space over the brain and spinal cord. Most of the CSF flows upwards through the gap in the tentorium cerebelli and then forwards and laterally over the inferior surface of the cerebrum.
- Finally, it ascends on the superolateral aspect of each cerebral hemisphere to reach the arachnoid villi and granulations which penetrate into the superior sagittal sinus. The CSF enters into the blood stream of the sinus through the mesothelial cell lining of these villi and granulations.
- Some of the CSF moves inferiorly in the subarachnoid space around the spinal cord and cauda equina.
- The flow of CSF is facilitated by the pulsations of cerebral and spinal arteries present in the subarachnoid space, and the movements of the head and spine.

Functions of CSF:

- It serves as a cushion between the CNS and surrounding bones.
- CSF acts as a shock absorber .
- It supports the brain and spinal cord, fluid.
- It nourishes the CNS.

- CSF removes the metabolites (waste products) from the CNS.
- It serves as a pathway for pineal secretions to reach the pituitary gland.

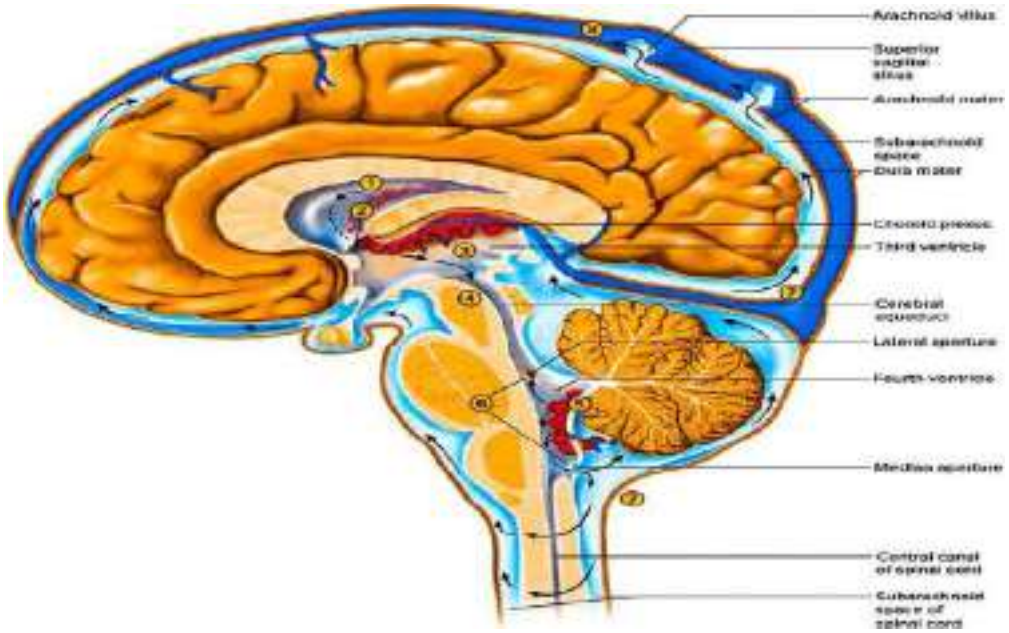


Figure (76): cerebrospinal circulation

Blood Supply of the Brain

- **Arteries of The Brain:**
- The brain is supplied by the paired internal carotid and vertebral arteries

The vertebral artery:

- Cerebral branches:
  - o meningeal
  - o posterior inferior cerebellar
  - o anterior ,posterior spinal artery
  - o Medullary arteries

- Basilarartery :
  - Basilar artery is formed by the union of two vertebral arteries at the lower border of the pons. It ascends in the basilar sulcus on the ventral aspect of the pons at the upper border of the pons by dividing into right and left posterior cerebral arteries.
  - Branches of basilar artery:

- Pontine branches are numerous short slender parame-dian vessels which pierce the pons to supply it.
- Anterior inferior cerebellar artery arises close to the lower border of the pons
- Labyrinthine artery is supply the internal ear.
- Superior cerebellar artery arises close to the superior border of the
- pons,
  Posterior cerebral artery

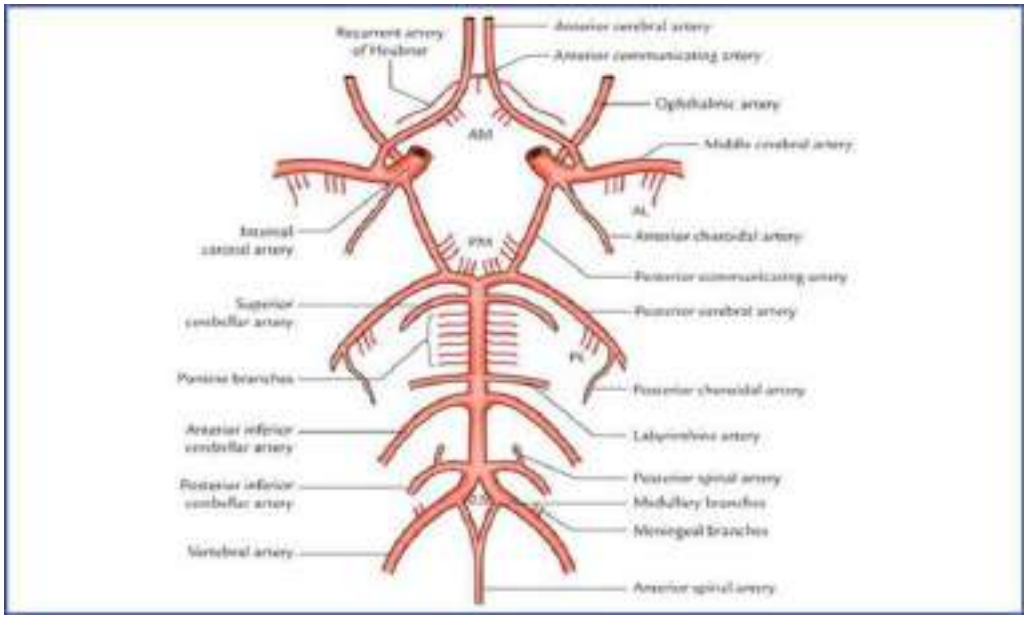


Figure (69): bloodsupply of brain

Internal Carotid Artery

- Theinternal carotid artery begins at the bifurcation of the common carotid artery
- Branches of the Cerebral Portion:
  - The ophthalmic artery arises as the internal carotid artery emerges from the cavernous sinus. It enters the orbit through
  - the optic
  - The posterior communicating artery
  - The choroidal artery,
  - The anterior cerebral artery is the smaller terminal branch of the internal carotid artery
  - middle cerebral artery

Circle of Willis (circulus arteriosus):

- **Definition**: interconnection between The major arteries supplying the cerebrum
- **Site**: The circle of Willis is formed around the interpeduncular fossa and lies in the interpeduncular subarachnoid cistern.
- **Formation**: Anteriorly, by the anterior communicating connecting two anterior cerebral arteries.
- **Posteriorly**, by the basilar artery dividing into two posterior cerebral arteries.
- **Laterally** on each side, by the posterior communicating artery connecting the internal carotid artery with the posterior cerebral artery.

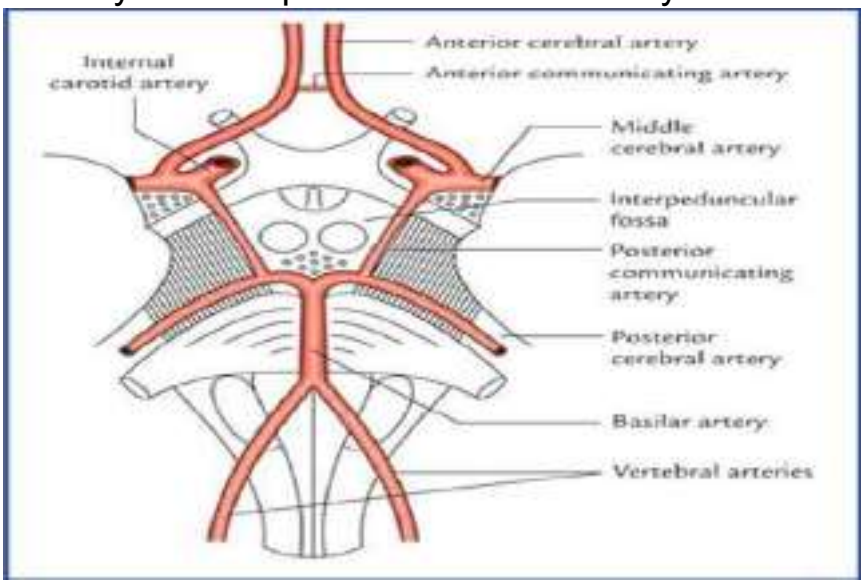


Figure (70): circle of Willis

Anterior cerebral artery:

- **Origin: terminal branch of the internal carotid artery.**

❑ Branches:

1) Cortical	2) central
• medial surface except occipital lobe • superior one inch of lateral surface • medial part of orbital surface	• Anterior part of caudate nucleus ,lentiform. and anterior half of anterior limb of internal capsule

middle cerebral artery:

- **origin: large terminal branch of internal carotid**
- branches:

2) Cortical	2) central
• Lateral surface (except upper and lower one inch, occipital lobe) • Inferior surface(lateral part of orbital surface, temporal pole)	• Post part of caudate ,lentiform nucleus and posterior half of anterior limb and posterior limp of internal capsule and genu

posterior cerebral artery:

- **origin: terminal branch of basilar artery**
- branches:

Cortical	central	Posterior choroidal a
• Lateral surface (lower one inch, occipital lobe) • Inferior surface(tentorial surface except temporal pole) • Medial surface(occipital lobe)	• Choroid plexuses of lateral and third ventricle	• Choroid plexuses of lateral and third ventricle

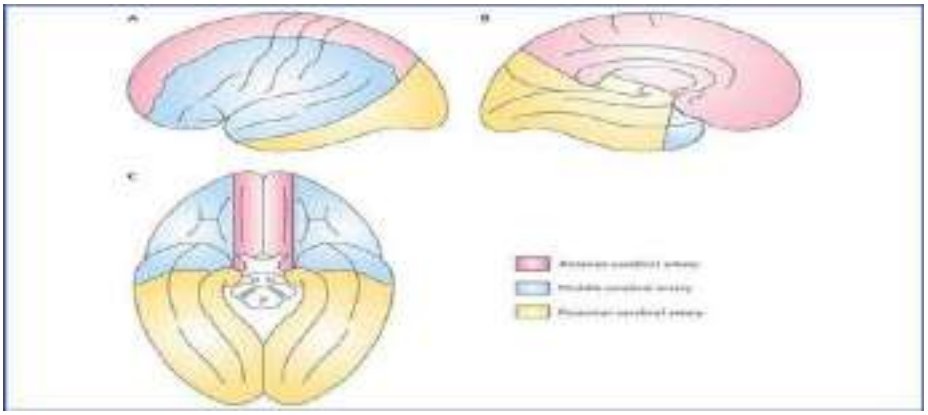


Figure (71): cerebral arteries

Veins of cerebrum:

Superficial veins of the cerebrum

- **Superior cerebral veins:**
  - draintheupper partsof thesuperolateral and medial surfaces of the cerebral hemisphere. They drain blood into the superior sagittal sinus
- **Inferior cerebral veins:**
  - return blood from the inferior aspect of the cerebral hemisphere into the transverse, superior petrosal, cavernous and sphenoparietal sinuses. Some also drain into the inferior sagittal sinus.
- **Superficial middle cerebral vein:**
  - receives blood from veins on the superolateral surface, as it lies superficially along the lateral sulcus. The posterior end of this vein is connected to the superior sagittal sinus by the superior anastomotic vein and to the transverse sinus by the inferior anastomotic vein. It terminates in the cavernous sinus.

Deep veins of the cerebrum

- **The Internal Cerebral Veins (veins of Galen; deep cerebral veins)**
  - drain the deep partsof thehemisphereandaretwointheyunitetoformashort trunk, the great cerebral vein
- **The Great Cerebral Vein (great vein of Galen)**
  - formedby the unionof thetwointernalcerebral veins end in the straight sinus.
- **The basal vein**
  - isformed atthe anterior perforated substance passes backward around the cerebral peduncle of midbrain, and ends in the internal cerebral vein

CEREBELLUM

- Is the largest part of the hindbrain and second largest part of the brain as a whole. It is located in the posterior cranial fossa underneath the tentorium cerebelli and behind the pons and medulla oblongata. It is separated from the pons and medulla by a cavity of the fourth ventricle

External Features

- The external features of the cerebellum comprise three parts, two surfaces, two notches, and three well-marked fissures.

❑ **Parts:**

o 1-two large lateral hemispherical lobes
o 2-called vermis. Connect two cerebellar hemisphere

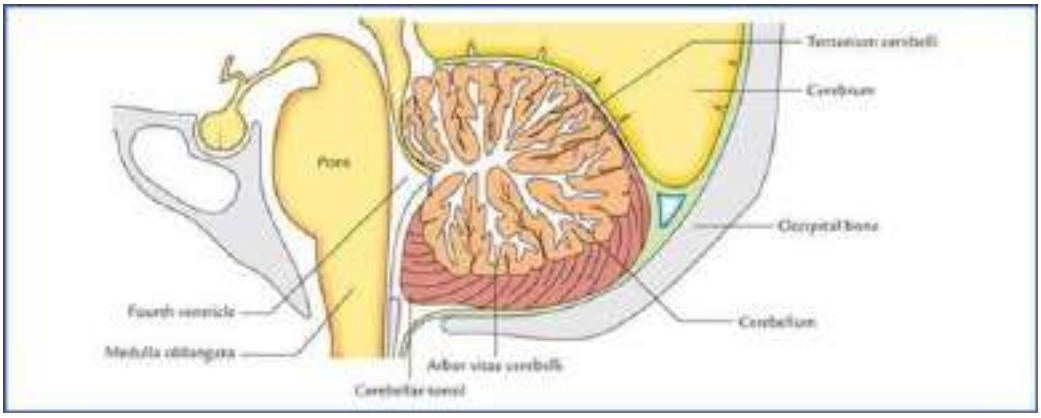


Figure (77): cerebellum

Surfaces:

- The superior surface of the cerebellum is convex. The two cerebellar hemispheres and ver
- mis are continuous with each other on this surface. 2-The inferior surface presents a deep median notch called vallecula which separates the two cerebellar hemispheres. The floor of the vallecula is formed by inferior vermis and is limited on each side by sulcus valleculae.

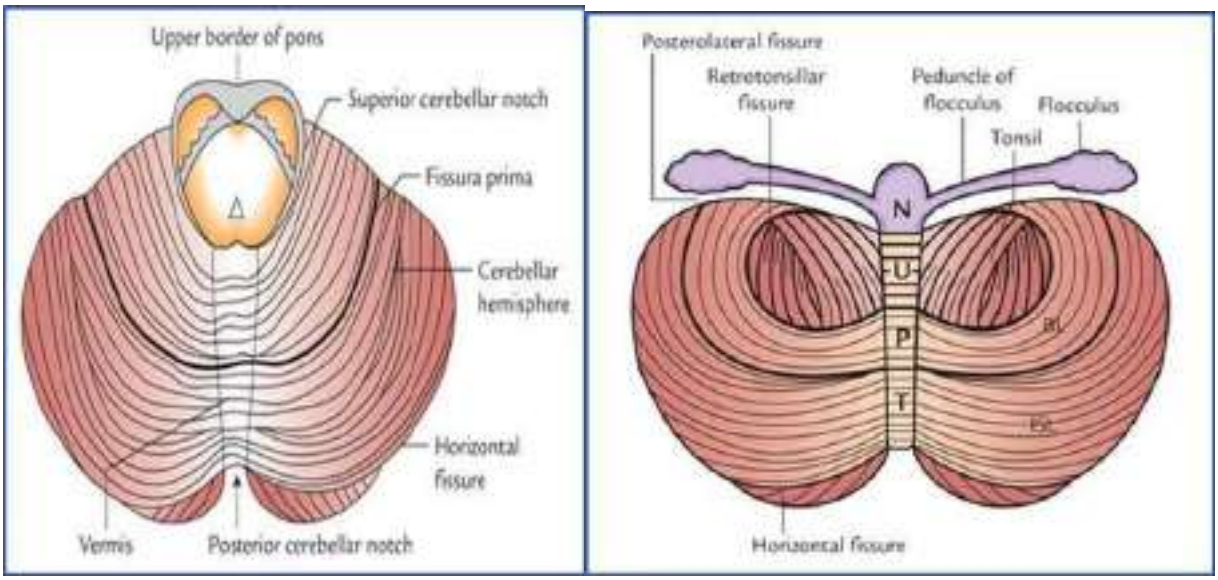


Figure (78): surfaces of cerebellum

Notches

- The anterior aspect of cerebellum is marked by a wide shallow anterior cerebellar notch which accommodates pons and medulla.
- The posterior cerebellar notch is deep and narrow, and lodges the falx cerebelli.

Fissures

- The horizontal fissure is most conspicuous and runs along the lateral and posterior margins of the cerebellum. It marks the junction between the superior and inferior surfaces of the cerebellum.
- The posterolateral fissure lies on the inferior surface of the cerebellum and separates the flocculonodular lobe from the rest of the cerebellum (corpus cerebelli).
- The V-shaped fissura prima on the superior surface cuts the superior vermis at the junction of its anterior two-third and posterior one-third. It divides the corpus cerebelli into anterior and posterior (middle) lobes.

Subdivisions of Cerebellum

- Anatomical subdivisions
- **Anatomically the cerebellum is divided into three lobe**
  - The anterior lobelies on the superior surfaceanteriorto the fissura prima.
  - The posterior/middle lobe lies between the fissure prima on the superior
  - surface and posterolateral fissure on the inferior surface.
    The flocculonodular lobe is smallest of all and lies on the inferior surface in front of the posterolateral fissure.
- **Morphological subdivisions**

- Based on phylogenetic and functional criteria the cerebellum is divided into three parts, archicerebellum, paleocerebellum, and neocerebellum.
- Archicerebellum (vestibular cerebellum)
- It consists of flocculonodular lobe and lingula.
- Function: is chiefly vestibular in connections and concerned with the maintenance of equilibrium, tone and posture of trunk muscles.
- Paleocerebellum (spinal cerebellum)
- It consists of vermal and paravermal part(include fastigial ,globose and embiliform nucleus)
- Function:is chiefly spinocerebellar in connections and concerned with the tone, posture and crude movements of the limbs.
- Neocerebellum (cerebral cerebellum)
- It is made up of lateral part of cerebellar hemisphere (it include dentate nucleus)
- The neocerebellum is chiefly corticoponto-cerebellar in connections and is concerned with smooth performance of skilled voluntary movements.

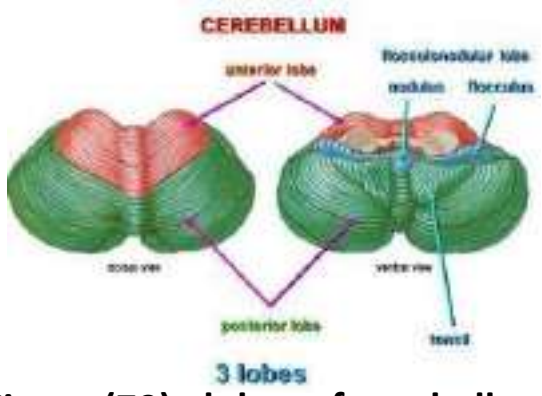


Figure (79): lobes of cerebellum

Internal Structure

- The cerebellum is made up of a thin surface layer of grey matter, the cerebellar cortex and a central core of white matter. Embedded within the central core of white matter are masses of grey matter called intracerebellar nuclei which are masses of grey matter embedded in the white matter of the cerebellum. On each side of the midline they are four in number. From lateral to medial side these are: (a) dentate nucleus, (b) emboliform nucleus, (c) globose nucleus, and (d) fastigial nucleus

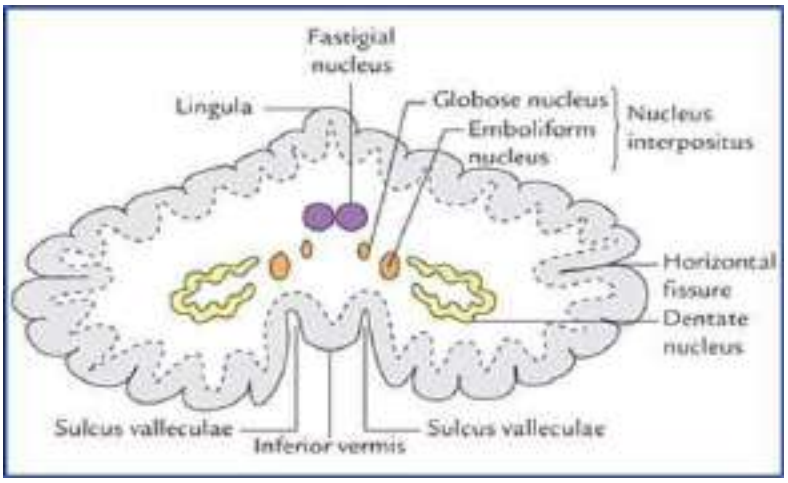


Figure (80): nucleus of cerebellum

Cerebellar Peduncles

- The afferent and efferent fibres of the cerebellum are grouped together on each side into three large bundles called cerebellar peduncles which connect cerebellum to different part of brain.
  - o The superior cerebellar peduncles connect the cerebellum to the midbrain
  - o the middle cerebellar peduncles to the pons
  - o the inferior cerebellar peduncles to the medulla oblongata.

functions of the cerebellum are:

- o Maintenance of posture.
- o Maintenance of muscle tone.
- o Coordination of voluntary motor activity.

Arterial Supply of the Cerebellum

- ❑ **Superior cerebellar artery,**
- **Anterior inferiorcerebellar artery,**
- ❑ a branch of basilar artery supplies the superior surface of the cerebellum.
- a branch of basilar artery supplies the anterior part of the inferior surface of the cerebellum.

Posterior inferior cerebellar artery,

- abranchof vertebralarterysuppliestheposterior part of the inferior surface of the cerebellum.

The cerebellar veins

The superior cerebellar veins

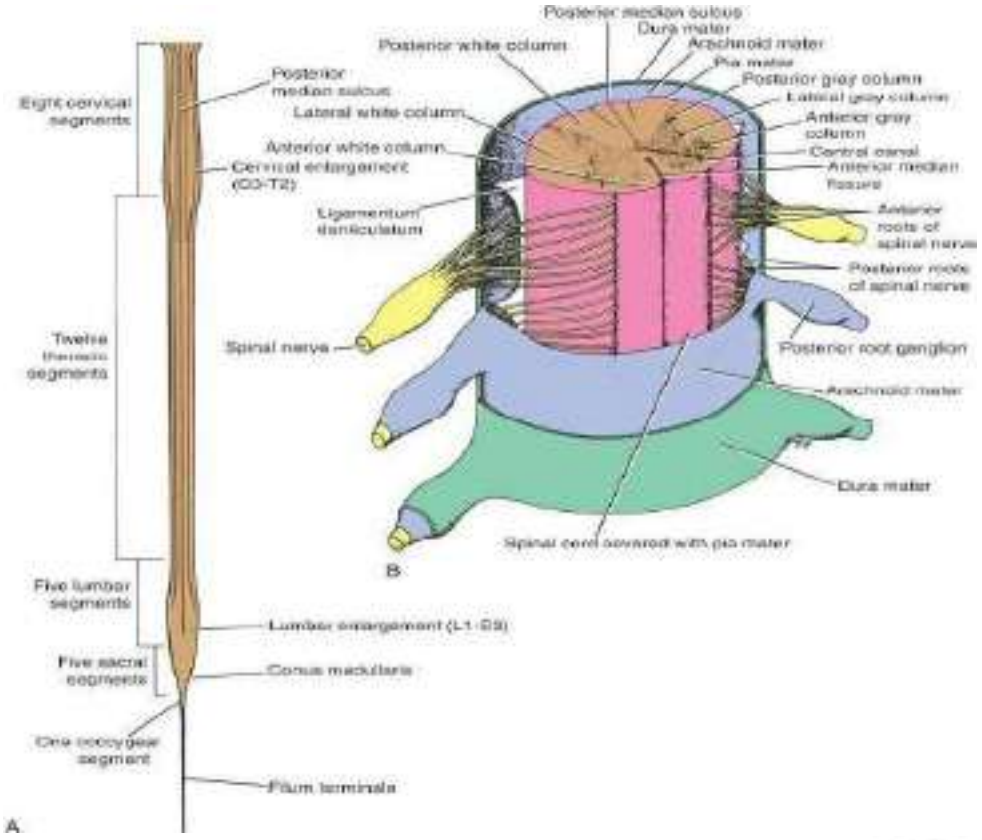
- endinthe straightsinus andthe internal cerebral veins, partly lateralward to the transverse andsuperior petrosal sinuses.

The inferior cerebellar veins

- endinthe transverse,superiorpetrosal, and occipital sinuses.

SPINAL CORD

- **Begin**: as a downward continuation of medulla oblongata at the upper border of the posterior arch of first cervical vertebra (C1)
- **End**: opposite to disc between the first lumbar vertebra (LI) and l2. Its lower tapering extremity is called conus medullaris. The apex of conus medullaris continues downwards as a thin, thread-like filament called filum terminale. Spinal cord provides attachment to 31 pairs of spinal nerves
- **Length** :45 cm



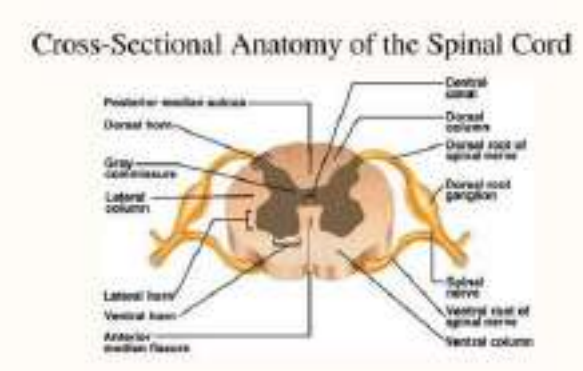
Gross appearance of spinal cord

External features of the spinal cord:

- The anterior aspect ofthe spinal cord presents anterior median fissure, and two anterolateral sulci
  - o The anterior median fissure is deep and extends along the entire length of the cord. The anterior spinal artery runs in it.
  - o Through the anterolateral sulcus emerge the ventral root fibers
- The posterior aspect presents: posteromedian sulcus, two posterolateral sulcus .
  - o Through posterolateral sulcus enter the dorsal root fibers of the spinal nerves.

Internal Structure of the Spinal Cord

- consists of an inner core of grey matter, and a peripheral zone of white matter.
- In cross-section of cord the grey matter is seen as H-shaped (or butterfly-shaped) fluted column, extending throughout the length of the spinal cord. Its part :anterior horn ,posterior horn, lateral horn
- The central canal of the cord passes through the centre of grey commissure
- peripheral zone of white matter divided into 3 part (anterior column. Lateral column. posterior column)
- the cord divided into two halves by anterior median sulcus &posterior median sulcus



Figure(81):external and internal structure of spinal cord

segment of spinalcord :

- thespinal cordis madeupof 31 spinal segments: 8 cervical, 12 thoracic, 5 lumbar, 5sacral, and one coccy-geal.

Enlargements:

- Thespinal cord is roughly cylindrical in shape
- opposite to the attachments of the nerve roots contributing to the formation of brachial and lumbosacral plexuses, the spinal cord presents definite fusiform swellings called cervical and lumbar enlargements respectively

- These enlargements are produced due to the presence of large number of large motor neurons in these regions to supply the musculature of the upper and lower limbs and associated girdles.
- **cervicalenlargements:**
  - o The cervical enlargement extends from C5 to T1 spinal segments .the cervical enlargement lies opposite C3 to T1 vertebrae
- **lumbar enlargement :**
  - o extends from L2 to S3 spinal segments. lumbar enlargement lies opposite T9 to L1 vertebra.

Attachment of Spinal Nerves

- 31 pairs of spinal nerves emerge from the sides of the cord.
- Of these, 8 are cervical, 12 are thoracic, 5 are lumbar, 5 are sacral, and 1 is coccygeal.
- The cervical nerves leave the vertebral canal above the corresponding vertebrae with the exception of eighth, which emerges between seventh cervical and first thoracic vertebrae. The remainder spinal nerves emerge below the corresponding vertebrae. Each spinal nerve is attached to the cord by two roots, anterior motor root and posterior sensory root.
- The anterior root formed by axon of cell in the anterior horn .
- The posterior root formed by the axon of spinal ganglia central pross enter the cord while peripheral process of ganglia unite with anterior root and form spinal nerve .
- Cauda Equina
- Since the cord is shorter than the vertebral column, The cauda equina consists of the roots of the lower four pairs of lumbar, five pairs of sacral and one pair of coccygeal nerves.

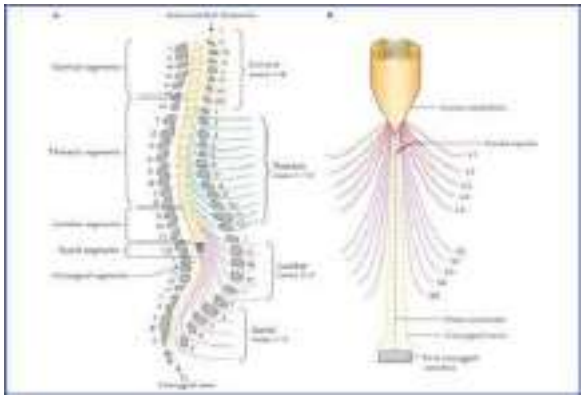


Figure (82): structure of cauda equine

Spinal Meninges:

- The spinal cord is surrounded by three protective membranes called spinal meninges. From outside inwards these are: dura mater, arachnoid mater, and pia mater

Dura Mater

- The spinal dura extends from foramen magnum to the lower border of second sacral vertebra (S2).
- The space between spinal dura and vertebral canal is termed epidural space. This space contains loose areolar tissue, semiliquid fat and internal vertebral venous plexus.

Arachnoid Mater

- Arachnoid mater is a thin delicate transparent avascular membrane which invests the spinal cord loosely. Above it is continuous with the arachnoid mater of the brain and below it extends up to the lower border of the second sacral vertebra (S2).
- The subarachnoid space between the pia mater and the arachnoid mater is filled with cerebrospinal fluid (CSF).

Piamater

is a thin highly vascular membrane that closely invests the spinal cord and continues below the spinal cord as a thin thread-like prolongation, the filum terminale which pierce dura ,arachanoid and attached to tip of .

Clinical Correlation

- Lumbar puncture
- Lumbar puncture :. The needle is introduce to spinal subarachnoid space below the termination of the cord, i.e. lower border of LI.
- Site: just above or below the tip of 4th lumbar spine

Supporting to spinal cord:

- filum terminalis
- denticulate ligment :extended laterally between anterior and posterior root of spinal nerve connecting pia matter to dura and arachnoid
- subarachnoid septum extend from post median septum to arachnoid matter spinal
- nerve roots
- dura matter itself attached to formen magnum ,laterally to inter vertebral foramen.

Tract in spinal cord :

- **Ascending tract:**
  - itsbundleofnervefiber that enter the spinal cord its end the cord or in higher center.
- **conscious:**
  - 1-gracil and cuneate tract 2-ventral spinothalamic tract 3-lateral spinothalamic
  - Unconscious :
- **spino cerebellar tract :**

Lateral spinothalamic tract(painandtemperature pathway)

- **Function**:it is the 2nd order neuron of pain and temperature sensations from opposite side of the body

Pathway of pain and temperature :

- **1st order neuron :**
  - The cell bodies of first order sensory neurons lie in the dorsal root ganglia of the spinal nerves. The central processes of these cells enter the cord through the dorsal root of the spinal nerves. In the cord the fibers ascend one or two segments in the dorsolateral tract at the tip of posterior horns and then relay in the posterior horn by synapsing with the cells of substantia gelatinosa.
- **2nd order neuron:**
  - The axonsof second order sensory neurons of substantia gelatinosa cross to the opposite side in the anterior white commissure and ascend up as lateral spinothalamic tract in the opposite lateral white column just lateral to the anterior horn. Theyterminate in the ventral posterolateral nucleus (VPL) of the thalamus.
- **3rd order neuron:**
  - (VPL) of thethalamus pass into posterior limb of internal capsule then reach cerebral cortex end in post central gyrus.

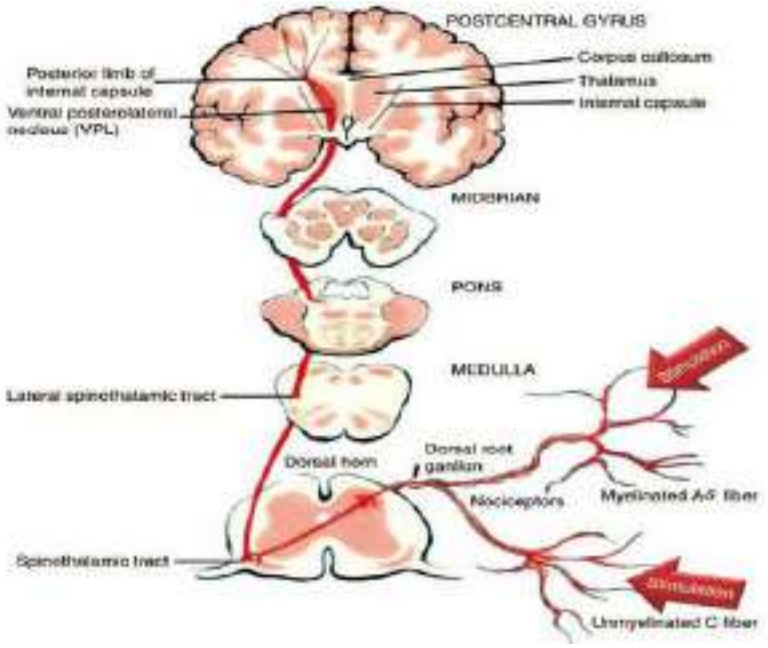


Figure (84):lateral spinothalamic tract

Anteriorspinothalamictract(crude&touchpathway)

- **Function**: 2ndorder neuron of crude touchand pressuretickle,anditch sensations fromthe opposite sideof thebody

1st neuron:

- The cell bodies of the first order sensory neurons of this tract lie in the dorsal root ganglia of the spinal nerves. The central process of these cells enters the cord relay in the nucleus proprius (main sensory nucleus) of the posterior horn.

2nd neuron:

- The axons of nucleus proprius cross the midline in the anterior white commissure and then ascend as ventral spinothalamic tract in the opposite anterior white column just in front of the anterior horn. These fibres terminate in the ventral posterolateral(VPLN) nucleus of the thalamus.

3rd neuron:

- axon of VPLN pass in posterior limb of internal capsule then reach sensory area

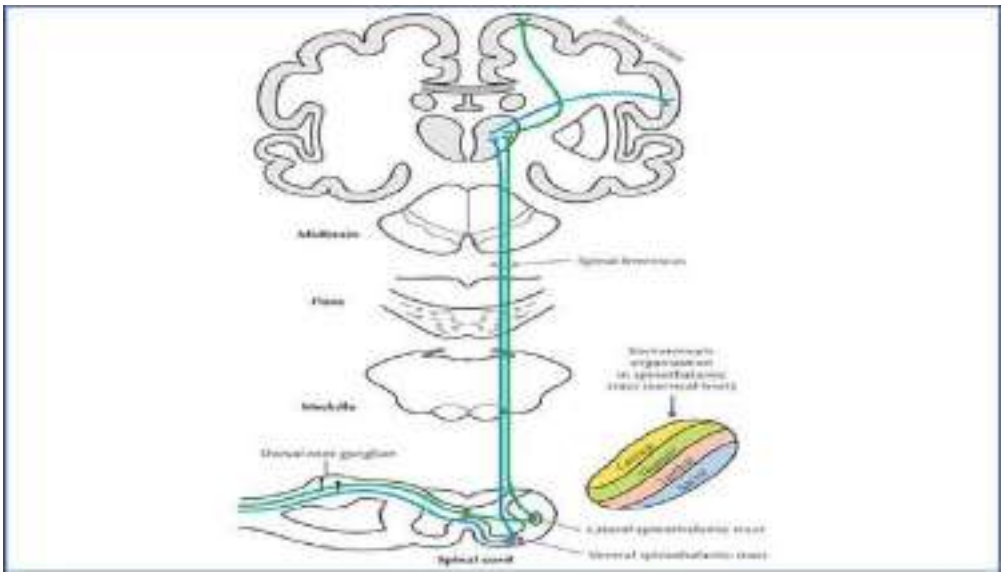


Figure (85): ventral and lateral spinothalamic tract

gracile and cuneate tracts

function:

- They carrysensations of conscious proprioception, two point tactile discrimination and vibration

1st order neuron:

- The cell bodies of the first order sensory neurons lie in the dorsal root ganglia of the spinal nerves. The central processes of these cells (thickly myelinated) enter the cord through the medial division of the roots of spinal nerves and continue in the posterior white column as gracile tract and cuneate tract ,gracile tract carry fiber from lower part of the body below t6 and ascend medially in the dorsal column while cuneate tract carry fiber from upper
- half of the body above t6 and pass lateral to gracile tract ,

2nd order neuron :

- gracile and cuneate tract terminate in gracile and cuneate nucleus respectively. The axons of second order sensory neurons from nucleus gracile and nucleus cuneatus The fibers decussate as internal arcuate fiber then ascend upwards as a flat tract called medial lemniscus which courses upwards to terminate in the ventral posterolateral nucleus (VPL) nucleus of the thalamus

3rd order neurons: from thalamus

- projectinto thecerebral cortexofthe cerebral hemisphere. This pathway carrying conscious proprioceptive sensations is termed ‘dorsal column—medial lemniscus-pathway’

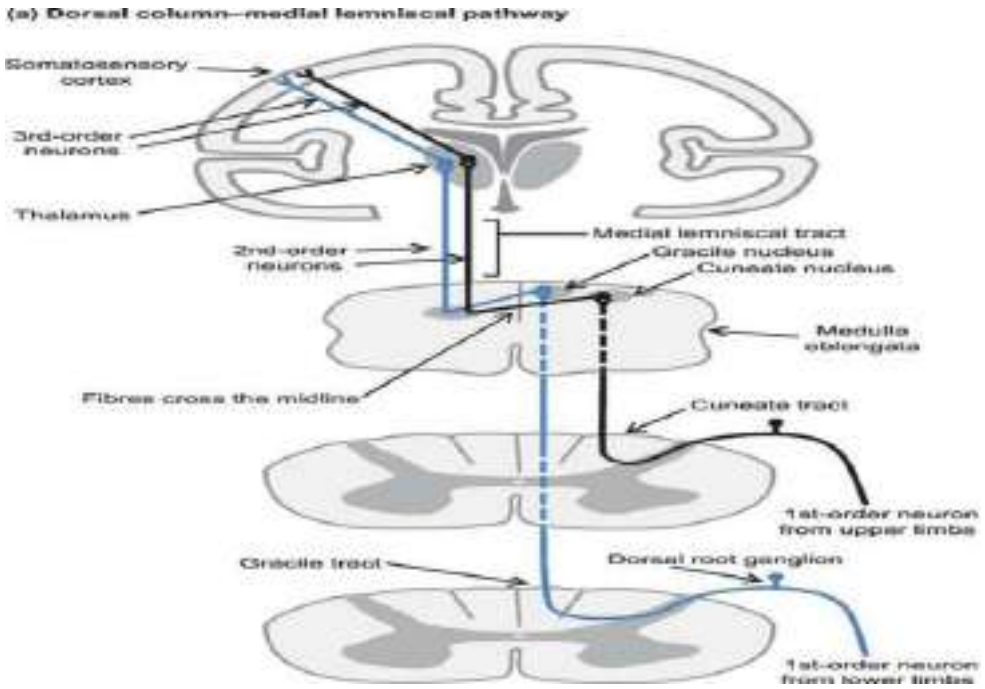


Figure (86): gracile and cuneate tract

spinocerebellar tracts :

- The spinocerebellar fibres are located in the lateral white column of the cord and are divided into two tracts: posterior spinocerebellar tract and anterior spinocerebellar tract according to their location in the cord.

Function:

- They carry unconscious proprioceptive sensations from cord to the cerebellum and play an important role in muscular coordination. both posterior and anterior spinocerebellarlar tracts carry impulses from the lower limb. The posterior tract is concerned with fine movements, whereas anterior tract is concerned with gross movements of the limb as a whole.

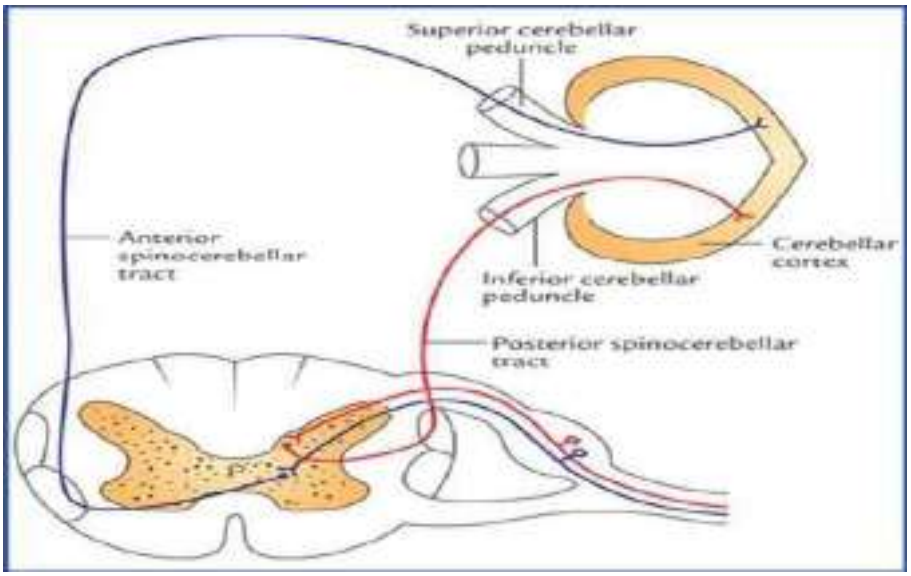


Figure (87): spinocerebellar tract.

descending tract:

The descending tracts conduct the impulses to the spinal cord from the brain

- **Pyramidal:**
  - arisein thecerebral cortex (upper motor neurons) and end on the nuclei of the cranial nerves as well as the anterior horn of the spinal cord
  - Include:Anterior corticospinal tract -Lateral corticospinal tract
- **Extra pyramidal :**
  - arise from the cerebral cortex as well as from other subcortical centres (e.g. corpus striaturn, red nucleus, etc.) which end below on the anterior horn cell.
- **function:**
  - responsible for adjusting muscle tone.posture and semiautomatic movement as swinging the arm during walking

(pyramidal tract)

- **Origin:**
  - The most of the fibres of arise from pyramidal cells (of Betz) of the motor area of the cerebral cortex. Some fibres arise from other parts of the cortex.
- **Course:**
  - The fibres pass through corona radiata, internal capsule, crus cerebri of cerebral peduncles, ventral part of the pons and pyramids of the medulla oblongata. In the

lower part of the medulla majority of fibres (about 75%) cross to the opposite side at the pyramidal decussation of the medulla and descend in the lateral white column of the spinal cord as the lateral corticospinal tract. The uncrossed fibers descend in the anterior white column of the spinal cord as the anterior corticospinal tract. .

Termination:

- **In spinal cord:**
  - The crossed fiber end on the lateral group of anterior horn cell which supply the limp
  - The uncrossed fiber end on the medial group of anterior horn cell supplying the trunk.
- **In brain stem:**
  - Some pyramidal fiber don't reach spinal cord but end on motor nuclei of cranial nerve that lie in the brain stem these fiber called corticonuclear tract which control muscle of the head ,the fiber end in contralateral and ipsilateral motor nuclei .except to nuclei receive fiber from contralateral motor area these nuclei:
    - o lower part of the facial nucleus that control lower part of the face
    - o part of hypoglossal nucleus that control genioglossus muscle

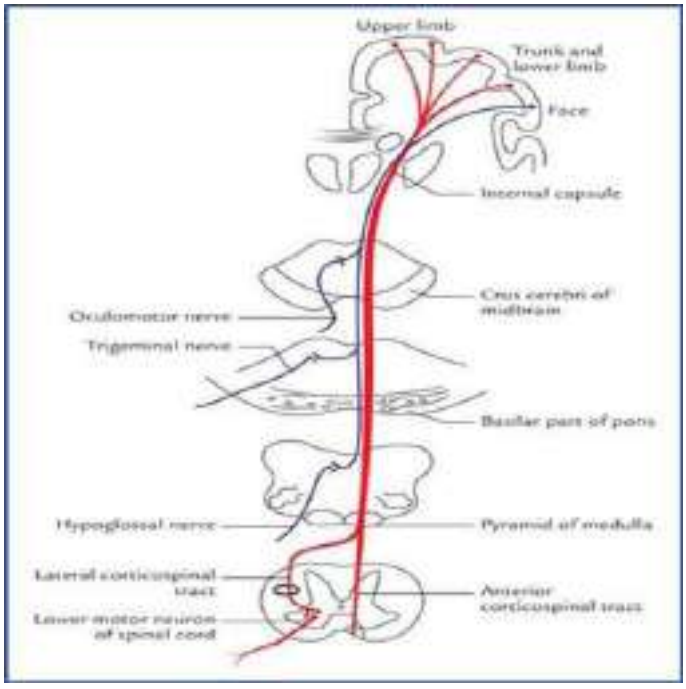


Figure (83): pyramidal tract

Blood Supply of The Spinal Cord:

Arterial Supply:

- **Anterior spinal artery:**
  - is formed by the union of two small spinal branches of the right and left vertebral arteries in the upper cervical canal. It runs caudally in the anterior median fissure of the spinal cord and terminates along the filum terminale.
  - Supply: medial part of medulla oblongata and anterior two third of spinal cord
- **Two posterior spinal arteries.**
  - each arising as a small branch from either the vertebral or posterior inferior cerebellar artery. Each posterior spinal artery runs down on the posterolateral aspect of the cord in the posterolateral sulcus along the line of attachment of posterior nerve roots and usually divides into two collateral arteries along the anterior and posterior sides of the posterior nerve roots.
  - Supply:posterior third of spinal cord

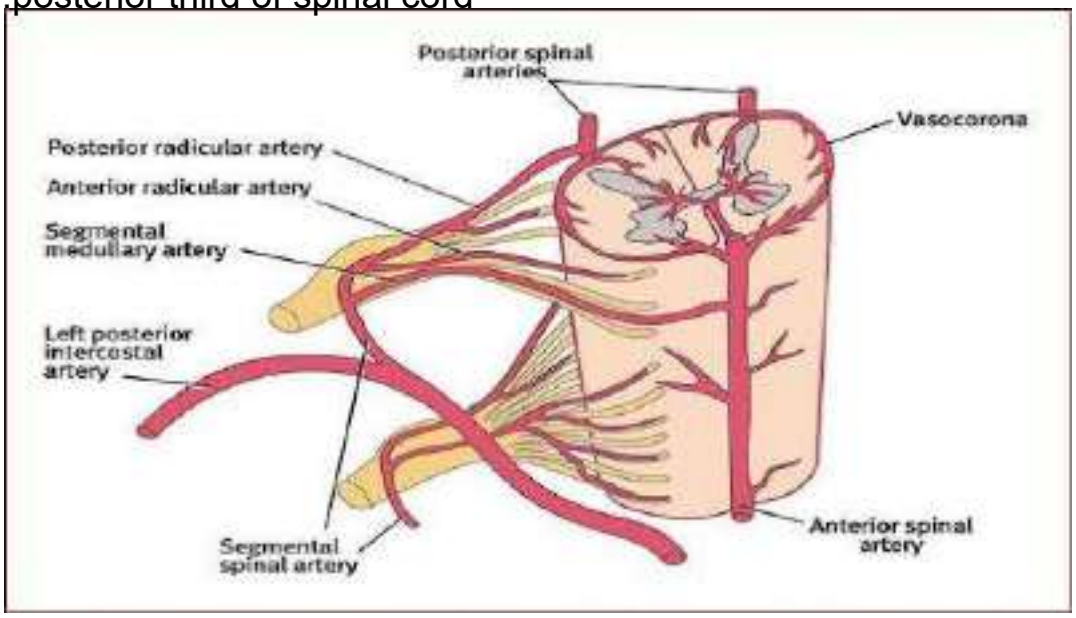


Figure (88): blood supply of spinal cord

- **Segmental arteries.**
  - Thearteries are branches of arteries outside the vertebral column (deep cervical, intercostal, and lumbar arteries). Having entered the vertebral canal, each segmental spinal artery gives rise to anterior and posterior radicular arteries that accompany the anterior and posterior nerve roots to the spinal cord.

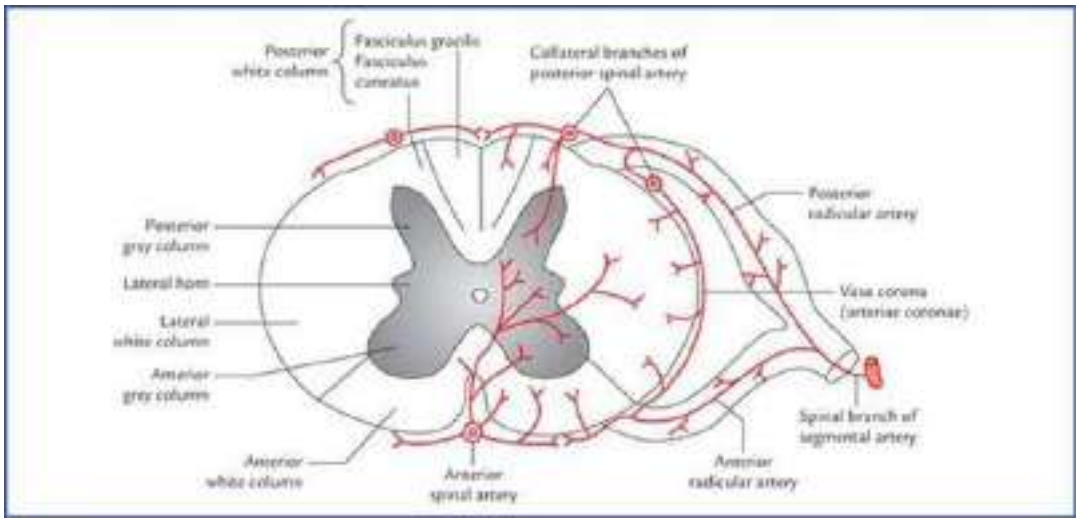


Figure (89): segmental arteries

Venous drainage of the spinal cord.

- Two median longitudinal vein, one in the anterior median fissure and the other in the posteromedian sulcus.
- Two anterolateral vein, one on either side, posterior to the anterior nerve roots.
- Two posterolateral, one on either side posterior to the posterior nerve roots.
- These longitudinal venous channels communicate with the internal vertebral venous plexus and drained by veins which leave through the intervertebral foramina to empty into the vertebral, posterior intercostal, lumbar and lateral sacral veins.

Peripheral nervous system

❑ peripheral spinalnerves

- There are 31 pairs of spinal nerves which arise from the spinal cord and pass through intervertebral foramina in the vertebral column. The spinal nerves are named according to the regions of the vertebral column with which they are

associated. There are 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, and 1 coccygeal nerves.

structure:

- A typical spinal nerve arises from spinal cord by two roots: an anterior root and a posterior root. The anterior root consists of bundles of nerve fibers which carry nerve impulses away from the spinal cord; these fibers are called motor (efferent) fibers. Their cells of origin lie in the anterior horn of the spinal cord.
- The posterior root consists of bundles of nerve fibers of cell bodies of posterior root ganglion these nerve fibers which carry impulses to the spinal cord, these are called sensory (afferent) fibers. The posterior root ganglion is located in the intervertebral foramen.
- The spinal nerve roots pass from the spinal cord to the intervertebral foramen, where they unite to form a spinal nerve. Thus, the spinal nerve is made up of both motor and sensory fibers.
- After emerging from the intervertebral foramen, each spinal nerve divides into a large anterior ramus and a smaller posterior ramus, each containing both motor and sensory fibres. The posterior ramus passes posteriorly around the vertebral column to supply the muscles and skin of the back. The anterior ramus runs anteriorly to supply the muscles and skin over the anterolateral part of the body wall.

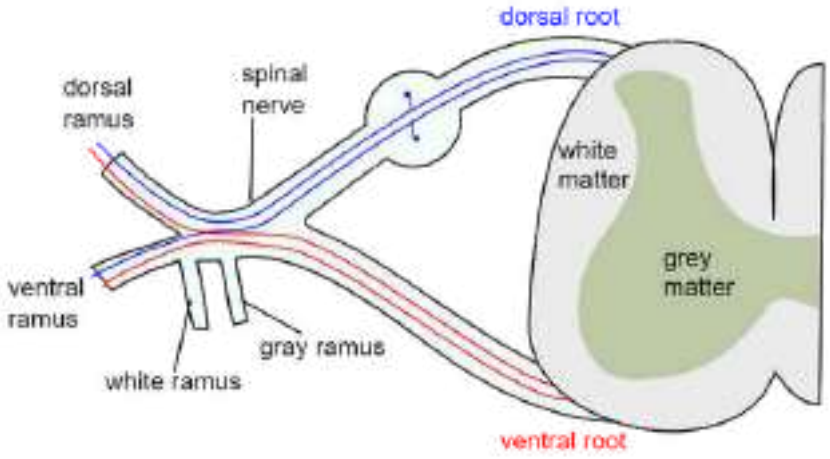


Figure (90): structure of spinal nerve

Plexus formation:

- The anterior primary rami of all the spinal nerves(except for thoracic nerves from T3 to T11) join together and form a network of nerves known as nerve plexus. There are three major plexuses:
- Cervical plexus :made up of C1 through C4 spinal nerves, innervates the muscles of neck and diaphragm.
- Brachial plexus : made up of C5 through T1 spinal nerves, innervates muscles of the upper limb.
- Lumbosacral plexus made up of L1 through S5 spinal nerves, innervates muscles of the lower limb. It is divided into two portions:

o Lumbar portion (lumbar plexus) L1 through L4, supplies mostly muscles of the thigh.
o Sacral portion (sacral plexus) L5 through S5, supplies mostly muscles of leg and foot.

Visual pathway

- 1st order neuron: formed by bipolar cell of retina
- 2nd order neuron:formed by ganglion cell of the retina their axon form the optic nerve fiber the two optic nerve join together in the optic chiasma.
  In the chiasma fiber from nasal half decussate into contralateral optic tract while from temporal half pass uncrossed to ipsilateral optic tract

N.B

- **nasal fiber carry impulse from temporal part of retina ,while temporal fiber carry impulse from nasal side .**

- Optic tract carry temporal fiber from the same side ,nasal fiber from opposite side and molecular fiber from both side
- It carry the fiber to lateral geniculate body(lGB).
- 3rd order neuron: axon of neuron of LGB form the optic radiation which project to the visual cortex(area 17 which present in occipital lobe below precalcarine sulcus).

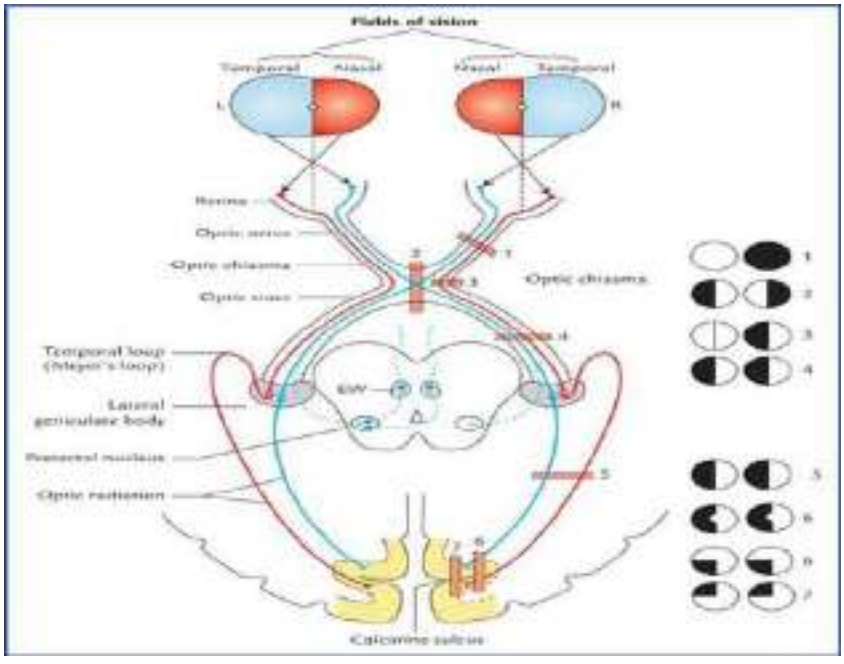


Figure (90): visual pathway

Visual Reflexes

- Light reflex or pupillary reflex:
- When light is shown to one eye, normally the pupils of both the eyes constrict. The constriction of pupil upon which the light is shown is termed direct light reflex and the constriction of the pupil of the other eye (even though no light fell upon that eye) is termed indirect or consensual light reflex.

Pathway of the reflex:

- The visual impulses travel through the optic nerve----optic chiasma --optic tract to terminate in the pretectal nucleus of both ipsilateral and contralateral side . Then the impulses from pretectal nucleus --- to the parasympathetic nuclei (Edinger-Westphal nuclei) of the IIIrd cranial nerve of both sides. The impulses from nucleus of oculomotor nerve ---- to nerve of inferior oblique ----to the ciliary ganglion. Then postganglionic fibres travel along the short ciliary nerves to ---- the sphincter pupillae muscle of the iris leading to constriction of pupils of the both eyes.

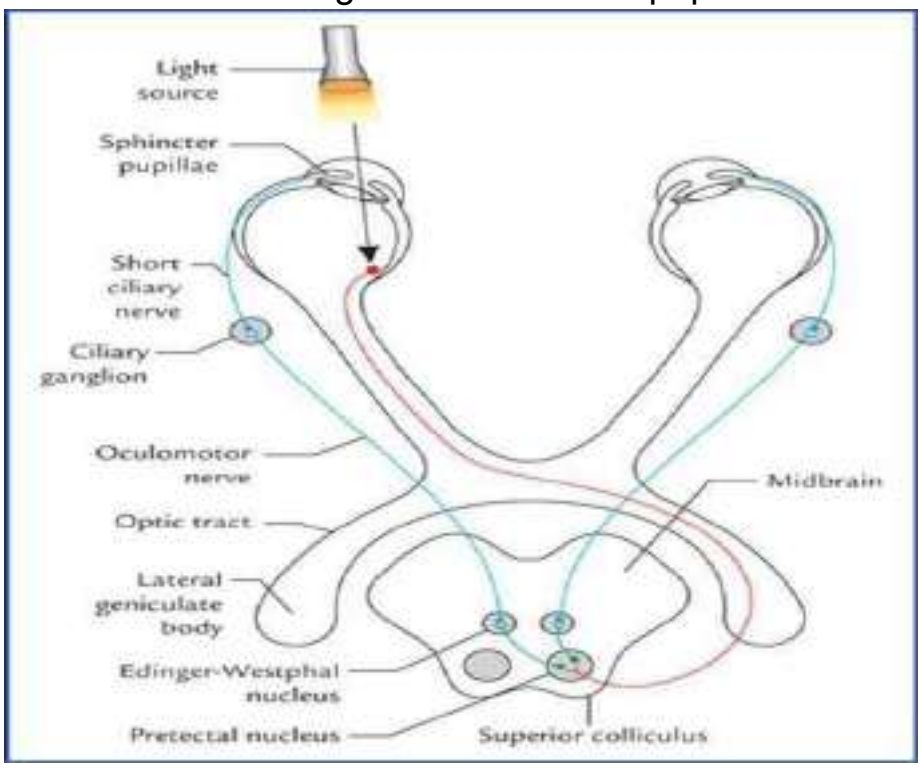


Figure (91):light reflex

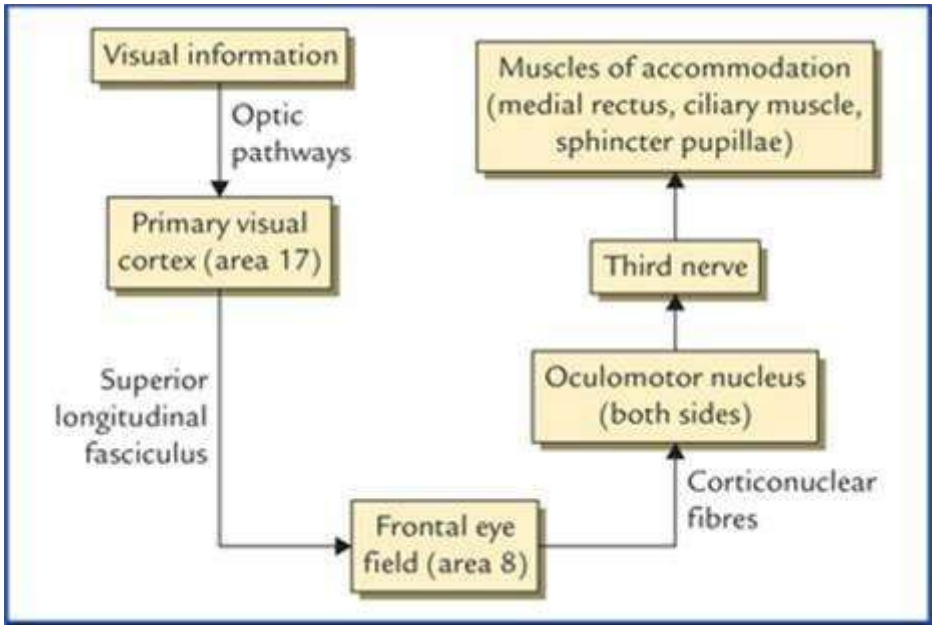
Accommodation reflex:

- **When the eyes are focused from a distant to a near object, three reactions take place intheeyes.**
  - There is constriction of the pupils due to action of sphincter pupillae to restrict the light waves to the thickest central part of the lens.

- o There is thickening of the lenses due to contraction of the ciliary muscles, to increase the refractive index of the lenses.
- o There is convergence of both the eyeballs, due to the contraction of medial recti, to bring about the convergence of the ocular axes.

Pathways of the accommodation reflex

- The visual impulses travel through the optic nerve---optic chiasma-- optic tract--
- lateral geniculate body-- the optic radiation to the visual cortex.
  From visual cortex the impulses reach the 'frontal eye field' through, the superior
- longitudinal bundle .
  From 'frontal eye field' the corticonuclear fibres project to the oculomotor nerve nuclei of both the sides. From here the impulses travel through the oculomotor nerves to supply, all the three muscles(sphincter pupillae, ciliary muscle and medial rectus of the eyeballs)whose active action is needed for accommodation.
  - o Since all the muscles responsible for accommodation are supplied of oculomotor nerve, the oculomotor nerve is often termed nerve of accommodation.
  - o Accommodation reflex is bilateral.



Corneal reflex:

- light touch of the cornea or conjunctiva result in blinking of eye lid
- Pathway :impulse pass along ophthalmic nerve -- main sensory nucleus of trigeminal nerve -- motor nucleus of facial nerve on both side to orbicularis oculi muscle

Auditory Neural Pathways

The first order sensory neurons

- The central process of cell in the spiral (cochlear) ganglion which is located within the cochlear modiolus. constitute the cochlear nerve, which enters the brainstem at the junction of medulla and pons. ending in the dorsal cochlear nucleus and the ventral cochlear nucleus.

The second order sensory neurons

- arise from the cells of cochlear nuclei, pass in the pons form trapezoid body end in the trapezoid and superior olivary nucleus of the same side as well as of the opposite side.

The third order sensory neurons

- arise from the dorsal nucleus of the trapezoid body and superior olivary nucleus of both sides and ascend as the lateral lemnisci and relay in the nucleus of inferior colliculus (center for auditory reflexes) which sends new sets of axons to the medial geniculate body through the brachium of inferior colliculus.
- However, most of the fibers of lateral lemniscus pass directly to the medial geniculate body without being relayed in the inferior colliculus.

4th order neurons

- The fibers arising from medial geniculate bodies form the auditory radiations which run through the sub lentiform part of the internal capsule to project into the upper surface of superior temporal gyrus (contains primary auditory cortex (areas 41 and 42 of Brodmann).

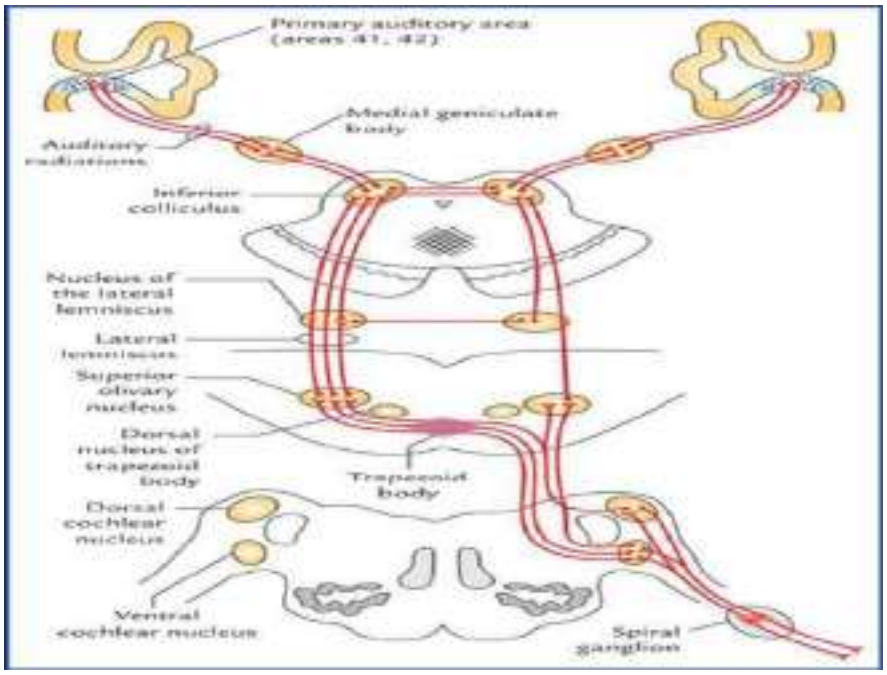


Figure (92): auditory reflex

Vestibular pathway :

- **first order sensory neurons.**
  - Thecentral processes of thebipolar cells of the vestibular ganglion(which receive impulse from receptor in semicircular canal and utricle and saccule) form the vestibular nervefibers that relay in the vestibular nuclei .
- **The second order sensory neurons vestibular nuclei project:**
  - o to the flocculonodular lobe of the cerebellum through inferiorcerebellar peduncle to affect equalilibrum
  - o to the motor nuclei of IIIrd, IVth and VIth cranial nerves for coordination of head and eye movement
  - o to the anterior horn cells of the spinal cord toControl of posture and balance
  - o The vestibular nuclei also project to the posteroventral nucleus of the thalamus to the vestibular area of the cerebral cortex in the postcentral gyrus. This provide sense of balance and information about body position

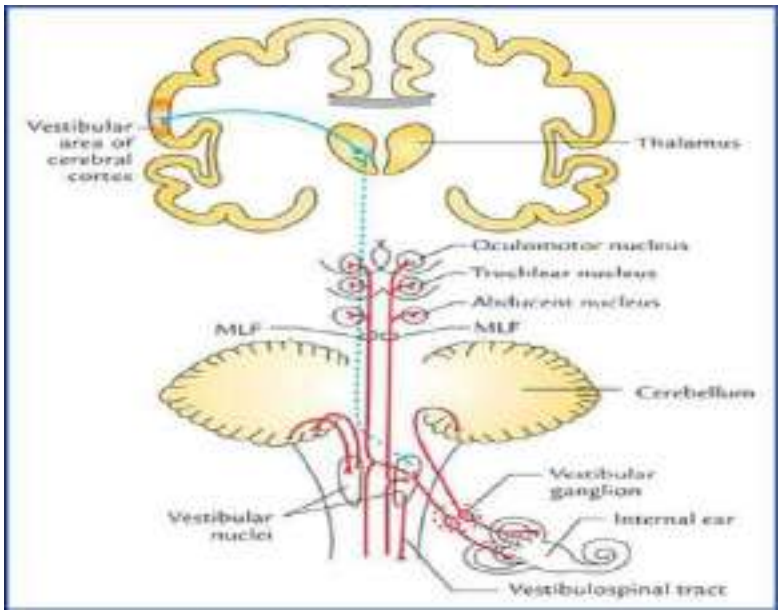


Figure (93): vestibular reflex

Taste pathway:

- **first order sensory neurons cell located in**
  - o the geniculateganglionof facial nerve receive Taste fibres from anterior two-third of the tongue .
  - o superior petrosal ganglion and inferior ganglion of glossopharyngeal receive taste from posterior third of the tongue

o inferior ganglion of the vagus nerves receive taste from posterior part of ganglia and epiglottis

The second order sensory neurons

- arise from nucleus tractus solitarius(include previous 3 nucleus) cross the midline, turn upwards to join the medial lemniscus and terminate in the posteroventral nucleus of the thalamus (VPMN).

The third order sensory neurons

- arise from posteroventral nucleus of thalamus and project to taste or gustatory area above posterior ramus of lateral sulcus

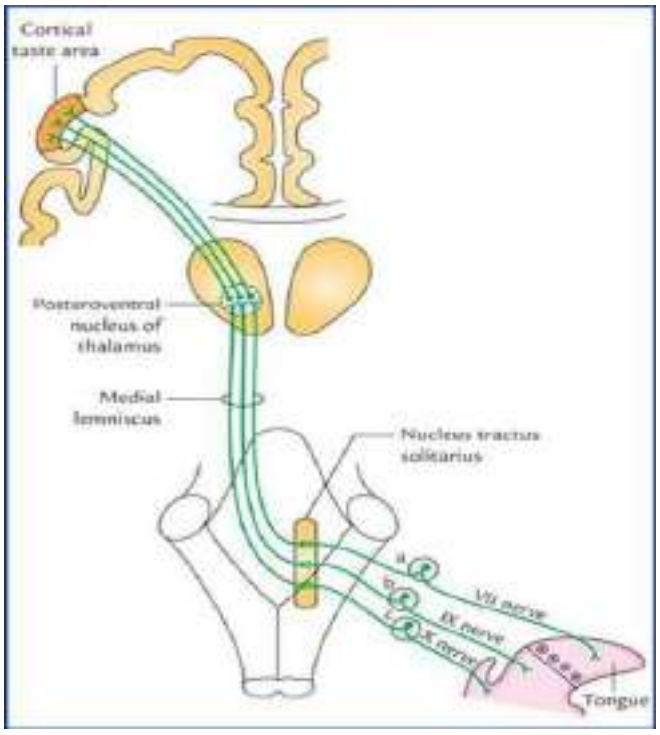


Figure (94): taste pathway

Olfactory pathway

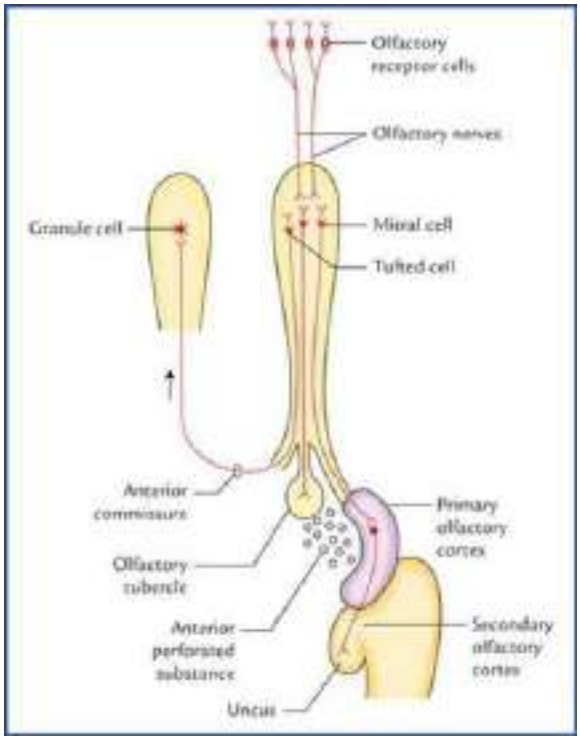
firstorder sensory neurons

- The axons of the olfactory receptor cells in olfactory epithelium carrying olfactory sensations collect to form small bundles (olfactory nerves) which pass through the cribriform plate of the ethmoid bone, to enter the olfactory bulb where they terminate in the mitral and tufted cell.

second order sensory neurons

- axons of receptor cells present in olfactory bulb -----run in the olfactory tract extend till the anterior perforating substance in front of the optic chiasma to form the lateral olfactory stria then run to the primary olfactory cortex and to 2ry olfactory area .

- The olfactory pathway consists of only two neurons whereas other conscious sensory pathways consist of at least three neurons.



N.B

- **It should be noted that in contrast to all other sensory pathways, the olfactory pathway only sensation reach the cortex directly without relay in one of the thalamic nuclei.**

Figure (95):olfactory pathway

Development of nervous system

❑ Development of the Central Nervous System:

Neurulation:

- Is the process of formation of neural tube,In the third week of development, median thickening in the overlying ectoderm called neural plate.
- The lateral edges of the neural plate then rise to form neural folds and the Plate invaginate form neural groove The neural folds move towards each other and meet in the midline, fusing to form the neural tube .fusion begin cervical and proceed cranially and caudally cranial end form anterior neuropore and fused at 25th day and caudal end form posterior neuropores and fused at 27th day

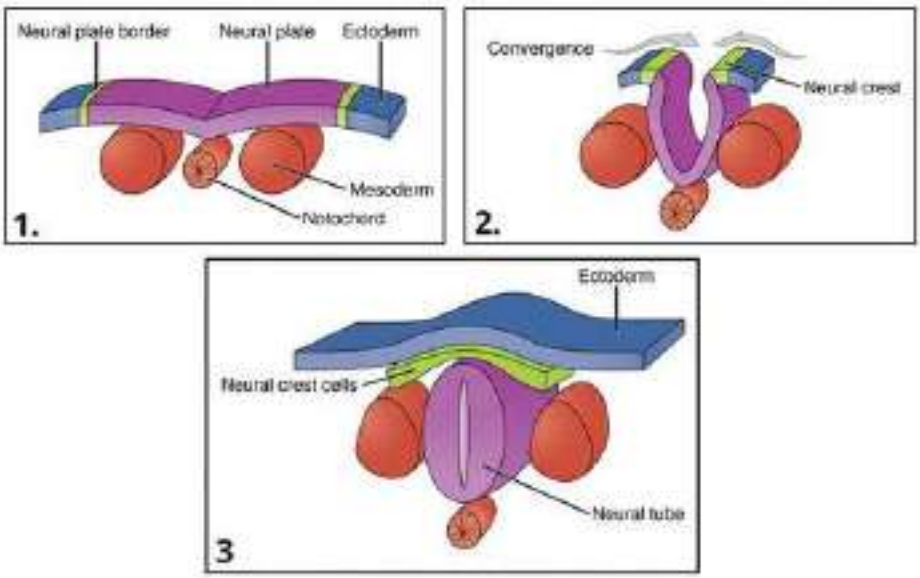


Figure (101): process of formation of neural tube

- During fusion ofthe neuralfolds, some cells within the edge of the folds migrate to form a distinct cell population – known as the neural crest. They give rise to a diverse cell lineage – including melanocytes, craniofacial cartilage and bone, smooth muscle, peripheral and enteric neurons and glial cell, adrenal medulla ,parafollicular cell, sympathetic and parasympathetic ganglia

❑ Devel

- In the
  tube.
- These
  well a

Pri			
Pros			
	Diencephalon	hypothalamus, and epithalamus	Third Ventricle
Mesencephalon	Mesencephalon	Midbrain	Cerebral aqueduct
	Metencephalon	Pons and cerebellum	Upper part of 4th ventricle
Rhombencephalon	Myelencephalon	Medulla	Lower part of 4th ventricle/central canal

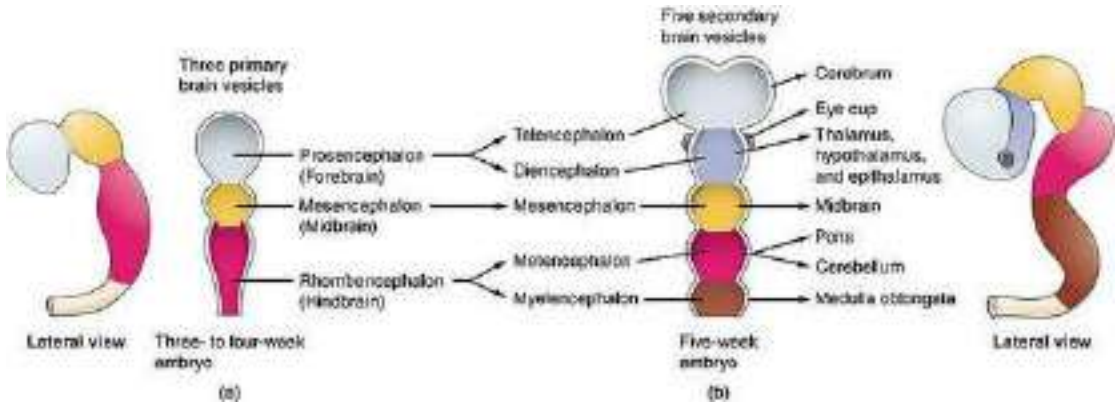


Figure (102): formationof the part of the brain

CONGENITAL ANOMALIES OF THE BRAIN:

Cranial meningocele

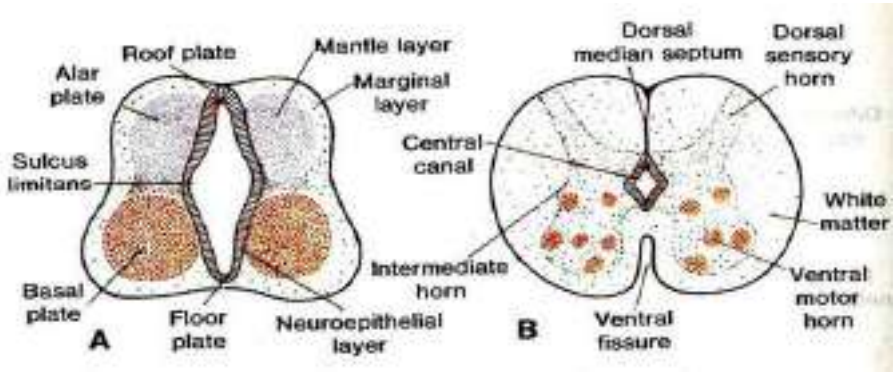
- Meningocele is the simplest form of open neural tube defects characterized by cystic dilatation of meninges containing cerebrospinal fluid without any neural tissue(defects in bone ossification).
- Meningo-encephalocele: abnormal sac of fluid, brain tissue, and meninges that extends through a defect in the skull.
- Meningo-hydroencephalocele : the condition including meninges, brain tissue and partof theventricular system in the herniation.

Hydrocephalus:

- is excessive accumulation of C.S.F. in the ventricles of the brain due to obstruction to the stream of C.S.F, usually at the foramen of Monro or at the cerebral aqueduct.

Development of Spinal Cord

- he spinal cord develops from the caudal elongated part of the neural tube. The neural tube increases in thickness due to repeated mitosis of its epithelial lining.
- By the middle of 5th week of embryonic development ,these neuroepithelial cells give rise to another cell type called nerve cells or neuroblasts.
- The neuroblasts form a zone which surrounds the neuroepithelial layer. It is known as mantle zone. Mantle zone later forms the grey matter of the spinal cord.



- The outermost layer of spinal cord contains the fibers emerging from the neuroblasts in the mantle layer and is known as marginal layer. Which give rise to the white matter of the spinal cord.
- On each side the wall of neural tube is demarcated into dorsal and ventral regions by an inner longitudinal sulcus called sulcus limitans.
- The cells of dorsal region called alar lamina are functionally sensory ,the cell of dorsal region called basal lamina are motor. The axons of cells of basal lamina leaving the cord as ventral roots join with the peripheral processes of dorsal root ganglia, to form the spinal nerves

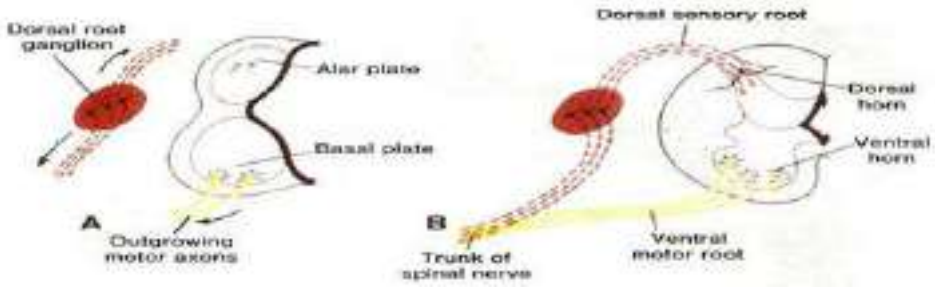


Figure (103): formation of the spinal cord

Positional changes of the developing spinal cord

- At month 6 of gestation, the end of the spinal cord lies at the level of S1.
- In the newborn infant, it lies at L 3 • In the adult, it lies at L 1-2.

Congenital anomalies of spinal cord

❑ Anencephaly(craniorachischisis)

- A failure of the cephalic part of the neural tube to close and associated defective development of the vault of the skull

Rachischisis (a severe form of spina bifida)

- Anincompleteclosureofcaudalneuroporeanddefective development of the associated vertebral arches

Spina bifida

- resultsfrom incomplete closure of the neural tube at the caudal end (most commonly in the lumbar region). There are three main types of spina bifida, of increasing severity:

Spina bifida occulta

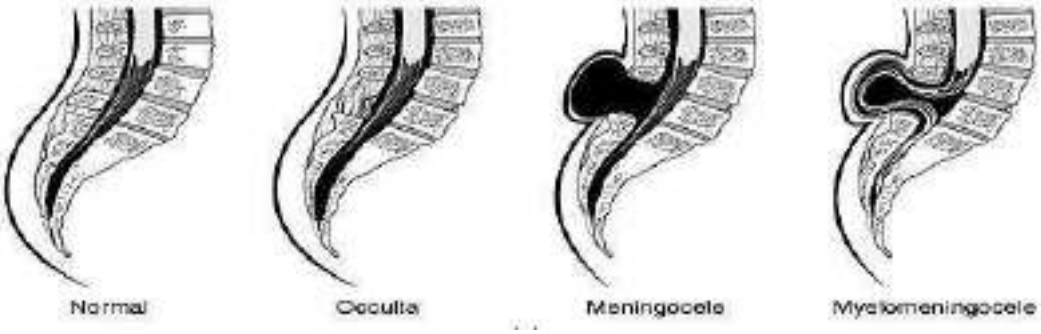
- is characterized by an incomplete closure of the vertebrae, without protrusion of the spinal cord. Most people with this form of spina bifida are unaware of having it, and its discovery is often incidental.

Meningocele (meningeal cyst)

- where the meninges protrude between the vertebrae posteriorly, but the spinal cord is undamaged.

Myelomeningocele

- the most severeform, where a portion of the spinal cord remains unfused and protrudes posteriorly through an opening between the vertebrae, in a sac formed by the meninges. This is associated with severe disability.



Scalp

Itis the **soft** tissue covering the skull cap; it is formed of **5 layers:**

1- **Skin:** **thick** and contains **hair follicles** and **sebaceous glands** 2- **Connective tissue layer:** formed of **dense fibrous** tissue and contains **numerous** blood vessels, **nerves** and **lymphatic**.

3- **Aponeurosis:** it is a **thin connective** tissue layer that units the **bellies** of **occipitofronatalis** muscle.

4- **Loose areolar tissue:** it lies in between the **epicranial** aponeurosis and the **periosteum** of the skull, it contains **emissary** veins that **connects scalp veins** with the **diploic** veins of **skull** and **intracranial** venous **sinuses**.

5- **Pericranium:** the **deepest** layer and it is called the **periosteum** which **covers** the **outer layer** of skull bones.

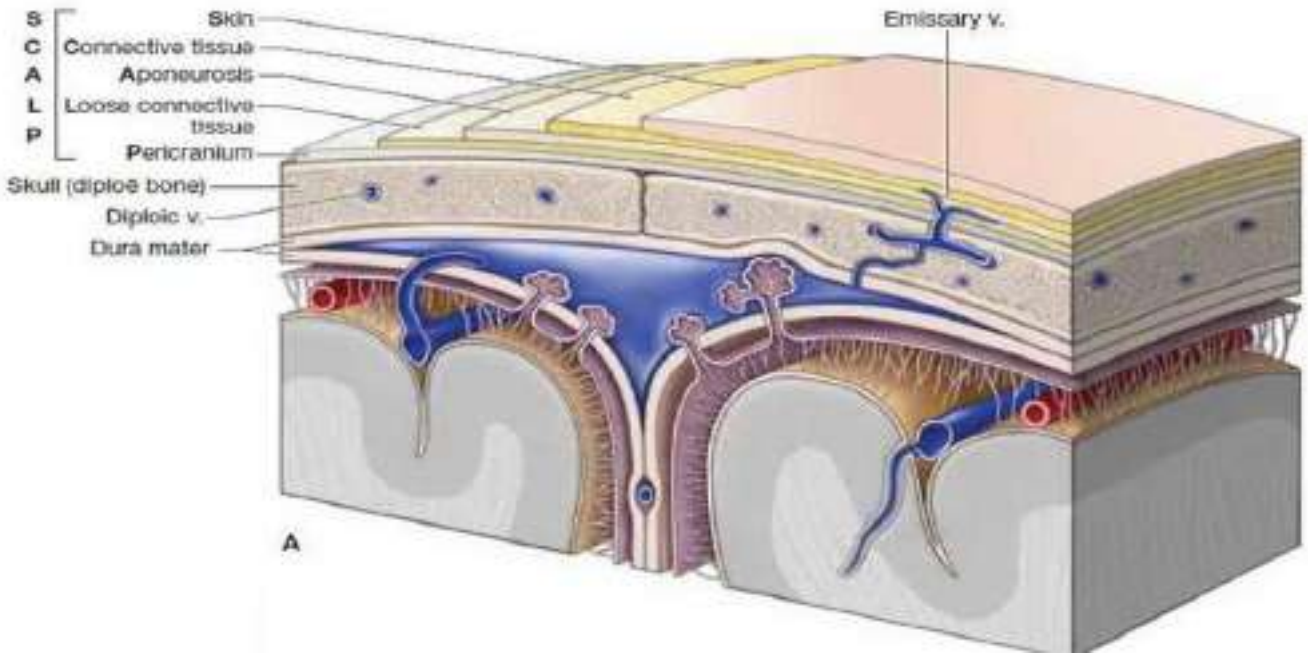


Figure (1): layers of the scalp

Occipitofrontalis muscle:

Thismuscleisformed of **2 frontal** bellies and **2occipital** bellies **connected** together by **epicranial aponeurosis** which is called **galeaaponeuorotica**

- **Action: frontal** bellies can raise the **eyebrows**(**facial expressions**).
- **Nerve supply:frontal** bellies are innervated by **temporal branch of facial nerve** and **occipital** bellies are innervated by **posterior auricular nerve of facial nerve**

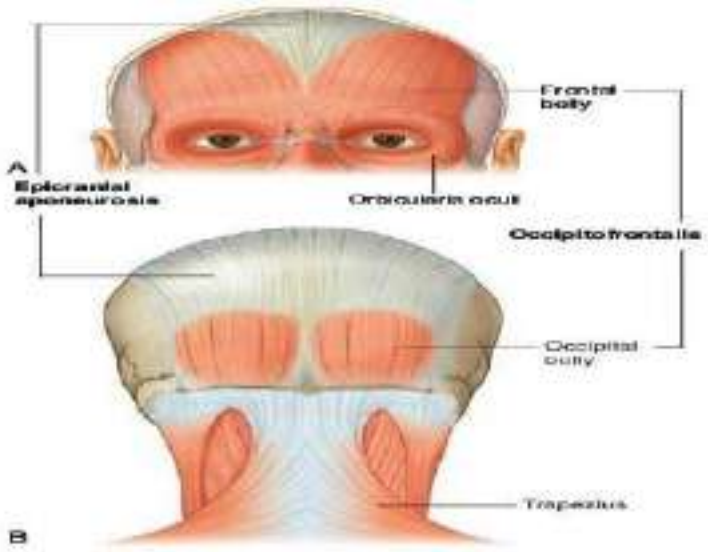


Figure (2): Occipitofantalis muscle

Face

Musclesofthe face:

- They are **numerous**,mainly **inserted** into **skin** of the **face**, also known as **muscles of expression.**
- They are **embedded** in the **connective tissue**, and connected with **bones** by **loose connective tissue.**

N.B.

There is no deep fascia In the face except buccal fascia

1.Muscles of the eyelids:

Orbicularis oculi, levator palpebrae superiors and occipitofrontalis muscles

Orbicularis oculi:

Itencircles the **orbit** and the **periorbital** area. It is composed of **three parts:**

1-Theorbitalpart: is the **most peripheral** part overlaying the **orbital rim.**

2- **Thepalpebralpart:** is the **central** part of the muscle that **comprises** the

3- **eyelids**.

Thedeeppalpebral(lacrimal) part: is the **deepest** part of the orbicularis oculi situated between the **medial palpebralligament** and the **sac**.

Action:

1- **the orbital part:** it **pulls** the skin of the **forehead and cheek towards** the **nose** and **tightly closes the eyes**, commonly for **protective purposes,**
2- while **the palpebral part: close** the **eyelids gently** during **blinking or sleeping.**
3- But **the deep part: pulls** the **eyelids** and **lacrimal papillae medially** and **dilates** the **lacrimal sac**, while compressing the **lacrimal gland** and **ducts**. These **actions facilitate** the **flow of tears** across the **lacrimal apparatus.**

Nerve supply: the **zygomatic** and **temporal** branches of **facial** nerve.

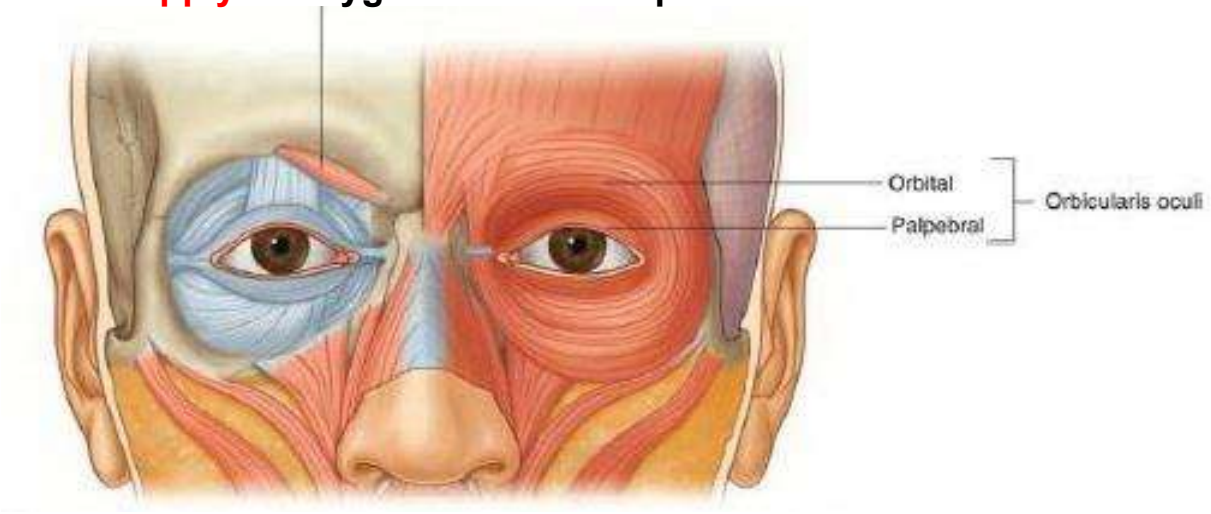


Figure (3): Orbicularis occuli muscle

2.Muscles of the lips andcheeks:

Orbicularisoris, buccinators and other muscles

Orbicularis oris:

Itsurroundsthe **oral orifice**
Action: Closes **mouth**, **compresses** and **protrudes lips** as(**whistling expression**).
Nerve supply: Buccal branch of **facial** nerve

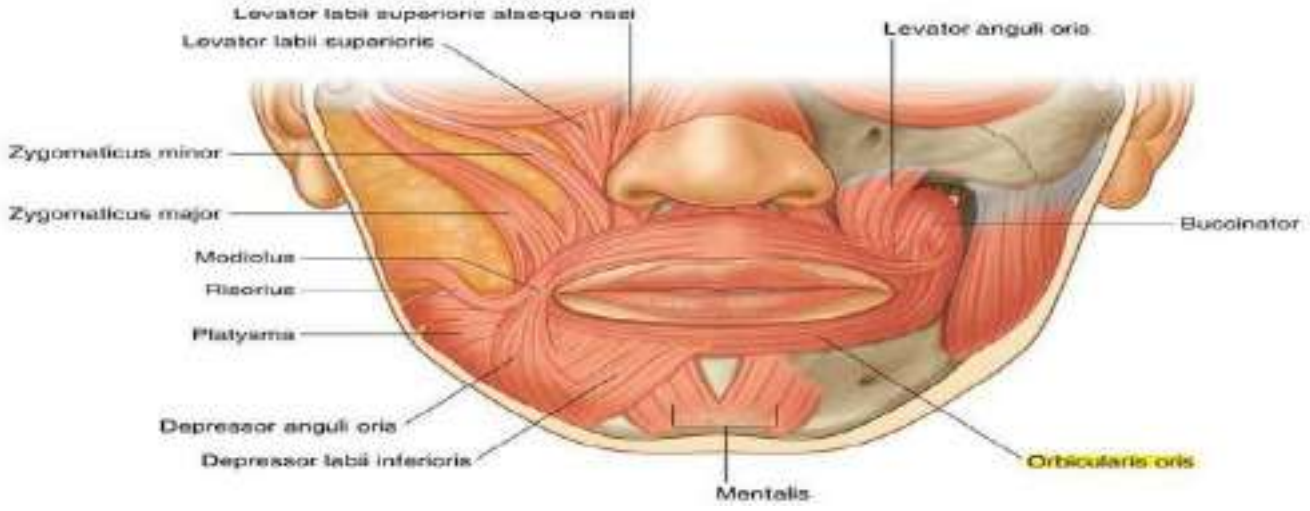


Figure (4): Orbiccularis oris muscle

Buccinator muscle:

Itforms the **muscular basis** of the **cheek**, **filling** the **interval** between the and **mandible**. **Action:** compress the **cheek** against the **molar teeth** keeping the **central** in the oral cavity and **preventing** it from **escaping** into the **Nerve supply: Buccal** branches of the **facial** nerve

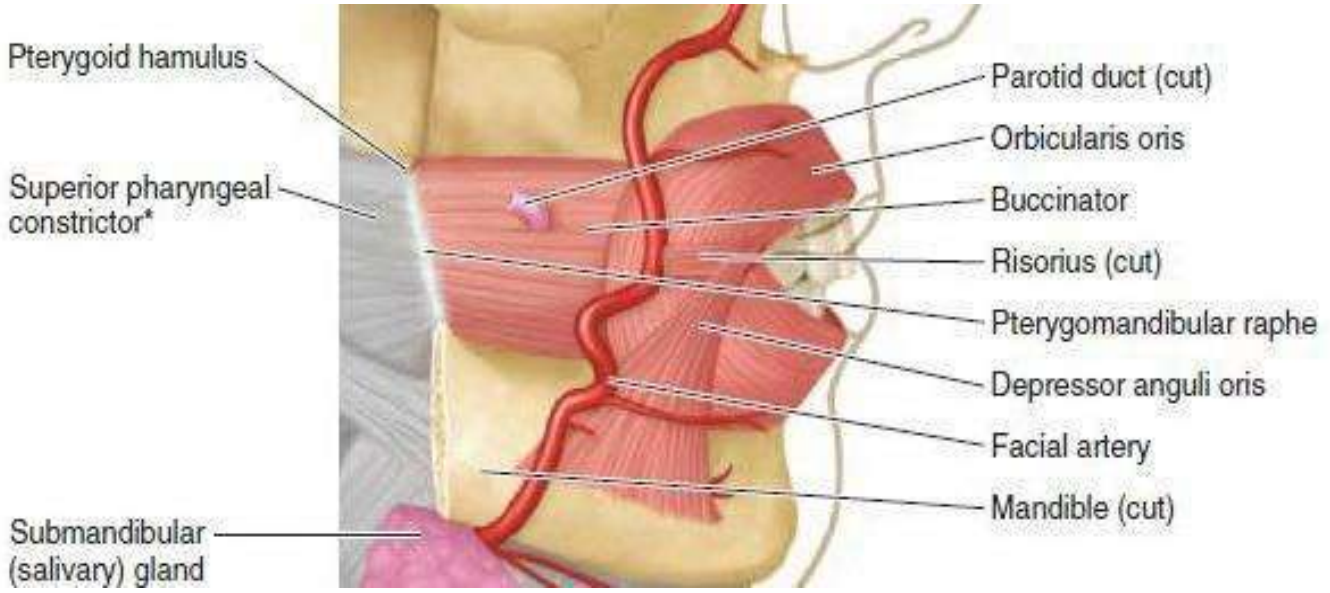


Figure (5): Buccinator muscle

Head

Musclesof the head:

It includes: **Muscle** ofthe **scalp**, **face, temporal** region muscle (**temporalis** muscle), **infratemporal** region, **palate**, **tongue**, **pharynx** and **larynx**.

Temporal Regionmuscle:

Is a **space** lies at the **lateral** sides of the skull **below** the **temporal lines**, it is covered **temporal fascia,**

Temporalismuscle:

It is a **fan-shaped**muscle that **fills** the **temporal fossa** with

A. **anteriorfibers**that have a **vertical** orientation,
B. **midfibers**havean **oblique** orientation,
C. and **posteriorfibers** have **more** of a **horizontal** orientation.

Action:

1. **Anterior** and **mid** fibersofthetemporalismuscle **elevate the mandible.**
2. **Posterior** fibers **retract the mandible**.It alsocontributesto **side to side grinding movement**.

Nerve supply: Itissuppliedbythe **deep temporal** nervewhich isa **branch** of **anterior division** ofthe **mandibular** nerve.

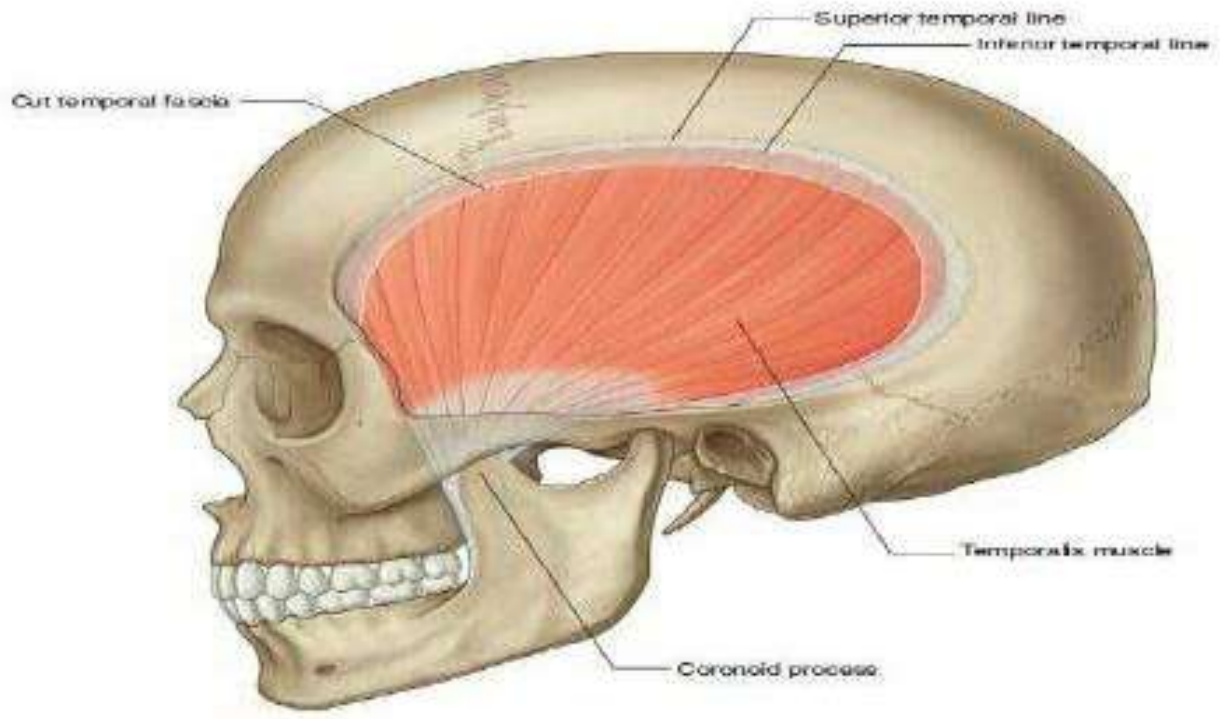


Figure (6): Temporalis muscle

Infratemporal Region Muscles:

It lies **inferior** tothetemporal region,between **ramus** of **mandible laterally** and **wall** of **pharynx medially,** it contains **2 muscles** (**medial pterygoid** and **laterl pterygoid** muscles).

Medial Pterygoid:

- Is a **thick rectangular** muscle with a **superficial** head and a **deep** head.
- The **deep** head of the medial pterygoid is **larger** than the **superficial** head and it lies **deep** to **masseter muscle.**

Action:

- The medial pterygoid muscle functions to **assist** with **elevation** and **protrusion** of the **mandible**.
- It also **assists** the **lateral pterygoid** muscle with **side to side mandibular** motion.

Nerve supply: It is innervated by the **branch** of the **main trunk** of the **mandibular** nerve.

Lateral Pterygoid:

It is a **short thick** muscle with **twoheads ,upper** and **lower** head.
Action: depression of the **mandible**, it also **assists** in **protrusion** (**protraction**) and **side to side** movement of the mandible with **help** of **medial** pterygoid muscle.
Nerve supply: it is supplied by a **branch** of the **anterior division** of the **mandibular** nerve.

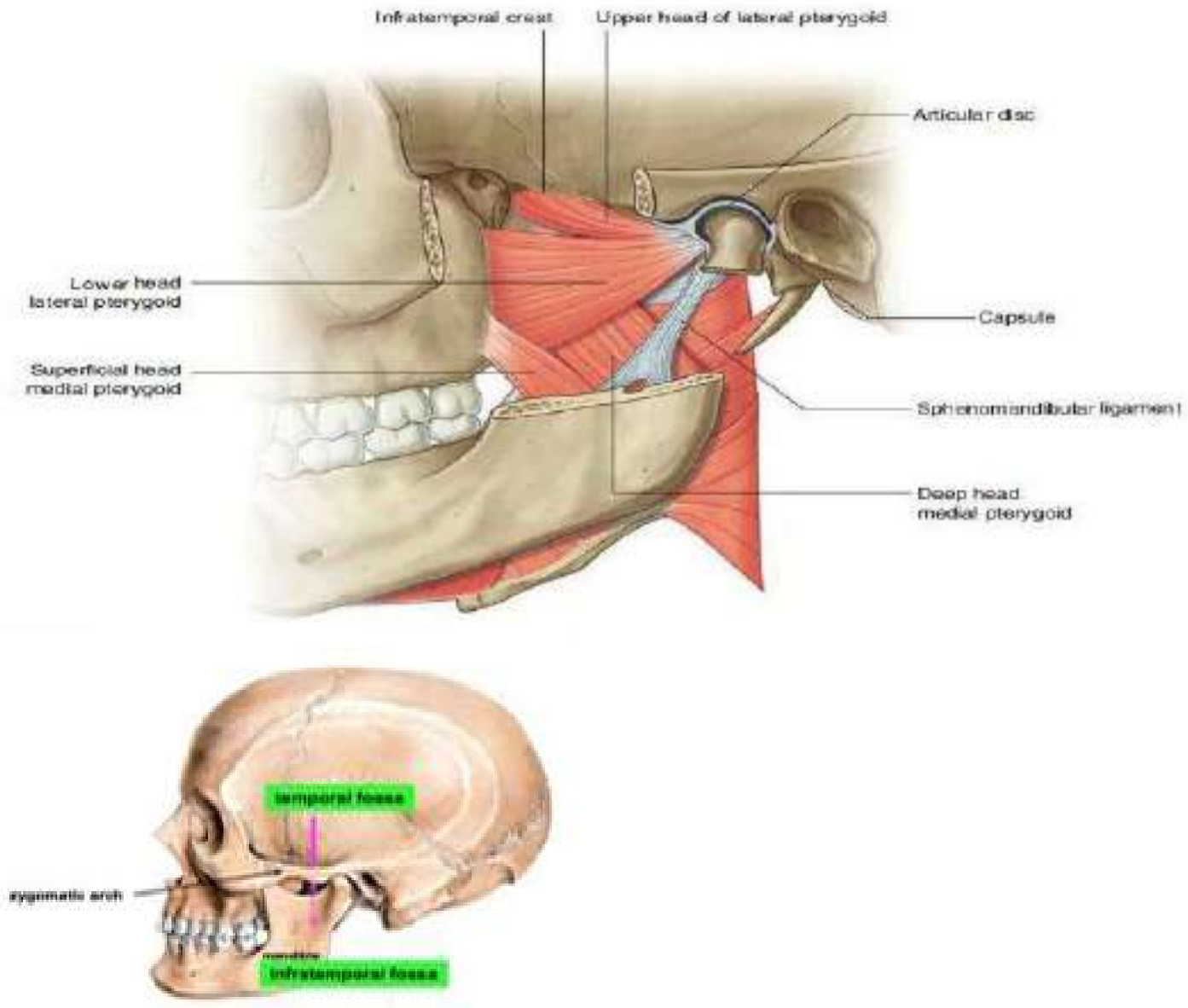


Figure (7): Medial and lateral pterygoid muscle

Masseter muscle:

- It lies at the **lateral** wall of the **infratemporal region**, with the **previous 3** muscles; they are called **muscles of mastication.**
- It is a **rectangular** muscle that **covers most** of the **lateral aspect** of the **ramus**.

Action:

- Its **major function** of the masseter muscle is to **elevate the mandible, approximate** the **teeth**.
- The **superficial** fibers **protrude** the mandible and the **deep** fibers are **important stabilizers** of the **TMJ** (**Tempromandibular joint**). **Nerve supply:** the **anterior division** of the **mandibular** nerve.

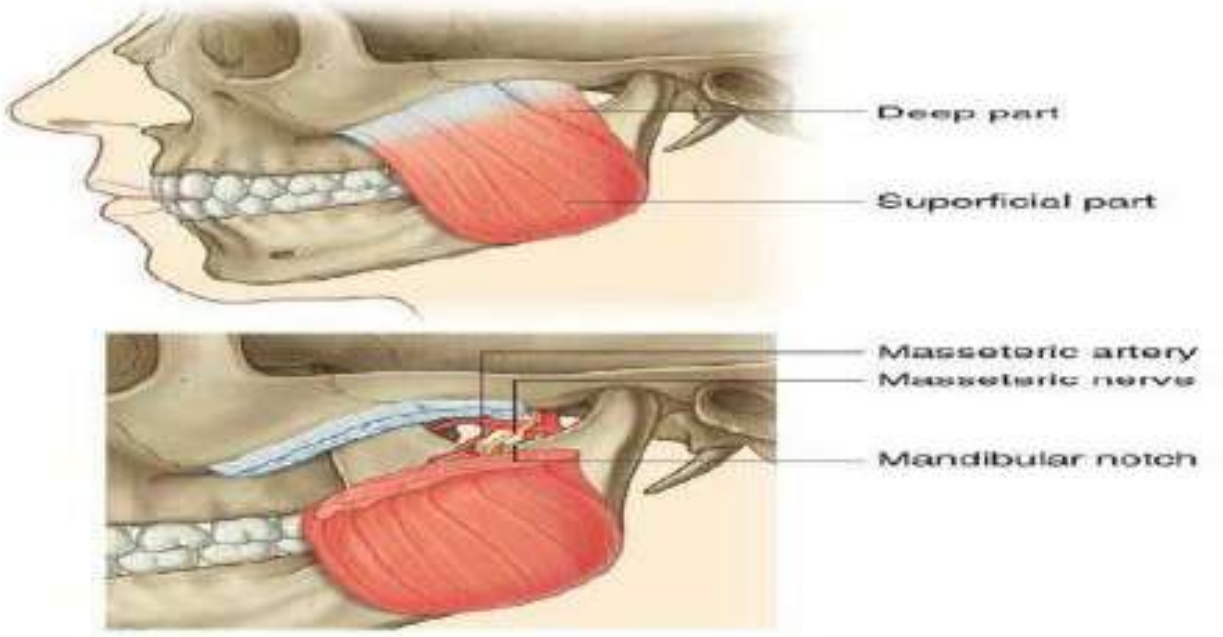


Figure (8): Masseter muscle

Tempromandibular joint (TMJ):

Type: synovial joint (**hing**)
Articulation: between **articular fossa** and **articular tubercle** of the **temporal bone** and **head of mandible**, **articular surfaces** are covered by **fibrocartilage**.
Articular disc: it divides the joint into **2 parts**, **upper** and **lower** parts, it **completely intervenes** between **temporal bone** and **head of mandible**, this disc moves **forward** and **backward** with the **head of mandible** during its **protraction** and **retraction**.
Capsule: it originates from the **border** of the **mandibular fossa, encloses** the **articular tubercle** of temporal bone and **inserts** at the **neck** of **mandible** above the **pterygoid fovea.**
Synovial membrane: it **underlines** the **capsule** of the **joint**.
Ligaments:

1. **The lateral temporomandibular ligament:**
   - it is located on the **lateral** aspect of the **capsule**;
   - its **function** is **preventing** the **lateral** or **posterior** displacement of the **condyle**.
2. **The stylomandibular ligament:**
   - it arises from the **styloid** process and attaches to the **mandibular angle.**
   - It is **responsible** for **allowing** the mandible to **protrude**.

3.**Thesphenomandibularligament:**

- it **stretches** between the **spine** ofthe **sphenoid** bone and the **lingula** of the **mandible**. of
- It **contributes** to the **limitation** **extensiveprotrusivemovements** and **jaw opening.** **masseteric**

Nervesupply:auriculotemporal nerveand branch of the **mandibular** nerve.

Movements: produced by the musclesof **mastication**, and the **hyoidmuscles** .

1- **Depression(openingofthemouth**):the **two-lateral** pterygoid, **digastric**,
2-
3- **geniohyoid** and **mylohyoids** muscles.
4- **Elevation (closureofthemouth temporalis masseter** and **medial** pterygoid. **Protrusion**

lateral and(**medial** pterygoids.), with help of **masseterdeep fibers**

and **digastric** muscle and **geniohyoid** muscle.

5-Sideto sidech(ewinglateral and **medial** pterygoids of **bothsides** when **Retraction temporalis posteriorfibers acting alternately.**

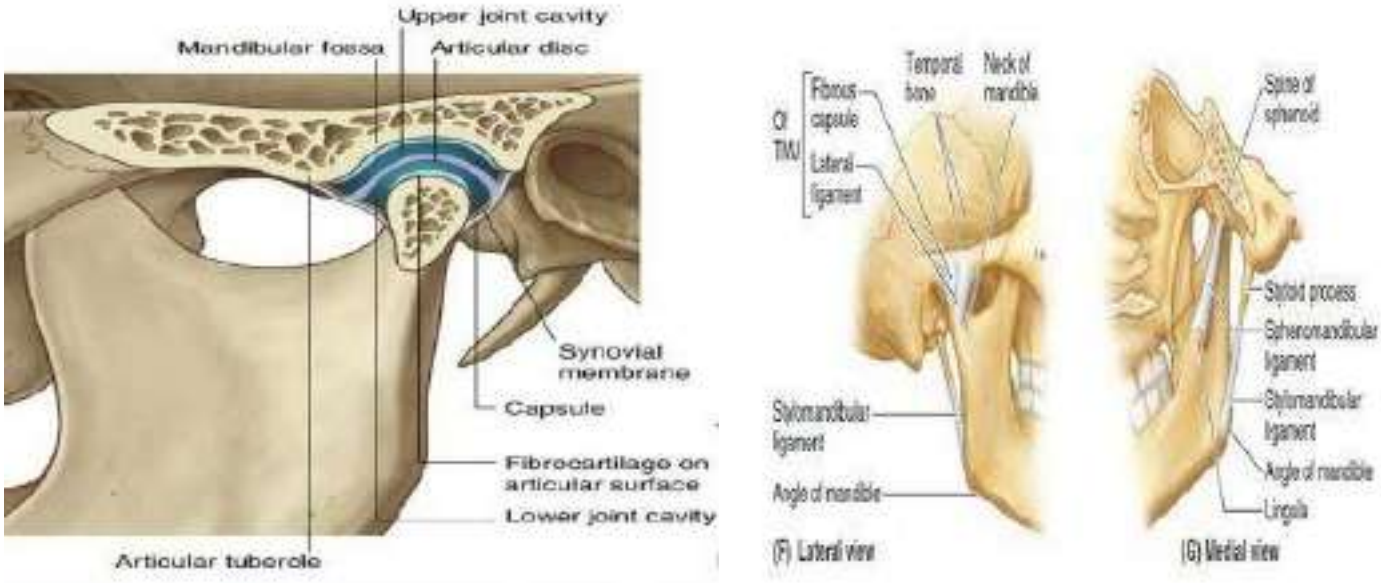


Figure (9): Ligaments of the TMJ

Neck

Triangles of the neck:

Neck isdividedinto **twomaintriangles anterior** and **posterior** by the **sternomastoid** muscle, which is a **large**,**two-headed** muscle present at the **side of neck** ,itsaction **depends** on whether the muscle **actsaloneor together.**

1- **Unilateral contraction:** produces **lateral flexion** of the neck on the **same** (**ipsilateral**) side and **lateral rotation** of the **head** to the **opposite** (**contralateral**) side.

2- **Bilateral contraction:** produces **flexion** of the **neck**, **drawing** the **head towards** the **chest**.

3- **When the head and neck are fixed:** it **elevates sternum** and **clavicle**.

Nerve supply: by the **accessory spinal nerve**, **ventral ramus** of **C2, C3.**

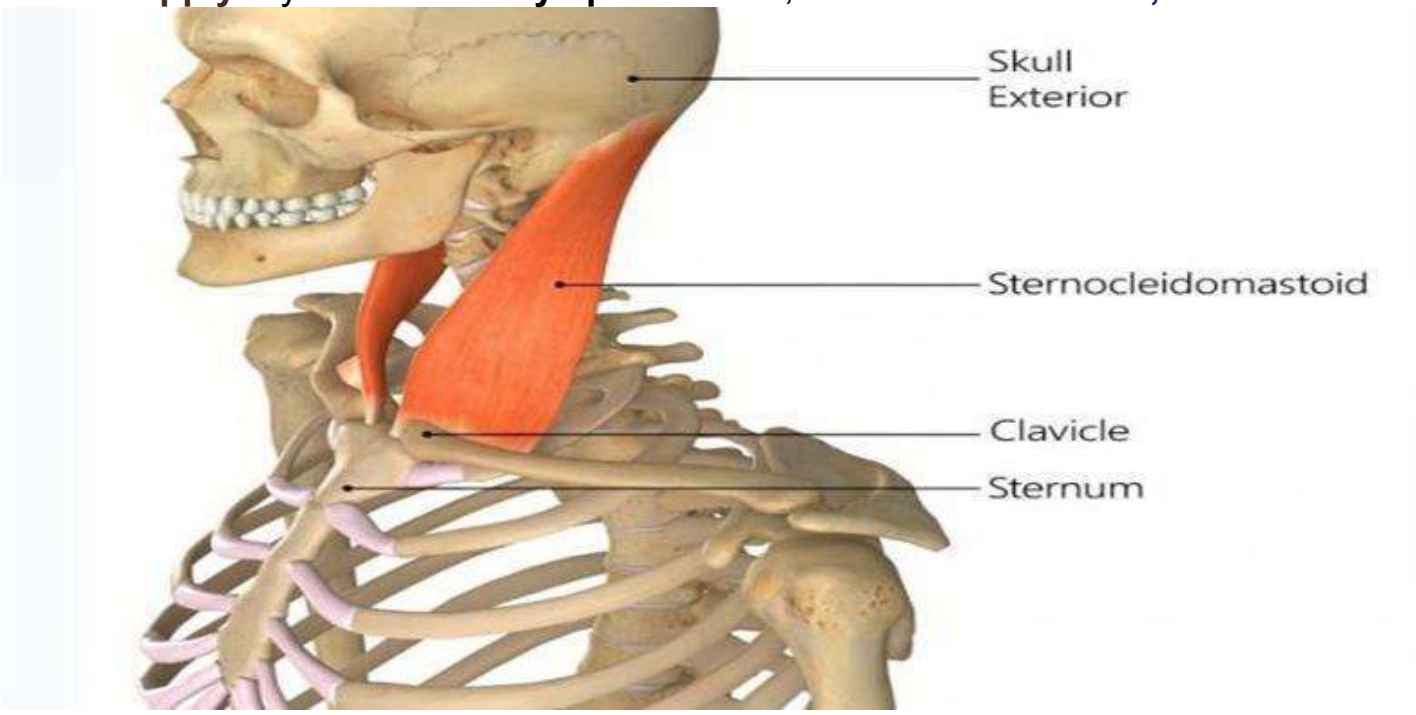


Figure (12): Sternocleidomastoid

Posterior triangleoftheneck:

- **Boundaries:** it is bounded **anteriorly** by the **posteriorborder** of the **sternocleidomastoid** muscle and **posteriorly** by the**anteriorborder** of the **trapezius** muscle, **Inferior** by the **middle third** of the **clavicle**.
- can be **divided** by **inferior belly of omohyoid** muscle into **2triangles** ; **omoclavicular** and **occipital triangles.**

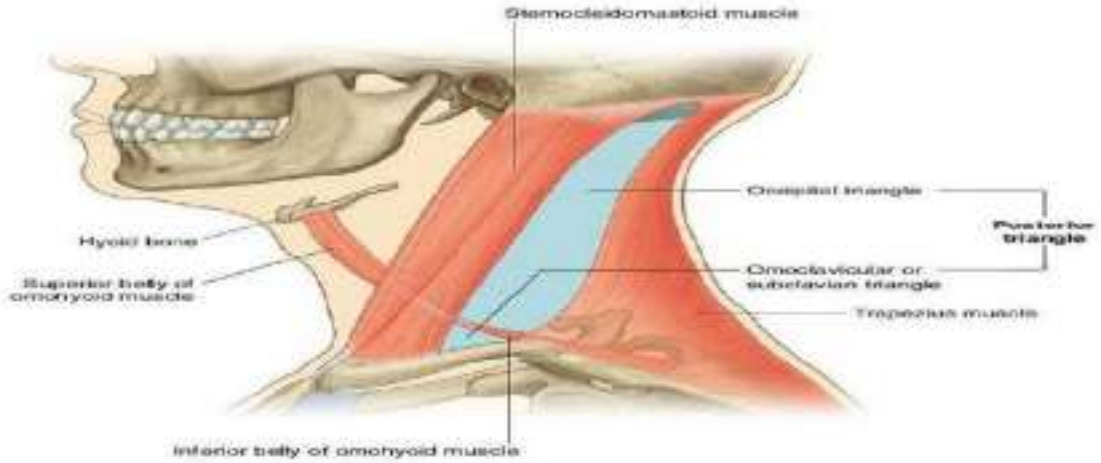


Figure (13): Posterior triangle of the neck

Anterior triangle of the neck:

- **Boundaries:** it is bounded **superiorly** by **inferior border** of **mandible** and **medially** by the **midline of neck, laterally** by **anterior border** of **sternocleidomastoid** muscle It is **divided** into: **muscular** triangle,
- **carotid** triangle, **submandibular** triangle and **submental** triangle.

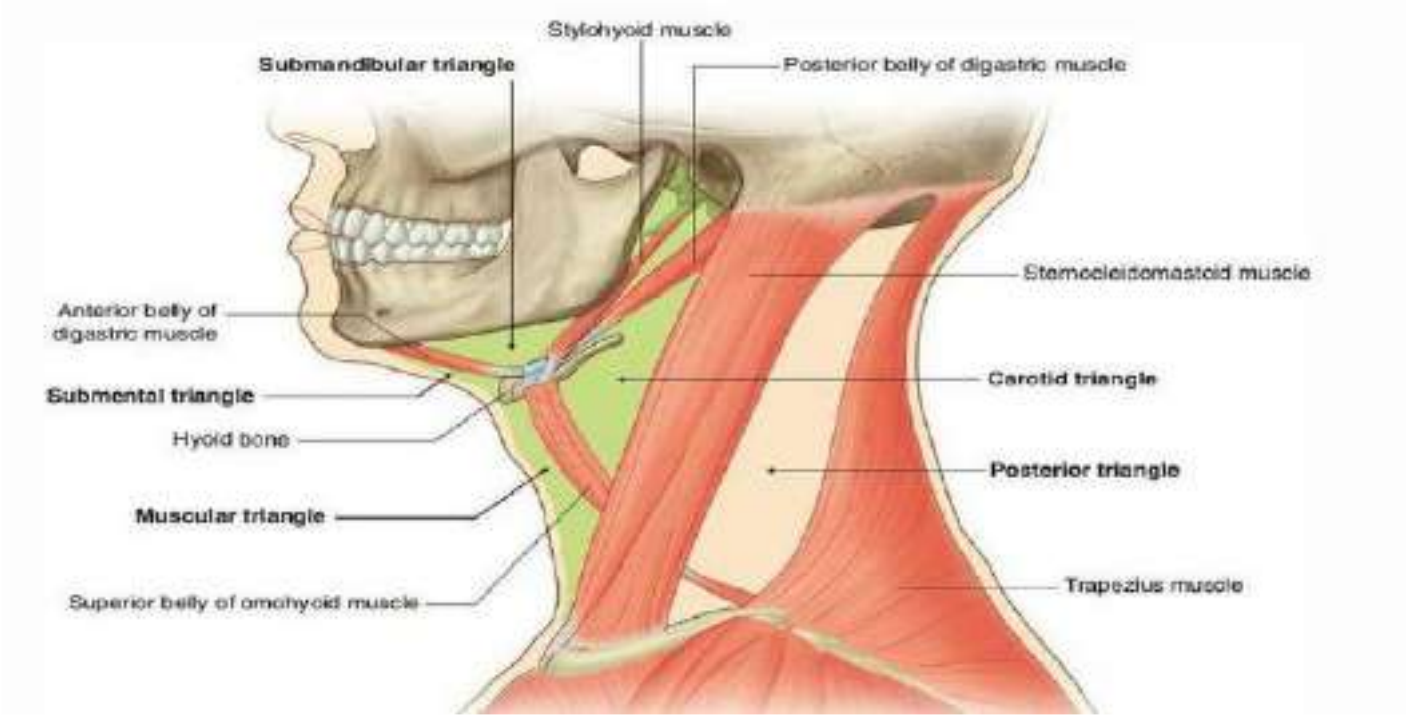


Figure (14): Anterior triangle of the neck

Muscles of the neck:

Can be classified into **2 groups; suprhyoid** and **infrahyoid** muscles

1- Suprahyoid muscles:

Muscles which located **above hyoid bone**; **digastric**, **mylohyoid**, **geniohyoid** and **stylohyoid** muscles.

Digastric muscle:

Itis a **small** muscle situated **below** the **mandible**,
Action: depresses mandible, **raises** and **steadies hyoid**
Nerve supply: it has **dual innervation**;

- the **anterior belly** is **innervated** by the **nerve to mylohyoid**, a **branch** ofthe **inferior alveolar nerve** (**mandibular nerve posterior division**),
- While the **posterior belly** receives innervation from the **digastric branch** of the **facial nerve.**

Mylohyoid muscle:

Action: it **elevates hyoid** to the **floor** of **mouth** and **tongue**.
Nerve supply: mylohyoid n.

Geniohyoid muscle:

Action: pulls hyoid **ant-sup**, **shortens** the **floor of mouth**, **widens pharynx** but If the **hyoid** bone is **fixed**, the **geniohyoid** muscle **depresses the mandible. Nerve supply: spinal nerve C1** via the **hypoglossal** nerve.

Stylohyoid muscle:

Action: it **elevates** and **retracts** hyoid
Nerve supply: facial n.

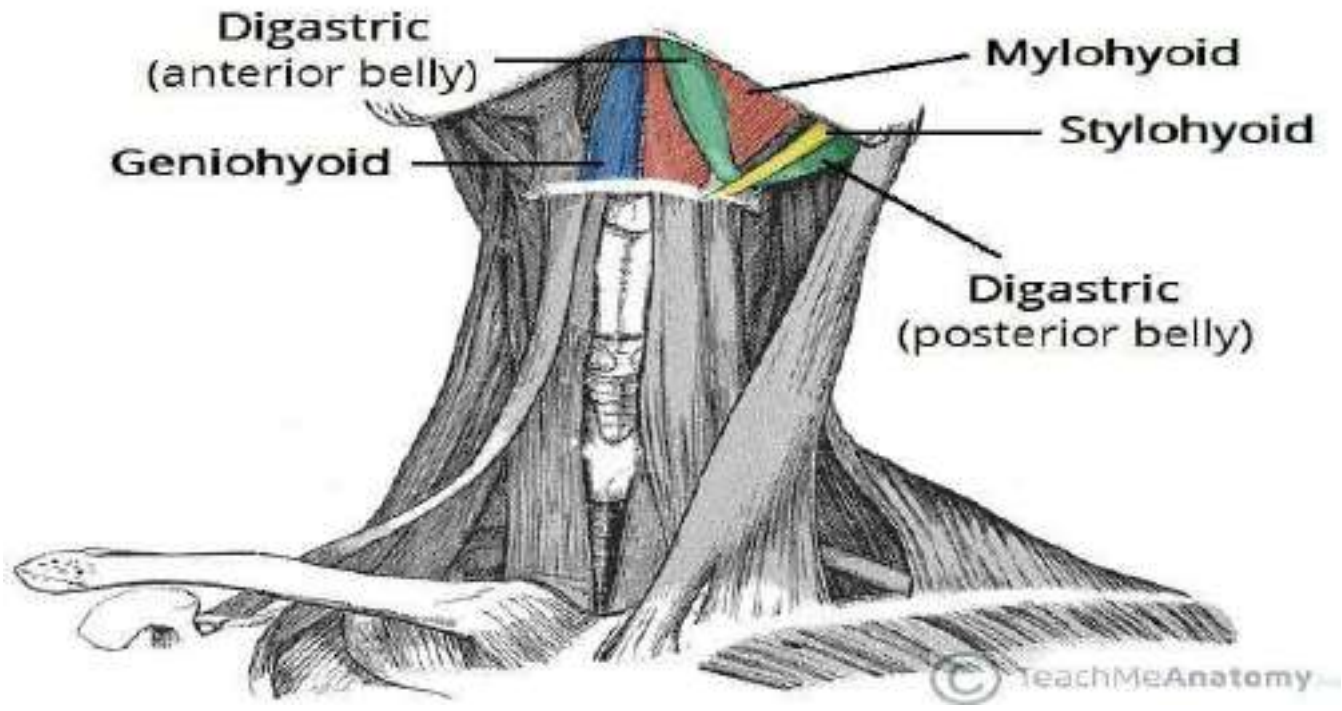


Figure (15): Suprahyoid muscles

2- Infrahyoid muscles:

Muscles which **located below** the **hyoid** bone; **sternohyoid**, **omohyoid**, **sternothyroid** and **thyrohyoid** muscles. **Action: all** the muscle **depresses** the **hyoid bone** and **larynx** after it has been **elevated** by the **suprahyoid muscles. Nerve supply: all** the muscle supplied by the **ansa cervicalis (C1-C3) EXCEPT thyrohyoid** supplied by **C1** joining **hypoglossal** nerve.

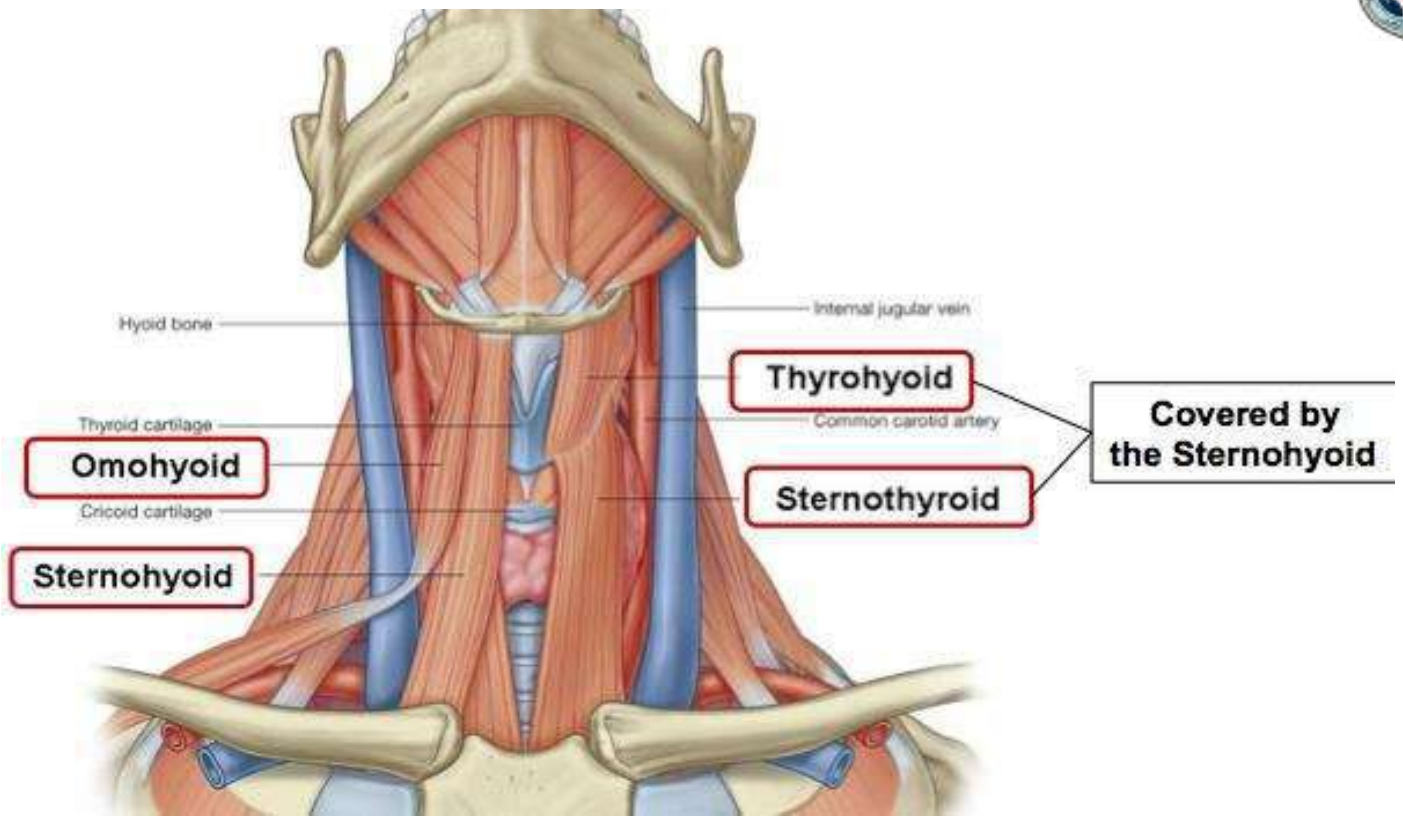


Figure (16): Infrahyoid muscles

5 Nerves of head and neck

- Innervation of the **scalp**
- Innervation of the **face**
- **Cervical** plexus
- **Cranial** nerves

Innervation of the scalp

Sensory:

1- Anterior to the ear and vertex: it is supplied by **supratroclear**, **supraorbital**, **zygomaticotemporal** and **auriculotemporal** nerves (branches of **trigeminal nerve**)
2- Posterior to the ear and vertex: it is supplied by **cervical** nerves (**great auricular**, **lesser occipital, greater occipital** and **third occipital** nerve).

Motor:

To **occiipitofrontalis** muscle: **two** branches of **facial nerve: temporal** and **posterior auricular** nerves.

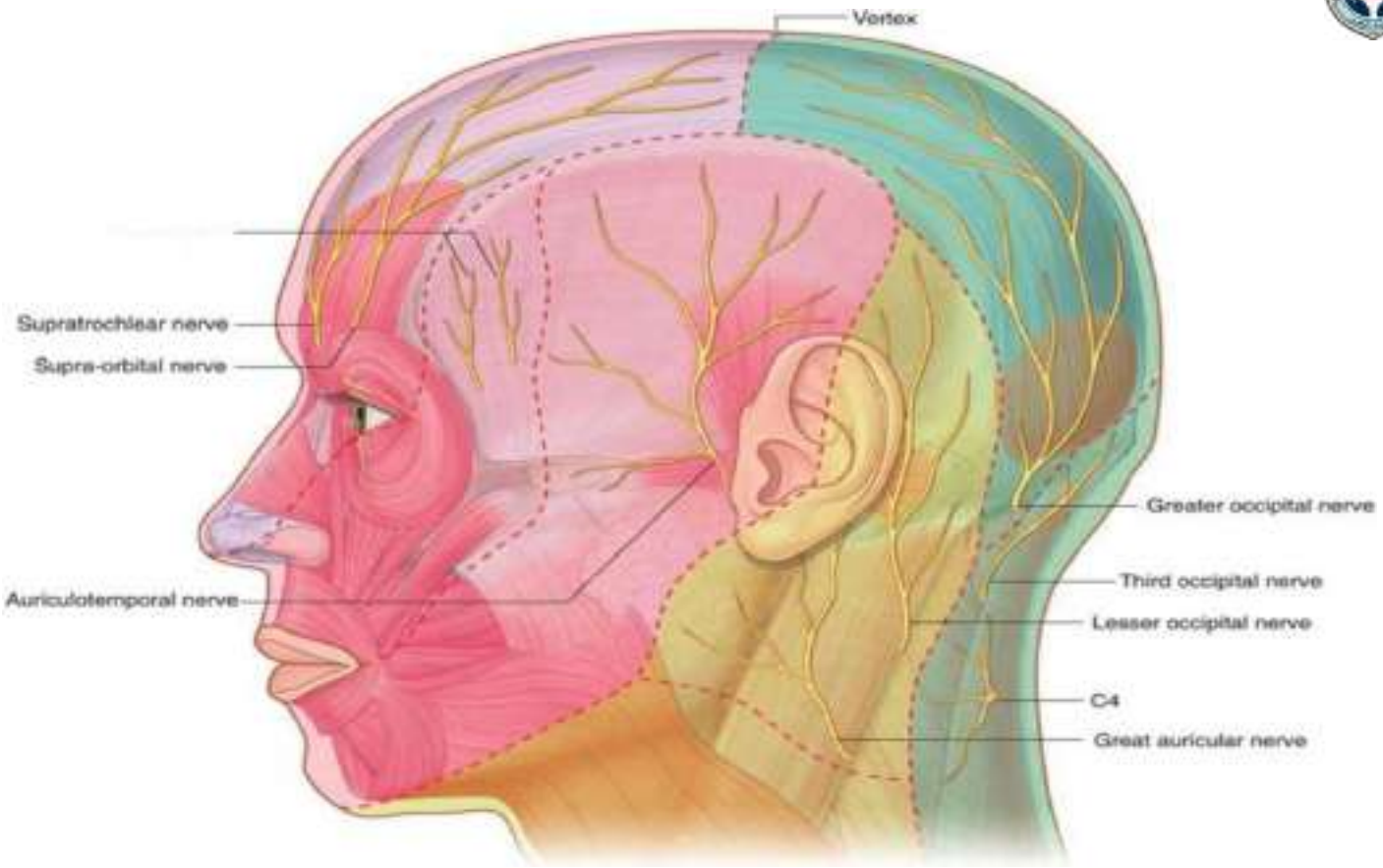


Figure (17): Sensory innervation of the scalp

Innervation of the face

Sensory:

Trigeminal nerve:

Itgives**3** branches (**ophthalmic**, **maxillary** and **mandibular**), it supplies **all** the **face EXCEPT** area over **angle** of **mandible** and **parotid gland** supplied by **greater auricular nerve (C2-3).**

A. Ophthalmic Nerve:

1- **Supraorbital nerve:** is a **terminal** branch of the **frontal** nerve.
2- **Supratrochlear nerve:**
3- **External nasal branches:**
4- **Infratrochlear nerve:**
5- **Palpebral branch of lacrimal nerve**

B. Maxillary Nerve:

Intermediate division of **trigeminal** nerve

1. **Infraorbital nerve:** it supplies the **skin** on **lateral** part **nose**, **upper lip** and **lower eyelid**
2. **Zygomaticofacial** n. (**small branch**) supply **skin** over **zygomatic prominence**
3. **Zygomaticotemporal** nerve supplies **skin** over **temporal region**

C. Mandibular Nerve:

Inferior division of **trigeminal** nerve

1. **Buccal** n. Supply **skin** of **cheek** over **buccinator**, **mucous membrane** lining and **post** part of **gingival surface.**
2. **Auriculotemporal** n. Supplies parts of **auricle**, **external acoustic meatus**, **tympanic membrane** and skin in **temporal** region
3. **Mental** (**chin**) supply skin of **chin** and **lower lip.**

Motor:

Facial nerve, it gives **5** branches **(temporal**, **zygomatic**, **buccal**, **marginal mandibular and cervical).**

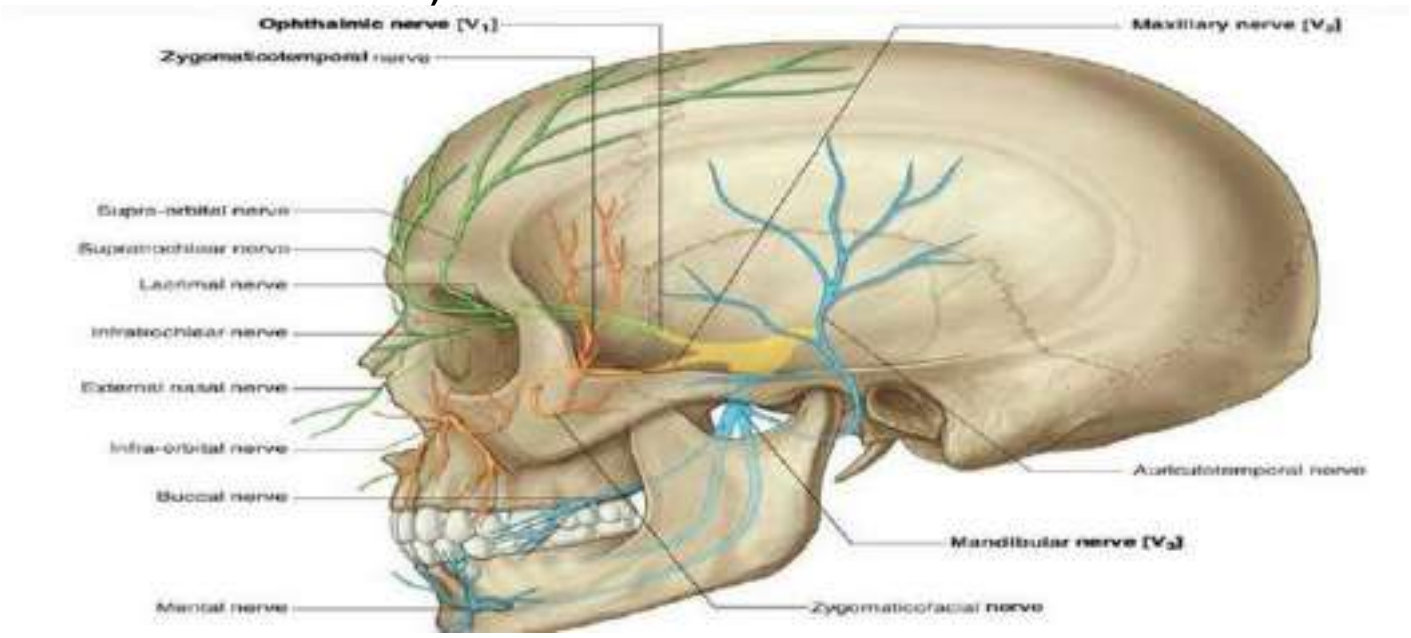


Figure (18): Sensory innervation of the face

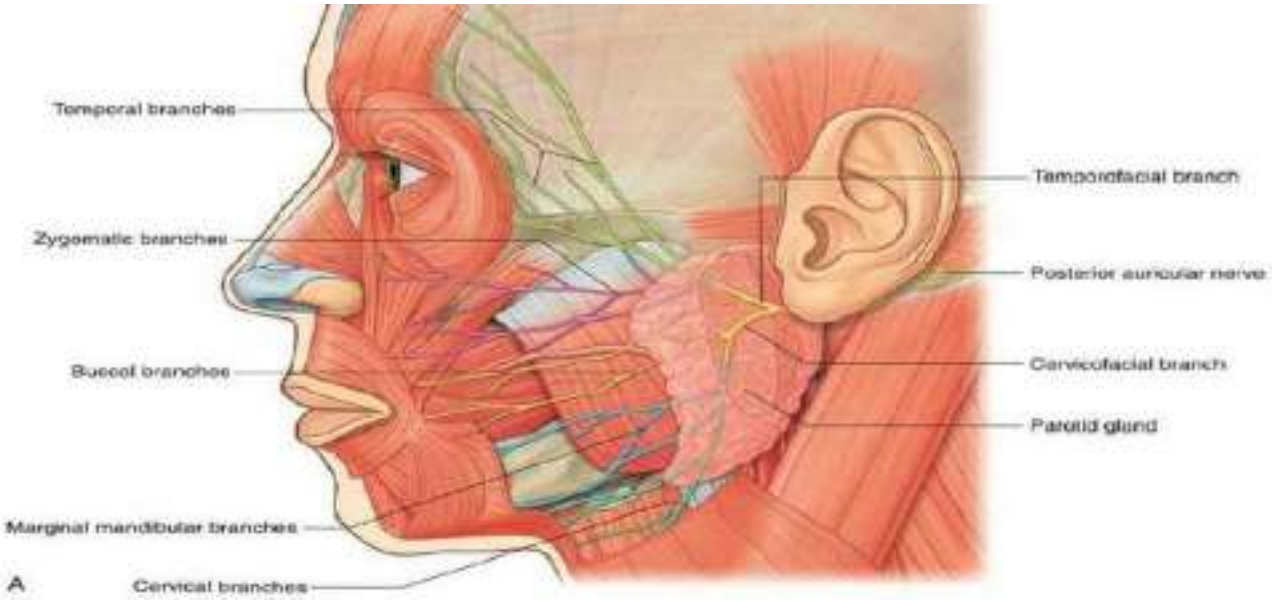


Figure (19): Motor innervation of the face (distribution of facial nerve)

Cervical plexus

Formation: The cervical plexus is formed in the **neck**, and lies **deep** to the **sternocleidomastoid** muscle, and **anterolateral** to the **levator scapulae** and **middle scalene muscles**
Roots: formed by **ventral** ramus of **upper 4 cervical** nerves
Branches:
A. Sensory branches:

1. **Lesser Occipital** nerve (formed by **C2**)
2. **Great Auricular** nerve (formed by **C2** and **C3**)
3. **Transverse Cervical** nerve (formed by **C2** and **C3**)
4. **Branches** from the **loop** between **C3** and **C4**
5. **Supraclavicular** nerves formed by (**C3** and **C4**)

B. Motor branches:

1. to **infrahyoid** muscle by **ansa cervicalis**
2. and **diaphragm** by **phrenic nerve**,
3. **prevertebral**, **scalenus** muscle and **levator scapula** by (**C3, C4**),
4. to **trapezius** (**C3, C4**)
5. and **sternomastoid** muscle (**C2, C3**).

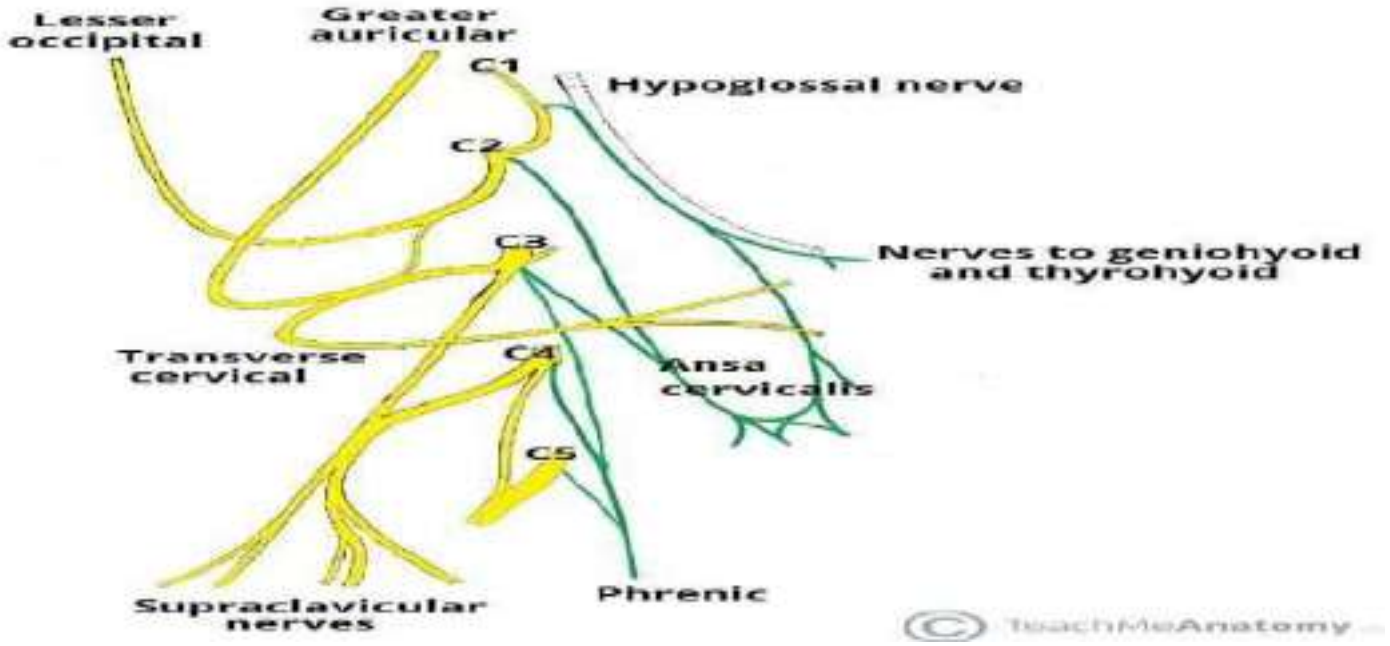


Figure (20): Distribution of cervical plexus

Cranial nerves (CN)

- They are part of **peripheral nervous system (PNS)** andpass through **foramina** and **fissures** ofthe **skull**.
  All cranial nervesarisefrom **brain** **EXCEPT** the **accessory** nerve.

1. Olfactory nerve:

- It arises from **olfactory receptor** nerve cells in the **olfactory mucous** membrane; it is **situated** in the **upper** part of the **nasal cavity above** the **level** of the **superior concha**, it **terminates** at **temporallobe of brain.**
- The **olfactory** nerves are the **only** cranial nerves to **enter the cerebrum directly.**

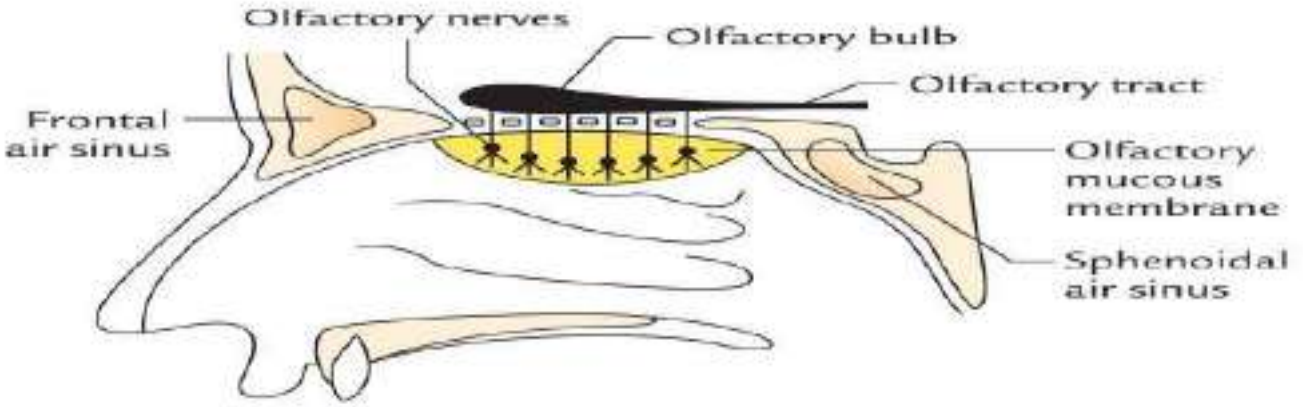


Figure (21): Olfactory nerve

2. Optic nerve:

- It is formed by **axon** of cell of **ganglionic** layer of the **retina**
- It enters **skull** through **optic canal,** later on, it is **fused** withthe other **optic nerve** to form **optic chiasm. Type of fibers: sensory (Vision).**

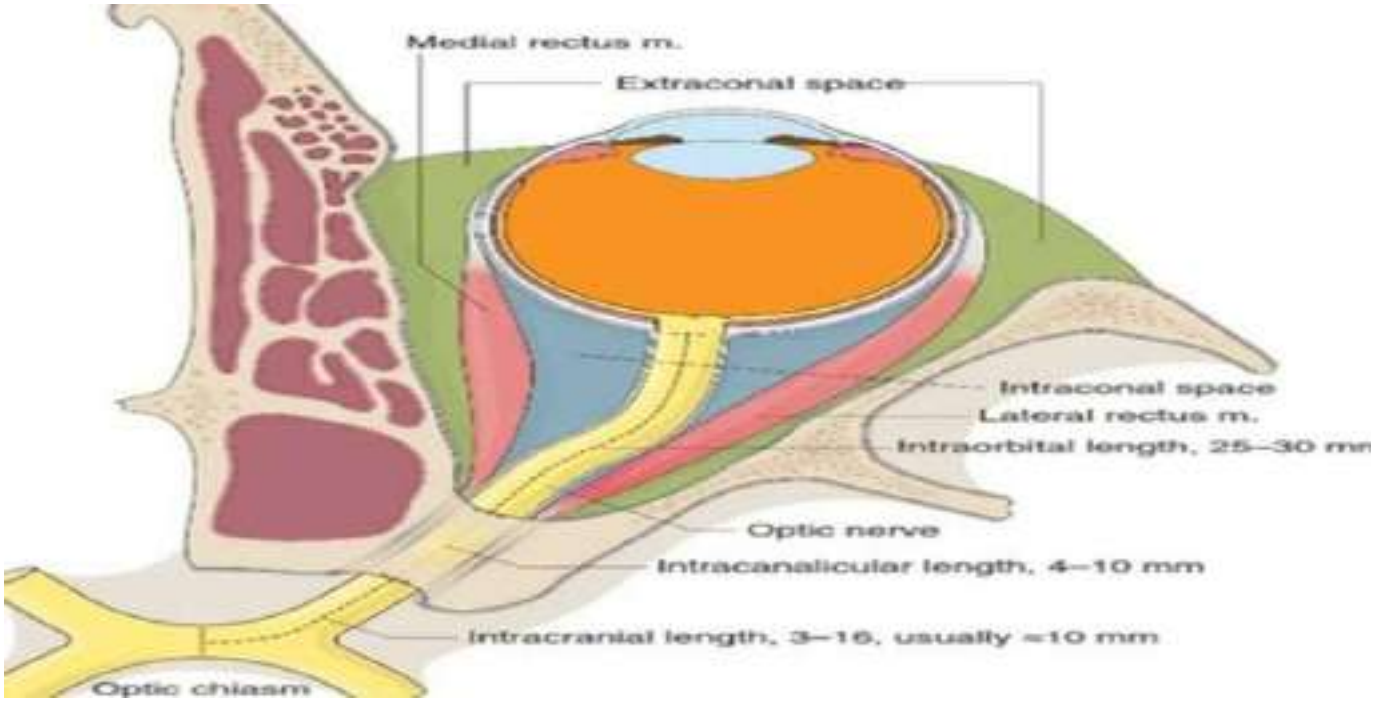


Figure (22): Optic nerve

3.Oculomotor nerve:

- It is originated at **mid brain,**
- It **leaves** cranial cavity through **superior orbital fissure**.
- **CN III** is the **chief motor** nerve to the **ocular** and **extra-ocular** muscles.

Type of fibers: Motor

It is **divided** into **two terminal** branches

1. **Superior division:** it supplies **superior rectus** muscle and **LPS** (**levator palpabrae superioris**) muscle.
2. **Inferior division:** it supplies **medial rectus** muscle, **inferior rectus** muscle and **inferior oblique** muscle, this division gives a **parasympathetic** branch to the **ciliary ganglia** which give **post ganglionic** fiber to **sphincter pupillae** muscle of **iris** and **ciliary** muscles.

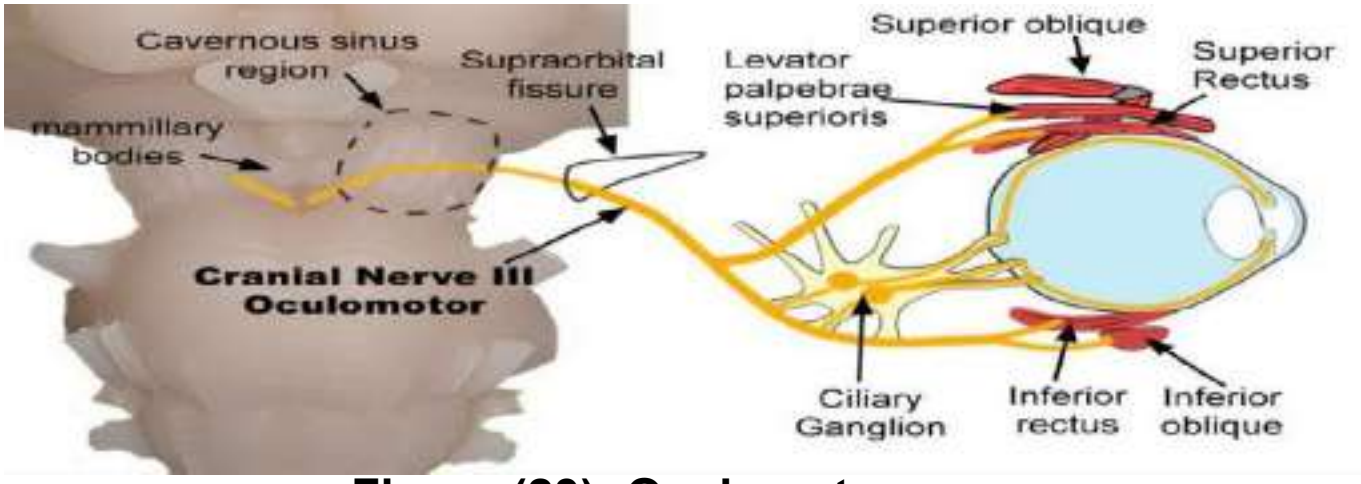


Figure (23): Oculomotor nerve

4. Trochlear nerve:

- It originates from the **trochlear nucleus** ; it is the **only** cranial nerve tœmerge from the **dorsal surface** of the **brain stem** , It enters the **orbit** through**superior orbital fissure**.
- It supplies the **superior oblique muscle.**

Type of fiber: Motor

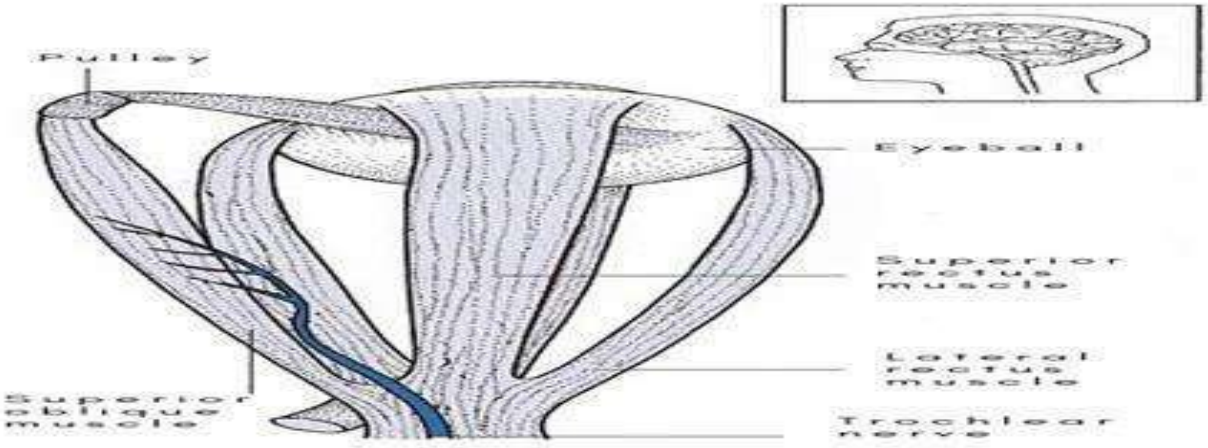


Figure (24): Trochlear nerve

5. Trigeminal Nerve:

Itis the **largest** cranial nerve; it emerges from the **lateral** aspect of the **pons** by a **large sensory** root and a **smallmotor** root.

Type of fiber: mixed(motor,sensory)

Roots:

1. **Sensory root:** it forms the **3 division** of **trigeminal** nerve (**ophthalmic**, **maxillary** and **mandibular**).
2. **Motor root:** it joins the **mandibular** nerve.

Branches:

1. Ophthalmic Nerve:

- The ophthalmic nerve is **purely sensory** it gives the **lacrimal**, **frontal**, and **nasociliary** nerves. It enters the **orbital cavity** through the **superior**
- **orbital fissure.**

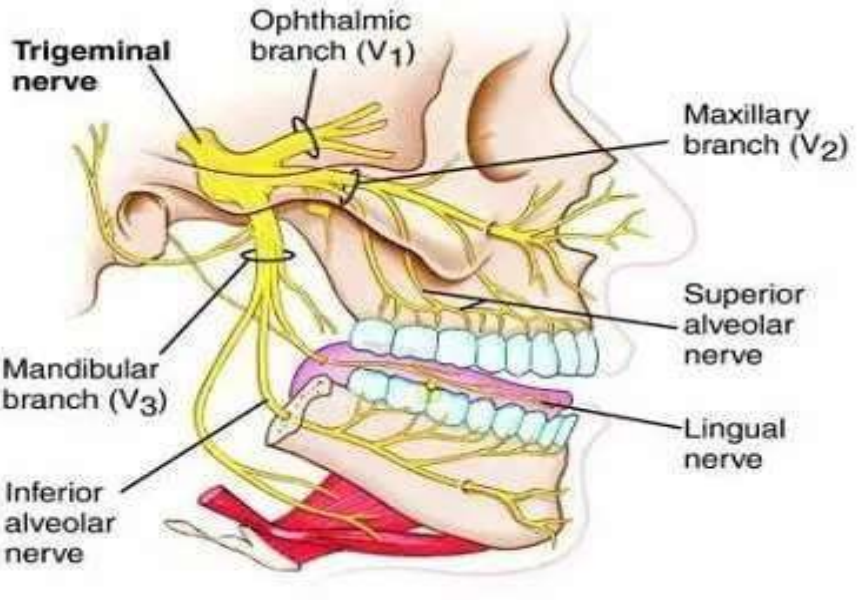


Figure (25): Trigeminal nerve

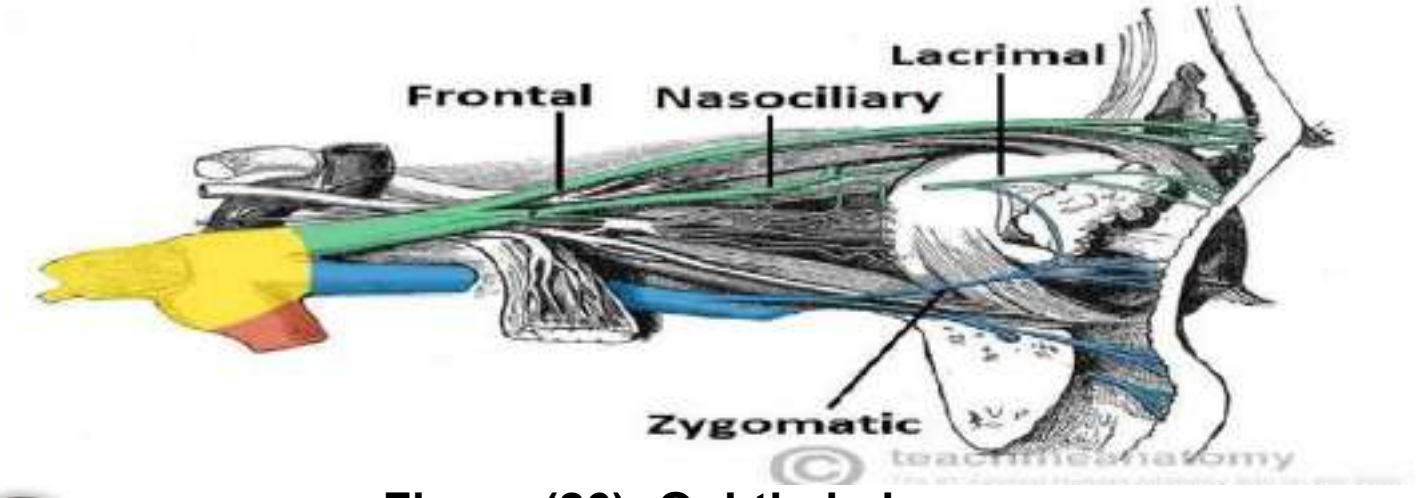


Figure (26): Ophthalmic nerve

2. Maxillary nerve:

It **leaves** the skull through the **foramen rotundum** and **continues** as **infraorbital** nerve.

Branches:

1. **Middle meningeal** nerve
2. **Zygomatic nerve:** it is divided into the **zygomaticotemporal** and the **zygomaticofacial**
3. **Ganglionic** branches
4. **Posterior superior alveolar** nerve
5. **Middle superior alveolar** nerve\
6. **Anterior superior alveolar** nerve:
7. **Branches** of **Pterygopalatine ganglia:**
8. **Infraorbital nerve: continuation** of **maxillary** nerve has **3 terminal branches: palperal, nasal and lacrimal**

3. Mandibular nerve :

Type of fibers: mixed nerve (motor and sensory)
Motor root and **sensory** root of the **trigeminal** nerve, it leaves the skull through the **foramen ovale**
Branches:

1. **From themaintrunk:Meningeal** branch, Nerve to the **medial pterygoid**
2. muscle:.

   From theAnteriorDivisionof the mandibular nerve:
   - **Masseteric** nerve, **deep temporal** nerves, Nerve to the **lateral pterygoid**
   - muscle,
     Buccal nerve, (it is the **only sensory** branch of the **anterior** division of the **mandibular** nerve).
3. **Branchesfromtheposterior division:** Auriculotemporal nerve, Lingual nerve, Inferior alveolar nerve:

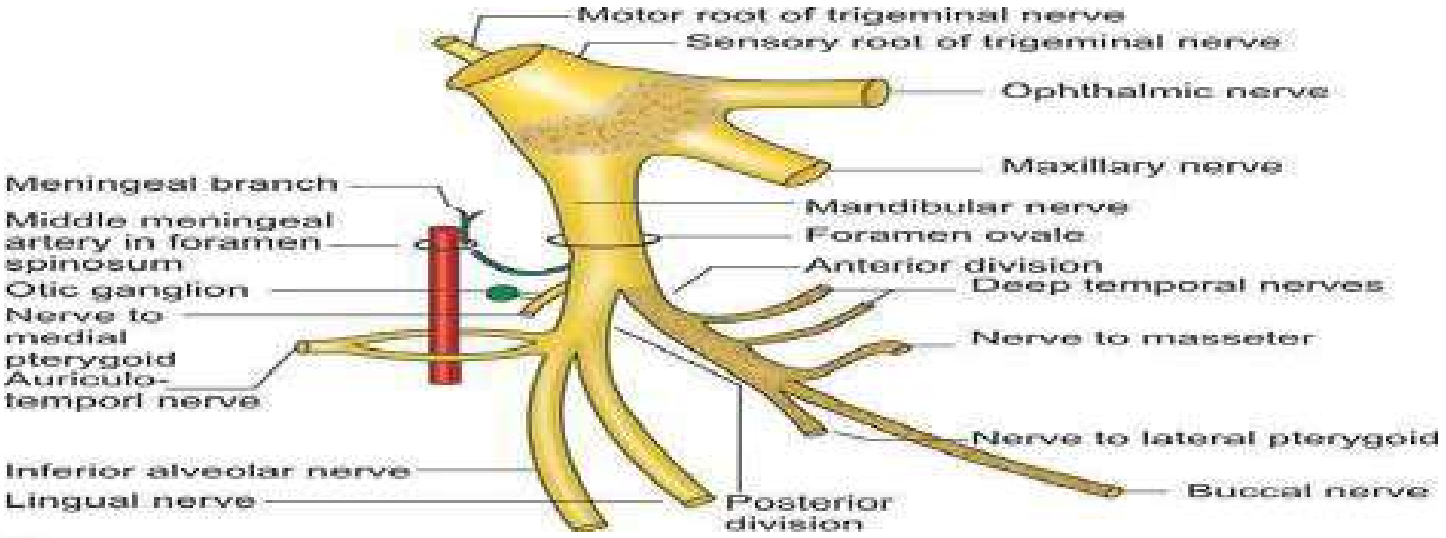


Figure (28): Mandibular nerve

6. Abducent nerve:

Type of fibers: motor
Nucleus: abducent nucleus in **pons**
Branches: it entersthe **orbit** through **superior orbital fissure** give **motor** branch to **lateral rectus** muscle.

7. Facial nerve:

Type of fiber: motor and sensory
Branches:

1- **Intracranial** branches:
- **Greater petrosal** nerve:
- Nerve to **stapedius**:
- **Chorda tympani:**

2- **Extra cranial** branches:
- **Posterior auricular** branch supplies **occipitalis** muscle.
- **Digastric** branch supplies **posterior belly** of the **digastric**.
- **Stylohyoid** branch supplies **stylohyoid** muscle.
- **Five terminal** branches to the muscles of **facial expression** (these are the **temporal**, **zygomatic**, **buccal**, **mandibular** and the **cervical branches**.

8. Vestibulocochlear nerve:

Type of fiber: sensory
Parts: 2 sets of fibers (**vestibular** and **cochlear**).
- The vestibulocochlear nerve **emerges** from the **junction** ofthe**pons** and **medulla** and enters the **internal acoustic meatus;** here it **separates** into the **vestibular** and **cochlear** nerves.
- For the **sense of equilibration** and **motion** , For the sense **hearing** ,

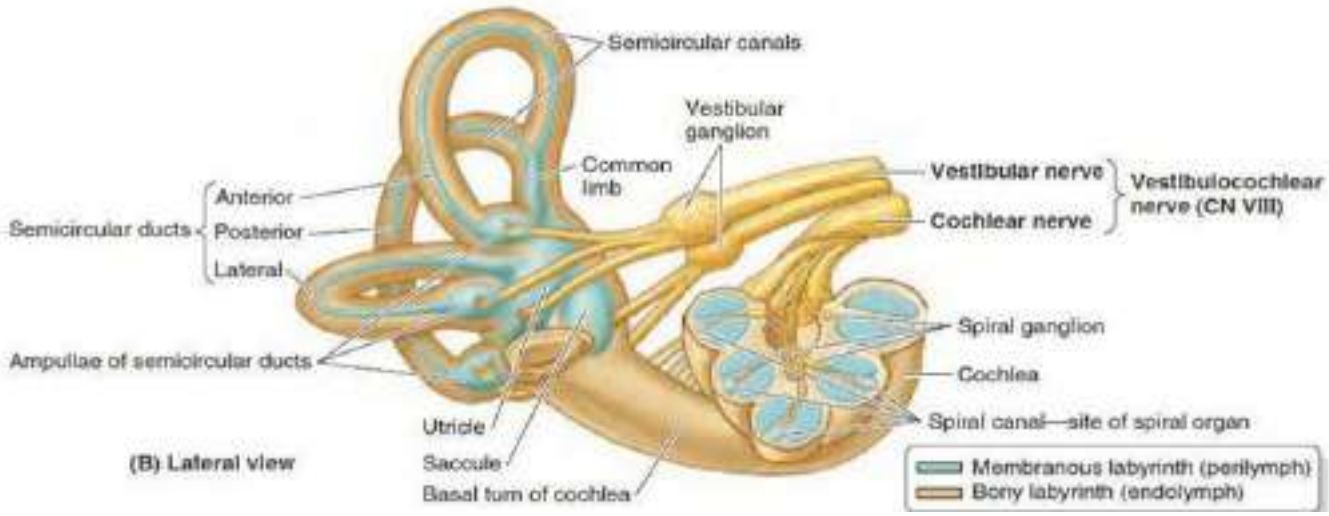


Figure (29): Vesibulocchlear nerve

9. Glossopharyngeal nerve:

Type of fiber: motor and sensory
Branches: tympanic branch: , **Carotid** branch , Nerve to the **stylopharyngeus** muscle, **Pharyngeal** branches, **tonsillar** branches:

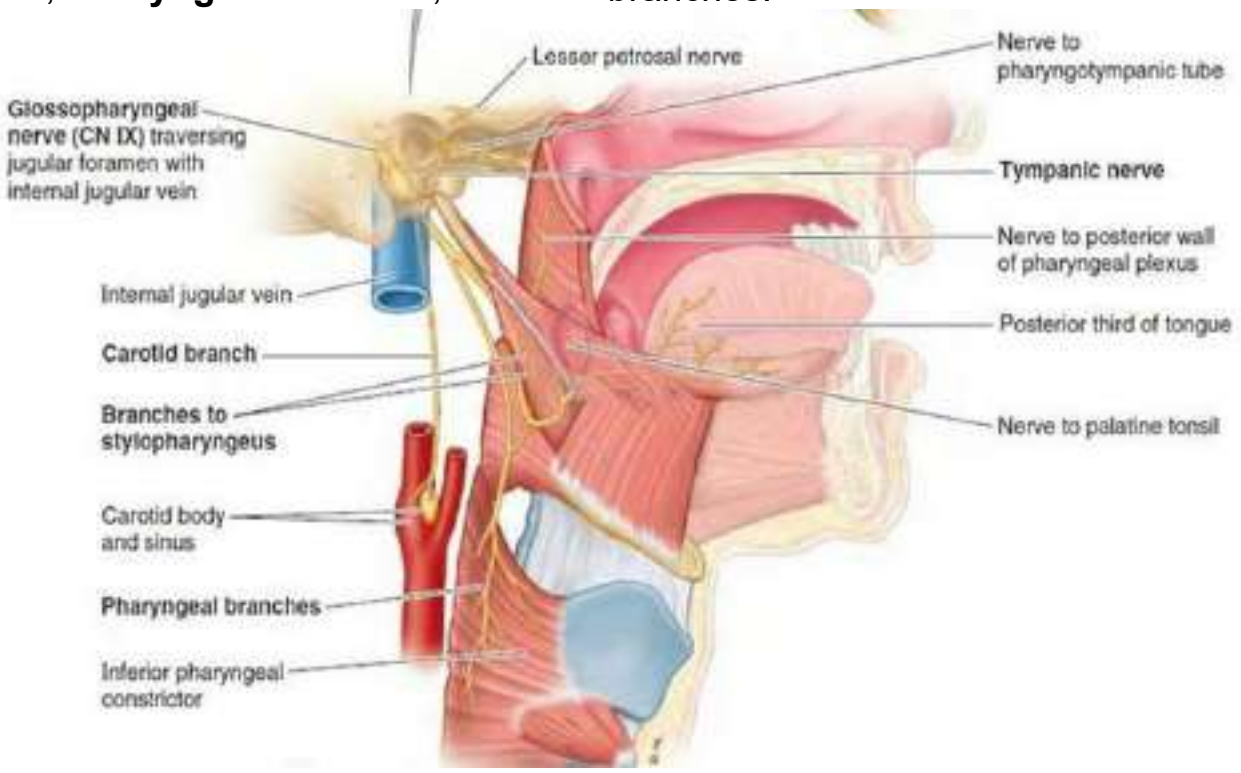


Figure (30): Glossopharyngeal nerve

10. Vagus Nerve:

Type of fibers: motor and sensory
Branches:

1- **Meningeal** and **auricular** branches,
2- **Pharyngeal** branchwhichcontains **nerve fibers** fromthe **cranial** partofthe
3- **accessory** nerve.
Superior laryngeal nervedividesinto: **internal** and **external laryngeal** nerves.
Recurrent laryngeal nerve, **Cardiac** branches, **Carotid body** branch:

The **vagus** nervethus **innervates**:

- the **heart** and **great vessels** withinthe **thorax**;
- the **larynx**,**trachea, bronchi,** and **lungs**;
- and **much** ofthe **alimentary tract** fromthe **pharynx** tothe **splenic flexure of**
- **the colon**.

Italsosupplies **glands** associatedwith the **alimentary** tract,suchasthe **liver**.

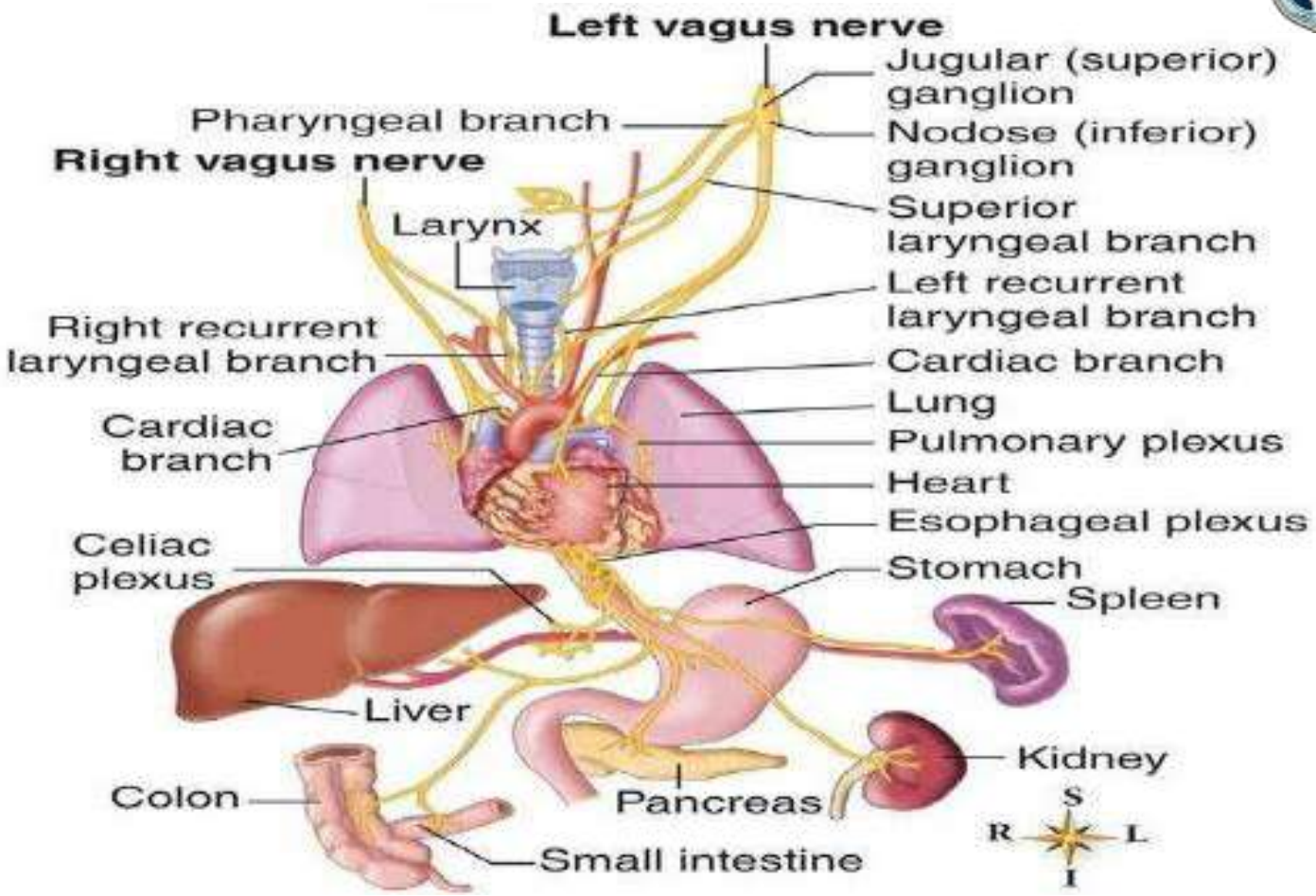


Figure (31): Vagus nerve

11. Accessory Nerve:

Type of fiber: is a motor nerve.
Roots: cranial and **spinal**, **both** cranial and spinal roots pass through the **jugular foramen** of the skull.
The **spinal accessory** nerve is **motor** to **both** the **trapezius** and the **sternocleidomastoid** muscles.

12. Hypoglossal nerve:

Type of fiber: motor
Branches:

1- **Meningeal branch:** it innervates the **dura mater** on the **floor** and **posterior wall** of the **posterior cranial fossa**
2- The **superior root** of the **ansa cervicalis (Descending hypoglossi):** it conveys **fibers** from the **first cervical nerve (C1).**
3- Nerve to the **thyrohyoid** muscle (**C1**).
4- **Muscular** branches to **all** the muscles of the **tongue EXCEPT** the **palatoglossus** which is supplied by (**pharyngeal plexus**)
5- Nerve to the **geniohyoid** muscle (**C1**).

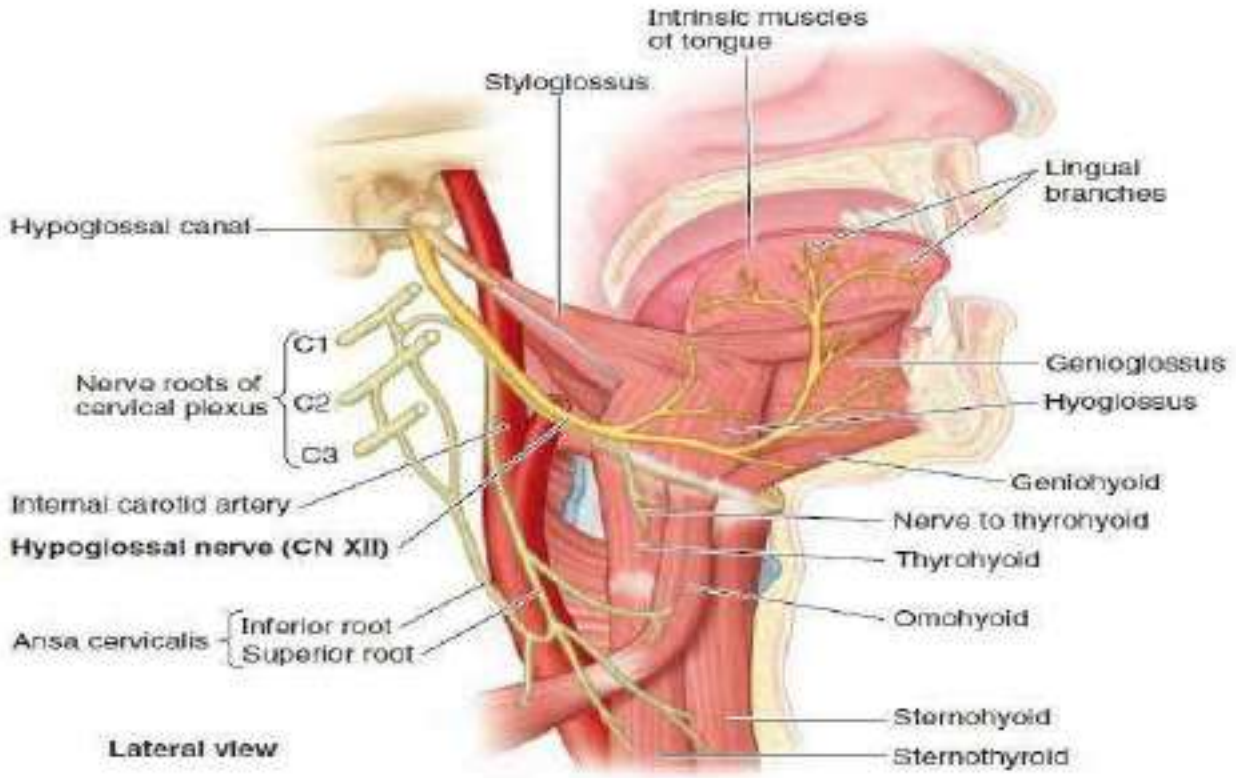


Figure (33): Hypoglossal nerve

Ansa cervicalis: nerve loopformed of**two limbs**: **superior**(**descending hypoglossi**) and **inferior** limb (**descending cervicalis**) (**C2 and C3**), it gives **muscular** branches to (**omohyoid** (**superior** belly and **inferior** belly) , **sternohyoid** and **sternothyroid** muscles).

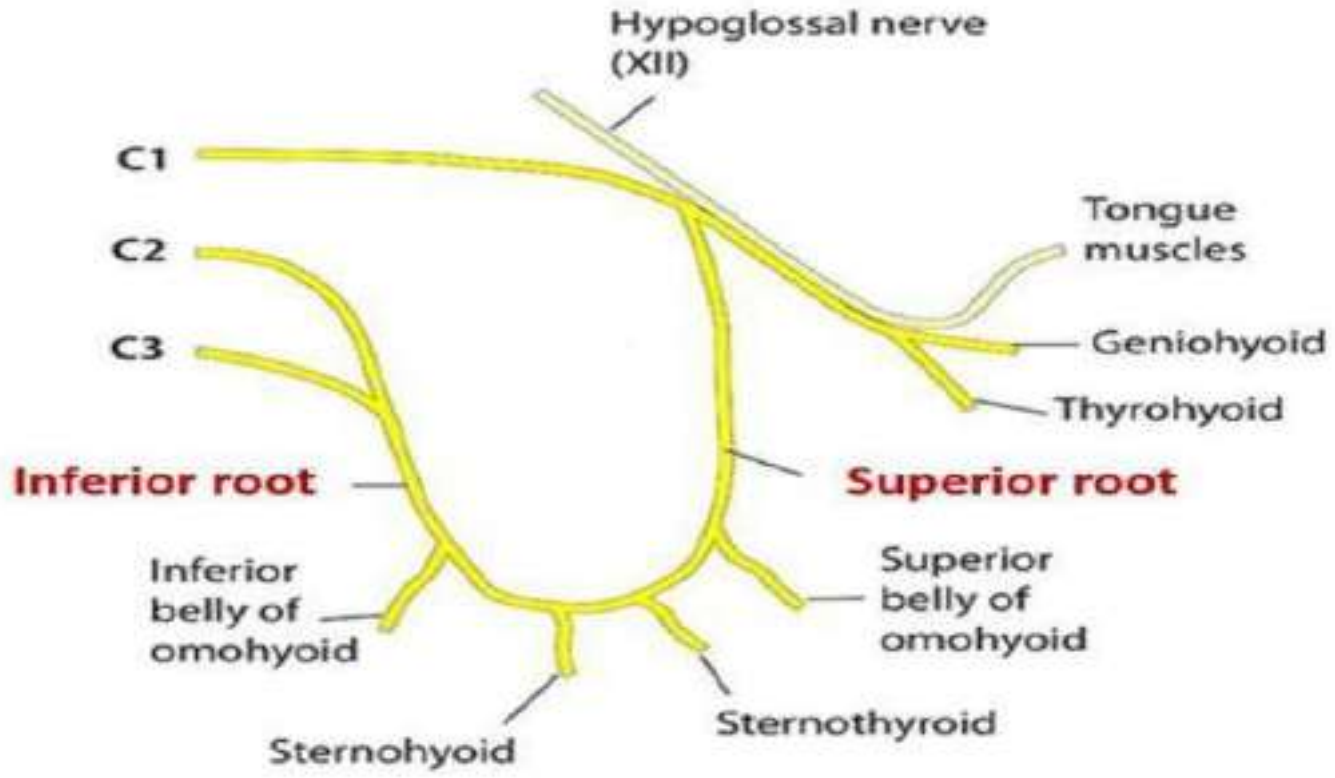


Figure (34): Ansa cervicalis

6 Blood supply of the head and neck

Arteries of head and neck:

1) Commoncarotidartery (CCA):

Itisa branch of **brachiocephalic** artery on the **right** side and branch of **arch of aorta** onthe **left** side, it is **enclosed** with the **internal jugular vein** and **vagus nerve** in the **carotid sheath.**

Branches: twoterminal branches

1. **External** carotid artery
2. **Internal** carotid artery

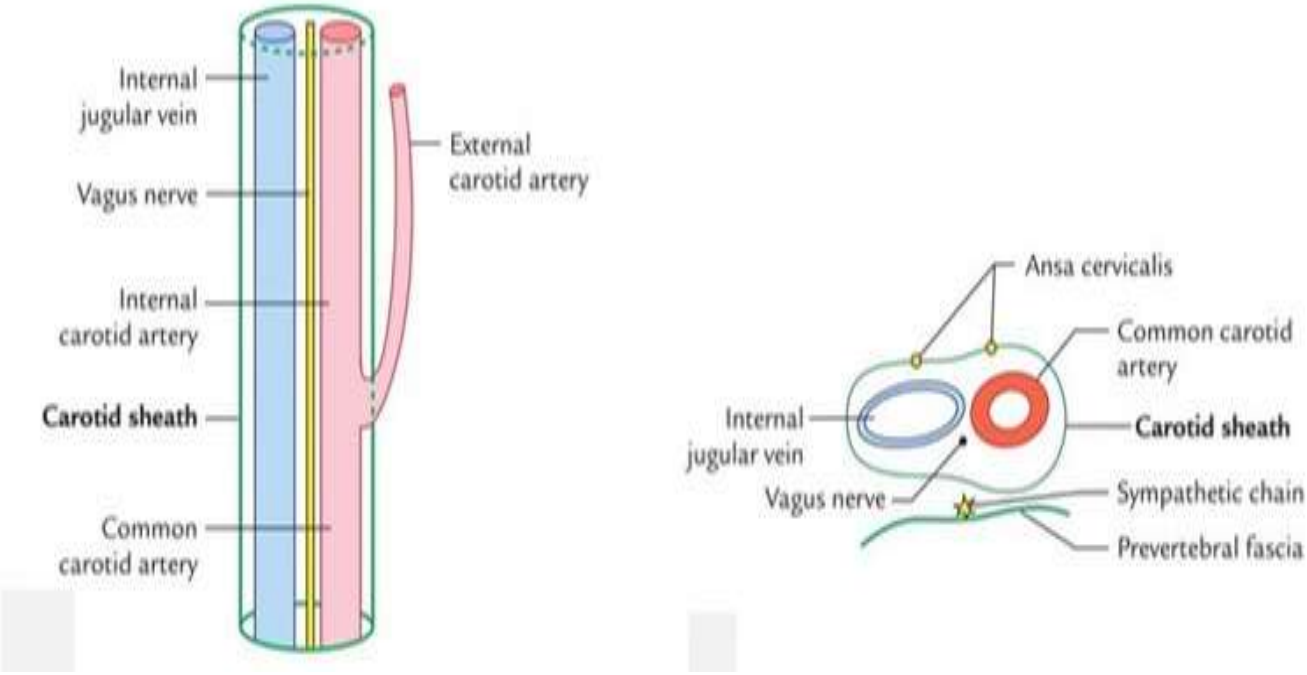


Figure (35): carotid sheath

1. External carotidartery(ECA):

Branches: 1. **Superior thyroid** 2. **Ascendingpharyngeal** artery artery

3. **Lingual** artery
4. **Facial** artery
5. **Occipital** artery
6. **Posteriorauricular** artery
7. **Maxillary** artery (**terminal** branch)
8. **Superficialtemporal** artery (**terminal** branch)

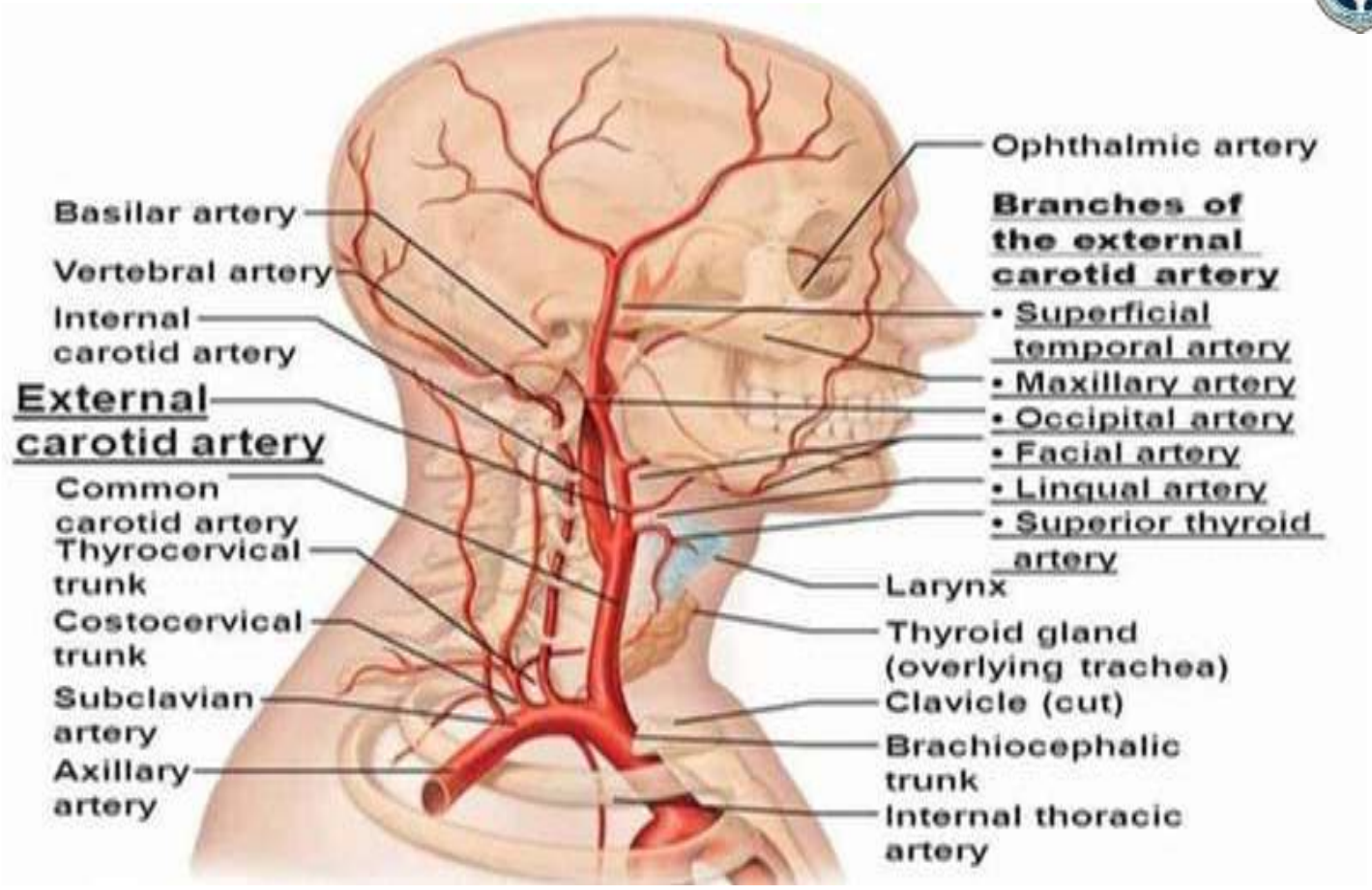


Figure (36): External carotid artery

Maxillary Artery

Branches:

A.Fromthe mandibular part:

1- **Deep auricular** artery,
2- **anterior tympanic** artery,
3- **middle meningeal** artery
4- **Inferior alveolar** arteryemerges,
5- **Accessory meningeal** artery

B. From the pterygoid part:

1- **Masseteric** artery,
2- **Pterygoid** artery,
3- **deep temporal** artery,
4- **buccal** or **buccinator** artery

C. From the pterygopalatine part:

1- **Sphenopalatine** artery
2- **Descending palatine** artery
3- **Infraorbital** artery
4- **Posterior superior alveolar** artery:
5- **Pharyngeal** artery
6- Arteryofthe **pterygoid** canal

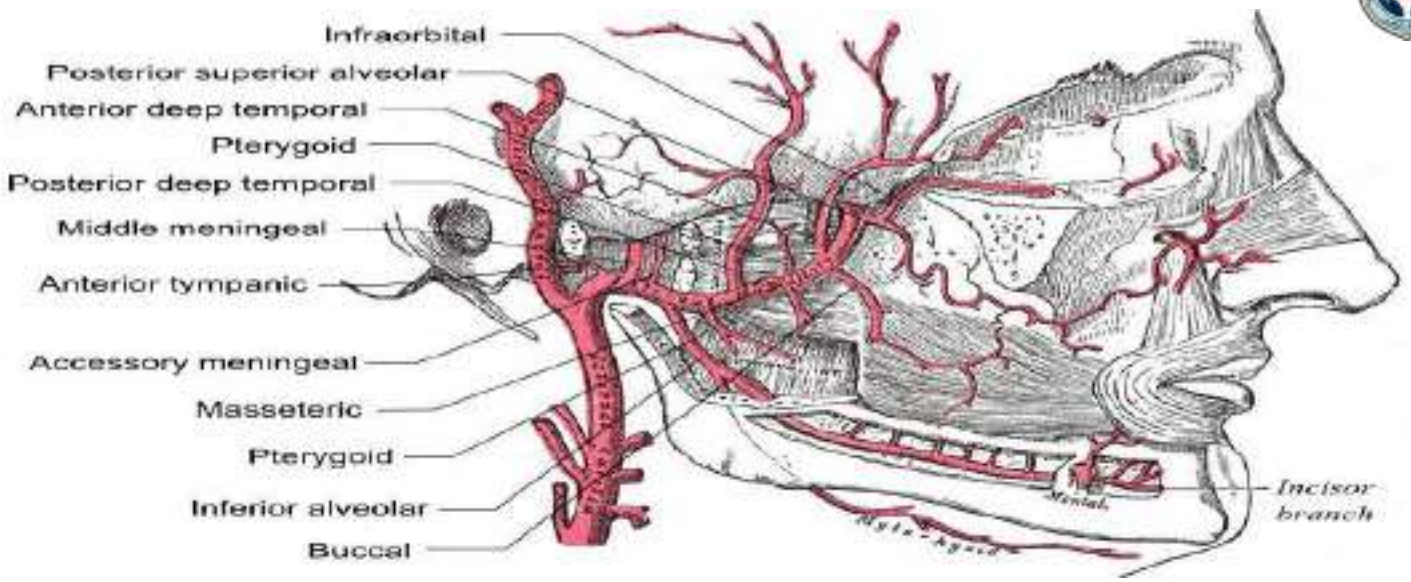


Figure (37): Maxillary artery

2. Internal CarotidArtery(ICA):

- It **enters** the skull trough the **carotidcanal,** there are **nobranchesin the neck.**
- **Intracranial** part of ICA which is divided into **three** parts:

1. **Petrouspart:** it gives **corticotympanic** artery and artery of the **pterygoid** canal
2. (**vidian artery**).
3. **Cavernouspart:** inside the **cavernous** sinus, **hypophyseal** artery, **meningeal**

A. artery

B. **Cerebral part:**
Ophthalmic artery
Posteriorcommunicating artery **anastomoses** with **posteriorcerebral** artery coming off the **basilarartery,** contributing to the formation of the **circle of Willis** (see blood supply of brain),

C. **Anterior choroidal** artery

D. **Two terminal** branches (**anterior** and **middlecerebral** arteries).

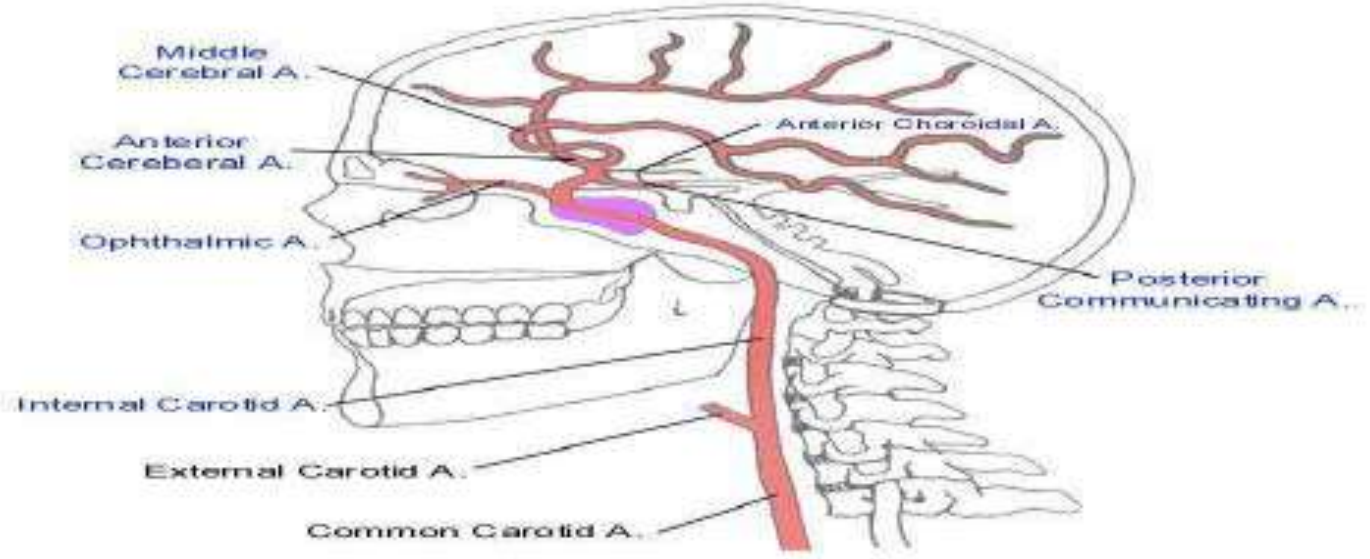
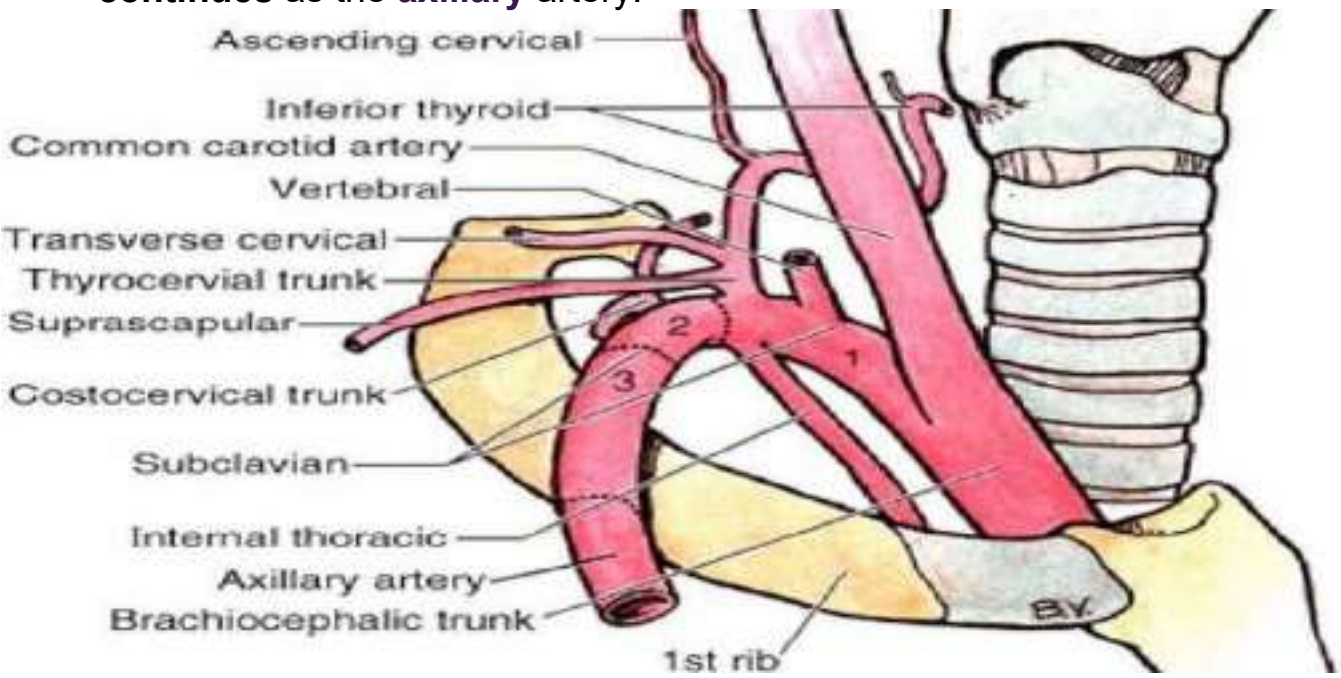


Figure (38): Internal carotid artery

2) Subclavian artery:

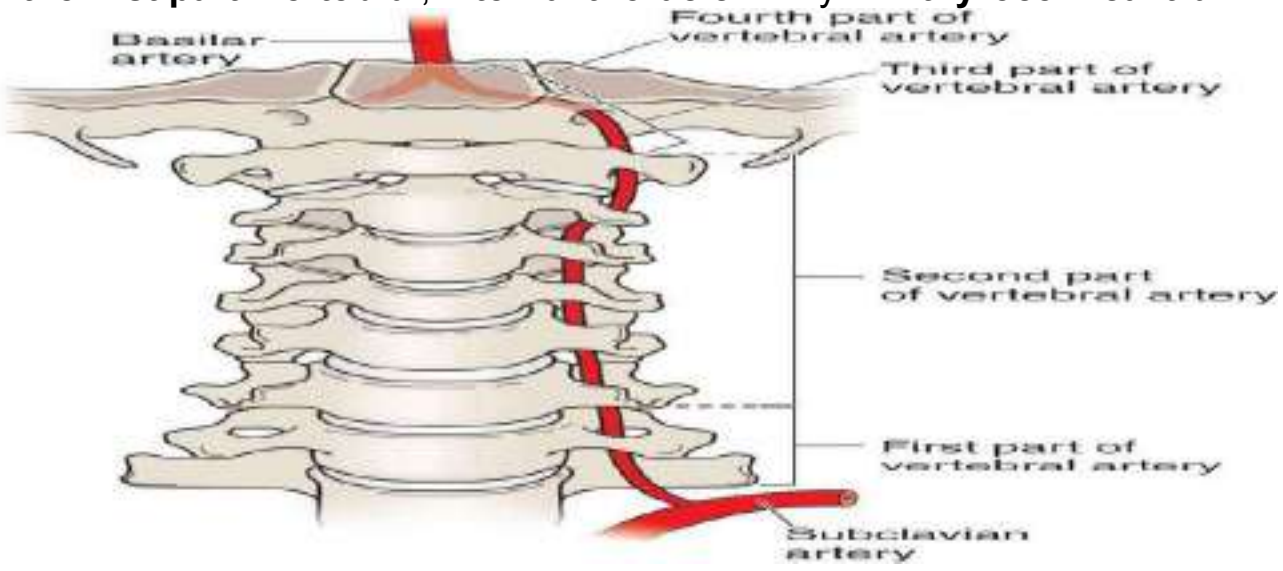
- On the **right side**, it is originated from the **brachiocephalic** artery
- while on the **left side** it is originated from the **arch of aorta.**
- It is divided into **3 parts**; it terminates at the **outer border** of the **first rib** and **continues** as the **axillary** artery.



Figure(39): Subclavian artery

Branches:

1. From the first part: vertebral, **internal thoracic** artery and **thyrocervical trunk**.



Figure(40):Vertebralartery

2. From the second part: costocervical trunk, it gives **deep cervical** artery and **superior intercostal** artery.

3. From the third part: dorsal scapular artery

Veins of the head and neck:

1) *Superficial veins:*

external jugular, **anterior jugular** and **posterior external jugular** veins

2) *Deep veins:*

internal jugular and **subclavian** veins.

1. External Jugular Vein:

- Itbegins **behind** the **angle** ofthe **jaw** bythe **union** of the **posterior auricular** veinwiththe **posterior division** ofthe **retromandibular** vein (which is formed by **union** of **superficial temporal** veinand **maxillary** vein);
- It has **multiple tributaries:**

1- **Anterior jagular** vein
2- **Posterior external jugular** vein
3- **Transverse cervical** vein
4- **Suprasacbular** vein

- Itis **drained** inthe **subclavain** vein.

2. Internal jugular vein:

1- **Inferior petrosal** sinus
2- **Common facial** vein (**facial** vein union with **anterior** division of
3- **retromandibular** vein)
4- **Pharyngeal** veins
5- **Lingual** vein
6- **Superior thyroid** vein
Middle thyroid vein

3. Subclavian vein:

- It **joins** with the **internal jugular** vein to form the **brachiocephalic** vein.
- **Tributaries: external jugular** veins.

4. Brachiocephalic veins:

- It is formedby **union** of **internal jugular** and **subclavian** veins.
- **Right** and **left brachiocephalic** veins **united** to form **superior vena cava.**

- **Tributaries:**
  Vertebral, **Internal thoracic, Inferior thyroid, Left superior intercostal** veins into **left** side**, 1st posterior intercostal** vein**, thoracic duct** in **right** side**, lymphatic duct** at **left** side**.**

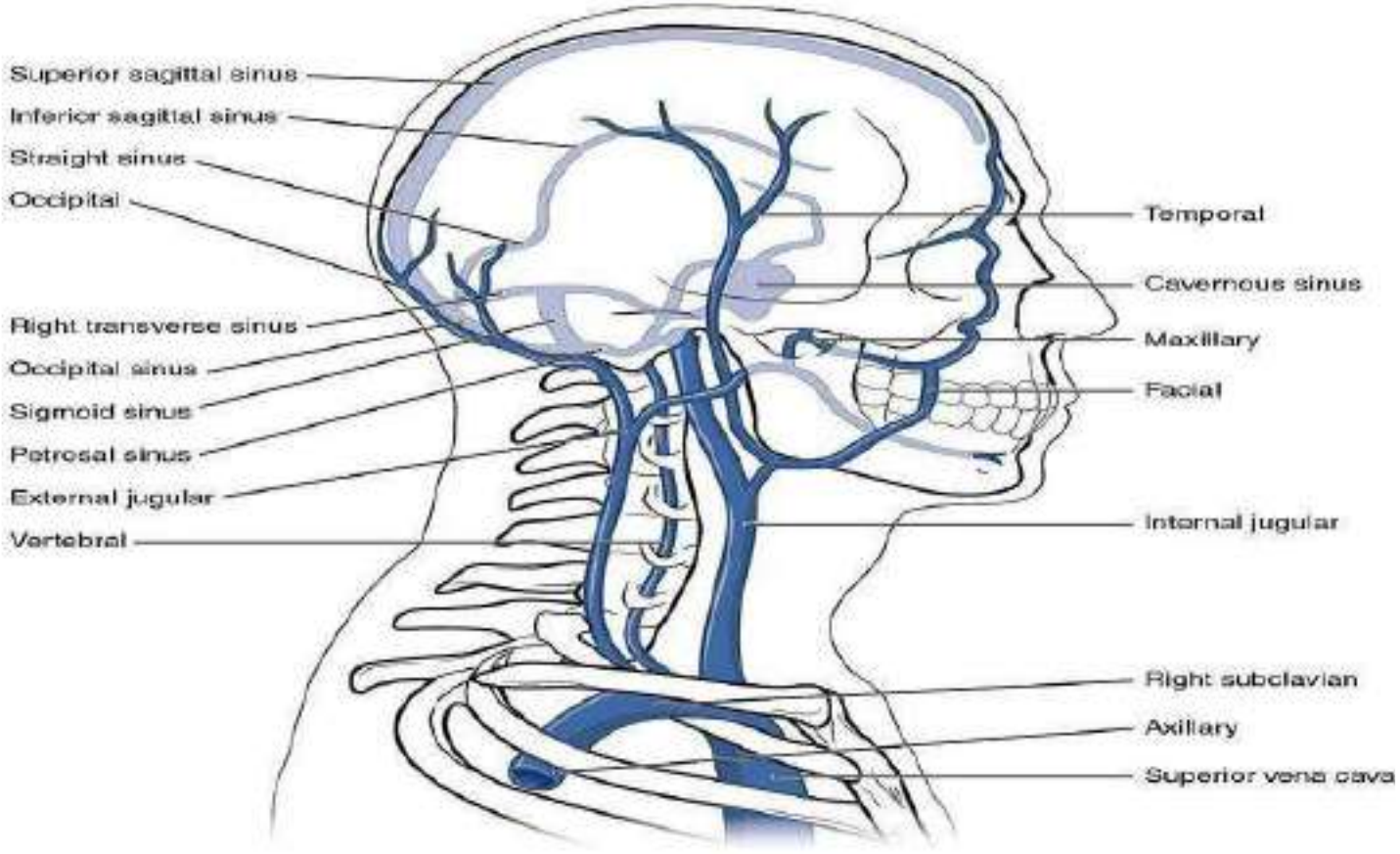


Figure (41): Veins of head and neck

7 *Lymphatic drainage of the head and neck*

1) *RegionalNodes:*

- **Occipital** nodes
- **Retroauricular**(**mastoid**) nodes:
- **Parotid** nodes:
- **Buccal**(**facial**)nodes:.
- **Submandibular** nodes:
- **Submental** nodes:
- **Anterior cervical** nodes:
- **Superficial cervical** nodes:
- **Retropharyngeal** nodes:
- **Laryngeal** nodes:
- **Tracheal** (**paratracheal**)nodes:

2) *Terminallymphnode(deepcervicallymphnode):*

1-The jugulodigastric node, whichislocated **below posterior belly of digastric**,is **mainly** concerned with **drainage** ofthe **tonsil** andthe **tongue**.

2-The juguloomohyoid node,

- which is situated **close** tothe **omohyoid** muscle,is **mainly** associatedwith **drainage** of the **tongue**.
- **Efferent:** from **deep cervical** lymphnodeto **jugular** lymphtrunkwhichdraininto **thoracic duct** (**left** side), **lymphatic** duct(**right** side).

8 The orbit

Orbits are **pair** of **bony cavities** thatcontains **eyeballs** and other **associating** structures

Bony orbit: seepracticalnotesofheadand neck

Contents:

1. **Orbital connective** tissue and **fat**
2. **Eyelids**
3. **Lacrimal** apparatus
4. **Extra ocular** muscles(**Extrinsic** muscles)
5. **Vessels:** seebloodsupplyofheadand neck
6. **Nerves** and **lymphatic**:seeinnervation and lymphatic drainage of H & N
7. **Eyeball**

1) The eyelids:

- They are **moveable folds** that **cover** the **eyeball anteriorly** when **closed**, thereby **protecting** it from **injury** and **excessive light**.
- They also **keep** the **cornea moist** by **spreading** the **lacrimal fluid.**
- They **keep eyes** from **dryness** during **sleeping**.
- The **eyelids** are **covered externally** by **thin skin** and **internally** by **transparent mucous membrane,** the **palpebral conjunctiva**.
- The **lines of reflection** of the **palpebral conjunctiva** onto the **eyeball** form **deep recesses**, the **superior** and **inferior conjunctival fornices.**

1. **The conjunctival sac:** is the **space** bound by the **palpebral** and **bulbar conjunctivae**; it is a **closed space** when the eyelids are **closed**.
2. **The palpebral fissure:** is **elliptical opening** between the **eyelids**.
3. **Eye lashes: short curved hairs** present on the **free edge** of eyelids.
4. **Layers of eyelid: skin** (**eye lashes follicles**, **sebaceous tarsal** glands), **subcutaneous** tissue, **orbicularis occuli**, **orbital septum**, **tarsal plate** and **palpebral conjunctiva.**

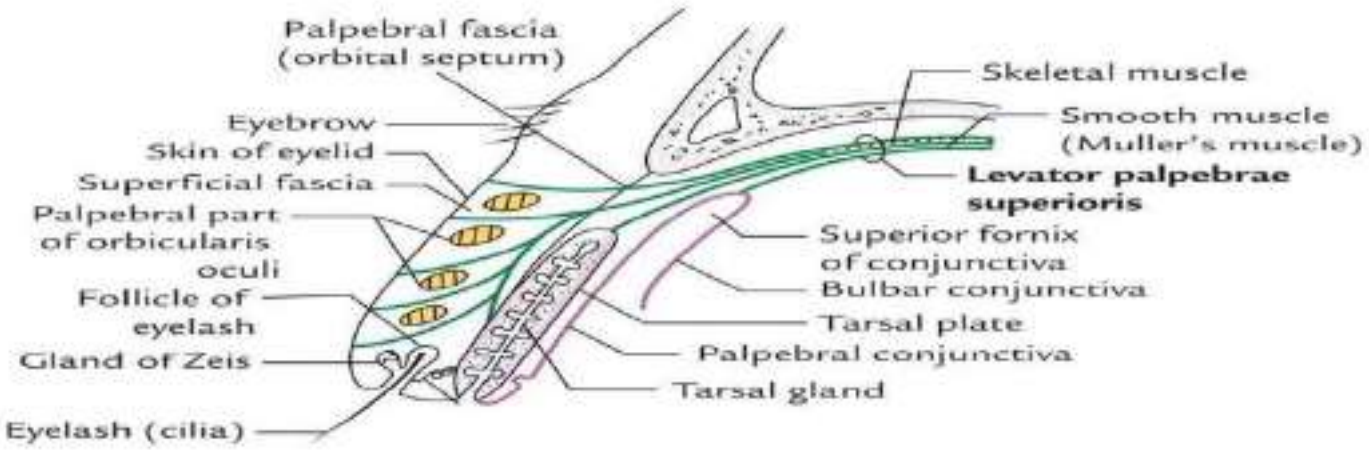


Figure (42): Eyelid anatomy

- **Tarsal plate:**
  - **thin dense fibrous tissue**, its **medial** and **lateral** end isattachedby **band** called **medial** and **lateral palpebral ligaments**.
  - **Insertion** of **levatpor palpaberae superioris** pierces the **orbital septum** to reach its **anterior** surface.

2) *Lacrimal apparatus:*

Itconsists of:

1-**Lacrimal gland** 2-**Conjunctival sac** 3-**Lacrimal puncti** and **canaliculi**
4-**Lacrimal sac** 5-**Nasolacrimal duct**

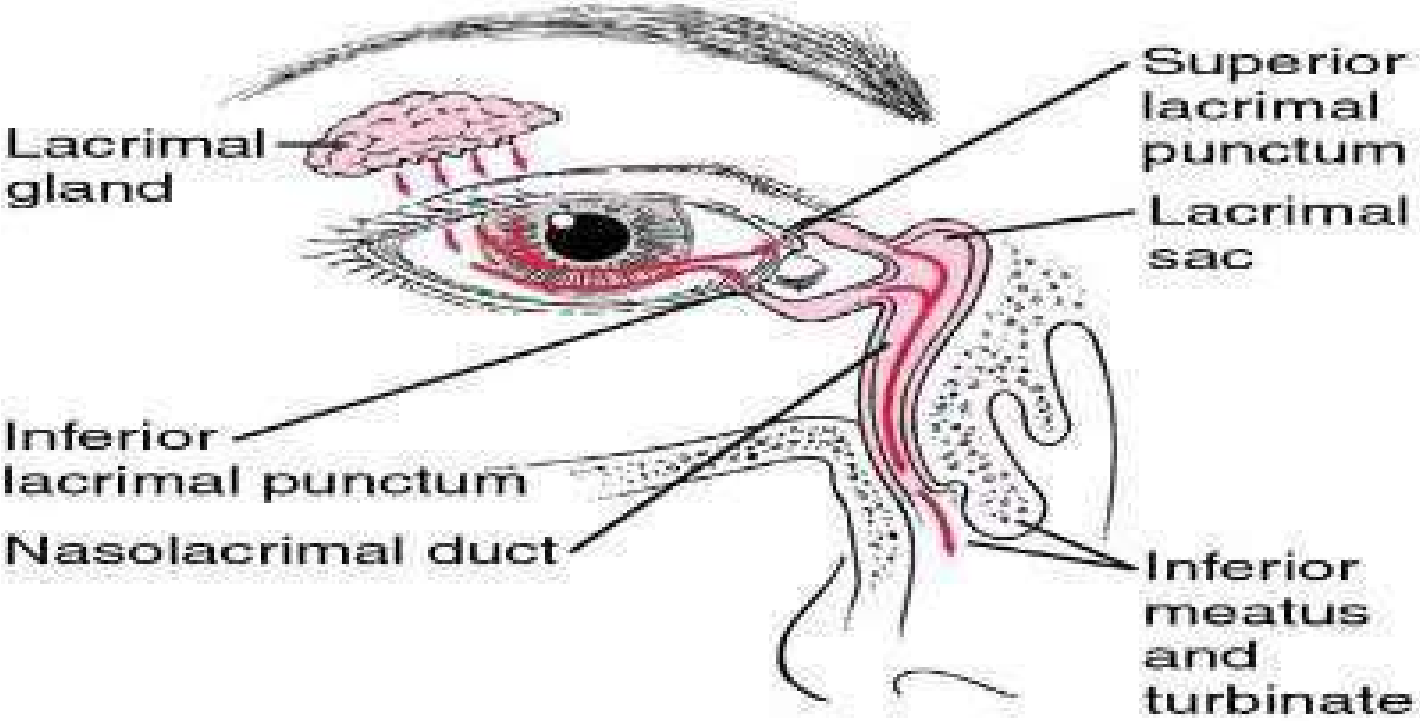


Figure (43): lacrimal apparatus

1. Lacrimal gland:

- is the **primary producer** of the **aqueous** componentofthe **tear** layer.Its **secretion** is a **watery** fluid.
- It is **almond shaped** and **approximately 2 cm long** ,itliesinthe **fossa** forthe **lacrimal gland** in the **superolateral part** of each orbit.
- The gland is **divided** into **superior orbital** and **inferior palpebral** partsbythe **lateral expansion** of the **tendon** of the **levator palpebrae superioris**.
- so, **palpebral** portion lies in **upper eye lid** whilethe **orbital** portionthatliein **lacrimal fossa** itself.

Ducts: it has about **12 ducts** pierce the **conjunctiva** of **upper** eyelidand **open** into **conjunctival sac.**

Arterial supply: **lacrimal** branch of **ophthalmic** artery

Nerve supply:

1. **sensory** nerve supply derived from **lacrimal nerve**
2. **Parasympathetic secretomotor** nerve supply is derived from the **lacrimal nucleus** of the **facial** nerve…… The **preganglionic** fibers from **great petrosal** branch nerve of the **pterygoid canal pterygopalatine ganglia**, the **postganglionic** fibers join the **maxillary** nerve then pass into its **zygomatic branch** and the **zygomaticotemporal** nerve **communicating** branch to **lacrimal** nerve the **lacrimal** gland.
3. **Sympathetic postganglionic** fibers are from **plexus** around **internal carotid artery.**

2. Lacrimal canaliculi/puncti:

Puncti small openings presentin **medial** end of eye margin each lead to **canaliculi** are **divided** into the **superior duct** andthe **inferior duct** that drain into the **lacrimal sac.**

3. Lacrimal sac:

Itlies in the **lacrimal groove behind** and **medial** to the **palpebral ligament**,it is **continuous** with **nasolacrimal duct.**

4. The nasolacrimal duct:

- Emerges from the **lower** end of the **lacrimal sac.**
- The duct **descends downward**, **backward**, and **laterally** in a **bony canal** and **opens** intothe **inferior meatus** of the **nose**.

5.Conjunctival sac:

see before

3) *Extraocular muscles:*

Extrinsic muscles of the eyeball (the orbit): are the **levator palpebrae superioris**, **4 recti** (**superior**, **inferior**, **medial** and **lateral**), and **2 obliques** (**superior** and **inferior**). These **7 muscles** work **together** to **move the superior eyelids** and **eyeballs**.

Levator palpebrae superioris

Nerve supply: **superior** division of **occulomotor** nerve.
Smooth muscle is **innervated** by **sympathetic** fibers from **superior cervical sympathetic ganglion.**
Action: **elevation** of **upper** eye lid.

N.B.

Instress conditions, upper eye lid is excessively elevated due to contraction of inferior lamellae due to sympathetic over stimulation, but when sympathetic innervation isparalyzed, partial ptosis will occur.

N.B.

Occulomotor paralysis leads to complete ptosis.

Recti muscles

Nerve supply: recti muscles are supplied by **occulomotor** nerve **EXCEPT lateral rectus** which is supplied by **abducent nerve.**

Action:

1. **Lateral** rectus**: abduction** ofeyeball
2. **Medial** rectus**: adduction** ofeyeball
3. **Superior** rectus: **elevation**,**adduction** ,**internal rotation** of eyeball
4. **Inferior** rectus: **depression**,**adduction** ,**external rotation** of eyeball

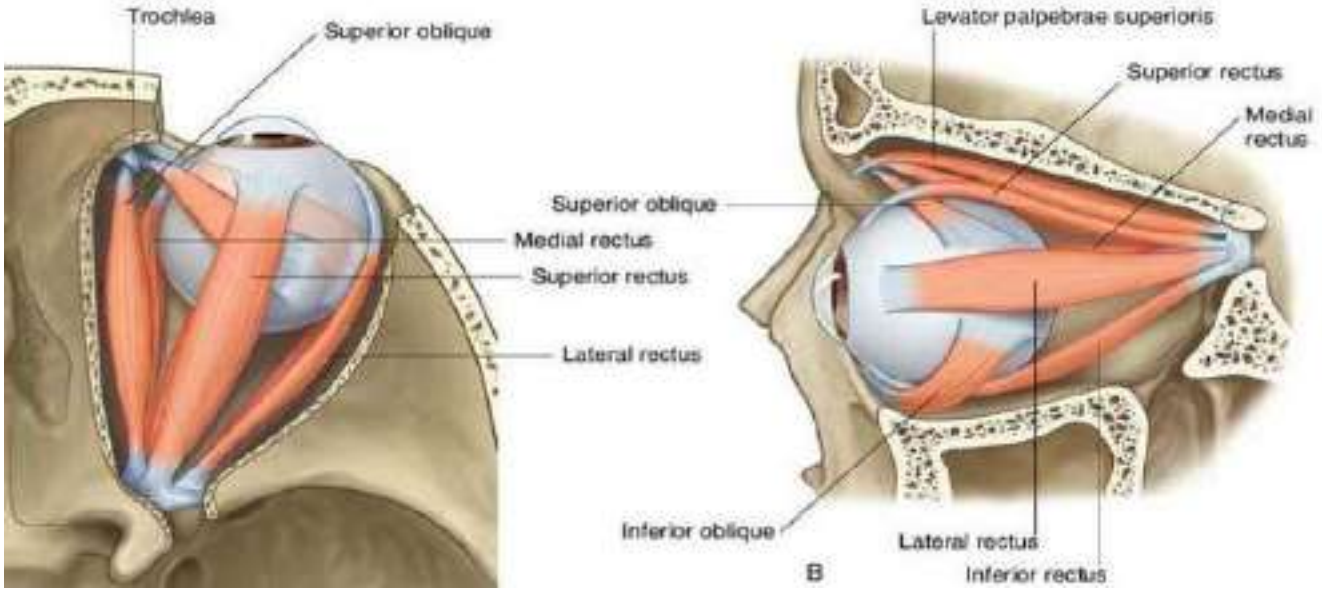


Figure (44): Extra ocular muscles

Superior oblique

Nerve supply: trochlear nerve

Action: depression and **abduction** (**rotates** the cornea **downward** and **laterally**)

Inferior oblique

Nerve supply: occulomotor nerve

Action: elevation, abduction (**rotates** cornea **upwards** and **laterally**).

Injury of nerves of extra ocular muscles:

see cranial nerves.

4) *Eyeball:*

- It lies in the **orbit** and **embedded** in the **orbitlfat** but **separated** rom by **fascial sheath** of the eye ball (**Tenon's capsule**).
- The **three** layers of the eyeball are:

1. **Fibrouslayer(outercoat),** consisting of the **sclera** and **cornea**.
2. **Vascularlayer(middlecoat),** consisting of the **choroid,ciliary body**, and **iris**.
3. **Innerlayer (inner coat),** consisting of the **retina**, which has **both optic** and **non-visual parts.**

1. The fibrous layer of the eyeball:

Is the **external fibrous** layer and formed of **two** parts:

1. **Cornea** covers the **anterior** part
2. and **sclera** covers the **posterior** part and **forms** about **five sixths (5/6)** of the surface

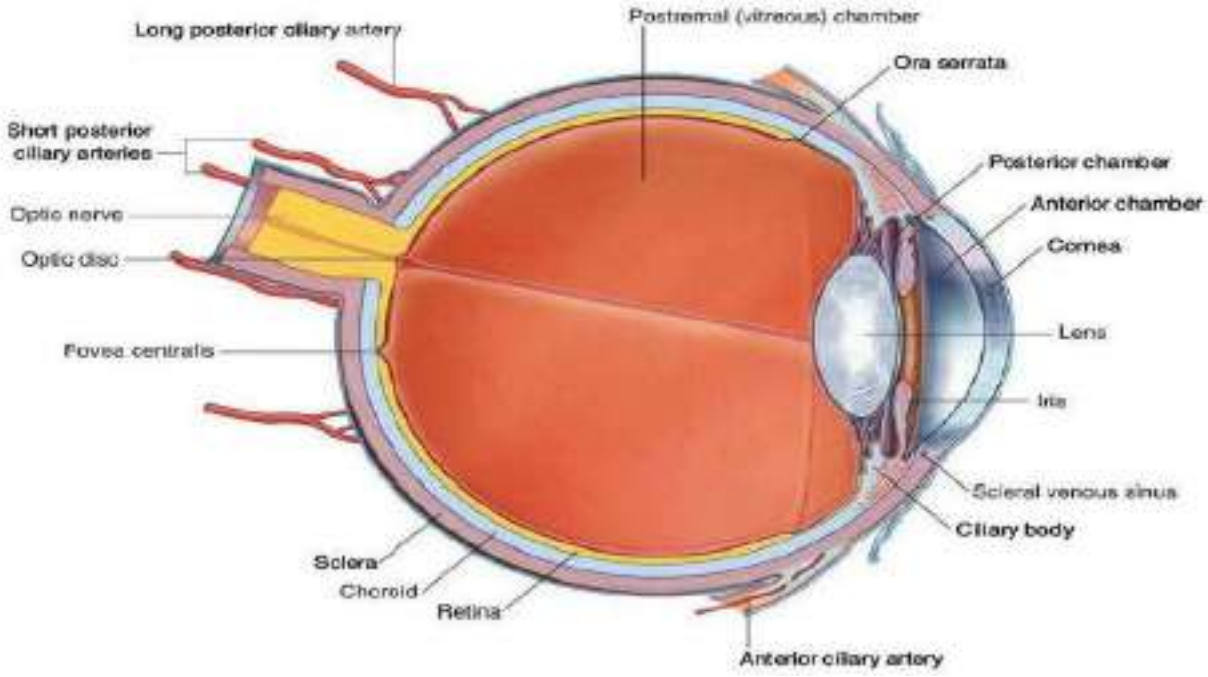


Figure (45): Eyeball

1. Cornea:

- It coversthe **anterior** part of the eyeball, it **represents one sixth (1/6)** of fibrous layer,
- **Transparent** and allows **passage of light.**

2. Sclera:

- It is an **opaque** layer "**white** part of the eye covered by the **conjunctiva**", it represents the **posterior five sixth (5/6)** of the fibrous layer.
  It gives **attachment** of the **ocular muscles**.

2. The vascular layer of the eyeball:

1. Choroid:

Itrepresents **two thirds (2/3)** of this layer, **thin highly vascular** and **pigmented** layer that attached to **retina firmly** but attached to **sclera loosely.**

2. Ciliary body:

Triangular shaped structure extends from **choroid** to the **iris**, it includes **cilliary muscle** (**intrinsic** muscle of the eyeball) and **cilliary processes** (**small projections** attached to the **lens**).

3. Iris:

Anterior part of the **vascular** layer, it represents the **colored** part of the eye with **central opening** called **pupil**.
Size of pupil opening is **controlled** by **smooth muscles** fibers of **iris** which are arranged in **two patterns**; **circular** (**sphincter pupillae** muscle) and **longitudinal** (**dilator pupillae** muscle).

Intrinsicmusclesof the eyeball are:

1. **CIliary** muscle
2. **Sphincter** pupillae
3. **Dilator** pupillae

Muscle	Location	Innervation	Function
Ciliary	Muscle fibers in the ciliary body	Parasympathetics from the oculomotor nerve [III]	Constricts ciliary body, relaxes tension on lens, lens becomes more rounded
Sphincter pupillae	Circularly arranged fibers in the iris	Parasympathetics from the oculomotor nerve [III]	Constricts pupil
Dilator pupillae	Radially arranged fibers in the iris	Sympathetics from the superior cervical ganglion (T1)	Dilates pupil

Table (1): Intrinsic muscles of the eyeball

3. The inner layer of the eyeball:

1. Retina:

- It is formed of **anterior non-visual** part and **posterior optic** part **joined** at point called **ora serrata.**
- **Optic disc:** where is the **optic nerve leaves** retina, **lateral** to it there is a **depressed** part called **macula lutea** with **central depression** called **fovea centralis** (area of **maximum sensitivity** to **vision** at retina).
- **Anterior chamber of the eye:** area **behind** the **cornea** and **anterior** to **iris**.
- **Posterior chamber of the eye:** area **behind iris** and **anterior** to the **lens**.
- **Both chambers** are **filled** with **aqueous humor** which is a **fluid continuously formed** and **absorbed** to the **scleral venous system**. It gives **nutrient** to **cornaea** and **maintains intraocular pressure.**

2. Lens:

transparent biconvex elastic disc attached to **cillary body** by **fibers** which change its **convexity** and **maintain visualacuity.**

3. Vitreous chamber:

area between **lens** and **retina**, filled with **transparent gelatinous** substance called **vitreous humor.**

9 The Ear

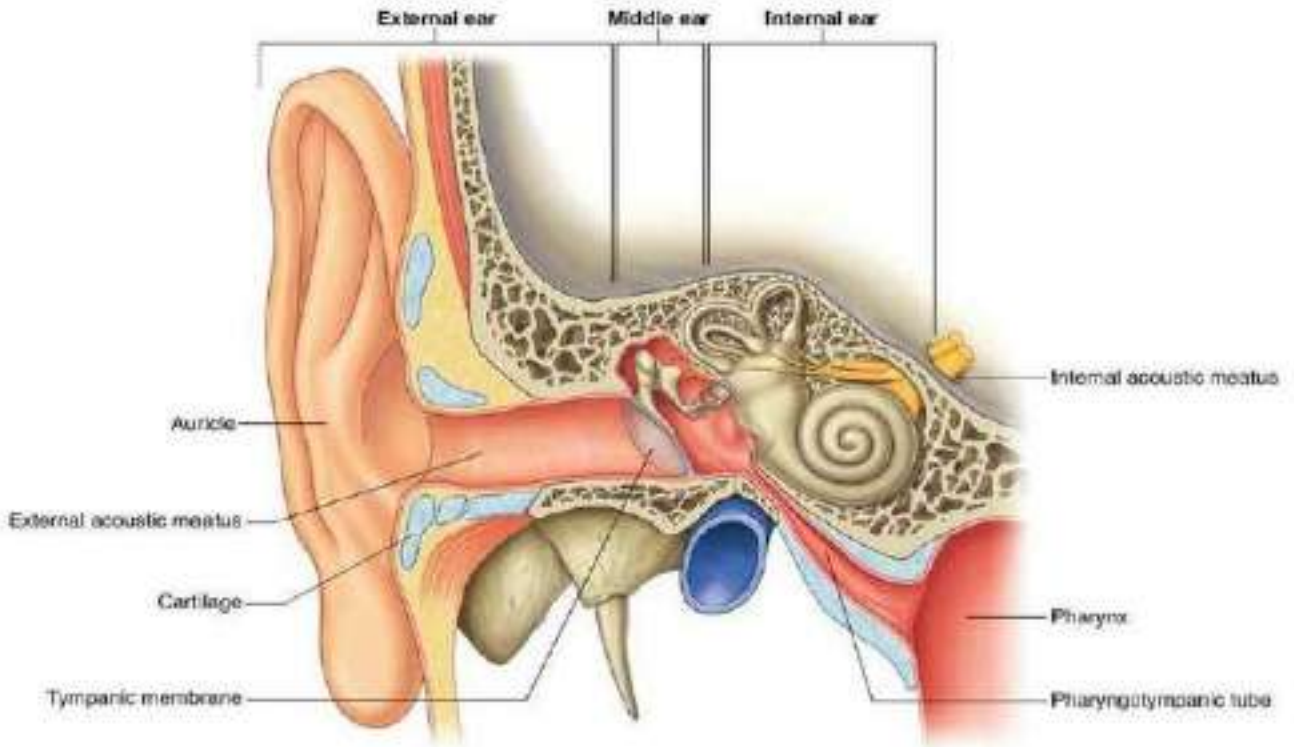
Itconsists of **external** ear, **middle** ear and **inner** ear.

1) External ear:

1- **Auricle:** it collects **air vibrations**, formed of **thin plate** of **elastic cartilage** covered by **skin**.

2- External auditory (acoustic) meatus:

- **passage** between the **auricle** and the **tympanic membrane**, **1 inch long.**



Figure(46):Ear

- **Lateral1/3** is **cartilaginous** and **medial 2/3 is bony**, connected at **isthmus**.
- **Meatus** is lined by **skin** which **contains hair** and**sebaceous glands.**
- **Sensorynervesupply:auriculotemporal** nerve,**auricular branch of vagus**
- nerve, **greatauricular** and **lesser occipital.**
- **Bloodsupply:posterioauricular**, **superficial temporal** and **deep auricular**
- branch of **maxillary** artery.

3-

- **Lymphaticdrainage:superficial parotid**, **mastoid**, and **superficial cervical** lymph nodes.

Thetympanicmembrane(eardrum):
thinmembrane separates the **external auditory meatus** and **middle ear cavity**; it is **convextowards** the **middle** ear cavity and **concave** towards the **outside**.

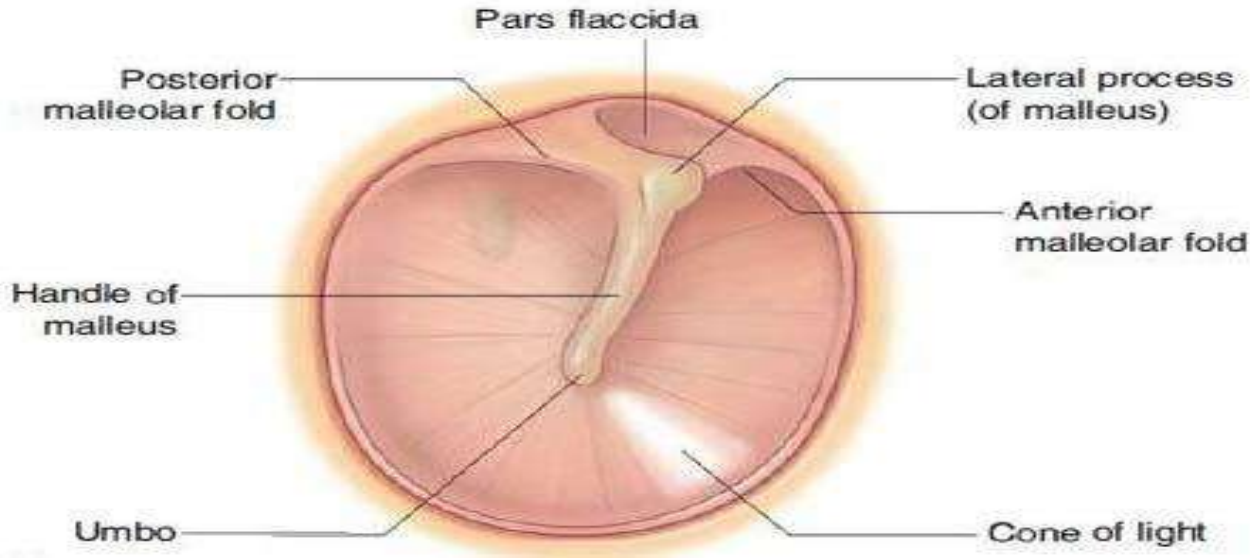


Figure (47): Tympanicmembrane

- **Umbo:** areaof**maximumconcavity**and producesby**attachment** of **handle** of **malleus**, around its **periphery** it is attached to **tympanic** part of **temporal bone.**
- Its wall is formed of **3 layers**: **outer (skin)**, **middle (fibrous)** and **inner (mucous)**
- **Nerve supply: outer** part **auriculotemporal** nerve, **inner** part **auricular branch** of **vagus**
- **Lymphatic drainage: preauricula**r, **retropharyngeal** lymph nodes.

2) *Middle ear (Tympanic cavity):*

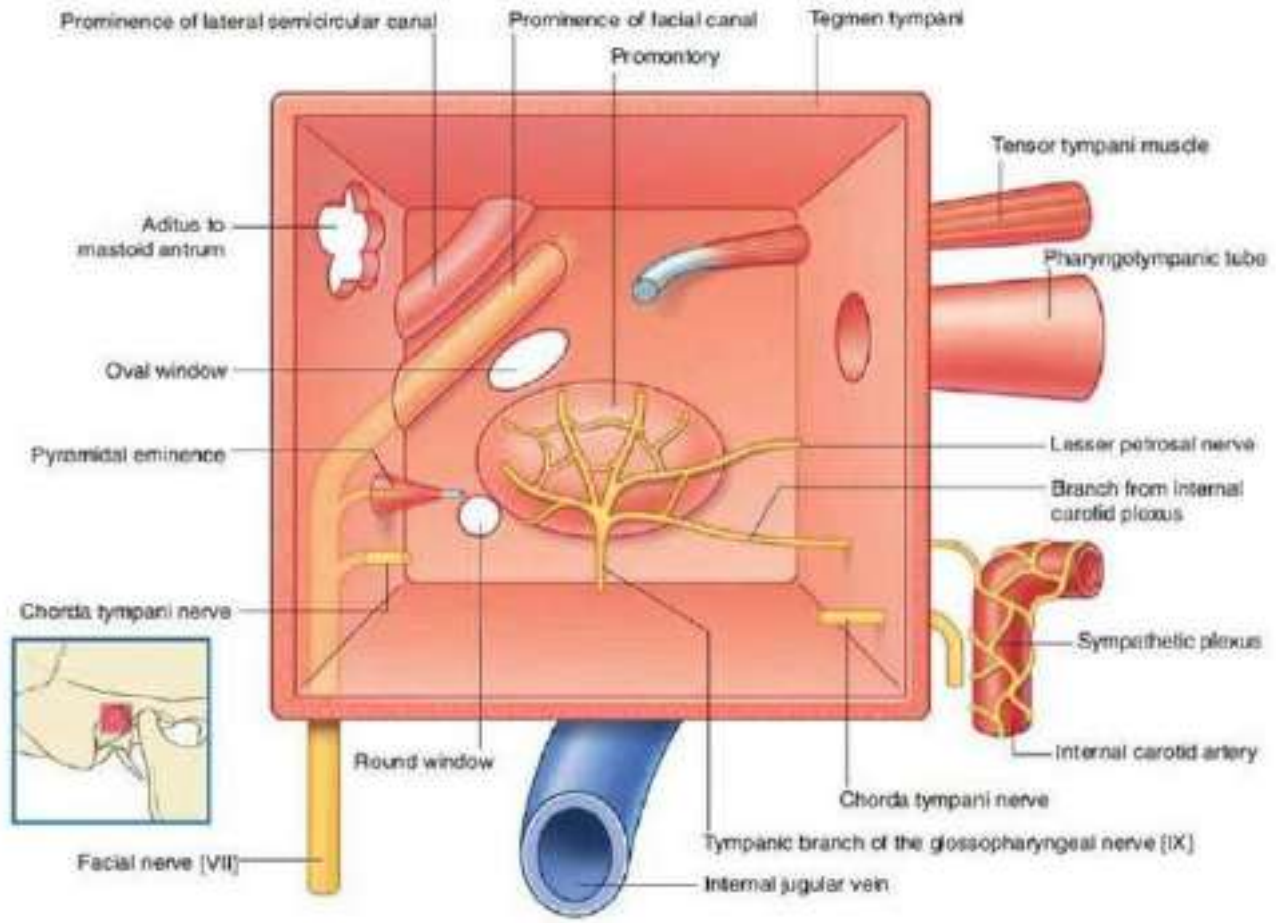
- The **middle ear** is an **air-containing cavity** in the **petrous** part of the **temporal** bone it lined with **mucous membrane**. It is **narrow** and **slit like cavity.**
- **Connections:**
  - **posteriorly** with **mastoid antrum** and **mastoid air cells**,
  - **anteriorly** with **nasopharynx** through **auditory** tube
- It has a **roof**, **floor**, **anterior** wall, **posterior** wall, **lateral** wall, and **medial** wall.

  1. The roof: thin plate of **bone**, the **tegmen tympani**, which is **part** of the **petrous temporal bone**, It **separates** the **tympanic cavity** from the **meninges** and the **temporal lobe** of the **brain** in the **middle cranial fossa.**

  2. **The floor: thin** plate of **bone**, It **separates** the **tympanic cavity** from the **superior bulb** of the **internal jugular** vein.

  3. **The anterior wall:** its **lower** part is formed by a **thin plate of bone** that **separates** the **tympanic cavity** from the **internal carotid artery**. It has **two** canals;

  i. **lower** and **larger** leads into the **auditory tube**
  ii. and **upper** and **smaller** for the **tensor tympani muscle**.



Figure(48):Middleearcavity

4. **The posterior wall:**its **upper** part shows a **large**, **irregular** opening, the **aditus** to the **mastoid antrum** and **below** this is a **small, hollow, conical projection**, the **pyramid** which is the **origin** of **stapedius muscle.**
5. **The lateral wall:** is **largely** formed by the **tympanic membrane**
6. **The medial wall:** is formed by the **lateral wall** of the **inner ear**.
   i. The **greater part** of the **wall** shows a **rounded projection,** calledthe **promontory**, which results from the **underlying first turn** ofthe **cochlea**.
   ii. **Above** and **behind** the **promontory**; there is the **fenestra vestibuli** (**oval window**) which is **closed** by the **base** of the **stapes** separating it from **perilymph** in the **cochlea of inner ear**.
   iii. **Below the posterior end** of the promontory lies the **fenestra cochleae** (**round window**), which is **round** and **closed** by the **secondary tympanic membrane** separating it from **perilymph** in the **cochlea of inner ear.**

Contents: **air**, **auditory ossicles**, **muscles** and **nerves**.

Auditory ossicles:

Malleus, incus and stapes: they **transmit vibrations** of the **tympanic membrane** the**perilymph** of the **inner** ear.

Muscles:

Tensor tympani:

- it **originates** from **wall** of **auditory** tube and **inserted** at **handle of maleus**,
- **innervated** by **mandibular nerve**.
- It **pulls** the **handle of malleus medially** thus **tenses tympanic membrane.**

Stapedius:

- it **originates** from **pyramid** and **inserted** at **neck of stapes**,
- **innervated** by **facial** nerve.
- It **pulls stapes posteriorly** thus **prevents excessive oscillation** (**attenuation reflex**).

Nerves:

chorda tympani and **tympanic plexus**

Auditory tube:

- **36 mm long**, it **connects** the **anterior** wall of the **middle ear** cavity to the **nasopharynx**.
- Its **posterior third (1/3)** is **bony**, and its **anteriortwo thirds(2/3)** are **cartilaginous**.
- It **serves** to **equalize air pressures** in the **tympanic cavity** and the **nasal pharynx.**

Mastoidantrum:

air sinus lies **behind** the **middle ear** cavity in the **petrous** part of **temporal bone, mastoid air cells** are a **series** of **cavities present** in it

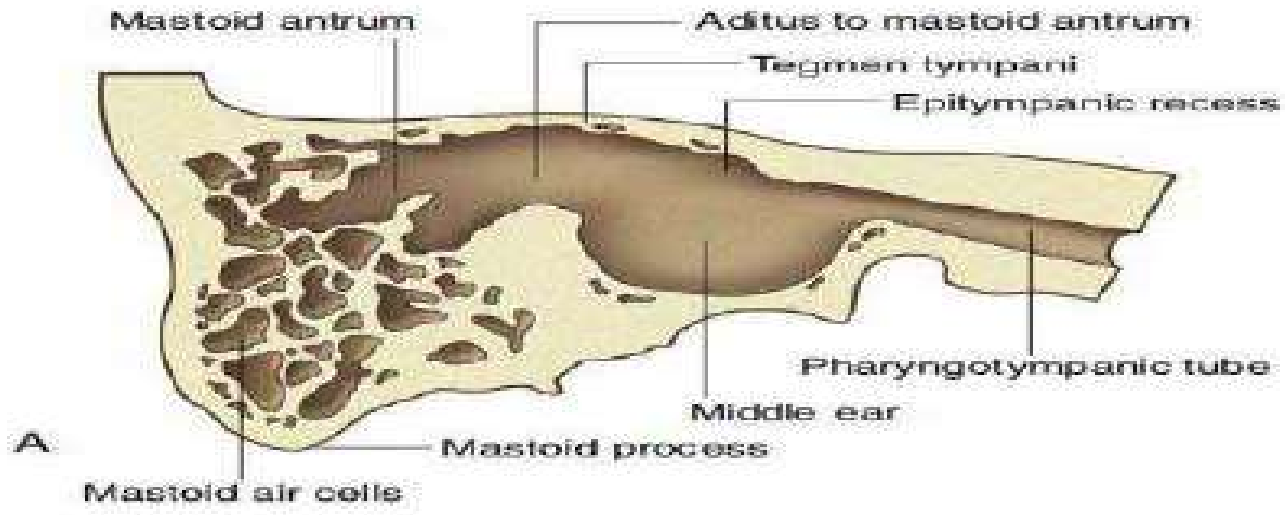


Figure (49): Mastiod antrum and mastoid air cells

3) *Inner ear:*

Itis formed of **bony labyrinth** and **membranous labyrinth**

Bony Labyrinth:

- It consists of **three** parts: the **vestibule**, **semicircular canals**, and the **cochlea**.
- They are **lined** by **endosteum** and **contain** a **clear fluid** called the **perilymph**.

The vestibule:

- is the **central** part of the **bony labyrinth**; it lies **posterior** to the **cochlea** and **anterior** to the **semicircular canals.**
- In its **lateral wall** there are the **fenestra vestibuli,** which is **closed** by the **base** of the **stapes** and the **fenestra cochleae** which is **closed** by the **secondary tympanic membrane.**

The three semicircular canals:

- **superior**, **posterior**, and **lateral open** into the **posterior** part of the **vestibule**.
- **Each canal** has a **swelling** called the **ampulla**.
- The **canals open** into the **vestibule** by **five orifices**, **one** of which is **common** to **two of the canals.**

The cochlea:

- it resembles a **snail shell;** it **opens** into the **anterior** part of the **vestibule**.
- It consists of a **central pillar,** the **modiolus**, around which a **hollow bony tube** makes **2andhalf spiral turns** .
- The **firstbasal turn** of the cochlea is **responsible** for the **promontory** seen on the **medialwall of the middleear.**

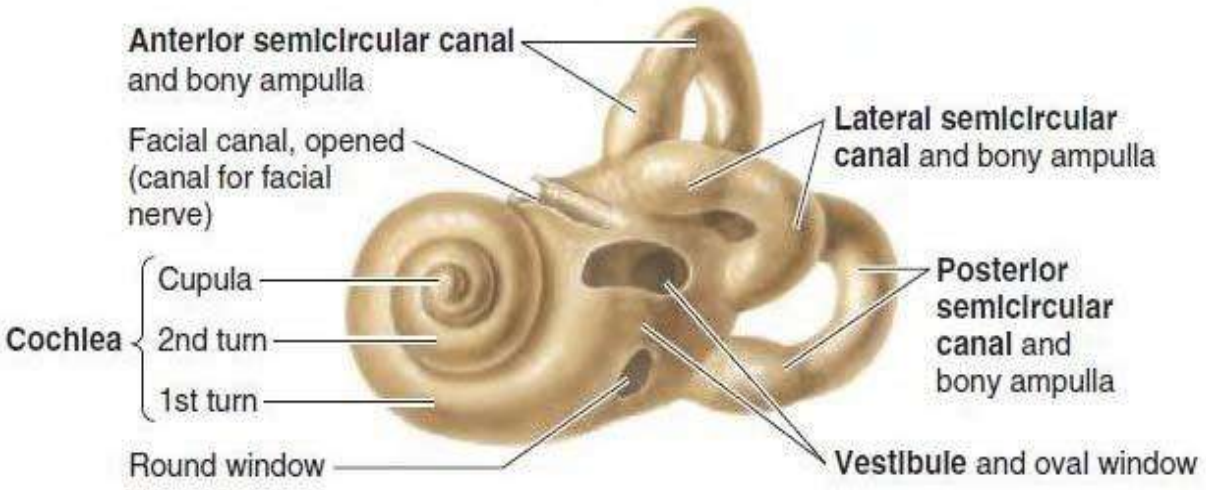


Figure (50): Bony labyrinth

Membranous Labyrinth:

itis **lodged** within the **bony labyrinth**.
Itis **filled** with **endolymph**.
It **consists** of the **utricle** and **saccule**, the **three semicircular ducts**, and **cochlear** duct.

The utricle and saccule:

- are **lodged** in the **bony vestibule; utericle** is **indirectly connected** to the **saccule** and the **ductus endolymphaticus** by the **ductus utriculosaccularis**,
- on the **walls** of the **utricle** and **saccule** there are **specialized sensory receptors** which **responsible** for **equilibrium**.

The semicircular ducts:

- are **much smaller** in **diameter** than the **bony semicircular canals** and have the **same shape**;
- they contain **sensory receptors** in the **ampullae** of the **semicircular ducts.**

Cochlear duct:

- duct is **triangular in cross section** and is **connected** to the **saccule** by the **ductus reuniens**,
- it contains **sensory receptors** for **hearing** (**Organ of corti**).

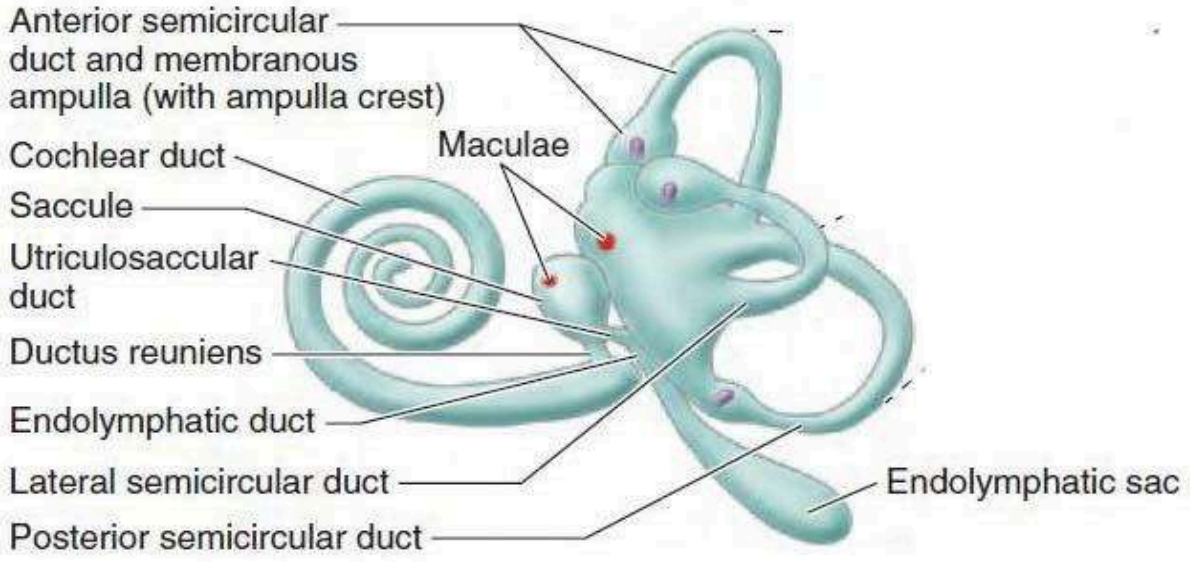


Figure (51): Membranous labyrinth

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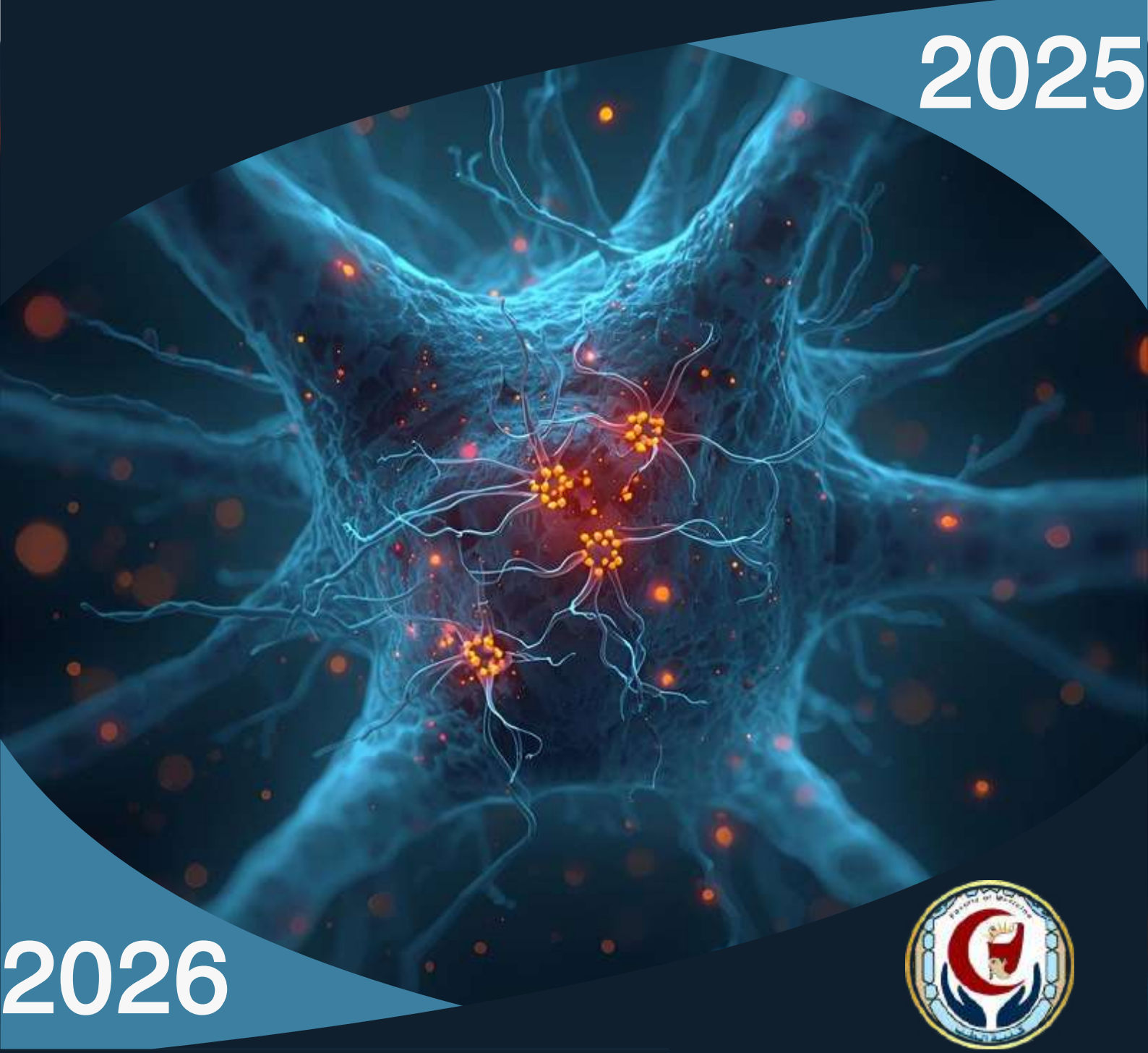

2026

SOHAG UNIVERSITY
FACULITY OF
MEDICINE

Neuroscience

Histology

NEU-316

page	Lecture
2	Histological structure of the eye and its accessory structures
13	Histological structure of sensory organs of hearing, balance, taste and smell
21	Types of neurons, synapses and neuroglia cells
27	Histological structure of meninges (pia arachnoid and dura matters) choroids plexus and different types of brain barriers
32	Myelination in PNS and CNS and Types of nerve fibers and nerve ending
39	Histological structure of the cerebral and cerebellar cortex

Histological structure of the eye and its accessory structures

Objectives:

- Bytheendof thelecture thestudent will beableto:

o Explain the histological structure of the eyeball.

Describe the ultrastructure of the retinal photoreceptors rods, cones and pigmented

epithelium.

o Describe the histological structure of accessory organs of the eye.

The eye

- It is the organ of vision. It consists of the eyeball and its accessory structures.
- The wall of the eyeball is made up of three layers (Fig. 1).
  - Outer fibrous layer.
  - Middle vascular layer.
  - Inner nervous layer (retina).

Microscopic structure of the eyeball:

[1] The outer fibrous

- Consists of the cornea and sclera.

The cornea (Fig. 2)

o Itisatransparentanterior portion and consists of:

- Stratified squamous non-keratinizing epithelium:
  - It has numerous naked nerve endings, the stimulation of which results reflexly in the blinking of the eyelids and in the flow of tears.

- **Bowman's membrane**
  - It is a protective membrane underlying the lining epithelium .It contains some collagenous fibers.
- **Substantia propria**
  - Lying beneath the Bowman's membrane it is a layer that forms 90% of the thickness of the cornea and formed of several lamellae of parallel collagenous bundles and flattened flbrobtasts in a connective tissue matrix rich in chondroitin sulfate. Cornea is always kept wet by tears and mucus from conjunctival glands (ulceration of the cornea occurs if it becomes dry).
- **Descement's membrane:**
  - It is an acellular smooth hyaline layer.
- **Descement's endothelium:**
  - Applied to the inner side of the Descement's membrane.

The sclera,Fig.3:

- It constitutes the posterior 5/6 of the outer layer of the eyeball and is composed of dense irregular white fibrous connective tissue containing some elastic fibers.

- **The sclera is opaque for the following reasons:**
  - The irregular arrangement of its fibers.
  - It has great water content.
  - Also its content of mucopolysacharides is probably different from those of the cornea.
- The sclera is perforated posteriorly by the fibers of the optic nerve at the lamina cribrosa.

[II] The middle vascular layer (Uveal tract)

- It is formed of the iris, ciliary body and the choroid.

The iris:

- It is the anterior portion of the uveal tract which has a hole at its center called the pupil. It is covered anteriorly with large fibroblasts and melanocytes, which is continuous with the corneal epithelium. Posteriorly, the iris is lined by two layers of pigmented epithelium beneath the anterior epithelium, there is a layer of poorly vascularized connective tissue followed by a more vascularized layer. These two layers form the stroma which contains muscle fibers i.e. the dilator and constrictor pupilae. The color of the iris is due to the presence of melanin pigment.

Ciliarybody:

- It is a triangular structure, the base of which faces the anterior chamber while its apex continues with the choroid, and it has the ciliary process medially. The bulk of the ciliary body consists of the ciliary muscle.

The choroid:

- The choroids are a thin highly vascular layer immediately beneath the sclera. It contains many blood vessels surrounded by loose connective tissue rich in fibroblasts, macrophages, lymphocytes and plasma cells. Melanocytes, present in limited numbers, and give this layer a brown color.

Aqueoushumor:

- It is a thin watery fluid containing most of the soluble constituents of the blood and formed mostly from the capillaries of the ciliary processes and the back of the iris.

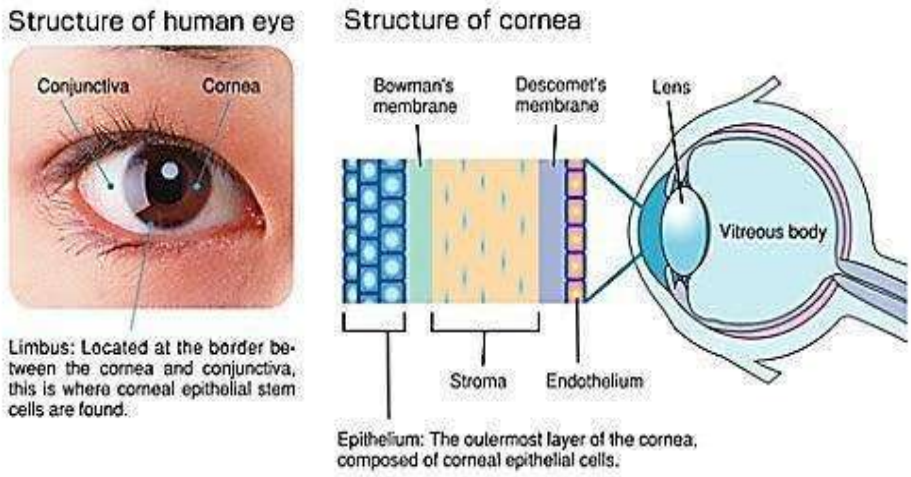

Fig. 1: Diagram showing the structure of the eye

- Vitreous humor: Is a mass of transparent gel, which is formed from water (99%) with hyaluronic acid and contains fine collagen like fibrils. It is limited by the inner limiting membrane of the retina and the lens.

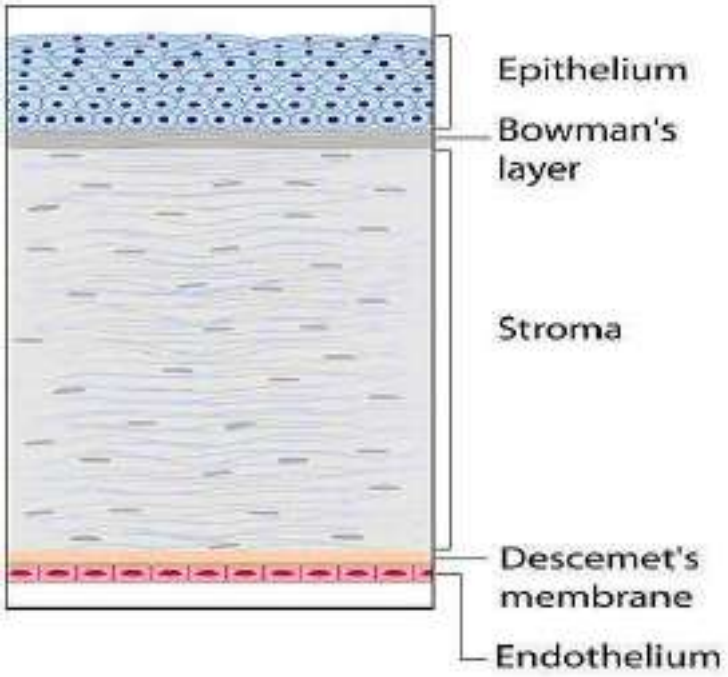

Fig. 2: The cornea of the eye.

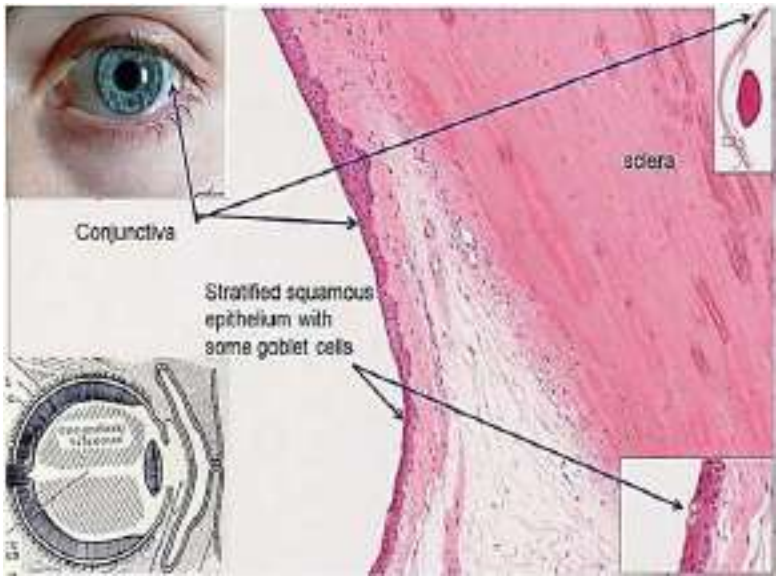

Fig. 3: The conjunctiva and sclera the eye.

❑ The lens. (Fig. 4)

- Isabiconvextransparent elastic structure. It is attached to the ciliary body by the zonule and surrounded by the lens capsule, which is a homogenous structureless membrane. This membrane is thicker anteriorly than posteriorly. Under the capsule there is a single layer of cells on the interior surface, which is thicker at the equator of the lens. At the sides of the lens the epithelium transforms into lens fibers which form the main substance of the lens, No cells are present on the posterior pole of the lens.

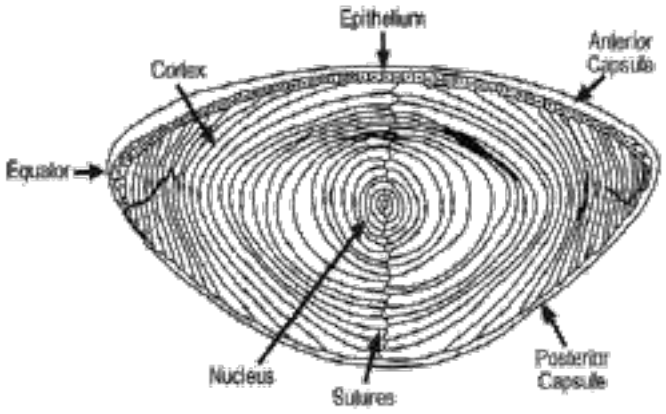

Fig. 4: The eye lens

[III] The inner layer (Nervous coat) (Fig. 5

- The innermost layer of the eyeball (The nervous coat) develops as an outgrowth from the forebrain (optic vesicle).

Being invaginated it divides into two layers the outer layer forms the pigmented epithelium of the retina.

❑ Structure

- The retina is formed of 10 layers from outward inward).

▪ Pigmented epithelium:

- It is a single layer of cuboidal cells with irregular apical and basal borders.
- Each cell has apical microvilli surrounding the outer segments of photoreceptors (rods and cones). There are numerous mitochondria around the spherical nucleus. The intermediate zone is filled with pigment granules, some granular and a granular endoplasmic reticulum and phagosomes containing lamellar inclusions which probably result from phagocytosis of membranous lamellae from the external ends of 33 the photoreceptor's outer segments. Lipofuscin granules also may be present representing residual bodies following process of phagocytosis.

▪ Function

- It absorbs light and prevents its reflection.
- They are concerned with the nutrition and support of the photoreceptors.

▪ **Layer of rods and cones:**

- Contains the modified dendrites of the photoreceptor cells (The rods and cones) which are crowded and appear as bacilli, thus, it is called "the bacillary layer". These are cells, which is sensitive to light. While the rod is sensitive to dim light, the cone cells are sensitive to bright light. Their number about 120 million per eye

- Outer limiting membrane:
- Outer nuclear layer:
- Outer plexiform (reticular) layer:
- Inner nuclear layer:
- Inner plexiform layer:
- Ganglion cell layer:
- Nerve fiber layer:
- Inner limiting membrane:

The Photoreceptors (Rode and cone cells) (Fig.7)

- These are cells, which is sensitive to light. While the rod is sensitive to dim light, the cone cells are sensitive to bright light. **Their number about 120 million per eye**

Fine structure of the rod cells:

- **L.M**

  This is a bipolar cell, which has a cell body, one dendrite and an axon. Its dendritic process is called the rod and is sensitive to dim light. It is located in the second layer of the retina i.e. "the layer of rods and cones"

- **E.M**

  The rod is composed of an outer thin segment, which is rod in shape, and an inner broad segment.

  - The two segments are joined together by a constriction, which contains a cilium.
  - The outer segment contains the transverse membranous discs, which are separated from each other by spaces and contain the visual purple (Rhodopsin). Not connected to the cell membrane and its continuously renewed.
  - The inner segment contains numerous mitochondria, many polyribosomes, Golgi bodies and rER. This inner segment is responsible for protein synthesis. The protein formed in the inner segment moves to the outer one by the cilium.
  - The cell bodies of the rod cells are present in the outer nuclear layer and are connected to the rods by the outer rod fibers, from the cell bodies, the inner rod fibers (Axons of the rod cells) end in the outer plexiform layer in rod spherules that synapse with the dendrites of the cells of the next layer.

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❑ Fine structure of the cone cells:

Cone cells are less numerous about 7 million per eye. It is similar to rod cells with the exception of the following point: 35 1. Their visual substance is called "iodopsin" which is sensitive to bright light. 2. The outer segment of the cone cells is cone-shaped. 3. Membrane discs are connected to the cell membrane and is not continuously renewed. 4. The outer cone fiber is short and thus the cell body is close to the outer limiting membrane and their nuclei are arranged in one row. The cone ends in an expansion called "the cone pedicle.

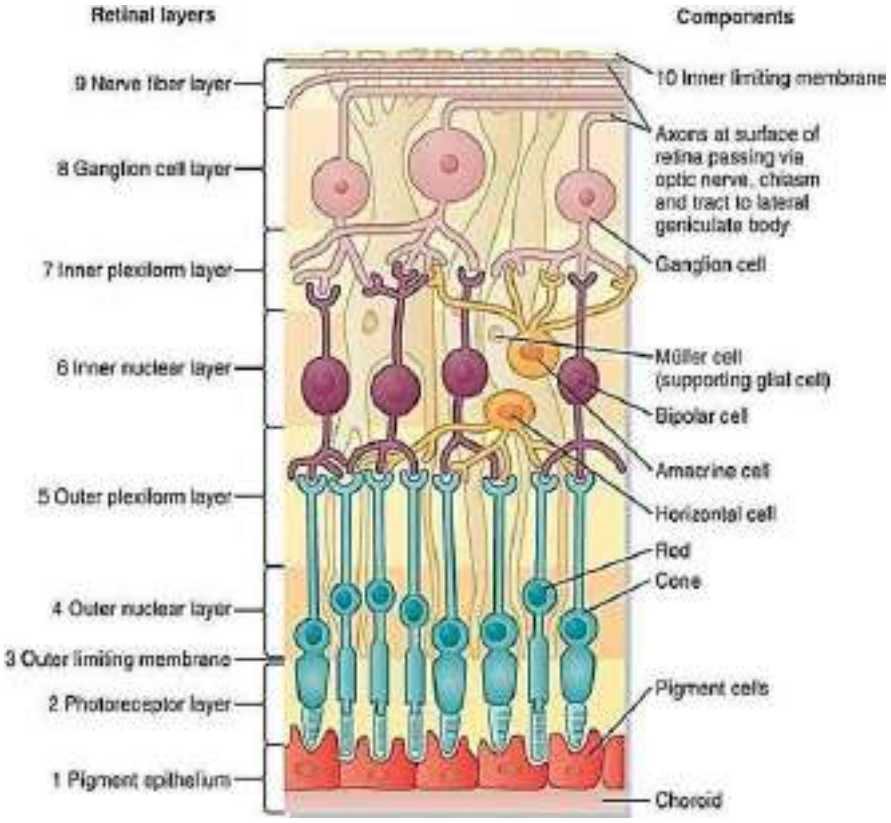


Fig5: The retina

The accessory organs of the eye:

- These include the eyelids, conjunctiva and the lacrimal apparatus.

The eyelids:(Fig. 6)

- There are two, upper and lower eyelids, which cover the opening of the bony orbit in which the eyeball is present. The two eyelids meet at the palpebral fissure. The

lids are important

for protection and lubrication of the anterior portion of the eyeball. Each lid is covered with

skin from the outside and lined with a mucous membrane from the inside.

- At the lid margin, the skin contains 3-4 rows of special hairs (Eyelashes), which penetrate

deeply into the dermis. Posterior to the eyelashes there are modified sweat glands (Glands of Moll) and sebaceous glands (glands of zeiss). Beneath the skin there are bundles of striated muscle fibers while under the mucous membrane there is a dense fibrous tissue containing the tarsal glands (Meibomian glands) which are special sebaceous glands the ducts of which open at the lid margin.

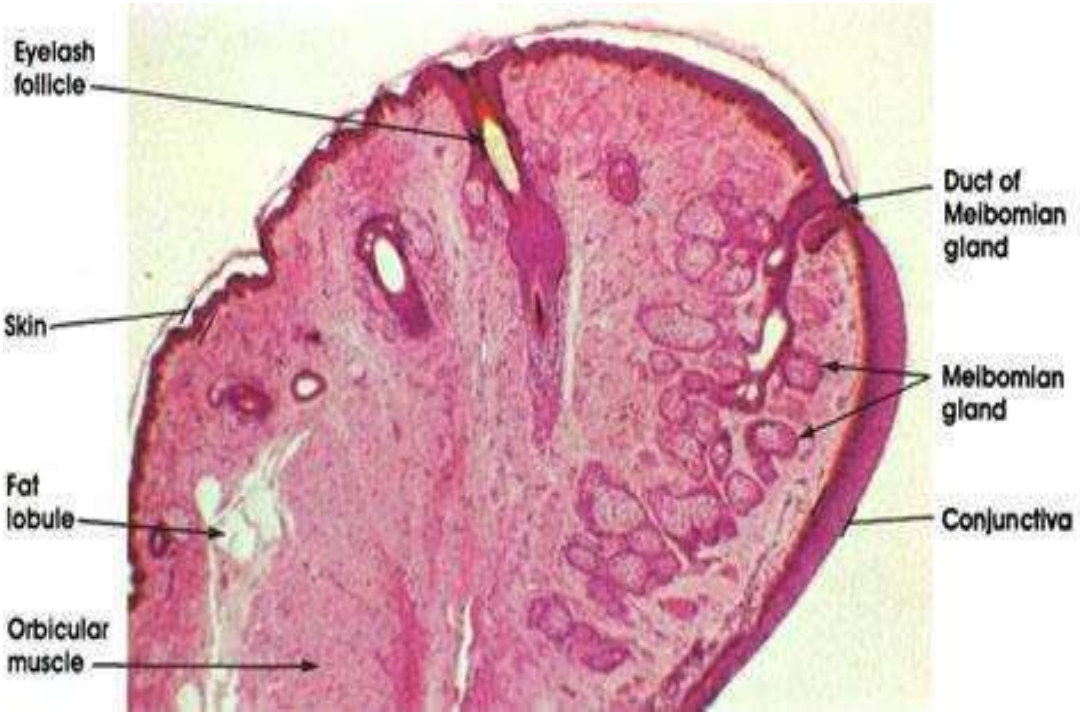


Fig.6 Histological structure of the eye lid

The conjunctiva:

- Is a thin transparent mucous membrane lining the inner surface of the eyelid "palpebral conjunctiva" and then reflected to cover the anterior surface of the eyeball "bulbar conjunctiva". The epithelium is stratified squamous over the bulbar conjunctiva near the limbs where it continues with that of the cornea. It becomes stratified columnar epithelium with goblet cells at fornix then stratified squamous epithelium at the palpebral conjunctiva and rests on a connective tissue rich in blood vessels, lymphatics and nerve fibers. The epithelium has a great capacity for reversible swelling and congestion.

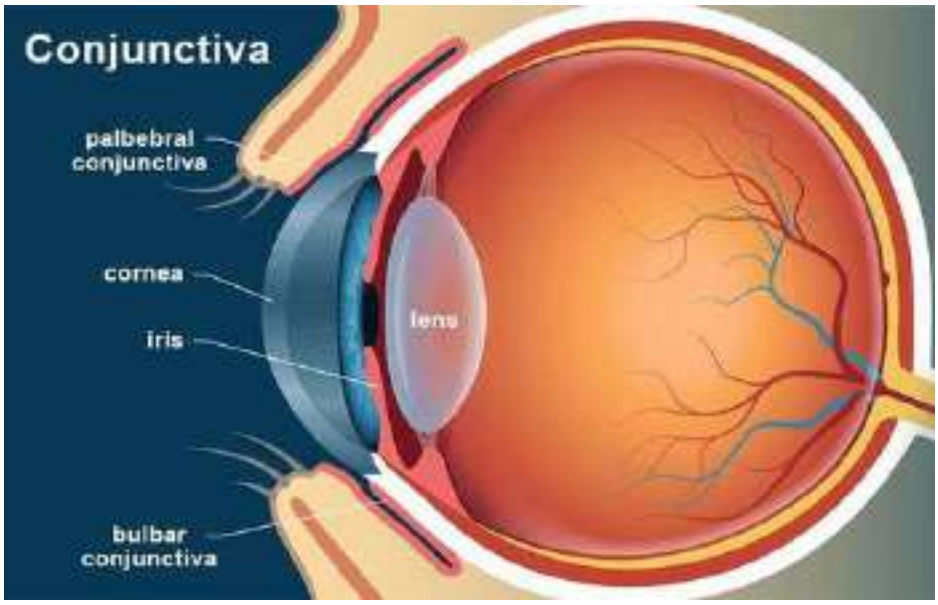


Fig., 6 conjunctiva

- The Lacrimal glands:
  - Itiscompoundtubuloalveolar glands present in the superolateral part of the orbit
- The optic nerve:
  - Isformedofthe non-myelinated axons of ganglion cells of the retina.

Histological structure of sensory organs of hearing, balance, taste and smell

Objectives:

- Bytheend of the lecture the student will be able to:

- Describe the structure of Organ of Corti, the vestibule and the semicircular canals.
- Identify the structure of taste buds and olfactory epithelium
- Explain the structure of olfactory epithelium

Histological structure of Sensory organ of hearing
(Organ of Corti)
(Fig. 1)

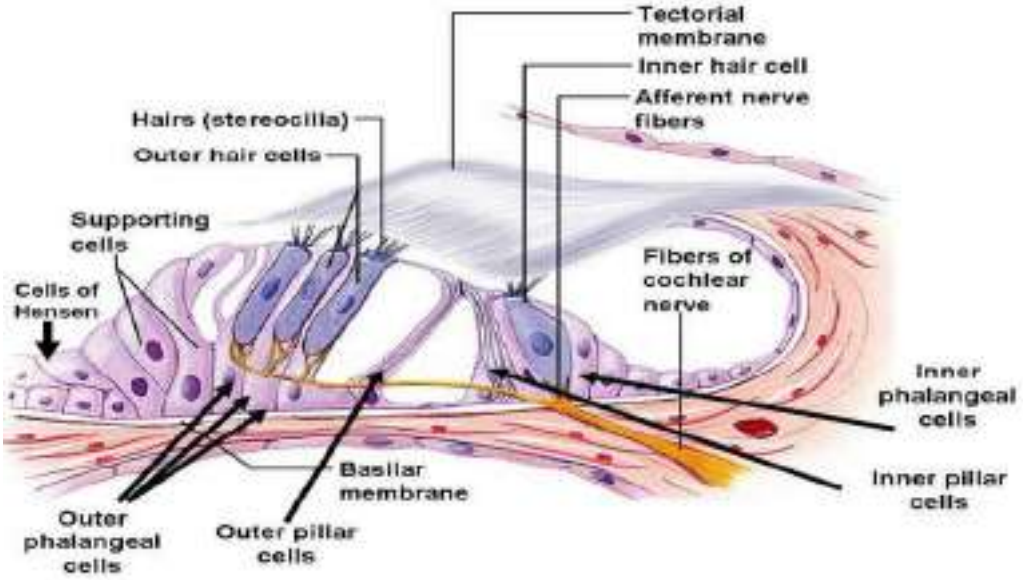


Fig. 1 organ of Corti

Definition:

- Organ of Corti is a structure in the cochlea of the inner ear that allows for the transduction of sound vibrations into neural signals

- Structure: Neuroepithelial structure rest on the basilar membrane. It is coverd with a gelatinous membrane called tectorial membrane.

Types of cells:

A: Supporting cells which are

- **Inner &outerpillarcells:**
  - Two rows of tall cells on either side of the Tunnel
- **Phalangeal(Supporting)cells:**
  - Tall columnar cells to support the hair cells & its nerve endings. They have concavities to hold the hair cells.
- **Hensencells:**
  - Supporting columnar cells outside the organ of corti (at the lateral part)

B: Sensory hair cells:

- **Thereare 2typesof haircells**
  - Outer hair cells arranged in 3 to 5 rows
  - Inner hair cells arranged in one row.

- The bases of these cells do not rest on the basilar membrane, but they are present on cup like processes of the phalangeal cells.

- **The apicalsurface**
  - of Hair cells have hairlike processes on their apical surface. These are steriocilia (non-motile cilia), they are long microvilli.
  - The basal surface of hair cells is surrounded by nerve endings. The nerve fibers are the dendrites of the bipolar nerve cells of the spiral ganglia in the inner ear. The

axons of these bipolar nerve cells form the **cochlear nerve** which passes first in the modiolus of the cochlea.

Sensory organ of balance
(Receptor cells are in 2 organs, the vestibule and the semicircular canals)

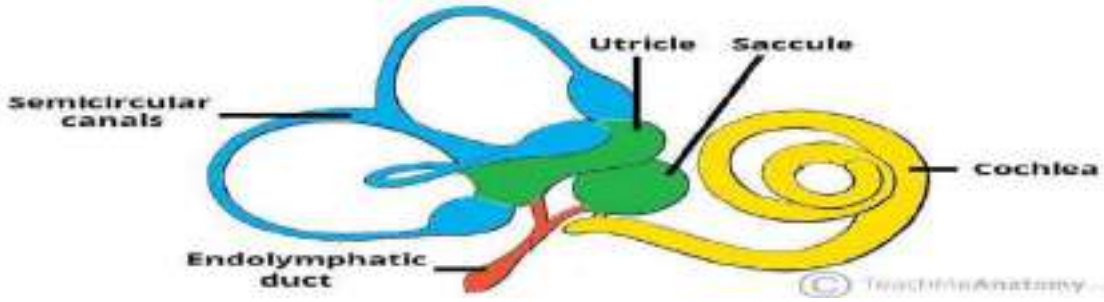


Fig.2: diagram for the semicircular canals, vestibule and cochlea

- The bony vestibule is filled with perilymph.
- It contains 2 sacs of membranous labyrinth; the utricle and saccule which are filled with endolymph. The utricle and saccule sacs are lined by simple squamous
- mesothelium. Each sac contains a neuroepithelial thickening in its wall called a macula.
  - In the saccule = Maculae sacculi.
  - In the utricle = Maculae utriculi

Structure of the macula (Fig. 3):

- The macula is formed of 2 main types of cells resting on a clear basement membrane and covered with a thick membrane (otolithic membrane).
  - These are.
    - Supporting cells
    - Sensory (Hair) cells containing stereocilia. The stereocilia are embedded in the otolithic membrane.

Functions:

The macula of the saccule is responsible for the sense of gravity (tilting or lifting the head) and linear acceleration or deceleration. The macula of the utricle is the organ of balance which regulates the position of the head in relation to the gravity.

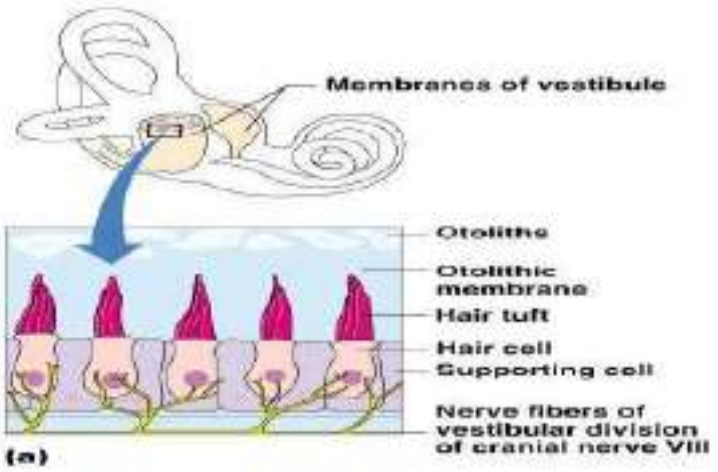


Fig. 3: diagram of structure of the Macula

The semicircular canals

- 3 perpendicular canals at right angle to each other
- Each canal formed of 2 parts
    - o Bony part.
    - o Membranous part: contain endolymph.
- They are suspended in the perilymph by C.T. strands. They are lined by simple
- squamous epithelium lying on a C.T. layer.
- Each semicircular canal has an expanded end called ampulla.
  The ampulla contains a neuroepithelial structure called crista ampullaris

The structure of crista ampullaris (Fig: 4):

- Neuro epithelial thickness include:
    - Two types of cells
        - o 1- Bipolar hair cells (the same structure as the maculae)
        - o 2- Supporting columnar cells.

- The hair cells are embedded in gelatinous mass called **"Cupula"** with no crystals.

Functions of crista ampularis

- It is stimulated by the movement of endolymph following angular acceleration of the head.

-The impulses are conducted along the vestibular nerve cells in the pons

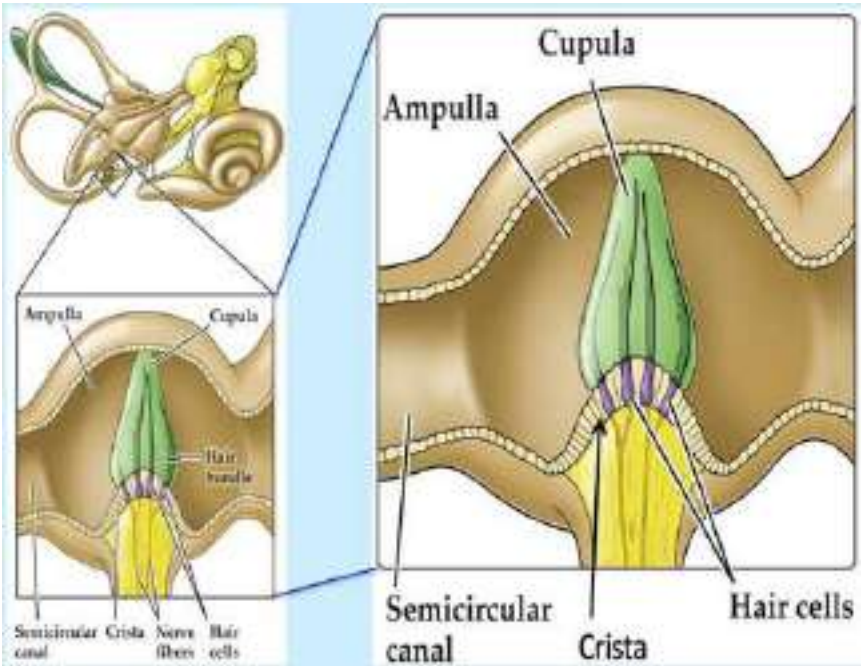


Fig.4: structure of tha Ampula containing crista ampularis

Olfaction – The Sense of Smell

- **Olfactoryreceptors** are in the roof of the nasal cavity
  - Neurons with long cilia
  - Chemicals must be dissolved in mucus for detection

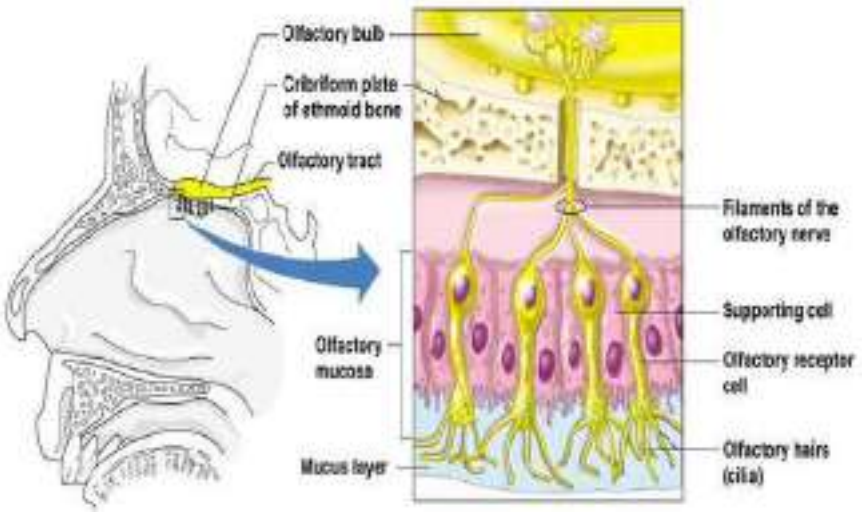


Fig. 5: The olfactory epithelium

- Impulses are transmitted via the olfactory nerve
- Interpretation of smells is made in the cortex (olfactory area of
- temporal lob(

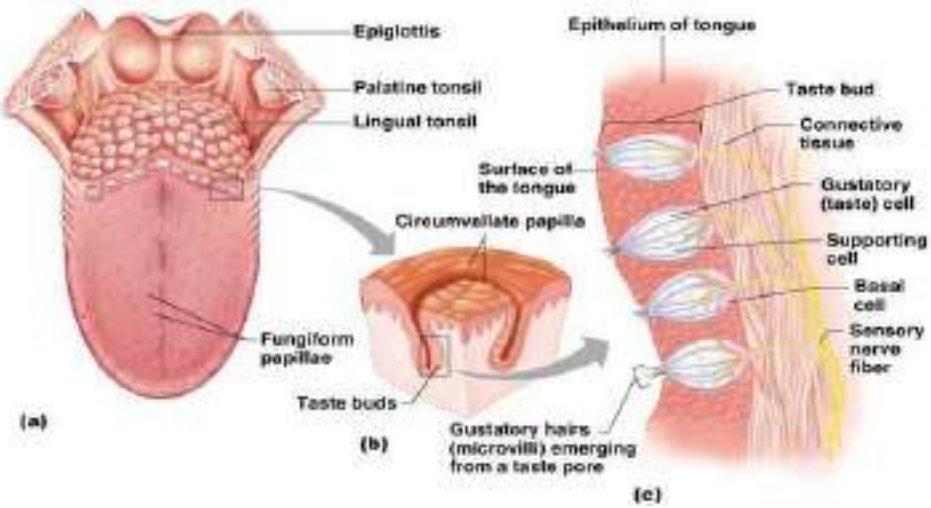
The Sense of Taste

- Taste buds house the receptor organs
- Location of taste buds:
  - The dorsum of the tongue
  - Soft palate
  - The posterior surface of the epiglottis.
  - Pharynx posterior wall down to the cricoid level Glossopalatine arch.

• Thetongue is covered with projections called papillae

- Filiform papillae – sharp with no taste buds
- Fungifiorm papillae – rounded with taste buds
- Circumvallate papillae – large papillae with taste buds

- Taste buds are found on the sides of papillae



(Fig. 6) Structure of Taste Buds

It lines the olfactory area of nasal cavity (upper part of nasal septum, the roof of nasal cavity, upper part of lateral wall of nasal cavity) The epithelium of olfactory region is **a** pseudostratified columnar epithelium composed of three types of cells:

1. The olfactory cells: they are the receptors of smell.Each is specialized for certain basic odor.

L.M

- They are bipolar neurons having an apical dendrite, which project between the tall columnar cells to reach the surface epithelium, and a centrally directed axon, which project deep to penetrate the basement membrane.

2. The sustentacular: or supporting cells are the most numerous types.

L.M: - They are tall columnar cells with a narrower lower parts and more expanded apical parts, which have microvilli. These sustentacular cells are secretory and supportive in function.

3. The basal cells: They are small, spherical or cone-shaped cells. It forms a single layer at the base of the epithelium. These cells are undifferentiated cells, which remain capable of division and transformation into other types of cells in the olfactory epithelium.

❑ Taste Buds (Fig. 6)

- The tastebuds aresmall neurosecretory bodies extending from the basal lamina of the epithelium almost to the surface. They open into the surface
- by taste pores. By LM appear as minute pale-stained barrel shaped
- structures present in the epithelium of mucous membrane covering lingual papillae.

- **Each bud is formed of:**
  - Dark supporting or sustintacularcells are long, crescentic in shape with pale nuclei. They form the wall of the bud
  - Light neuroepithelial or Gustatory cells are distributed between the supporting cells. They are long slenderical cells with central elongated nuclei .The cells have long microvilli called taste hair which project into the taste pore. Hairs are stimulated by chemicals dissolved in saliva.

- Basal cells are present close of the basement membrane and give rise to other two types.

References:

Text Books:

- Integrated system

- Integrated medical sciences

- Clinical Neuroanatomy

- Basic Histology

-Ross, Michael H Histology: a text and atlas: with correlated cell and molecular biology/ Michael H. Ross, Wojciech Pawlina.-6 th ed.

MEDICAL HISTOLOGY

Histological structure of the neurons and neuroglia cells

Objectives

➢ By the end of this unit, the student should be able to:

1. Mention the different types of neurons in the body
2. Describe the structure of neurons and nerves
3. Identify different types, structure and function of neuroglia

Contents

- Structure of the neuron (LM & EM) - cell body - axon – dendrites
- Structure of nerve fibers
- Neuroglia

NEURONS

- Neurons are highly differentiated cells, of various sizes and shapes with the highest degree of conductivity. The neuron consists of a cell body (Perikaryon) and processes axon and dendrites

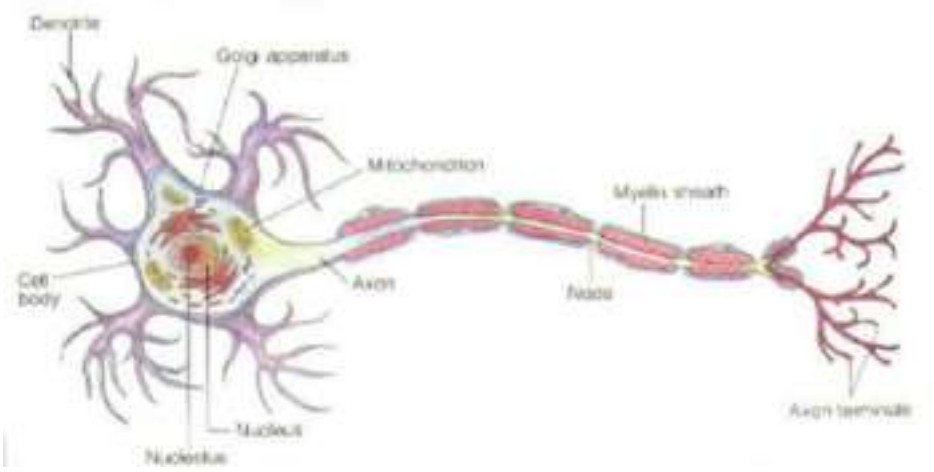


Fig. (1): Schematic drawing of neuron

The cell body

- Includes the neuroplasm (cytoplasm) and nucleus. The processes attached to it are the axon and dendrite(s).
- Nerve cells consist of nucleus and neuroplasm

L.M

- The nerve cells have different shape with pale acidophilic cytoplasm with deep basophilic masses of Nissl granules or bodies and oval vesicular pale-staining nuclei with a large and clear nucleoli
- The staining of nerve cell body by silver impregnation causes aggregation of neurofilaments and forming neurofibrils that appears brown or black

E.M

- The nerve cell is protein-producing cell
- Nucleus is large, euchromatic with distinct nucleolus
- Cytoplasm contains Nissl granules (aggregation of RER and polyribosomes), neurofilaments, microtubules, Golgi apparatus, mitochondria, lysosomes and lipofuscin pigment. The latter is more numerous in older individuals

❑ NEURAL PROCESSES

➢ There are twotypes:

1. **Dendrite**
2. **Axon**

Dendrite

- One or many in each cell
- Thick at the base, then branching repeatedly to increase area receiving impulse and becoming tapering toward the apex
- Some may have short spine like processes
- the cytoplasm contains mitochondria, ribosomes, RER and neurofilaments and there is no Golgi apparatus

Axon (nerve fiber)

Structure

- Each cell has only one axon The point of axon origin is called axon hillock
- Straight, regular in diameter and smooth surface Axoplasm contains
- mitochondria, transport or synaptic vesicles, large number of
- neurofilaments and microtubules. It is free of Nissl granules and depends on the perikaryon for maintenance The terminal segment of an axon branches into telodendria The distal end of the telodendrium enlarges into axon
- terminals
-

Organization

- The whole nerve (e.g. sciatic nerve) is enclosed by the connective tissue sheath called 'epineurium'
- Each nerve is made of fasciculi which are enclosed by the connective tissue sheath called perineurium
- Each fasciculus is made of many nerve fibers which are enclosed by the connective tissue sheath called 'endoneurium

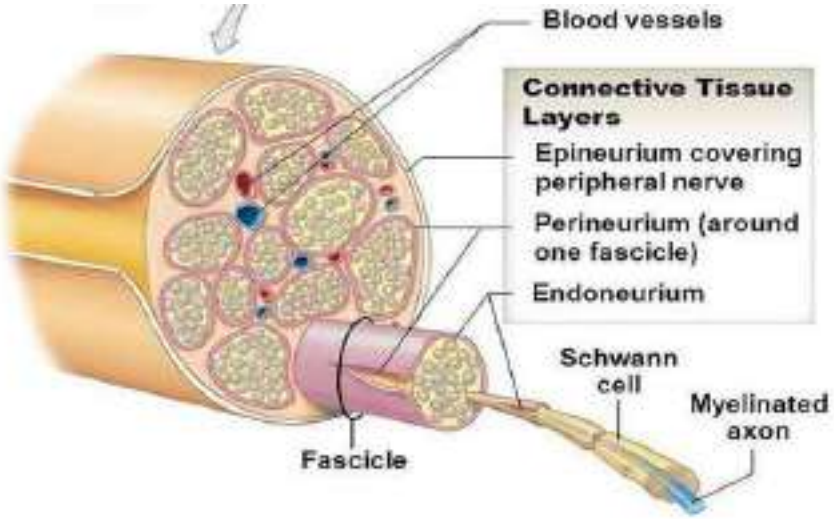


Fig.2 Organization of nerve fibers

Types of Neurons

I-According to the number of dendrites

1. Pseudo-unipolar (unipolar)
2. Bipolar
3. Multipolar

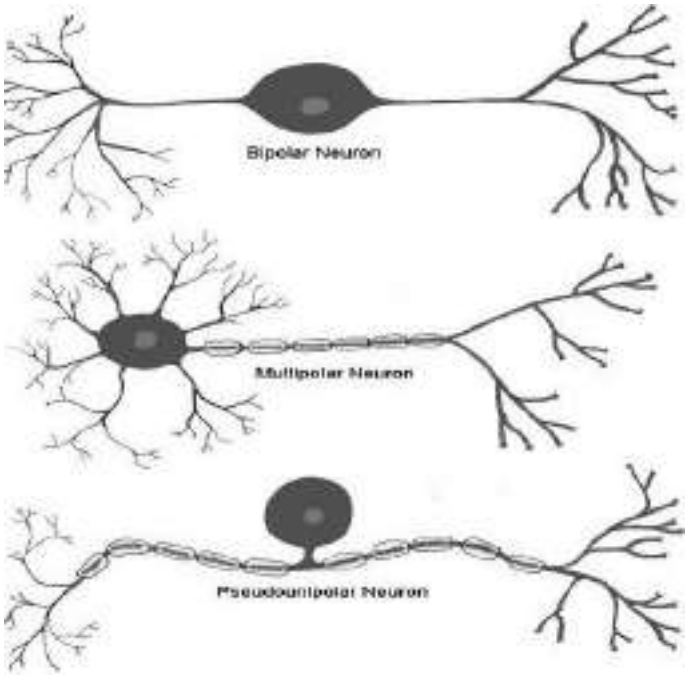


Fig.(3): Schematic drawing of different types of neurons

- Multipolar neurons have a variety of shapes of their cell bodies:
  - **A. Stellate neurons**: star-shaped and make the majority of multipolar cells
  - **B. Purkinje cells**: have flask-shaped or fusiform cell bodies
  - **C. Pyramidal cells**: have cell bodies in the form of pyramids

II-According to the function of fibers

1. **Sensory neurons**
2. **Motor neurons**
3. **Interneurons**

III-According to the length of fibers

1. **Golgi type I :** the axons are long
2. **Golgi type II :** the axons are short

❑ Neuroglia (glial cells)

- In CNS, there are astrocytes, oligodendroglia, microglia, and ependymal cells
- In PNS, there are Schwann cells and satellite cells

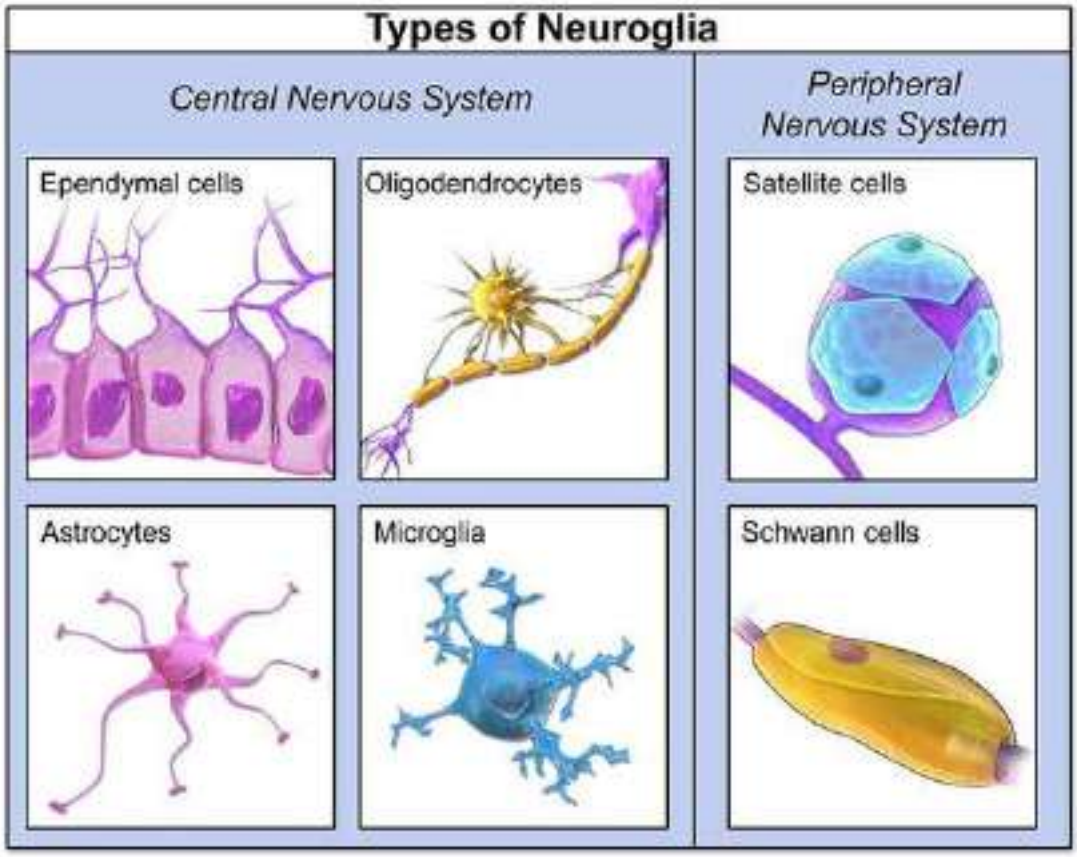


Fig.4 glial cells

Astrocytes

- They are two types fibrous and protoplasmic astrocytes
- They are star-shaped cells with long branching processes which extend between neurons and form end feet on the wall of capillaries
- They contain glial fibers (bundles of neurofilaments) in cytoplasm
- Nuclei are quite large, round or ovoid shape
- They share in blood-brain barrier
- After injuries or inflammation of CNS they proliferate and form fibrous scar

Oligodendrocytes (oligodendroglia)

- Found in white matter between nerve axons
- In gray matter they associated with axons or neural bodies.
- They are small cells with a few short cytoplasmic processes and denser cytoplasm
- Have an oval, dark, eccentrically located nucleus
  Responsible for the formation of myelin sheath in CNS

Microglia Cells

- Monocyte in origin
- They are small, ovoid with cytoplasmic processes have dense oval nuclei
- Cytoplasm contains numerous lysosomes
- They remove debris in injury or inflammation as they become motile and phagocytic

Ependymal Cells

- Similar to epithelia in that they line the ventricles of the brain and the central canal of the spinal cord, and their lateral borders bounded by a junctional complex
- Consists of cuboidal or low columnar cells with a large basal nucleus
- The apical surface provided with motile cilia to move the CSF

Schwann cells

- They are responsible for myelination in the peripheral nervous system One Schwan
- cell forms myelin around a segment of one axon
- **L.M:** Osmic acid is used to stain the myelin, allowing the dark (osmiophilic) myelin of Schwann cells to be clearly distinguished

E.M: the lamellar structure and Schwann cell cytoplasmic content can be visualized clearly, microfilaments.

Satellite cells

- They form one layer of flattened or cuboidal cell with round, ovoid and dark nucleus.
- They are surrounding the Neuron in ganglion

Histological structure of meninges, choroid plexus and different types of brain barriers

Objectives

➢ By the end of lecture student should be able to:

1. Define the meninges
2. Mention the different types of meninges and their structure
3. Explain the structure and importance of the choroid plexus
4. Describe the different brain barriers

Meninges

Definition

- Are the membranous covering of the brain and spinal cord

Types

➢ They are 3 layers from outer to inner

1. Dura mater
2. Arachnoid mater
3. Pia mater

- The arachnoid and pia mater are linked together and are considered a single membrane called pia-arachnoid

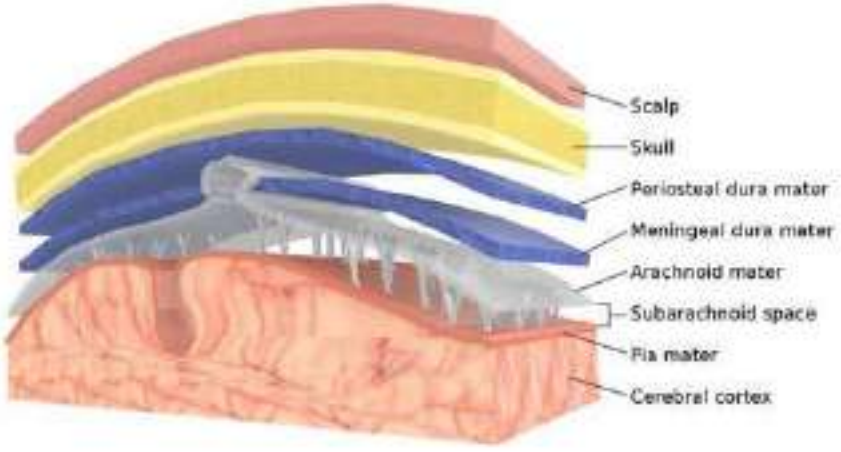


Fig.1: Meninges

❑ Histological structure of meninges

1) Dura mater

- It is composed of dense connective tissue continuous with the periosteum of the skull. The dura mater that envelops the spinal cord is separated from vertebra by epidural space that contains blood vessels, loose connective tissue and adipose tissue (Fig.2)

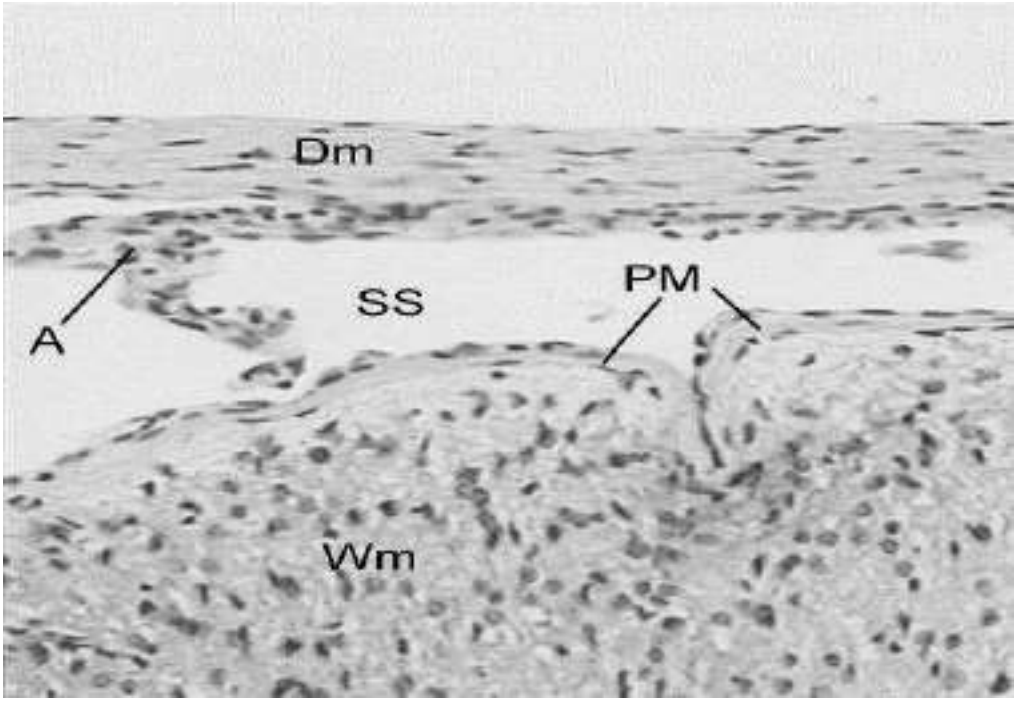


Fig.2: Histological structure of meninges

2) Arachnoid mater

➢ Ithas 2 layers:

- One is in contact with the dura mater and a layer of trabeculae connecting the arachnoid with pia mater
- There are cavities between these trabeculae forming the subarachnoid space filled with cerebrospinal fluid
- The subarachnoid space communicates with the ventricles of the brain
- This space protects the central nervous system from trauma. It acts as a hydraulic cushion
- The arachnoid is composed of connective tissue devoid of blood vessels and covered with simple squamous epithelium on its surfaces
  In some areas, the arachnoid perforates the dura mater, forming protrusions that terminate in venous sinuses in the dura mater and called arachnoid villi. Its function is to reabsorb CSF.

3) Pia mater

- It is composed of a layer of loose C.T containing many blood vessels (fig.2)
- The pia mater follows the irregularities of the surface of CNS and penetrates it to some extents along with the blood vessels and disappears before blood vessels transformed into capillaries. Its surface is covered with squamous epithelium
- The tunnel through which blood vessels penetrate CNS with pia mater is called perivascular space

Choroid plexus and cerebrospinal fluid (CSF)

Structure

- It is formed of invaginated folds of pia mater that penetrates deep in the interior of the brain ventricles (fig.3)
- It consists of loose c.t covered with simple cuboidal or low columnar epithelium (ion transporting cells)

Function

- Formation of CSF that fills the brain ventricles, central canal of the spinal cord, subarachnoid space and perivascular space (fig.4)
- CSF is important for the metabolism of CNS and act as shock absorber.

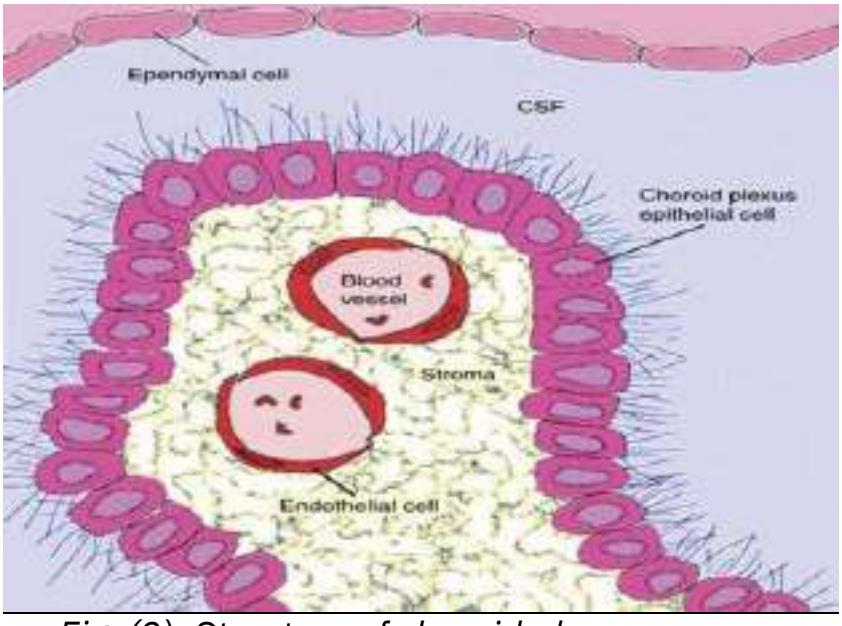


Fig. (3): Structure of choroid plexus

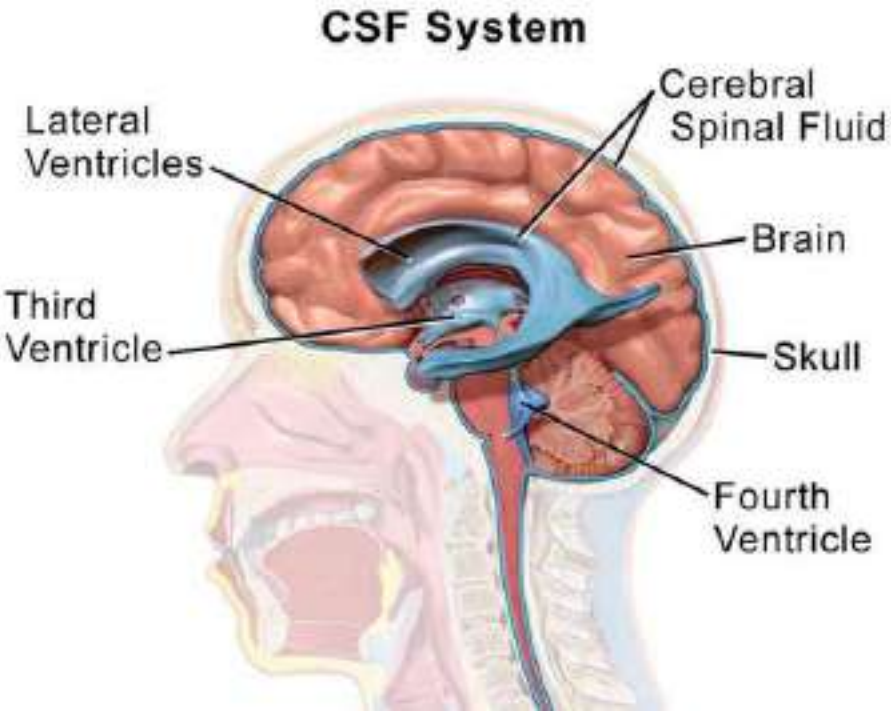


Fig. (4): CSF system

Brain barriers

1. Blood brain barrier
2. Blood CSF barrier
3. CSF brain barrier

1) Blood brain barrier

Structure

a) Endothelial cells of brain capillaries with tight junctions
b) Basement membrane of blood capillaries
c) Astrocyte processes.Fig(5)

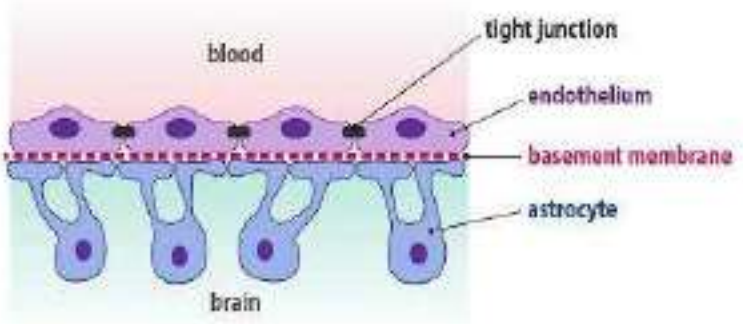


Fig. (5): Blood brain barrier

2) Blood CSF barrier

Structure

a) Endothelium of blood capillaries of choroid plexus
b) Basement membrane of capillary endothelium
c) Choroidal epithelium

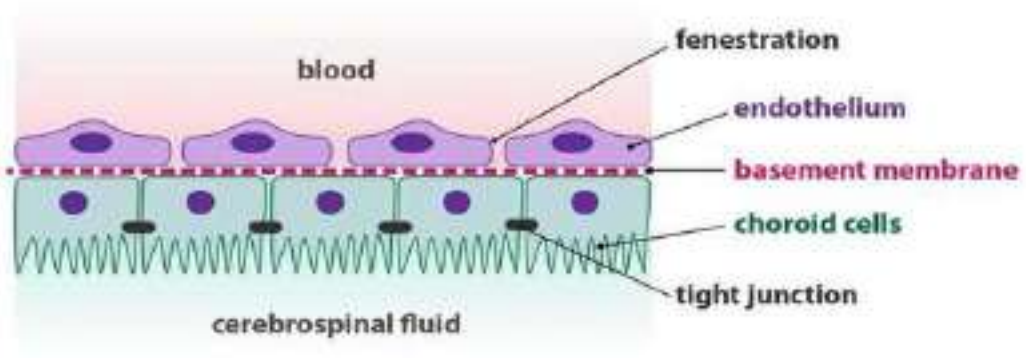


Fig. (6): Blood CSF barrie

2) Brain CSF barrier

Structure

- Gap junction between ependymal cells lining brain ventricles foot processes of astrocyte. fig.(7)

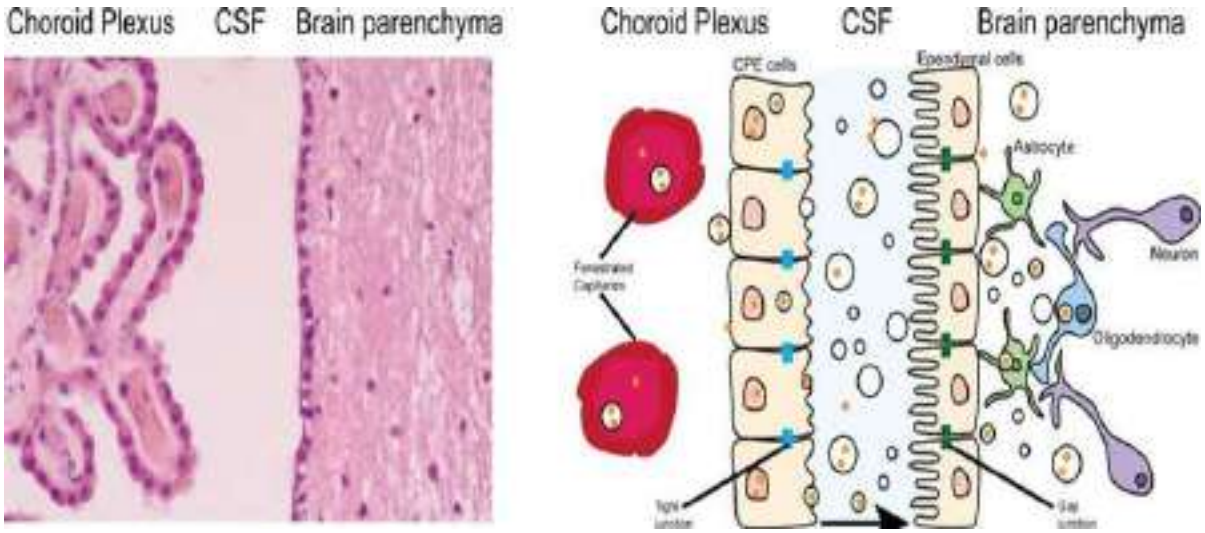


Fig. (7): Brain CSF barrier

Myelination in PNS and CNS and Types of nerve fibers

Objectives

By the end of lecture student should be able to:

4. Identify the mechanism of myelination inperipheral and central nervous system
5. Compare between myelination in peripheral and central nervous system
6. Mention the characters of different typesof sensory and motor nerve endings

Contents

- Myelination
- Types and structure of nerve endings (receptors & effectors)
- Types of nerve fibers
- Structure of synapse

Myelination in PNS

- As the plasmalemma of the Schwann cell encircles the fiber it forms a membranous sheath enclosing it called the myelin sheath. So myelin is composed of Schwann cell membrane wrapped spirally up to fifty times around the axon
- Myelin sheath is deficient at certain points along the axon known as nodes of Ranvier. They correspond to the spaces between adjacent Schwann cells
- The distance between two nodes is the internodal segments
- The thickness of myelin is directly proportional to the conductivity of the nerve as it faster with large diameter and thick myelin sheath

❑ Morphological Characteristics of Myelin Sheath in PNS

- Myelin sheath is seen in H&E-stained slide as an empty (vacuolated) layer between the axolemma and neurolemma (cell membrane of Schwann cell) and can be demonstrated using fat stains
- Myelin sheath is formed by overlapping the Schwann cell membrane that push the cytoplasmic contents to the periphery
- One Schwann cell can share in the myelination of only one axon
- At the EM level, it displays many concentric lamellae with alternate light and dark lines
- Unmyelinated fibers are included inside only one layer of the Schwann cells membrane without nodes of Ranvier. As many as 12 axons may be seen within one Schwann cell

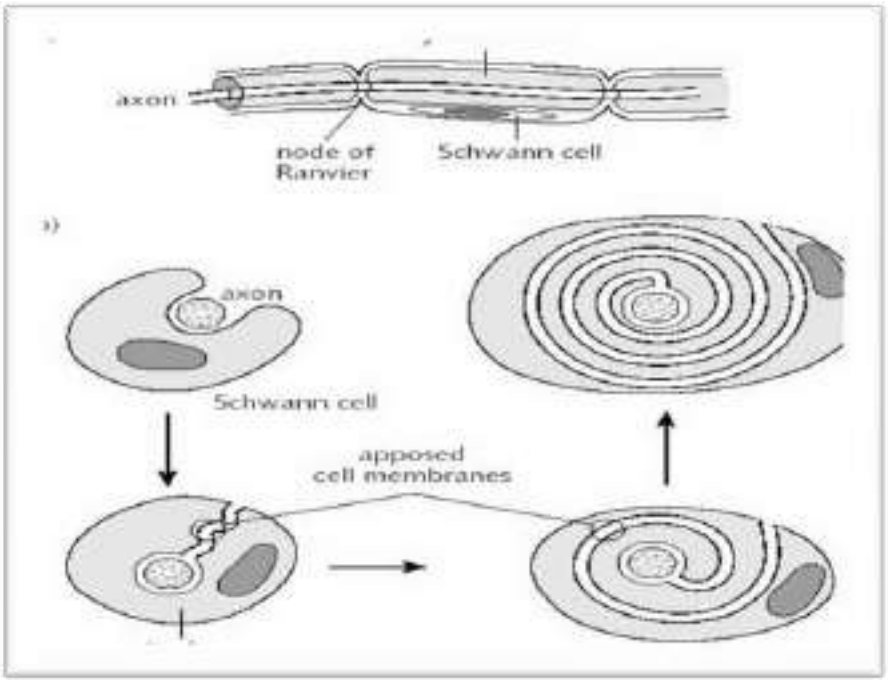


Fig. 1: Myelination in peripheral nerves

Myelination in CNS

➢ It differs from that in PNS:

- the oligodendroglia, is responsible for myelin formation. It sends short processes away from cell body which become flattened and wrap the axons resulting in various degrees of myelination the myelinated fiber is located far from the body of the oligodendroglia and is therefore considered to be lacking a neurolemmal sheath
- One oligodendroglia can share in the myelination of more than one axon
- The smaller axons of the CNS are nonmyelimated. Non myelinated fibers of the CNS are not particularly related to the oligodendrocytes.

❑ Types of nerve fibers:

According to the myelin and the neurolemmal sheath, the nerve fibers are classified into:

A. Myelinated with neurolemmal sheath: in thePNS
B. Myelinated without neurolemmal sheath: intheCNS
C. Non-myelinated with neurolemmal sheath
D. Non-myelinated without neurolemmal sheath

According to the thickness of myelin sheath the nerve fibers areclassified into:

A. Heavily myelinated
B. Moderately myelinated fibers
C. Lightly myelinated fibers

Nerve endings

- Nerve endings are always in contact with a transducer cell that converts a mechanical, thermal or chemical stimulus to an electrical potential in the nerve ending. There are three main types of nerve endings:

1. Synapses – on other neurons
2. Motor nerve endings – on skeletal muscle
3. Sensory nerve endings – on receptors

SYNAPSES

- A synapse is a specialized membranous contact between the axon of one cell and the dendrite, axon or soma of another cell. It may be axosomatic, axodendritic or axoaxonic (Fig. 2)

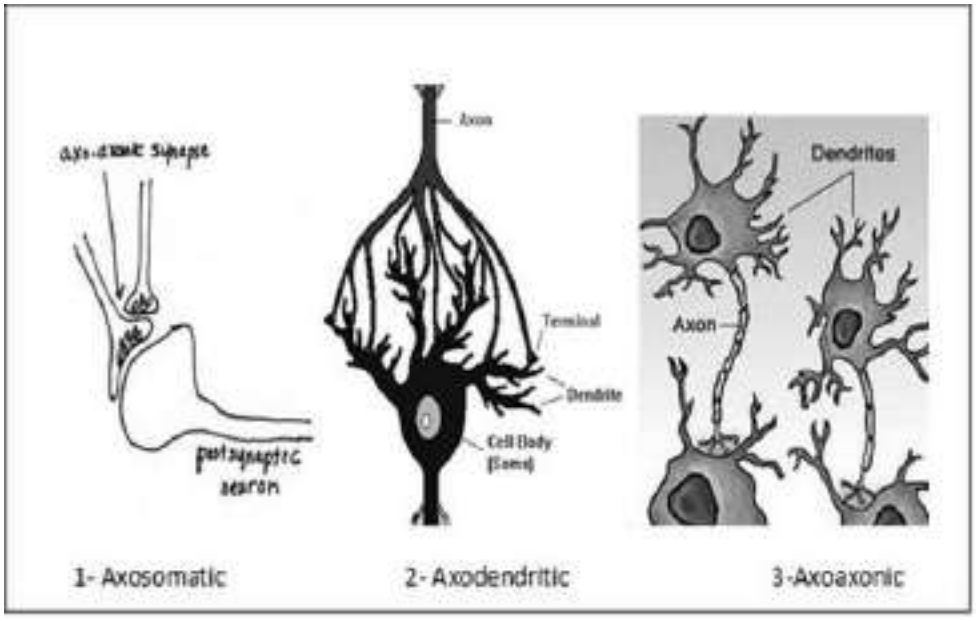


Fig. 2 Types of synapses.

Structure(Fig. 3)

1. Bulbous enlargement of an axon (Axon terminal) containing
   a)Mitochondria
   b)Synaptic vesicles filled with acetylcholine
   c)Thickened presypnactic membrane.
2. Synaptic cleft separates one neuron from another
3. Thickened post-synaptic membrane, the site for cholinesterase activity.

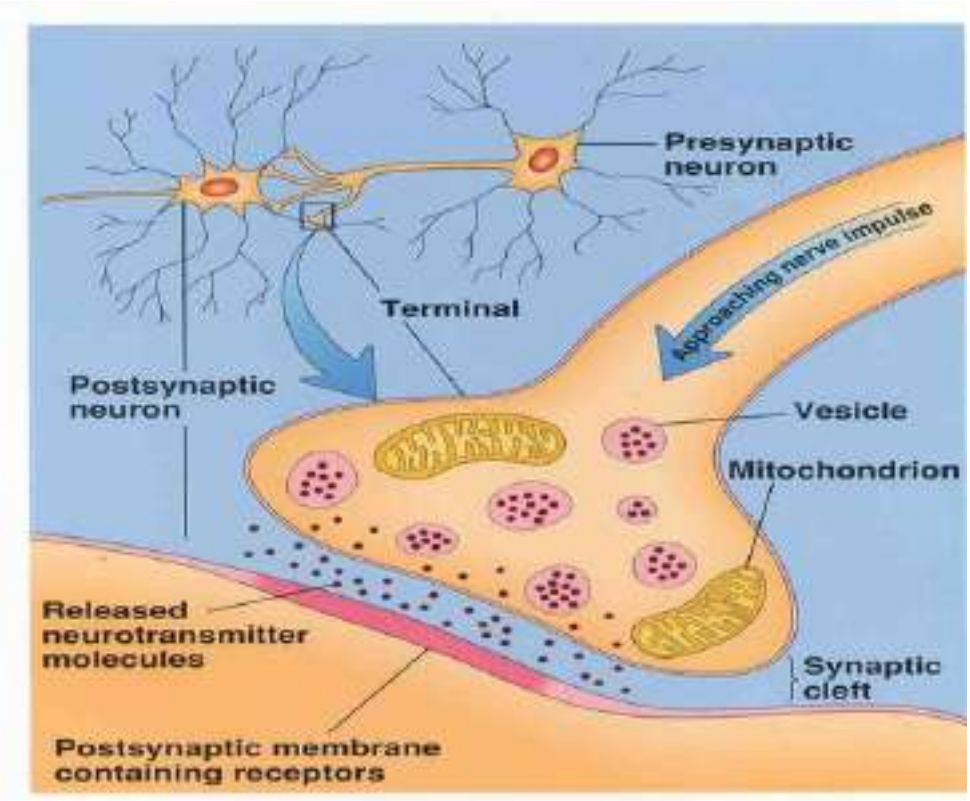


Fig.3 Structure of synapses

Sensory nerve endings (receptor)

➢ **The receptors are classified based on their morphology into:**

1.Non-encapsulated

a. Free nerve ending. They are the branched terminations of the axons. Myelinated nerves lose their myelin sheath, and end in a number of branches that penetrate the area being innervated.

b. Merkel's discs - The axon terminals in the epidermis end also in small expansions that are in contact with specialized epidermal cells (Merkel's cells).

f. Muscle Spindle

➢ They consist of a group of 6 to 12 intrafusal muscle fibers, surrounded by connective tissue continuous with the perimysium.
➢ The intrafusal muscle fibers are modified skeletal muscle fibers that have the characteristic striations but do not have peripheral nuclei.
Morphologically they are of two types:
1. Nuclear bag fibres contain a cluster of nuclei at the centre of the muscle.
2. Nuclear chain fibres have their nuclei arranged as a longitudinal chain at the centre of muscles.
➢ Intrafusal muscles have a motor supply from the gamma efferent, like those in extrafusal fibers.
➢ Their sensory supply is of two types that differ in the arrangement of their endings:
(a) Anulospiral endings
(b) Flower spray endings Other sensory receptor:
• Carotid and aortic body:
• Carotid sinus:
• In the lung, gut, and bladder 20
• In the hypothalamus

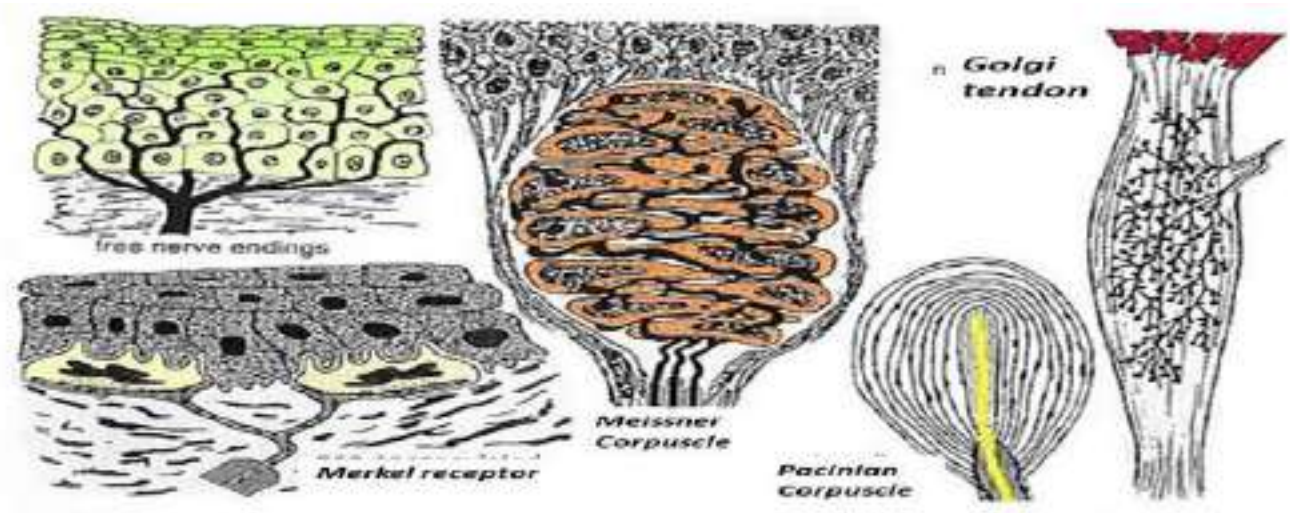


Fig. 4 Sensory receptors

2.Encapsulated

- They consist of branched axon enclosed in connective tissue capsule. The nerve fiber loses its myelin sheath inside the capsule.

a) Pacinian corpuscle

➢ It is large structures
➢ Have branched, unmyelinated nerve terminals in the core of the corpuscle surrounded with concentric lamellae of flattened cells arranged longitudinally along the corpuscle and densely packed centrally.
➢ it is enclosed within an thin connective tissue sheath

b) Meissner Corpuscle

➢ It is oval structure
➢ They consist of thin connective tissue capsule surrounding a mass of flattened, transversely arranged modified Schwann cells.
➢ The non-myelinated terminal branches of one or two large myelinated axons ramify within the core between the transverse cells.

c- Ruffini Corpuscle

➢ It has a core of nerve endings and collagen bundles
➢ It is surrounded by capsule

d. Krause Corpuscle

➢ It consists of thin connective tissue capsule surrounding highly coiled non-myelinated terminal branches that form a ball-like mass (glomerulus).

e. Golgi tendon

➢ It is enclosed within thin connective tissue capsule
➢ It has a core of intracapsular tendon bundles
➢ it has unmyelinated axon that forms spray-like branches ending in small dilatations twisted around the tendon core

Motor nerve endings

1. In skeletal muscle; Motor end plates (Fig.5)

➢ They are axon terminals that end on skeletal muscle similar in structure as synapse with two modifications at post synaptic membrane

- The first one is the presence of junctional folds, which are deep invaginations or grooves of the sarcolemma that provide a large surface area where the acetyl choline from the synaptic end bulb can interact
- Secondly, the sarcolemma of the junctional folds contains 30 to 40 million acetylcholine receptors. These receptors act as an ion channel once activated

2. Nerve endings on smooth muscle, cardiac muscle and secretory cells

- They are always free nerve endings and do not have motor end plates or other specialised endings

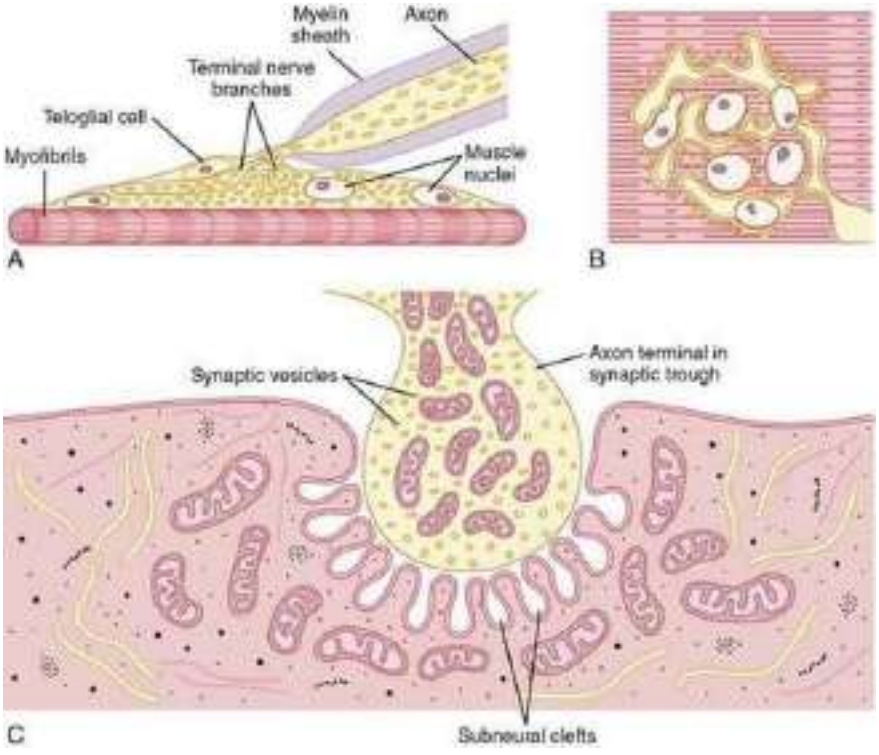


Fig. 5: Motor end plate.

Histological structure of the cerebral and cerebellar cortex

Objectives

➢ By the end of lecture student should be able to:

1. **Describe** the cytoarchitecture of the cerebellum
2. **Explain** the cytoarchitecture of the cerebral cortex

Grey matter of the cerebellum

➢ **The cerebellar cortex**: formed of nerve cells, non-myelinated nerve fibers, neuroglia and blood vessels. The nerve cells are arranged into three layers:

1. The molecular layer: the most superficial layer

➢ It is formed of:

- Dendrites of purkinje cells
- The dendrites of the Golgi neurons
- The cell bodies, axons and dendrites of basket cells
- The cell bodies, axons and dendrites of stellate cells
- Recurrent collaterals from the axons of purkinje cells to synapse with the dendrites of Golgi, basket and purkinje cells
- The axons of the granular cells
- The climbing fibers which are the terminals of the olivo-cerebellar tract that contacts the dendrites of purkinje cells.

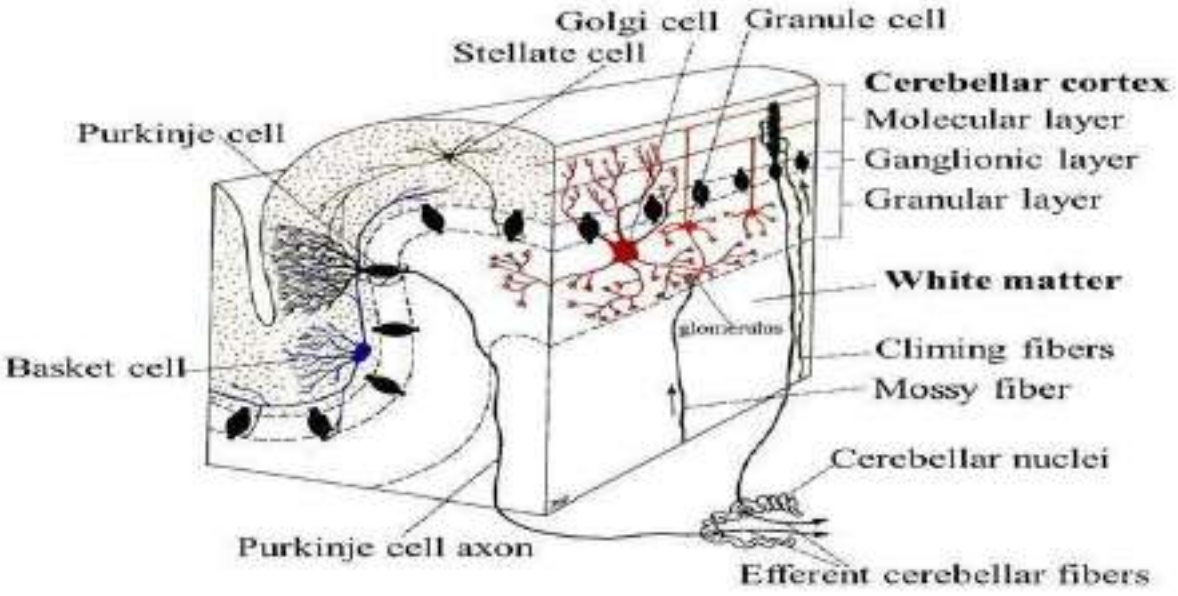


Fig. (1): A diagram of layers of cerebellar cortex

2. The purkinje cell layer: (Fig. 2)

- A thin layer formed of single row of the cell bodies of the purkinje cells widely separated from each other
- These cells are pyriform in shape
- Their dendrites are extensive and tree like present in the molecular layer

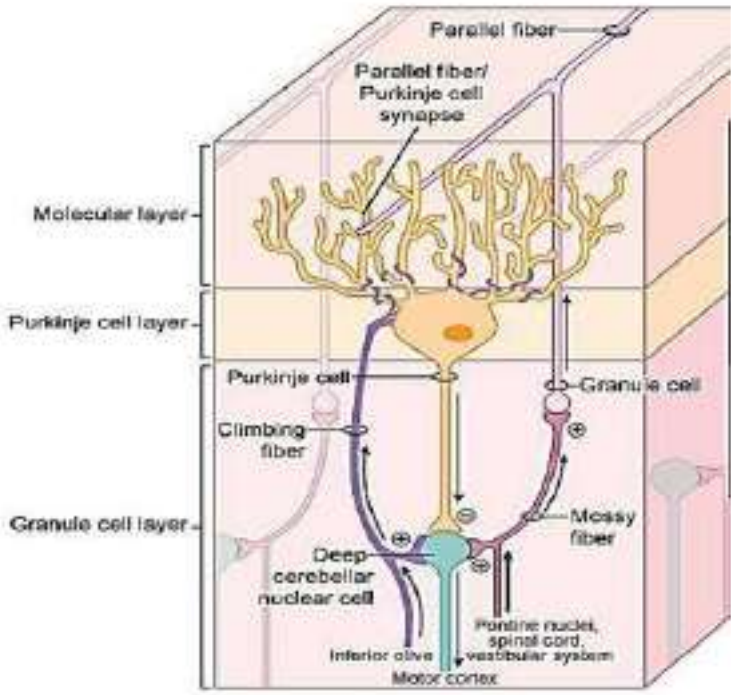


Fig. 2: Layers of cerebellar cortex

3. The granular layer (Fig. 2):

➢ **It is the deepest layer; it is formed of:**

- The cell bodies and dendrites of the granular cells which have large dense nuclei and thin cytoplasm as lymphocyte
- The Golgi type II cells have vesicular nuclei, and their dendrites extend all over 3 layers
- The axons of the purkinje cells passing to the white matter
- The climbing fibers passing to the molecular layer
- Cerebellar glomeruli: in area free from granule cells. It is formed of:
  1. The terminals of the afferent mossy fibers (rosette appearance) forming its center
  2. Axon of Golgi type II cells
  3. Surrounded by sheath of glial cells
  4. Terminal dendrite of granule cell

:

The white matter of the cerebellum:

- It is formed of myelinated nerve fibers forming the afferent and efferent fibers of the grey matter

THE CEREBRUM

Grey matter of the cerebrum (Fig.4)

- **The cerebral cortex**: formed of nerve cells, non-myelinated nerve fibers, neuroglia and blood vessels. The nerve cells are arranged into six layers:

a) The molecular L. (10%)
b) The outer granular L. (10%)
c) The outer pyramidal L. (30%). Betz cells are large pyramidal cells present in this layer and their axons form the cortico-spinal T
d) The inner granular L (10%)
e) The inner pyramidal L. (20%)
f) The polymorphic L. (20%)

NOTE

-The layers c & e are well developed in the motor cortex
-The layers b & d are well developed in the sensory cortex.

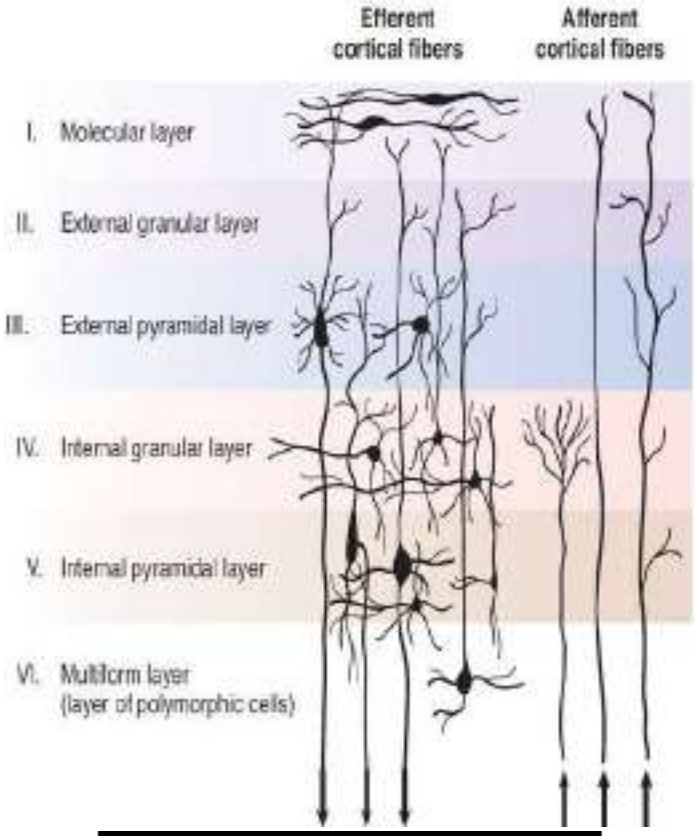


Fig. (4): A diagram of layers of cerebral cortex

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Neuroscience

Physiology

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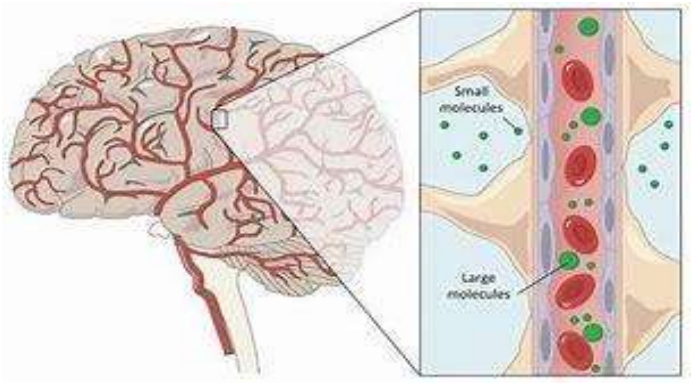
165	smell & Taste

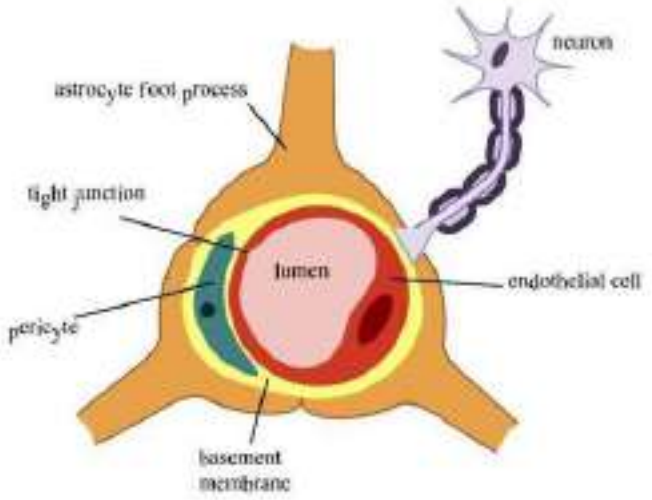
BLOOD–BRAIN BARRIER AND CEREBROSPINAL FLUID (CSF):-

- **Anatomy of the blood–brain barrier**

Itis the barrier between cerebral capillary blood and the CSF. CSF fills the ventricles and the subarachnoid space.

Itconsistsof the **endothelial cells of the cerebral capillaries** and **the choroid plexus epithelium.**





- **Permeability of BBB:**

√ A.The blood-brain barrier is highly permeable to water, O2, CO_2 & lipid soluble substances as alcohol & anesthetics.

√ B. It is impermeable to plasma proteins, cholesterol or bile pigments.

√ C. Other substances cross the barrier with varying degrees of difficulty, ie; other substances are transported by carriers in the choroid plexus epithelium. They may be secreted from blood into the CSF or absorbed from theCSF into blood

- **Functions of the blood–brain barrier**

1.It maintains a constant environment forneurons intheCNS and protects the brain from endogenous or exogenous toxins.

2. It prevents the escape of neurotransmitters from their functional sites in the CNS into the general circulation.

3. Drugs penetrate the blood–brain barrier to varying degrees. For example, nonionized (lipid-soluble) drugs cross more readily than ionized (water soluble) drugs.

NOTE

Inflammation, irradiation, and **tumors** may destroy the blood–brain barrier and permit entry into the brain of substances that are usually excluded (e.g., antibiotics, radiolabelled markers).

➢ **Formation of CSF by the choroid plexus epithelium**

Lipid-solublesubstances (CO2 and O2) andH2Ofreelycross the blood–brain barrier and equilibrate between blood and CSE.

Other substances are transported by carriers in the choroid plexus epithelium.

They may be secreted from blood into the CSF or absorbed from the CSF into blood.

Protein and **cholesterol** are excluded from the CSF because of their large molecular size. The composition of CSF is approximately the same as that of the interstitial fluid of the brain but differs significantly from blood (Table1).
CSF can be sampled with a **lumbar puncture**.

CSF ≈ Blood	CSF < Blood	CSF > Blood
Na^+	K^+	Mg^{2+}
Cl^-	Ca^{2+}	Creatinine
HCO_3^-	Glucose	
Osmolarity	Cholesterol* Protein*	

Table 1Comparison of Cerebrospinal Fluid (CSF) and Blood

- **General information**

√ CSF fills ventricles and subarachnoid space.
√ Volume = 150 ml
√ Rate of production =550 ml/d, so it turns 3.7 times/day.
√ Lumbar CSF pressure = 80-180 mm water
√ Absorption of CSF occurs by bulk flow is proportionate to CSF pressure.
√ At pressure of 112 mm water(normal average): filtration and absorption are equal.
√ Below pressure of 68 mm water, CSF absorption stops

- **Formation**

√ Secreted by choroid plexuses into each ventricle .
√ Choroid plexus are areas where the lining wall of the ventricle is very thin and has a large amount of capillaries

- **Drainage**

√ From the roof of the 4th ventricle CSF flows through foramina into the subarachnoid space and completely surrounds the brain and spinal cord
√ When CSF pressure is higher than venous pressure CSF passes into the blood and when the venous pressure is higher the arachnoid villi collapse, preventing the passage of blood constituents into the CSF.

- The CSF passes back into blood through tiny diverticula of arachnoid mater called arachnoid villi (arachnoid granulations), which project into the venous sinuses. Some reabsorption of CSF by cells in the walls of the ventricle occurs.
- Movement of the CSF is by pulsating blood vessels, respiration and changes of posture.
- CSF is secreted continuously at a rate of about 0.5ml per minute i.e. 600 ml per day
- CSF pressure can be measured by attaching a vertical tube to the lumbar puncture needle – 10 mmHg.

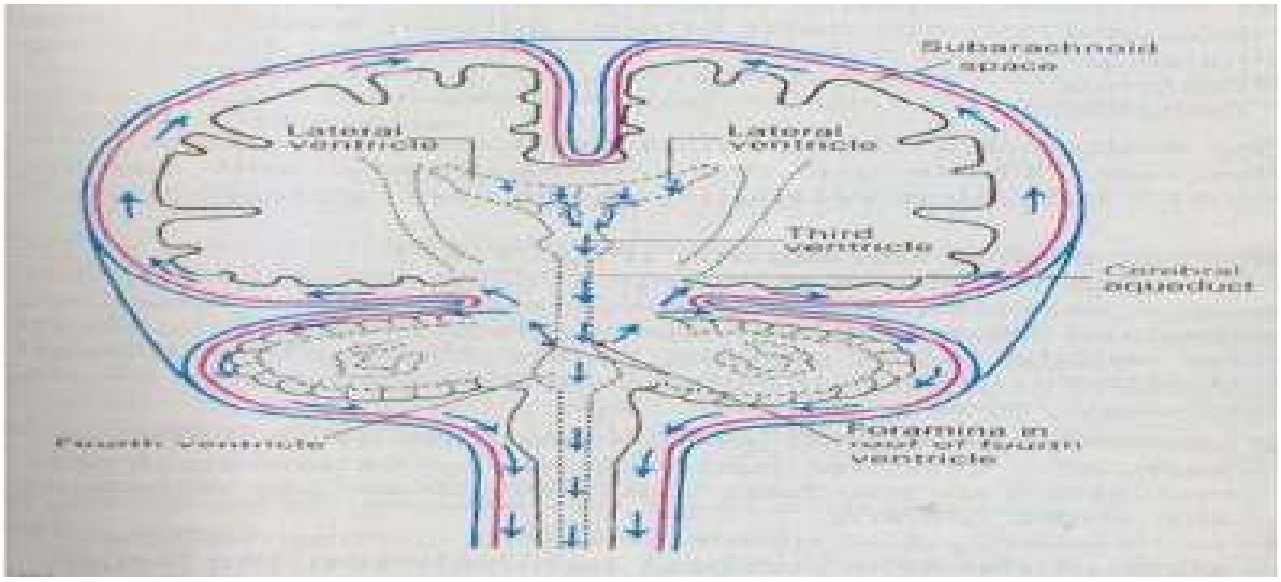
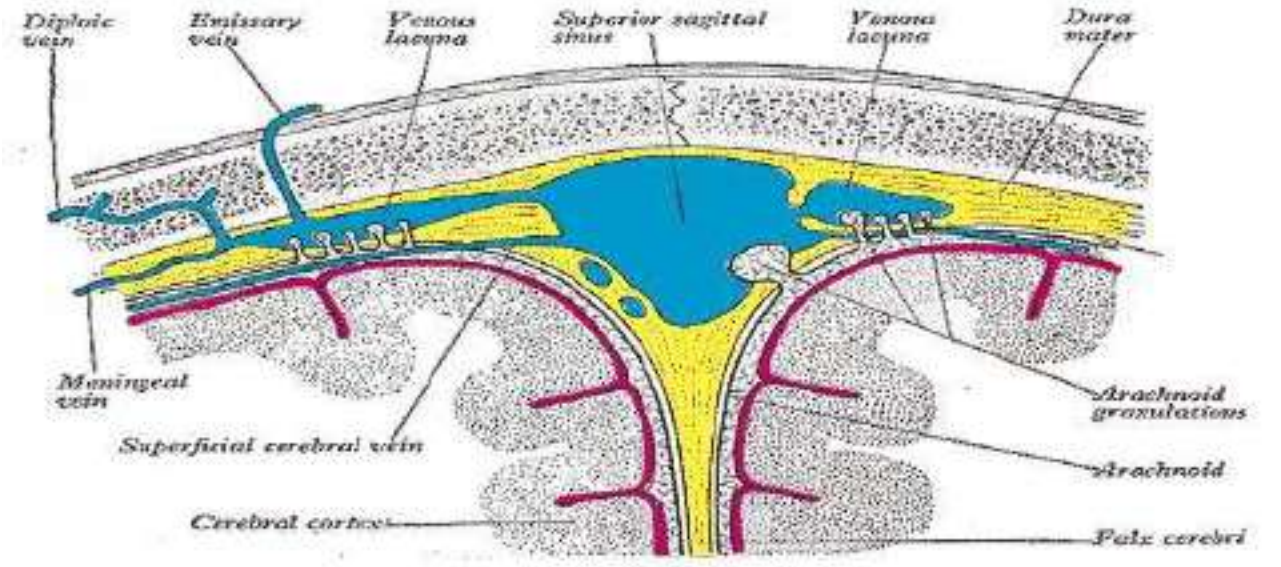


Figure 1circulation of CSF, arrows showing the flow of cerebrospinal fluid



Warwick R, Williams PL. Gray's Anatomy, 35th British ed. Philadelphia: W.B. Saunders, 1973. fig. 7.166, p. 991.

Figure 2drainge of the CSF into the arachnoid granulations and villi

- **Composition of CSF:**

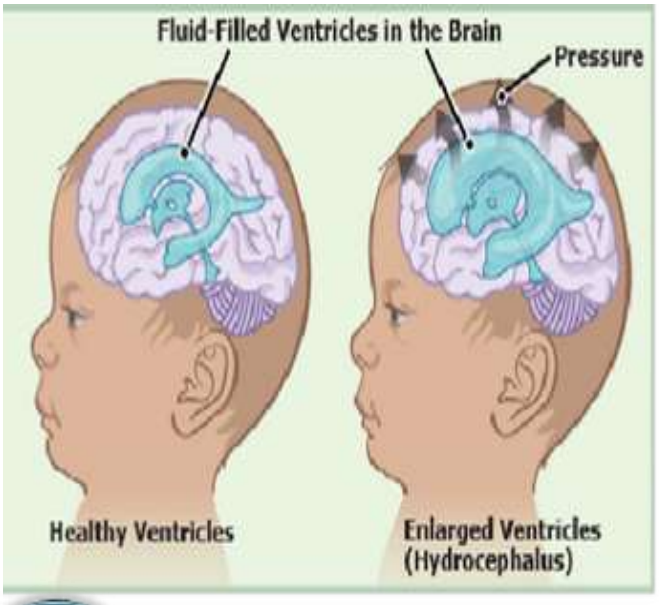
√ The compositionofCSFisapproximately the same as that of the interstitial fluid
√ of the brainbutdifferssignificantlyfrom blood (Table1).
√ CSF canbesampledwithalumbarpuncture**.**
√ **Protein**and**cholesterol**areexcluded from the CSF because of their large molecular size.

- **Functionofcerebrospinalfluid:**

1) Protectivefunction:Thebrainissupported within the arachnoid by the blood vessels,nerverootsandthearcahnoid trabiculae. When the head receives a blow,thearachnoidslidesonthedura and the brain moves, but its motion is gentlycheckedbytheCSFcushionand by the arachnoid trabiculae. Removal of CSFduringlumbarpuncturecancause severe headache.
2) Facilitationofpulsatilecerebralblood flow,
3) Distributionofpeptides,hormones,neuroendocrine factors and other nutrients and essentialsubstancestoblood,
4) Wash awaywasteproducts.
5) Cardiovasculardynamicsarealsoaffected by CSF pressure, as the flow of blood mustbetightlyregulatedwithin the brain to assure consistent brain oxygenation. (Cushing reflex).
6) level of oxygen,CO2andhydrogenion concentrations in CSF regulate respiratoryandcardiovascularcenters in medulla oblongata

Hydrocephallus:

1)External hydrocephallus	2) Internal hydrocephallus:
• Large amounts ofCSF accumulates when the reabsorptive capacity of arachnoid villi decreases.	• occurs when foramina of Luschka & Magendie are blocked or obstruction within ventricular system, resulting in distention of the ventricles.



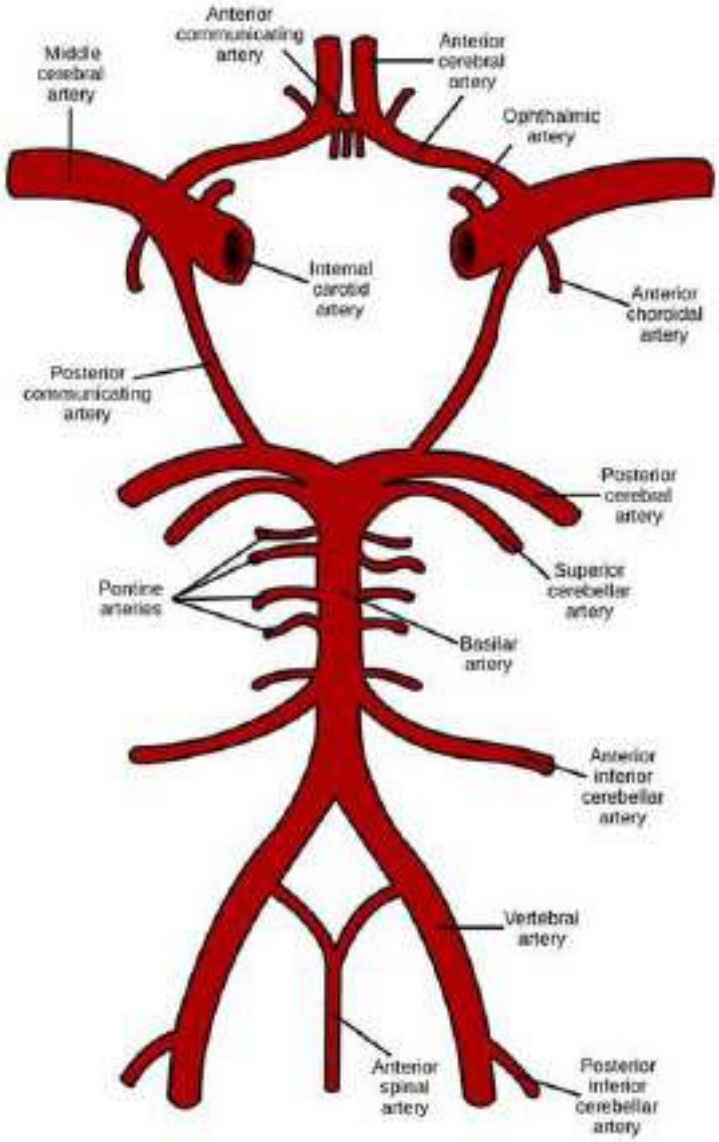
Cerebral Circulation

- **Anatomical Consideration:**

❖ The principal arterial inflow tothe brain in humans is via 4 arteries that form the circle of Willis:
  - √ 2 internal carotid arteries.
  - √ 2 vertebral arteries unite to form the basilar artery.

❖ The 2 internal carotid arteries + the basilar artery ---> circle of Willis ---> 6 large cerebral arteries (2 anterior, 2 middle and 2 posterior cerebral arteries) ---> supply the brain.

❖ Venous drainage from the brain is mainly via the deep veins and dural sinuses that empty into the internal jugular vein.



- **Regulation of cerebral blood flow:**

❖ Cerebral blood flowis regulatedbythe following mechanisms:

1. Peripheral mechanisms
2. Central Mechanisms
3. Physical Mechanism

1- Peripheral mechanisms:

A) Chemical (metabolic) mechanism:	B) Myogenic Mechanism:
• **At least 3 metabolic factors control the cerebral blood flow:** - PCO2, - H+ concentration - PO2	• **Increased arterial blood pressure** tends to increase the cerebral blood flow. Increased arterial blood pressure ---> contraction of the smooth muscles in the wall of arterioles ---> vasoconstriction of cerebral vessels ---> decreased cerebral blood flow toward normal
• **Increased PCO2** increased H+ concentration and decreased PO2 in arterial blood perfusing the brain ---> vasodilation of cerebral blood vessels ---> increased cerebral blood flow.	• **Decreased arterial blood pressure** tends to decrease the cerebral blood flow. Decreased arterial blood pressure relaxation of the smooth muscles in the walls of cerebral arterioles ---> cerebral vasodilation ---> increased blood flow toward normal.
	• **In the brain**, the myogenic mechanism maintains normal cerebral blood flow with arterial pressure of 65-140 mmHg. Fig (130)
	• **This process** tends to maintain the cerebral blood flow within normal range despite of changes in arterial blood pressure

2- Central Mechanisms:

A) Nervous Mechansim:	B) Hormonal Mechanism:
• The cerebral blood vessels receive sympathetic nerve fibers from the superior cervical ganglion along with the cerebral arteries.	• Angiotensin II and vasopressin cause cerebral vasoconstriction----> decreased cerebral blood flow.
• Neither transection of these nerves nor mild to moderate sympathetic stimulation of them causes significant change in cerebral blood flow because the autoregulation mechanism can override the nervous mechanism. However, strong sympathetic stimulation ---> cerebral vasoconstriction.	
• When the arterial pressure rises to a high level during strenuous exercise and during other states of excessive circulatory activity. The sympathetic nervous system ---> vasoconstriction of middle and large cerebral blood vessels --> prevent the high arterial pressure from reaching the smaller blood vessels of the brain.	

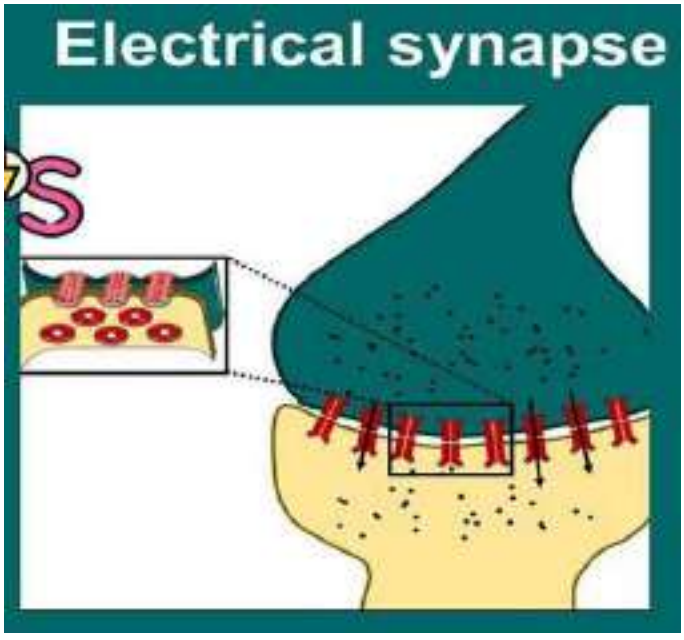
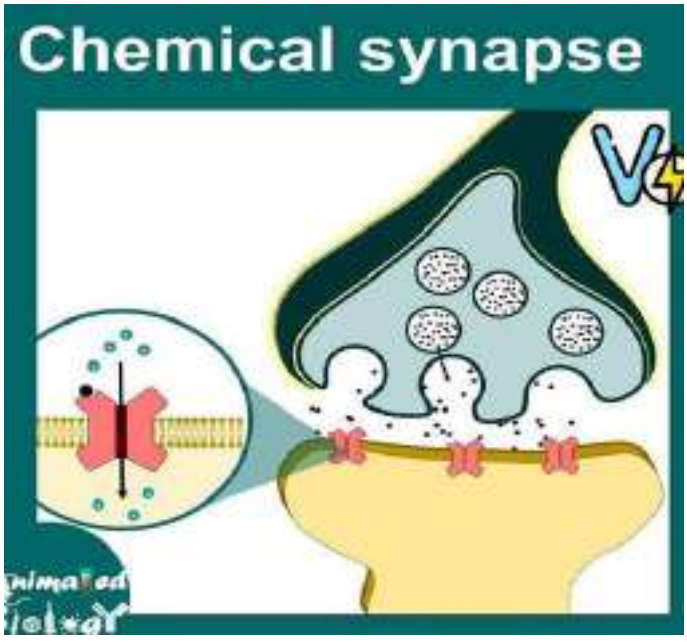
3-Physical Mechanism:

Increased intracranial tension ---> direct compression on cerebral blood ---> decreased cerebral blood flow.
• Decreased intracranial tension ---> opposite effects
• Cushing Reflex (CNS Ischemic Response):
• When I.C.P. ↑ more than 30 mm Hg (normally it is about 11 mm Hg) Marked reduction of cerebral blood flow cerebral ischemia. (+)VCC hypertension.
• The increase of ABP helps to maintain cerebral blood flow in spite of ↑ I.C.P.

The synapse

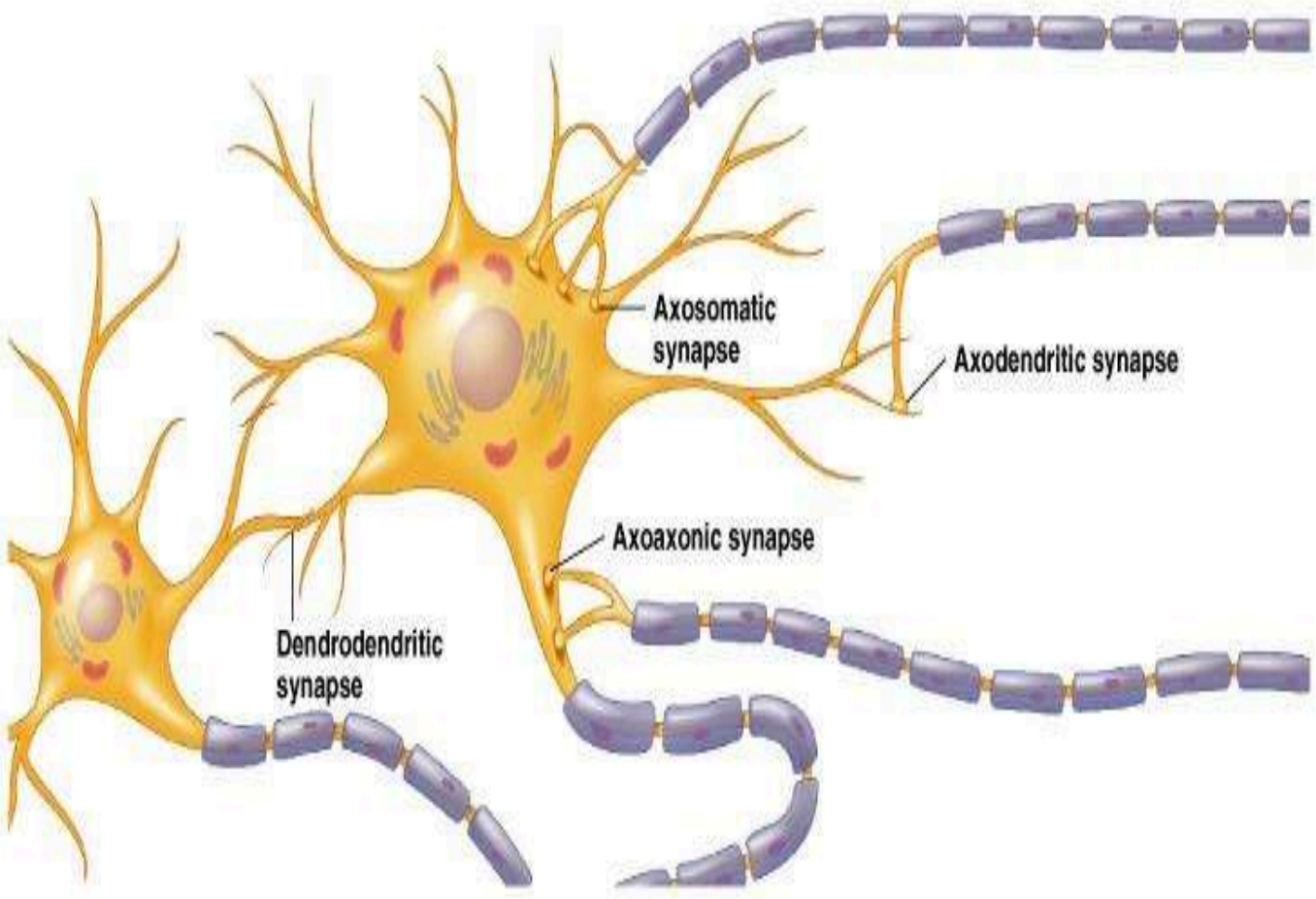
- **Definition:**
  Itthesiteofcontact between one neuron (presynaptic neuron) and another (postsynapticneuron).
- **Types:**
  1) AccordingtoMethod of Transmission
  2) Accordingtothe area of postsynaptic neuron involved
  3) Accordingtothe effects produced on the postsynaptic neuron, the synapses can be: a- Excitatory. b- Inhibitory.

1) According to Method of Transmission

A. Eectrical synapse:	B.chemical synape:
• The electrical current is transmitted directly from cell to cell through direct channels called gap junction as in the smooth and cardiac muscles.	• In which the activity is transmitted from the presynaptic to the postsynaptic neuron through release of chemical transmitter from vesicles.
• It is very rare in CNS, only it is important in arousal from sleep, mental attention emotions and memory	
• Rapid bidirectional transmission	• Slow unidirectional transmission
	

2) According to the area of postsynaptic neuron involved:

- a- Axodendritic – synapses between the axon of one neuron and the dendrite of another . It is least excitable but is most common (80-95%)
- b- Axosomatic - synapses between the axon of one neuron and the soma of another.
- c- Axoaxonic (axon to axon) most excitable but least common.
- d- Dendrodendritic (dendrite to dendrite).
- e- Dendrosomatic (dendrites to soma).



(b) Locations of neuron-to-neuron synapses

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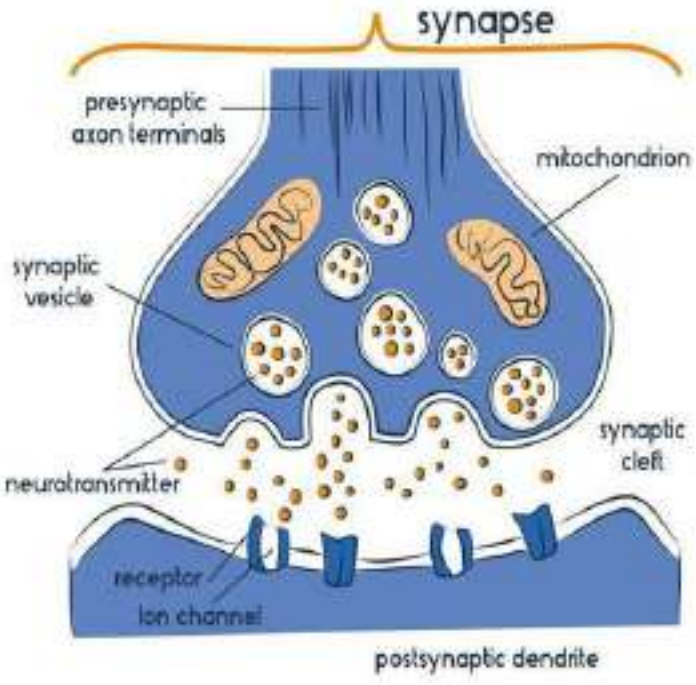
Anatomy of the synapse:-

The presynaptic terminals:

- at asynapse , the endofthenerveaxon is called end bulb or synapticknob.The synaptic knob contains2important structures:
  - √ Synaptic vesiclescontainthe chemical transmitteroftheneuron
  - √ Mitochondria provideenergy required for synthesisofthe chemical transmitter.
- **Thesynaptic cleft**: There is a gap between the pre and postsynaptic neuronal membranes
- **The postsynaptic neuronal membrane:** contains the receptors upon ligand gated ionic channels and second messenger receptor system



Mechanism of Synaptic Transmission::-

- **1- Release of chemical transmitters**

√ Action potential along axon cause depolarization of axon terminal→ voltage – gated Ca++ channels on axon terminal to open.

√ synapsin proteins bind synaptic vesicles to the cytoskeleton which prevents them from migrating to the presynaptic membrane and releasing transmitter

√ Ca++ influx into axon terminal → a- Ca++ will bind with calmodulin→ formation of Ca++ / calmodulin that activates protein kinase II → phosphorelation of synapsin protein → detachment of vesicle . b- Ca++ binds to synaptotagmin→ fusion of vesicular and neural membrane → release of transmitter.

√ A chemical transmitter diffuses across the synaptic cleft to bind to its receptors on postsynaptic membrane.

- **Types of postsynaptic receptors:**

A) Ligandgated ionic channels.

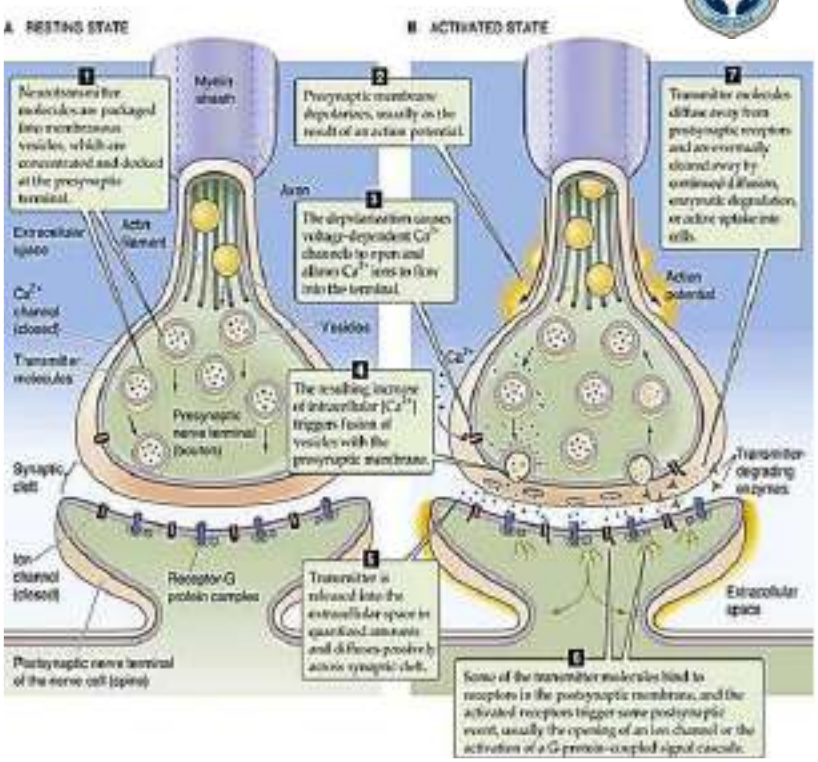
B) Second messenger receptor system

This type of receptors is provided by a specific membrane receptor protein called G-protein

This mechanism is important where prolonged post-synaptic changes are needed to stay for days, months, years (memory).

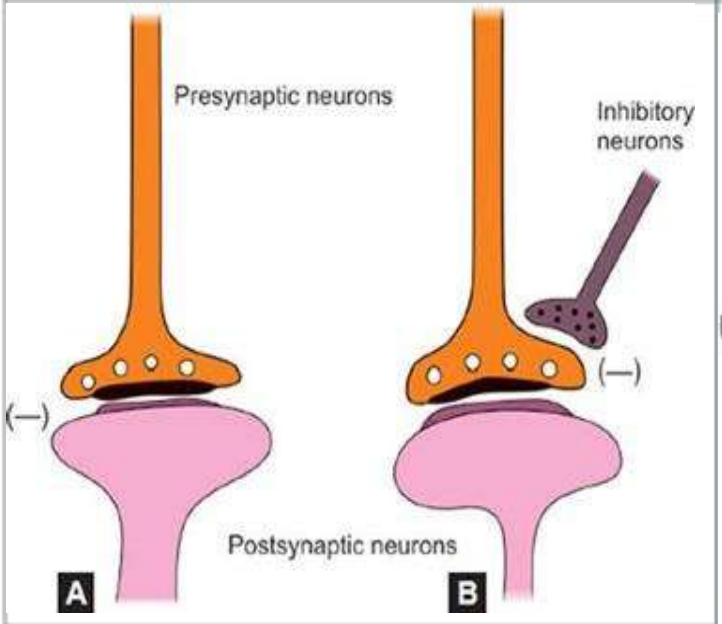
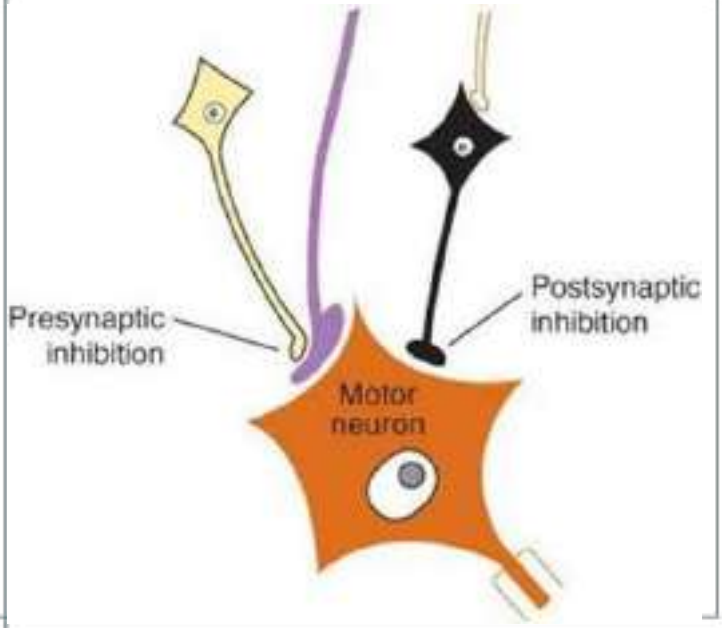
2- Action of transmitters:

√ This binding triggers either opening or closure of specific chemical – gated ion channels in that membrane → local electric changes called postsynaptic potentials → excitation or inhibition of the postsynaptic neuron, depending on the nature of the released transmitter.

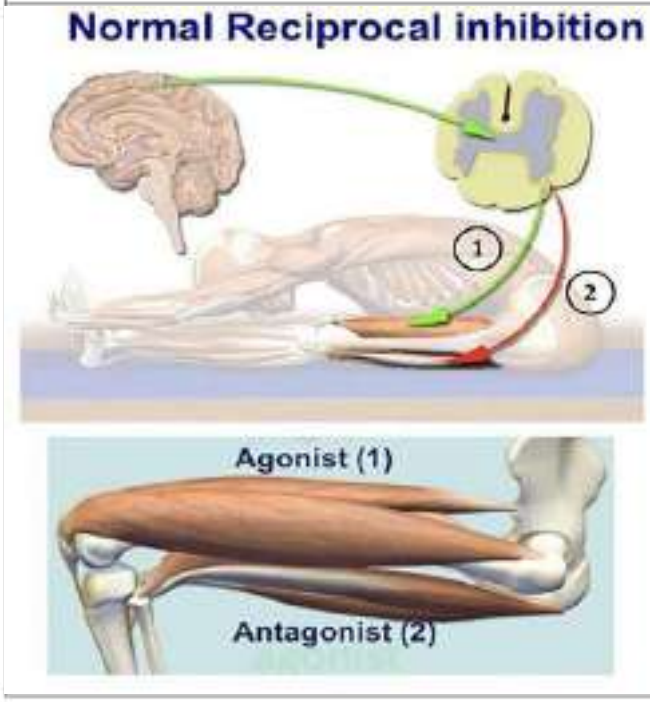


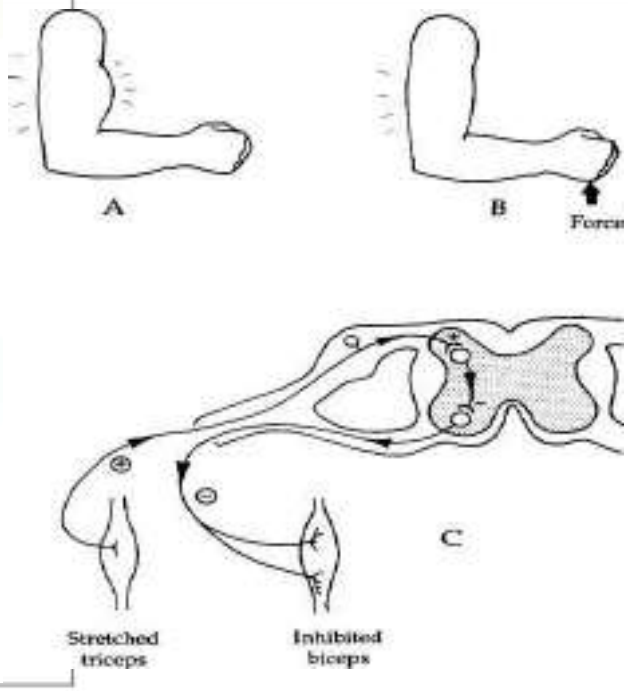
1) Electrical events inexcitatory synapses:	2) Electrical eventsininhibitory synapses:
• **synapses:** Binding of theexcitatory transmitter (e.g. acetylcholine noradrenaline and serotonin) to postsynapric receptors→ • a-↑ the permeability of the postsynaptic membrane to Na+ by the opening of ligand-gated Na+ channels. • b- Depressed conduction throygh K+ and CL- channels. • - a and b →slight depolarization of the postsynaptic neuron which will increase its excitability and bring it closer to threshold. This depolarization is called excitatory postsynaptic potential (EPSP). • The EPSP is localized and lasts for about15 milliseconds . A single EPSP is not enough to increase the neurnal potential to thereshold value (-70 mv →-59 mv) instead the effect of several EPSP must add up or summate to trigger an action potential.	• Binding of the inhibitorytransmitter (e.g gamma aminobutyric acid(GABA & glycine) to postsynapticreceptors →↑ the postsynaptic membrane to CL- influx and k efflux (↓ the Na+)→↑ the degree of intracellular negativityleading to a state of hyperpolarizationat the postsynaptic neuron causing its inhibition. • This hyperpolarization state is called the inhibitory postsynaptic potential (IPSP) (or direct inhibition)

Types of Synaptic Inhibition::-

1) pre-synaptic inhibition (Indirect Inhibition):	2) Post-Synaptic Inhibition (Direct Inhibition):
• **Def:**Occurs when inhibitoryneuron releases inhibitory neurotransmitter to inhibit the release of the excitatory neurotransmitter in response to normal impulse coming in presynaptic neuron	• Occurs when the pre-synaptic terminal releases an inhibitory transmitter→ opening of Cl- or K+ channels in the excitatory postsynaptic membrane → ↑ Cl influx and K+ efflux → hyperpolarization.
	
• **Mechanism** **a)** Opening of CL- or K+ channels in the presynaptic membrane →↑ CL- influx and K efflux→ hyperpolarization → decresase Ca++ influx → decrease release of neurotransmitter. **b)** Closure of Ca++ channels in presynaptic terminals → decrease Ca++ influx → decrease release of neurotransmitter.	
• **Example:** inhibition of pain (pain analgesia system)	

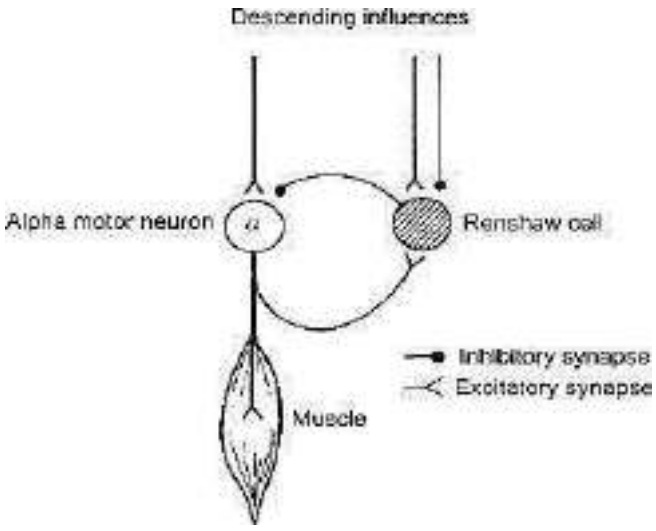
3) Reciprocal inhibition:	4) Lateral inhibition:
• Inhibition of antagonist activity isinitiated in the spindle in the agonist musele. Impulses pass directly to the motor neurons supplying the same muscle and via branches to inhibitory interneurons that end on motor neurons of antagonist muscle	• Lateral inhibition: A singleinputsignal can cause stimulation of centralneurons and inhibition of surroundingneurons in the field. • **Mechanism** A nerve fibergivescollatera fibers to stimulate inhibitoryinterneurons the lateral in turn inhibitsurrounding neurons in the field, thereare2chemical transmitters one excitatoryandone inhibitory e.g. Renshaw cell.





Inhibitory interneuron (Renshaw cells)

- They are present in spinal cord.
- When motor nerve is stimulated, it sends collateral fiber (releasing acetylcholine) activating the Renshaw cell (that release glycine) which inhibits this motor neuron (negative feedback mechanism) and surrounding neuron (lateral inhibition).
- Significance of Reshow cell:
  1) Negative feedback prevents after discharge to prevent prolonged and repeated activity of the motor nerve.
  2) Lateral inhibition helps increase of sharpness of sensation by decrease lateral spread of stimulus.



Clinical importance of synaptic inhibition:

- the main function of synaptic inhibition is to select exact number of impulses and to omit or block the excess ones.
- When a poison like strychnine is introduced into the body, it destroys the inhibitory function at synaptic level resulting in continuous and convulsive contraction even with slight stimulation.
- In the nervous disorders like parkinsonism, the inhibitory system is impaired resulting in rigidity

Properties of Synaptic Transmission:

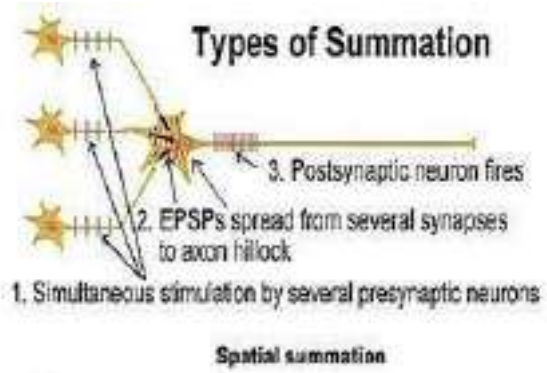
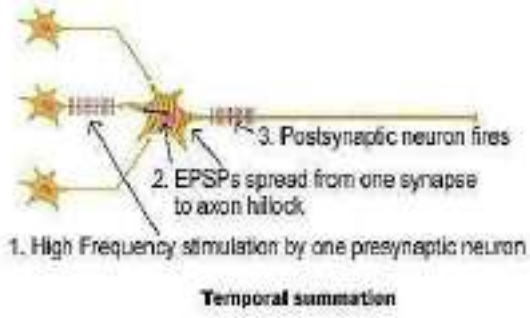
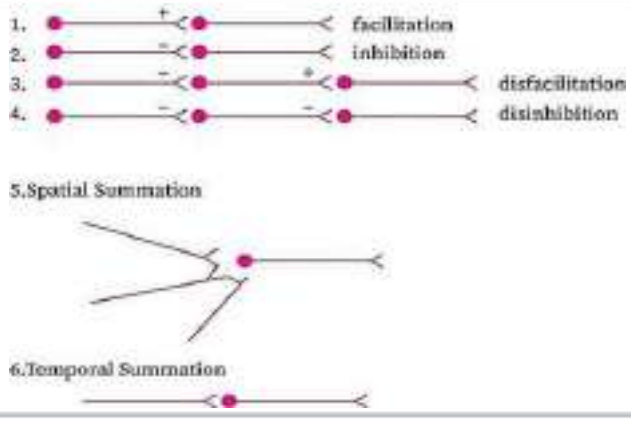
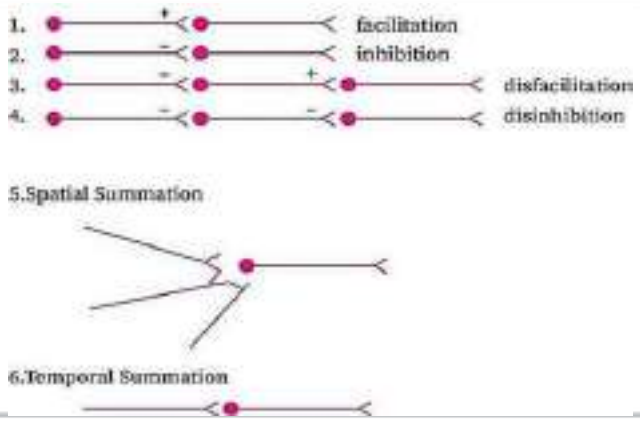
1) Forward direction (one way conduction):

Transmission in synapses is unidirectional i.e. from presynaptic neuron and not the reverse. This is because the postsynaptic neuron cannot release a chemical transmitter at the synapse.

2) Summation:

A singleEPSPisnot sufficient to stimulatethepostsynaptic neuron. Therefore,theeffects of several (EPSP) mustadduporsummate to increase the excitabilityofthepostsynaptic neuron to thethresholdlevelin order to trigger a propagatedactionpotential.

- There are two types of summation:

A) Spatial summation:	B) Temporal summation:
• It occurs by stimulation of several presynaptic fibers that convergeon the postsynaptic neuron at the sametime, so that the produced EPSPS summate together and raise the postsynaptic membrane potential to the threshold level.	• It occurs by stimulation of a single presynaptic fiber repetitively, provided the intervalinh.b between impulsesdoes not exceed 15 milliseconds thus successive EPSPs can summate andraise the postsynaptic membrane potential to threshold level.
	
	

3) Synaptic delay

is the minimumtimerequired for transmissionacrossthesynapse.

- It is about0.5msec.This time is taken by:

√ Releaseofchemicaltransmitter.
√ Diffusioninsynapticcleft
Action oftransmitteron its reeeptor in
√ postsynapticmembrane.
IncreaseddiffusionofNa+ to ↑ post
√ synaptic potential.
Summationofexcitatory postsynaptic potentialsandgeneration of an action potential.

4) Synaptic fatigue:

√ **Definition:** this means decrease in the response of postsynaptic neuron after a long period of high frequency stimulation of the presynaptic neuron.
√ **Causes of fatigue**: Exhaustion of presynaptic terminals chemical transmitter (because in repetitive stimulation , the rate of release of neurotransmitter is more than the rate of its reformation).
√ **Significance:** It is a protective mechanism. It prevents over excitation CNS (e.g in epileptic fits , fatigue stops convulsion)

5) synaptic plasticity:

√ **Definition:** it theability to change the function of synapse (mainly in learning and memory) according to the demand (i.e.increased or decreased for short or long duration)

√ **Types:**

1- short term plasticity:

2- long term plasticity (lasts for days to few weeks)

6) Factors affecting synaptic transmission: (3h + drugs):

A) H ion concentration (pH) :	B) Hypoxia:
• Alkalosismarkedly ↑ neuronal excitability and synaptic transmissione.g. in persons predisposed to epileptic attacks , hyperventilation (which washes out CO2 and raises blood ph) leads to convulsions . • acidosis greatly ↓ neuronal excitability and synaptic transmission e.g. acidosis that occurs in diabetic and uremic patients often leads to coma.	• It ↓ synaptictransmissionand prolongs reflex time due to inhibitonof chemical transmitter synthesis andrelease. loss of • consciousness occursif the cerebral circulation is temporarily interrupted for 3-5 seconds.
C) Hypoglycemia:	**D) Effect of drugs:**
• it depresses synaptic transmission as glucose is the main fuel of brain	• **Drugs that increasesynaptic transmission:** Caffeine, theophylline, Theo bromine which are found in coffee, tea and coca ⇒ enhance synaptic transmission by increasing neuronal excitability and decreasing the threshold for neuronal excitation. • **Drugs that decrease synaptic transmission:** Hypnotics and anesthetics depress synaptic transmission. (↑ Cl influx) ⇒ hyperpolarization ⇒ neuronal inhibition.

SENSORY SYSTEM

Functions of sensations:

1- detection and perception of sensation

2- reflexes based on these sensation

3-memory

General sensory functions

Sensory system

Definition: It is the part of the nervous system which is concerned with awareness of the external and internal environment.

Includes

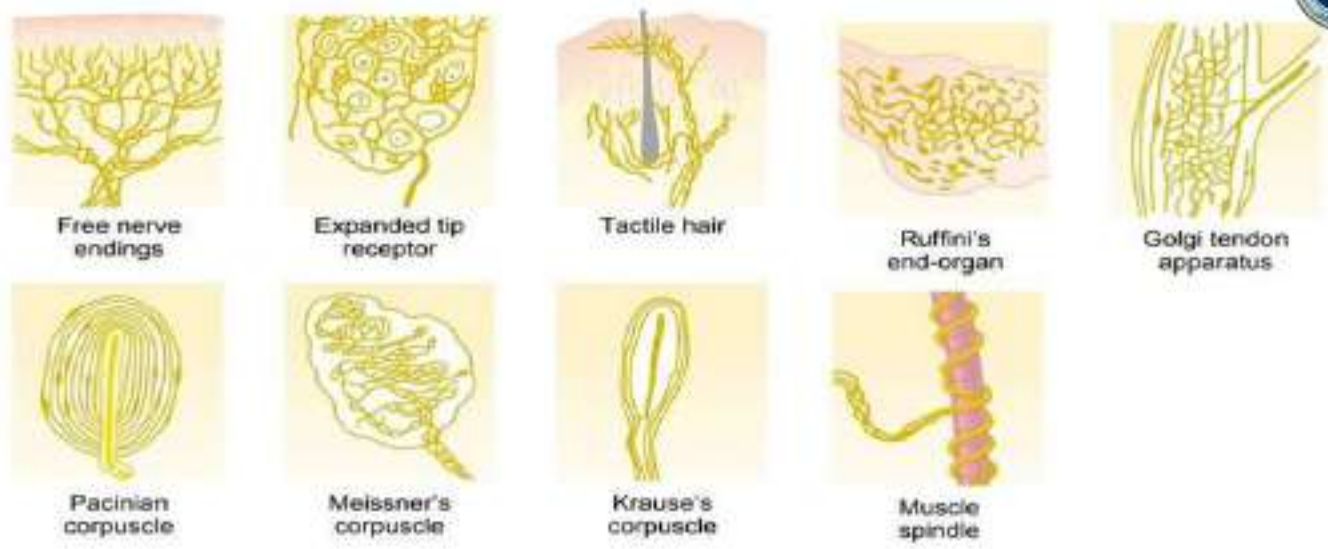
- Sensory Receptors
- Sensory Code
- Sensations
- Sensory Pathway
- Somatic Sensory Cortex
- Lesions of Sensory System

Sensory Receptors

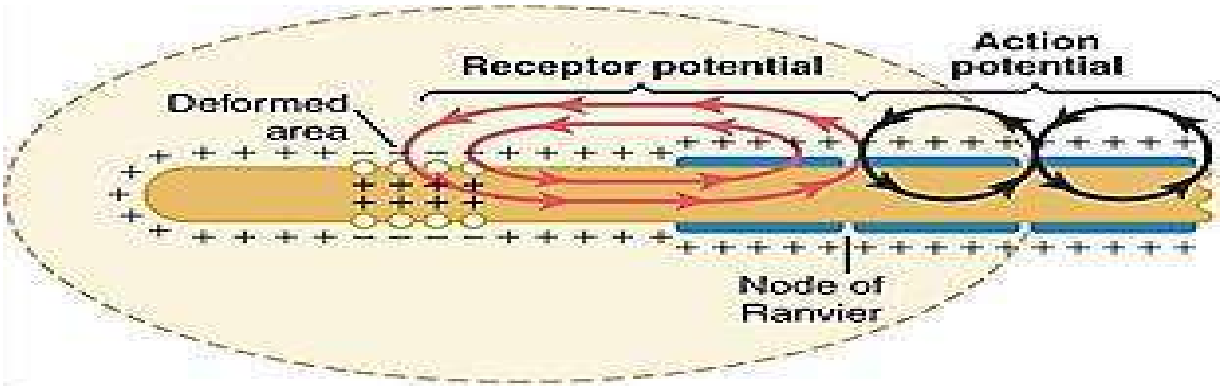
Definition:		They are modified nerve endings of afferent fibers which receive (detect) different stimuli & convert (transform) them into action potentials
Functions:	**1-Detectors:**	Detect changes in the surrounding environment
	2-Generator:	generate receptor potential.
	3-Transducers:	Transform any form of energy in the stimulus into action potentials

		Initiate reflexes which regulate body responses to different changes in internal and external environment.
Classification According to::	**Site :**	Superficial “Cutaneous”
		Deep
		Visceral
		Special sense

Physiological: Mechanoreceptors:	-Detect mechanical deformity in the receptors as: 1- Touch 2- Pressure 3- Sound (cochlear receptors) 4- Acceleration (vestibular receptors) 5- Blood pressure (baroreceptors) - They are found in: 1- Skin 2- Mucous membrane 3- Muscle 4- Tendons 5- Joints (Proprioceptors) 6- Blood vessels 7- Lung
Thermoreceptors	-In hypothalamus for body temperature regulation -In skin, mucous membrane: Detect cold & warm energy
Chemoreceptors	1- Taste 2- Smell 3- Osmoreceptors 4- Glucoreceptors 5- CO2 6- O2 7- H+
Nociceptors:	Detect pain
Electromagnetic	Detect light



Properties	Specificity:	(Muller's law):
		Each receptor is **sensitive to one type of sensation** called **adequate stimulus** eg light for photo receptors.
		Other forms of energy can stimulate the receptor, but it needs stronger stimulus than the adequate one.
	Excitability:	Generator potential "Receptor potential" [RP]:
		Def : It is a state of **partial depolarization** in the receptor membrane which occurs when the receptor is stimulated
		Receptor potential **spreads passively** with decreasing its magnitude to: ▯ Adjacent part of sensory nerve (unmyelinated) or ▯ 1stnode of Ranvier (myelinated nerves)
		If depolarization **reaches the firing level** → it leads to action potential which is propagated by salutatory conduction
		As long as there is receptor potential with enough magnitude to generate action potential → there is train of nerve impulses
		Ionic basis of RP: The energy of the stimulus causes **non specific opening of Na+ channels** → Na+ entry leading to partial depolarization
		Number of opened channels: Is directly proportional to intensity of stimulus.



Excitation of a sensory nerve fiber by a receptor potential produced in a Pacinian corpuscle.

ReceptorPotential	ActionPotential
1) A local state of **partial depolarization** which spreads passively .	1) It is **complete depolarization** followed by overshoot, reversal of polarity .
2) Due to non specific increase in permeability to Na + .	2)It Is due to increase in Na + & K+ permeability.
3) Not obey all or non law. It has a variable magnitude.	3)Obey or non law. It has a fixed magnitude
4)Can be graded & its amplitude is increased by increasing intensity of stimulus.	4)Can not graded.
5) absolute refractory period	5)Followed by **absolute** refractory period
6)Can be summated	6) Can not be summated
7) Its duration is 5- 10 msec (longer than AP) → allow repetition of AP	7) Its duration of spike:2 msec
8) Not blocked by local anesthesia	8) Blocked by local anesthesia

1- Adaptation of receptors:

Definition:Adaptationisdeclineinthereceptor potentia1 and frequency of impulses **inspite of** constant maintained application of the stimulus.

Classification According to **rate of adaptation**, receptors are classified into:

1- Rapidly adapting receptors (phasic)

1- touch receptors **eg**

2- They adapt rapidly to continuously applied stimuli but respond rapidly if change take place

2- Moderately adapting receptors

1- Temperature **As**

2- Smell

3- Taste

3- Slowly adapting receptors (tonic)

1- Pain receptors **eg**

2- Muscle spindle

3- Alveolar stretch receptors

4- Pain receptors are slowly adapting or they do not adapt at all

5- Their function is to keep the brain continuously informed about **dangerous** changes in environment.

3- Mechanism:

Each receptor has its own property of adaptation

1- Rods&cones : Adapt by changing the concentration of their pigment

2- Mechanoreceptors: Adapt by:

1- Remodeling (readjustment) of the receptor structure: Where maintained pressure → steady displacement in outer lamellae but inner lamellae of nerve fiber slip back to original. Position ending distortion → decline in generator potential

2- Inactivation of Na+ channels in terminal nerve fiber

3- Inactivation of Na+ channels in 1st node of Ranvier

2- Sensory Code

Definition: It is the ability of CNS to recognize type “Modality”, site “Locality” & strength “Intensity” of sensation.

1- Code for Type of Sensation: -Each receptor is most sensitive & specialized to one type of stimuli -The sensation perceived to brain will be the same whatever the method of stimulation -This known as MULLER LAW

2- Code for Site of Sensation:

-Each part of bodysend sensory signal to particular area of the brain & the brain project the sensation to the same part of the body
-This known as LAW OF PROJECTION

3- Code for Strength of Sensation:

-Braindependonfrequencyofaction potential to determine the strength of sensation
-Increase frequency mean increase in strength.

3- Somatic Sensations

Definition: Feeling produced by application of stimulus
Classification : Classified into:
1- General Sensations:
1- Somatic sensations
2- Organic Sensation (Thirst, hunger, sexual desire)
2- Special sensation:
1- Hearing
2- Vision
3- Taste
4- Smell
3- Emotional: As fear

1-Somatic sensations

1- Mechanoceptive Sensation
1) Tactile sensation:
1- Touch Types 1- Crude touch:

- **Receptors**: Free nerve ending
- Hair end organs

-Afferent: Aδ fibers
-Pathway: Ventral spino-thalamic tract
-Poorly localized
-eg feeling of clothes & hair comb
-Tested by: A piece of cotton passed on skin while eyes are closed

2- Fine touch:

-**Receptors**: Merkels & Meisner's
-**Afferent**: Aβ fibers
-**Pathway**: Dorsal column
-Poorly localized
-Well localized

1) Tactile localization : It is the ability to localize the point touched while eyes are closed .

2)Tactile discrimination: It is the ability to feel two points touched simultaneously as two separate points while eyes are closed.

-The distance between the two touched points should be above the threshold distance.

-Threshold distance is shorter if:

1- Number of receptors is: More
2- Number of afferent fibers is: More
3- Area of cortical representation is: Greater
4- Central convergence of afferent is: Less

1- Texture of materials: It is the ability to know the texture of materials eg silk or wool by touching them while eyes are closed

1-Tickling & itching:

- **Receptors**: Free nerve endings
- **Afferent**: : C fibers (different from pain fibers)
- **Pathway**: Ventral spino-thalamic tract
- Tickling is ability to feel light moving things on the skin as insects which cause local repeated mechanical stimulation .
- Itching is the sensation caused by chemical substance secreted near the receptors as histamine, Kinins & proteolytic enzymes .

- Itching is the sensation caused by chemical substance secreted near the receptors as histamine, Kinins & proteolytic enzymes .

1- Stereognosis:

- **Receptors**:Mixture of receptors of different sensations
- **Afferent**:Aβfibers
- **Pathway**:Dorsal column
- It is the abilityto recognize the nature of familiar objects put in hand with both eye closed
- It dependson:All cutaneous & deep sensation
- Previousknowledge about the object

1- Pressure:

- **Receptors**:Pacinian corpuscle
- **Afferent**: Crude: Aδ fibers - Fine: Aβ fibers
- **Pathway**: Dorsal column
- It enables the person to know the weights of objects & discriminates between different weights

2-Vibration:

- **Receptors** :Meissner's corpuscle: Responds to vibration up to 80 cycles/sec
- Pacciniancorpuscle: Responds to vibration up to 500 cycles/sec

Thesereceptorscontribute to sense of roughness when hand is passed over ➢ rough surface

- **Afferent** : Aβfibers
- **Pathway :** Dorsal column

-Vibrationisrhythmic repetitive pressure sensation
-Itisfeltwhentuning fork put on bony prominences (as leg malleoli) to allow magnificationofstimulus and avoid its damping by soft tissues
-Impairedvibration sense: is an early diagnostic sign in degeneration of posterior columnofspinalcord
-eg:1-Uncontrolled Diabetes mellitus “DM“
-2-Perniciousanemia “PA”

1)Kinetheticsensation : (Proprioceptive sensation) ■

- **Receptors:** Rapidly adapting: Pacinian corpuscle (for rate of movement)

Types:

Sites:

A)In:1)Smalljoints as fingers 2) Large joints as knee
Inligaments&tendons)B

Pathway : Dorsal column

They inform the CNS about:
1) Static: **Sense of position** of different parts of body
2) Kinetic (dynamic): **Sense of movements & Rate of movements** of different parts of body.

NBJoints: - For each degree of angulations, there is specific receptors in joints which discharge to specific area in cortex.
- In thalamus, there is specific neuron for slow rate of movement and other for high rate of movement .

Proprioceptive impulses go to:
1) Cerebellum: Via spino-cerebellar tract to keep Equilibrium
2) Cerebral cortex: Via dorsal & ventro-lateral tract

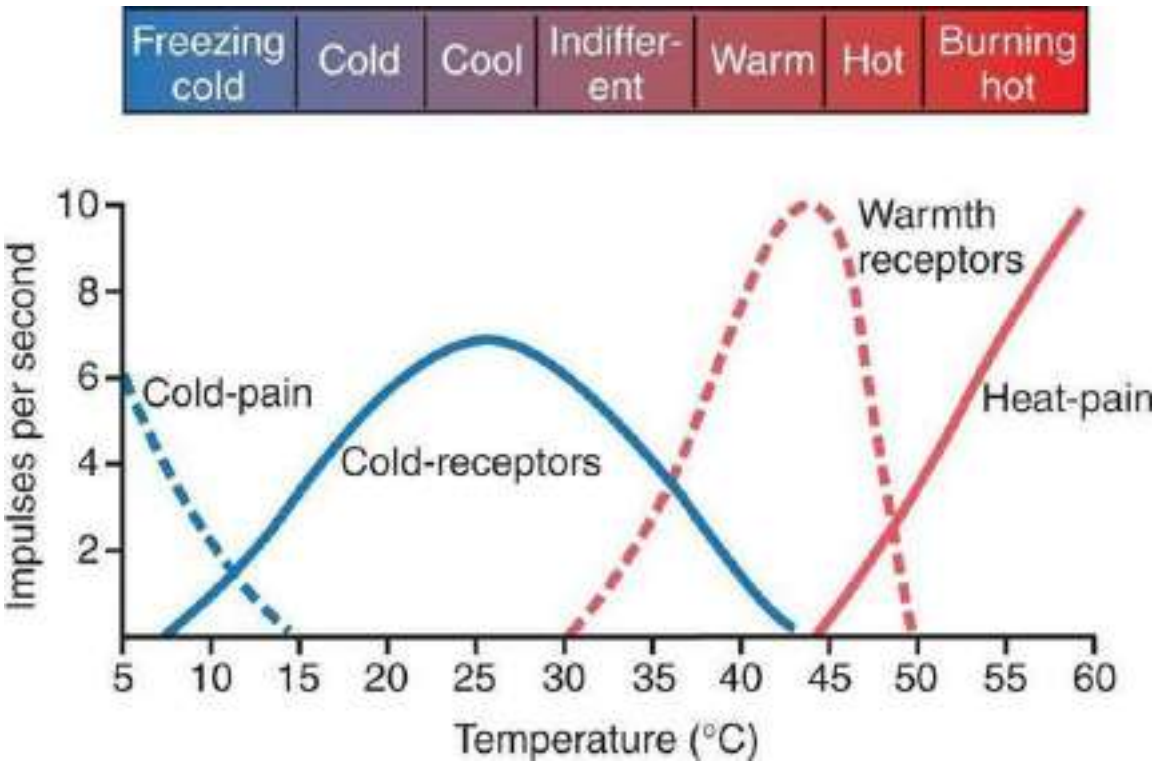
2-ThermalSensation

- **Definition:** - Sense of:
1) Warm
2) Cold

Thermo receptors:

1) Types:
1- Cold receptors: Free nerve ending attached to: C & Aδ fibers
2- Warm receptors: Free nerve endings attached to: C fibers
3- Two subtypes of pain receptors (high threshold receptors):
   1. Cold pain receptors: For freezing cold
   2. Heat pain receptors: For burning hot sensation

1) Characters:
1- Are located immediately under the skin
2- Number: Cold receptors: are more numerous than warm receptors
   Gratest in: Lips
3- Distribution: Moderate in: Fingers tips Least in: Trunk
4- Mechanism of stimulation: Stimulated chemically by:
   - Accumulated metabolites (due to change in metabolic rate)
   - Each 10° Change increase the concentration of metabolites two folds
5- Adaptation:
   - Warm receptors:
Adapt more rapidly than cold receptors (but both aremoderately adapting receptors).



Discharge frequencies at different skin temperatures of a cold-pain fiber, a cold fiber, a warmth fiber,and a heat-pain fiber.

1] Detection of thermal sensation:

According to range of temperature theydetect:

1- Cold receptors: Stimulated from 10-43°C
- Maximum rate of discharge at 25°C

2- Warm receptors: Stimulated from 30-50°C - Maximum rate of discharge at 45°C

3- Cold **pain** receptors : Stimulated from 5-10°C
- Maximum rate of discharge at 5°C

4- Warm **pain** receptors: Stimulated above 45°C

NB: At 0°C: Degree Centigrade "Celsius" there is no action potentials ie anesthesia

At 35°C: Comfort zone exist where awareness of temperature disappears

1- Temperature sensation perceived depends on:

1- The original skin temperature
2- The rate of temperature change
3- The surface of the skin exposed to temperature change.

Pain sensation

Defined as: Unpleasant protective sensation caused by noxious = nocuous stimuli.

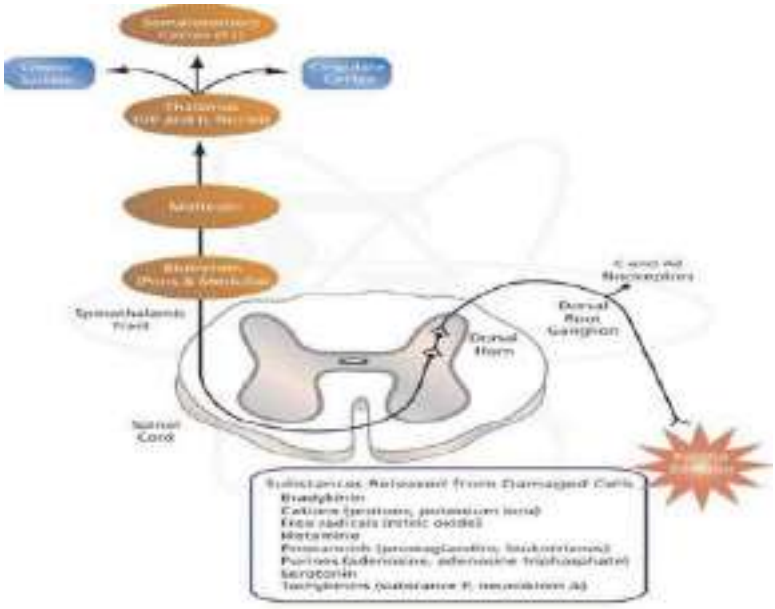
 Receptors are free nerve endings called nociceptors., Distributed allover the body, common in skin, periosteum, walls of blood vessels, peritoneum, parietal pleura, meninges less common in viscera, absent in kidney, liver,brain and lungs although very common in the capsuls covering them.

Classificationof pain

According to time of onset: fast and slow ■
According to duration: acute and chronic ■
According to the site: cutaneous, deep and visceral. ■

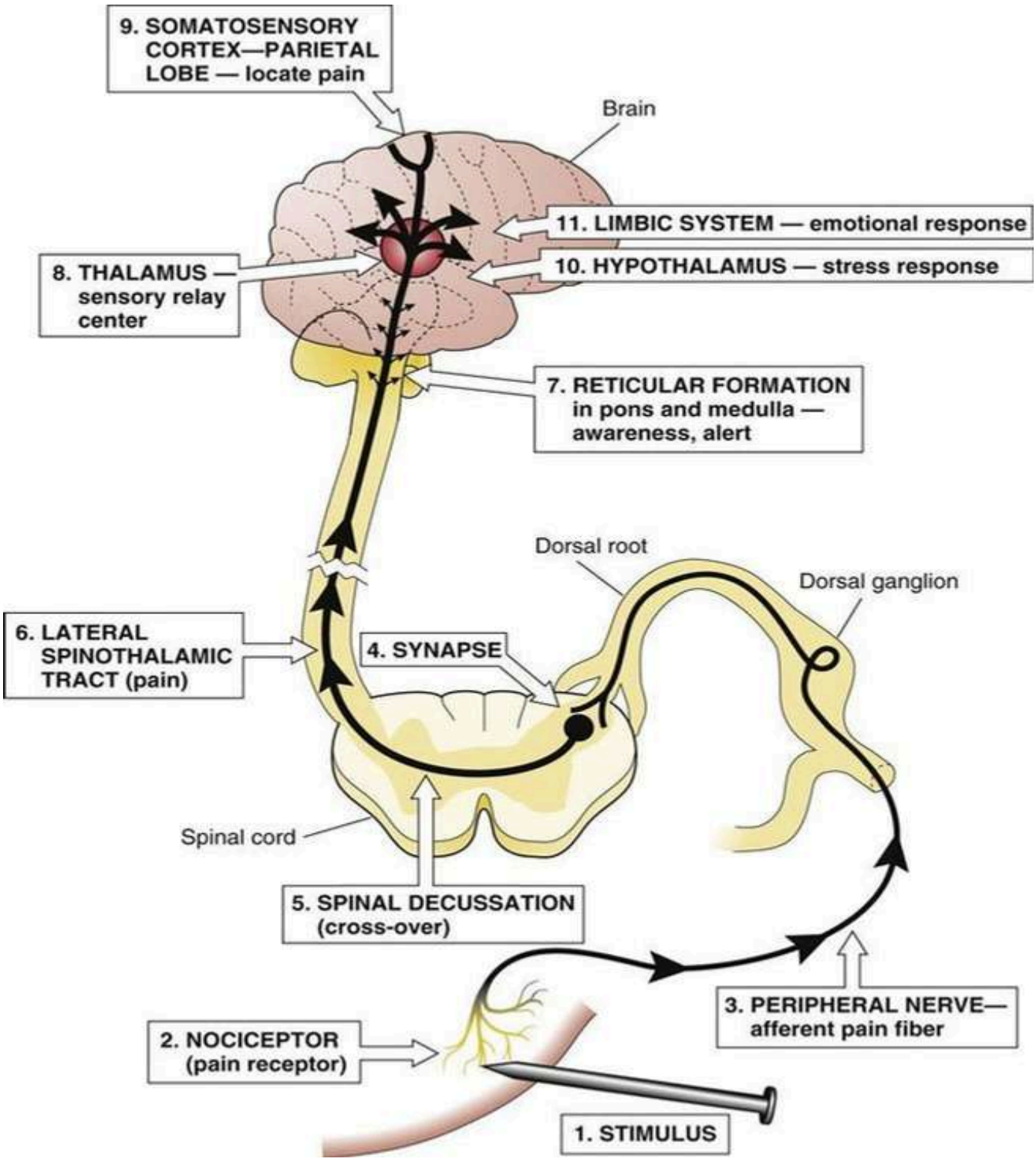
Quality of pain

 Pricking: pin prick or cutting lesions ➢
 burning: burn , inflammation, hyperacidity in stomach ➢
 dull aching : diffuse pain from deep tissues ➢
 throbbing : causes by abcess ➢



Mechanism of stimulation of nociceptors and pathway for pain sensation
Tissue injury leads to release of substance p, K ions histamin……which stimulate nociceptors, firing across the afferent sensory neurons to CNS

pathway for pain sensation



Fast pain	Slow pain
-Cutaneous only -Pricking, sharp, or acute -Well localized -Perceived within 0.1 sec after application of stimulus -Continue for short duration just seconds -can't be summated	-Cutaneous, deep, or visceral -Burning, aching, or chronic. -Poorly localized -Perceived within one sec or more after the stimulus -Continue for longer duration -Marked summation become intolerable

According to site of origin

1- cutaneous pain
2- deep pain
3- visceral pain

CUTANEOUS PAIN

it is either fast well-localized pricking pain or slow diffuse burning pain
It* is *not referred,
It is associated with

(1)Somatic effects (the withdrawal reflex)
(2) Autonomic effects (usually sympathc e.g.: V.C., tachycardia & rise of the arterial blood pressure. but parasymp. effects may occur if pain is severe e.g. V.D., bradycardia & hypotension)
(3) cutaneous hyperalgesia

DEEP PAIN

originates from muscles, joints, periosteum, tendons, bursa etc.

characterizedby

)1(It is slow pain that is conducted by non myelinated C fibres
)2(It is diffuse and dull -aching.
(3) It is often associated with parasympathetic effects
(4) It initiates reflex contraction of near muscles
(5) It may radiate to other sites**,.**

Deep pain may result from trauma to the deep structures, bone fractures, Inflammation; arthritis (inflammation of joints) and muscle ischemia.

ISCHEMIC PAIN

- Pain due to ischemia, decrease blood supply.
- May occur in skeletal, cardiac muscle (angina pectoris) or in viscera

- It takes 2 forms

1- intermittentclaudication pain: partial occlusion of blood vessels by astherosclerosis. Appears during walking or exercise and relieved by rest.
2- muscle cramps: occlusion of blood vessels by severe contraction of the surrounding muscles.

Mechanism ofischemicpain

- Pain does not occur during rest. ■
- It occurs after a period of muscle exercise or cramps. ■
- Due to accumulation of pain producing factors like K ions, lactic acid, ■ bradykinin, histamine causing the ischemic pain.
- After a period of rest, blood supply improves, wash the pain metabolites and pain disappears

VISCERAL PAIN

Viscera contain few numbers of pain receptors.
■ Liver, lung, kidney are pain insensitive.
But their covering membranes are highly sensitive to pain.

Characters:

- - diffuse, dull-aching, coliky or spasmodic ✓
- -Parasympathetic reflexes are often present in form of nausea, vomiting, bradycardia, and hypotension…… √
- - Somatic reflexes in form of contraction of the overlying abdominal muscles (guarding, or rigidity). √
- -Usually referred to an area of skin supplied by the same dermatom ✓
- - Secondary hyperalgesia may be present ✓

Causes ofvisceralpain

(1)mechanicalstimuli:dueto :

a-Overdistension of hollow viscera (e.g. the urinary bladder). Pain in this case is

due to both direct stimulation of the pain receptors and the ischemia that occurs due to collapse of the vessels that encircle the viscus.

b- Spasm of certain viscera (e.g. the small intestine. bile duct and ureters) in which pain occurs by the same mechanisms of overdistension. c- Traction on the mesentery (this has little clinical importance) .

(2) Visceral ischemia (due to excessive release of' the Lewis factor (K ions).
(3) Chemical stimuli (e.g. irritation of the parietal peritoneum by the gastric juice that leaks from a perforated peptic ulcer), gastric burning pain due to peptic ulcer

Visceral afferentnerves ➢

(l) **Autonomic nerves** : Generally. sympathetic and parasympathetic nerves transmit pain from the viscera.
(2) **Somatic nerves** : Pain from the *gall bladder is* transmitted by the *phrenic nerves* (when irritating the central part of the dinphragm).
Also pain from the *parietal layers* or the pleura, peritoneum & pericardium is transmit ted by *somatic spinal nerves* (so it is sharp. well-localized and not referred).

HYPERALGESIA

It is an exaggerated pain sensation = abnormal hypersensitivity to painful stimuli.√

It is of 2 types: ✓

Primaryhyperalgesia

Secondary hyperalgesia

(1) Primary hyperalgesia	(2) Secondary hyperalgesia
This occurs in the injured skin area and the surrounding area of flare (= V.D.), in which the pain threshold is lowered, so that non-noxious stimuli become painful (allodynia). It is due to 1-**sensitization of the pain receptors** by some substances released from the damaged tissues (histamine. kinins. K+. certain enzymes or prostaglandins). 2- **release of substance P** from nerve endings as a result of local axon reflex through antidromic impulses. which sensitizes the pain nerve endings even more, and also produces V.D. (i.e. flare) or the triple response.	occurs in the healthy skin area beyond that of primary hyperalgesia. the pain threshold is not lowered but it is normal. however, the pain aroused from such area is prolonged and exaggerated. This is explained by ***the convergence facilitation mechanism*** which occurs as follow: The central neurons are facilitated by impulses discharged from the area of injury (1ry hyperalgesia). and afferent fibres from the area of secondary hyperalgesia converge on the facilitated neurons. thus, the pain aroused at this area becomes exaggerated. **B: Secondary hyperalgesia** can .N occur in absence of primary hyperalgesia • if there is central facilitation . This occurs in certain, thalamic and spinal cord lesions, and also in some cases of visceral pain (in which the facilitatory impulses are discharged from the diseased viscus). • Opioid induced hyperalgesia

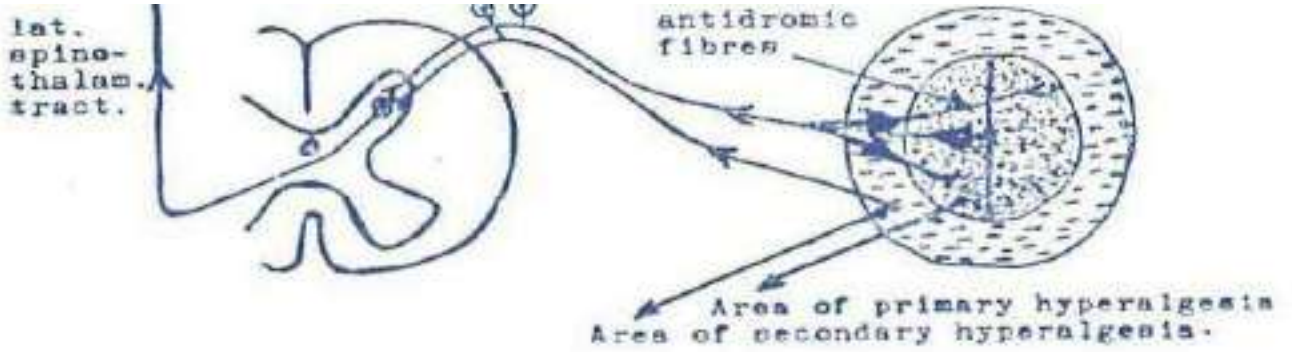


Figure 9 : Areas of primary and secondary cutaneous hyperalgesia.

Cutaneous Hyperalgesia

➢ ***Definition:*** Increased skin sensitivity to pain.

➢ ***Types:*** 1- Primary hyperalgesia. 2- Secondary hyperalgesia.

1- Primary hyperalgesia	2- Secondary hyperalgesia
- Develop 30-60 min. after injury.	- Develops later.
- Lasts for several hours or days.	- Shorter duration than 1ry.
- In the area of redness.	- In healthy skin surrounding red area.
- Non-painful stimuli (as touch) becomes painful.	- Pain is felt more sever than normal.
Mechanism:	
Decreased pain threshold due to local axon reflex releasing substance P	Central sensitization explained by ***convergence-facilitation theory.***

Referred pain

- It is Pain felt away from its original site.
- It is characterized by:
- Often visceral pain in its origin, cutaneous pain is not referred.
- The organ in which pain originates and the organ to which pain is referred develop from the same embryonic segment,
- Frequently, theyare far away from each other due to migration of the various organs during development.

- EXAMPLES OF REFERRED PAIN

(1) Pain of an inflamed gall bladder is transmitted by afferent phrenic nerve fibers (due to irritation the diaphragm) is usually referred to the tip of the right shoulder.

(2) Cardiac pain is usually referred to the left shoulder and inner side of the left arm (less frequently to the right shoulder or the epigastrium).

(3) Pain from the kidneys and ureters is referred to the testicular region.

(4) Gastric pain is referred to the abdominal surface above the umbilicus.

(5) Pain of appendicitis is referred to the umbilical region(T 10)

- Mechanism of referred pain by Convergence projection theory

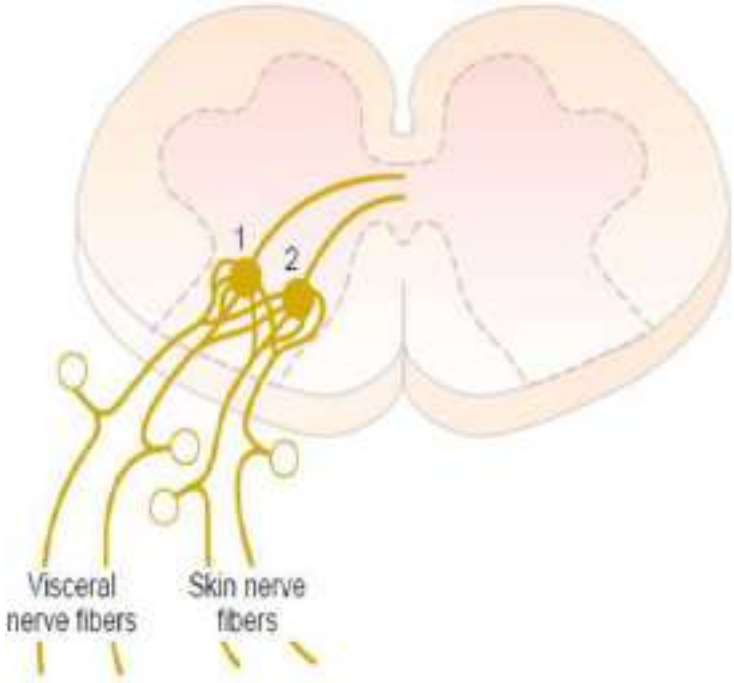
The main cause ofreferred pain is **convergence** of cutaneous andvisceral pain fibers on the same spinothalamic neurons or athigher levels (thalamus) along their pathway to the cortex, that project to the higher centers in the brain.

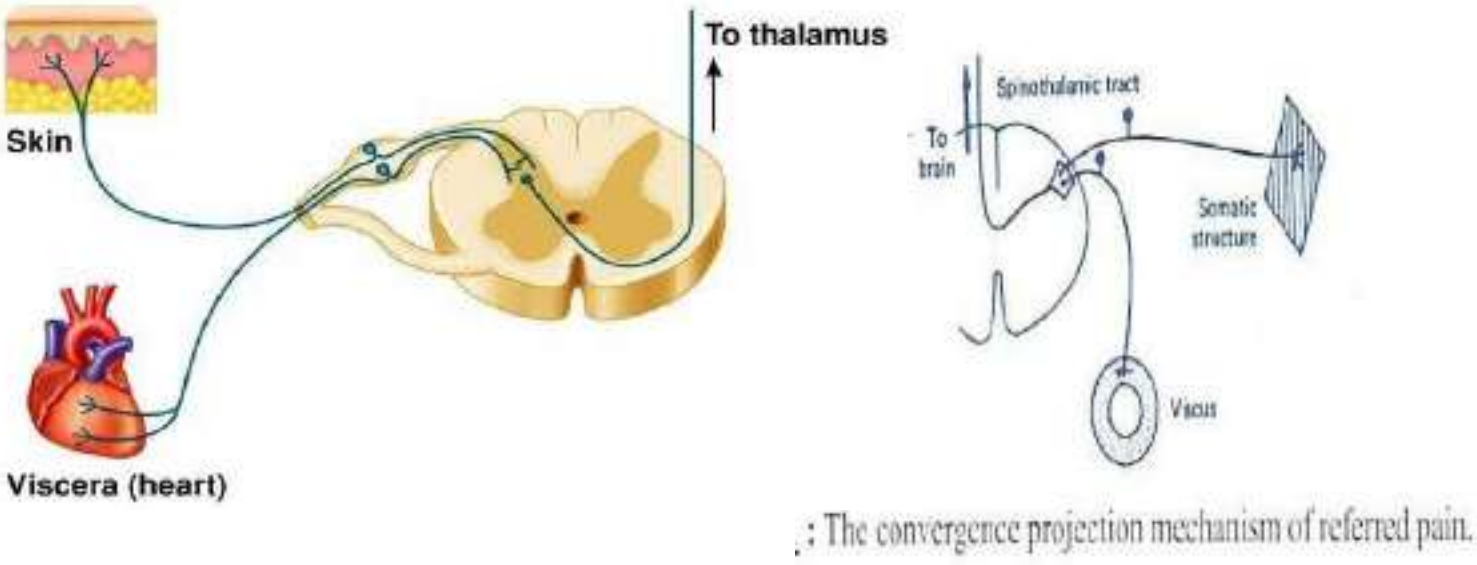
Pain fibers from skin and viscera finally activate the same neurons in the cerebral cortex.

The cerebral cortex project (interpret) this sensation as coming from the skin itself not from the viscerabecausebrain is accustomedto receive pain mainly from skin and not the viscera.

N.B.: in most cases local pain is also felt at the site of the diseased viscera but the referred pain may be more prominent.

- branches of visceral pain fibers make synapse in the spinal cord on the same second-order neurons that receive pain signals from the skin.
- When the visceral pain fibers are stimulated, pain signals from the viscera are conducted By the same neurons which conduct pain signals from the skin, and the person has the feeling that the sensations originate in the skin itself.





: The convergence projection mechanism of referred pain.

Convergence projection theory

Pain analgesia system

- Referred to as descending analgesia system or corticofugal fibers.
- it blocks the synaptic transmission of pain sensation in spinal cord and thus attenuates the experience of pain.
- analgesic drugs such as opioids act by activation of this system and provide a controlled pain relief.

- Role of Analgesic Pathway in Inhibiting Pain Transmission

1. Fibers of analgesicpathway arise from frontallobe of cerebral cortex and hypothalamus

2. These fibers terminate in the gray matter surrounding the third ventricle and aqueduct of Sylvius (periaqueductal gray matter)

3. Fibers from here descend down to brainstem and terminate on:

i. Nucleus raphe magnus, situated in reticular formation of lower pons and upper medulla

ii. Nucleus reticularis, paragigantocellularis situated in medulla

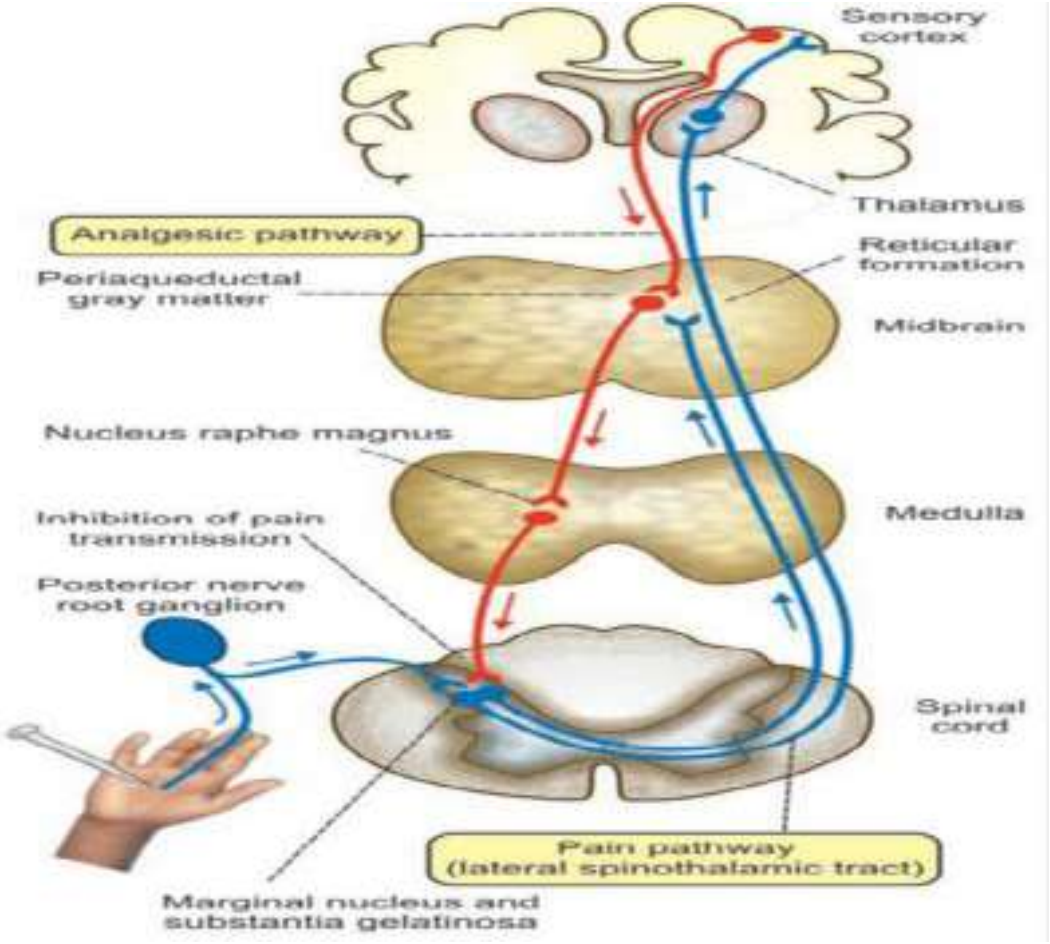
4. Fibers from these reticular nuclei descend through lateral white column of spinal cord and reach the synapses of the neurons in afferent pain pathway situated in posterior gray horn

- Synapses of the afferent pain pathway are between :

i. Aδ type afferent fibers and neurons of marginal nucleus

ii. C type afferent fibers and neurons of substantia gelatinosa of Rolando.

5. At synaptic level, analgesic fibers release neurotransmitters and inhibit the pain transmission before being relayed to brain.



Pain pathway andanalgesia pathway

- Neurotransmitters of Analgesic Pathway

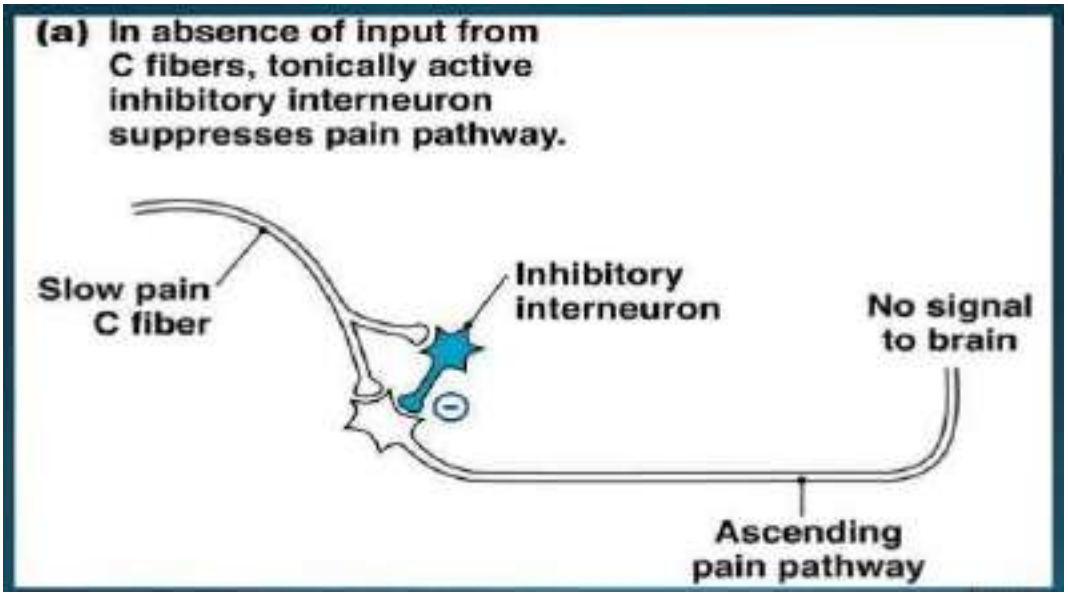
Neurotransmitters released by the fibers of analgesic pathway are serotonin and opiate receptor substances, namely enkephalin, dynorphin and endorphin. These substances act on specific receptors along the pain pathway.

GATE CONTROL THEORY

thepain stimuli transmitted by afferent pain fibers are blocked by gate mechanism which is located at the posterior gray horn of spinal cord.

- **If the gate is opened, pain is felt. If the gate is closed, pain is suppressed.**
- **Gates also may be found at thalamic neurons.**

Gate control theory



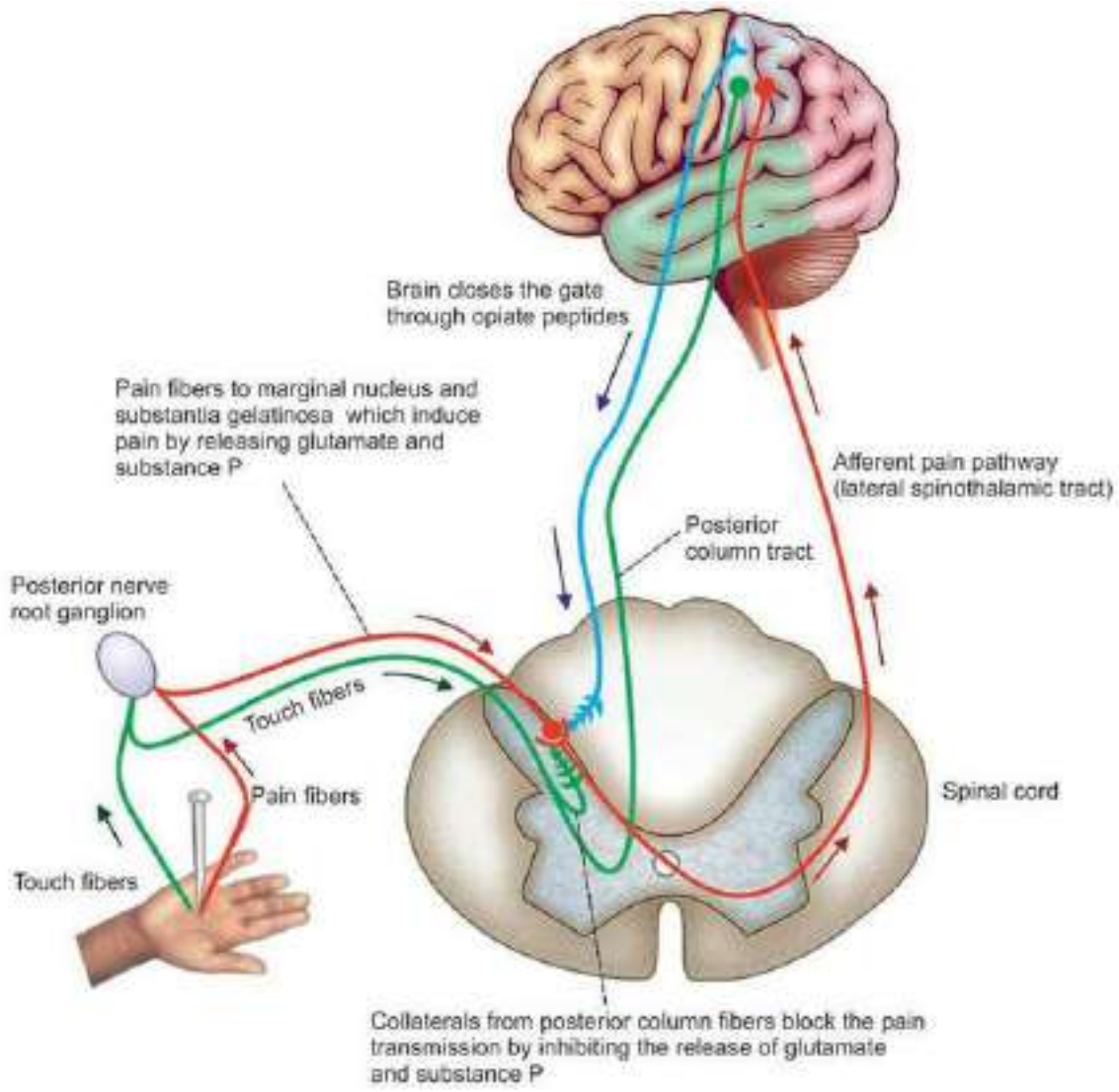
-Mechanism of Gate Control at Spinal Level

1. When pain stimulus is applied on any part of body, besides pain receptors, the receptors of other sensations such as touch are also stimulated

2. When all these impulses reach the spinal cord through posterior nerve root, the fibers of touch sensation (posterior column fibers) send collaterals to the neurons of pain pathway, i.e. cells of marginal nucleus and substantia gelatinosa

3. Impulses of touch sensation passing through these collaterals inhibit the release of glutamate and substance P from the pain fibers

4.This closes the gate and the pain transmission is blocked



- Role of Brain in Gate Control Mechanism

1. If the gates in spinal cord are not closed, pain signals reach thalamus through lateral spinothalamic tract
2. These signals are processed in thalamus and sent to sensory cortex
3. Perception of pain occurs in cortical level in context of the person's emotional status and previous experiences
4. The person responds to the pain based on the integration of all these information in the brain. Thus, the brain determines the severity and extent of pain.
5. To minimize the severity and extent of pain, brain sends message back to spinal cord to close the gate by releasing pain relievers such as opiate peptides
6. Now the pain stimulus is blocked and the person feels less pain.

- Significance of Gate Control

Thus, gating of pain at spinal level is similar to presynaptic inhibition.

It forms the basis for relief of pain through rubbing, massage techniques, application of ice packs, acupuncture and electrical analgesia.

All these techniques relieve pain by stimulating the release of endogenous pain relievers (opioid peptides), **which close the gate and block the pain signals.**

- Stress analgesia

certain stress conditions are associated with analgesia e.g. during the stress of a battle, severely-wounded soldiers frequently feel no pain till the battle.: is over. Such analgesia is produced by impulses discharged from the cerebral cortex and hypothalamus, which excite the central pathway of pain inhibition

Chapter 3
The motor system

The motor Nervous system

The skeletal muscles are controlled by many levels of thecentral Nervous system (CNS) including motor cortex, basal ganglia (BG), cerebellum, brain stem and spinal cord

The reflex arc : Is the functional unit of the nervous system.

A sensory stimulus initiates a motor response directly in a simple neural circuit

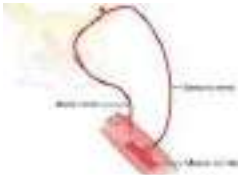
The reflex arc consists of :

1. Receptor: The sense organ
2. An afferent neuron
3. One or more synapses
4. An efferent neuron
5. An effector organ

Spinal Reflexes

[A] Monosynaptic reflex :

Stretch reflex



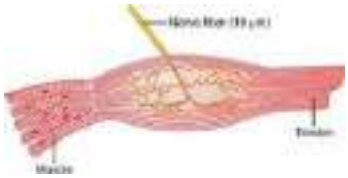
Neuronal circuit of the stretch reflex

[B] Bisynaptic reflex:

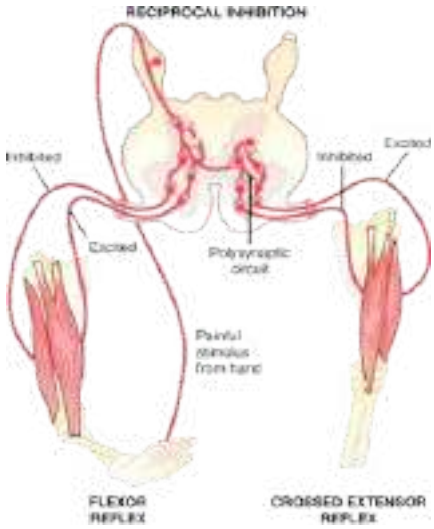
Inverse stretch reflex (Golgi tendon reflex)

[C] Polysynaptic reflexes:

- **Flexor withdrawal reflex**
- **Crossed extensor reflex**
- **Reflexes ofposture &locomotion**
  1. Positive supportingreaction
  2. Stepping and walkingmovement
- **Superficial skin reflexes**
  1. Abdominal reflexes(upper & lower)
  2. Cremasteric reflex
- **Autonomic reflexes in the spinal cord**
  1. Micturition reflex
  2. Defecation reflex
  3. Changes in vasculartone
  4. Sweating: Resultingfrom local skin heat.



Marked stretch → Reflex Relaxation Golgi tendon organ

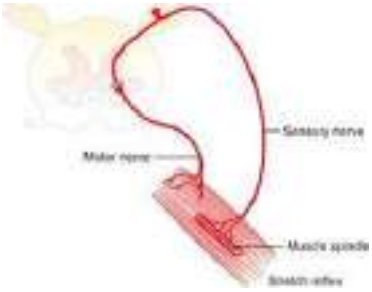


Flexor reflex, crossed extensor reflex, and reciprocal inhibition

Spinal Reflexes

1) Monosynaptic reflexes

- There is only one synapse.
- No interneuron



• Stretch reflex

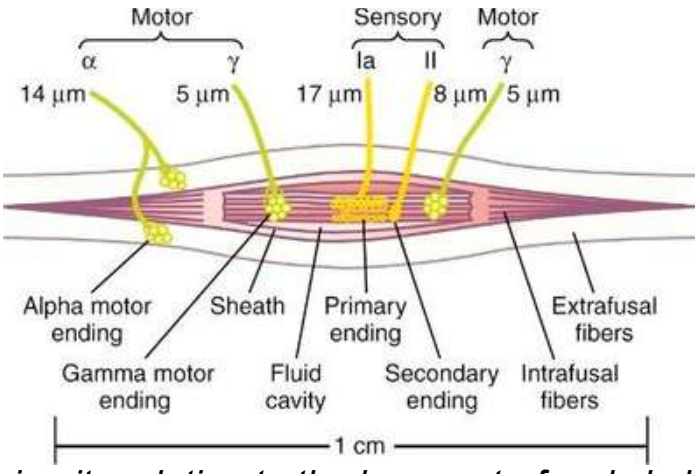
- Is the only monosynaptic reflex
- It is a spinal reflex

❖ **Def:** It is reflex contraction of muscle in response to stretch

❖ **Pathway:**

- Stimulus : Stretch of muscle (extrafusal fibers) → stretch of midportion of muscle spindle (intrafusal)
- Receptors: Muscle spindle (central portion) ↓
- Afferents: Thick myelinated fibers (primary & secondary)
- Center: Alpha motor neuron in spinal cord (AHCs) .It is monosynaptic spinal reflex
- Efferent: Thick myelinated A **α** fibers
- Response: Muscle contraction

Structure of the muscle spindles



Muscle spindle, showing its relation to the large extrafusal skeletalmuscle fibers. (Note also both motor and sensory innervation of the musclespindle)

muscle spindles

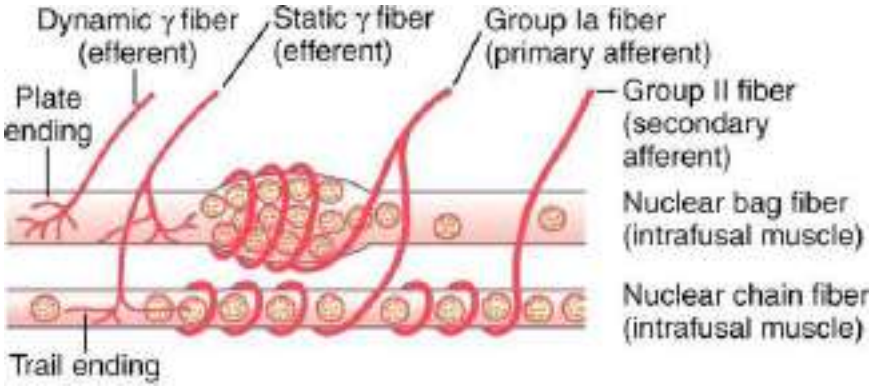
- They are mechanoreceptors
- They monitor muscle length or rate of change of muscle length.
- They are : - Spindle shaped

  - Encapsulated (ends of the capsule are attached to the tendons end of the muscle)
  - Distributed throughout the belly of muscle (Each muscle contain spindles)
  - Each muscle spindle contains 2-10 **intrafusal muscle** fibers:

    - Characteristics:
      - o Smaller & less striated than the ordina muscle fibers
      - o Parallel with the rest of muscle fibers
    - Types: According to the position of the nuclei :

Nuclear bag fibers	Nuclear chain fibers
- 1:3 nuclei in each spindle - They are thicker & longer	spindle - They are thinner & shorter

- The central part of the intrafusal fibers is the receptor area:
  - o It is stretchable
  - o It is not contractile, It contains no actin or myosin
- The peripheral part of the intrafusal is the contractile part



Details of nerve connections from the nuclear bag and nuclear chain muscle spindle fibers

❖ **Innervation:**

Sensory innervations	
1ry ending	2nd ending
Encircle central part of bag & chain	Encircle nuclear chain only

Motor Innervations	
Peripheral contractile part of receptor receive motor supply from gamma motor	
Dynamic	Static
For peripheral part of bag	For peripheral part of chain

Stimulation of receptor

Stretch the whole muscle	Contraction of peripheral part of receptor

❖ **Types:** Therearetwocomponentsofstretch reflex:

1. Static stretch reflex:

- **The stimulus is**: Slow sustained stretch
- **The receptor is**: Nuclear chain which is type of muscle spindle
- **The afferent is**: Primary & secondary nerve endings
- **The response is**: Continuous contraction of the muscle
- **It is the basis of**: Muscle tone.

2.Dynamic response:

- **The stimulus is**: Sudden quick stretch
- **The receptor is:** Nuclear bag which is type of muscle spindle
- **The afferent is:** Primary nerve endings
- **The response is:** Sudden strong contraction followed by sudden relaxation within a second
- **It is the basis of:** Tendon Jerk

Properties of stretch reflex :

1- It is monosynaptic
2- It is the fastest reflex in body
3- It is local (restricted to the stretched muscle)
4- It is more in the antigravity muscle
5- It shows reciprocal innervation (inhibition of antagonistic muscle)
6- It does not show fatigue

Supraspinal regulation of stretchreflex:

Stretchreflexisaspinalreflexwhichiscontrolled by Supraspinal centers :

Supra spinal facilitatory area	Supra spinal inhibitory area
Impulses fromthesesareaIncrease activity of gamma motor neuron **Facilitatory RF "reticular formation" which**: - Send impulse to gamma through ventral reticulo spinal tract. - It has intrinsic activity **Vestibular nucleus**: Send impulse to gamma through vestibulospinal tract **Neo cerebellum**: Send impulse to gammaindirectly through facilitatoryRF **Primary motor area 4**	Impulses fromthesearea decrease activity of gamma motor neuron **Inhibitory RF "reticular formation" which**: - Send impulse to gamma through lateral reticulo spinal tract. - It has no intrinsic activity **Red nucleus**: Send impulse to gamma through rubrospinal tract **Paleocerebellum**: Send impulse to gamma indirectly through inhibitoryRF **Basal ganglia**: Sendimpulse indirectly **Cortical suppressor area**

Functions:

- Static stretch reflex is the basis of muscle tone.
- Stretch reflex has a role of in controlling voluntary motor activity
  - Damping function of stretch reflex: Brain discharges irregularly to motor neurons: Leading to jerky (oscillation) movements

- Stretch reflex:
  - o Prevents oscillations of movement & makes them smooth because:
  - o There is simultaneous discharge to **γ** (gamma) motor neuron (coactivation of **α** (alpha) & **γ** (gamma)motor neurons → causing extra & intrafusal to contract at same time)

- Servo-assist function during muscle contraction:
  - o During a normal movement (eg lifting a weight):
  - o The motor cortex sends signals to both **α** (alpha) & **γ** (gamma) motor neurons
  - o (ie there is coactivation **α** (alpha) & **γ** (gamma) motor neurons) & this maintains constant position of contracted muscle in spite of application of different loads.

NOTE

1.The impulses sent to **α** (alpha) motor neurons stimulate the **extrafusal** fibers
2.The impulses sent to **γ** (gamma) motor neurons stimulate **intrafusal** fibers

❖ Clinical importance

Muscle tone:Static stretchreflex

- It is observed as a degree of resistance when the neurologist flexes and extends the muscles alternatively

- **Definition**: - It is reflex sub tetanic alternating contraction of skeletal muscle fibers especially the antigravity muscle (extensors of lower limb & back and flexors of upper limbs)

- During rest, the muscle spindle is **continuously stretched** because:
  - **1-** The muscle length is shorter than the distance between origin & insertion
  - **2-** There is continuous **γ** (gamma) efferent discharge (from higher centers) during normal condition (which keeps the muscle spindle stretched)

- Skeletal muscle tone is maintained **without fatigue** because:
  - **1-** The reflex contraction is sub tetanic
  - **2-** The alternating activity of different motor units

- **Functions of muscle tone**:

1) Keeps body position against gravity
2) Keeps viscera in position
3) Helps to maintain body temperature
4) Helps venous return & lymph drain.

- **Tendon jerk (Deep reflexes)**: Dynamic stretch reflex
  Sudden striking of tendon produces sudden stretch of muscle resulting in reflex rapid contraction followed by rapid relaxation.

- It is facilitated by:

1. Putting the muscles in a stretched position
2. Increasing **γ** (gamma) efferent discharge by:
   a) Clinching the teeth
   b) Pull the hand apart while fingers are flexed & hooked together

Examples of tendon jerks:

1- Biceps jerk:
A sudden tap on biceps tendon of semiflexed elbow → reflex contraction of biceps & flexion of elbow
Center: C5 & C6

2- Triceps jerk:
A sudden tap on triceps tendon → reflex contraction of triceps & extension of elbow
Center: C6 & C7

3- Knee jerk:
A sudden tap on patellar tendon → reflex contraction of quadriceps & extension of knee
Center: L2, L3, L4

4- Ankle jerk:
A tap on Achilles tendon → reflex contraction of gastrocnemius & plantar flexion of ankle
Center: S1, S2

Deep tendon reflexes of upper limb	Deep tendon reflexes of lower limb
Biceps jerk C5 (C6)	**Knee jerk (center: L2, L3, L4)** (legs must not be incontact with each other)
Triceps jerk C6, C7	**Ankle jerk of recumbent patient (S1)**

Clinical significance of skeletal muscle tone & tendon jerk

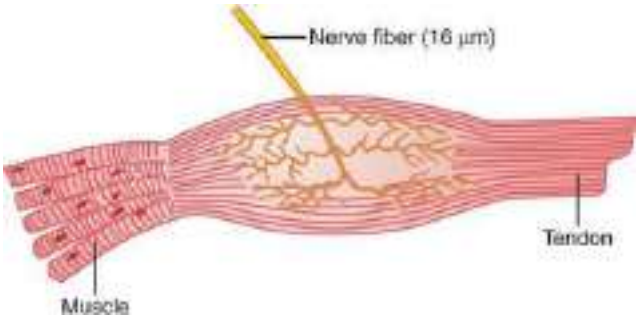
- Muscle tone and tendon jerks areaffectedtogetherinthe same manner (ie they are absent, inhibited or facilitated together)
- Exaggerated tendon jerk & hypertonia: Occurs in UMNL (upper motor neuron lesion)
  Weak tendon jerk & hypotonia: Occurs in LMNL (lower motor neuron lesion)

Inverse stretch reflex (Golgi tendon reflex)

Pathway:

- Stimulus: Increased tension in muscle & its tendon e.g. marked stretch or strong active contraction
- Receptor: Golgi tendon organ:
  - Is a high threshold receptor which is stimulated by: High tension in the muscle
  - Found in the tendon in series with muscle fibers
- Afferent: Large myelinated nerve fibers (Ib)
- Center: Spinal cord, it is **bisynaptic spinal reflex**
- Efferent: Alpha motor neuron, which inhibits the muscle fibers.
- Result: Reflex relaxation of the muscle → decrease muscle tone

- **Aim:** Itpreventsthedevelopment of too much tension on the muscle
- **Clinical application:**
  - Lengthening reaction (in clasp knife rigidity)
  - NB: Clasp knife means: initial resistance then sudden release.



Marked stretch → Reflex Relaxation Golgi tendon organ

Mechanism:

- If upper limb is passively flexed:

- Extensors are stretched & lengthened

- Stimulation of their muscle spindle → increasing the tone and resistance

The high tone & tension in the muscle → stimulates Golgi tendon receptor → inhibit **α** (alpha) motor neuron → leading to muscle relaxation

2) *Polysynaptic reflexes*

A.Flexor withdrawal reflex: Nociceptive reflex

- **Pathway:**

- Stimulus : Injurious (painful) stimulus applied to a limb
- Receptors : Free nerve endings
- Afferent : A delta & C
- Center : In spinal cord. It is polysynaptic containing 4 to 5 interneurons at least
- Efferent : Alpha motor neuron
- Response : Flexion & withdrawal of the limb

- **Properties:**

1-The pattern of the reflex **depends on the locality of the stimulus** ie stimulation of inner side of the limb produces: Flexion & abduction

2- **Divergence** : Is present to allow the spread of excitation to several flexor muscles

3- **Reciprocal innervation**:contractionof flexors is accompanied by relaxation of extensors
This is done throughlateralinhibitionofthe extensors

4- **Recruitment**:Thetensionincontracting muscle rises gradually (due to gradualincreaseinnumberof activated motorunits)

5- **Tetanus**: Isofreflexnature

- Reflex tetanus:

- Means : Continuouscontractionofmuscledue to repeated stimulationofafferentnerve
- Characterizedby:

1- Long latentperiod(duetoimpulse conduction in afferent nerve & interneuron)
2- Thetensionrisesgradually(ie recruitment) then declines due to synapticfatigueandpersistsforashort time after the stimulus is removed (after discharge)

NOTE

Motor tetanus:

- Is due to: Repeated stimulation of motor nerve of muscle (efferent)
- Is characterized by:

1- Short latent period
2- The tension rises rapidly, maintained & then drops rapidly after the stimulus is removed

B.Crossed extensor reflex

-Applicationofinjuriousstimulustoalimbproduces:

1-Reflexflexionoftheipsilaterallimb
2-Reflexextensionofthecontralateral limb to support the body weight

-Properties:

1- The latent period is longer than the flexor reflex

2- The after discharge is longer than the flexor reflex.

3- Reciprocal innervation is present, as excitation of the extensor is associated with inhibition of the flexor muscles.

C.Reflexes of posture & locomotion:

1. Positive supporting reaction:

- Deep pressure on the sole of the foot produces reflex Contraction of both flexors & extensors to convert the limb into a rigid column to support the body weight (the reflex shows no reciprocal innervation)

2. Stepping and walking movement:

Rhythming stepping of the limb & reciprocal stepping of the other limb indicating that pattern of walking is in spinal cord.

D.Superficial skin reflexes :

1. Abdominal reflexes (upper & lower):

- Stroking (scratching) the skin of the abdomen leads to reflex contraction of:

-Upper abdominal muscle (Center: T7 to Tl0)

-Lower abdominal muscle (Center: T10 to T12)

2.Cremasteric reflex: Scratching the skin of the inner aspect of the thigh → contraction of Cremasteric muscle & elevation of testis

-Center: L1

E.Autonomic reflexes in the spinal cord

1. Micturition reflex : Distension of bladder results in:

- Reflex contraction of bladder
- Reflex relaxation of urethral sphincters
- Center: S2, S3,S4

2. Defecation reflex: Distension of rectum results in:

- Reflex contraction of rectum
- Reflex relaxation of anal sphincters
- Center: S2, S3, S4

3. Changes in vascular tone: Resulting from local skin heat or cold

4. Sweating: Resultingfromlocal skin heat.

➢ **NB :** Decerebrate rigidity: Caused by transverse section at brain stem between superior & inferior colliculi.

-This section :1- Removes the inhibitory effect of Supraspinal centers & red¬ nucleus

2- Leaving the strong facilitatory effect of facilitatory reticular formation and vestibular nuclei

-Effect: The tone increases markedly in the antigravity muscle so the animal shows extended neck, back, limbs & elevated tail.

Asimilar condition occurs in man when there is lesion of internal capsule.

The vestibular system

Equilibrium

-This is maintained by afferent pathways from:

1- Pressure receptors in the feet: Which detect the distribution of support

2- Body proprioceptors: Which detect parts of the body & its relative to another

3- Neck proprioceptors: Which detect position of head relative to body

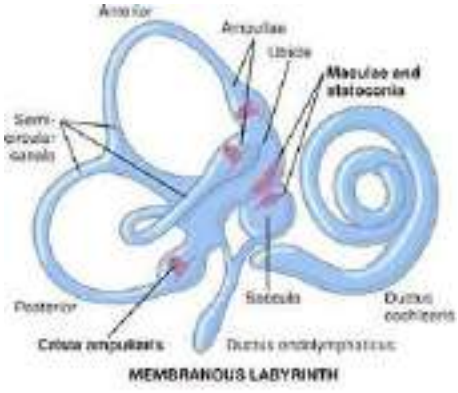
4- Visual system: Which detects head position in visual space

5- Vestibular system: Which detects position & motion of head in space

Vestibular system & maintenance of equilibrium

The membranous labyrinth is :

- Present inside the bony labyrinth
- Composed of 2 functional units:



- Auditory labyrinth ie cochlea concerned with hearing
- Non auditory labyrinth :

- Important in equilibrium

: Three :

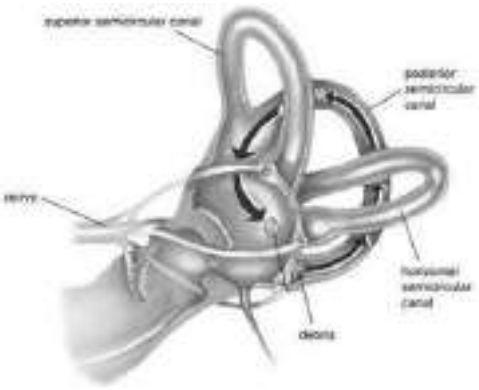
- Semicircular canals
- Utricle
- Saccule

- Itiscalledvestibular apparatus
- Itisfilledwithendolymph & surrounded by perilymph

1) *TheSemicircularCanals(SCCs)*

-There are SCCs in each vestibular apparatus:

- Anterior vertical (superior)
- Posterior vertical
- Horizontal (external)



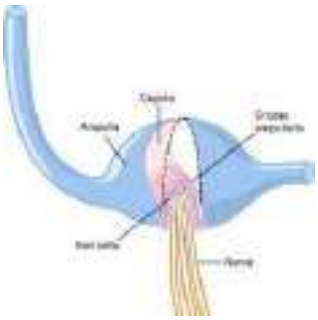
• The 3 canals are perpendicular to each others.

• Act in copairs

• horizontal canals share the same horizontal planeplane.

• posterior canal shares plane with contralateral anterior canal.

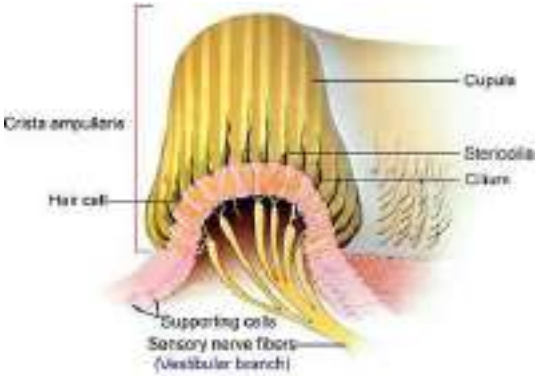
- Each SCCs has an enlargement at one of its end called ampulla.



Movement of the cupulaand its embedded hairs at theonset of rotation

- **Receptors:** Cristsa ampullaris which are specific mechanoreceptors located in each ampulla. It consists of a ridge of hair cells

- **Stimulus**: Crista ampullaris detect Angular Acceleration



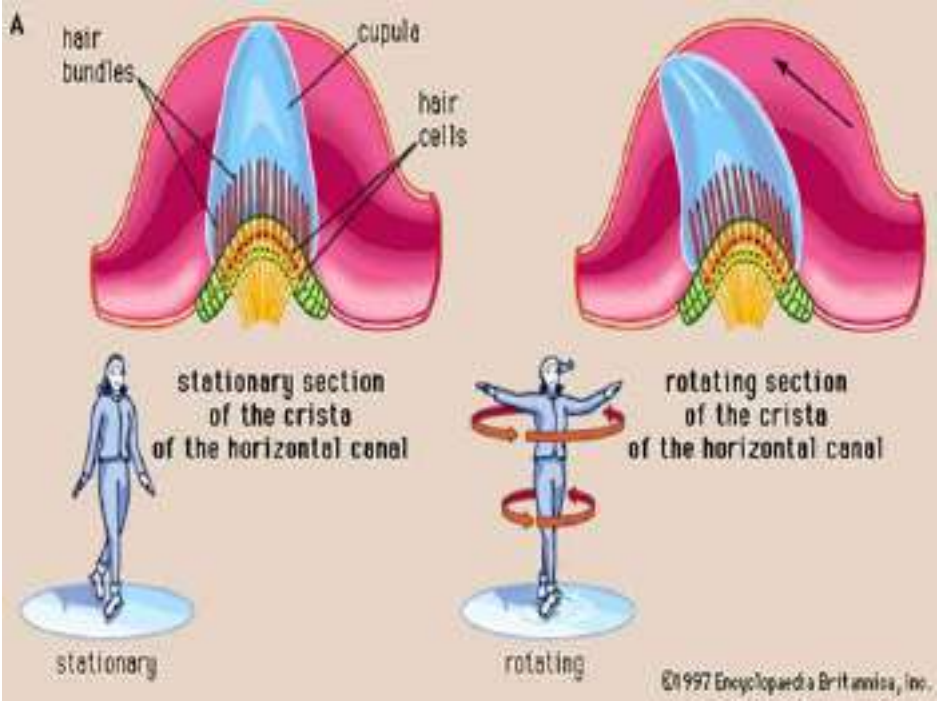
- Structure of crista:

Each hair cell has :

- Basal part: -In contact with sensory nerve
  -Surrounded with perilymph
- Apical part: -From which cilia rise
  -Surrounded with endolymph
- Cilia type: -Sterocilia: Short & Numerous
  -Kinocilium: Single & Large
- The membrane potential of hair cell = -60
- The endolymph: Surround apical part contain high amount of K
- The perilymph :Surround basal part contain high amount of Na
- Movement of Sterocilia : Toward Kinocilium open ionic channel lead to K influx produce partial depolarization in receptor (stimulation)

- All All kinocilia in a single canal are oriented in the same direction, so each canal respond to one direction (unidirectional receptor)
- Crista in each pair of canals respond inversely to each other e.g. when rt horizontal canal is stimulated, the left horizontal canal is inhibited.

Hair cell of the equilibrium apparatus and its synapses with the vestibular nerve



- Mechanism of stimulation of SCCs

1. At the start of rotation (To the Right)
   - The endolymph by its inertia rotates to the left, so both cristae are bent to the left
   - The right crista bends towards the utricle, so it is stimulated
   - The left crista bends away from the utricle, so it is inhibited
   - The unbalanced discharge give the sensation of rotation to RIGHT
2. With continuation of rotation
   - With constant speed, the inertia is overcomed
   - Both cupula regain their normal resting position within 30 sec
   - The sensation of rotation **DISAPPEARS**
3. At the end of rotation(deceleration)
   - The endolymph by its momentum continues to rotate to the right after stoppage of rotation, so, both crista are bent to the right
   - The left crista bends towards the utricle, so it is stimulated
   - The right crista bends away from the utricle, so it is inhibited
   - The person feels a false sensation of rotation to LEFT, despite the stoppage of rotation which is called Vertigo

- Effect of stimulation of SCCs: -

Can be stimulated by : rotation, caloric test, and electric stimulation -

Effects of stimulation include:

1. Nystagmus

- Definition: Oscillatory (jerky) eye movements consisting of
  - Slow component opposite to direction of rotation
  - Fast component in same direction of rotation

Other causes of nystagmus: 1- Optokinetic: Looking from window
2- Menier disease: Enlargement of SCCs
3- Neocerebellar lesion

2. Vertigo:

- Definition: It is a false sensation of rotation in absence of actual rotation which leads to loss of balance

Other causes of vertigo:

a. Toxicity by: -Alcohol -Streptomycin

b. Menier disease

c. Motion Sickness: Due to Excessive, repetitive labyrinth stimulation caused by Conflict between visual and Vestibular information

3. Postural reflexes:

• Definition: - Change in muscle tone due to stimulation of SCCs to maintain balance during angular acceleration

- Thus, stimulation of SCCs produces :
- Increase muscle tone on ipsilateral side
- Decrease muscle tone on contralateral side

4. Autonomic changes:

- Bradycardia
- Hypotension
- Increase rate of respiration
- Nausea
- Vomiting
- Pallor
- Sweating

2) *Theutricleand saccule*

1-Maculae (otolith organ)

- Maculae are mechanoreceptors located inside each utricle & Saccule
- The macula of the utricle lies in a horizontal plane
- Macula of the Saccule lies in a vertical plane

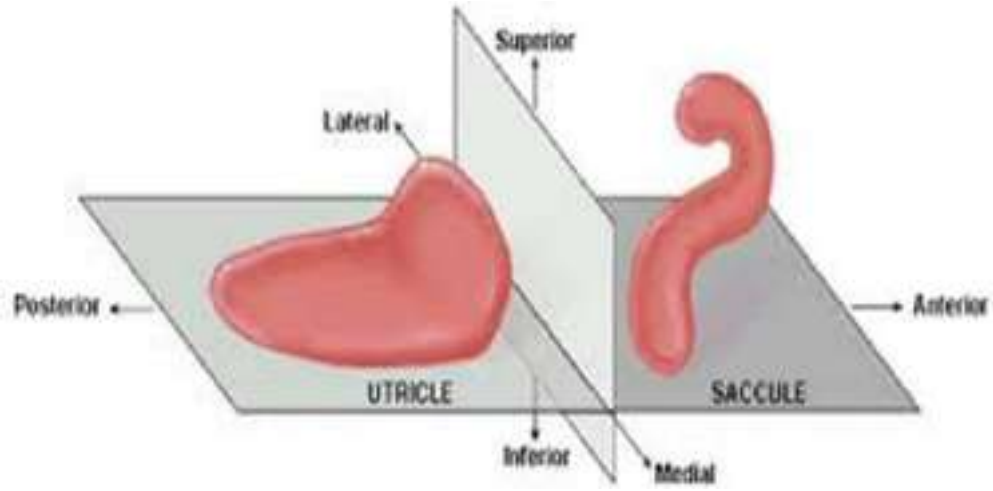
- **Structure of the macula :**
  - The sensory organ in both the saccule and utricle is the maccula. hair cells sit with their cilia embedded in a gelatinous membrane containing otoliths/otoconia/ stratoconia (calcium carbonate crystals. Cilia in one macula are arranged in differen position so one macular receptor can respond to multiple directions of movement (multidirectional receptor)

- **Functions of otolith organs** (maculae) or mechanism of stimulation of maculae:

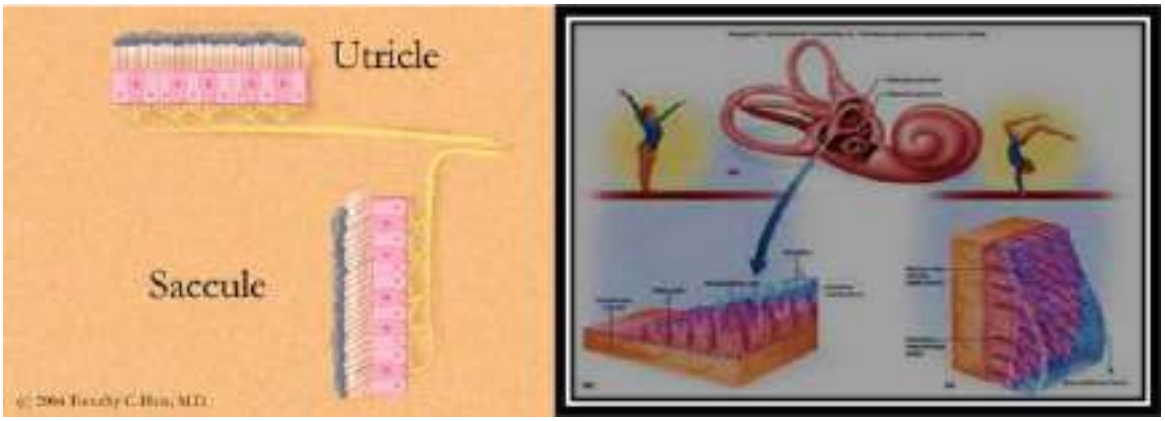
1- Maculae can detect changes in the position of head relative to forces of gravity
   - ie they detect orientation of head in space & its tilting front to back or side to side
   - The macula of utricle detects the orientation of head when person is upright
   - The macula of Saccule detects it when a person is lying down
   - Macula are important in swimming under deep water where vision & proprioceptive sensation is absent

2- Maculae detect linear acceleration:
   - Utricle responds to horizontal acceleration
   - Saccule responds to vertical acceleration

	Otolith organ	Semicircular canals
organ	Utricle, saccule	Anterior, posterior, horizontal
receptor	Macula	Crista ampularis
directionality	multidirectional	Unidirectional
function	Detect Head position and linear acceleration	Detect angular acceleration(rotation) in different 3 dimensional planes



Macula of the utricle is mainly horizontal but macula of thesaccule is mainly vertical



VESTIBULA R PATHWAY

Signals from the vestibular receptors pass across the vestibular nerve(branch of the vestibulocochlear nerve) to the vestibular nuclei in the medulla oblongata. From the vestibular nuclei fibers may be conveyed to

- **Vestibulo-Ocular Pathways**:
  - – Direct: to cranial nerve nuclei (III IV and VI) responsible for movement of the eye
  - – Indirect: via reticular formation to oculomotor nuclei (III IV and VI)
- **Vestibulo-Spinal Pathways**:
  - – Lateral V-S-throughout spinal cord
  - – Medial V-S-cervical & thoracic
  - – Reticulospinal tract-via brainstem reticular formation
- Cerebellum
- Toreticular formation, thalamus, and then to cerebral cortex, equilibrium center located in the temporal lobe.
  - **Function of the vesibular receptors**

Predict ahead of time that mal-equilibrium is going to occur

Send impulses to CNS for corrective measures before the start of the fall

Chapter 4:

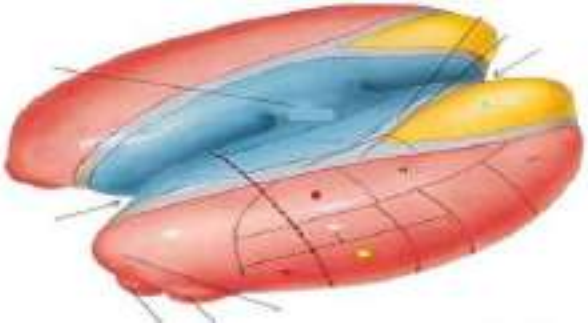
The higher centers and higher intellectual functions

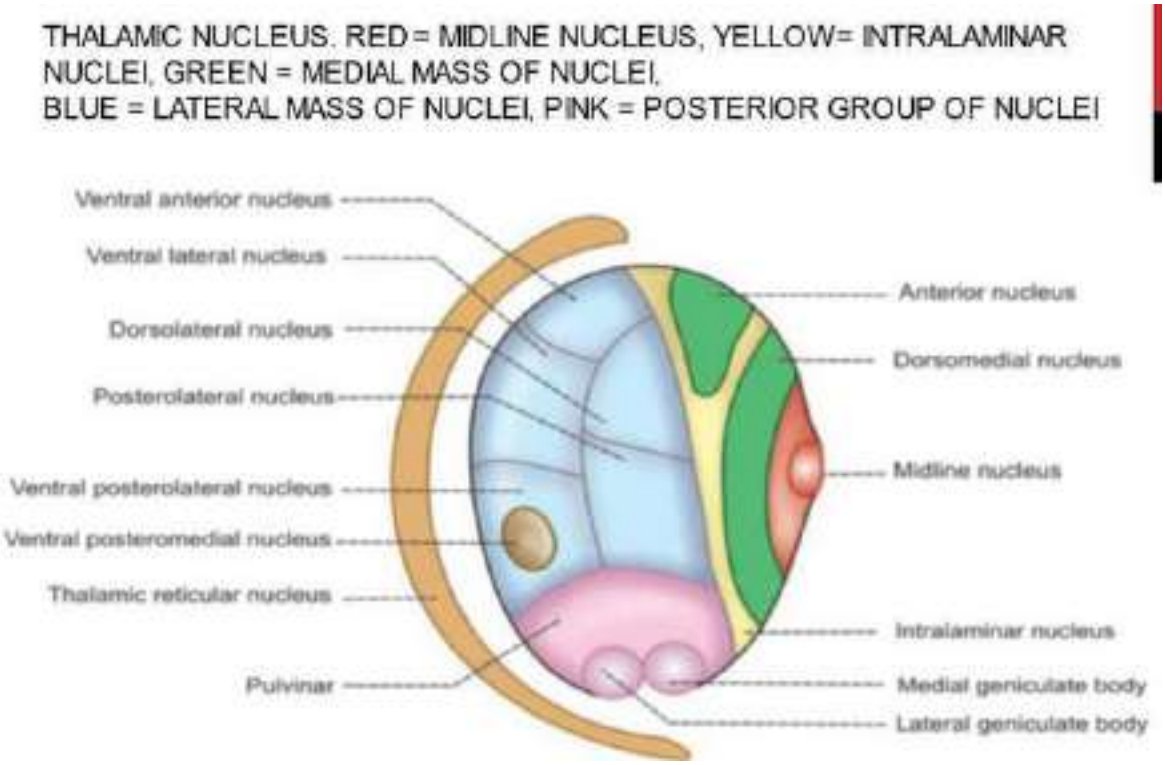
1 The higher sensory centers

1. thalamus
2. Primary somatosensory cortex
3. Secondary or association sensory cortex

1) The thalamus

- Thalamus is a large ovoid mass of gray matter, situated bilaterally in diencephalon.
- Both thalami form 80% of diencephalon.
- Thalami on both sides are connected in their rostral portions.

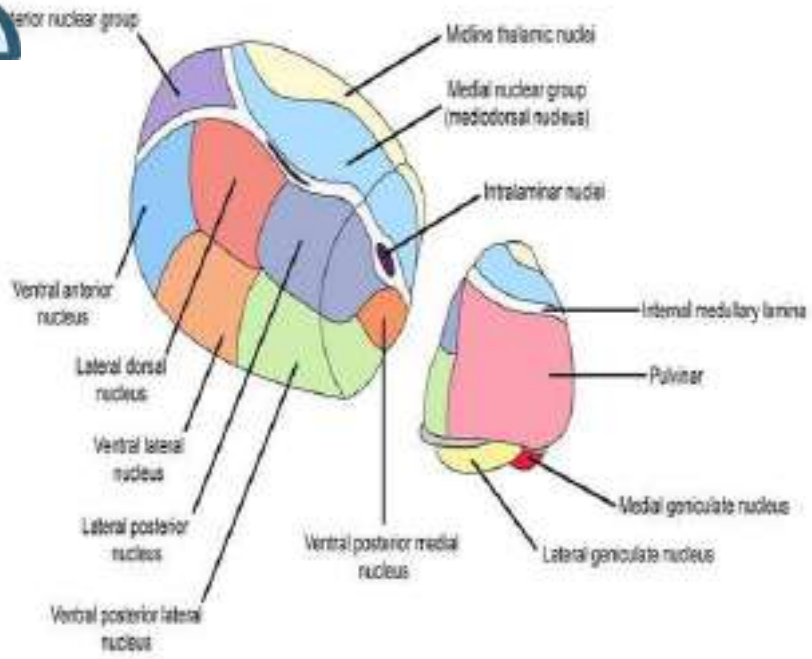




A natomical classification

- Thalamus on each side is divided into five main nuclear groups by **'Y'-shapedinternal medullary lamina**.

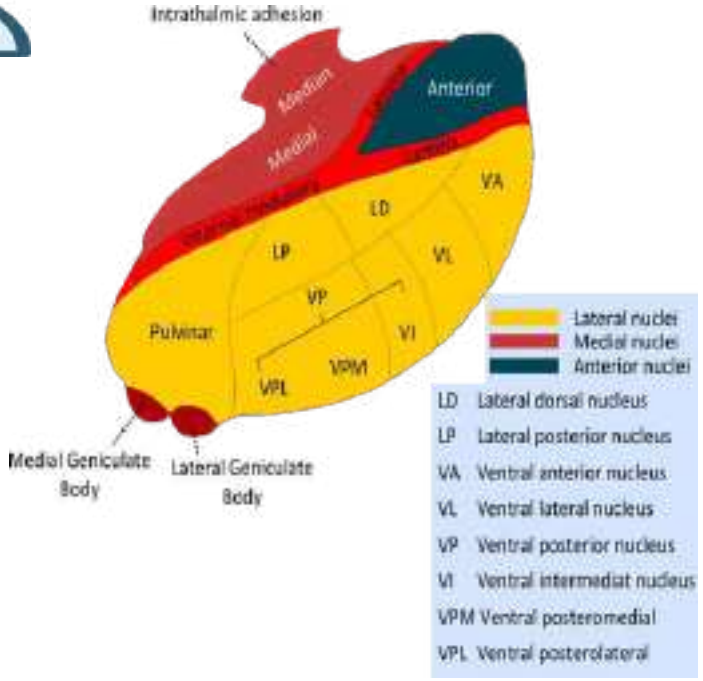
1. Midline nuclei
2. Intralaminar nuclei
3. Medial mass of nuclei
4. Lateral mass of nuclei
5. Posterior group nuclei.



PHYSIOLOGICAL CLASSIFICATION

- On the basis of functions and their projections, thalamic nuclei are classified into five groups.
- groups of thalamic nuclei are:

1. Specific sensory relay nuclei
2. Specific motor nuclei
3. Association or less specific nuclei
4. Non-specific nuclei
5. Limbic system nuclei.



Group	Nuclei	Functions
1. Specific sensory relay nuclei	i. Ventral posterior nucleus ii. Medical geniculate body iii. Lateral geniculate body	Project sensory signals to distinct (specific) areas of cerebral cortex
2. Specific nuclei	i. Ventral anterior nucleus ii. Ventral lateral nucleus	Receive signals controlling motor activities from cerebellum and corpus striatum and send these signals to motor areas in the cerebral cortex to complete the feedback system of motor control mechanism
3. Association or less specific nuclei	i. Dorsolateral nucleus ii. Posterolateral nucleus iii. Pulvinar	Send information to association areas of cerebral cortex
4. Non-specific nuclei	i. Midline nuclei ii. Intralaminar nuclei iii. Reticular nucleus	Project signals to diffused areas of cerebral cortex
5. Limbic system nuclei	i. Anterior nucleus ii. Dorsolateral nucleus	Project into limbic cortex

Functional classification

Functions of thalamus

- Thalamus is primarily concerned with somatic functions and it plays little role in the visceral functions.

1. RELAY CENTER

- Thalamus forms the relay center for the sensations. Impulses of almost all the
- sensations reach the thalamic nuclei, particularly in the ventral posterolateral nucleus. After being processed in the thalamus, the impulses are carried to cerebral cortex through thalamocortical fibers.

2. CENTER FOR PROCESSING OF SENSORY INFORMATION

- Thalamus forms the major center for processing the sensory information. All the peripheral sensory impulses reaching thalamus are integrated and modified before being sent to specific areas of cerebral cortex. This function of thalamus is usually called the processing of sensory information.
- 'functional gateway' for cerebral cortex= secretary of the cortex

3. CENTER FOR DETERMINING QUALITY OF SENSATIONS = emotional function

- Thalamus determine the affective nature of sensations. Usually the sensations havetwo qualities:

I. Discriminative nature II. Affective nature.

I.Discriminative Nature :

√ Discriminative nature is the ability to recognize the type, location and other details of the sensations and it is the function of cerebral cortex.

II. Affective Nature:

√ Affective nature is the capacity to determine whether a sensation is pleasant or unpleasant and agreeable or disagreeable. Determining the affective nature of sensations is the function of thalamus.

4. **ROLE IN AROUSAL AND ALERTNESS** of cerebral cortex Because of its connections with nuclei of reticular formation,

5. MOTOR function

- Through the connections with cerebellum and basal ganglia, thalamus serves as a center for integration of motor functions.

Thalamic syndrome

- 1. Loss of Sensations on the opposite side Loss of all sensations (anesthesia) occurs as the sensory relay system in thalamus is affected. 2. Astereognosis Astereognosis is the loss of ability to recognize a known object by touch with
- closed eyes. It is due to the loss of tactile and kinesthetic sensations in thalamic syndrome. 3. loss of kinesthetic sensation.

  This leads to loss of sense of position, muscle movement, and sensory ataxia due to loss of sensation is called sensory ataxia. It is very common in thalamic syndrome.
- 4. Thalamic Phantom Limb:

  The patient is unable to locate the position of a limb with closed eyes. The patient may search for the limb in air or may have the illusion that the limb is lost. This is
- called thalamic phantom limb. 5. Spontaneous Pain and Thalamic Over-reaction

  Spontaneous pain occurs often. Pain stimulus is felt more acutely than in normal conditions (hyperalgesia). Pain may be so intense, that it even resists the action of powerful sedatives like
- morphine. Threshold for pain is very much reduced. Even the light touch may be unpleasant. Sometimes, the patient feels pain even in the absence of pain stimulus. It becomes worst in conditions such as emotional disturbance and

exposure to cold or heat. Pain is due to over activity of medial mass of nuclei of thalamus, which escape the lesion.

6. Abnormal reaction to various stimuli is called thalamic over-reaction.

7. *Involuntary Movements*

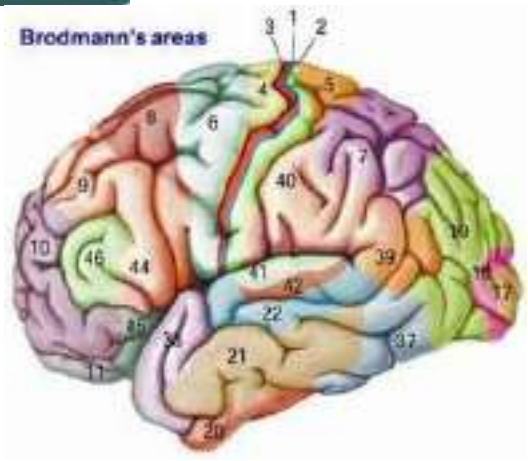
2) Primarysensory cortex = areas 1,2,3

- **Body representation is**

1. crossed or contralateral
2. inverted or upside-down
3. the area of representation is related to the sensory function of the organ and not to its size.

Functions of primary Somatosensory cortex

1. Primary area I is responsible for perception and integration of cutaneous and kinesthetic sensations.
   - It receives sensory impulses from cutaneous receptors (touch, pressure, pain, temperature) and proprioceptors of opposite side through of the body.
   - Area 1 is concerned with sensory perception.
   - Areas 3 and 2 are involved in the integration of these sensations.



2. This area sends sensory feedback to the premotor area

3. Discriminative functions like:
   a) Spatial recognition: Tactile localization, two point discrimination and recognition of positionand passive movements of limbs
   b) Recognition of intensity of different stimuli
   c) Recognition of similarities and differences between the stimuli.

4- localization of pain and temperature.

Effect of lesion in 1ry sensory cortex

1. Inability to perceive fine cutaneous sensation and kinethetic sensations.
2. Crude localization of various sensations
3. Defect in motor performance especially inability to excute learned sequences of movement. Like writing , cutting.

NOTE

Somatic sensory area 11: Potentiate area the primary motor cortex. Body representation is different

3) *Association somatic area = secondary sensory cortex*

- Lie behind area 1
- receive impulses from 1ry sensory cortex and thalamus.
- Concerned with:
  - ❑ Interpretation of sensation
  - ❑ Center for stereognosis
  - ❑ Sensory memory
  - ❑ Orientation of position of each part of the body relative to each other and relative to surroundings.

Effect oflesion of the association sensory area

1. Autotopagnosis: person loses the ability to recognize one part or one side
2. Astereognosis: inability to identify common objects
3. Impaired memory

Syringomyelia

- Syringomyelia is spinal cord disorder characterized by the presence of **fluid-filled cavities** in the spinal cord.
- Gray matter around the central canal is the most affected part. So the sensory disturbances are more pronounced than the motor disturbances.

Cause

- Stringomyelia occurs due to the over growth of neuroglial cells in spinal cord accompanied by cavity formation and accumulation of fluid. Initially, a cavity appears in gray matter near the central canal of spinal cord. In later stages, the cavity extends and involves the surrounding white matter to a variable degree. The
- disease usually starts in one or two segments. Then, it extends up and down for considerable distances. **Lower cervical and upper thoracic regions are affected the most.**
- Characteristic features of this disease are the loss of pain and temperature sensations and muscular weakness. Severity of the loss of sensations depends upon the extent of disease in spinal cord. Fine touch sensation is not affected because the fibers of fine touch pathway are in the posterior white column.

Tabes dorsalis

Tabes dorsalis is another disease of the spinal cord. It is a slowly progressive nervous disorder affecting both the motor and sensory functions of spinal cord.

Cause

- It occurs due to the **degeneration of posterior (sensory) nerve roots. It usually occurs in syphilis. Posterior nerve roots are affected proximal to the posterior root ganglia**. As the fibers of posterior root, the fibers of lateral division are affected much. Along with lateral fibers of posterior root, the fibers in posterior white column of spinal cord are also affected

Features

- In tabes dorsalis, both sensory and motor functions are affected. Following are the features:
  1. During the **onset of degenerative changes, there is exaggeration of pain sensation**
  2. Then, there is **impairment and loss of all sensations**

Reflexes

- Both superficial and deep reflexes are lost in tabes dorsalis mostly because of loss of sensations.

Voluntary movements

- There is lack of coordination of movements **(ataxia).** Normal movements like
- walking also become clumsy. The gait is unbalanced. Every movement of the limb is exaggerated while walking. Patient keeps the leg apart, raises the leg very high and stamps it down forcibly. This is called **stamping gait.**

Urinary bladder

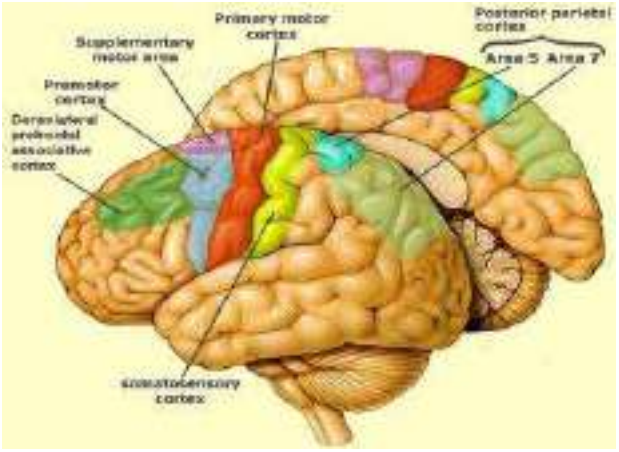
- If sacral segments are affected in tabes dorsalis, the smooth muscles of the urinary bladder become hypotonic. Micturition reflexes are lost. And the urinary bladder becomes **atonic bladder** .

2 The motor cortex

- **Location and composition :**

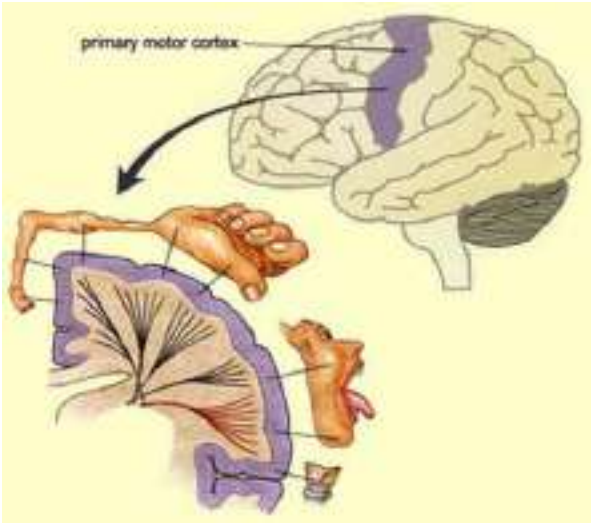
It is located in the frontallobe in front of the central fissureand **Divided into :**

- Primary motor cortex= area 4
- Premotor cortex= area 6
- Supplimentary motor area
- Area 4 suppressor



1) The primary motor area = area 4

- located at the precentral gyrus
- controls muscle activity in the opposite side. = crossed representation
- The body is represented in an inverted manner= upside down
- and the area of representation of each part is proportionate to its motor function not to its size. so the areas-. that control the movements or the hands and fingers are much larger than the area that controls the trunk muscles



- It contains the large pyramidal Betz cells
- Upper half of the face, respiratory and abdominal muscles are bilaterally represented.

Functions of area 4

1. It is the only area that initiates the voluntary fine skilled movement of distal limb specially the hands and fingers.
2. it shares area 6 in initiating the gross movements. Of the proximal limbs and trunk.
3. it facilitates the anterior horn cells= facilitation of stretch reflex
4. It is necessary for occurrence of the superficial reflexes.

Effects of lesions of area 4

- manifestations appear in the **opposite** :

1. flaccid Paralysis or the skeletal muscles, commonly monoplegia on the opposite side.= loss of voluntary muscle contraction
2. permanent loss of the fine and skilled movements (specially of the hands and lingers)
3. loss of superficial and deep reflexes
4. babeneski sign without fanning

NOTE

NB: lesion restricted to area 4 causes flaccid paralysis, hypotonia and hyporeflexia with babeniski sign (dorsiflexion of big toe without fanning)

2) *the premotor cortex area 6*

1. Located anterior to the area 4, less excitable than area 4
2. It works with the help of basal ganglia, thalamus, posterior parietal cortex, cerebellum in planning for motor actions
3. It initiates gross movement of the arms, trunk and hip.
4. Inhibit stretch reflex and muscle tone.
5. It produces subconscious automatic movements. Swinging of the arm during walking.
6. Inhibit grasp reflex
7. It shares in controlling autonomic functions.

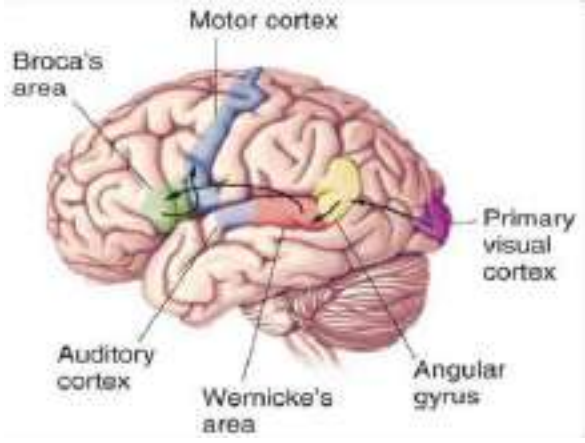
Function of area 6

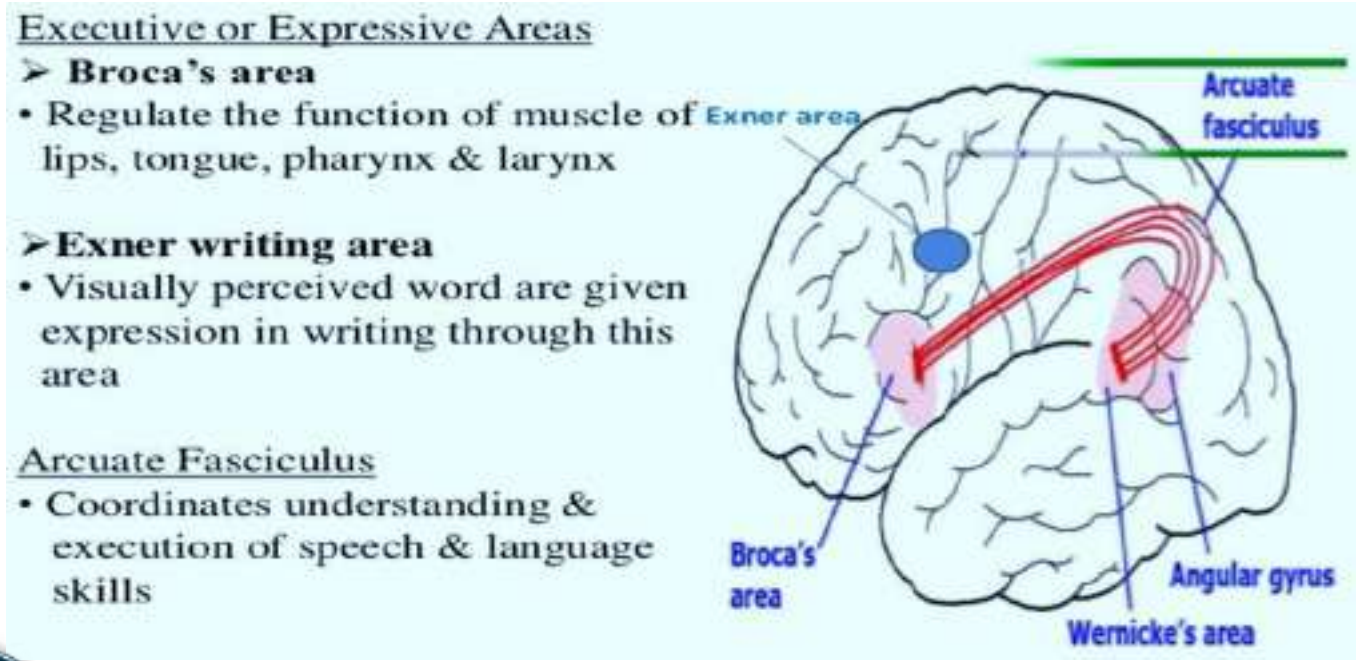
- It contains special areas:

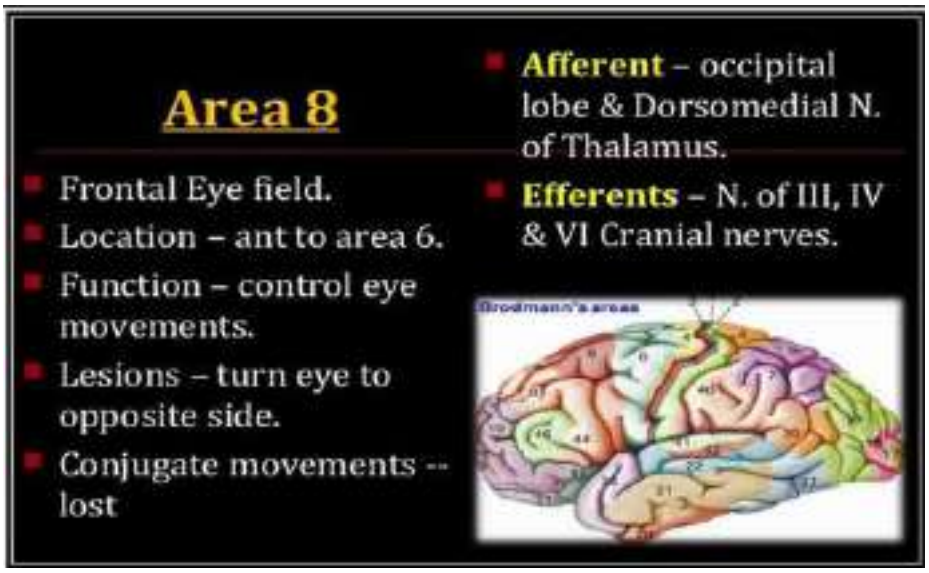
a) Broca's area: word formation area= area 44.

b) Exner's area= memory of hand skills area.

c) Eye movement of area = frontal eye field area=area 8: it causes conjugate deviation of the eye to the opposite side.







Effect of lesion of area 6

1. Contralateral muscle paresis
2. Motor apraxia: inability to do skilled movements in absence of paralysis due to loss of memory.
3. Motor aphasia= inability to speak
4. Muscle spasticity on the affected side.
5. Exaggerated deep reflexes
6. Babeniski sign: fanning of the outer toes.
7. Appearance of grasp reflex
8. Autonomic disturbances

3) Supplementary motor area

- **Site**: medial surface of the frontal lobe, anterior to area 4 and superior to area 6.
- Representation is in the horizontal plane with face anteriorly and leg posteriorly.
- **Function**: it supplements the premotor cortex in:
- planning and programming of some complex movements
- Control of gross postural movements

4) Area 4 suppressor 4S

- Located between area 4 and area 6
- It inhibits area 4 and area 6
- It inhibits anterior horn cells.
- The descending motor tracts include pyramidal and extra pyramidal tracts.

3 the pyramidal system

- Thepyramidal system is Composed of 3 tracts

1.corticospinal tract	2. corticobulbar tract	3. corticonuclear tract

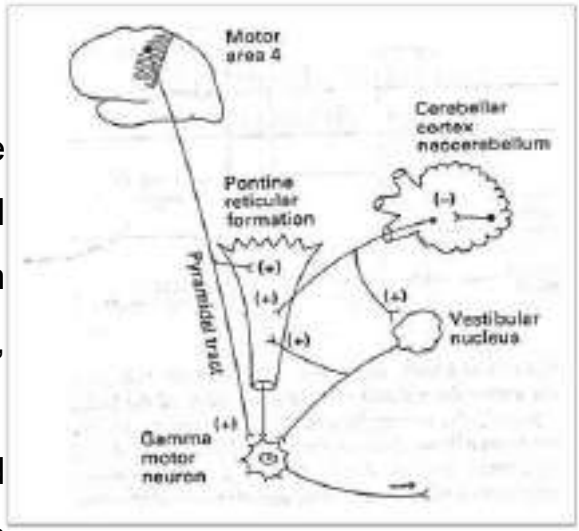
1) corticospinal tract

Origin

1- from area 4: 30 % of corticospinal fibers. They terminate mainly on interneurons.
2- area 6 and supplementary motor areas: 30% of the fibers
3- somatic sensory areas: 40% of the fibers.

course

- Fibers descend in corona radiata then in the internal capsule



1) In the medulla oblongata: 90% cross to the opposite side and called the lateral corticospinal tract. These fibers terminate on motor interneurons then to anterior horn cells, some fibers terminate directly on AHC.

2) 8% of the fibers don't cross in the medulla and descend as ventral corticospinal tract they cross in the spinal cord

3) uncrossed corticospinal tract: 2% terminate on ipsilateral neurons.

Function of the corticospinal tract

1) the crossed lateral corticospinal tract:

a) initiate the voluntary descrete skilled fine movement of fingers and hand

b) facilitate motor neurons and stretch reflex

c) necessary for superficial reflexes

2) the uncrossed tract:

a) supply the respiratory and abdominal muscles which are bilaterally represented.

b) allow for partial recovery of voluntary movement in hemiplegia.

2) *Corticobulbar tract*

- terminate on cranial nerve nuclei of 5, 7, 9, 10, 11, and 12 .
- they supply the head muscles, tongue, pharynix,

3) *Corticonuclear tract*

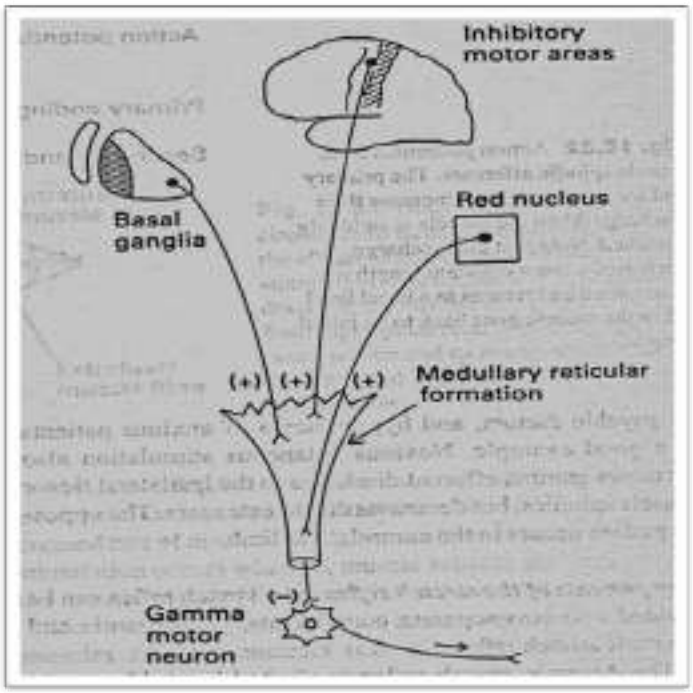
- Supply cranial nerve nuclei of 3, 4, 6 of the opposite side.

4 The extrapyramidal system

- Include descending tracts other than the pyramidal
- Originate mainly from area 6 then relay in the basal ganglia and brainstem
- Include :
  1. rubrospinal tract
  2. reticulospinal tract
  3. Vestibulospinal tract
  4. Tectospinal tract
  5. Olivospinal tract

Function of the extrapyramidal system

1. control gross movement
2. produce subconscious movement
3. mediate movement toward visual and auditory stimuli
4. Weak alternative route for fine movement (ruprospinal tract) .
5. adjust muscle tone



5 The cerebellum

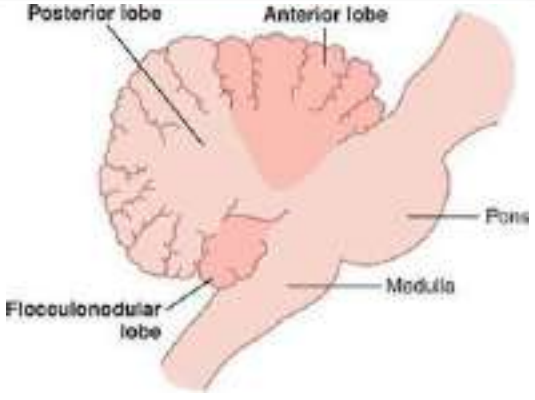
- Cerebellum is vital to the control of rapid muscular activities as:
  - Writing
  - Running
  - Typing
  - Talking
  - Playing piano

Functional anatomy:

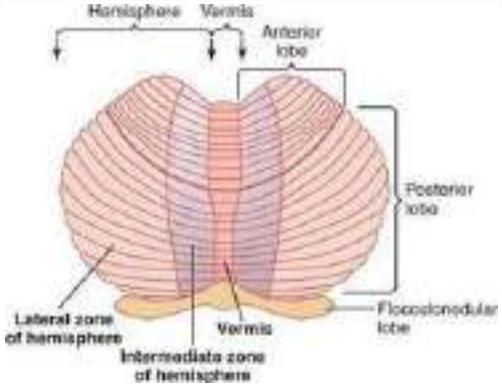
- Cerebellum comprises the following:
  - Vermis (midline structure)
  - Two cerebellar hemispheres
  - Flocculonodular lobe

- From the functional point of view, cerebellum is divided into 3 parts:

part	Connected To	Concerned with
• Archicerebellum = Vestibulo-cerebellum (Flocculonodular lobe)	• Vestibular nuclei	• Equilibrium
• Paleocerebellum = Spino-cerebellum (Intermediate zone of hemisphere)	• AHCs + • Cerebral cortex	• Responsible for coordination of movements • It is inhibitory to the muscle tone
• Neocerebellum = Cerebro-cerebellum (lateral zones of hemisphere)	• Cerebral cortex	• Planning & programming of movements • It is facilitatory of the muscle tone



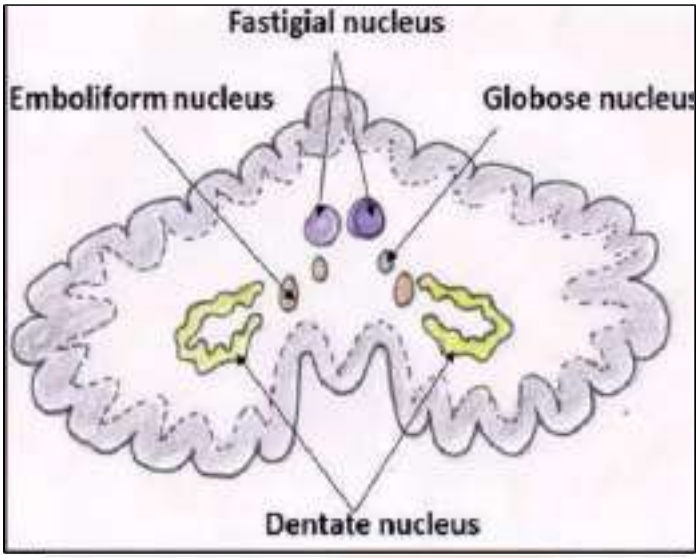
Anatomical lobes of the cerebellum
(as seen from the lateral side)



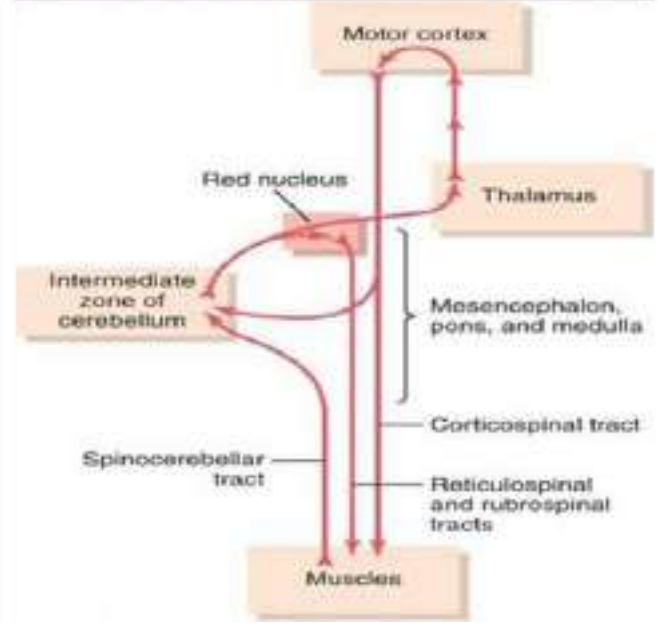
Functional parts of the cerebellum
(as seen from the postero inferior view)
with the inferior most portion of the cerebellum rolled outward to flatten the surface

Connections of cerebellum

- The cerebellum has:
  - External layer of gray matter (cerebellar cortex)
  - Inner white matter: Embedded in it 3 deep nuclei:
    - √ Fastigeal (medially)
    - √ Interpositus (Globose & emboliform)
    - √ Dentate (laterally)



- Input pathways to cerebellum (afferent)
  - **Afferents**: It receives from:
    - √ Cerebral cortex
    - √ Vestibular apparatus
    - √ Spinal cord
- Output pathways (efferent)
  - Efferents: It sends to:
    - √ Cerebral cortex
    - √ Basal ganglia (BG)



Cerebralandcerebellarcontrol ofvoluntary movements,involvingespeciallytheintermediate zoneof thecerebellum

Functions of cerebellum:

- Control of posture & equilibrium:

1. During rapid movement:

The vestibular apparatus send signals to the **archicerebellum** (flocculnodular lobe) → output signals → brain stem → reticulospinal & vestibulospinal tracts → maintain equilibrium through changes in tone of muscles

2. Effect on muscle tone (stretch reflex):

- Neocerebellum :
  Is facilitatory to stretch reflex (through Facilitatory reticular formation) → increase muscle tone
- Paleocerebellum: Is inhibitory to stretch reflex → decrease muscle tone

3. Function of cerebellum in control of voluntary movements:

•**Servocomprator** : The Paleocerebellum (spinocerebellum):
Isinformed about:

- √ The intended plan of motor cortex
- √ The performance of movement (via spinocerebellar tracts)
- √ Compares performance of muscles with motor cortex intention
- √ If not appropriate it sends corrective signals

NOTE

NB: Thespinocerebellum:

- **At the onset of movements :**

Itturns onthe agonist muscles & turns off the antagonist muscles

- **At the end of movements :**

It turns off the agonist muscles & turn on the antagonist muscle

- The point at which reversal of excitation between agonist & antagonist depends on the rate of movement

- **The braking effect of the cerebellum (Prevent overshoot) (Damping movement)**

√ All movements of our body are pendular & they have a momentum & tend to
√ overshoot The cerebellum assesses the rate of movement calculates the length of time needed to reach the intended point and then transmits inhibitory impulses to the motor cortex to inhibit the agonist and excite the antagonist muscles thus brakes are applied to stop the movements at the precise intended point. If the cerebellum is damaged: There is overshooting
√ (oscillatory movement around the intended point) Resulting in kinetic tremors

- **Planning and timing function of the cerebellum:**

√ The lateral zones of cerebellar hemisphere plan for rapid movements
√ The ability to progress smoothly from one movement to the next depends on cerebellum that plans for the next movement at the same time the present movement is occurring
√ The cerebellum also provides appropriate timing for each movement
√ The neocerebellum is important for the smooth progression of movement to the next movement especially rapid movements eg running

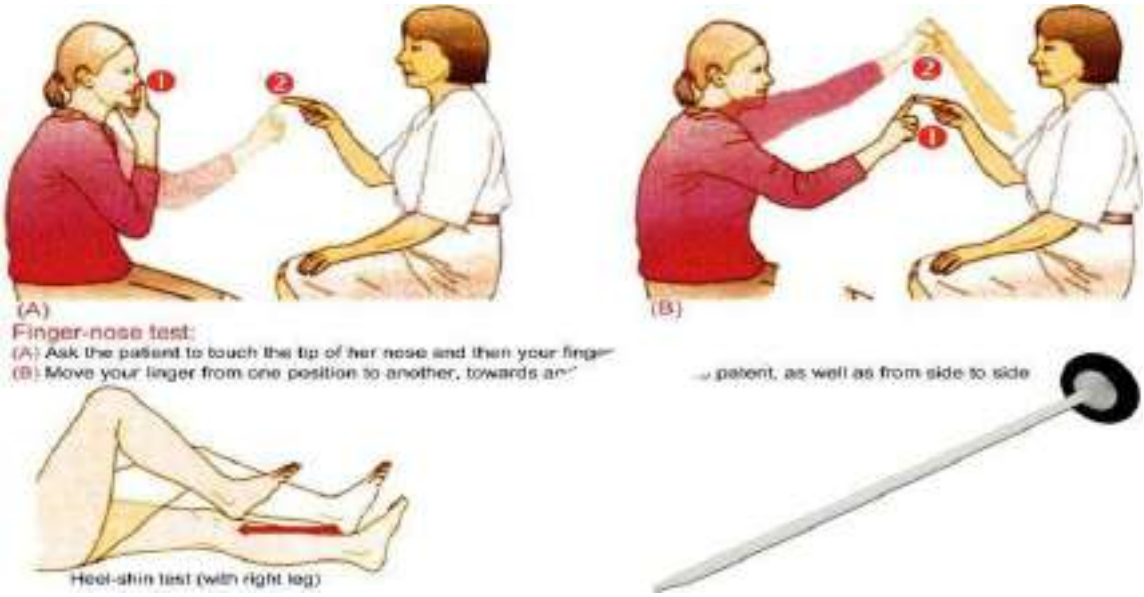
Clinical abnormalities of Cerebellar lesion

- Lesion in deep cerebellar nuclei → neocerebellar syndrome
- **Manifestations:**
  - Ipsilateral (ie they appear on the same side of the lesion):
  - Hypotonia: Due to loss of facilitatory effect of neo cerebellum on muscle tone
  - Asthenia: The patient feels muscle weakness (as there is difficulty in maintaining musclecontraction)
  - Cerebellar ataxia:

√ **Definition**: Incoordinationofvoluntary movements in absence of motor lesion(ie UMNLorLMNL)

√ **Manifestations:**

1.Dysmetria:Movementsovershoottheir intended point
2. Decomposition of movements
3.Intentionkinetictremors:Dueto:Absence of damping function of cerebellum
NB:1,2,3areshownby**fingerto nose test**
4. Dysdiadokinesia:Inabilitytodorapid alternating opposite movements eg repeated pronation & supination
5. Dysarthria:StaccatospeechDifficulty in producing clear correct speech
Dueto: Failureofsmoothprogression of movements in muscles of larynx,mouth&respiratorymuscles
6. CerebellarnystagmusieEyeballtremors
7. Reboundphenomenon:Inabilitytostop the motor act in the proper time
Dueto:Absenceofcerebellarbrake
8. Staggeringgait:Thepatientwalkson a wide base swinging from side to side & mayfallonthediseasedside(drunken gait)



Ataxia

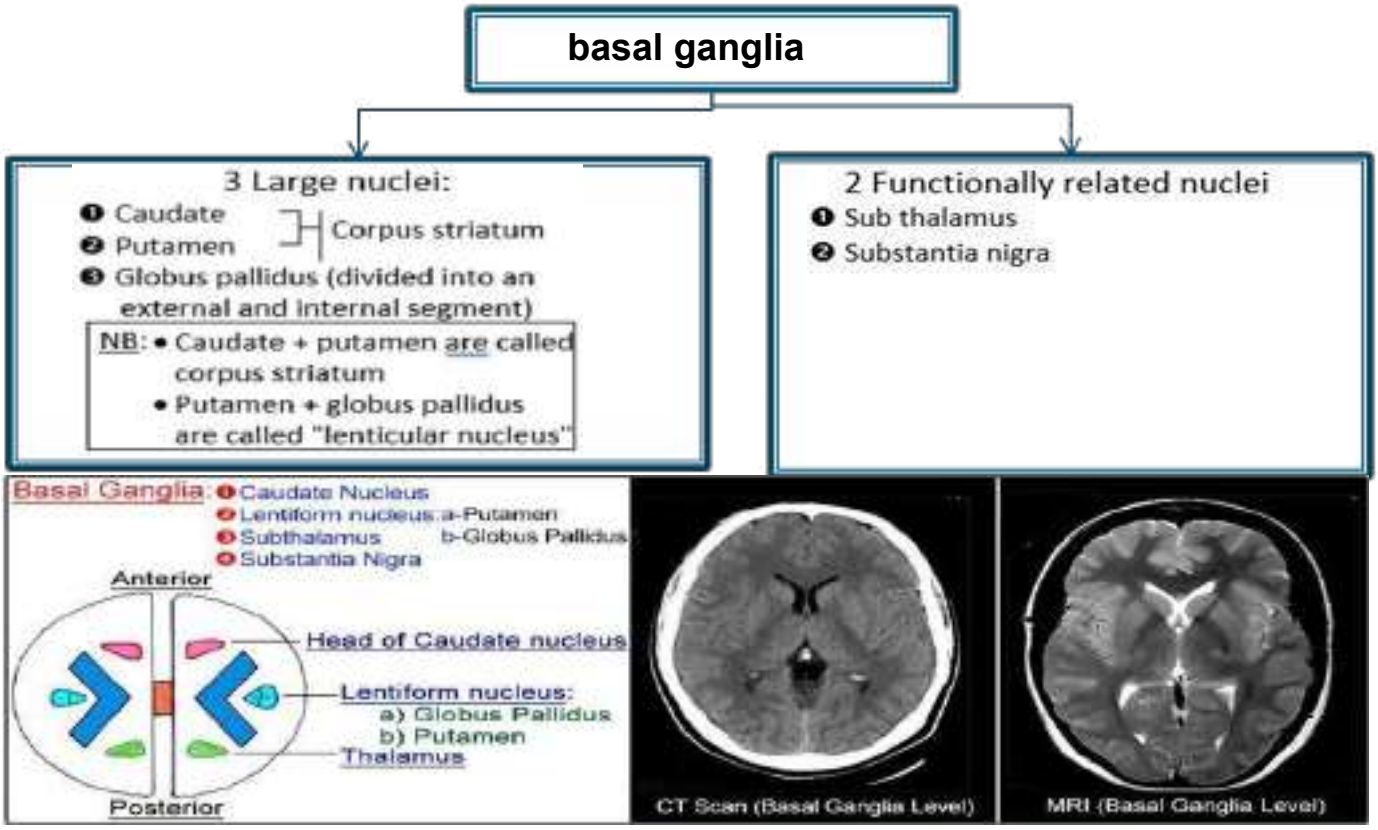
- **Definition:** Incoordination of voluntary movements in absence of motor lesion (UMNL or MNL)
- **Types:**
  a. Sensory ataxia: Due to lesion in proprioceptive pathway eg Tabes dorsalis, pernicious anemia
  b. Motor ataxia: Cerebellar ataxia: Due to lesion in neocerebellum
  c. Mixed ataxia: Thalamic syndrome (sensory & motor)

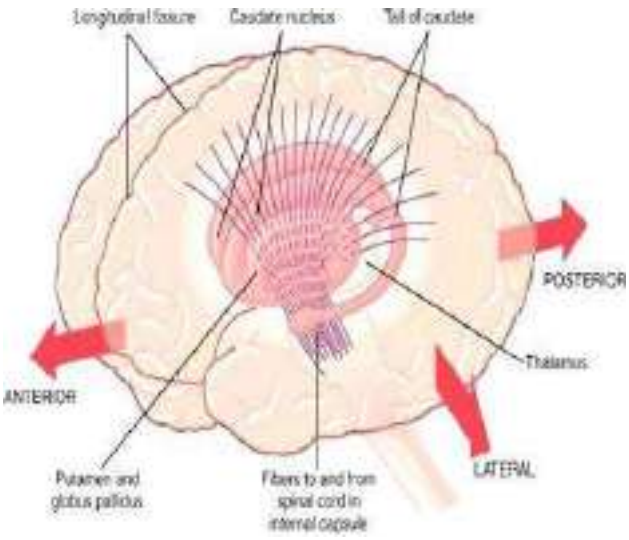
- **Organization of motor cortex, cerebellum & basal ganglia:**
  - Planning:
    a. The idea of voluntary movement: Originate in cortical association area
    b. Basal ganglia: Puts the plan for the slow gross movements
    c. Lateral portions of cerebellar hemisphere puts the plan for the rapid movements
    d. Plans are sent to motor & premotor
  - Execution:
    a. The motor cortex sends the final motor commands through pyramidal tracts
    b. The intermediate cerebellum (spinocerebellar) monitors the performance of movements & corrects any deviation from the plan

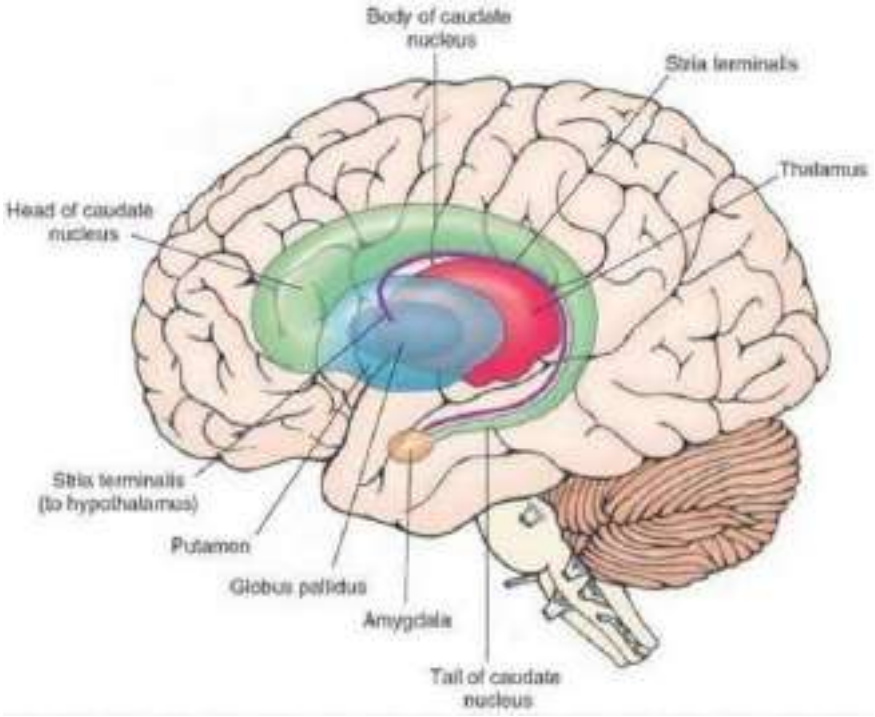
CHAPTER 5
BASAL GANGLIA

1 Basal ganglia

- **Definition:** Inter connected nuclei found Deep to the cerebral cortex and Lateral tothalamus
- **They include**: 5 structures on each side of the brain:







Anatomical relations: of the basal ganglia to thecerebral cortex & thalamus (shownin three-dimensional view)

Nervous connection of basal ganglia:

- Cortical connections:
  - √ Caudate Circuit: Concerned with planning the sequence of pattern of movement
  - √ Putamen Circuit: Execution patterns of movement
- Efferent pathway from the basal ganglia:
  - √ There are interconnection between different nuclei of basal ganglia
  - √ Outflow tracts from BG to brain stem
  - √ Globus pallidus:
    - Is the chief efferent pathway of basal ganglia.
    - It sends impulses to the sub thalamic nuclei and substantia nigra → reticular formation in brain stem → motor neurons of spinal cord

Neurotransmitters in basal ganglia:

- Excitatory Neurotransmitters: 1- Acetyl choline 2-Noradrenaline
- Inhibitory Neurotransmitters: 1-Dopamine 2-GABA

Functions of BG:

- Function of BG depends on balance between excitatory & inhibitory neurotransmitters:
  1) Inhibit muscle tone: Diffuse stimulation of BG reduces muscle tone
  2) Control voluntary movements: BG & motor cortex control complex pattern of movements
     - √ Caudate Circuit: Important for
       - Planning sequences of patterns of movements ie caudate circuit
       - Converts thought into plans and pattern

- Modifying the timing of these patterns of movement occurs slowly of rapidly
- Modifying their spatial dimensions eg one can write very small or very large letter
- NB: Damage of caudate: Inability to write or draw figures with a fixed scale

√ Putamen circuit:

- Helps the cortex to execute subconscious learned pattern of movement: they store programs of familiar automatic pattern of voluntary activity e.g writing the letters of alphabet, cutting with scissors or hammering nails.
- Damage of putamen: leads to motor apraxia i.e. inability to do familiar movements in absence of muscle paralysis e.g. one's writing becomes crude.

√ Globus pallidus: May be responsible for:

- Posture taken by the body to perform a particular voluntary movements
- Thus facilitate the fine movements of the hand

3) BG: Initiation and regulation of the gross intentional movements of the body eg swinging of the arms while walking and facial expressions

- NB: In birds & fish: BG perform all motor functions & Voluntary movements

Metaboliccharacteristicsof BG:

- BG has high oxygen consumption.
- BG (especially the substantia nigra) has high cupper content.

Lesions of BG:

- All are characterized by the presence of involuntary movements:

	Athetosis	Parkinsonism	Chorea
• Site of lesion	• Globus pallidus		• Caudate & putamen
• Involuntary movements	• Continuous slow snake like movements • Hypertonia		• Rapid involuntary dancing jerky movements • Hypotonia

Parkinsonism (Paralysis agitans)

- Itiscaused by lesion in substantia nigra which leads to loss of dopaminergic neurons leading to loss of dopaminergic inhibition of putamen

Causes:

1. Idiopathic: It occurs in elderly
2. Cerebral atherosclerosis (old age): Loss of dopamine receptors
3. Phenothiazine (group of tranquilizers): Block D2 dopamine receptors
4. Repeated head trauma

- **Manifestationsofparkinson'sdisease :**

√ **Akinesia:** Difficulty in initiating voluntary movements

- Decrease in associative movements
- **Characters:** Mask face (expressionless)
- Slow monotonous speech
- Gait: Shuffling gait (ie patient walks in short steps)
- Decrease in associative movements ie walking without swinging his arms
- NB: The patient is usually bent forward as rigidity is more in flexors muscle

√ **Rigidity:** (ie resistance all through limb bending)

- Lead pipe or cog wheal rigidity
- It occurs in both antigravity & prograviy muscles but more in flexors, so, the patient develops the generalized flexion attitude
- Mechanism: Due to excess impulses transmitted along the corticospinal tract to both alpha AHCs and gamma ACHs

√ **Hyperkinesia:** Involuntary Static tremor:

- Involuntary
- Static (present at rest)
- Disappears: During voluntary movement & sleep
- Rhythmic
- Rapid (At a rate of 4-8 cycles/sec)
- Affecting distal joints
- Alternating contractions of the antagonistic muscles resulting in:

- Pill rolling movements of hands
- Up & down movement of mandible

Treatment:

√ Anticholinergic drugs: To reduce Acetylcholine activity

- L-dopa (Levodopa): Is a Dopamine precursor that cross the BBB (Blood Brain Barrier)
  We cant give Dopamine itself as it does not cross the BBB

√ Implantation of dopamine-secreting tissue in near the BG

Chapter 6
Limbic system and hypothalamus

Physiology of memory and learning

- The understanding of brain function in humans has been increased by the development and availability of **positron emission tomographic (PET), functional magnetic resonance imaging (fMRI), computed tomography (CT) scanning,** and other imaging and diagnostic techniques.
- PET imaging can measure local glucose metabolism, blood flow, and oxygen.
- fMRI can measure local amounts of oxygenated blood.
- PET and fMRI provide an index of the level of the activity in various parts of the brain in healthy humans and in those with pathologies or brain injuries.
- They are used to study not only simple responses but also complex aspects of learning, memory, and perception.

Learning

Itis acquisition of the information that give humans their ability to alter behavior based on experience.

Memory

Itis the retention and storage of that information.

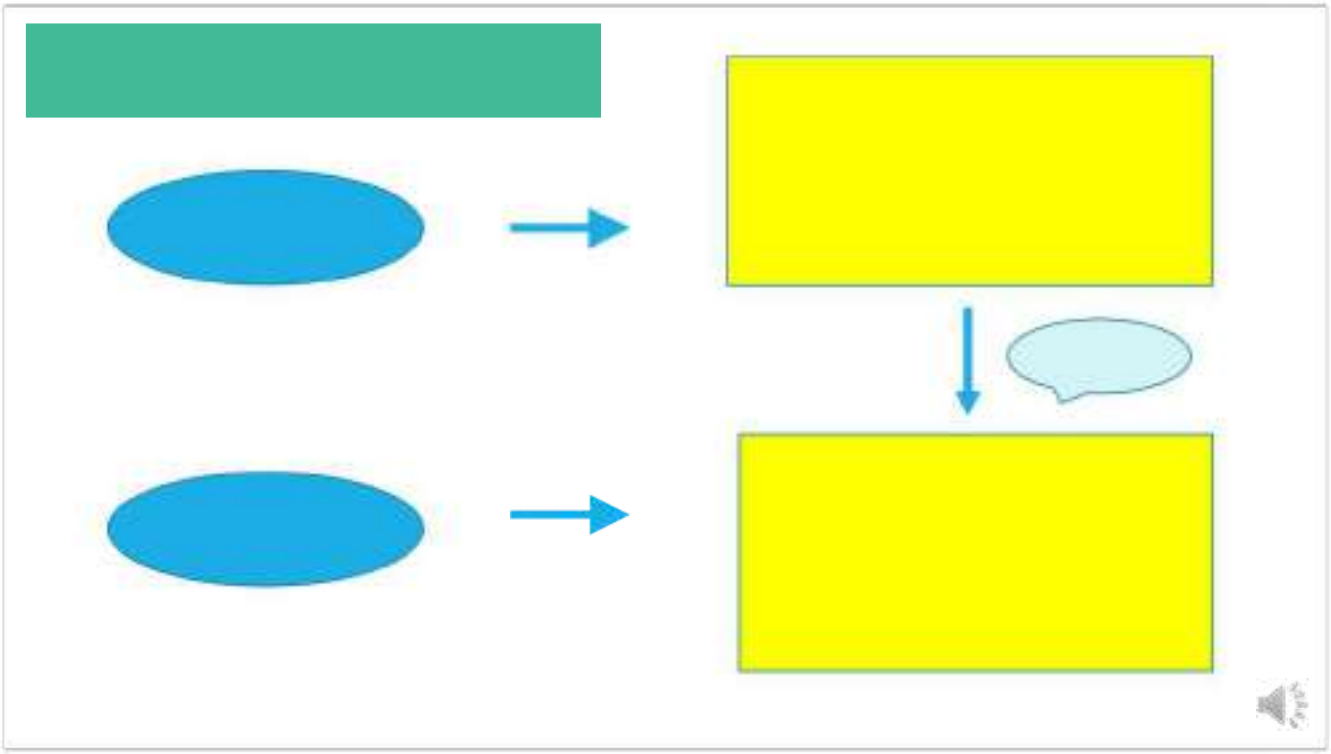
The two are obviouslyclosely related and are considered together.

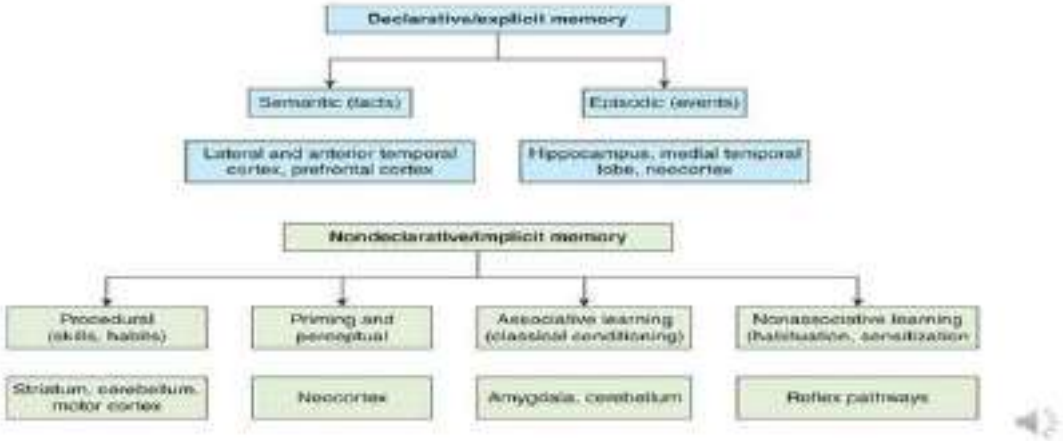
Forms of memory

1- **working memory:** keeps incoming information available for a short time while deciding what to do with it. It is that form of memory which permits us, for example, to look up a telephone number, and then remember the number while we pick up the telephone and dial the number.

2- **short-term memory:** which lasts seconds to hours, during which processing in the hippocampus and elsewhere lays down long-term changes in synaptic strength.

3- **long-term memory:** which stores memories for years and sometimes for life. During short-term memory, the memory traces are subject to disruption by trauma and various drugs, whereas long-term memory traces are remarkably resistant to disruption.



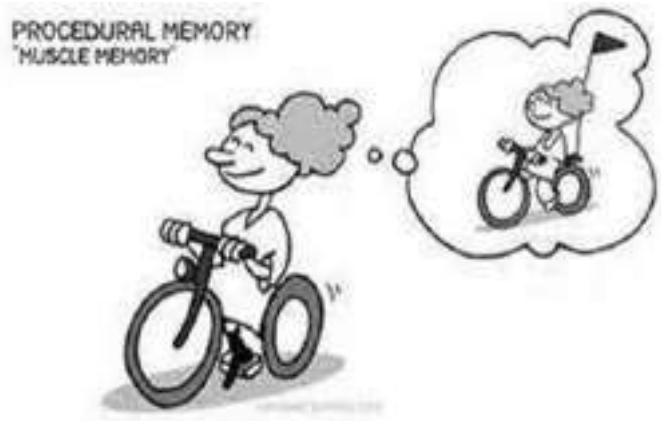


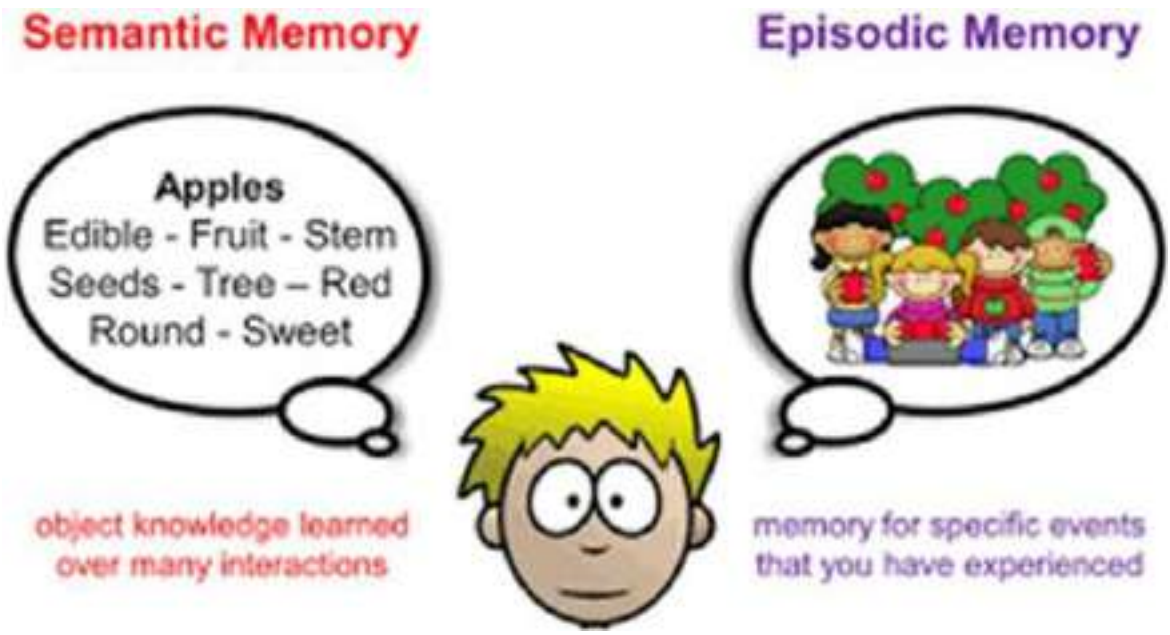
Explicit memory is for facts or knowledge about people, places, and things.

Explicit memories that are initially required for activities such as riding a bicycle can become implicit once the task is thoroughly learned

semantic memory for facts (eg, words, rules, and language) and

episodic memory for events.





Implicit memory

Implicit memory is important for training reflexive motor or perceptual skills and is subdivided into **four** types.

Priming is the facilitation of the recognition of words or objects by prior exposure to them and is dependent on the neocortex. An example of priming is the improved recall of a word when presented with the first few letters of it.

Procedural memory includes skills and habits, which, once acquired, become unconscious and automatic. This type of memory is processed in the striatum.

Associative learning relates to classical and operant conditioning in which one learns about the relationship between one stimulus and another. This type of memory is dependent on the amygdala for its emotional responses and the cerebellum for the motor responses.

Non-associative learning includes habituation and sensitization and is dependent on various reflex pathways.

Centers for memory

- Declarative memory involves the hippocampus and the medial temporal lobe for retention.
- Priming is dependent on the neocortex.
- Procedural memory is processed in the striatum.
- Associative learning is dependent on the amygdala for its emotional responses and the cerebellum for the motor responses.
- Non-associativelearning is dependent on various reflex pathways.

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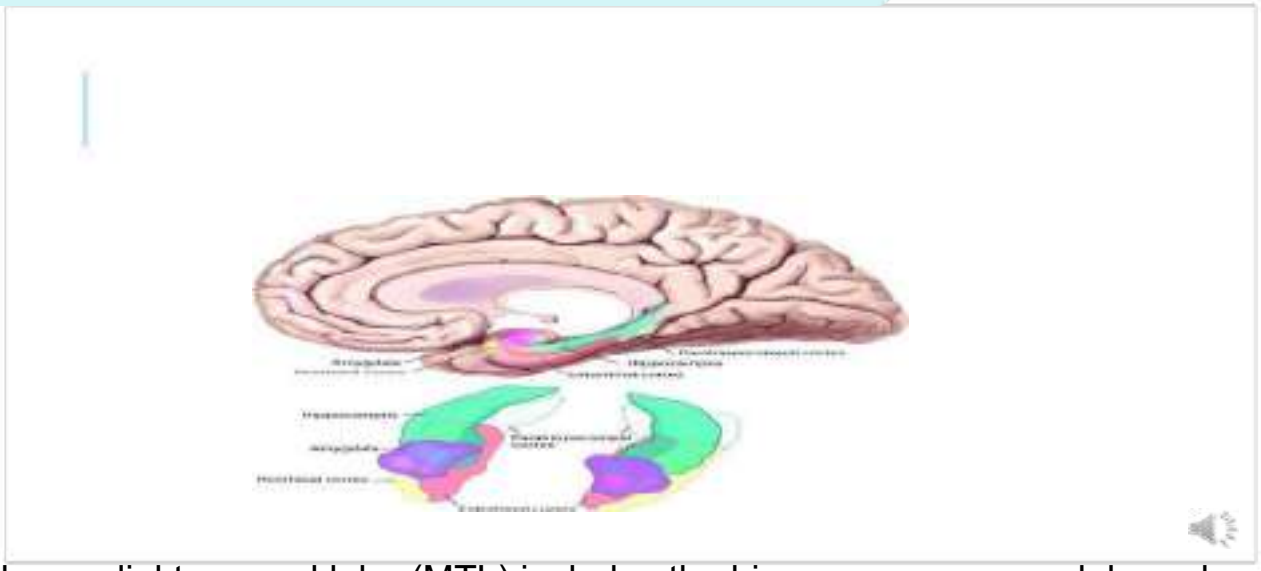
CONDITIONED REFLEX

ASSOCIATIVE LEARNING: CONDITIONED REFLEXES

Aconditioned reflex is a reflex response to a stimulus that previously elicited little or noresponse, acquired by repeatedly pairing the stimulus with another stimulus that normally produce the response.

In***Pavlov's classic experiments***, the salivation normally induced by placing meat inthe mouth of a dog was studied. A bell was rung just before the meat was placed inthe dog's mouth, and this was repeated several times until the animal would salivate when the bell was rung even though no meat was placed in its mouth. Inthis experiment, the meat placed in the mouth was the unconditioned stimulus (US), the stimulus that normally produces an innate response. The conditioned stimulus (CS) was the bell ringing. After the CS and US had been paired a sufficient number of times, the CS produced the response originally evoked only by the US**.**

HIPPOCAMPUS & MEDIAL TEMPORAL LOBE (MTL)



- The medial temporal lobe (MTL) includes the hippocampus, amygdala and para-hippocampal regions, and is crucial for episodic and spatial memory.
- MTL memory functionconsists of distinct processes such as **encoding, consolidation and retrieval**.

- **Encoding** is the process by which perceived information is transformed into a ***memory trace***. After encoding, are stabilized by **consolidation**.
- **Memory retrieval (recall)** refers to the process by which memory tracesare reactivated to access information previously encoded and stored in thebrain.

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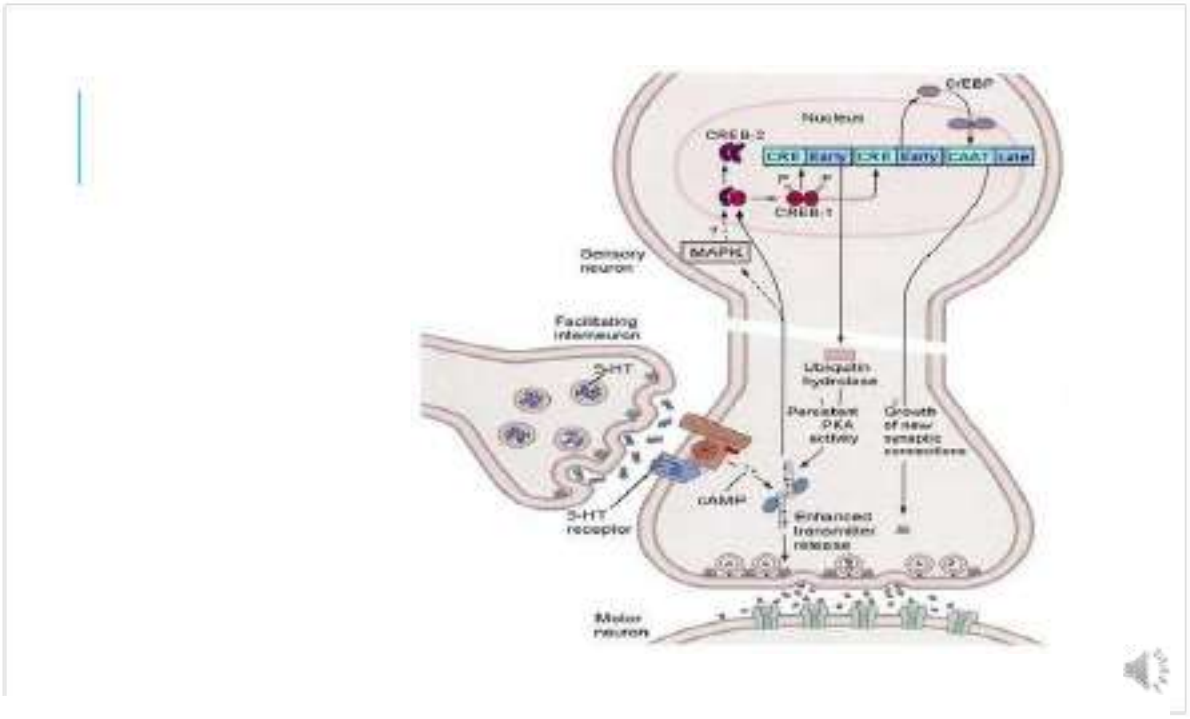
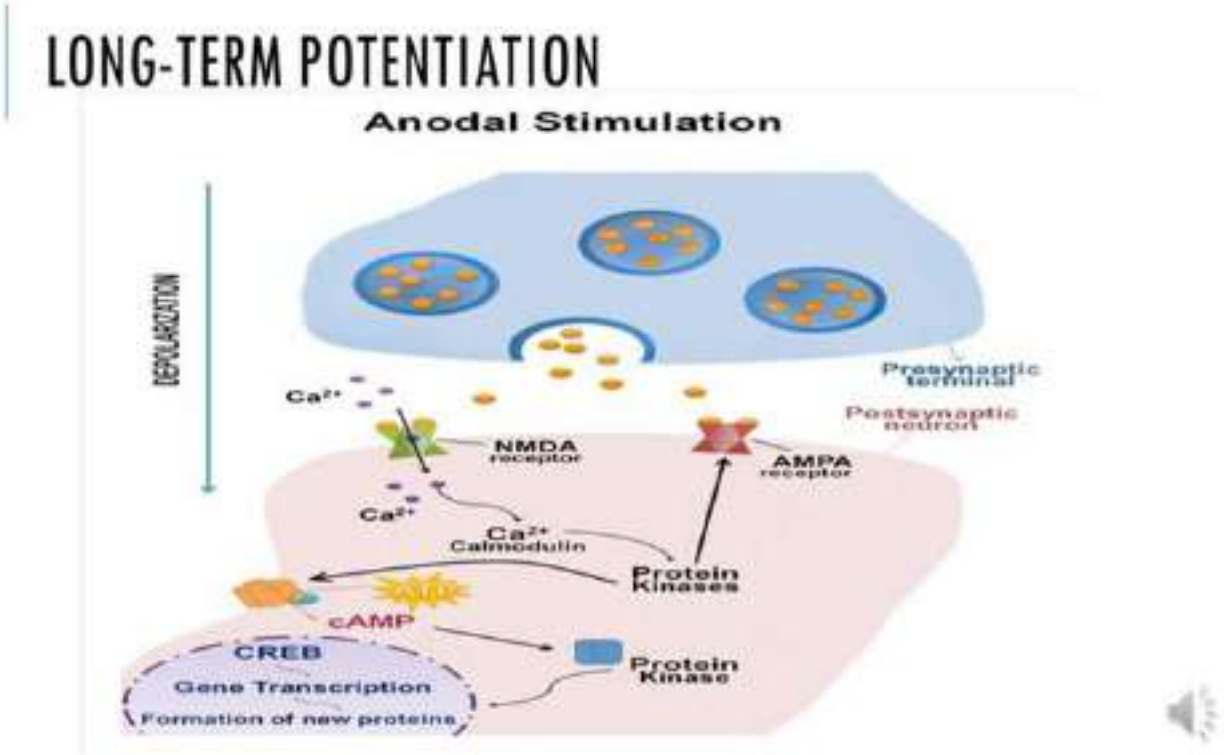
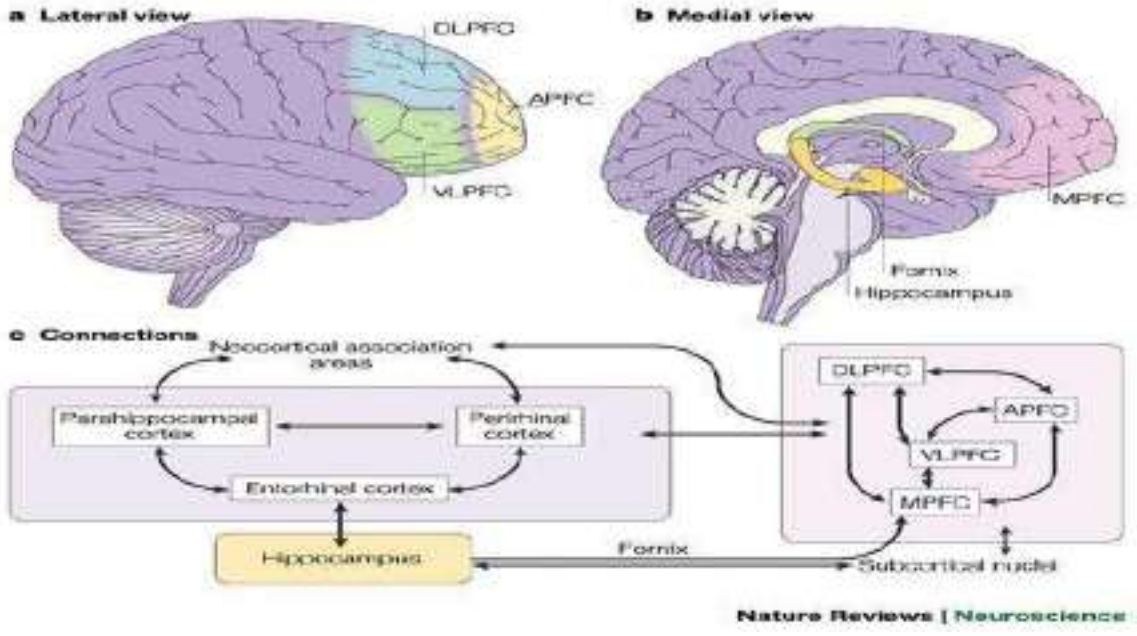




Consolidation of long-term memory andformation of memory engrams

- Once memory is encoded, consolidation occurs to maintain and strengthen the memory.
- Synaptic plasticity mechanism **(long-term potentiation)** is involved in long-term memory consolidation, with structural changes in the presynaptic terminals (increased numbers of synaptic terminals, stored and released vesicles, increased enzymes and protein synthesis.
- Memory is redistributed from the MTL to the neocortex during systems consolidation which occurs ***during sleep and quiet waking***.
- Presentation of a ***sensory cue*** which was present during memory encoding would facilitate memory consolidation.
- The same cue is introduced to help recall or retrieval of memory engrams.



Alzheimer disease

- Alzheimer disease is the most common age-related neurodegenerative disorder. Memory decline initially manifests as a loss of episodic memory, which impedes recollection of recent events.
- Loss of short-term memory is followed by general loss of cognitive and other brain functions, agitation, depression, the need for constant care, and, eventually, death. Bilateral destruction of the ventral hippocampus, or
- Alzheimer disease and similar disease processes that destroy its CA1 neurons, can cause striking defects in short-term memory. Severe loss of cholinergic fibers from the basal fore brain occurs in Alzheimer disease.
-

Amnesia

- Loss of memory.
- Types:

  1- retrograde amnesia : loss of memory of recent events (after brain concussion).

  2- anterograde amnesia: inability to form new long-term memory.

  3- hysterical amnesia: with psychological stress.

  4- global amnesia: loss of retrograde and anterograde, excessive brain lesion.

Ascending Reticular Activating System (ARAS) or (RAS) and sleep

- RAS is the ascending branches of neurons of the facilitatory reticular area
- The fibers of RAS extend upwards to all areas of cerebral cortex either:
    1- Directly
    2- Through non-specific thalamic nuclei (reticula-thalamo-cortical pathway)

Functions of RAS:

1- Responsible for the:
    - Alert
    - Conscious
    - Wakefulness state of person
2- Plays an important role in control of sleep
3- Its depression leads to sleep while its damage leads to coma

Factors affecting activity of RAS:

Factors that increase RAS activity

1- All sensory signals from sensory pathway tracts:
    - Auditory are more effective than visual
    - Pain and proprioceptive signals are more effective

2- Signals from cerebral cortex through Corticofugal fibers to RAS:

eg:

- Motor signals from motor cortex → so performance of voluntary movements help person to resist desire for sleep

- Signals from temporal & frontal lobe in emotions

3- Drugs: Sympathomimetics eg :

1- Adrenaline
2- Noradrenaline
3- Dopamine
4- Amphetamine
5- Aminophylline
6- Caffeine

Factors that decrease RAS activity:

1- Reduction of signals from sensory pathways

2- Reduction of signals from cerebral cortex

3- Stimulation of sleep centers

4- General anesthesia & drugs eg alcohol, morphine, narcotics

5- Extensive damage of RAS eg tumors

Excitatory-activating system of the brain. Also shown is an inhibitory area in the medulla that can inhibit or depress the activating system.

Electro-encephalogram (EEG)

- It is recording of the electrical activity of the brain in conscious subjects through intact skull

Types of normal EEG waves:

	Alpha	Beta	Theta	Delta
1- Frequency	8 – 13 Hz	18 – 30 Hz	4 – 7 Hz	1 – 3 Hz
2- Amplitude	50 µV	20 µV	100 µV	100 µV
3- Age	Adult	Adult	Child	Infant
4- Recorded from	Parito-occipital region	Frontal	Parietal & temporal	----------
5- State of person	Physical & mental res Awake but with closed eyes	Intense activation of the CNS during thinking & tension	In adults during emotional stress	In adults during sleep

Different types of brain waves in the normal electroencephalogram

- Their frequency is inversely proportional to amplitude

Alpha waves (α) Have:

- Low frequency
- High amplitude
- Recorded in adults who are relaxed with eye closed

Beta waves (β) Have:

- High frequency
- Low amplitude
- Recorded in adults with open eyes, at anxiety, alert & sensory stimuli

Sleep

Sleep:

- It is a state of unconsciousness from which a person can be aroused by proper sensory stimuli

Sleep is two types that alternate with each other:

Slow wave sleep (non-rapid eye movement) (non REM):

Which is the first to occur when the person falls sleep

1- It is orhodoxical sleep

2- It is the first to occur at start of sleep

3- It is about 80% of sleep

4- It persists for 90 minutes

5- It is associated with non rapid eye movement

6- Decrease: - Heart rate
   - Arterial blood pressure (ABP)
   - Respiratory rate
   - Basal metabolic rate

7- The muscle tone is reduced

8- No dreams or teeth grinding

9- No swallowing, erection or ejaculation

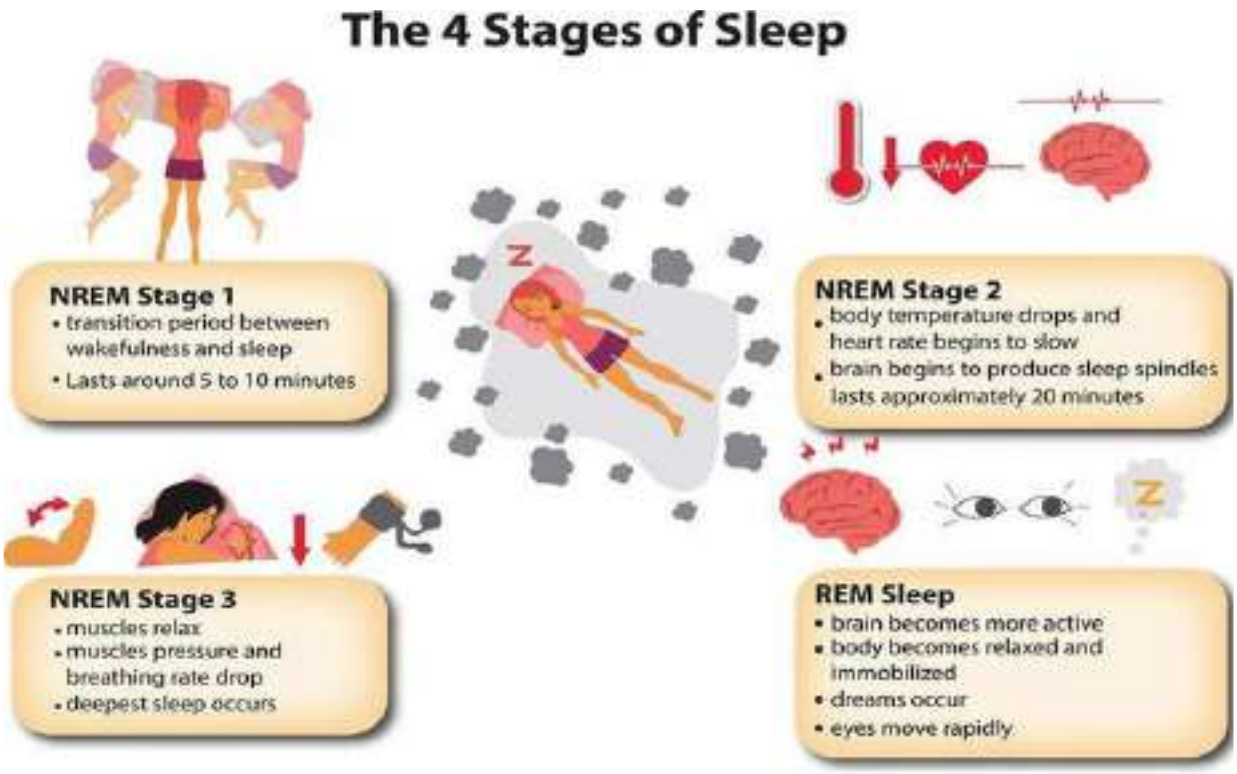
10- EEG show four stages:

Stage I: Waves are characterized by: - Low amplitude
- High frequency

Stage II: The waves are spindle shaped (light sleep)

Stage III: Waves are characterized by:
- High amplitude, Low frequency (moderate sleep)

Stage IV: Very large waves (deep sleep)



The first three stages are known as non-rapid eye movement (NREM) sleep, whereas the fourth stage is known as rapid eye movement (REM) sleep. The brain transitions through four unique stages of sleep several times through the night.

Rapid eye movement (REM) or paradoxical sleep

1- It is paradoxical sleep

2- It occurs after 4th stage of Non REM

3- It is about 20% of sleep

4- It persists for 20 minutes

5- It is associated with rapid eye movement

6- Increase: Heart rate & Arterial blood pressure (ABP) & Respiratory rate

7- The muscle tone is reduced due to activation of inhibitory reticular area

8- The dreams occurs in this stage

9- It is associated with: Teeth grinding, Swallowing, Erection & Ejaculation

10- EEG shows small rapid irregular β waves:

It like an alert person. So, it is termed paradoxical as the waves are like that of alert person while he is difficult to be aroused

	Non- REM	REM
(1) Occurrence	At the start of sleep	After the 4th stage of non REM
(2) Duration of sleep time	80 % of sleep time	20% of sleep time
(3) duration/cycle	About 90 min	About 20 min
(4) Eye movement	Eyes deviate up	Rapid eye movement
(5) ABP HR, RR, MR	Decreased	Increased
(6) Muscle tone	Hypotonia	Marked hypotonia
(7) Dreams	Present but not remembered	Present & remembered
(8) Talking & walking	Present	Absent
10)Threshold of awaking	Low (easily awaked)	High (difficult to awaken)

Mechanism of Sleep:

Sleep is secondary to inhibition of RAS by the following mechanisms:

Passive mechanism (theory)

PassiveinhibitionofRAS by:

1- Fatigueofits neurons after a period of wakefulness

2- Eliminationof its exciting stimuli eg:

-Visual - Auditory

Active mechanism (theory)

- Active inhibition of RAS by signals discharged from subcortical centers
- The chemical transmitters of sleep are:

1- Serotonin
2- Prostaglandin
3- Acetyl choline
4- Noradrenaline

Language and speech

Memory and learning are functions of large parts of the brain, but the centers controlling some of the other “higher functions of the nervous system,” particularly the mechanisms related to language, are more or less localized to the neocortex. Speech and other intellectual functions are especially well developed in humans, the animal species in which the neocortical mantle is most highly developed.

PHYSIOLOGY OF LANGUAGE

Language is one of the fundamental bases of human intelligence and a key part of human culture. The primary brain areas concerned with language are arrayed along and near the sylvian fissure (lateral cerebral sulcus) of the categorical hemisphere. A region at the posterior end of the superior temporal gyrus called the **Wernicke area** is concerned with comprehension of auditory and visual information. It projects via the **arcuate fasciculus** to the **Broca area** in the frontal lobe immediately in front of the inferior end of the motor cortex. **Broca** area processes the information received from Wernicke area into a detailed and coordinated pattern for vocalization and then projects the pattern via a speech articulation area in the insula to the motor cortex. This then initiates the appropriate movements of the lips, tongue, and larynx to produce speech. The probable sequence of events that occurs when a subject names a visual object. The angular gyrus behind the Wernicke area appears to process information from words that are read in such a way that they can be converted into the auditory forms of the words in Wernicke area. In individuals who learn a second language in adulthood, the portion of Broca area concerned with it is adjacent to but separate from the area concerned with the native language. However, in children who learn two languages early in life, only a single area is involved with both. Children acquire fluency in a second language more easily than adults.

Speech

- Itis an extremely complex sensorimotorfunction
- Speech function depend more on **categorical** hemisphere
- The two cerebral hemispheres are:
  - **Categorical hemisphere**

    Dominant hemisphere

    Concerned with categorization & symbolization

    - *Lesion* :
      - Language disorders
      - Depression
    - Site:Itisthelefthemisphere in 96% of right handed persons
  - **Representational hemisphere**

    Non dominant hemisphere

    Concerned with recognition of faces & objects

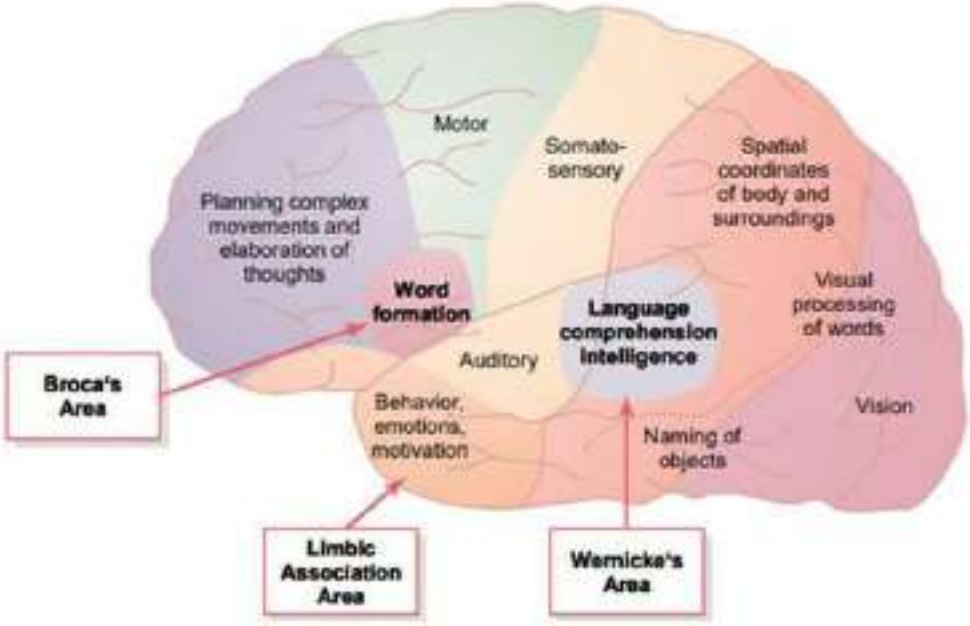
    - Lesion - :
      - No language disorders
      - Euphoria
      - Agnosia: Ignore stimuli from contralateral part ofbody
- **Centers of speech:**
  - **Wernicke's area**
    - Site: Posteriorendof superior temporal gyrus in the **categorical hemisphere**
    - Function: Comprehension of auditory & visual information
  - **Broca's area**
    - Site: Frontal lobe in front of the motor cortex in the **categorical hemisphere**
    - Function: Coordinate vocalization
  - Hand skills area

    - Present in premotor area
    - **Its lesion**: Cause motor apraxia ie incoordination of hand movements
  - **Auditory area**

    - Primary auditory area (41, 42)
    - **Auditory association areas**:
      - It is around the primary auditory area in superior temporal gyrus
      - Concerned with: *understanding the meaning of spoken words*
  - **Visual areas**

    - Primary visual area (17)
    - **Visual association areas (18-19)**:

Concerned with: *understanding the meaning of written words*



Aphasia

- Definition: Abnormalities of language functions that are **not due to**: Visual, hearing or motor defects .
- Cause: They are due to lesions in the categorical hemisphere
- Types of aphasia: Fluent aphasia
  - Lesion: In Wernicke's area
  - Effects:
    - **Speech:**
      - Itself is normal & sometimes the patients talk excessively
      - However, what they say is full of jargon & new irrelevant words (neologism)
      - The patient fails to understand the meaning of spoken or written words
  - **Subtypes of fluent aphasia:**
    1- Auditory aphasia (word deafness):
    - Lesin **auditory association area**
    - Effect : Inability to understand the meaning of spoken words

    2- Visual aphasia (world blindness)
    - Lesion : In **visual association area**
    - Effect : Inability to understand written words

- Non fluent aphasia (Broca's aphasia)
  - Lesion: In Broca's area
  - Mild damage: Speech is slow & words are hard to come
  - Severe damage: Speech is limited to 2 or 3 words
- Global aphasia
  - Lesion: Extensive lesion of categorical hemisphere
  - Effect: - Impaired production & comprehension of both spoken & written language
    - Speech is few & non fluent

Chapter 5
Special senses
vision

INTRODUCTION

The eyes are complex sense organs that have evolved from primitive light sensitive spots on the surface of invertebrates. They gather information about the environment; the brain interprets this information to form an image of what appears within the field of vision. Within its protective casing, each eye has a layer of photoreceptors that respond to light, a lens system that focuses the light on these receptors, and a system of nerves that conducts impulses from the receptors to the brain. A great deal of work has been done on the neurophysiology of vision; in fact, it may be the most studied and best understood sensory system. This chapter describes the way the components of the visual system operate to set up conscious visual images.

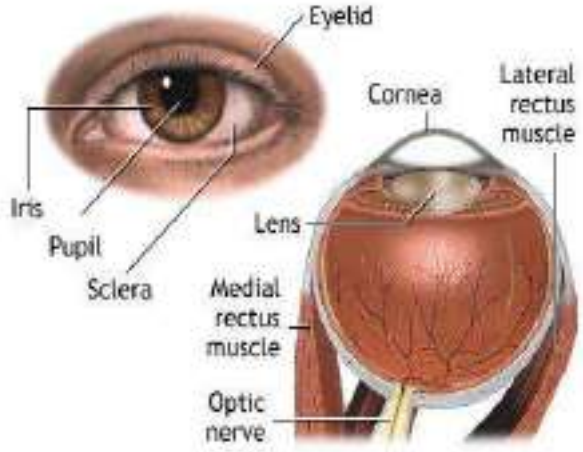
Basic Structure of the Eye and Supporting Structures

The eye has three layers or coats, three compartments and contains three fluids:

1. The three coats of the eye are as follows:

(a) Outer fibrous layer:

- cornea
- sclera
- lamina cribrosa.



(b) Middle vascular layer ("uveal tract"):

- iris
- ciliary body.
- choroids.

(c) Inner nervous layer:

• pigment epithelium of the retina • retinal photoreceptors • retinal neurons.

2. The three compartments of the eye are as follows:

(a) Anterior chamber – the space between the cornea and the iris diaphragm.

(b) Posterior chamber – the triangular space between the iris anteriorly, the lens and zonule posteriorly, and the ciliary body.

(c) Vitreous chamber – the space behind the lens and zonule.

3. The three intraocular fluids are as follows:

(a) Aqueous humour – a watery, optically clear solution of water and electrolytes similar to tissue fluids except that aqueous humour has a low protein content normally.

(b) Vitreous humour – a transparent gel consisting of hyaluronic acid molecules and water. It fills the space between the posterior surface of the lens, ciliary body and retina.

(c) Blood – in addition to its usual functions, blood contributes to the maintenance of intraocular pressure. Most of the blood within the eye is in the choroid. It might be that the choroid serves as a heat exchanger for the retina, which absorbs energy as light strikes the retinal pigment epithelium.

Function of different parts of the Eye

The Cornea

The main function of the cornea is refraction. It is 0.5 -1 mm thick, and about 11 mm in diameter.

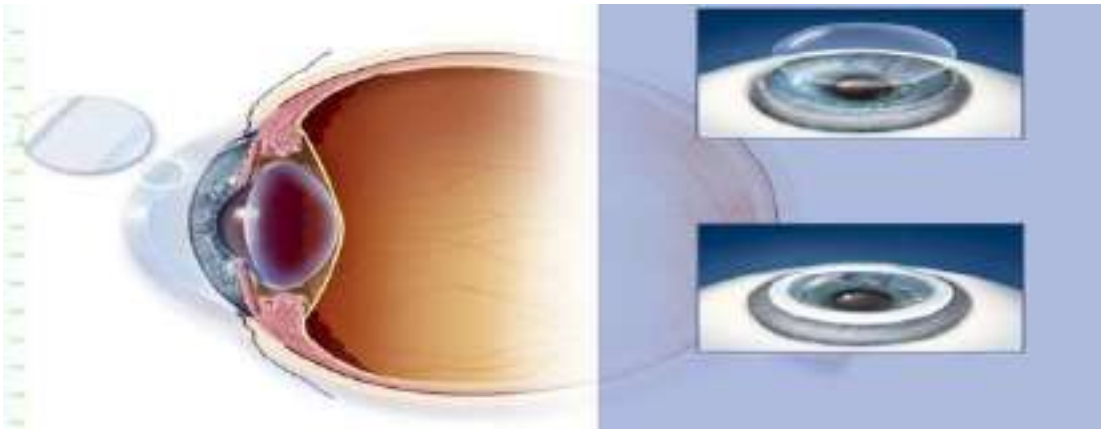
In order to perform this function, the cornea requires the following:

• Transparency: It allows entry of light into the eye.

• Smooth and regular surface: It is the most important refractive medium in the eye, since it acts as a convex lens having a power of 39-43 diopters.

• Spherical curvature of proper refractive power leads to formation of sharp retinal images.

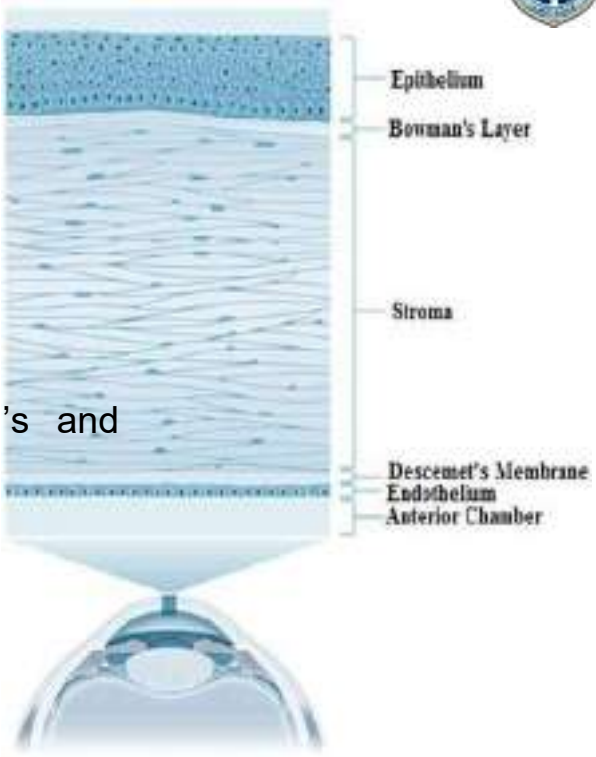
• Appropriate index of refraction.



Corneal transparency is contributed to by anatomical and physiological factors:

1. Anatomical:

- **Absence of keratinization of epithelium.**
- **Tightpacking of epithelial cells.**
- **Mucous layer** providing smoothlubricated surface.



- **Homogeneity of membranes** – Bowman's and Descemet's.
- **Regular arrangement of corneal lamellae** (parallel collagen fibers within each lamella, with adjacent lamellae being perpendicular).
- **Interspaces – absence of blood vessels**: This is the main cause of maintaining corneal transparency.

2. Physiological

• active dehydration of the cornea through Na+/HCO3 -

Metabolic pump

located in the corneal endothelium. This dehydration is supplemented by the physical barrier provided by the corneal epithelium and endothelium.

-Osmotic pump:
The osmolarities of the fluids that surround the cornea (the aqueous humour posteriorly and the tears anteriorly) are relatively greater than that of the cornea. Accordingly, fluids move down an osmotic gradient from the cornea outwards (which contributes to its relative dehydration).

The Tear Film

The cornea is always covered by a thin film of tears (which is secreted by the lacrimal glands) that gives the eye its particular luster (= brightness). and it also contains extremely sensitive non-myelinated nerve endings (which transmit pain as well as other sensations from the cornea).

The tear film consists of three layers: the mucoid, aqueous, and oily layers.

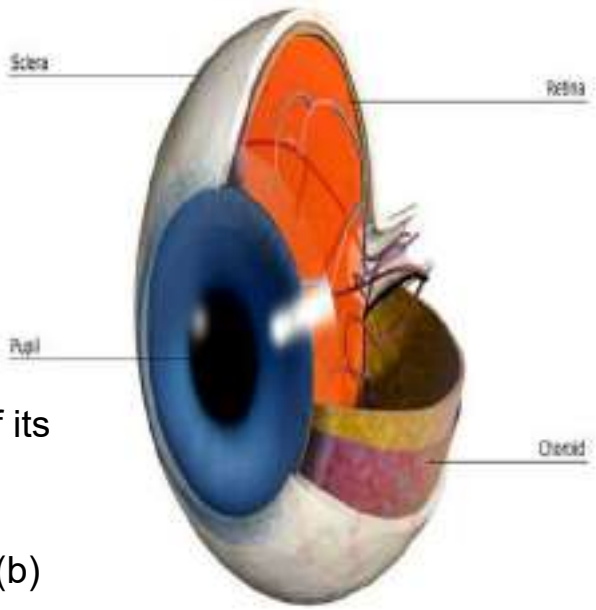
- **The mucoid layer** lies adjacent to the corneal epithelium. It improves the wetting properties of the tears.

The watery (aqueous) layer is produced by the main lacrimal gland in the superior temporal part of the orbit and accessory lacrimal glands found in the conjunctival stroma.

The oily layer (superficial layer of the tear film), This oily layer helps maintain the vertical column of tears between the upper and lower lids and prevents excessive evaporation.

The Sclera

- The sclera constitutes the posterior 5/6 of the outer layer of the eye. It is formed of hard fibrous tissue and normally, it is whitish in adults and bluish in infants and young children. However, its color can be changed in disease (e.g., it becomes yellowish in jaundice).

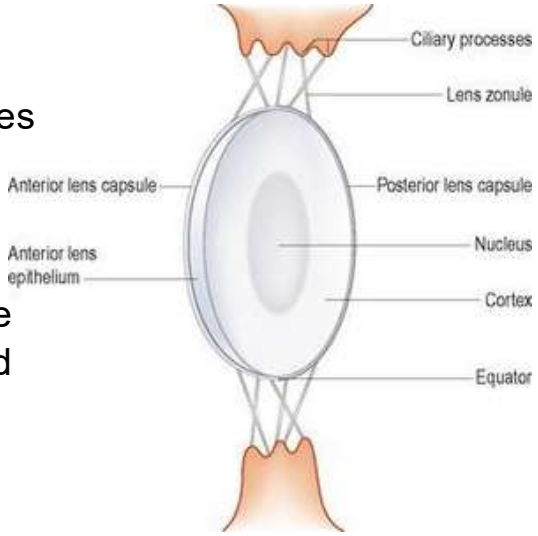


- It is opaque (i.e., it does not transmit light rays) due to the marked irregularity of its connective tissue lamellae. However, it performs the following functions (a) It protects the delicate inner eye structures (b) It gives attachment to the external ocular muscles (which move the eyeball in various directions).

The Lens

■ The lens, like the cornea, is transparent. It is avascular and depends on the aqueous for nourishment. It has a thick elastic capsule, which prevents molecules (e.g., proteins) moving into or out of it. During rest, its posterior surface has a greater curvature than its anterior surface.

■ T the lens is comprised of 65% water and 35% protein. The water content of the lens decreases with age and the lens becomes less pliable.



Causes of transparency of the lens

a- Absence of blood vessels and nerves_ Because of its a vascularity, the lens obtains its oxygen and nutrients from the aqueous humour.

b- The uniform arrangement of the different lens & fibers. c- The nearly equal refractive indices of the various lens constituents.

Functions of the lens

(1) It is an. important refractive media providing about 1/3 of the total refractive power of the eye optical system during rest.

(2) Its elasticity allows it to be more spherical (with a subsequent increase in its power), which is essential for the process of accommodation.

(3) It absorbs (with the cornea) a considerable amount of ultraviolet rays (thus protecting the retina from their harmful effects).

The Ciliary Body

The ciliary muscle (within the ciliary body) is a mass of smooth muscle, which runs circumferentially inside the globe and is attached to the scleral spur anteriorly. It consists of two main parts:

1. Longitudinal (meridional) fibers – form the outer layers and arise from the scleral spur and insert into the choroid. Contraction of this part of the muscle exerts traction on the trabecular meshwork and the choroid and retina.
2. Circular fibers – form the inner part and run circumferentially. Contraction moves the ciliary processing inwards towards the center of the pupil leading to relaxation of the zonules.

Functions ofthe ciliary body :

- This contains the ciliary muscle and the ciliary processes.

(1) The ciliary muscle is essential for accommodation.

(I) The ciliary processes secrete the aqueous humour into the posterior chamber and give attachment to the suspensory ligament of the lens.

The Iris

- This is a circular diaphragm-like opaque structure that lies anterior to the crystalline lens. Its margins rest on the anterior surface of the lens, and in front of it there is the anterior chamber while behind it there is the posterior chamber.

It has a central round aperture (opening) called the pupil, the diameter of which is greatly variable (it may be as small as 1.5 mm or as large as 8 mm). The iris contains the pigment that gives the eye its characteristic color (particularly in its posterior layer). It also *contains thefollowing 2 typesof smooth muscle which control the size of the pupil:*

(I) Circular muscle fibers, which constitute the constrictor (or sphincter) pupillae muscle.

(I) Radial muscle fibers, which constitute the dilator pupillae muscle.

Functions of the Iris:

(1) It regulates the entry of light rays into the eye through the pupil.

(I) It prevents both spherics and chromatic aberrations by cutting out most of the peripheral rays falling on the eye.

(3) It protects the retina (by preventing excessive entry of ultraviolet rays into the eye).

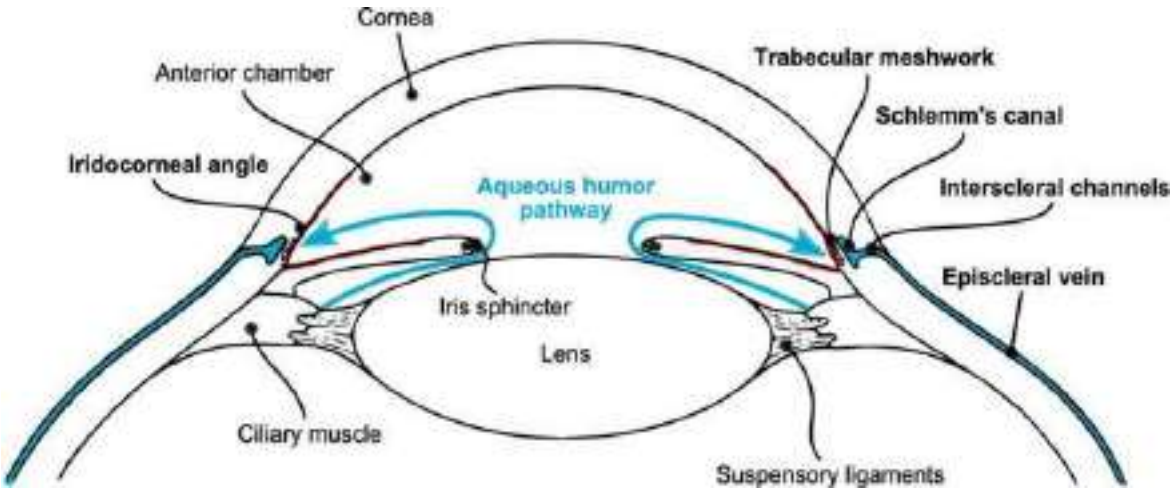
(4) The pupillary reflexes are important in the diagnosis of many nervous lesions and diseases.

The Aqueous -Humour and Intraocular Pressure

- The aqueous humour is an alkaline clear fluid which fills both the anterior and posterior chambers of the eye.
- The aqueous humour is formed at a rate of 2-3 mm3/minute mainly by the epithelium of the ciliary processes of the ciliary body. Na is first actively secreted by the ciliary epithelium, and this is followed by passive diffusion of Cl, HC03- and waterby osmosis.

Circulation and drainage of the aqueous humour:

- The aqueous humour enters the posterior chamber and flows through the suspensory ligament of the lens, then it reaches the anterior chamber through the pupil.
- The fluid then flows towards the iridocorneal angle (= anterior chamber angle or filtration angle) where it passes through the spaces of Fontana and drains into the canal of Schlemm.
- This canal is a thin-walled venous channel that is extremely permeable (even to large protein molecules). From the canal of Schlemm, the aqueous humour drains into the aqueous veins then finally into the extraocular veins.



Functions of the aqueous humour:

(1) It gives nutrients to the avascular structures (lens and cornea).

(l) It maintains the intraocular pressure. (3) It plays a role as a refractive medium in the eye. (4) It is a route for removal of waste products from the eye.

THE INTRAOCULAR PRESSURE (IOP):

- The IOP is normally 12-20 mmHg (average 15 mmHg) and it shows a diurnal variation up to 4 mmHg (being highest in the morning and lowest in the evening).
- It is maintained nearly equal to the choroidal capillary blood pressure.
- Clinically, it is measured by an apparatus called the tonometer, which is applied to the cornea (after anaesthetizing it by cocaine drops). It can also be roughly assessed by palpating the eyeball with the fingers.

Importance of the IOP:

- The IOP is important for the norma/focusing mechanism of the eye. If the IOP decreases, the suspensory ligament of the lens relaxes and the lens becomes more

spherical, resulting in an increase in its power (so images are focused in front of the retina as occurs in myopia).

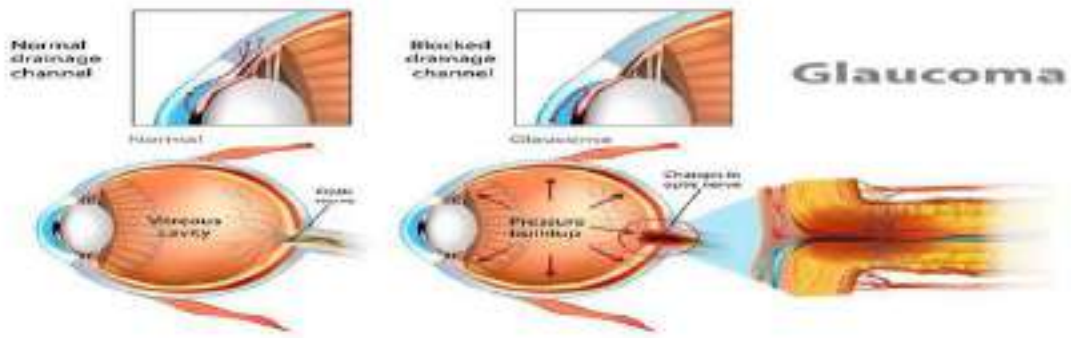
- On the other hand, if the IOP increases, the suspensory ligament is stretched and, consequently, the accommodation mechanism for near vision will be impaired.

GLAUCOMA

Glaucoma is a group of eye diseases that lead to damage of the optic nerve, which is important for transmitting visual information from the eye to the brain. This damage is often caused by increased pressure within the eye, known as intraocular pressure (up to 60-70 mmHg). and may cause vision loss if left untreated.

It is due to (a) Obstruction of the canal of Schlemm (b) Decreased permeability through the trabeculae of the spaces of Fontana due to an inflammatory process (open-angle glaucoma) (c) Infections of the iris that cause its sticking to the lens (d) Forward movement the lens, which may obliterate the iridocorneal angle (angle- closure glaucoma).

The increased IOP leads to headache and severe eye-pain, mydriasis, misty vision, impairment of the focusing mechanism of the eye, cupping of the optic disc and pressure on both the retinal vessels and optic nerve fibers against its rim (which may finally lead to blindness).



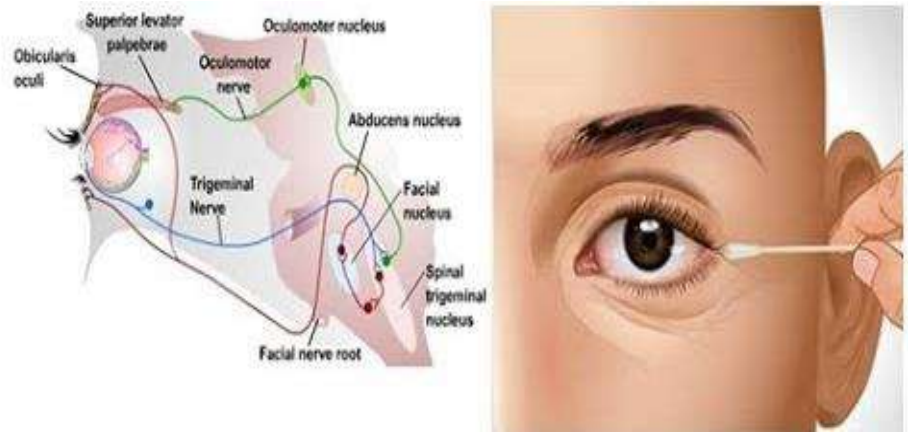
PhysiologicalEyereflexes

➢ The Corneal Reflex:

Touching the cornea of one eye by a foreign body (e.g., cotton) causes blinking at both eyes. It is a superficial reflex.

➢ Afferent impulses are transmitted from the cornea via the ophthalmic division of the trigeminal (5th cranial) nerve to the pons, where they stimulate the facial (7th cranial) nuclei at both sides. Efferent impulses are then transported by the facial nerves and lead to contraction of the orbicularis oculi muscles at both sides resulting in bilateral blinking.

➢ This reflex is primarily protective.However, it is commonly tested clinically to check (a) The integrity of the 5th nerve(b) The depth of anesthesia.



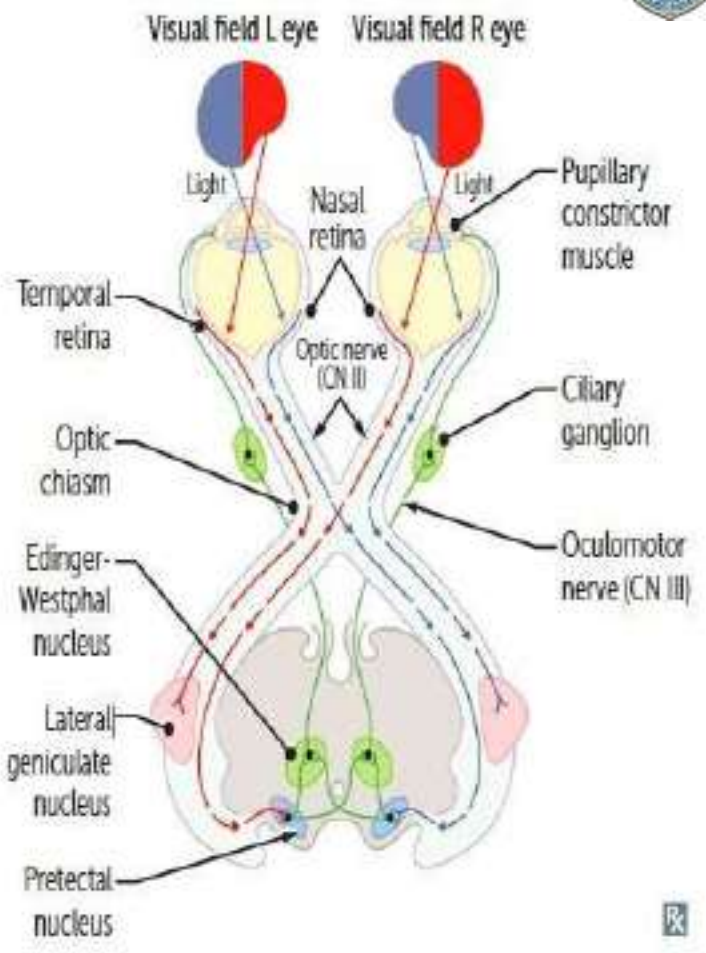
The pupillary light reflex

When one eye is exposed to light, Pupillo constriction occurs in this eye **(= direct reflex)** as well as in the other eye **(=consensual reflex).** The response in the stimulated eye protects its retina from excess light. **Testing for this reflex is important clinically in:**

(a) Diagnosis of certain nervous diseases.

(b) Localization of the sites of lesions in the visual pathway.

(c) Determination of the depth of anesthesia.



➢ Nervous pathway of the pupillary light reflex :

• The photoreceptors of the stimulated retina generate signals that are transmitted by afferent fibers in the **optic nerve** to the optic chiasma then to the optic tract.

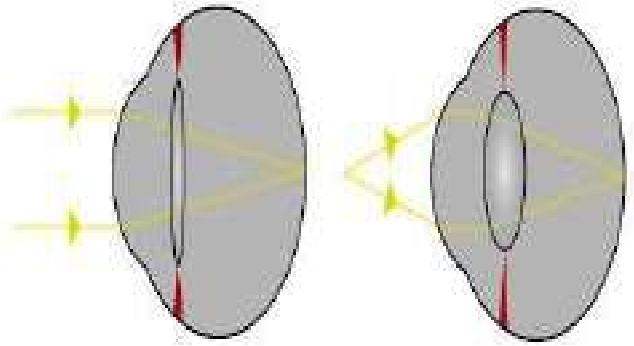
•

● Collateral fires leave the optic tract and terminate at the **pretectal nucleus** in the midbrain *(center of the reflex)*. Fibers then arise from the pretectal nucleus (= tecto-nuclear tract) and terminate at the **Edinger Westphal nuclei of both sides** (the fibers that reach the opposite side cross close to the aqueduct of Sylvius).

• Fibers arise from the Edinger Westphal nuclei and relay in the ciliary ganglia, from which **postganglionic fibers (short ciliary nerves)** arise and enter the eyes, where they lead to contraction of the constrictor pupillae muscles, resulting in **bilateral miosis**.

➢ Accommodation (=Near Response or Reflex):

- Accommodation is a process by which the optical system of the eye is adjusted to see near objects clearly. Normally, during distant vision, the ciliary muscle is relaxed, and parallel rays (= those coming from objects placed more than 6 meters away from eye) are brought into a sharp focus the retina. This normal optical condition is called **emmetropia.**



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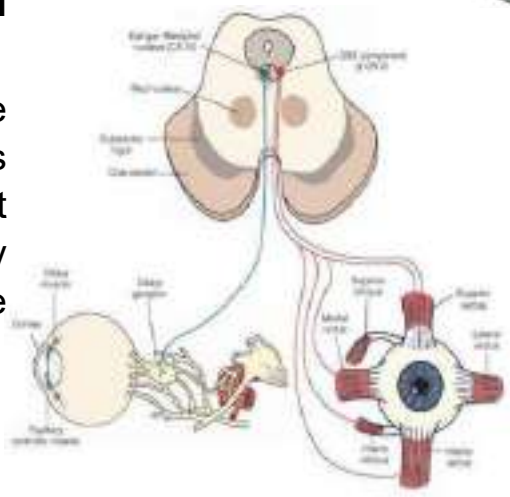
- Light rays coming from objects placed at nearer distances than 6 meters from the eye are divergent, and they form a focus behind the retina and a blur (or a diffusion circle) on the retina. Accordingly, these objects are indistinct, and in order to see them clearly, their images should be brought on the retina. This is produced through the process of accommodation (= near response or near reflex). The change is greatest in the anterior surface of the lens. In young individuals, the change in shape may add as many as 12 diopters to the refractive power of the eye.

➢ Eye changes that occur during accommodation :

1. Convergence of both eyes for visual axes:

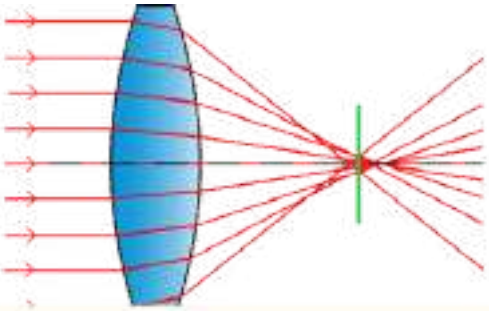


This is produced mainly by contraction of the medial recti muscles of both eyes, and its function is to shift the images of the near object to the fovea centra/is of both retinae where they can be seen clearly, this will prevent double vision.

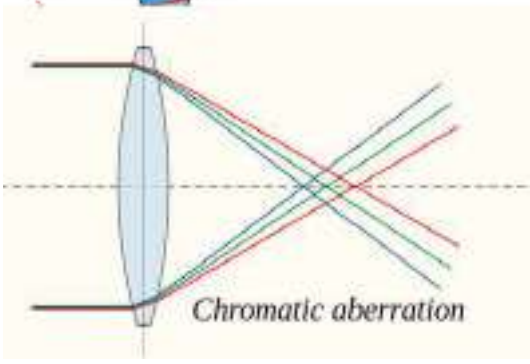
2. Bilateral miosis (or Pupillo constriction):

This cuts off a considerable amount of the peripheral rays, thus allowing only the central rays to enter the eyes. This effect prevents both spherical and chromatic aberrations resulting in clear vision and also increases the depth of focus.

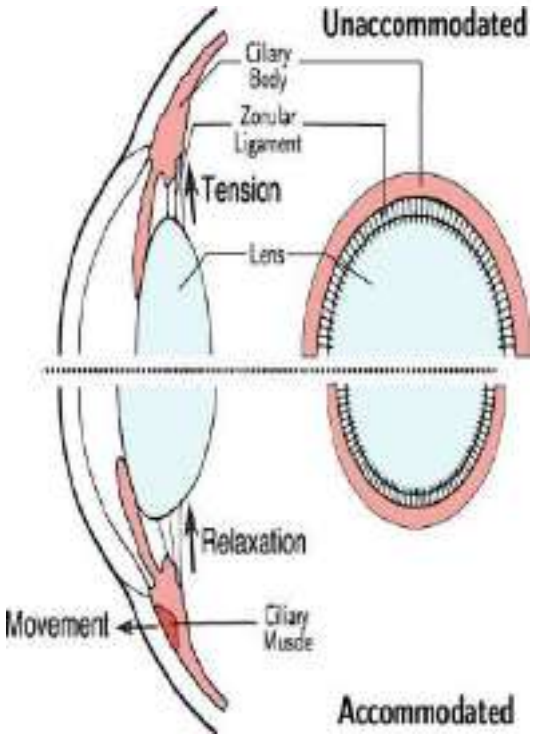
- **Spherical aberration**: This is formation of blurred images as a result of inability of the convex lens to collect all rays into a single focus due to differences in the refractive powers of its peripheral & central parts.



- **Chromatic aberration:** This is formation of images that are surrounded by halos of the various colors of the spectrum (which also causes blurring). This occurs as a result of analysis of white light at the peripheral parts of the lens (which, act as prisms). The spectral colors are refracted by different degrees according to the wavelength of each color.



3. Increasing the power of the lens:



The ciliary muscles act on the crystalline lens through the **zonular fibers** that make up the **suspensory ligament**. The crystalline lens and zonular fibers are both elastic. When the eye is observing **distant objects**, the ciliary muscles are **relaxed**, the zonular fibers are under tension, and the elastic **crystalline lens is maximally flat**, resulting in minimal dioptric power and maximal focal length. However, when the eye is observing nearby objects, the ciliary muscle **contracts** and pulls itself forward, releasing the tension on the zonular fibers. This causes the **crystalline lens to become more curved**, and the dioptric power to increase and the focal length to decrease. Note that a camera lens fine-focuses nearby objects by moving the lens closer to the object and farther from the CCD, instead of changing the shape of the lens.

- The nearest point to the eye at which an object can be brought into clear focus by accommodation is called **the near point of vision**. The near point recedes throughout life, slowly at first and then rapidly with advancing age, from approximately 9 cm at age 10 to approximately 83 cm at age 60. This recession is due principally to increasing hardness of the lens, with a resulting loss of accommodation.

- By the time a healthy individual reaches age 40–45 years, the loss of accommodation is usually sufficient to make reading and close work difficult. This condition, which is known as **presbyopia**, can be corrected by wearing glasses with convex lenses. In addition to accommodation, the visual axes converge, and the pupil constricts when an individual looks at a near object.

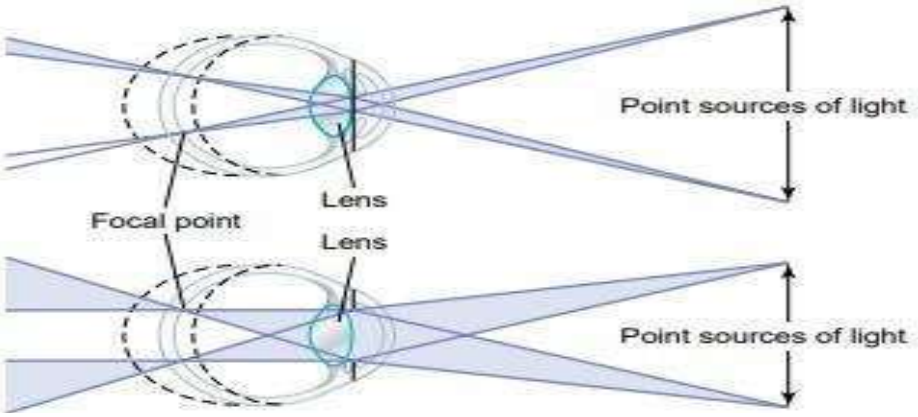
Principles of optics :

√ Light rays are bent when they pass from a medium of one density into a medium of a different density, except when they strike perpendicular to the interface**.** The bending of light rays is called **refraction** and is the mechanism that allows one to focus an accurate image onto the retina.

√Parallel light rays striking a biconvex lens are refracted to a point **(focal point** or **principal focus)** behind the lens. The focal point is on a line passing through the centers of curvature of the lens, the **principal axis.** The distance between the lens and the principal focus is the **focal length.**

√Light rays from an object that strike a lens more than 6 m away are, for practical purposes, parallel. The rays from an object closer than 6 m are diverging and are therefore brought to a focus farther back on the principal axis than the principal focus. Biconcave lenses cause light rays to diverge.

√Refractive power is greatest when the curvature of a lens is greatest. The refractive power of a lens is measured in **diopters,** the number of diopters being the reciprocal of the principal focal distance in meters. For example, a lens with a principal focal distance of 0.25 m has a refractive power of 1/0.25, or 4 diopters.



- Focal length is the distance between the retina and the lens=17 mm
- Total refractive power of the eye=60D
- The power of the cornea = +42 D.
- The power of the lens = +18 D.

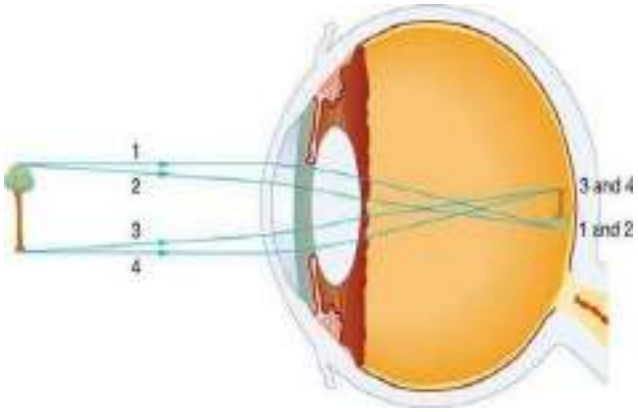
➢ The Refractive index:

- The difference between the densities of the two media as indicated by their refractive index.
  - Light rays pass from the **cornea** to **aqueous humor** then through the **pupil** to the **lens** to **vitreous humor.**

Refractive index= Light velocity in air/ Light velocity in medium

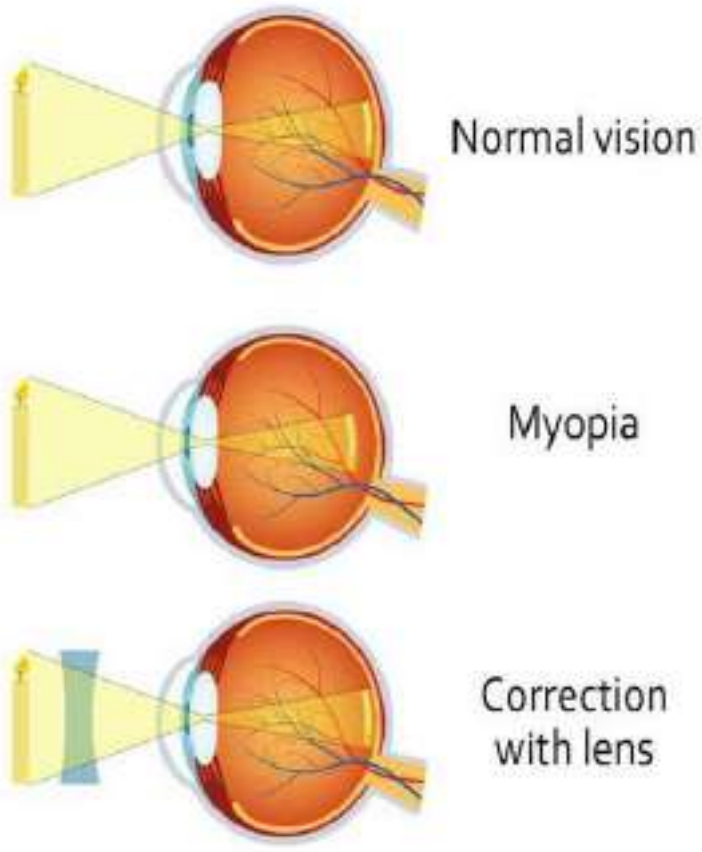
➢ Refractive indices:

- Cornea 1.38
- Aqueous humor 1.33
- Crystalline lens 1.4
- Vitreous humor 1.34



NB: The image is formed on the retina inverted.

Errors of refraction :



1-Myopia:

Focus parallel light rays Infront of retina, while accommodation is relaxed. ***Cause:*** • Axial myopia: The myopic eye is longer than the normal eye, each 1 mm difference causes -3D myopia (Long eyeball). • Refractive myopia: Increase in the refraction power of the lens or the cornea due to the increase in their convexity.

Effect: • Far vision: light rays are focused Infront of the retina. • Near vision: not affected, accommodation is at rest.

Treatment:

- Wearing glasses with **concave lens** –ve (divergent lens).

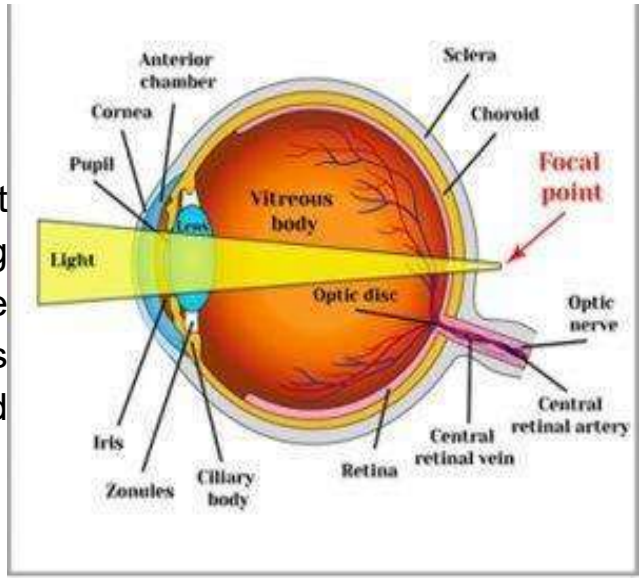
2- Hypermetr opic eye:

Focus parallel light rays behind the retina, while accommodation is relaxed. ***Cause:*** • Axial hypermetropia: The eye is shorter than the normal eye, each 1 mm difference cause +3D myopia (short eyeball) • Refractive myopia: Decrease in the refraction power of the lens or the cornea due to the decrease in their

• convexity.

Effect:

- Far vision: not affected.
- Near vision: focus behind retina.



In hypermetropia: • The patient continuously uses accommodation to bring the focus on retina and so medial eye muscles are continuously contracting. This led to headache, eye fatigue, and Convergent squint(in children).

Treatment: • Wearing glasses with **convex lens** +ve (convergent lens)

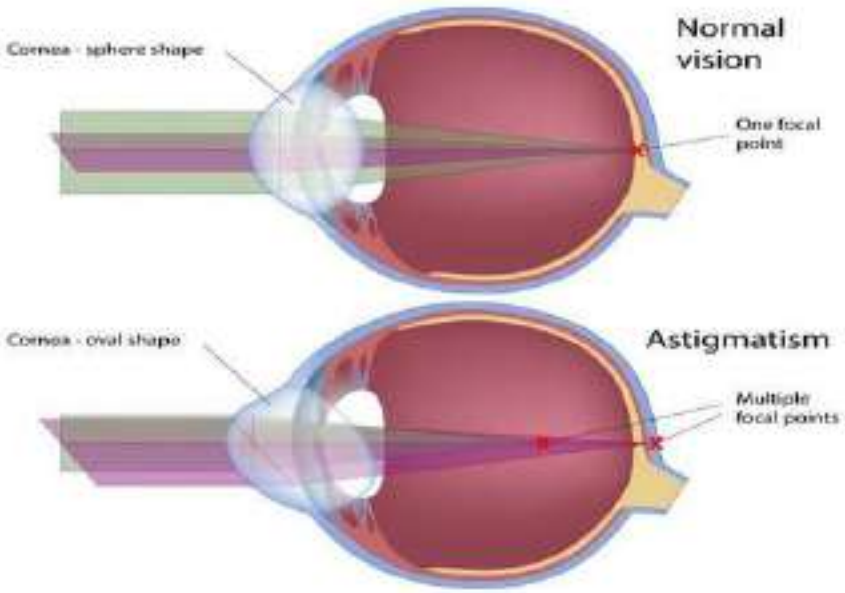
3-Astigma t ism:

Parallel rays come into different foci, blurred image.

Cause:

• The refractive power of the eye is not the same in different meridia.

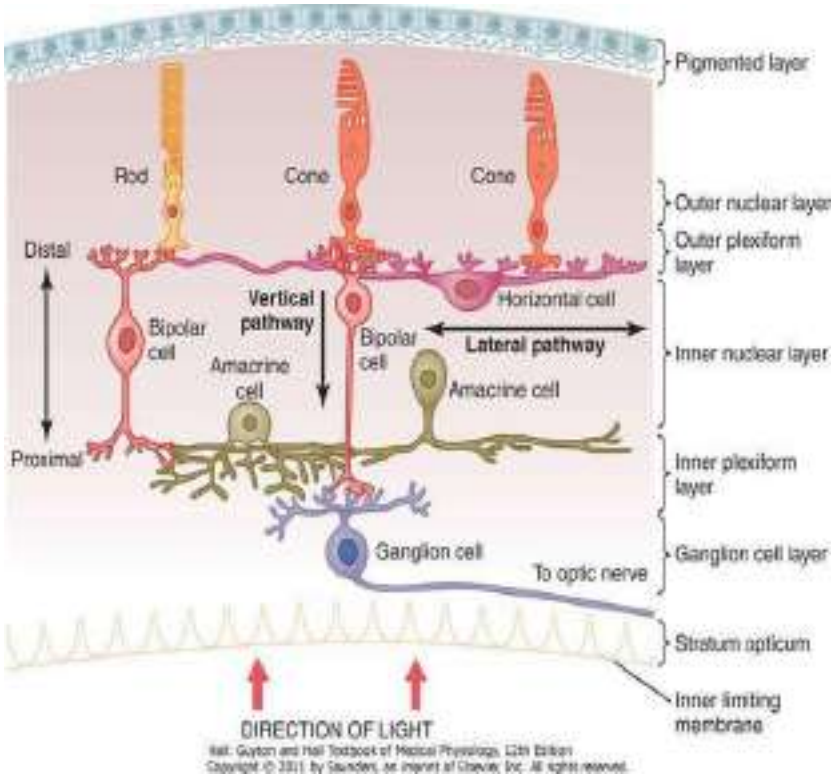
• Curvature of cornea and lens is not equal or uniform.



1 Physiology of Retina

Layers of the Retina:

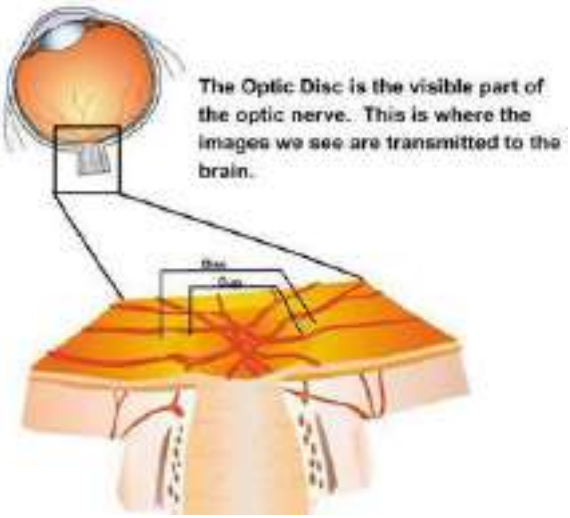
1-Retinal pigmentepithelium
2-Layer of rods and cones
3-External limiting membrane
4-Outer nuclear layer containing the cell bodies of the rods and cones
5 -Outer plexiform layer
6-Inner nuclear layer
7 -Inner plexiform layer
8 -Ganglionic layer
9 -Layer of optic nerve fibers
10 -Internal limiting membrane



Regions of the Retina:

1-Optic Disc :

- **Site:** 3mm medial to posterior pole .
- **Size:** 1.5 mm in diameter.
- **Properties:** It is the site of Optic nerve exist and Retinal blood vessels entry.
- No visual receptors {Blind spot.}



2 -Maculae lutaea :

- **Site:** located at the center of the retina .
- **Size:** yellowish spot (less than 1 mm) yellow color is due to special type of melanin pigments .
- **Properties:** Highest visual acuity.

Why does the maculae have the highest visual acuity?

- It contains mainly cones that are responsible for the detection of the details and colors of the objects. It doesn't contain blood vessels.

3 -Photoreceptors {Rods and cones}

a) They convert light energy to electric activity transmitted through retinal layers to optic nerve to brain.

b) They contain specific photosensitive pigments that are degraded or chemically changed on exposure to light a process called (Bleaching), but reformed on the dark

c) They contain large amounts of vitamin A

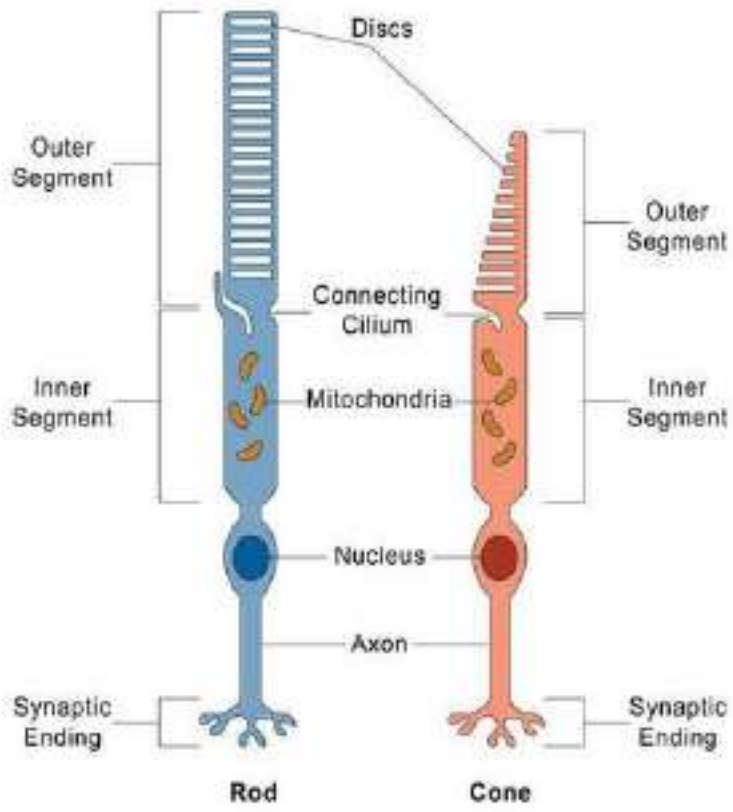
d) They are 2 types: Rods and Cones.

Rods and Cones

Rods (Scotopic vision)	Cones (Photopic vision)
100 – 120 million	5 - 8 million
Peripheral vision	Central vision
Located everywhere except fovea	High density in the macula and fovea
Very sensitive to light	Less sensitive to light
Used in low light situations	Most normal lighting conditions
One type	Three types
Highly convergent	Non-convergent
Black and White	Color vision

Structure Of The Rods And Cones

- Each rod or cone consists of an outer and an inner segment (the latter forms the outer nuclear layer and is rich in mitochondria) as well as a synaptic body orterminal .



- Theouter segments are modified cilia andare thin rod-like in rods but conical in cones. They are made up of regular flattened saccules or discs (about 1000 in each rod or cone .(

- In cones, each disc is actually an in folded shelf of cell membrane but in rods the discs separate from the membrane and form flat sacs lying totally inside the cell .

- These discs and sacs contain photosensitive compounds that react to light. Such compounds are so concentrated that they constitute 40-90% of the mass of the outer segments .

- Theold (outer) discs are continuously phagocytized (by cells of the pigment epithelium) andare constantly renewed (by formation of new discs at the inner edges of the segments.(

- Therods have a uniform shape all over the retina while the cones vary in shape (they arelong & thin in the fovea and shorter & broader in the periphery of the retina(.

- The inner processes of the rods & cones synapse with the dendrites of the bipolar cells, th axons of which synapse in tum with the dendrites of the ganglion cells. The axons of the ganglion cells collect and form the optic nerve fibers which leave the eye at the optic disc.

Neural Function Of The Retina

Photoreceptor potentials:

- Inthedark,theinnersegmentsofthe retinal photoreceptors (rods and cones) continuously pump Na+ outwards (by activity of Na+- K+ ATPase) and the membranes of their outer segments are very leaky to Na + by effect of cyclic GMP .

- Therefore, Na + continually re-enters these receptors (via their outer segments), leading to decreased negativity inside the receptors, and resting membrane potential is about –40 mV. It is because of influx of sodium ions (Dark current).

- Onthe other hand, on exposure to light, the retinal receptors are excited, and a receptor potential develops. However, the developed potential is normally hyperpolarizing (and not depolarizing as in most other sensory receptors) .

This occurs through a cascade of reaction as follows:

(1) .When light strikes the outer segments of the receptors, it is absorbed by the photosensitive pigments leading to their activation (i.e., a change of their structures.)

(2) .The activation of the photosensitive pigments leads to formation of a certain substance (unknown in the cones, and meta rhodopsin II in rods) which activates a G protein (called transducin in rods and a similar protein in cones)

(3) The activated G proteins, in turn, activate the phosphodiesterase enzyme, which catalyzes breakdown of cyclic GMP. Accordingly, the intracellular content of cyclic GMP has decreased.

(4) The decreased cyclic GMP leads to closure of the Na + channels in the outer segments of the receptors, while Na+ is continuously pumped out from their inner segments. This creates marked negativity inside the receptors and marked positivity outside them, leading to a state of hyperpolarization, and the potential becomes -70 to -80 m V at maximal light intensity.

(5) The release of transmitter from the synaptic terminals of the photoreceptors (which is probably glutamate) is steady in the dark, but on exposure to light, the resulting hyperpolarization reduces the release of the synaptic transmitter, and this generates a signal that leads to action potentials in the ganglion cells.

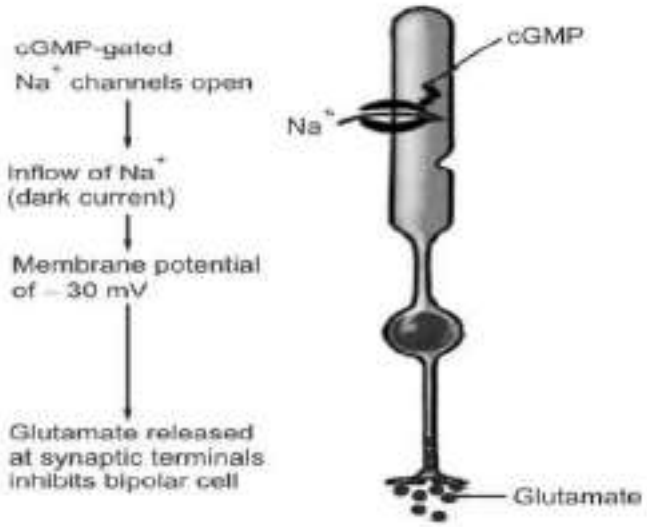


Fig. : Operation of photoreceptor in darkness

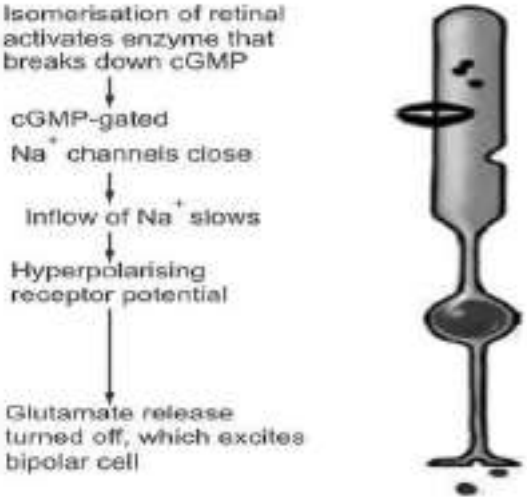


Fig. : Operation of photoreceptor in light

Photochemical changes in the retina

- Light rays cannot directly stimulate the nerve fibers. However, when light is absorbed by the photosensitive compounds (or pigments) present in the photoreceptors, the structure of these compounds' changes, and such change triggers a sequence of events that leads to development of the receptor potential. Photochemical changes in the visual receptor cells are called Wald visual cycle.

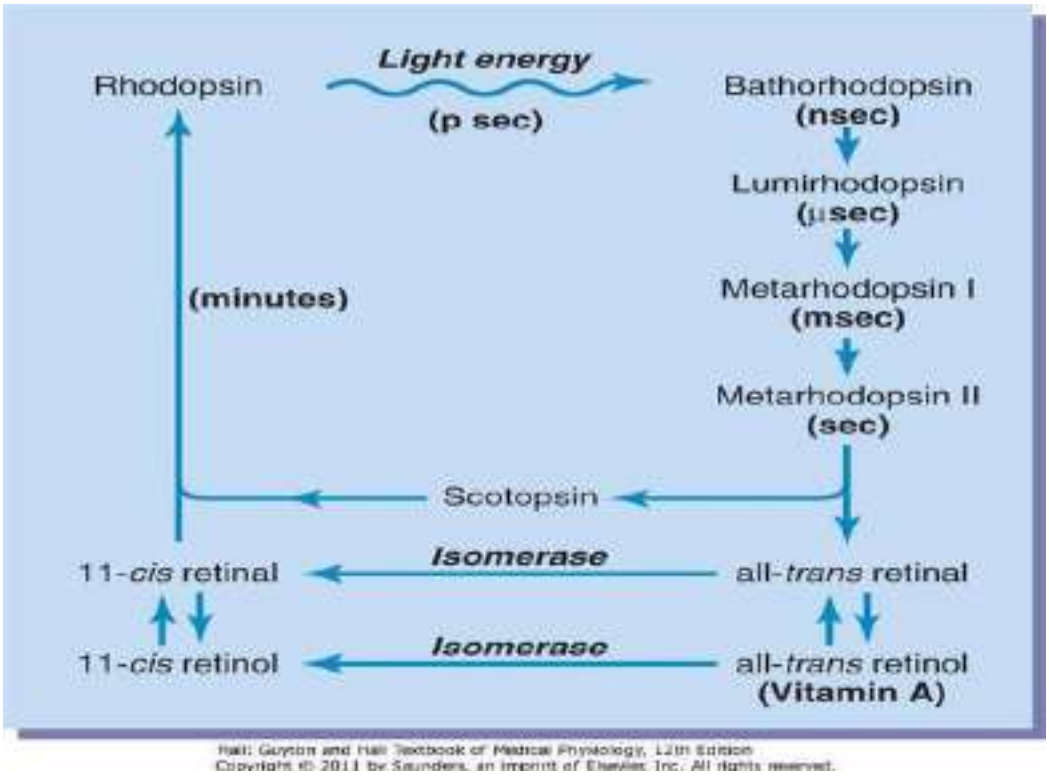
- Accordingly, the photoreceptors act as a transducer; that convert light energy into neural activity, and such process of photo-transduction involves closure of Na+ channels in their outer segments .

- Thephotosensitive pigments are made up of a protein called opsin and a carotenoid pigment called retinenel or retinal (*whichisthealdehyde of vitaminA*).
- Thepigmentintherodsiscalled rhodopsin (or visual purple). Its protein is called scotopsin, and in the dark its retinene, molecule is in the 11-cisform (which has a curved or angulated configuration). The 3 cone color pigments are almost structurally similar to rhodopsin, except that their proteins are slightly different and are called photopsins.
- When rhodopsin is exposed to light, it is immediately decomposed. The only action of light is to change the shape of 11-cis retinene1, converting it into the all-trans isomer (which has a straight molecule), and the resulting compound is called Batho-rhodopsin (which is
- extremely unstable). It is very rapidly converted to Lumi-rhodopsin which is also rapidly converted to metarhodopsin I then to metarhodopsin ll, and finally complete bleaching occurs i.e., separation of all-trans retinene1 -from opsin .

- Itis metarhodopsin II (which is also called the activated rhodopsin) that initiates the electric changes in rods.

Reformation of rhodopsin in the dark:

- Sincethe 11-cis retinene1isthe only formthatcan bind to scotopsin, the all-trans retinene1 is first re-converted to the 11- cis form (a process that is catalyzed by the enzyme retinal isomerase), which then automatically combines with scotopsin to form rhodopsin.



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Role of Vitamin A for Formation of Rhodopsin

- Note in Figure above that there is a second chemical route by which all-trans retinal can be converted into 11-cis retinal .
- This is by conversion of the all-trans retinal first into all-trans retinol, which is one form of vitamin A. Then the all-trans retinol is converted into 11-cis retinol under the influence of the enzyme isomerase .
- Finally, the 11-cis retinol is converted into 11-cis retinal, which combines with scotopsin to form new rhodopsin.
- Vitamin A is present both in the cytoplasm of the rods and in the pigment layer of the retina. Therefore, vitamin A is normally always available to form new retinal when needed .
- Accordingly, vitamin A deficiency decreases the formation of rhodopsin, and since the latter is essential for night vision, vitamin A deficiency leads to night blindness (=nyctalopia).

Neural Function of the Retina

Different NeuronalCellTypesAreAs Follows :

(1) The photoreceptors themselves-the rods and cones-which transmit signals to the outer plexiform layer, where they synapse with bipolar cells and horizontal cells .

(2) .The horizontal cells, which transmit signals horizontally in the outer plexiform layer from the rods and cones to bipolar cells .

(3) The bipolar cells, which transmit signals vertically from the rods, cones, and horizontal cells to the inner plexiform layer, where they synapse with ganglion cells and amacrine cells

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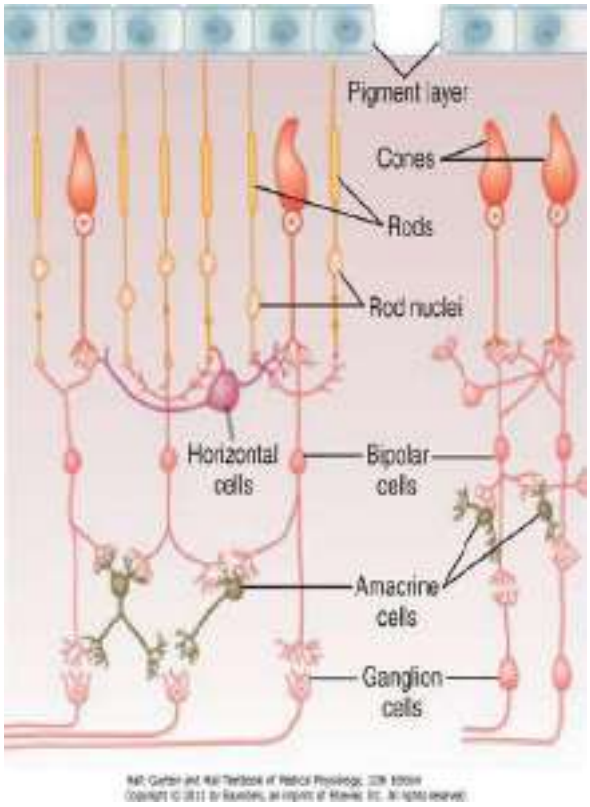


(4) The amacrine cells, which transmit signals in two directions, either directly from bipolar cells to ganglion cells or horizontally within the inner plexiform layer from axons of the bipolar cells to dendrites of the ganglion cells or to other amacrine cells .

(5) The ganglion cells, which transmit output signals from the retina through the optic nerve into the brain .

Retinal adaptation to varying light intensities

Dark adaptation:

Definition:

- Darkadaptation is theprocess bywhichthepersonis able to see the objects in dim light. If a person enters a dim-lighted room (darkroom) from a bright-lighted area, he is blind for some time, Maximum duration for dark adaptation is about 20 minutes.

Causes for Dark Adaptation:

Darkadaptationisduetothefollowing changes in eyeball:

(1)Increasedsensitivityofrodsasaresultofresynthesis of rhodopsin:

- Time required for dark adaptation is partly determined by the time for resynthesis of rhodopsin.
- Within 20 minutes in the dark, most of the retinene1 in the rods is converted into rhodopsin, and vitamin A is withdrawn from the pigment cells of the retina to the rods where it is converted into retinene1, which forms more rhodopsin.

(2) Dilatation of pupil:

- Dilatation of pupil during dark adaptation allows more and more light to enter the eye.

Light Adaptation

Definition:

- Lightadaptationisthe process in which eyes get adapted to increased illumination .

Causes of Light Adaptation

- **Thereare twocauses of lightadaptation:**

(1) Reduced sensitivity of rods:

During light adaptation, the sensitivity of rods decreases. It is due to the breakdown of rhodopsin.

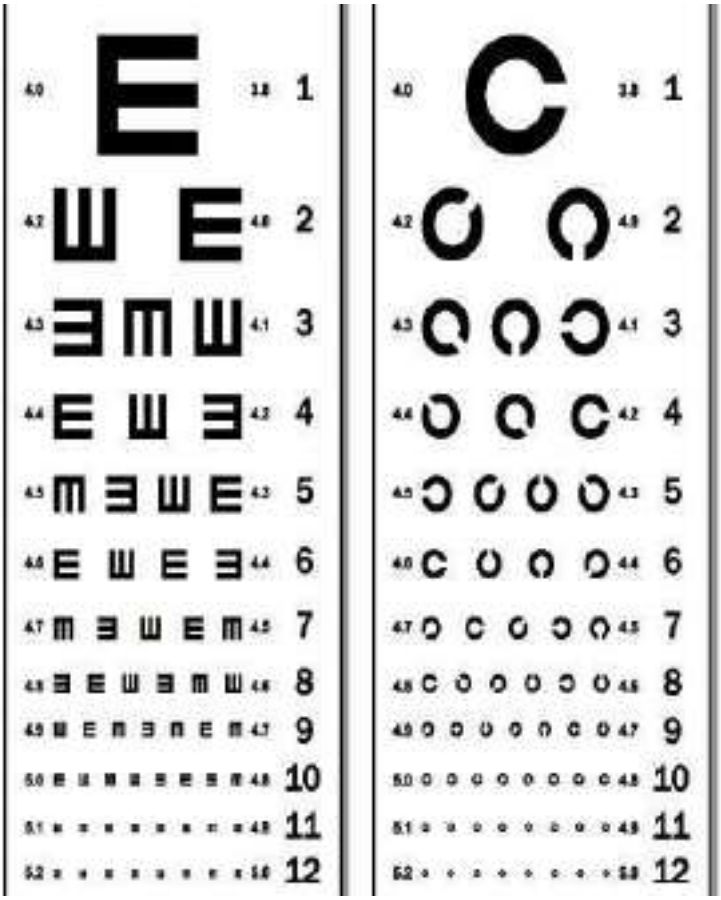
(2) Constrictionof pupil:

Constriction of pupil reduces the quantity of light rays entering the eye, thus decreasing excessive stimulation of the retina (which helps the rapid return of acute vision).

Visual Acuity

- This is the degree of perception of the details and contours of objects by the eye. It is usually determined by finding the minimum separable (= the shortest distance by which 2 lines can be separated and still seen as 2 lines).
- Normally, the detection of 2 adjacent lines (or objects) requires that their retinal images shouldstimulate 2 non-adjacent cones.
- Thisisexplained as follows: The light rays from 2 adjacent objects form an angle at the nodalpoint of the lens known as the visual angle.

- Inthe normal eye, the minimal visual angle produced by light rays from 2 adjacent objects which the images of the 2 objects are perceived as two is one minute (=1/60 degree.)



- Based on this fact, the various charts used for clinical determination of the visual acuity are constructed. The commonest of these charts are the Snellen's and Landolt's charts, which depend on reading letters drawn in black onawell-illuminated white background.
- Thepatient is tested on the chart from above downwards, and the smallest row of letters he can distinguish is determined. The

numerators of which is 6 (the distance between the patient and the chart) while its denominator is the distance at which the smallest letters distinguished by the person can be seen by the normal eye e.g., if the person was able to see only the letters that a normal eye can see at a distance of 12 meters, then his visual acuity will be 6/12.

Factors that affect the visual acuity:

(1) Age: The visualacuity is most sharp in children, then it gradually declines till it becomes markedly decreased in old age due to presbyopia.

(2) The refractive power oftheeye: The visual acuity is decreased if there are errors in the refractive system of the eye e.g., due to myopia, hypermetropia or astigmatism (which are the commonest causes of reduction of the visual acuity);

(3) The degrees of illumination of the test chart and the contrast between the letters and the background.

(4) Diseases of the eyes: Visual acuity is decreased in various diseases of the eye e.g., keratitis (corneal inflammation) iridocyclitis (inflammation of the iris), cataract, glaucoma, retinal detachment and retinitis pigmentosa.

Colour Vision

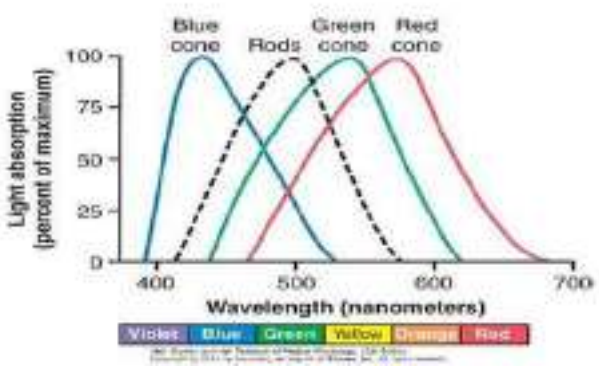
- Colour vision is the function of the retina in a light-adapted eye because it depends on cones. The dark-adapted eye sees different colours as grades of grey.

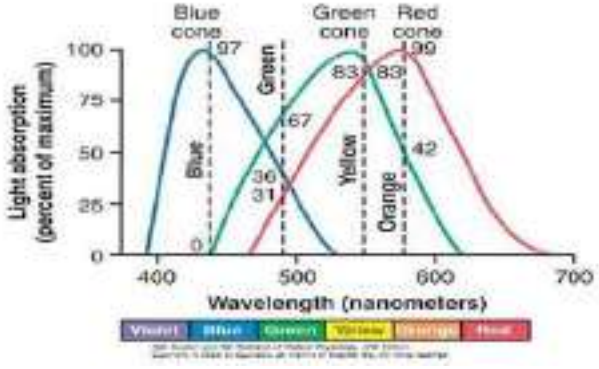
Tricolor Mechanism ofColor Detection:

- All theories of color vision are based on the well-known observation that the human eye can detect almost all gradations of colors when only red, green, and blue monochromatic lights are appropriately mixed in different combinations .

Interpretation of colour inthe C.N.S:

- A coloured light stimulates the 3 kinds of cones unequally depending on its wavelength.
- Accordingly, the frequency of impulses discharged from each kind will be different and when these reach the visual cortex, a sensation of the colour will be perceived.
- This is demonstrated in the colour sensitivity curves, which show the degree of stimulation of the 3 colour sensitive cones by different lights .





Forexample:

(1) An orange light with a wavelength 580 nm stimulates the red cones by an intensity of about 99 %. the green cones by 42% and the blue cones by 0 %. When the ratios 99:42 : 0reach the visual cortex, it interprets the sensation of orange.

(2) Ablue light with a wavelength 450 nm stimulates the red, green and blue cones by ratios 0:0: 97, which is interpreted by the visual cortex as a sensation of blue.

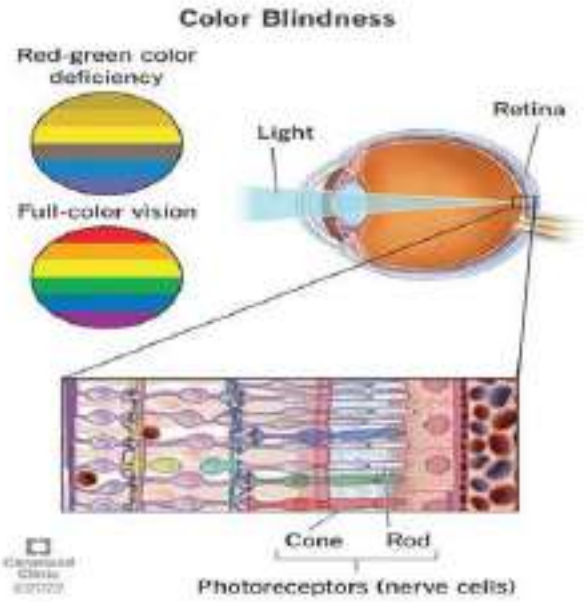
(3) Aratio of 83: 83: 0 is interpreted as yellow.

(4) Aratio of 31 :67: 36 as green.

In this sense, it can be concluded that colour perception starts in the retina.

Colour blindness

- Colour blindness may be a complication of certain eye diseases or a manifestation of certain psychological disorders, but commonly it is inherited as a recessive X-linked characteristic (due to presence of an abnormal gene on the X chromosome) especially red blindness .



- Males are more affected than females (8% and 0.4% respectively) because in males, colour blindness occurs if their single X chromosome has the abnormal gene while females develop the defect only when both X chromosomes contain the abnormal gene.

Types of colour blindness:

- Weaknessof a colour is termed anomaly while blindness to a colour is termed anopia.
- On the other hand, the primary colours, (red, green and blue) are referred to by the terms prot, deuter and trit respectively .

(1) **Trichromats:** These patients have the 3 cone systems, but one is weak, so they may suffer either Protanomaly (= weakness of the red colour), Deuteranomaly (= weakness of the green colour) or Tritanomaly (= weakness of the blue colour).

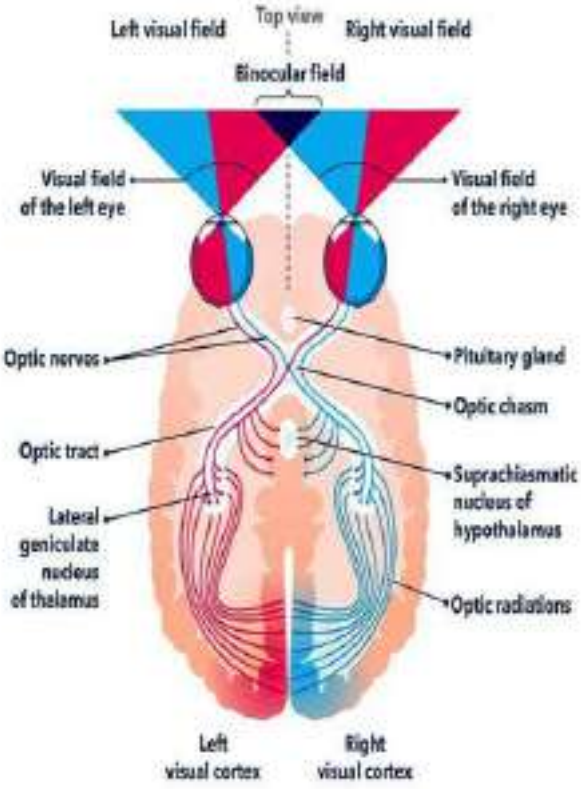
(2) **Dichromats:** These patients have 2 cone systems only (i.e. one system is missing), so they may suffer either protanopia (= red blindness in which red is seen green), deuteranopia (= green blindness, in which green is seen red or blue) or tritanopia (= blue blindness in which blue is seen green) .

(3) **Monochromats:** These patients have only one cone system (i.e. 2 systems are missing) or no cone activity at all. In the first condition (cone monochromatism), day vision and visual acuity are normal while in the second condition (rod monochromatism), there is day blindness in addition to colour blindness (however, these conditions are extremely rare)

Central Neurophysiology of Vision

Visual Pathways:

- The visual nerve signals leave the retinas through the optic nerves.
- At the optic chiasm, the optic nerve fibers from the nasal halves of the retinas cross to the opposite sides, where they join the fibers from the opposite temporal retinas to form the optic tracts .
- The fibers of each optic tract then synapse in the dorsal lateral geniculate nucleus of the thalamus, and from there, geniculocalcarine fibers pass by way of the optic radiation to the primary visual cortex in the calcarine fissure area of the medial occipital lobe .



Visual fibers also pass to several older areas of the brain :

(1) From the optic tracts to the suprachiasmatic nucleus of the hypothalamus, presumably to control circadian rhythms that synchronize various physiologic changes of the body with night and day.

(2) Into the pretectal nuclei in the midbrain, to elicit reflex movements of the eyes to focus on objects of importance and to activate the pupillary light reflex.

(3) Into the superior colliculus, to control rapid directional movements of the two eyes.

(4) Into the ventral lateral geniculate nucleus of the thalamus and surrounding basal regions of the brain, presumably to help control some of the body's behavioral functions .

The visual cortex:

- This is present in the occipital lobe, and it receives fibres from the temporal side of the ipsilateral retina and the nasal side of the contralateral retina. It contains areas 17, 18 and 19.

(A) Area 17:

- This is the primary visual centre and it is also known as the visuosensory area. lt occupies the calcarine fissure and the 2 gyri above and below it on the medial surface of the occipital lobe .
- Area 17 performs the following functions:

(1) Perception of visual sensations (but without understanding their meaning)

(2) fusion of the 2 images formed on both retinae.

(3) Localization of objects in space in relation to each other.

(4) Perception of colour vision .

(B)Area 18:

- This area is also known as the visuopsychic area, and it lies anterior to area 17.
- It performs the following functions:

(1) It recognizes the nature of objects and correlates their colours.

(2) It is concerned with interpretation of visual sensations (so it is also called the visual association or interpretative area). Secondary signals are transmitted to these areas for analysis of visual meanings .

(3) It is concerned with localization of objects in space in relation to the observer (i.e., depth perception).

(C) Area 19:

- This area is also known as the occipital eye field area, and it lies anterior to area 18.
- It performs the following functions :

(1) It shares area 18 of its functions .

(2) It sends information to other parts of the cerebral cortex e.g., the frontal eye field area (area 8) during accommodation.

(3) **It exerts motor functions, including :**

(a) Conjugate deviation of both eyes to the opposite side

(b) Fixation movements by which the eyes are fixed on an object.

Lesions of the visual pathway

[A] Unilateral lesionsintheopticnerve

- These lead to blindness in the corresponding eye.
- No light reflex is obtained if the blind eye is stimulated, but miosis occurs in the blind eye if this reflex is initiated from the other eye (because the oculomotor nerve is intact).

[B]Lesions in the center of the optic chiasma

- These lesions are usually due to anterior pituitary tumours.
- They lead to bitemporal hemianopia (i.e., loss of vision in the temporal parts of the visual fields of both eyes) because the crossing nasal fibers of both retinae are injured.

[C] Lesions io the optic tracts

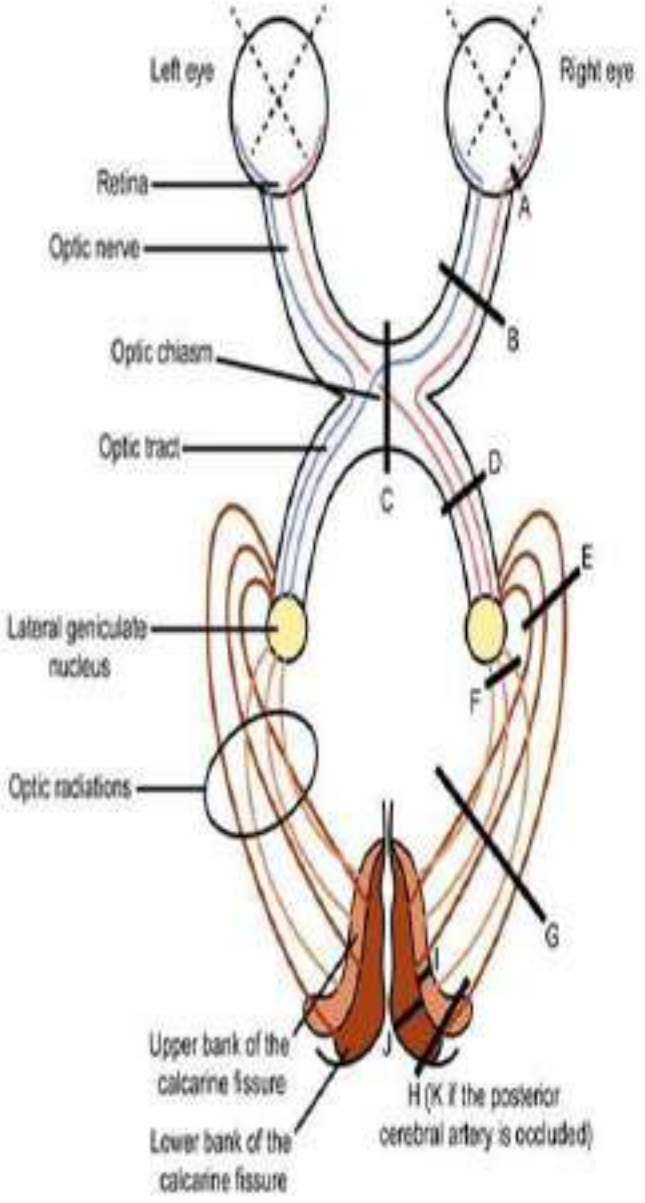
- These lesions cause blindness in the opposite parts of the visual fields of both eyes i.e., contralateral homonymous hemianopia e.g., a lesion in the right optic tract leads to blindness in the left part of both fields of vision (i.e., the temporal field of the left eye and the nasal field of the right eye).

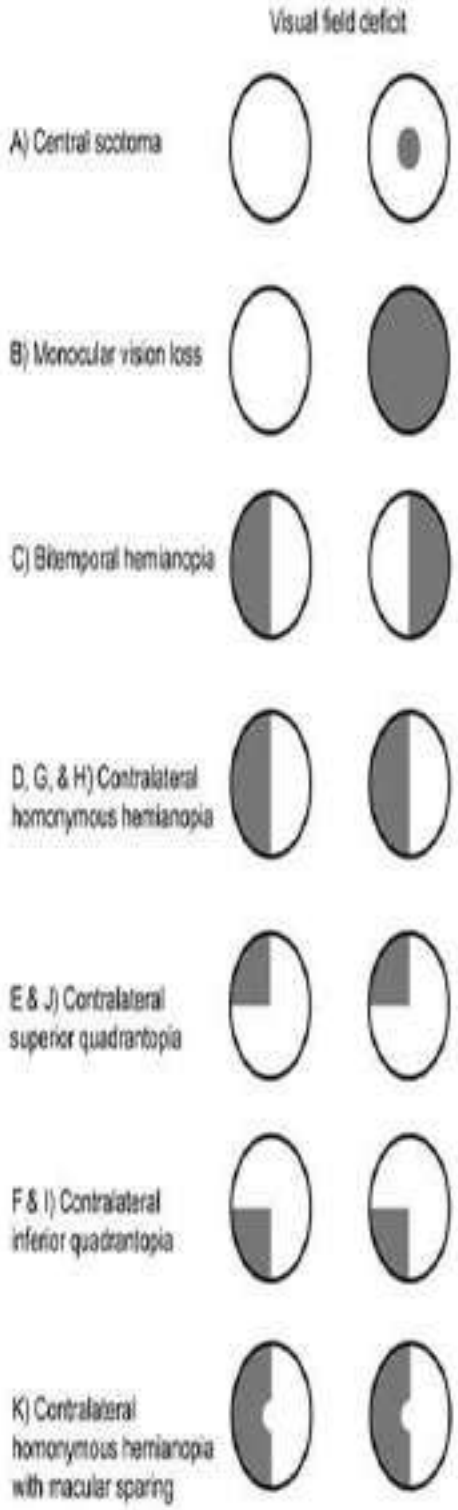
[D] Lesions io the lateral geniculate body, optic radiation or the occipital cortex

- These lesions also lead to contralateral homonymous hemianopia.
- However, the light reflex is still obtained from the blind sides of both retinae (because the optic tract is not affected in such lesions).

Sparing of the fovea = macular sparing

Unilateral lesions of the occipital lobe led to contralateral homonymous hemianopia, but macular vision is usually selectively spared i.e., not affected.


Left eye
Right eye
Retina
Optic nerve
Optic chiasm
Optic tract
Lateral geniculate nucleus
Optic radiations
Upper bank of the calcarine fissure
Lower bank of the calcarine fissure
A
B
C
D
E
F
G
H (K if the posterior cerebral artery is occluded)
I
J


Visual field deficit
A) Central scotoma
B) Monocular vision loss
C) Bitemporal hemianopia
D, G, & H) Contralateral homonymous hemianopia
E & J) Contralateral superior quadrantopia
F & I) Contralateral inferior quadrantopia
K) Contralateral homonymous hemianopia with macular sparing

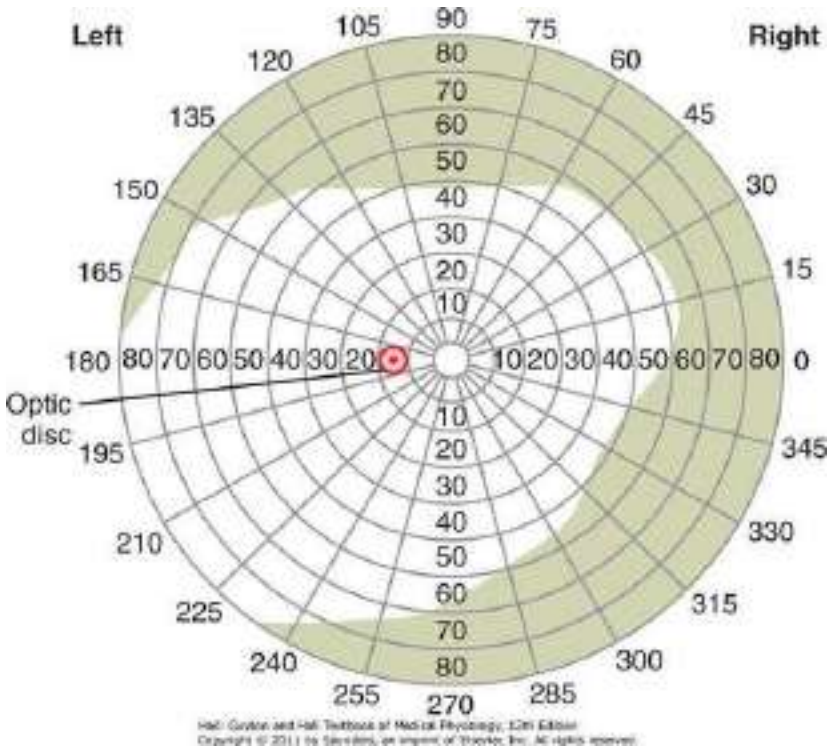
visual field

- The visualfield of an eye is the maximal area of space that can be seen by this eye.
- It is not circular because it is cut off medially by the nose, superiorly by the roof of the orbit (and eyebrows) and downwards by the cheeks (so its nasal side is smaller with an irregular outline while its temporal side is larger with a regular outline)

- Normally, the visual field of each eye extends about 50 degrees upwards, 80 degrees downwards, 60 degrees nasally and 100 degrees temporally .

- The visual fields can be mapped by an instrument called the perimeter and the process of measurement is called perimetry.

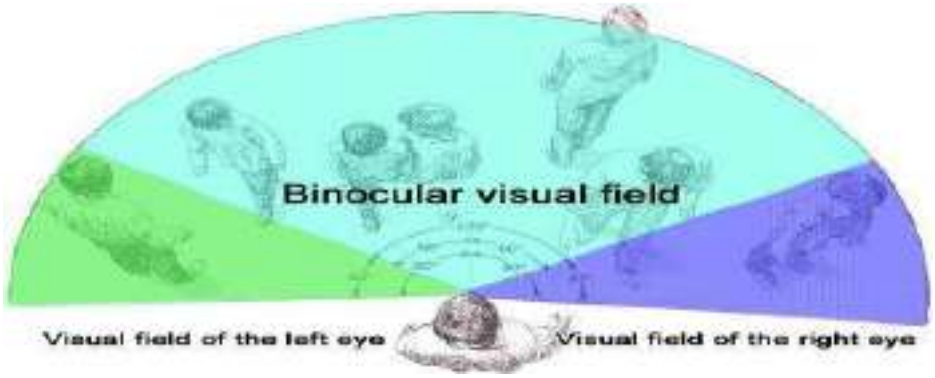
- The visual field can also be determined by the confrontation test.



- In this test the patient's field is compared with that of the examiner (who is supposed to be normal).

Binocular vision

- This is seeing an object with 2 eyes at the same time.
- It is due to the overlap of the central parts of both visual fields, so objects placed in the overlapping area form images on the retinae of both eyes.



- The 2 images of an object placed in the area of binocular vision (one from eachretina) are fused at the cortical level (area 17) into a single image.
- The retinal points at which the images of an object must fall to be seen binocularly as a single image are called the corresponding points.
- They are located at the nasal side of one retina and the temporal side of the other .

- If the external ocular muscles are not well coordinated (or the eye is gently displaced by the finger), diplopia occurs i.e., the object's image is doubled (double vision) because the images in such cases fall outside the corresponding points .

Advantages of binocular vision :

(1)Itgivescompensationfor aweak eyee: g. if there are defective spots in the retina of one eye, they can be masked through binocular vision .

(2) It provides a more accurate perception of depth (distance of objects from the eyes) than that provided by a single eye. Accordingly, binocular vision is important for stereoscopic vision.

Stereoscopic vision

This depends on determination of the 3 dimensionsofobjects(height,widthand depth).

2 Physiology of HEARING

Function Of Hearing

1-Communication

2-speaking

Sound: is the sensation produced when longitudinal vibrations of air molecules strike the tympanic membrane .

Sound waves has 3 characters :

1 -Wave length:

2 -Frequency is measured in hertz (Hz):

- The sound frequencies audible to humans range from about 20 to a maximum of 20,000 cycles per second (Hz).
- The threshold of the human ear varies with the pitch of the sound (Figure 11–9), the greatest sensitivity being in the 1000- to 4000-Hz range.
- The pitch of the average male voice in conversation is about 120 Hz and that of the average female voice about 250 Hz. pitch of a sound is correlated with the frequency of that sound.

3 -Intensity is the loudness of voice: it correlates with the amplitude of the sound wave. measured in decibels (Db) ,a log scale .

where:

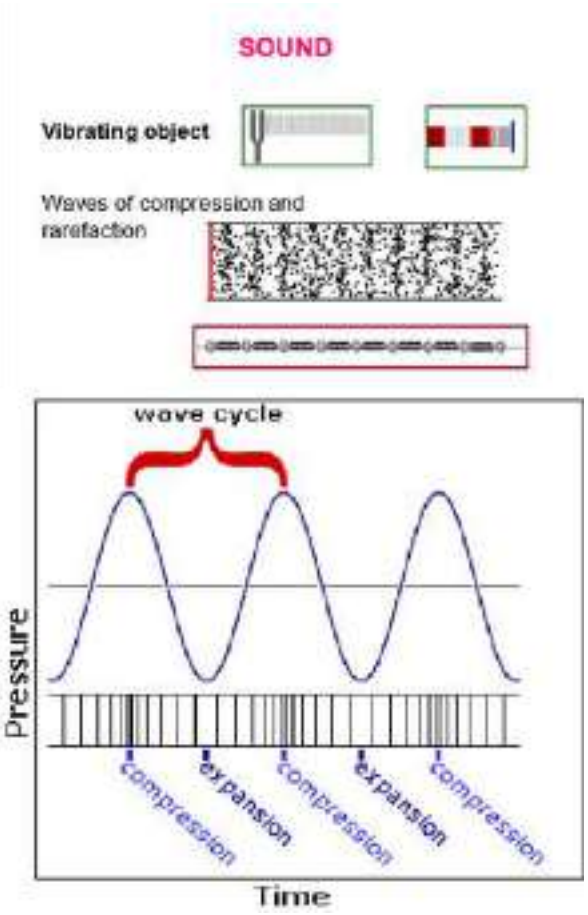
dB = decibel

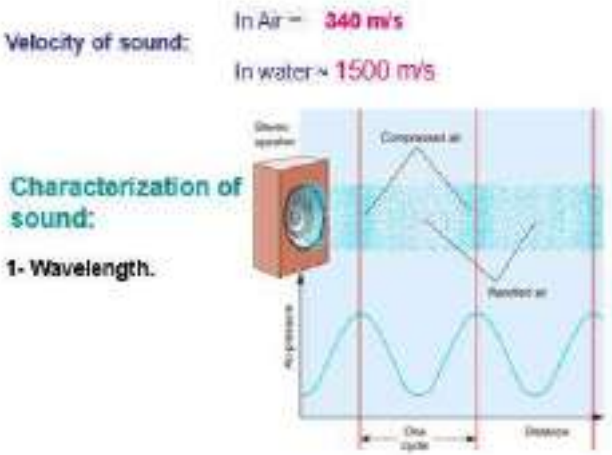
P = sound pressure being measured

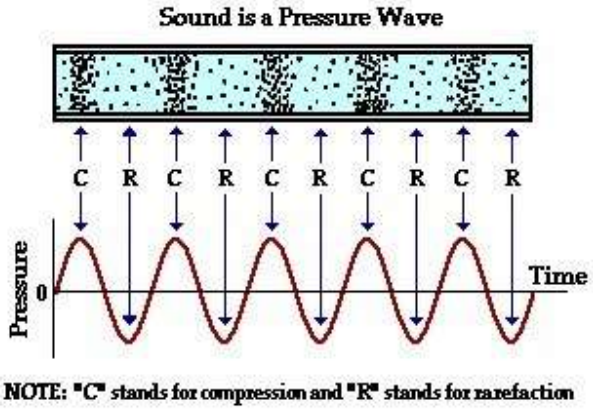
P0 = reference pressure measured at the threshold frequency

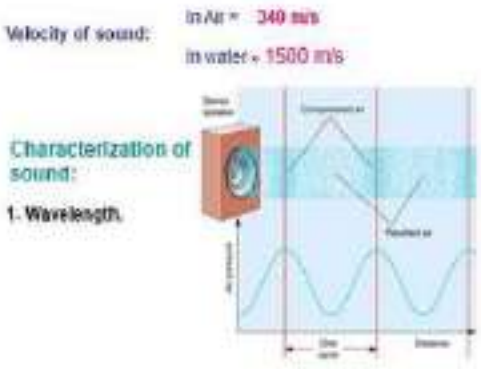
$$dB = 20 \log \frac{P}{P_0}$$

Loudness is correlated with the amplitude of a sound wave, pitch is correlated with frequency, and timbre with harmonic vibrations









Function of the External ear:

1-Collectionofsoundstimuli .

2-Increase sound pressure at the tympanic membrane.

3-Sound localization .

4-Protective function: Wax -Keep airmoist& warm. .

Function of the tympanic membrane

- It vibratesin responsetosound.
- Its vibration is transmitted to the middle ear
- **It has the following characteristics:**

  (1) Elastic

  (2) tense

  (3) aperiodic (resonator(

  (4) damped.

Middle ear

- Air-filled
- Connected to nasopharynx via Eustachian tube.
- **Three ossicles :**Malleus , Incus , Stapes
- **Two muscles**: Tensor tympani., Stapedeus

Function Of The Middle Ear

a) Couplessound energy to thecochlea .

b) Application of sound preferentially to the oval window.

c) Impedance Matching:

√ Movement of stapes is only three fourths that of the malleus so lever system does not amplify movement instead increase the force of movement by 1.3.

√ This system provide impedence matching by 60-70%

√ In absence of this mechanism , transmission decrease by about 20-30 dB or more .

d) Middle ear muscles:

√ Tensor tympani: keeps the tympanic membrane tense

√ .Stapedeus: limits the movement of the stapes. Acousti

√ c reflex. Latency is 40-160 msec .

- **The tympanic reflex (= attenuation reflex(:**

√ This is reflex contraction of both the tensor tympani & stapedius muscles in response to excessively loud sounds.

a) The strong contraction of the tensor tympani causes excessive stretch of the tympanic membrane, which decreases its vibrations .

b) The simultaneous contraction of both muscles causes the entire ossicular system to be more rigid (which greatly reduces ossicular conduction).

√ the latent period (reaction time) of this reflex is only 40-80 (up to 160) milliseconds ,so it cannot protect the inner ear from brief intense stimulations (e:g. ·that produced by a gunshot), because they will be transmitted to the inner ear before reflex contraction of these muscles occurs.

Function of the Eustachian Tube :

- Itconnectsthe middle earwiththe nasopharynx.
- Usually Closed, but it opens during chewing, swallowing & yawning.

a) Equalizes pressure on tympanic membrane.

b) Lined by ciliated epithelium: drainage of middle ear .

- If the tubes are closed for a long time (e.g. as a result of common cold), the air in the middle ears will be gradually absorbed and the produced negative pressure sucks the eardrums inwards, leading to their excessive stretch, which decreases the auditory acuity (so the patient will temporarily suffer partial deafness).

- **During ascending in an airplane**,

the atmospheric pressure is decreased and the tubes open automatically, allowing air in the middle ear cavities to go out (thus decreasing the pressure within the middle ears, and an equal pressure is maintained at both sides of the ear drums).

- **During Descending In An Airplane** the atmospheric pressure increases and this would lead to rupture of the eardrums (which may occur in sleeping and unconscious individuals).

- Therefore, during the descent of airplanes, one should swallow periodically to open the Eustachian tubes, thus allowing air to enter the middle ears. In this way, the pressure inside the middle ears will also increase, and an equal pressure will be maintained at both sides of the eardrums (which prevents their rupture

Sound Transmission to Inner Ear

1-Ossicularroute:

- Via tympanic membrane and ossicles.

2 -Air route:

- Through secondary tympanic membrane.
- Only when tympanic membrane and ossicles are damaged.

3 -Bone route:

- Via skull bones

Inner Ear (Labyrinth)

- is fluid filled .
- consists of a bony labyrinth (**semicircular canals, cochlea, and vestibule**) and a series of ducts called the membranous labyrinth.
- The fluid outside the ducts is **perilymph**; the fluid inside the ducts is **endolymph**.

[A] Structure of the cochlea:

Cochlea is a coiled structure like a snail's shell (cochlea = snail's shell). **It consists of two structures:**

a) Central conical axis formed by spongy bone called modiolus.

b) Bony canal or tube, which winds around the modiolus .

Three Tubular Canals

a) The scala vestibuliandscala tympani contain **perilymph**, which has a high **[Na.]+**

b) The scala media contains **endolymph,** which has a high **[K.]+**

c) The scala media is bordered by the basilar membrane, which is the site of the organ of Corti .

[B] Location and structure of the organ of Corti

- The organ of Corti is located on the basilar membrane.
- It contains the **receptor cells** (inner and outer hair cells) for auditory stimuli.
- **Cilia** protrude from the hair cells and are embedded in the tectorial membrane.
- **Inner hair cells** are arranged in single rows and are **few in number.**
- **Outer hair cells** are arranged in parallel rows and are **greater in number** than the inner hair cells .
- **The spiral ganglion** contains the cell bodies of the auditory nerve (cranial nerve [CN] VIII), which synapse on the hair cells .

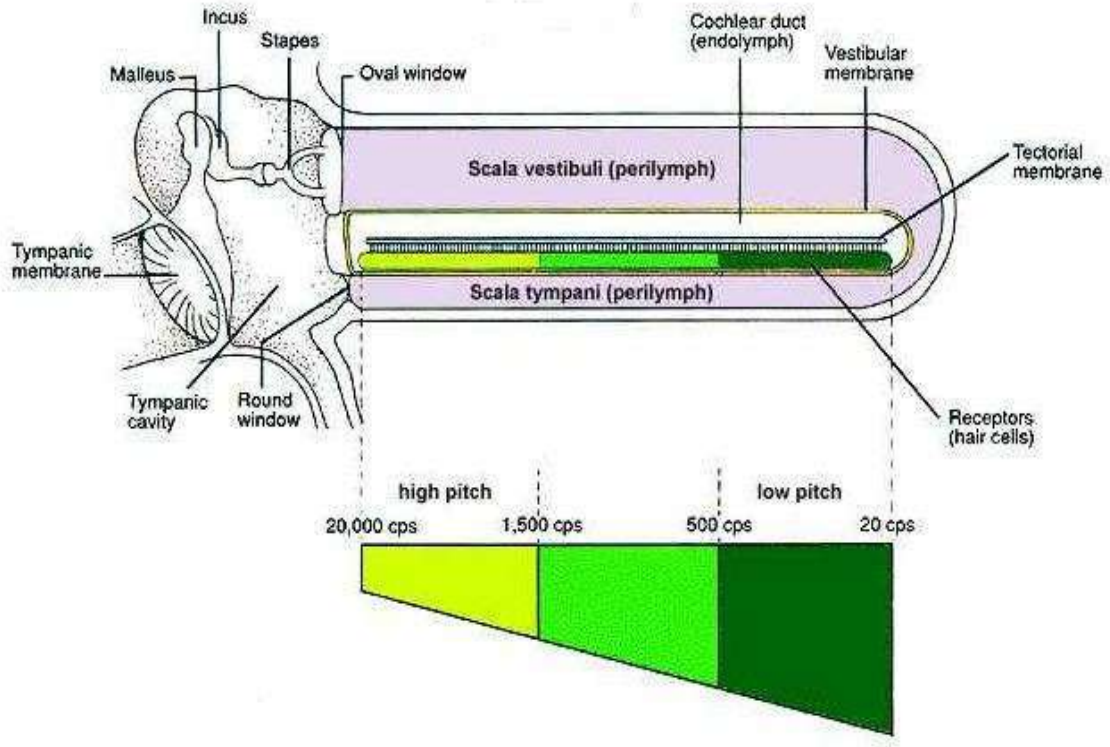
[C] Steps in auditory transductionby the organ ofCorti

- The cell bodies of hair cells contact the **basilar membrane.**
- The cilia of hair cells are embedded in the **tectorial membrane.**

a) Sound waves cause **vibration** of the organ of Corti. Because the basilar membrane is more elastic than the tectorial membrane, vibration of the basilar membrane causes the hair cells to bend by a shearing force as they push against the tectorial membrane.

b) **Bending of the cilia** causes changes in **K+ conductance** of the hair cell membrane. Bending in one direction causes depolarization; bending in the other direction causes hyperpolarization. The oscillating potential that results is the **cochlear microphonic potential.**

c) The oscillating potential of the hair cells causes intermittent firing of the cochlear nerves.



[D] How sound is encoded

- The frequency that activates aparticularhaircelldependsonthe location of the hair cell along the basilar membrane.

a) The base of the basilar membrane (near the oval and round windows) is narrow and stiff. It responds best to **high frequencies.**

b) The apex of the basilar membrane (near the helicotrema) is wide and compliant. It responds best to **low frequencies**.

[E] Central auditory pathways

- Fibers ascend through the laterallemniscustothe **inferior colliculus** to the medial geniculate nucleus of the thalamus to the **auditory cortex.**
- Fibers may be **crossed or uncrossed**.
- As a result, a mixture of ascending auditory fibers represents both ears at all higher levels.

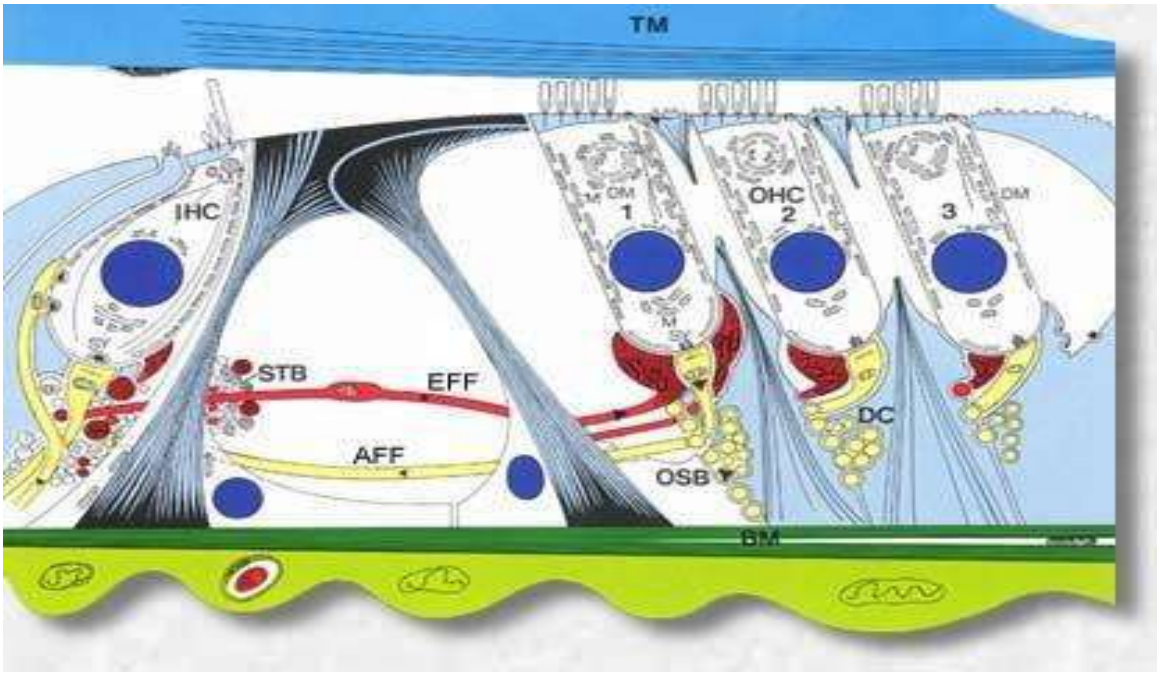
- Therefore, lesions of the cochlea of one ear cause unilateral deafness, but more central unilateral lesions do not.
- There is **tonotopic representation** of frequencies at all levels of the central auditory pathway. Discrimination of complex features (e.g., recognizing a patterned sequence) is a property of the cerebral cortex.

Resting membrane potential and receptor potential of hair cells:

- Haircellarethereceptorsforhearing.Theyareattachedtothefibersoftheauditory nerve.
- The resting membrane potential of these cells is called **endolymphatic potential**. It is positive potential and equals 120mv.

- Thereceptor potential is called **cochlear microphonic** potential and follows the nature of thestimulating sound in frequency and amplitude.

- Figure 7inner and outer hair cells and afferent and efferent nerves



Functions Of The Outer Hair Cells:

- Theinner hair cells are theprimary sensory receptors that generate action potentials in the auditory nerves and are stimulated by the fluid movements.
- On the other hand, the outer hair cells respond to sound like the inner hair cells, but depolarization makes them shorten and hyperpolarization makes them lengthen.

- They do this over a very flexible part of the basal membrane, and this action increases the amplitude and clarity of sounds.

- Thus, outer hair cells amplify sound vibrations entering the inner ear from the middle ear.
- These changes in outer hair cells occur in parallel with changes in a membrane protein prestin, the motor protein of outer hair cells.

- The olivocochlear bundle is a prominent bundle of efferent fibers in each auditory nerve that arises from both ipsilateral and contralateral superior olivary complexes and ends primarily around the bases of the outer hair cells of the organ of Corti.

- Theactivity in this nerve bundle modulates the sensitivity of the hair cells via the release of acetylcholine.

- Theeffect is inhibitory, and it may function to block background noise while allowing other sounds to be heard.

FUNCTIONS OF CORTICAL

[A] Auditory Centers

- Cortical auditory centers are concerned with the perception of auditory impulses, analysis of pitch and intensity of sound and determination of source of sound.
- Areas 41 and 42 are concerned with the perception of auditory impulses only. However, analysis and interpretation of sound are carried out by Wernicke area, with the help of area.

APPLIED PHYSIOLOGY –EFFECT OF LESION:

a) Lesionof cochlear nervecausesdeafnessof the ear.

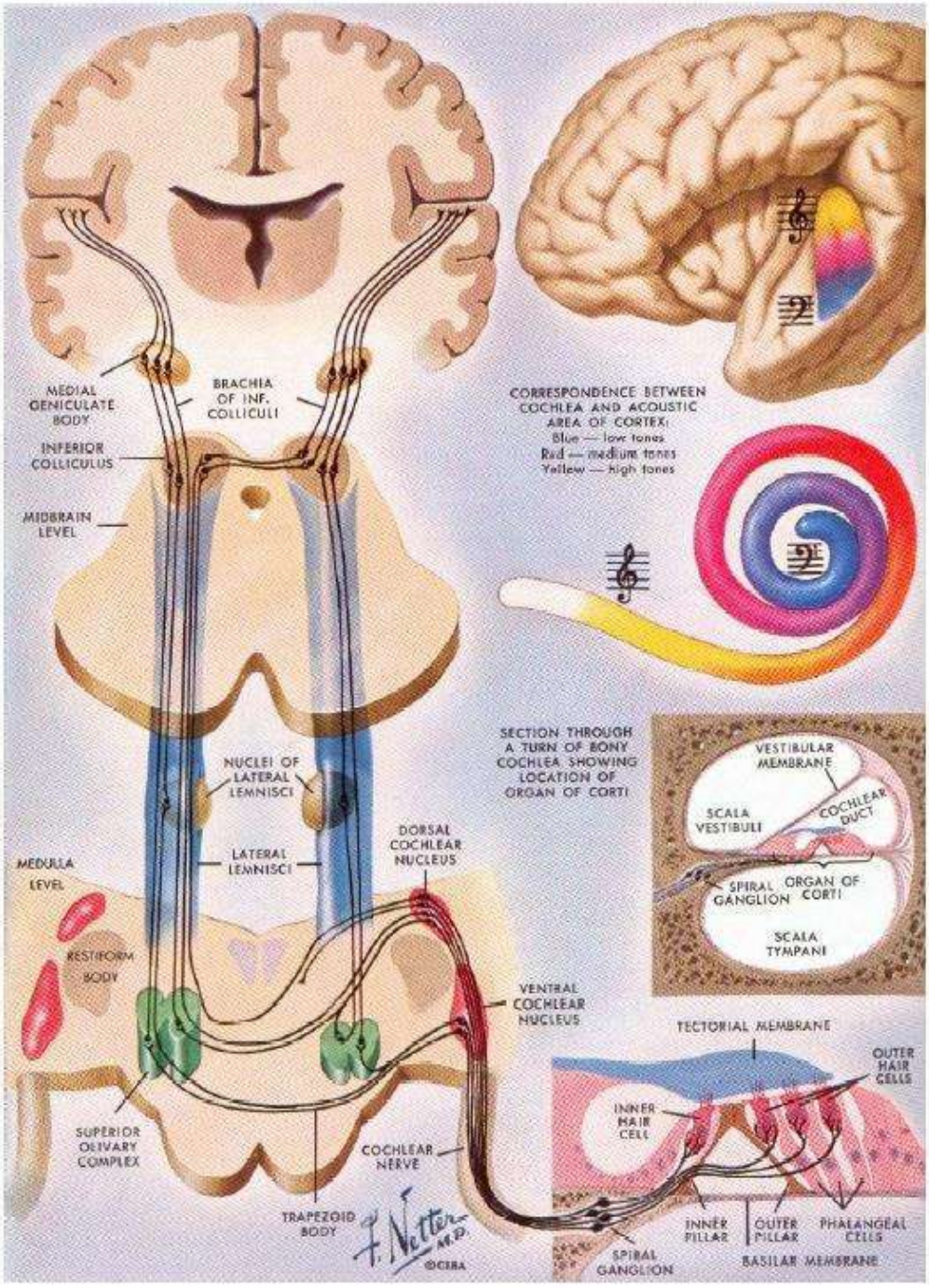
b) Unilateral lesion of auditory pathway, above the level of cochlear nuclei causes diminished hearing.

c) Degeneration of hair cells in the organ of Corti leads to presbycusis. Presbycusis is the gradual loss of hearing. It is common in old age.

d) Lesion in superior olivary nucleus results in poor localization of sound.

	Inner hair cells	Outer hair cells
Arrangement	• single row on cells inner to rods of corti	• 3-5 rows outer to rods of corti
Connection	• Connected mainly to afferent neurons	• Connected mainly to efferent neurons
Function	• Main receptor of hearing which transduce mechanical sound energy into electrical action potential signals	• Regulate sensitivity of the coclea toward certain sound, as it amplifies the movement of basilar membrane at specific regions according to the stimulating sound, a function called fine tunnig or filtration of signals or noise cancellation

Auditory pathway



3 Physiology of Smell & Taste

INTRODUCTION

- Smell (**olfaction**) and taste (**gustation**) are examples of visceral senses because of their close association with gastrointestinal function.
- Physiologically, they are related to each other as the flavor of food is a combination of its taste and smell .
- This is why food may taste "different" if one has a cold that depresses the sense of smell. Smell and taste receptors are chemoreceptors that are stimulated by chemical molecules in solution in mucus in the nose (**odorants**) and saliva in the mouth (**tastants**).
- The sensations of smell and taste likely evolved as protective mechanisms to avoid the intake of potentially harmful substances.

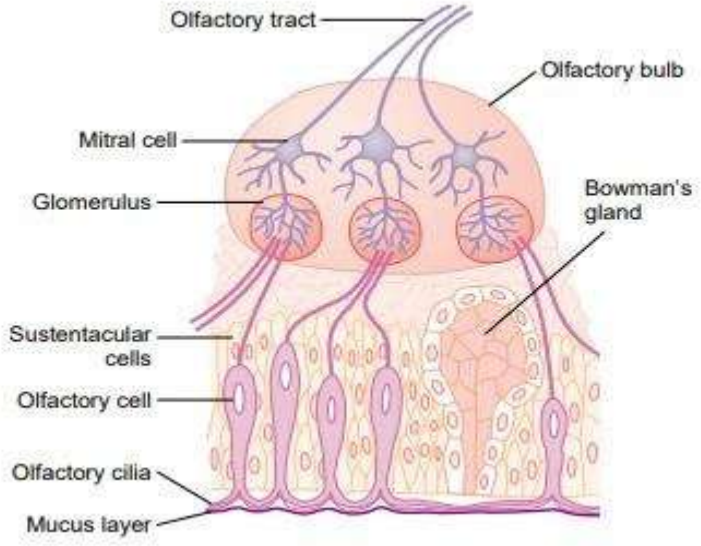
SMELL OLFACTORY EPITHELIUM:

- The yellowish pigmented **olfactory epithelium** is a specialized portion of the nasal mucosa that covers an area of 10 cm2 in the roof of the nasal cavity near the septum in humans.

- The olfactory epithelium is the place in the body where the nervous system is closest to the external world.

- **It contains three types of neurons that are important for olfaction:**
  olfactory sensory neurons, supporting cells, and basal stem cells. at the base of the epithelium.
- Each olfactory sensory neuron has a dendrite that projects to the epithelial surface.
- Numerous cilia protrude into the mucus layer lining the nasal lumen.
- Odorants bind to specific odorant receptors on the cilia and initiate a cascade of events leading to generation of action potentials in the sensory axon.

- Each olfactory sensory neuron has a single axon that projects to the olfactory bulb, a sma ovoid structure that rests on the cribriform plate of the ethmoid bone .



- Thebipolar olfactory sensory neurons (also called olfactory receptor cells) are responsible forolfactory transduction.

- They have a short, thick dendrite that projects into the nasal cavity where it terminates in a knobcontaining **6–12 cilia** that protrude into the thin layer of mucus overlying the epithelium.

- Theaxons of the olfactory sensory neurons (ie**, olfactory nerve**) pass through the cribriform plate of the ethmoid bone to enter the **olfactory bulbs.**

- Thesupporting cells secrete the mucus that provides the appropriate molecular and ionic environment for odor detection in the olfactory epithelium.

- Odor-producing molecules (**odorants)** dissolve in the mucus and bind to **odorant receptors** on the cilia of the olfactory sensory neurons.

- **Odorant- binding proteins** in the mucus may facilitate the diffusion of odorants to and from the odorant receptor.

- Basal stem cells undergo mitosis to generate new olfactory sensory neurons as needed to replace those damaged by exposure to the environment; olfactory sensory neurons generally survive for only 1–2 months .

ODORANT RECEPTORS & SIGNAL TRANSDUCTION :

- The olfactorysystem candiscriminate perhapsmorethan 1 million distinct odors due in part to the existence of many different functional odorant receptors

- There are about 1000 olfactory genes in humans, accounting for 3% of the human genome; approximately 400 of these genes function as odorant receptors.

- The amino acid sequences of odorant receptors are very diverse, but all are **G protein-coupled receptors (GPCRs).**

- When an odorant molecule binds to its receptor, the G-protein subunits (α, β, γ) dissociate

- Theα-subunit activates adenylyl cyclase to catalyze the production of cAMP which acts as a second messenger to open cation channels, increasing the membrane permeability to Na+,K−, and Ca2+.

- Thenet effect is an inward-directed Ca2+ current which produces the **graded receptor potential.**

- Thisthen opens Ca2+- activated Cl− channels, further depolarizing the cell due to the high intracellular Cl− levels in olfactory sensory neurons.

- If the stimulus is sufficient for the receptor potential to exceed its threshold, an action potential in the olfactory nerve (1st cranial nerve) is triggered.

OLFACTORY SENSORY PATHWAY:

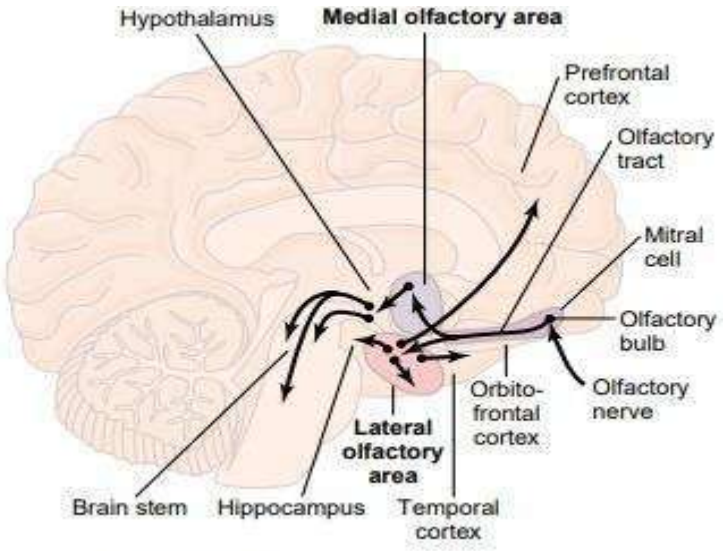
- Intheolfactory bulb,theaxonsof theolfactory sensory neurons synapse on the primary dendrites of the **mitral cells and tufted cells** to form **olfactory glomeruli.**

- Each olfactory sensory neuron expresses only one of the 400 functional olfactory genes, but each odorant can bind to a large pool of odorant receptors.
- Each olfactory sensory neuron projects to only one or two glomeruli.

- This provides a distinct two-dimensional map in the olfactory bulb that is unique to the odorant.

- Themitral cells with their glomeruli project to different parts of the olfactory cortex.

- Thecentral olfactory system is able to decode the pattern of receptor-cell activity that signals the identity of the odorant.



- The olfactory bulbs also contain **periglomerular cells,** which are inhibitory neurons connecting one glomerulus to another, and **granule cells**, which have noaxons and make reciprocal synapses with the lateral dendrites of the mitral and tufted cells.

- Atthese synapses, the mitral or tufted cells excite the granule cell by releasing **glutamate,** andthe granule cells in turn inhibit the mitral or tufted cell by releasing **GABA.**

- Thelateral inhibition mediated by periglomerular and granule cells functions to sharpen andfocus olfactory signals.

- Theaxons of the mitral and tufted cells pass posteriorly through the **lateral olfactory stria** to terminate on apical dendrites of pyramidal cells in **five regions of the olfactory cortex**: **anterior olfactory nucleus, olfactory tubercle, piriform cortex, amygdala, and entorhinal cortex** .

- From these regions, information travels directly to the frontal cortex or via the thalamus to theorbitofrontal cortex.

- Conscious discrimination of odors is dependent on the pathway to the orbitofrontal cortex. The pathway to the amygdala mediates the emotional responses to olfactory stimuli, and the pathway to the entorhinal cortex is concerned with olfactory memories. epithelium located along the nasal septum in the **vomeronasal organ** that is concerned with the perception of **pheromones.**

- Vomeronasal sensory neurons project to the **accessory olfactory bulb** and from there to regions of the amygdala and hypothalamus that are concerned with reproduction and ingestive behavior.

- Thevomeronasal organ has about 100 G-protein-coupled odorant receptors that differ in structure from those in the rest of the olfactory epithelium.

TASTE

TASTE BUDS:

- Thespecialized sense organ for taste (gustation) consists of about **5000 taste buds** located primarily on the **papillae** of the dorsal surface of the tongue in humans.

- The **fungiform papillae** are rounded structures most numerous near the tip of the tongue; the **circumvallate papillae** are prominent structures arranged in a V on the back of the tongue; the **foliate papillae** are on the posterior edge of the tongue.

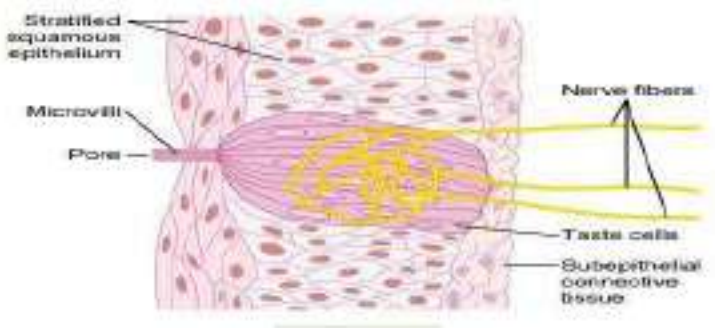
- Each fungiform papilla has up to five taste buds, mostly located at the top of the papilla, while each circumvallate and foliate papilla contain up to 100 taste buds, mostly located along the sides of the papillae.

- Taste buds are also located in the soft palate, epiglottis, and pharynx.

- Each taste bud contains **50–100 taste receptor cells** and numerous **basal cells and support** cells . The taste receptor cells are modified epithelial cells that respond to chemical stimuli or tastants.

- Theapical ends of taste cells have **microvilli that project into the taste pore**, a small opening on the dorsal surface of the tongue where tastes cells are exposed to the oral contents. Saliva in the oral cavity acts as a solvent for tastants; after dissolving, the chemical diffuses to the taste receptor sites.

- Saliva may also function to cleanse the mouth to prepare the taste receptors foranew stimulant.



- Each taste bud is innervated by about 50 nerve fibers, and conversely, each nerve fiber receives input from an average of five taste buds. **The basal cells** arise from the epithelial cellssurrounding the taste bud.

- Theydifferentiate into new taste cells as taste cells survive for only about 10 days. If the sensory nerve is cut, the taste buds it innervates degenerate and eventually disappear.

TASTE PATHWAYS:

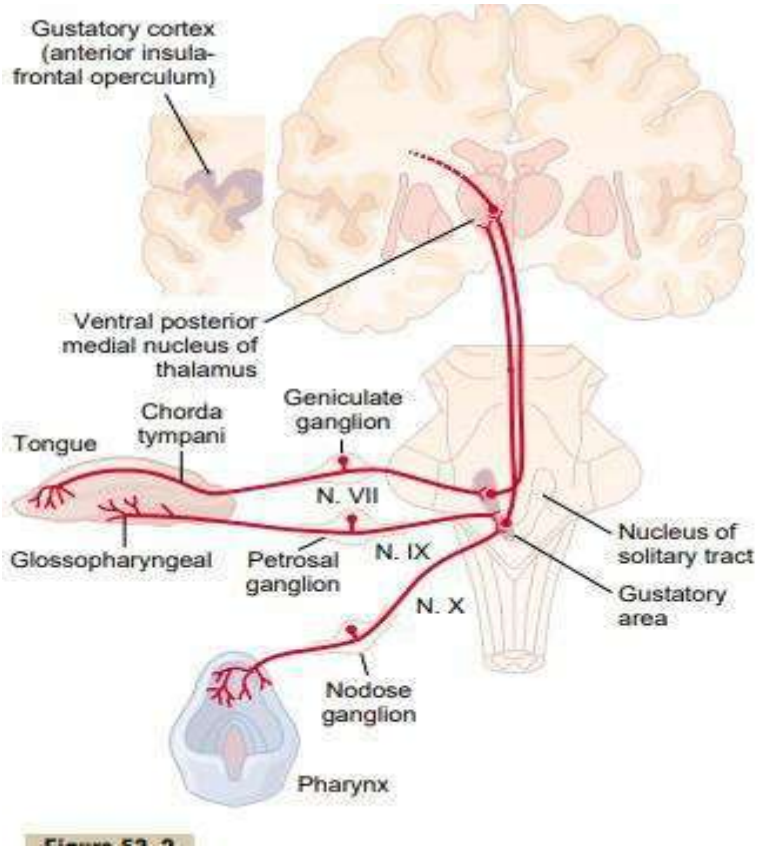
- Thesensory nervefibers from the taste buds on the anterior two-thirds of the tongue travel in **the chorda tympani branch of the facial nerve**, and those from the posterior third of the tongue reach the brainstem via the **glossopharyngeal nerve**.

- The fibers from areas other than the tongue (eg, pharynx) reach the brain stem via the **vagus nerve**. On each side, the myelinated but relatively slowly conducting taste fibers in these three nerves unite in the gustatory portion of the **nucleus of the tractus solitarius (NTS) in the medulla oblongata.**

- From there, axons of second-order neurons ascend in the ipsilateral medial lemniscus and project directly to the ventral **posteromedial nucleus of the thalamus**. From the thalamus, the axons of the third-order neurons pass to neurons in the **anterior insula** and the frontal operculum in the ipsilateral cerebral cortex.

- Thisregion is rostral to the face area of the postcentral gyrus, which is probably the area thatmediates conscious perception of taste and taste discrimination. Sensory fibers in the trigeminal (5th cranial) nerve also innervate the tongue and contribute to the burning sensation experienced when we eat foods containing capsaicin.

- Taste buds are surrounded by TRPV1 receptors on trigeminal nociceptive fibers that are activated in response to eating spicy foods.



TASTE MODALITIES, RECEPTORS, & TRANSDUCTION:

- Humans havefive basictastemodalities:**salt, sweet,sour,bitter, andumami.** Common
- stimuli for these sensory modalities are sodium chloride, sucrose, hydrochloric acid, quinine, and monosodium glutamate, respectively.

- All tastants are sensed from all parts of the tongue and adjacent structures. Afferent nerves to the NTS contain fibers from all types of taste receptors, without any clear localization of types. An individual taste receptor cell may respond to more than one type of tastant.

- The central nervous system can distinguish the various tastes from one another because each type of taste receptor cell connects to a particular gustatory axon. the putative receptors for the five modalities of taste.

- They include the two major types of receptors: **ligand-gated channels** (ionotropic receptors) and GPCRs (metabotropic receptors). Salt and sour tastes are triggered by

activation of ionotropic receptors; sour, bitter, and umami tastes are triggered by activation of metabotropic receptors. Many GPCRs in the human genome are taste receptors (T1Rs, T2Rsfamilies.) Salt-sensitive taste is mediated by an **epithelial sodium channel** (ENaC.)

- Theentry of Na+ into the salt receptor depolarizes the membrane, generating a receptor potential. The sour taste is triggered by protons (H+ ions). ENaCs permit the entry of protons and may contribute to the sensation of sour taste.

- TheH+ ions can also bind to and block a K+-sensitive channel. The fall in K+ permeability candepolarize the membrane. Also, a hyperpolarization-activated cyclic nucleotide-gated cation channel (HCN) and other mechanisms may contribute to sour transduction.

- Substances that taste sweet are detected by at least two types of GPCRs, T1R2 and T1R3. Sugars taste sweet, but so do compounds such as saccharin that have an entirely different structure.

- Natural sugars such as sucrose and synthetic sweeteners may act on gustducin via different receptors. Sweet-responsive receptors act via cyclic nucleotides and inositol phosphate metabolism.

- Bitter taste is produced by a variety of unrelated compounds. Many of these are poisons, and bitterness serves as a warning to avoid them. Some bitter compounds (eg, quinine) are membrane permeable and bind to and block K+- selective channels.

- Many bitter tastants (eg, strychnine) bind to GPCRs (T2R family) that couple to the heterotrimeric G-protein, gustducin. **Gustducin l**owers cAMP and increases the formation ofinositol phosphates (IP3), which releases Ca2+ to trigger depolarization.

- Umami tastants activate a receptor comprised of T1R1 and T1R3 .

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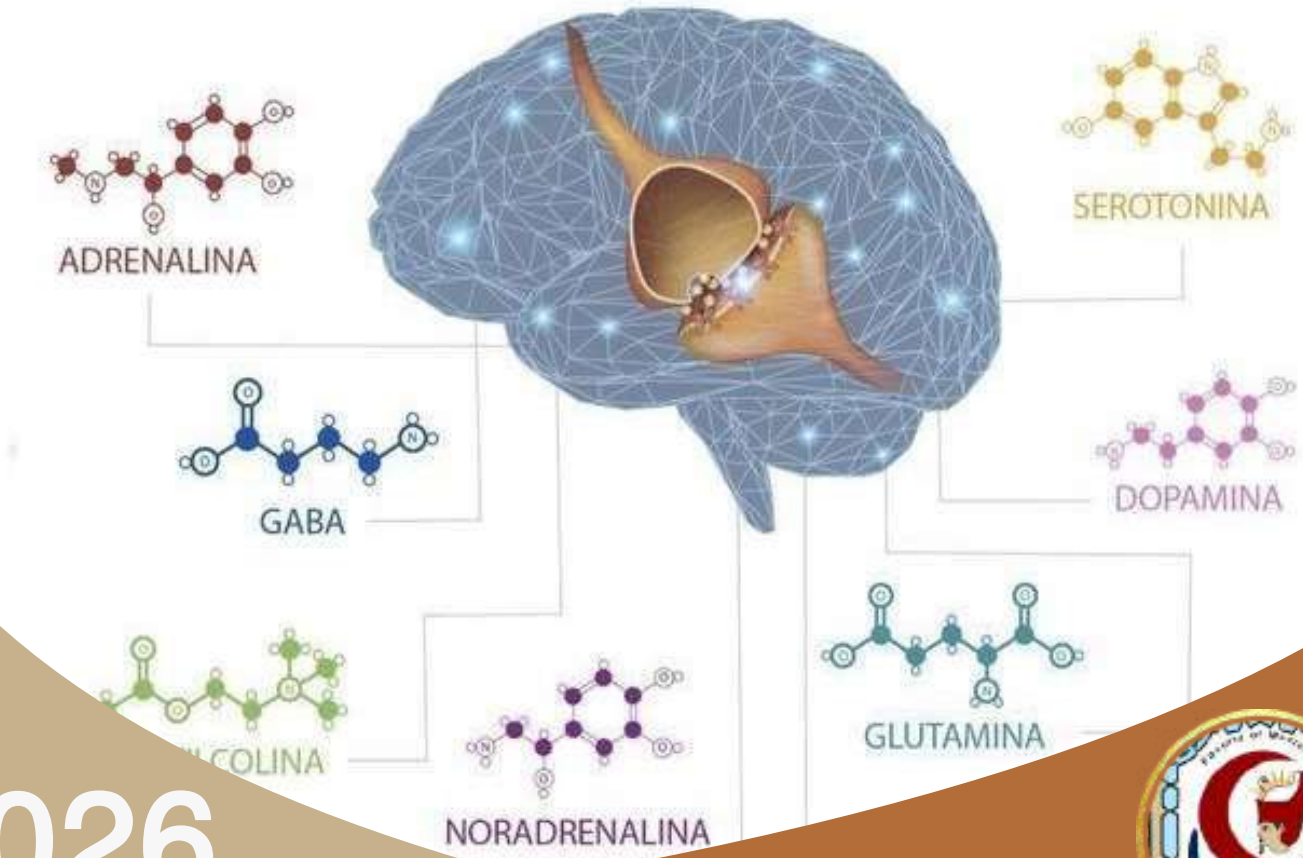
SEVENTH EDITION

2025

Neurotransmisores



2026

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Neuroscience

Biochemistry

NEU-316

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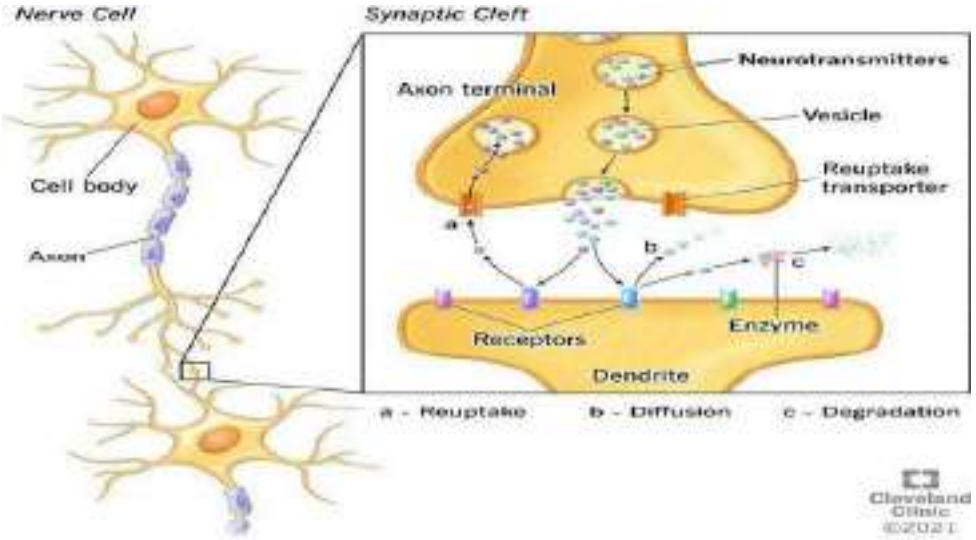
1 Medical Biochemistry

1) Neurotransmitters

Aneurotransmitter is a signaling molecule secreted by a neuron to affect another cell across a synapse. The cell receiving the signal, or target cell, may be another neuron, but could also be a gland or muscle cell. The response produced by a neurotransmitter is mediated by a neurotransmitter receptor capable of recognizing it.



Classification

Inhibitory	Excitatory
• GABA	• Acetylcholine
• Serotonin	• Epinephrine
• Dopamine	• Norepinephrine
	• Histamine
	• Glutamate
	• Dopamine

Dopamine is bothexcitatory and inhibitory.

Excitatory neurotransmitters function to activate receptors on the postsynaptic membrane and enhance the effects of the action potential, while inhibitory neurotransmitters function to prevent an action potential. In addition to the above classification, neurotransmitters can also be classified based on their chemical structure:

Amino acids	Monoamines
• GABA • Glutamate	• Serotonin • Histamine
	Catecholamines (subcategory of monoamines): • Dopamine • Norepinephrine • Epinephrine

Properties

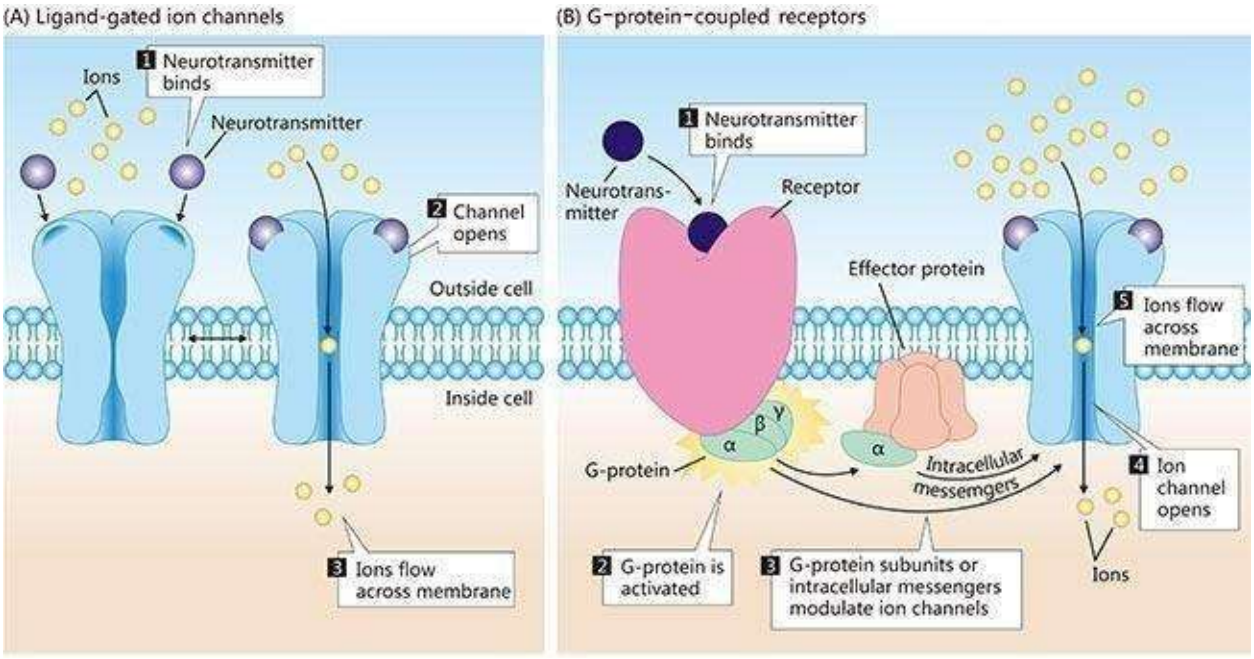
Neurotransmitters are usually considered to be endogenous substances that are released from neurons, act on receptor sites that are typically present on membranes of postsynaptic cells and produce a functional change in the properties of the target cell. Over the years general agreement evolved that several criteria should be met for a substance to be designated a neurotransmitter:

1. A neurotransmitter must be synthesized by and released from neurons. Thus, the presynaptic neuron should contain both the transmitter, and the appropriate enzymes needed to synthesize that neurotransmitter.
2. The substance should be released from nerve terminals.
3. A neurotransmitter should reproduce at the postsynaptic cell the specific events (such as changes in membrane properties) that are seen after stimulation of the presynaptic neuron.
4. The effects ofneurotransmitter can be blocked by competitive antagonists of the receptor for thattransmitter. In addition, the effects of presynaptic stimulation can be inhibited byinhibiting the synthesis of the transmitter.
5. There should be active mechanisms to terminate the action of the neurotransmitter.Among such mechanisms are the uptake of the transmitter by the presynapticneurons or glial cells through specific transporter molecules and enzymatic inactivation of the chemical messenger.

Types of neurotransmitter receptors

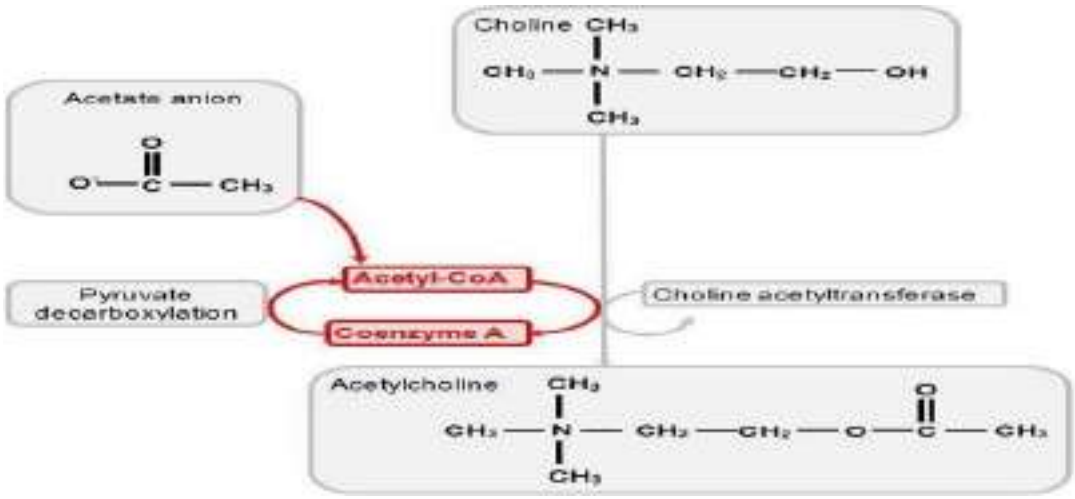
Ionotropic means that ions can pass through the receptor

metabotropic means that a second messenger inside the cell relays the message (i.e.metabotropic receptors do not have channels).

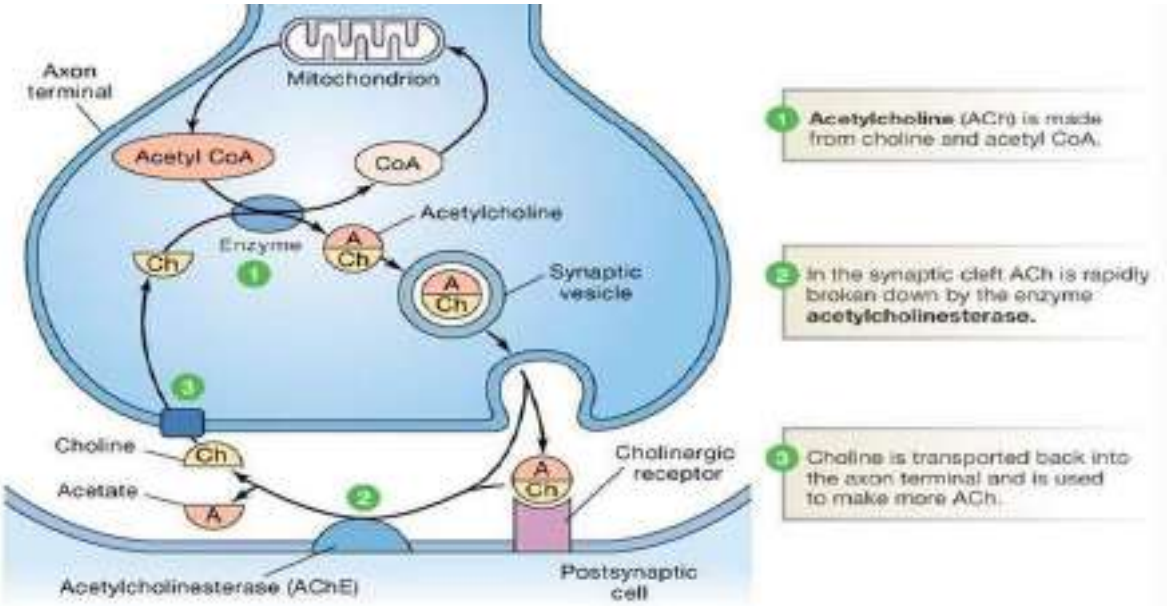


Acetylcholine

- Acetylcholine is an organic chemical that functions in the brain and body of many types of animals, as a neurotransmitter.
- Acetylcholine is a neurotransmitter of the parasympathetic autonomic nervous system and of the sympathetic ganglia
- Its name is derived from its chemical structure; it is an ester of acetyl coenzyme A and choline.
- The synthesis is a single step reaction catalyzed by choline-acetyltransferase.



Storage: After its synthesis, acetylecholine is packaged into vesicles by means of a carrier protein on the vesicle membrane called vesicular Ach transport (VACHT). Once packaged in vesicles, acetylcholine is stored at the nerve ending until an action potential arrives and allows for its release into the synaptic cleft. **Degradation:** Ach is released from the nerve into the synaptic cleft and binds to Ach receptors on the post- synaptic membrane, relaying the signal from the nerve. Ach-esterase, is located on the post-synaptic membrane, terminates the signal transmission by hydrolyzing Ach into choline and acetate. The liberated choline is reuptaken by the pre-synaptic membrane and used for resynthesis of Ach.



Catecholamines

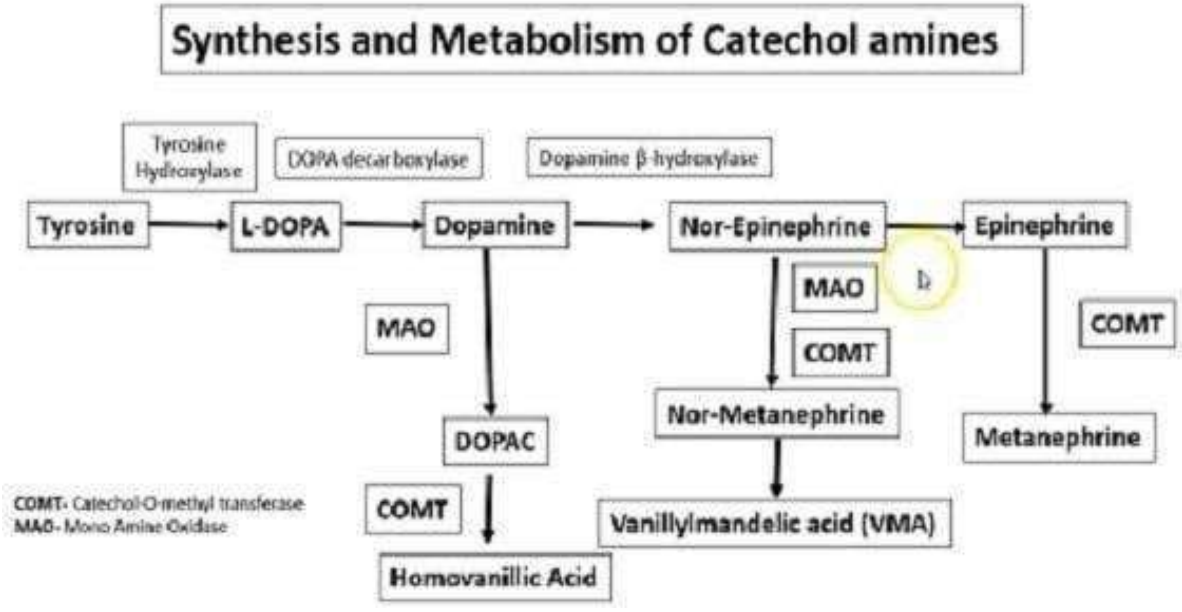
The class of endogenous catecholamines includes dopamine, norepinephrine, and epinephrine. These molecules have key roles in mediating the sympathetic nervous response and several neurologic pathways. Norepinephrine produces many effects in the body, the most notable being those associated with the **'fight or flight'** response to perceived danger.

Dopamine is hypothalamic neurotransmitter derived from tyrosine amino acid which is obtained from the diet. Dopamine is a precursor for noradrenaline and adrenaline.

The rate limiting step in catecholamines synthesis is hydroxylation of tyrosine by tyrosine hydroxylase which is found only in the cytosol of catecholamine-containing cells; in the adrenal medulla and in a few brain regions.

Norepinephrine is stored in vesicles in the nerve terminals, which concentrate it and protect it from metabolism until it is released following nerve stimulation. The major mechanism by which the effects of norepinephrine are terminated is reuptake back into the nerve terminal. Norepinephrine can also be metabolized to inactive products.

Norepinephrine is metabolized by the enzymes monoamine oxidase and catechol-O- methyltransferase.



The degradation pathways have clinical significance. **Parkinson disease:** is associated with insufficient dopamine. When pharmacological therapy is necessary, MAO & COMT inhibitors drugs decrease the rate of dopamine degradation. **Schizophrenia:** as opposed to Parkinson disease, is associated with excess dopamine. The primary treatment is with D2 receptor blockers. **Pheochromocytomas:** are adrenal medullary tumors associated with the production of large amounts of catecholamines. They can manifest as episodes of tachycardia, hypertension, and headache. The diagnosis of pheochromocytomas and other catecholamine-producing tumors requires detecting metabolites HVA and VMA in the plasma and urine.

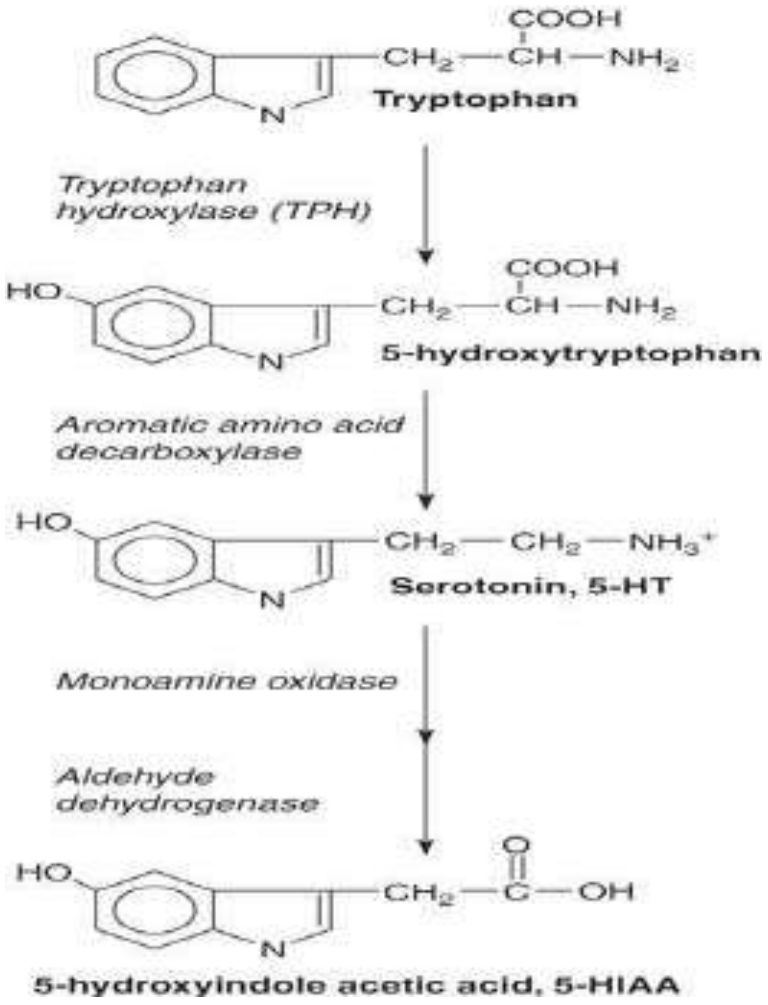
Serotonin

Serotonin, also known as 5-hydroxytryptamine (5-HT), is a monoamine neurotransmitter. It also acts as a hormone. It is principally found stored in three main cell types:

i) serotonergic neurons in the CNS and in the intestinal myenteric plexus.
ii) enterochromaffin cells in the mucosa of the gastrointestinal tract.
iii) and in blood platelets.

Serotonergic neurons and enterochromaffin cells can synthesize serotonin from its precursor amino acid L-tryptophan, whereas platelets rely upon uptake of serotonin for their stores. Serotonin mediatessatisfaction, happiness and optimism. Serotonin levels are reduced in depression, and most modern anti-depressant drugs, known as serotonin reuptakeinhibitors (SSRIs), act by increasing the amount of serotonin available to braincells. But too much serotonin causes signs and symptoms that can range from mild(shivering and diarrhea) to severe (muscle rigidity, fever and seizures). Severeserotonin syndrome can cause death if not treated.

Synthesis of serotonin: The biochemical pathway for serotonin synthesis initially involves the conversion of L- tryptophan to 5-hydroxytryptophan by the enzyme L-tryptophan hydroxylase (TPH), which has been found both in cytosolic and particulate brain cell fractions. This enzyme provides the rate- limiting step for serotonin synthesis. **Metabolism of serotonin:** Metabolism of serotonin is carried out primarily by the outer mitochondrial membrane enzyme monoamine oxidase (MAO).



2) Brain Metabolism

Energy is needed for:

1- Excitation and Conduction, which are reflected in the unceasing electrical activity of Cerebral tissue.

2- Electrical energy is derived from chemical processes.

3- Energy consumption used for active transport of ions to sustain and restore membrane potentials.

Substrates used for energy production in cerebral tissue

- Glucose is the major substrate for energy production in cerebral tissue. Cerebral tissue utilizes glucose directly from arterial blood. About 0.1% Glycogen is stored in Cerebral tissue. Insulin is not required for uptake of glucose by cerebral tissue. When blood glucose is low, Glycogen stored in the brain is used to maintain cerebral metabolism for a very short duration. Mannose can easily cross the blood–brain barrier (BBB) and be converted to Fructose-6- Phosphate then joins glycolysis Mannose is not normally present in the blood and cannot therefore be considered as a substrate for cerebral energy metabolism. Fructose, Galactose, Lactate and Pyruvate have limited permeability across the BBB, therefore cannot directly serve as substrates for cerebral energy metabolism; Lactate and Pyruvate when formed within the BBB are useful metabolic substrates for cerebral metabolism.

Metabolism of carbohydrates in cerebral tissue

- Aerobic and Anaerobic Glycolysis occurs in cerebral tissue. HMP shunt occurs in cerebral tissue mainly for production of NADPH, required for biosynthesis of Fatty acids and Steroids. Major substrate used in Blood glucose. Other Carbohydrates (Maltose, Fructose, Galactose, Hexose-phosphates) and Intermediate metabolites (Lactate, Pyruvate and Glyceraldehydes) are used only after their conversion to Glucose via Gluconeogenesis in the liver. These compounds act by raising Blood Glucose Level

Significance of O2 supply to brain energy metabolism

- Brain represents 2 to 3% of total body weight of an average adult, however it utilizes 20 to 25% of the total Oxygen consumed.
- In children up to 4 years of age, the brain utilizes about 50% of the total Oxygen consumed.
- Cerebral tissue utilizes Oxygen more than other tissues.
- Example, it utilizes about 20 times more Oxygen than muscle tissue at rest.
- Oxygen consumption varies throughout the brain:
- Grey matter utilizes about twice more Oxygen than White matter
- Sleep reduces cerebral Oxygen uptake by only 3%.
- Amount of Oxygen stored in the brain is extremely small compared to the rate of utilization.
- Brain requires continuous replenishment of its Oxygen via the circulation.
- Consciousness is lost when Cerebral blood flow is interrupted.

Reduced cerebral Oxygen uptake occurs under certain conditions that lead to depressed consciousness

- Insulin inducedhypoglycemia.
- Diabetic coma.
- Cerebral tumors.
- Uremia, hepaticcoma.
- Exposure to depressant drugs used during surgery.

Uses of O2 consumed by cerebral tissue

- Energy metabolism via Oxidative Phosphorylation
- Maintenance of energy components in BBB
- Impulse transmission (Ion pumps).
- Signal transduction.
- Functioning of specific enzyme systems: Mixed Functional Oxygenases used in the biosynthesis of Neurotransmitters and other biologically active compounds.

Theamino acid content in cerebral tissue

- Very high conc. of free amino acids compared to plasma; Highest amount of free Glutamate, compared to any other mammalian tissue.
- Some unusual amino acids: e.g. Gamma Aminobutyrate (GABA), N- Acetyl-Aspartate and Cystathione.
- GABA is an inhibitory neurotransmitter that acts by increasing the passage of Chloride ions through the Postsynaptic membrane of Neurons.
- Glutamate is involved in several metabolic processes: Biosynthesis of GABA, Detoxification of Ammonia and as Neurotransmitter.

Lipids in brain

The brain has the second highest lipid content behind adipose tissue, and brain lipids constitute 50% of the brain dry weight the brain. The brain utilizes acylated lipids to generate phospholipids for cell membrane.

1-Fatty Acids of the brain: rich in long-chain polyunsaturated fatty acids (LCPUFAs), particularly arachidonic acid (AA), eicosapentaenoic acid, and docosahexaenoic acid (DHA).

2-Cholesterol: an essential component of neuronal physiology during both development and adulthood. Cholesterol depletion in neurons impairs vital functions, including synaptic vesicle exocytosis, neuronal activity, and neurotransmission and results in synaptic loss and neurodegeneration . Clinically, deficits in cholesterol homeostasis in the CNS manifest as severe primary neurological disorders such as Neimann–Pick C disease and Parkinson's disease.

3-Apolipoprotein E-containing lipoproteins: ApoE play a major role in glia neuronal lipid metabolism and acts as a ligand for multiple receptors in neurons, which interact with a number of downstream physiological processes.

Effects of hypoxia on cerebral metabolism

- After a brief period of hypoxia:
- There is a drastic slowdown in Oxidative Phosphorylation.
- Rate of Glycolysis is increased.
- Lactic acid production is increased, which can consequently lead to intracellular acidosis.
- These changes are due to the Pasteur Effect.
- Inhibition of Glycolysis in the presence of oxygen.
- Pasteur effect reflects the increased energy yield obtained via Aerobic metabolism of glucose as compared to Anaerobic metabolism.
- Hypoxia causes an increase in Glucose utilization from cerebral blood stream, followed by a decrease in cerebral glucose concentration.
- Resulting is an increase in Lactic acid production in cerebral tissue.
- Gradual increase in Spinal fluid. Lactate level occurs during hypoxia.

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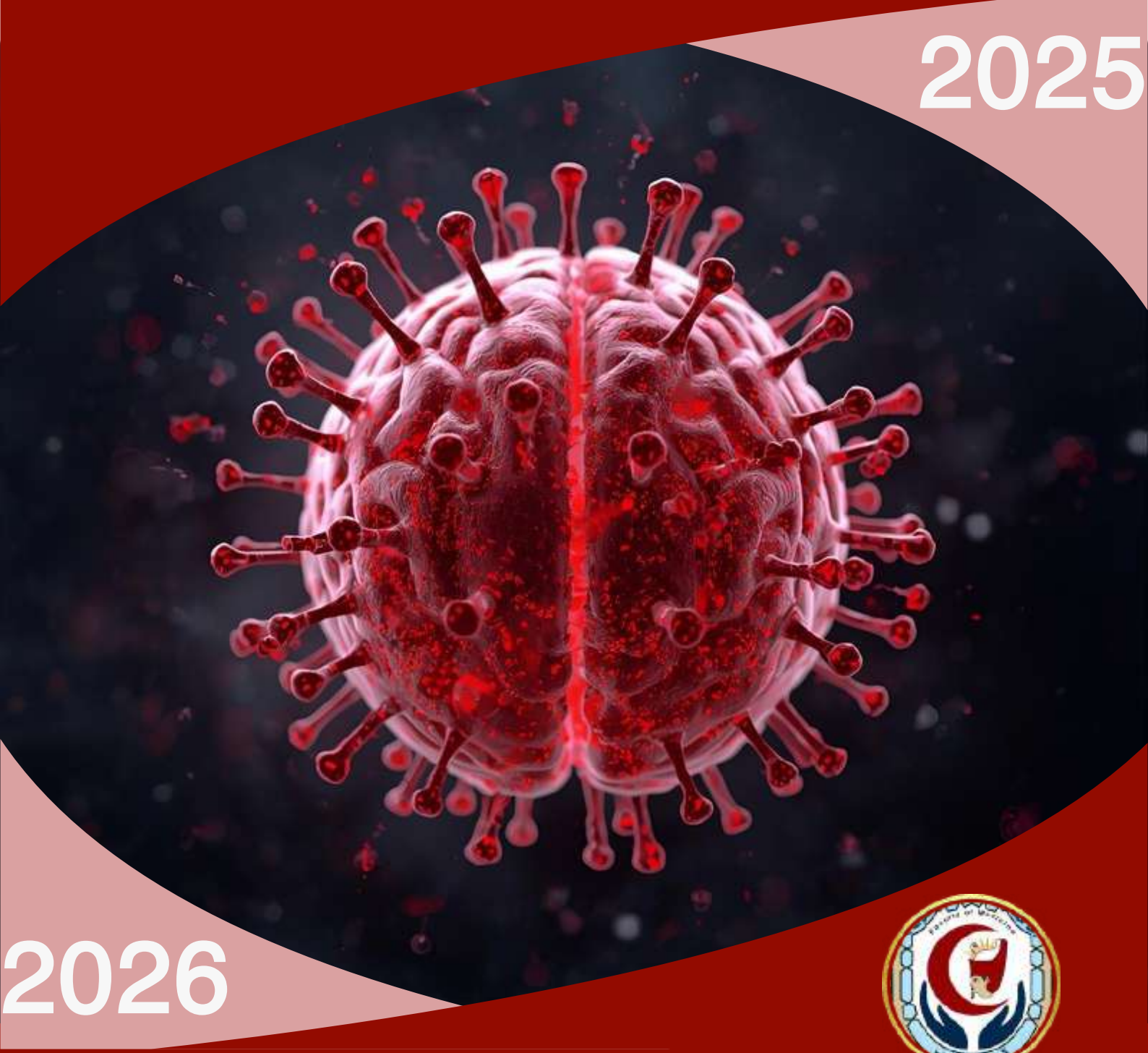

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Neuroscience

Microbiology & immunity

NEU-316

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Encephalitis

1 Definition

Encephalitis is an infection of the brain parenchyma predominantly caused by viruses. Sometimes both the brain and the meninges are involved, a condition called meningoencephalitis.

2 Causes

- **1- Viruses are the main causes of encephalitis:**
  - **- Human herpes viruses:**
  a) Herpes simplex type I and 2
  b) Varicella-zoster virus (VZV)
  c) Cytomegalovirus(CMV)
  d) Epstein-Barr- virus
  - **- Animal bite** (dog, cat, bat): Rabies virus.
  - **- Arthropod-borne viruses (Arboviruses):** West Nile virus (WNV).
  - **-Rash-causing viruses:** mumps, measles and rubella.

- **2- Bacteria and fungi can cause infectious encephalitis more rarely.**
  - -Bacteria: mycoplasma, meningococci, pneumococci andlisteria.
  - -Fungi: histoplasma, cryptococcus and candida.

1- Human herpes viruses

a) Herpes simplex virus

Approximately 15% are caused by HSV-1. HSV-1 probably reaches the temporal lobe by travel down sensory neurons following activation of latent infection in the trigeminal ganglion. Neonates acquire HSV-2 during passage through the birth canal. HSV-2 then reaches the brain by hematogenous spread. Mothers with visible vesicular lesions are much more likely to have newborns with serious HSV-2 infections because the amount of virus present is significantly greater.

b) Varicella-Zoster virus (VZV)

VZV can cause encephalitis during the primary infection (varicella or chickenpox) or during the reactivation infection (zoster or shingles). VZV also causes post infectious encephalomyelitis involving the brain and spinal cord after resolution of the primary infection.

c) Cytomegalovirus (CMV)

It causes encephalitis primarily in immunocompromised individuals such as AIDS patients and those receiving drugs to prevent transplant rejection.

d) Epstein–Barr virus (EBV)

Encephalitis caused by Epstein–Barr virus (EBV) is a rare complication of infectious mononucleosis.

2- Rabies virus

Rabies virus also reaches the brain by axonal travel from the site of the animal bite.

3- Arbovirus encephalitis

Different encephalitis arboviruses in the togavirus and flavivirus groups are endemic in many parts of the world. In Egypt, two such viruses exist; the West Nile virus and the Sindbis virus which are transmitted by mosquitoes. Infection occurs through the bite of the infective mosquitoes and then travel to the brain via the bloodstream.

3 Clinical manifestations

The most characteristic clinical manifestations of encephalitis include fever, headache, and altered mental status, as well as seizures and focal neurologicdeficits. Rabies encephalitis has two clinical manifestations. Most cases of rabies (80%)

present with hyperactivity, agitation, delirium, hydrophobia, and seizures (called furious rabies). The other 20% of cases have paralytic symptoms in which an ascending paralysis without hyperactivity is the predominant feature (called dumb rabies). Coma and death is the final common pathway in both forms.

4 Diagnosis

1- In contrast to meningitis, CSF findings in encephalitis are more variable. A mild elevation in CSF lymphocytes can be seen along with an elevation of protein and a normal glucose. A normal CSF pattern can also be seen in encephalitis.

2- PCR-based testing of CSF is commonly used to determine a specific etiology, such as with HSV and VZV.

3- Serology: WNV encephalitis is often diagnosed by finding WNV-specific IgM in the spinal fluid. Rabies can be diagnosed by direct fluorescent antibody staining of a biopsy of skin from the nape of the neck.

5 Treatment

HSV-1, HSV-2, and VZV encephalitis: Intravenous acyclovir is the treatment of choice.

6 Prevention

- **Pre-exposure prophylaxis**with the killed vaccine should be given to veterinarians and others at risk of exposure.
- **Post-exposure prophylaxis** consists of both the killed vaccine and the hyperimmune globulins that contain a high titer of anti–rabies virus antibodies.
- **To reduce the transmission of HSV-2** to the neonate, pregnant women with active lesions late in pregnancy should receive acyclovir and should be considered for Cesarean section.

Meningitis

1 Definition

Meningitis is an infection of the meninges, the membranes that line the brain and spinal cord. Meningitis is a devastating disease with a high case fatality rate, which can lead to serious long-term complications.

2 Classification

- **1-Septicmeningitis (bacterial):**
- N. meningitidis, H. influenzae (the commonest cause in children aged 2-5 years)and Pneumococci are the commonest cause of pyogenic meningitis.
- Staphylococci, streptococci, pneumococci, E. coli, klebsiella, proteus, pseudomonas or anaerobes may be the cause, after trauma or surgery.
- Spirochetes: syphilis and leptospirosis are rare causes.
- Neonatal meningitis is caused by group B haemolytic streptococci(Str. agalactia), E. coli and other intestinal flora e.g. klebsiella and proteus. Listeria monocytogenes is a less common cause.

- **2-Aseptic meningitis:**
- M. tuberculosis meningitis.
- Viral meningitis: Enterovirusesi.e. coxsackieviruses, echoviruses, poliovirus and mumps virus arethe main viral causes. Arboviruses, herpesviruses, rubella, CMV, rabies and many other viral agents can cause meningitis and encephalitis.
- Fungal meningitis: Cryptococcus neoformans.

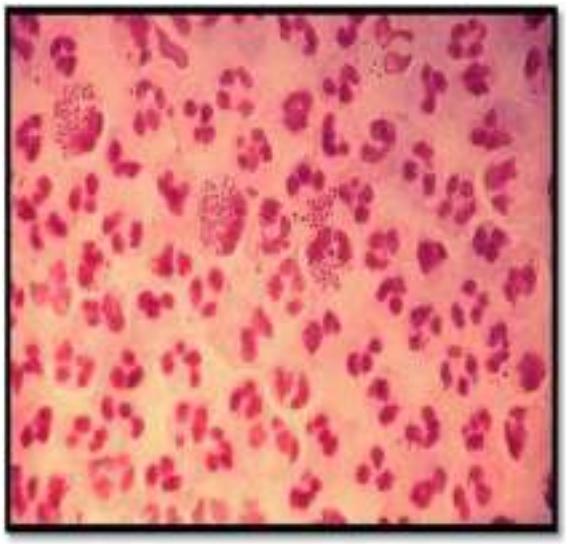
Neisseria meningitides (Meningococci)

Humans are the only natural hosts for whom meningococci are pathogenic.

Morphology:

Gram negative cocci arranged in pairs with the adjacent sides flattened; "kidney shaped appearance". In pathological specimens, N. meningitidis occurs intracellularly in pus cells as well as extracellularly. N. meningitides is capsulated (polysaccharide capsule).

■Cultural characters:



N.meningitidis is aerobeand requires for growth enriched media containing heated blood e.g. chocolate agar or the selective Modified Thayer-Martin (MTM) medium. Cultures are incubated in a humid atmosphere containing 5-10% CO2 at 35-37°C. Colonies are convex, glistening, non-hemolytic and non-pigmented.

■Biochemical Activities:

Oxidase test:All Neisseria species give a positive oxidase reaction. Sugar fermentation with acid production: Sugar fermentation action of Neisseria species, can be used for differentiation of the pathogenic and commensal Neisseria as shown by the following table.

	Glucose	Maltose	Sucrose
N. meningitides	+	+	-
N. gonorrhoeae	+	-	-
N. flavescens	-	-	-
N. sicca	+	+	+

➢ Antigenic composition and virulence factors:

1-Capsularpolysaccharideantigens; according to whichthe meningococci are classified into 13 serogroups. The most important serogroups associated with disease in man are A, B, C, Y and W-135

2- **Pili mediate** attachment to the nasopharyngeal mucosa.

3- **Outer membrane proteins** play a role in attachment.

4- **Lipopolysaccharide** (endotoxin) is responsible for the septic shock due to septicemia associated with meningococcal diseases.

5- **IgAi protease hydrolyzes** secretory IgAi and helps the attachment of the organism to the mucous membrane (colonization).

MENINGOCOCCAEMIA and MENINGITIS

The organism occurs in the nasopharynx of healthy carriers. The carrier rate is 3-30%, this rate increases up to 80% during epidemics. It causes meningitis which may be preceded by meningococcaemia and usually occurs in epidemics among young adults.

1 Mode of transmission

The infection is transmitted by airborne droplets from cases or carriers.

2 Clinical presentations

 Infection starts in the nasopharynx where it may remain silent or gives rise to exudative pharyngitis.
Meningococcaemia: From the nasopharynx, the organism may invade the blood stream and spread to specific sites, such as the meninges or joints, or be disseminated throughout the body.
Meningococcaemia occurs usually in persons deficient in complement components mainly C6-C9.
Waterhouse-Friderichsen syndrome is the most severe form meningococcaemia. It is the life threatening and characterized by high fever, shock, widespread purpura, DIC and adrenal insufficiency.
Meningitis is the most common complication of meningococcaemia. It usually begins suddenly, with severe headache, fever, vomiting and rigidity of the neck and back muscles. It may progress to coma within few hours.

2 Diagnosis

Specimens include CSF, blood, and puncture material from petechiae or jointfluids. Throat swabs are collected for detection of carrier state.
1- The CSF is withdrawn by lumbar puncture under complete aseptic conditions.
 Physical examination: In septic meningitis, the CSF is under tension and turbid due to the large number of pus cells; 20000/cmm.

√ **Chemical examination:** The proteins are elevated, and glucose is reduced.

√ **Bacteriological examination**:

a. The CSF is centrifuged, and the deposit is examined microscopically after staining with gram. The presence of intracellular capsulated Gram-negative diplococci in pus cells is diagnostic.

b. Detection of meningococcal polysaccharide antigens in CSF by coagglutination test. Latex agglutination kits for the detection of antigens in the CSF, for the most important 3 causes of meningitis i.e. meningococci, pneumococci, and H. influenzae type b, are very useful for rapid diagnosis.

Since meningitis is a serious condition; the detection of intracellular Gram- negative diplococci in CSF smears, as well as, direct detection of meningococcal antigens should be immediately reported to the treating physician, who should start treatment immediately.

c. The deposit is cultured on chocolate agar and incubated at 35-37°C in a humid atmosphere containing 5-10% CO_2. Colonies appear in 2-3 days and are identified by:

- Morphology as gram negative diplococci.
- Biochemical reactions; oxidase positive and acid production from glucose and maltose.
- Agglutination with anti-meningococcal serum.

2- Blood cultures commonly give positive results.

3- PCR test has been developed for the detection of meningococcal DNA in blood orCSF

➢ Diagnosis of N. meningitidis carriers:

Commensal Neisseria e.g. N. sicca, N. flavescens, N. lactamica...etc, are normal inhabitants of the throat and nasopharynx and very rarely cause disease.

For diagnosis of meningococcal carriers nasopharyngeal swabs are cultured on enriched selective media (MTM) as described above. Isolated Gram-negative cocci should be differentiated from commensal Neisseria by the difference in their cultural characters, biochemical reactions and serologic identification with specific anti-meningococcal serum.

3 Treatment

willdepend on the type of organism. Prompt treatment with a combination of I.V. antibiotics should be used. Ampicillin + third generation cephalosporins e.g., cefotaxime or ceftriaxone are recommended as a first line of empirical treatment. Vancomycin with or without rifampicin may be added to cover resistant pneumococcal strains.

√ **Chemoprophylaxis:** Rifampicin, 600 mg orally twice daily for 2 days, oral ciprofloxacin or intramuscular ceftriaxone are used for contacts.

√ **Vaccines:** Two meningococcal vaccines are available:
A meningococcal polysaccharide vaccine (MPSV4) and a meningococcal conjugate vaccine (MCV4), both are tetravalent and immunize against types A, C, Y and W-135. Both vaccines are protective, reduce the carrier rate and are effective in preventing epidemics. However, MCV4 is expected to give better, longer-lasting protection.

MCV4 is recommended for:
-Adolescentsand adults 11-55years who are at increased risk for infectioni.e. military recruits, travelers to crowded areas (e.g. Hajj) and asplenic patients.

-MCV4 has been licensed in 2007 for children 2-10 years and for adults over 55 years who are at risk. MPSV4 can be used as an alternative if MCV4 isnot available.

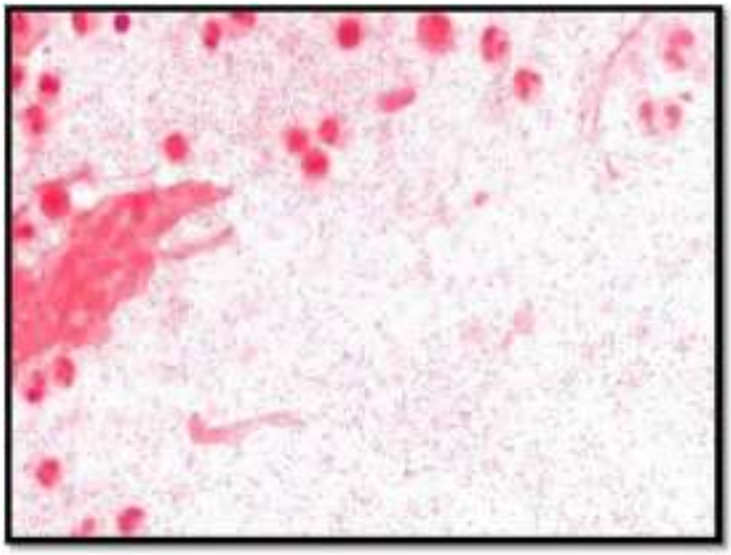
Haemophiles influenza

Members of the genus Haemophilus require certain growth factors present in blood (hemophilic) for their growth.

■ **Morphology:** Gram negative pleomorphic small cocco-bacilli. Long filamentous forms occur. Many strains are capsulated.

■ **Culturalcharacters :**

Facultative anaerobes require heated blood containing media for their growth. They grow on chocolate agar which provides X factor (hematin) and V factor (NAD) that are essential for their growth. Around Staph, aureus colonies, the colonies of H. influenzae are much larger and denser, since staphylococci produce V factor. This is called satellitism.



■ **Serological characters**: Smooth capsulated strains can be classified into 6 types (af), depending on the capsular polysaccharide. H. influenzae type b (Hib)is the most pathogenic. **The capsule is a major virulence factor.**

➢ **Pathogenesis,virulence factors and diseases:**

The organism enters the upper respiratory tract by inhalation or droplets. The polysaccharide capsule is the major virulence factor that facilitates invasion. Other factors include fimbriae and IgA protease that degrades secretory IgA and facilitates attachment to respiratory mucosa, resulting in colonization. Outer membrane proteins and lipopolysaccharide contribute to invasion. Capsulated strains mainly Hib cause invasive diseases in children under 5 years including meningitis, septic arthritis, pneumonia and bacteraemia. Non-capsulated strains cause non-invasive diseases; otitis media and upper respiratory infections, initiated by viral infections. They may cause pneumonia and bacteraemia in adults in presence of predisposing factors e.g. viral infections, malignancy or old age.

Diagnosis

Specimens are pus, sputum or CSF.

1- Smears are stained with gram. When present in large numbers in specimens, the organism can be directly detected by Quellung reaction, immunofluorescence or by PCR.

2- Direct detection of H. influenzae polysaccharide in CSF by latex agglutination commercial kits.

3- Culture on chocolate agar at 37°C in 5% CO2. Colonies are identified by; their morphology, inability to grow in absence of X-V growth factors, serologically with specific antisera or by DNA probes. □

5 Treatment

Cefotaxime given intravenously, or ceftriaxone are the drugs of choice for Hib meningitis. □

6 Prevention

Haemophilus b conjugated (Hib) vaccines containing the capsular polysaccharide conjugated to a carrier protein is recommended for children at 2, 4, 6 months and at 12-15 months. The vaccine may be given in combination with DTP i.e. DTPH.

√ **Chemoprophylaxis**: Rifampin is given to non-immune children in close contact with meningitis patients.

Streptococcus agalactiae

◆ These belong to Lancefield group B and are β-haemolytic but bacitracin resistant. They possess a polysaccharide capsule, which is antiphagocytic. They hydrolyze hippurate and are CAMP factor positive.

◆ They inhabit about 25% of the normal adult vagina. They are acquired by the infant from the birth canal during labour and cause neonatal septicemia, meningitis and pneumonia. The main predisposing factor is delayed delivery after rupture of membranes -longer than 18 hours. Premature babies are at higher risk. Women identified as carriers should receive intravenous ampicillin at least 4 hours prior to delivery.

◆ A rapid test for detection of group B streptococci in vaginal samples is available. It detects the DNA of the organism and gives results in one hour.

◆ In adults Str. agalactiae may cause pneumonia, endocarditis, postpartum endometritis and osteomyelitis. Diabetes is the main predisposing factor.

◆ CAMP test is used for identification of Str. agalactiae cultures: Single straight streaks of the streptococcus to be tested and a p-lysin-producing Staph, aureus strain are made perpendicular to each other, and 4 mm apart on the surface of a sheep blood agar plate. After 24-48 hrs. incubation at 35°C, a positive test result will appear as an arrowhead-shaped zone of complete hemolysis in the area into which both staphylococcal P-lysin and CAMP factor have diffused. The haemolysin is enhanced in the vicinity of the P-lysin producing Staph. aureus.

Cryptococcus neoformans

C. neoformans are yeast cells with a gelatinous capsule. It is found in soil contaminated with the excreta of birds, specially pigeons' feces. It is an opportunistic pathogen affecting mainly immunosuppressed individuals, specially AIDS patients. Infection occurs by inhalation where it causes subclinical lung affection or pneumonia. It may spread systemically to the CNS causing meningoencephalitis or to the skin, adrenals, bone or eyes.

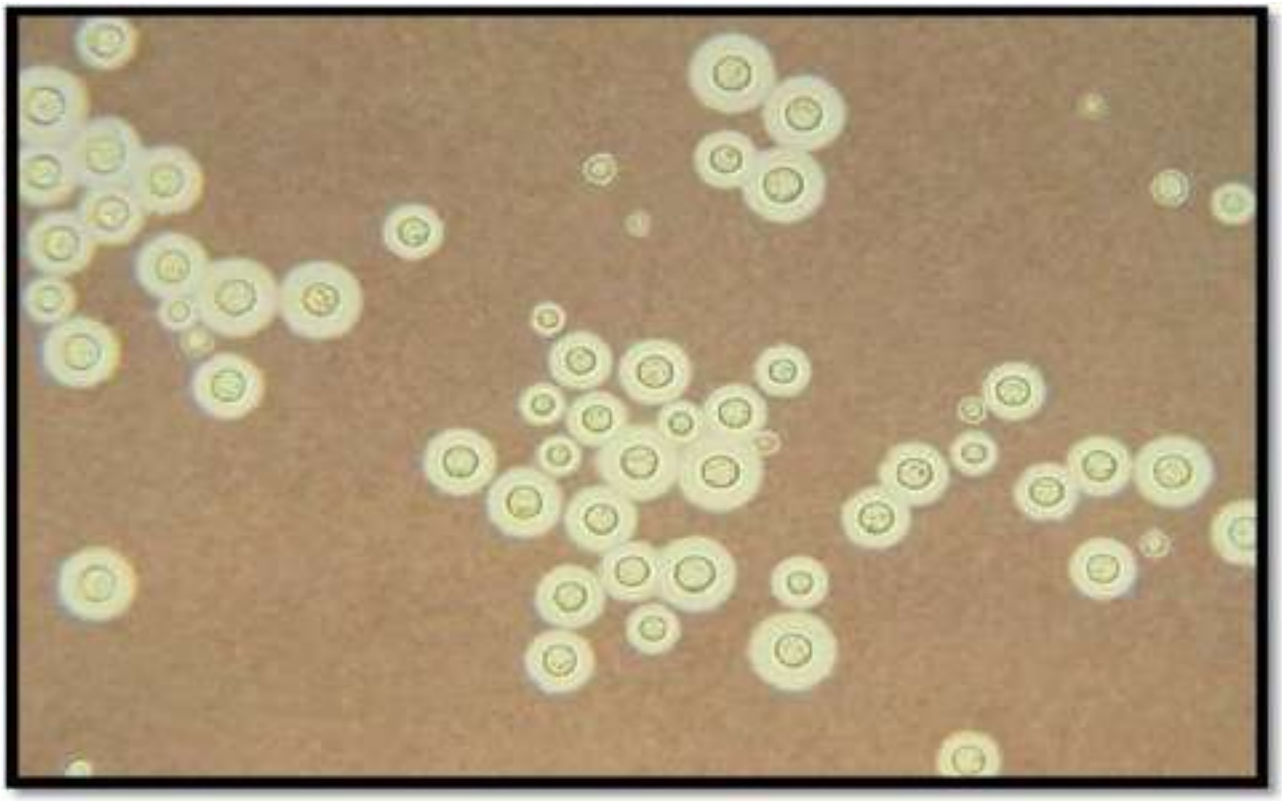
❖ Laboratory diagnosis:

1- Direct microscopic examination of sputum and CSF after staining with India ink, reveals large gelatinous capsule around budding yeast cells.

2- Cultures done on SDA without actidione, at 20-37°C show mucoid colonies which are identified by India ink staining, biochemical reactions (urease positive), pathogenicity to mice and DNA probes.

3- Capsular antigens are detected in CSF, serum or urine by using anticapsular antibodies in a latex agglutination test or enzyme immunoassays.

4- Antibodies are detected in patients' sera.



3 Tetanus (Lockjaw)

1 Definition

Tetanus is an acute, often fatal, disease caused by an exotoxin produced by the bacterium Clostridium tetani characterized by strong muscle spasm (spastic paralysis).

Clostridium tetani

The organismisfound in the intestine of man and animals and in manured soil. It causes tetanusinman and animals.

Morphology:

Gram positivebacilli swollen at one end due to terminal spherical projectingspores; "drum-stick appearance". They are motile by peritrichous flagella.

Cultural characters:

Strict anaerobes grow on nutrient agar, but better growth occurs on blood agar on which colonies are surrounded by a clear zone of hemolysis due to its tetanolysin toxin. The organism grows readily on Robertson cooked meat medium.

Toxin Production.

Cl. tetani producestwotypes of toxins which may be plasmid encoded:

1. **Tetanospasmin**:Apotent polypeptide neurotoxin of one antigenic type.

Mechanism of action:

• The toxin is releasedfrom the site of the wound and diffuses via the blood stream to the peripheralnervous system and reaches the CNS by retrograde axonal transport.
• The toxin bindstoganglioside receptors and blocks the release of inhibitory mediators (glycineand-aminobutyric acid).
• Motor neuronsareleftunder no inhibitory control and undergo sustained excitatory discharge,causing the characteristic motor spasms of tetanus.
• The toxin exertsitseffect on the spinal cord, the brain stem, peripheral nerves, at neuromuscular junctionsand directly on muscles. Thus, leading to, generalized muscular spasm,hyper-reflexia, seizures and spastic paralysis.

2.Tetanolysin(haemolysin) that causes hemolysis on blood agar and has cardiotoxic effects.

2 Mode of transmission

- Infection occurs by contamination of wounds with street dust (containing the spores).
- The injury may be a trivial nail prick, a contaminated surgical wound or a gunshot wound.
- The spores can also be introduced by "skin-popping," a technique used by drug addicts to inject drugs into the skin.
- Infection via umbilical stump leads to "tetanus neonatorum".

3 Pathogenesis

√ C. tetani is not an invasive pathogen. The infection remains localized to the area of devitalized tissues where spores have been introduced.

√ At the local site of infection, the spores germinate releasing the toxin which exerts its action.

√ Germination is enhanced by the presence of necrotic tissue or associated pyogenic infections. The Incubation period varies from 4 days to several weeks.

4 Clinical manifestations

1-Generalized tetanus

The disease is characterized by convulsive tonic contractions of voluntary muscles including:

Jaw muscles lead to trismus (lock jaw).
Facial spasm (sardonicgrin).
Muscles of the back leading to pronounced arching of the back

(opisthotonos).

* Death rate is high and occurs due to respiratory or cardiac failure.

2- Neonatal tetanus follows contamination, of the umbilical stump or circumcision wound of the newborn infants, by tetanus spores. It is a major problem in developing countries. It occurs if the mother is not immune.

3-Localized tetanus:

Persistent painful muscle contractions are localized in areas surrounding injury sites.

* An attack of tetanus does not provide immunity to any future attacks. This is due to the fact that the toxin is potent and minimal amount will induce the symptoms of the disease. Such amounts are too small to trigger immune mechanisms.

5 DIGNOSIS

Treatment with antitoxin should be started on clinical suspicion and history of contaminated wounds without waiting for laboratory diagnosis, which is done for confirmation as follows:

- **Wound exudate - aspirated from deep sites of the wound - is examined:**

1. Microscopically for the presence of Gram-positive bacilli with drum-stick appearance.
2. Direct immunofluorescent staining is used when available.
3. Culture on blood agar incubated anaerobically to detect haemolysis and on Robertson cooked meat medium or other anaerobic media. Proof of isolation of Cl. tetani must rest on production of toxin and its neutralization by specific antitoxin in laboratory animals.

6 Treatment

1- **Antitoxin**

should be given at once to suspected cases to neutralize any toxin that did not fix to the CNS. Human tetanus immune globulin (HTIG) is given in a dose of 3000-10,000 IU divided into three equal portions, given intramuscularly in three different sites at the same time. However, the efficacy of antitoxin treatment is doubtful except in neonatal tetanus, where it may be lifesaving.

2- **Proper care of the wound** and surgical debridement are very important.

3- **Penicillin** or metronidazole is given in big doses to eliminate the organism from tissues and to treat associated pyogenic infections.

4-An **adequate airway** must be maintained, and respiratory support given.

5- **Benzodiazepines**, e.g. valium, should be given to prevent spasm.

7 Prevention

Tetanus is a totally preventable disease. The results of treatment of tetanus are unsatisfactory. Prophylactic immunization is the only way to control tetanus.

Active immunization:

-Alumprecipitated tetanus toxoid is given in combination with diphtheria toxoid andpertussis vaccine "DTP" or DTaP in three intramuscular injections at the age of2,4and 6 months. Booster doses are given at 18 month and upon school entry.

-Abooster dose of tetanus and diphtheria (DT) is recommended every 10 years, to maintain a good serum antitoxin level.

-Boosters are recommended for pregnant women to guard against labour infection andtoprovide maternal immunity for the newborn.

Passive immunization:

-Antitoxin is given to wounded persons without previous history of vaccination or those immunized more than 10 years ago. Intramuscular injection of 250-500 IU HTIG. That will give protection for 2-4 weeks.

-Antitoxin prophylaxis should be accompanied by active immunization with tetanus toxoid at different sites i.e. passive-active immunization.

4 Botulism

Definition:

Botulism is a paralytic disease (flaccid paralysis) caused by a potent neurotoxin elaborated by *Clostridium botulinum.*

Cl. botulinum is essentially a saprophyte in soil. It is frequently present in the intestinal tract of domestic animals.

Morphology:

Gram positive large straight rods, motile, non-capsulated. Spores are oval, central or subterminal.

Cultural characters:

Strict anaerobes, grow on simple media. Colonies areirregular with fimbriated edge. Most strains are beta-hemolytic on blood agar.

Botulinum toxin:

Cl. botulinum produces a very potent **neurotoxin.** It is apolypeptide encoded by a lysogenic phage. The toxin is absorbed from the gut andis carried *via* the blood then bindsirreversibly to the presynaptic nerve endings ofthe peripheral nervous system andcranial nerves, where it blocks the release ofacetylcholine at the neuromuscular junction leading to **flaccid paralysis.** There are 8 antigenic types (A-H); types A, B, E and occasionally F cause botulism in man.The toxin is heat labile; can bedestroyed by boiling for 20 minutes.

Botulinum toxin is considered to be a major agent for bioterrorism and biologic warfare.

Clinical forms of botulism:

1- **The classical type(Food.borne),** which is an intoxication resulting from ingestion of canned food containing the preformed neurotoxin produced by *Cl.botulinum.* The toxin has an affinity to the cranial motor nerves causing bulbar paralysis manifesting as diplopia, dysphagia and respiratory muscle failure.Symptoms begin 18-24 hrs. after the ingestion of the toxic food. Death results from respiratory or cardiac failure. There is no diarrhea, vomiting or fever. Foods responsible include canned green beans, mushrooms, tuna fish, liver paste, home preserved vegetables, meat or ham and others.

2- **Wound botulism,** in which spores contaminate a wound, germinate and producetoxins at the site. This type is associated with drug abuse, especially skin-popping with black tar heroin.

3- **Infant (Intestinal) botulism** 'floppy baby' occurs due to ingestion of spores inthe baby's food, where they germinate in the gut and produce the toxin, which is absorbed into the blood. Honey contaminated with the spores is the most frequent vehicle of infection. It affects infants below 6 months when the colonization resistance of the gut is poor. The infants developpoor feeding, weakness orparalysis and may need respiratory supportbut usually recover spontaneously.However, it may be one of the causesof sudden infant death syndrome.

Diagnosis

Diagnosis is a clinical one but may need laboratory confirmation:

1- The toxin can be directly detected by ELISA or PCR in patient's serum, stool orfood remnants.
2- In infant botulism, *Cl. botulinum* and toxin can be detected in bowel contents, but not in serum.
3- The toxin can be demonstrated in patient's serum or food remnants by injection into mice, which will die with generalized flaccid paralysis, unless protected byantitoxin. Typing of the toxin can be done by neutralization with specific antitoxinin mice, i.e. mouse protection tests.
4- Cultures of foodremnants are rarely done.

Treatment:

1- Trivalent antitoxin (types A, B, and E) is promptly administered I.V. within 12 hours after ingestion i.e. before the toxin binds to the tissues.
2- Supportive Care: Mechanical ventilation, body positioning and parenteral nutrition.

Prevention:

1- Botulinum Vaccine:Pentavalent toxoid (antigen types A, B, C, D, and E) is available for those at high riskofexposure as laboratory workers or in the event ofa bioterrorist attackofbotulism toxin.
2- Careful sterilizationoffood before canning. Swollen cans should be discarded.
3- Honey should notbegiven to infants.

5 Prion diseases

Definition:

Prions are abnormal forms of a cellular protein (PrP) that causes rapidlyprogressive neurodegenerative disorders that affect both humans and animals;called prion diseases or transmissible spongiform encephalopathies (TSEs).

Pathogenesis:

When these proteins are in the normal, alpha-helix configuration PrPC, they arenon-pathogenic, but when their configuration changes to a beta-pleated sheet PrPSc(Sc refers to scrapie), they aggregateinto filaments that disrupt neuronal functionsand result in the symptoms of thedisease.

Characters of prion-mediated diseases:

1- They are confined to CNS causing neurodegeneration (amyloid) and spongiform changes due to neuronal vacuolation and neuronal loss.
2- They have a long incubation period and a chronic progressive course with dementia and end fatally.
3- There is no inflammatory or immune response to these diseases as they arenor mal human proteins.
4- They are transmissible by ingestion of infected tissues mainly the brain,tran splanted tissues and contaminated surgical instruments.

Slow diseases caused by prions:

A-Humanprion-mediated diseases:

They are called (transmissible spongiform encephalopathies) because they cause spongy, Swiss cheese-like holes in thebrain parenchyma.These diseases include Creutzfeldt Jakob disease (CJD), Variant CJD, fatal familial insomnia and kuru.

B- Animal prion diseases:

They include scrapie in sheep, bovine spongiform encephalopathy (BSE) of cattle (mad cow disease).

6 Rabies

Important Properties

- Member of the rhabdovirus family.
- Enveloped, single-stranded RNA, Bullet-shaped capsid
- Thevirion contains an RNA-dependent RNA polymerase
- Single antigenic type. The antigenicity resides in the envelope glycoproteinspikes.

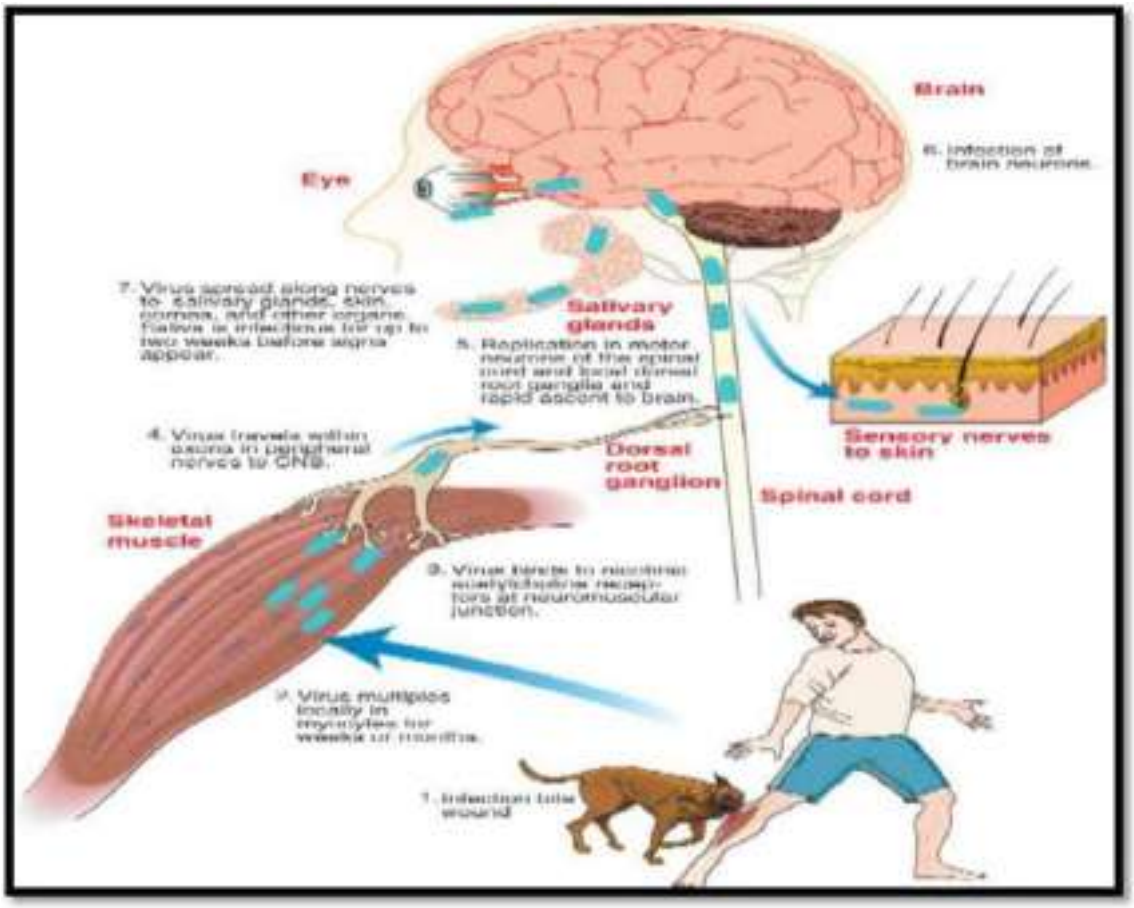
Transmission

- Itistransmittedbythe**bite**ofarabidanimal;bite of **wild animals** such asraccoons,dogs,cats,jackals,mongooseand bats.
- Rodents andrabbitsdonottransmitrabies.
- Human rabies has also occurred inpeople who have not been bitten, so- called"nonbite"exposures;byexposuretoaerosols of bat secretions containingrabies virus.
- Transplantsofcorneastakenfrompatientswho died of undiagnosed rabies.

Pathogenesis& immunity

- Thevirus multiplies locallyatthebitesite,infects the sensory neurons, and **moves by axonal transport to the central nervous system.**
- Thevirus multiplies in thecentralnervoussystem and **then travels down theperiphe ralnerves** to the salivaryglandsandotherorgans.
- From **the salivary glands**,itentersthesalivato be transmitted by the bite.
- Thereis **no viremic stage**.
- Withinthe central nervoussystem, **encephalitis** develops, with the death of neurons and demyelination.

Infected neurons containaneosinophiliccytoplasmic inclusion called a **Negribody,** which is importantinlaboratorydiagnosis



Figure(4):Pathogenesis of rabies

Clinical Findings:

✓Risk ofdevelopingrabiesafterabite: 5- 80%. depending upon:

- Severity of exposure
- Location of the bite
- The bitinganimal

The incubationperiodvariesfrom2to 16 weeks according to location of the bite.

✓ Aprodromeofnonspecificsymptoms such as fever, anorexia, and changes

✓ insensation at the bite site.

✓ Withinafewdays,signssuchas confusion, lethargy, and increased salivation development. Mostnotableisthe painful spasm of the throat muscles on swallowing.Thisresults in **hydrophobia**.

✓ Within several days, the disease progresses to seizures, paralysis, and coma, withdeathafterfewdays.

Laboratory diagnosis

In humans can be madeby

- Fluorescent antibodystaining of a biopsy specimen, usually taken from the skinof the neck at the hairline.
- Demonstration of Negribodies in corneal scrapings and in autopsy specimens oft he brain.
- Isolation of the virus fromsources such as saliva, spinal fluid, and brain tissue.

In the animal is usuallymade by examination of brain tissue by using eitherfluorescent antibodytorabies virus or histologic staining of **Negri bodies** in thecytoplasm of hippocampalneurons.

Treatment:

There is noantiviral therapy for a patient with rabies. Only supportive treatment is available.

Prevention:

The **rabies vaccine** contains inactivated virus grown in human diploid cells.

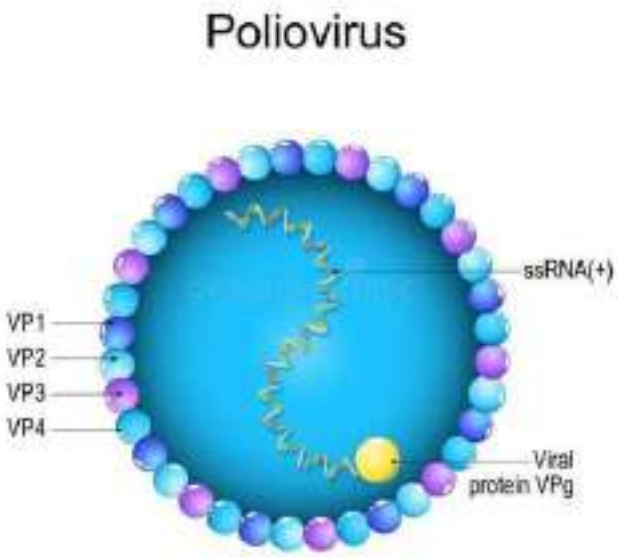
- **Preexposure immunization** should be given to individuals in high-risk groups,suchasveterinarians, zookeepers, and travelers to areas of hyperendemic infe ction.

Preexposure immunization consists of three doses of **rabies vaccine** given ondays 0, 7, and 21 or 28. Booster doses are given as needed to maintain an antibodytiter of 1:5.

- **Postexposure immunization** involves the use of both the **vaccine and humanrabies immune globulin** (RIG, obtained from hyperimmunized persons) plusi mmediate cleaning of the wound.

7 Poliovirus

It causes poliomyelitis which in its full-blown picture affect the CNS and causes flaccid paralysis. Fortunately, most poliovirus infections are subclinical.



Important Properties

1- It is an icosahedralnaked virus, the genome is single stranded RNA, 25-30 nm diameter.

2- There are 3 antigenic types. little cross-reaction.

3-The virus infects only primates. The RNA can bypass the cell membrane receptor ,i.e., it is "infectious RNA."

Pathogenesis & ClinicalFindings

1- Infection occurs bytheingestionoffoodor drink contaminated by stools of **cases or carriers.**

-The organism multipliesinthe **oropharynx (tonsils) and the peyer's patches intheintestine** andisexcretedinthestools. Infection may stop at this stage i.e **in apparent infection.**

2- Infection may continueandtheviruspasses to the deep cervical and deepmesenteric lymphnodes.Thenitinvades the blood stream. **viraemia** is associatedwith mildsymptomsoffever,malaise, headache, nausea and vomiting, the diseasemay be stoppedatthisstagei.e **abortive infection.**

3- Aseptic meningitis mayoccurandmanifests by stiffness and pain in the backand neck. It usuallyrecoversbutitmay progress to paralysis**.**

4- Paralytic poliomyelitis in 0.1- 1 % of cases.

-In CNS, poliovirusreplicatesinthe **anterior horn cells** of the spinal cord. Deathof these cellsresultin **flaccid paralysis** of the innervated muscles.

-The virus also affects the **brain stem**,leading to "bulbar" poliomyelitis, Deathmay occur from **respiratory paralysis.**

Transmission & Epidemiology:

Poliovirus istransmitted by the **fecal–oral** route.It replicates in the oropharynx and intestinaltract.

Humans arethe only natural hosts.

As a resultofthe success of thevaccine, poliomyelitis caused by naturally occurring "wildtype" virus has been**eradicated from the entire Western hemisphere.**

Immunity:

Immune response consists of both

intestinal IgA

humoral IgG whichprovides lifelong type-specific immunity.

Laboratory diagnosis

1- **Isolation of thevirus** from stool, throat washing is done on tissue culture.Character istic **CPE** appears and can be neutralized by specific antibodies

2- **Paired serum samples** are tested to demonstrate **a rising antibody titer** byneutral ization or complement fixation tests.

Treatment:

·There is **no** antiviral therapy.

·Treatment is limited to**symptomatic** relief and respiratory support

· **Physiotherapy** for theaffected muscles is important.

Prevention:

•Poliomyelitis canbe prevented by both:
·The **killed** vaccine (Salk vaccine, IPV).
·The **live, attenuated** vaccine (Sabin vaccine, OPV).
•**Gamma globulins** given early to unimmunized contacts are effective in preventing paralytic form of the disease.
•The current version of the inactivated vaccine is called **enhanced polio vaccine,**or **eIPV.** It induces some mucosal immunity IgA, making it capable ofinterrupti ng transmission.
• The currently approved vaccine schedule consists of four doses of inactivatedvaccine administered at **2 months**, **4 months**, **6 to 18 months**, and at **4 to 6years**. One booster (lifetime) is recommended for adults who travel to endemic areas .

Table (1): Poliovirus vaccines

Table 40–2 Important Features of Poliovirus Vaccines		
Attribute	**Killed (Salk)**	**Live (Sabin)**
Prevents disease	Yes	Yes
Interrupts transmission	No	Yes
Induces humoral IgG	Yes	Yes
Induces intestinal IgA	No	Yes
Affords secondary protection by spread to others	No	Yes
Interferes with replication of virulent virus in gut	No	Yes
Reverts to virulence	No	Yes (rarely)
Coinfection with other enteroviruses may impair immunization	No	Yes
Can cause disease in the immunocompromised	No	Yes
Route of administration	Injection	Oral
Requires refrigeration	No	Yes
Duration of immunity	Shorter	Longers

8 Brain abscess

- It is a collectionof pus enclosed in a brain tissue caused by bacterial or fungalinfection.
- It is caused byinflammation and collection of infected material coming fromlocal (earinfection, dental abscess, infection of paranasal sinuses, epidural abscesses) orremote (lung, heart, kidney) infectious sources within the braintissue.

Symptoms:

Usually develop slowly over several weeks, but they can also come on suddenly.

❑Differences in mental processes, such as increased confusion, decreasedresponsiveness, and irritability.
❑Decreased speech-decreased sensation-
decreased movement due to loss ofmuscle function-changes in vision.
❑ Changesinpersonalityorbehavior-vomiting-fever-chills-neck stiffness- sensitivity to light.

Diagnosis:

❑ Aneurological exam; increased pressure within the brain.

❑ CTand MRI scans can also be used to diagnose a brain abscess.

❑ Insome cases a lumbar puncture, or spinal tap.

Treatment:

❑ Broad-spectrum antibiotics or antifungal drugs.

❑ Surgery is often the next step if an abscess doesn't get smaller with the use of drugs.

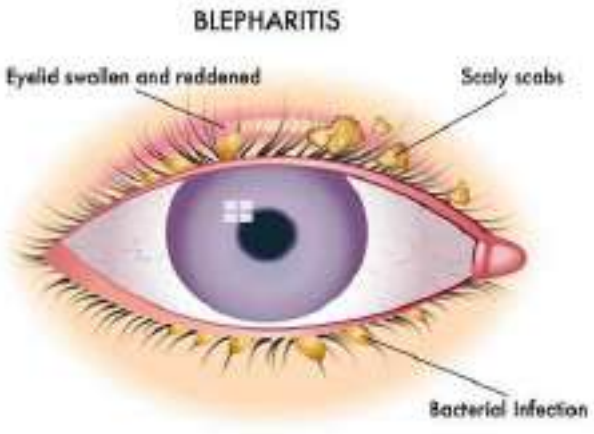
9 Infection of the eye & ear

1) Eye infection

- **Eye lid infection:** Blepharitis, Stye.
- **Conjunctiva:** Conjunctivitis.
- **Cornea:** Keratitis. Endophthalmitis.

Eyelid infection

Blepharitis: it is an infection involved the lid margins.



Stye: it is an infection involved the eyelid glands or follicles.

The most common causes of eyelid infections are:

1. ***Staphylococcus aureus.***
2. ***pneumococci.***
3. ***Haemophilus influenza.***

Conjunctivitis

The conjunctiva is a thin layer of tissue that covers the whites of the eyes andinsides of the eyelids.

Conjunctivitis (Pink eye): is an infection or inflammation of the conjunctiva which may be unilateral or bilateral. It is the most common type of eye infections.

Causes:

(A)Bacteria:

Staph.aureus, Pnemococci, H. influenza, Moraxella catarrhalis, Neisseriagonorrhoeae,P.aeruginosa,Chlamydiatrachomatis; types A-C (trachoma) andtypes D-K (inclusion conjunctivitis).

(B) Viruses:

Adenoviruses, Enteroviruses, Herpes simplex virus, Varicella-zoster virus, Measles andZikavirus.

(C)Fungi: Candida and Aspergillus.

Clinical presentation:

Eyelid swelling, Itching,Pain, Glued eye, Vessel dilatation “red eye”, Pusformation, Purulent discharge, Eyelids are crusted together upon waking up (Lidshut).

Keratitis

Itis inflammation of the cornea which is the clear outer layer on the colored part oftheeye.

Causes:

A. Bacteria: *Staph. Aureus, P. aeruginosa* (contact lenses), *N. gonorrhoeae and H.infl uenzae.*

B. Viruses: HSV.

C. Fungi: *Aspergillus*

Symptoms:

Redness andirritation of the eyes, pain in the eyes, sensitivity to light, blurredvisionand discharge or watery eyes.

Endophthalmitis

Endophthalmitis is an infection of fluids or tissue inside the eye.

It requires immediate medical treatment, or it may cause blindness. Most cases develop after intraocular surgery as cataract surgery is the most common cause of this condition.

Types

❖ **Two common types of endophthalmitis include:**

1- Exogenous endophthalmitis: This most commontype can occur after aneye injury, surgery, or injection. When the eye's protective surface iscompromised, bacteria or fungi get inside the eye.

2- Endogenous endophthalmitis: This infection stems from an infection inanother part of the body that spreads to the eye.

❖ **Organisms involved** are *Staph. aureus, Staph. epidermidis and pneumococci*.

Symptoms:

❖ **Symptomsofendophthalmitis include** eye pain, redness, change in vision andsensitivity to light.

2) *Ear infection*

Types of ear infections:

1. **Outer ear infection (Otitis externa):** Infectionoftheexternal auditory canal.
2. **Middle ear infection (Otitis media):** It's caused by fluidtrapped behindtheeardrum, which causes the eardrumtobulge.
3. **Mastoiditis:** Mastoiditisisan infection of the mastoid bone of theskull.Themastoid bone is located just behindtheear.

Otitis externa

An outer ear infection often starts as an itchy rash. The ear may become painful,tender, red, and swollen.

Causes:

There is predominance of *Staph.epidermidis, S.aureus* and *Corynebacteria* and lesser extent anaerobic bacteria like *Propionibacterium acne.* Others as pneumococci,*Haemophilusinfluenza, Moraxella catarrhalis* , gram negative bacilliand *Pseudomonas aeruginosa*.

Otitis media

A middle ear infection often starts from common cold or other respiratory problems. The infection moves to one or both ears through the eustachian tubes.

An infection can irritate the eustachian tubes and cause them to swell. Swelling canprevent them from draining properly. When fluid inside these tubes can't drain, itbuilds up against the eardrum.

Peak incidence occurs in the first 3 years of life. The disease is less common in the school aged children, adolescents and adults.

Clinical presentation:

Non-specific symptoms and signs such as fever, irritability and lethargy.

Specific symptoms and signs such as ear pain, ear discharge, hearing loss andtinnitus.

Causes:

The mostcommon causes are *Streptococcus pneumonia, Haemophilus infleunza Staphylococcus aureus, Group A streptococcus and Moraxella catarrhals.*

Mastoiditis

Mastoiditis is most often caused by a middle ear infection (acute otitis media). Theinfection may spread from the ear to the mastoid bone.
The condition is most common in children.
Specific features of mastoiditis include swelling, redness and tenderness over themastoid bone; pinna is displaced outward and downward, and a purulent dischargemay be seen after perforation of the tympanum.

2025

2026

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Neuroscience

Parasitology

NEU-316

page	lecture
1	*Parasites that affect the CNS*
15	*Parasites that affect the Eye*

1 Parasites that affect the CNS

1) Helminthes

1. ***Larval*** stage of ***Multiceps*** multiceps (***Coenurus cerebralis***)
2. ***Larval*** stage of ***Echinococcus*** granulosus (***Cerebral Hydatid cyst***).
3. ***Taenia solium*** (***cerebral cysticercosis***)
4. ***Diphyllobothrium*** mansoni and Diphyllobothrium proliferum (***sparganosis***)
5. ***Visceral*** larva migrans (VLM)
6. ***Heterophyes heterophyes*** egg (embolic phenomenon)
7. ***Cerebral schistosomiasis***

2) Protozoa

1. ***Free living*** amoebae
   - ***Naegleria fowleri***- primary amoebic meningoencephalitis (PAM)
   - ***Acanthamoeba spp***.- Granulomatous amoebic encephalitis (GAE)
2. ***Entamoeba histolytica*** (Amoebic brain abscess)
3. ***Trybanosoma brucei*** (African sleeping sickness)
4. ***Plasmodium falciparum*** (cerebral malaria) 5. ***Toxoplasma gondii***

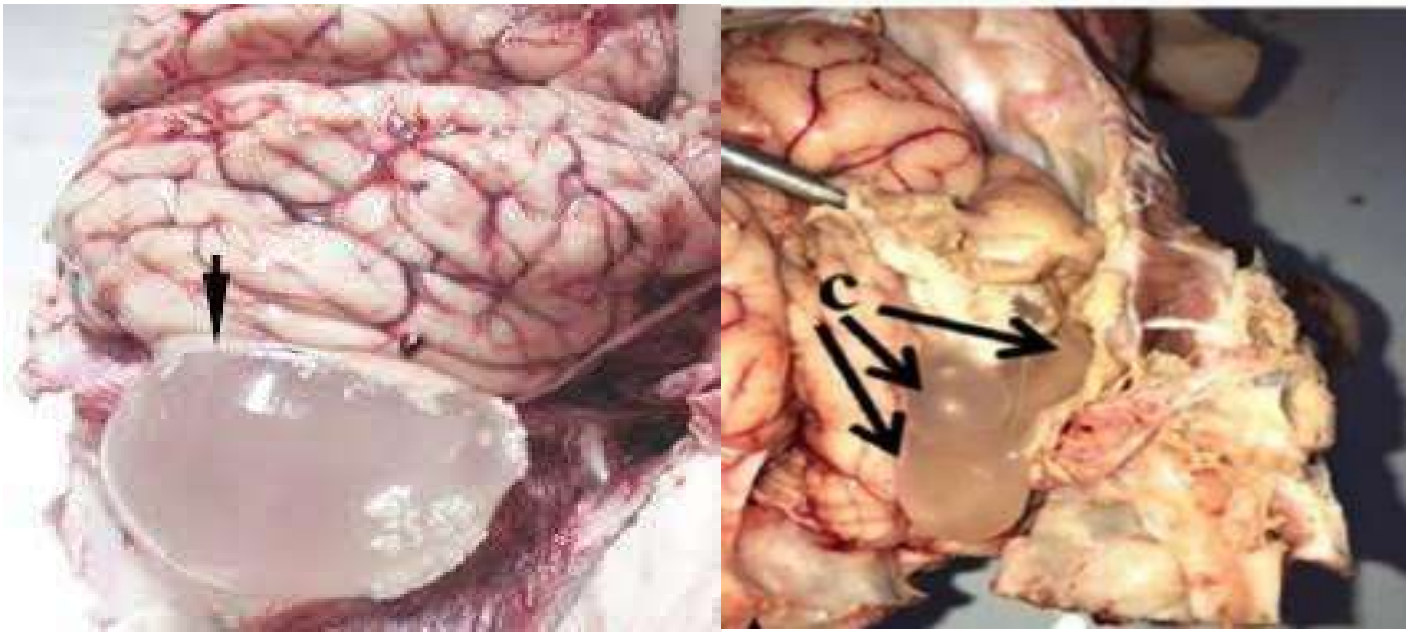
1) Helminthes

1.Cerebral coenurosis

- ***Caused by Coenurus cerebralis:*** the ***larval*** stage of Taenia multiceps (***multiceps multiceps***).
- ***Definitive host:*** dogs.
- ***Intermediate host:*** sheep, cattle, man may be ***accidentally*** infected.
- ***Habitat:*** adults live in the ***small intestine*** of dogs.
- ***Mode of transmission:*** ingestion of ***eggs*** by sheep, cattle and man and ***oncospheres*** carried by blood to the ***brain*** or ***spinal cord*** where each develops into the ***larval*** stage, Coenurus cerebralis.

- The ***coenurus*** takes about ***eight months*** to mature in the central nervous system.

- ***Clinical picture:*** Coenurus cerebralis causes neurological manifestations in ***animals*** and can cause ***tumour like symptoms*** in the brain and lower motor neurone lesions in the spinal cord according to its ***location***.
- ***Diagnosis:***
  - ***Diagnostic imaging techniques*** like MRI and CT scans.
  - ***Serological*** tests like ELISA.
  - ***Histopathological examination*** of brain tissue.
- ***Treatment:*** Surgical removal, where ***feasible*** is the ***only*** mode of ***treatment***.



2.Cerebral Hydatid cyst

- ***Caused*** by the larval stage of ***Echinococcus granulosus*** (***unilocular*** hydatid cyst and ***Echinococcus multilocularis*** (***multilocular*** hydatid cyst)
- ***Definitive host:*** dog, wolf in ***E. Granulosus***, and silver fox in ***E. Multilocularis***
- ***Intermediate host:***
  - Herbivorous ***animal*** and ***man*** in ***E. Granulosus***
  - ***Rodent*** and ***man*** in ***E. Multilocularis***
- ***Habitat: adults*** live small intestine of dog
- ***Mode of transmission: Ingestion*** of the eggs of ***E. Granulosus*** or ***E. Multilocularis*** in ***contaminated raw vegetables*** or ***water*** or on

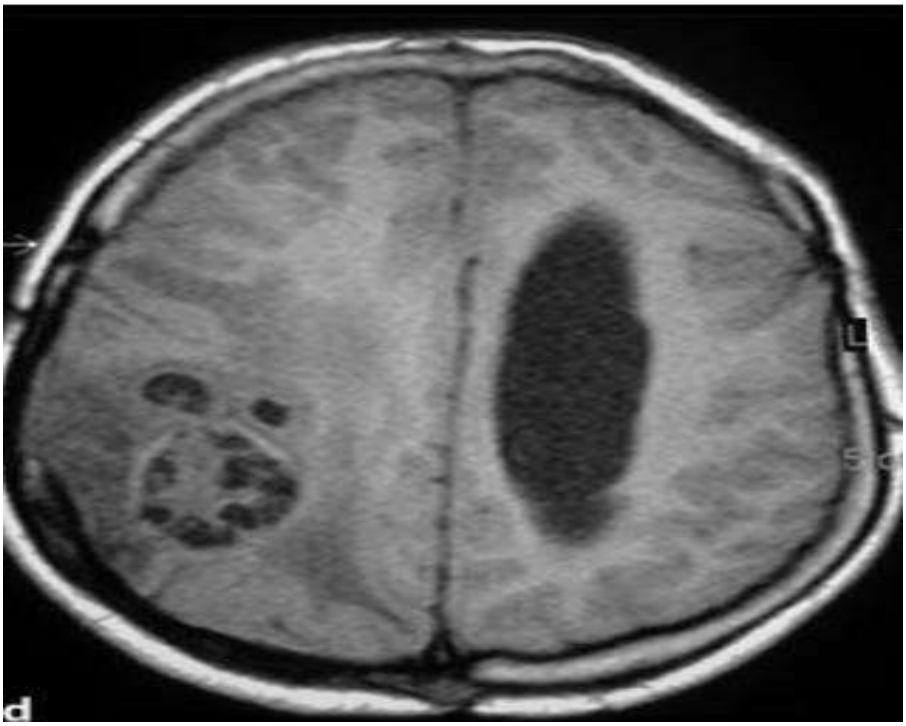
contaminated hands, especially in ***children*** playing, fondling and kissing ***infected dogs*** or ***polluted soils Clinical picture:***

- 1. Asymptomatic.

2. ***Cysts*** cause pressure symptoms and tumour-like mass depending on the location liver, lung, brain and any other organs.

3. ***Complications:*** rupture of the cyst causing ***secondary cysts***, ***anaphylactic shock*** and ***secondary bacterial infection.***

- ***Diagnosis:***
  - ***Blood film examination:*** high eosinophilia.
  - ***Serological:***
    - ***Casoni test*** (intradermal test) intradermal injection of ***0.5 ml*** of hydatid fluid in ***one forearm*** and ***saline*** in the ***other*** and ***wheel*** appears within ***30 mins*** in ***positive*** cases.
    - ***ELISA, IHT, IFT*** by finding the ***antibody*** to ***cysticerci*** in cerebrospinal fluid
  - ***Imaging techniques:*** X-ray, ultrasonography and CT scan show the lesionsin the brain.



- ***Treatment:***

1. ***Single, accessible cysts:*** Surgical removal of cysts is still the ***method of choice***, ***Care*** should be taken to remove the ***cyst intact*** to avoid ***anaphylactic reaction*** and ***dissemination*** by microscopic fragments of

germinal layer or scolices which may form ***new cysts.***

- 2. ***Multiple cysts, inaccessible cysts:***
- ***Albendazole*** can be given in cases like those in ***bone or brain***. It can be given ***prior*** to surgical removal to ***avoid dissemination.***
- ***Sonographic-guided aspiration***, then intracystic injection of ***10% formalin*** or ***70% alcohol*** and its ***re-aspiration.***

3. C erebral cysticercosis

- ***Caused*** by cysticercus cellulosae; the ***cystic larval*** stage of Taenia solium.
- ***Definitive host:*** man
- ***Intermediate host:*** pig, man
- ***Habitat: adults*** live in the small intestine of ***man***
- ***Modes of transmission:***

1. ***Ingestion*** of the eggs of ***Taenia solium*** in contaminater

a vegetables or water.

2. ***Autoinfection***

A. Internal autoinfection: as result of ***regurgitation*** of eggs released by ***disintegration*** of the gravid segment into the ***stomach*** due to ***antiperistalsis***.

B. External autoinfection: as a result of hand-to-mouth

• ***Diagnosis:***

A. ***Direct method:*** usually post-operative microscopic examination of biopsy specimens.

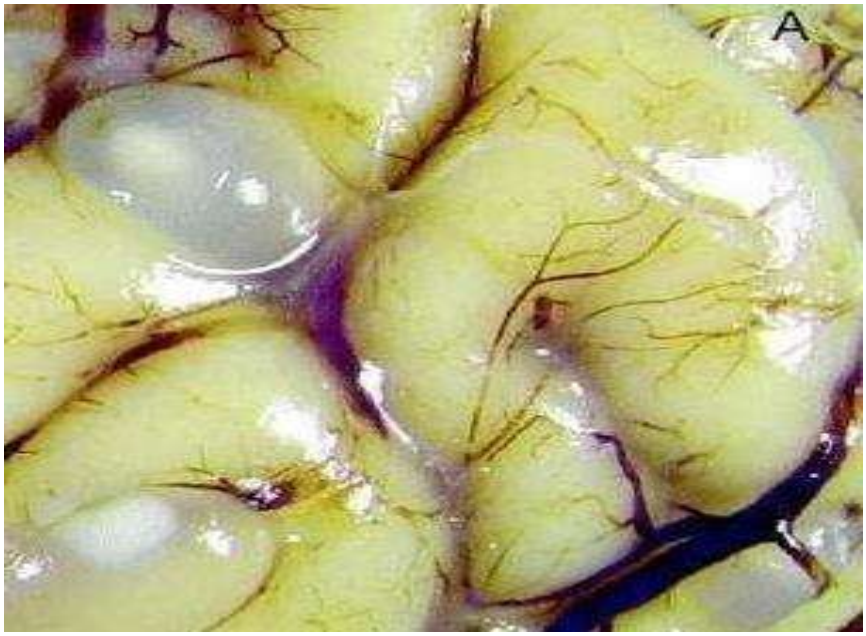
B. ***Indirect method:*** ELISA, IHT, IFT by finding the antibody to ***cysticerci*** in cerebrospinal fluid.

C. ***Imaging techniques: X-ray*** (may show calcification present in old cases), Ultrasonography, C.T., and magnetic resonance can show the lesions in the brain.

- ***Treatment:***

A. ***Surgical removal*** of the cyst is frequently successful.

B. ***Anthelmintic treatment*** can be used in the early stages when the ***oncospheres*** are migrating.



4. C erebral Sparganosis

Caused by ***larval*** stages of Dipyllobothrium ***mansoni*** & Dipyllobothrium ***proliferum***.

A. Dipyllobothrium mansoni

- ***Definitive host:*** dog and cat.
- ***Intermediate hosts:***

1. ***1st intermediate host:*** waterflea (***Cyclops***).
2. ***2nd intermediate host:*** frog, snake, birds and man.

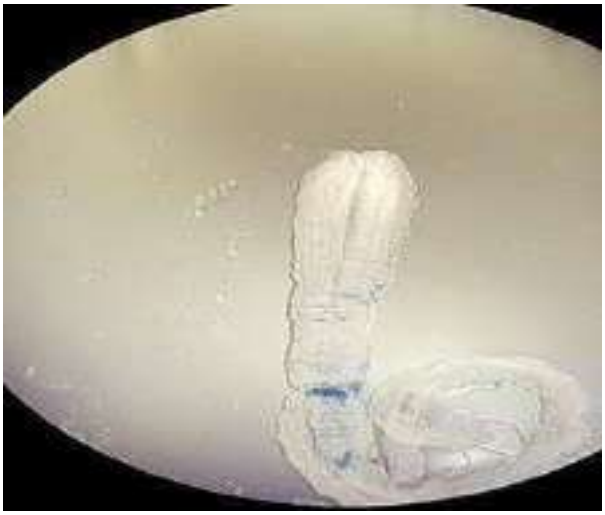
- ***Habitat: adults*** live in small intestine of dog and cat.

- ***Mode of transmission:***

1. Swallowing Cyclops containing ***procercoid*** with ***drinking water.***
2. Eating raw or ***undercooked flesh*** of infected ***2nd intermediate host*** as in the Far East infected with ***plerocercoid (sparganum).***
3. Folk medicine in some areas uses the ***flesh*** of these animals as ***foments*** or ***poultices*** for wounds, eyes or inflamed parts of the body. When the flesh is ***infected***, the ***sparganum*** migrates to human tissues.

- ***Clinical picture:***
  - They depend on the ***sites*** affected in the ***brain*** and the ***degree*** and ***duration*** of infection.
  - ***Fever*** is a common finding. Patients may develop neurological disturbances and endophthalmitis.
- ***Diagnosis:***
  - ***History*** of using foments or drinking infected water.
  - Biopsy and removal of the swelling.
  - Serological tests.

- ***Treatment:***
  - ***Surgical removal*** whenever ***possible*** with an ***anti-inflammatory*** drug.



B. Dipyllobothrium proliferum

- The infection is ***present*** in ***Japan*** and ***USA***.
- The ***adult*** has an obscure life cycle.

- The ***infecting sparganum*** gives bizarre lateral branches in the human tissue.
- It was reported from ***subcutaneous***, ***kidney***, ***lung*** and ***brain***.

5.Visceral larva migrans (VLM)

- This results from the ***ingestion*** of the ***embryonated eggs*** of ***dog*** and ***cat ascarids***, ***Toxocara canis*** and ***Toxocara cati***, respectively.
- They may cause ***epilepsy***.

6.Heterophyes heterophyes (embolic phenomenon)

- The ***pumping action*** of ***eggs*** by ***genital sucker*** helps in their ***passage*** into ***mucosal blood vessels*** and ***lymphatics*** acting as ***egg emboli*** and are ***lodged*** in the ***parenchymatous*** organs and ***muscles***, mainly ***manifest*** in ***heart*** and ***brain***.

7.Cerebral schistosomiasis (egg emboli)

- ***Cerebral manifestations*** of schistosomiasis are ***most commonly*** seen in ***S. Japonicum*** infections arising from ***egg embolism*** due to its ***egg*** is ***smaller ovoid*** with a ***rudimentary lateral spine*** or ***protuberance***. Therefore, ***eggs*** are ***easily flushed towards*** the ***liver*** and other organs.
- It may also be caused by ***S. Haematobium*** and ***S. Mansoni.***

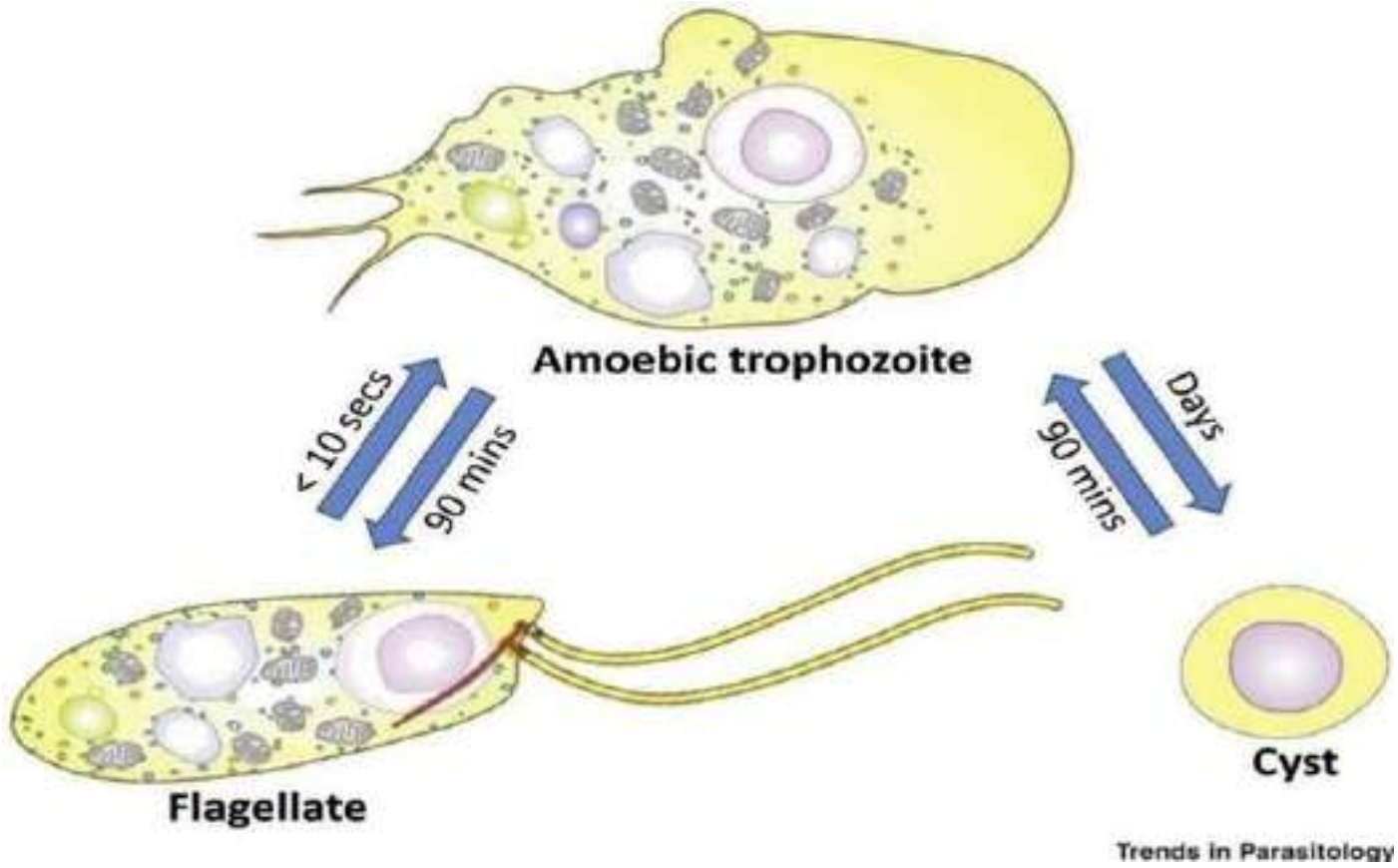
2) Protozoa

1.Free living amoebae

A. Primary amoebic meningoencephalitis (PAM)

- ***Causative parasite*** is ***Naegleria fowleri.***
- It is an ***inhabitant*** of ***soil*** and ***stagnant water*** especially ***warm water*** of canals, lakes, tanks for water storage, and ***heated swimming*** pools.
- ***Morphology:*** Naegleria fowleri has ***3 morphologic*** forms:

1. ***trophozoite*** (ameoboid form),
2. ***flagellated form***
3. ***cyst*** which is ***never*** found in ***host tissue***



- ***Mode of transmission:***
  1. The ***flagellated forms*** are forced ***deeply*** into the nasal passages while ***swimming*** in the ***stagnant water*** of pools, lakes or canals.
  2. ***Amoebae*** usually migrate along the olfactory tract through the cribriform plate into the ***cranium*** leading to ***brain destruction*** and ***death***. They ***do not form cysts*** in the host tissues.

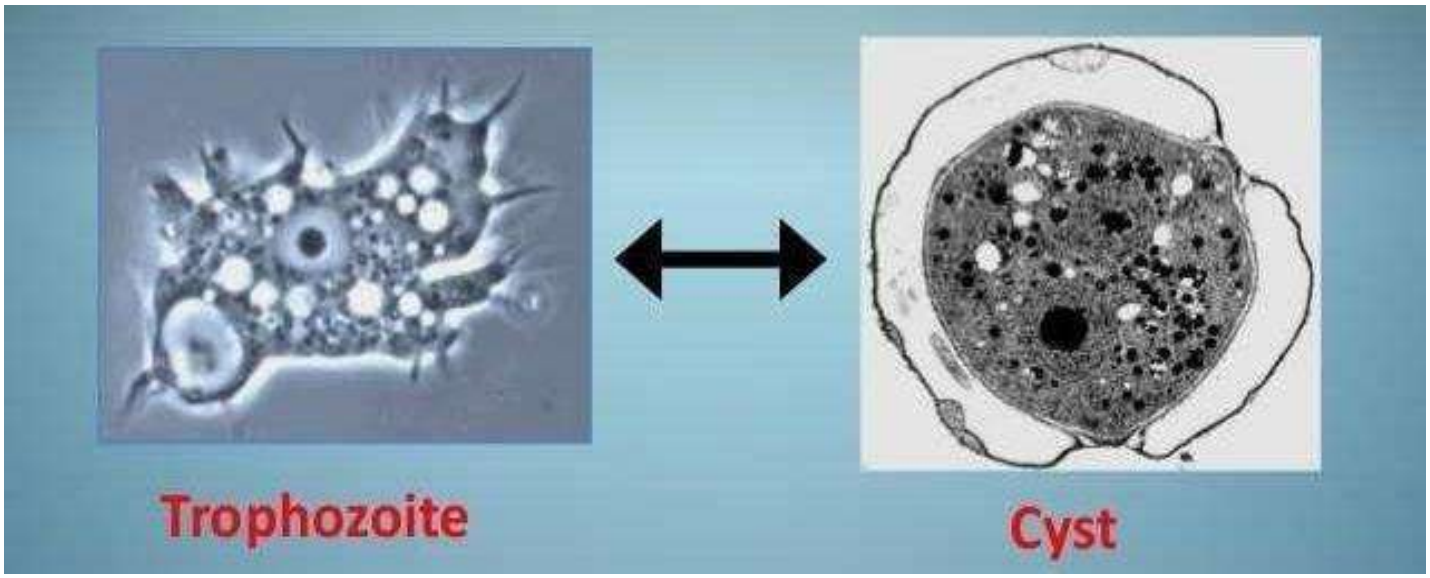
- ***Clinical picture:***
  1. It causes ***primary amoebic meningoencephalitis*** (***PAM***), in the ***healthy young*** person.
  2. ***Early symptoms*** include fever, headache, ***disturbances*** in ***smell*** and ***taste***.
  3. ***Illness rapidly progresses*** to signs of ***meningoencephalitis***
  4. ***Death*** occurs within ***weeks*** of the ***onset*** of symptoms
- ***Diagnosis:***
  1. ***Clinically*** there is rapid acute meningitis. There is headache, nausea, vomiting, fever and ***stiff neck***. The illness may ***progress*** within ***days*** to ***coma***.

2. ***CSF*** examination shows the ***ameboid stage.***
3. ***Culture*** of ***CSF*** on non-nutrient agar smeared with ***Gram-ve bacilli*** at 37Cº gives ***trophozoite*** and ***cyst stages*** in 48 hours.

- ***Treatment:***
  Intravenous and ***intercaecal*** injection of ***Amphotericin B*** and ***Rifampicin***.

B. Granulomatous amebic encephalitis (GAE)

- The ***causative parasite*** is ***Acanthamoeba spp.***
- It is ***free-living soil amoeba***, present as ***trophozoite*** and ***cyst stages*** in ***moist soil***, ***fresh*** and ***brackish water, air condition filters*** and in ***dialysis units.***
- ***Morphology:*** Acanthamoeba spp. Has ***two forms:***
  cyst and ***trophozoite*** ***both*** are found in host tissue.



- ***Mode of transmission:*** The ***trophozoite*** are the ***infective stage*** and entry the body through ***lower respiratory tract***, ***ulcerated skin*** and ***invade CNS*** through ***hematogenous dissemination***

- ***Clinical picture:***

A. Granulomatous amebic encephalitis (GAE):

- It causes ***GAE*** in ***chronically ill***, ***immunocompromised*** patients.
- The ***course*** of the disease has an ***insidious onset*** and the ***manifestations*** occurring ***weeks*** to ***months*** after ***exposure***.
- ***Signs*** and ***symptoms*** include seizures, headache, neck rigidity, ataxia cranial

nerve palsies, hemiparesis and other deficits.

• ***Fever*** is low-grade and intermittent.

B. Amebic keratitis:

- Occurs in ***immunocompetent*** persons through ***contamination*** of water or contact ***lens*** solution.
- ***Corneal abrasions*** may ***enhance*** the infection.
- ***Keratitis*** leads to visual affection with ***severe pain*** and the ***lesion*** is ***irreversible***.
- ***Sometimes*** the infection results in ***corneal perforation.***

• ***Diagnosis:***

- ***Corneal smear*** in case of amebic keratitis.
- ***CSF examination*** in case of brain affection may be ***helpful*** in addition to ***CT*** or ***MRI***.
- ***Culture*** of corneal ***scrapings*** or ***CSF***. On ***agar*** as in ***Naegleria***.

2.Secondary amoebic brain abscess

- The ***causative parasite*** is ***Entamoeba histolytica.*** • ***Geographical***
- ***Distribution:*** Cosmopolitan
- ***Host: Mainly*** a human parasite.
- ***Extraintestinal amebiasis: Brain***-causing amoebic brain abscess
  Cerebral amebiasis is rare; it occurs in ***less*** than ***1%*** of patients with ***dysenteric amebiasis***. However, it is the ***second commonest*** cause of ***death*** from ***parasites*** and accounts for ***4.2–8.5% of deaths*** due to ***amebiasis***.
- ***E. Histolytica*** arrives in the ***brain*** by ***hematogenous*** spread from the ***colon*** through the ***hepatic*** or ***pulmonary*** veins.
- ***Diagnosis:***
  - ***Blood picture:*** Leukocytosis
  - ***Imaging investigations:*** X-ray, CT scan and MRI.
  - ***Finding*** the parasite in ***stools*** (***trophozoites*** in ***diarrheic stools*** and ***cysts*** in ***formed stools***).
  - Serology as ELISA.
- ***Treatment:*** Flagyl (***Metronidazole***), Tinidazole.

3.African trypanosomiasis (sleeping sickness)

T. b. Gambiense & T. B. Rhodesiense

- These are the ***causative*** agents of African sleeping sickness. ***Both*** differ in ***some physiological aspects***, in ***pathogenesis*** and ***biology***.
- T. b. ***Gambiense*** causes ***chronic*** type of the disease and is present in ***central***, and ***west central Africa***. It is ***transmitted*** by ***G. Palpalis.***
- T. b. ***Rhodesiense*** causes ***acute*** sleeping sickness and occurs in ***central*** and ***east central Africa***. It is ***transmitted*** by ***G. Morsitans.***
- ***Life cycle:***

A. In the vector:

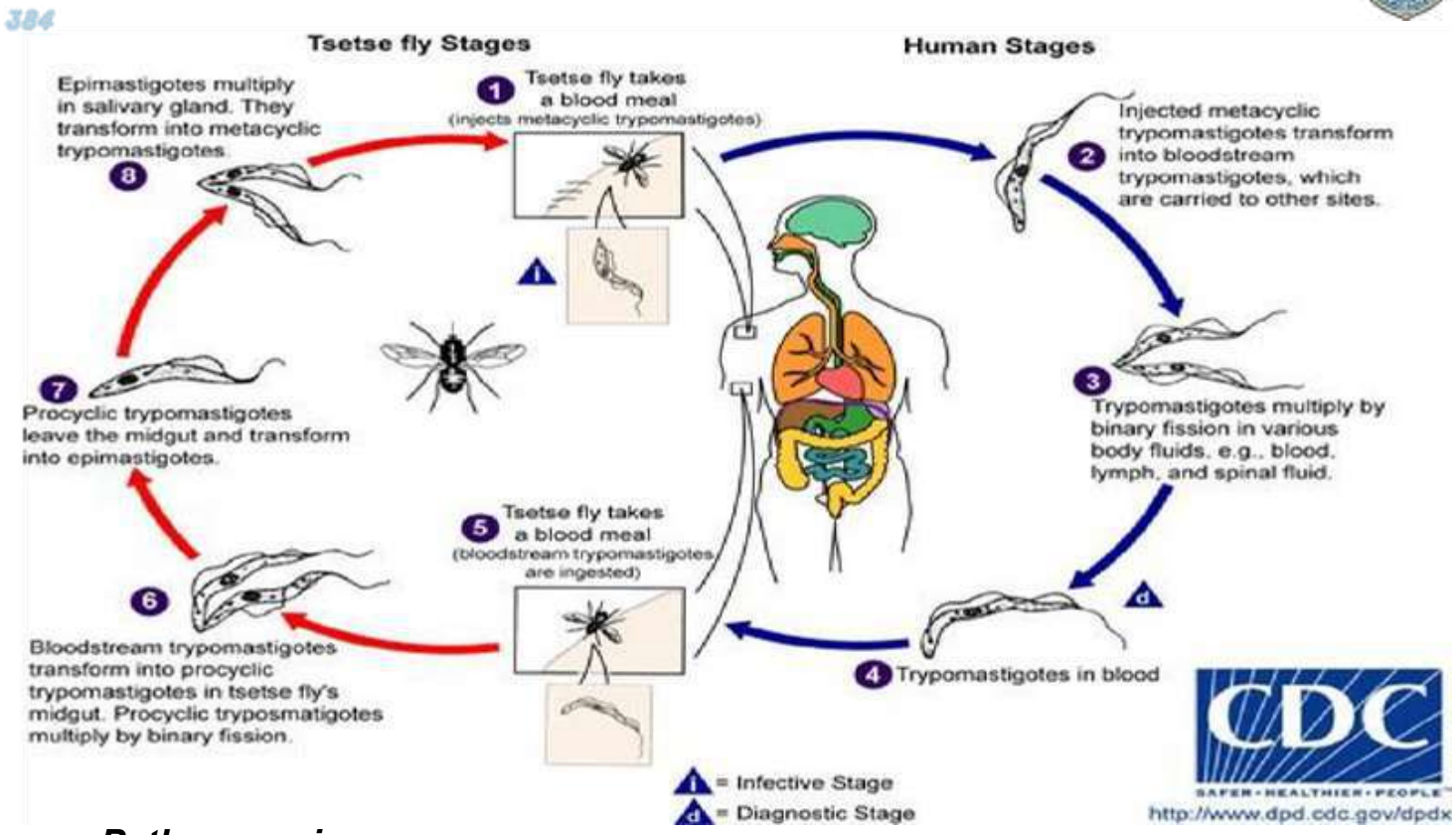
1. When the ***trypanosomes*** are ingested by a tsetse fly, they are ***localized*** in the posterior part of the ***midgut***, change into ***epimastigotes*** and ***multiply*** in this form for about ***10 days.***

2. Then they migrate ***slowly*** under the ***peritrophic membrane*** (coating the stomach epithelium) towards the ***proventriculus***. They migrate ***forward*** into the ***oesophagus***, ***pharynx***, ***hypopharynx*** and ***finally*** into the ***salivary glands.***

3. They change into ***slender*** forms and ***multiply***, then transform into ***metacyclic trypanosomes*** (the ***infective stage***), which is a ***short stumpy trypanosome***.

B. In man:

1. When the ***insect bites*** a ***healthy*** person, it ***injects*** the ***metacyclic trypanosomes*** into the ***blood***, which is the infective stage.
2. The ***trypanosomes*** spread ***rapidly*** and ***multiply*** in blood and lymph nodes.
3. Later in the disease they ***migrate*** to the ***cerebrospinal fluid.***



- ***Pathogenesis:***
  - In ***human*** infections with T. B. Rhodesiense or T. B. Gambiense a ***small sore*** develops at the ***site of bite*** (***Trypanosomal chancre***).

  - In ***blood*** and ***lymph***, the parasites ***multiply rapidly,*** they produce ***parasitemia*** and invade ***all the organs*** of the ***body***. In this ***stage***, ***defence mechanisms*** of the body ***induce*** fever, malaise and body aches.
  - ***Invasion*** of the lymph nodes by the parasite causes ***generalized lymphadenopathy***, ***most observed*** in the groin, axilla and cervical lymph nodes especially the ***posterior cervical (Winterbottom"s sign).***

- Infection with ***T. B. Rhodesiense*** is a severe illnessofrapidprogress to ***weight loss, heart*** involvement. ***Death*** may occur***before******neurological involvement***. Early neurological symptoms can occurwith***more rapid*** progress to ***coma*** than in ***T. B. Gambiense.***
- Infection with ***T. B. Gambiense*** is less severe, andoflongduration. Parasites usually invade the ***C.N.S.*** and replicate in the ***C.S.F***.and***inter-cellular reticular*** tissue. This causes what is known as ***chronicsleeping sickness*** stage. ***Signs*** and ***symptoms*** of the disease appearasbehavioural changes, confusion, ***sensory disturbances*** and ***poor coordination***.***Disturbance*** of the ***sleep cycle***; gives the disease its ***name***.

- ***Diagnosis:***
  1. Finding the ***trypanosomes*** in the ***punctures*** from the ***chancre***, during the ***febrile stage*** in ***blood***, ***bone marrow*** or ***lymph node aspiration***. The films are best stained with ***Giemsa***.
  2. ***Examination*** of the ***C.S.F.*** in early neurological signs. There is an ***increase***in ***WBC*** and ***protein*** content with a ***decrease*** of ***sugars***.

3. Animal inoculation of patient's blood or CSF.
4. Culture of the blood, CSF and lymph node aspirates on ***N.N.N.*** or ***GLSH*** (glucose, lactalbumin, serum and haemoglobin) media. 5) ***Serological tests.***

- ***Treatment:***
  - ***Suramin*** and ***pentamidine*** are used for ***early cases.***
  - ***Melarsoprol*** is the ***only drug effective*** in ***late cases.***

4.Cerebral malaria

- It is ***one*** of ***Plasmodium falciparum*** complications. These are caused by the ***adhesive phenomena*** of the ***infected erythrocytes.***
- The ***onset*** is usually ***sudden***, with ***headache*** progressive to ***coma***, ***uncontrollable*** rise of temperature to ***above 40º C***, and ***psychotic symptoms*** or convulsions, especially in ***children***. ***Death*** occurs within ***hours***.

5.Toxoplasma gondii

Toxoplasmosis in ***immunocompromised*** patients: in which ***affection*** of the ***brain*** is ***more common***, with ***meningoencephalitis***, and ***neuropsychiatric manifestations.***

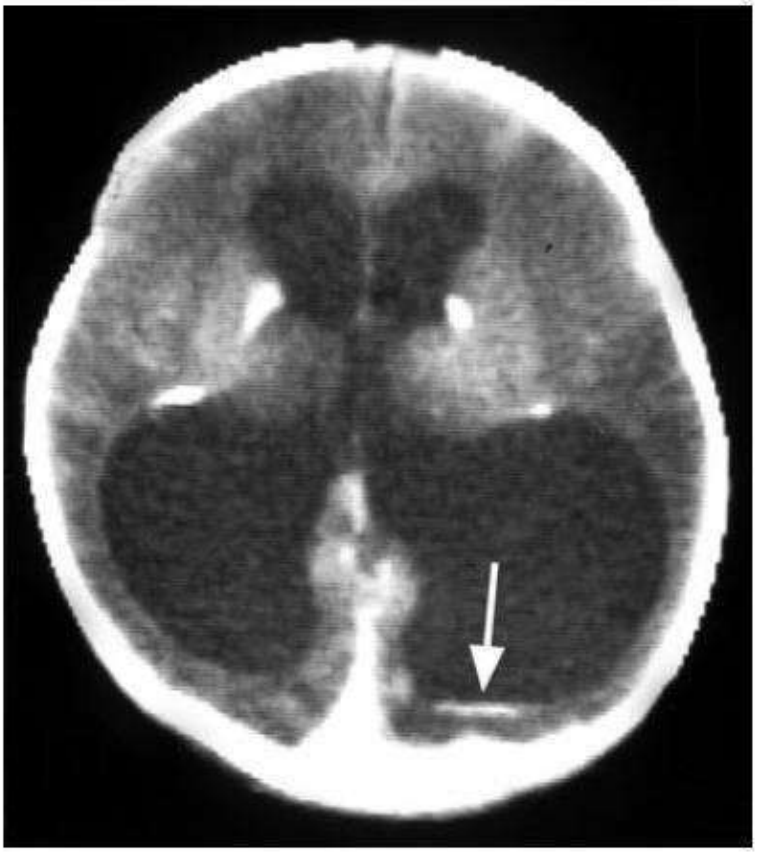


Fig. 2.1 Congenital toxoplasmosis. Axial non-enhanced CT scan of the brain shows periventricular calcifications and encephalomalacia, hydrocephalus, and intraventricular hemorrhage (*arrow*)

2 Parasites that affect the Eye

1) Helminthes

1. ***Taenia solium*** (cysticercosis)
2. ***Diphyllobothrium*** mansoni and D. Proliferum (sparganosis)
3. ***Trichnella spiralis***
4. ***Loa loa*** (the African eye worm)
5. ***Onchocerca volvulus***
6. ***Visceral larva migrans*** (VLM)
7. ***Hydatid*** cyst

2) Protozoa

1. ***Free living amoeba*** (Acanthamoeba spp).
2. ***Trypanosoma cruzi*** (Romana′ s sign)
3. ***Toxoplasma gondii***

3) Arthropods

1. ***Ocular Myiasis***
2. ***Phthirus pubis***

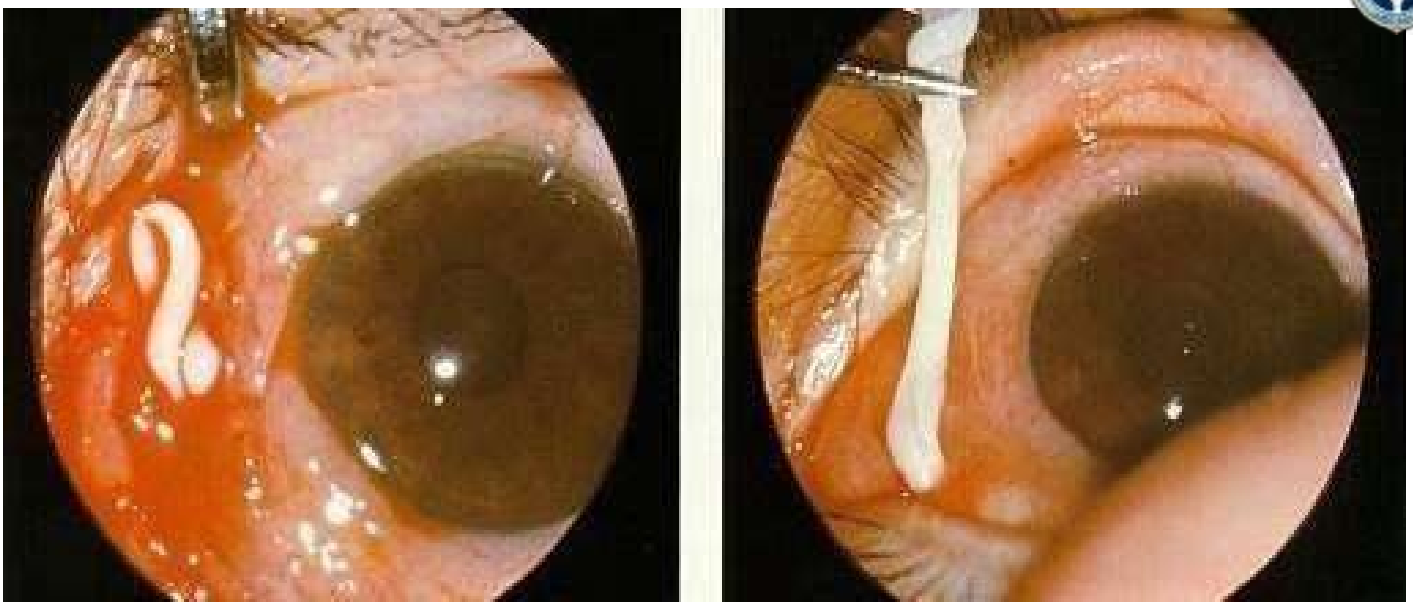
1) Helminthes

1-Taenia solium (cysticercosis)

Caused by Cysticercus cellulosae; the ***cystic larval*** stage of Taenia solium.

2-Diphyllobothrium mansoni & D. Proliferum (sparganosis)

Caused by the ***larval*** stages (***Sparganum***) of Dipyllobothrium ***mansoni*** & Dipyllobothrium ***proliferum***.



3-Trichinella spiralis

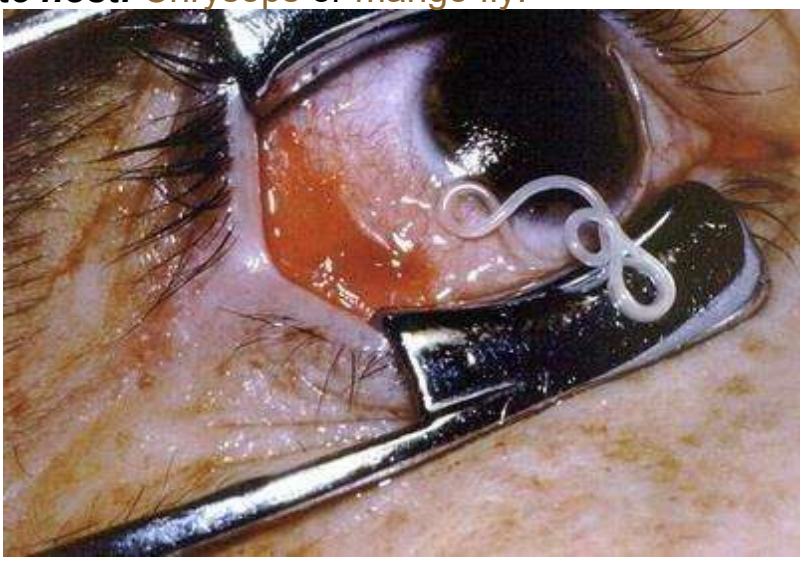
Affecting ***eyelids*** and causing ***periorbital edema.***

Conjunctivitis and periorbital edema caused by *Trichinella* sp.
(Credit: Dr. T.F. Sellers, Emory Univ., CDC, PHIL.)

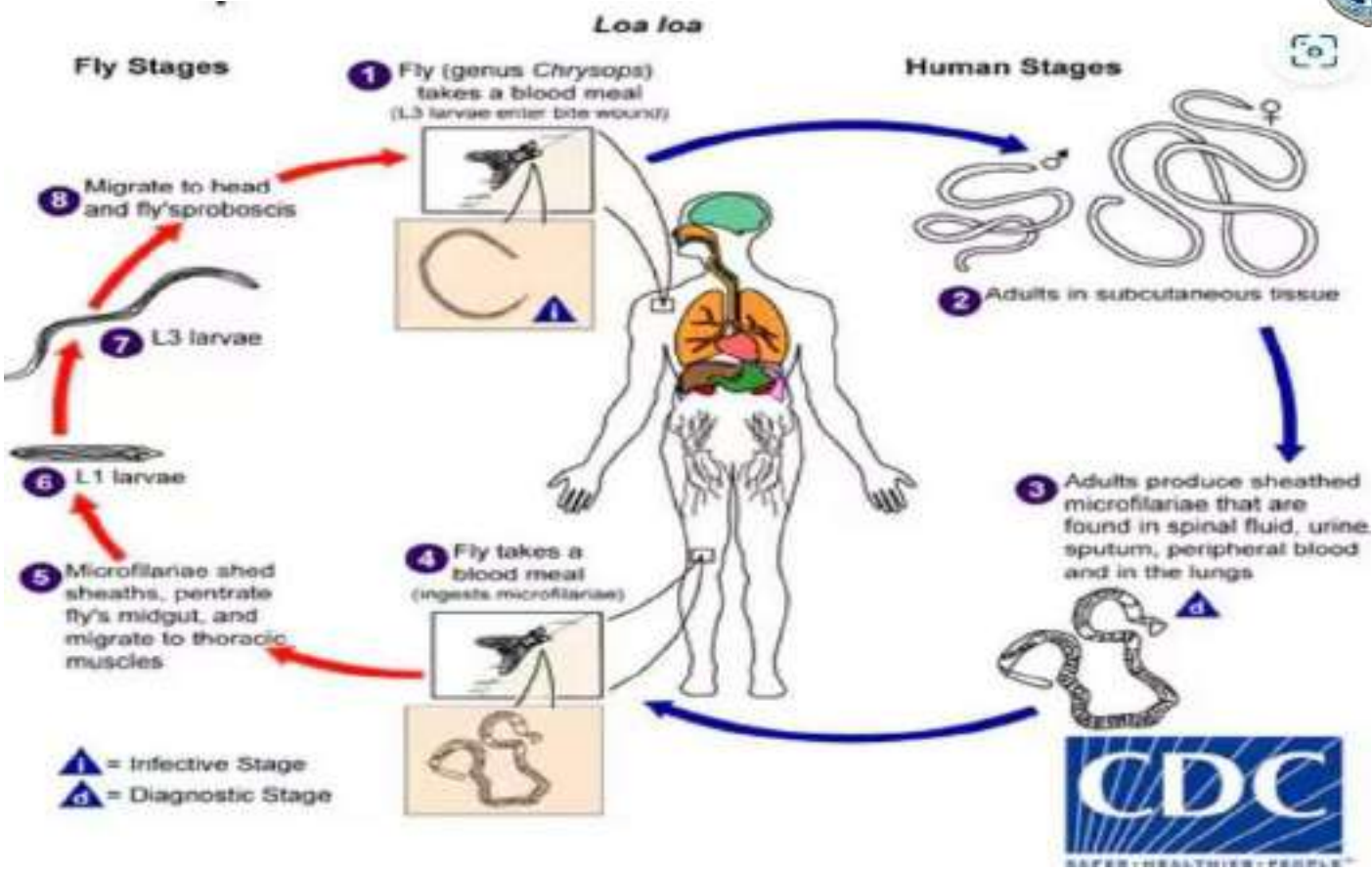
4-Loa loa (the African eye worm)

Disease: Loaisis
Distribution: Tropical Africa.
Hosts:

- ***Final host:*** Humans
- ***Intermediate host:*** Chrysops or mango fly.



Life cycle: Nearly similar to Wuchereria bancrofti .



Clinical picture:

- ***Adults*** live subcutaneously migrating ***freely*** from place to place and may appear ***under the conjunctiva*** (***eye worm***).
- The presence of the worm may lead to acute inflammation from ***mechanical injuries***, the effect of metabolites and the host immune reactions.
- As a result, from these ***acute inflammations***, transient ***cutaneous swellings*** (***Calabar or fugitive swellings***) appear. They are about the ***hen"s egg***, ***slightly*** painful, itchy, non-pitting, and appear and disappear within a few days.
- There may be fever, urticaria and eosinophilia.

Diagnosis:

- In ***endemic areas***, provisional diagnosis is ***possible*** from the symptoms.
- ***Diurnal***, thick blood film showing ***microfilariae*** is a sure way for diagnosis.

Treatment: DEC + General and surgical measures.

5-Onchocerca volvulus

Disease: Onchocerciasis or river blindness.

Distribution: Tropical Africa, Asia, and Central America, near fast running water and

waterfalls.

Hosts:

- ***Final host:*** Humans.
- ***Intermediate host:*** Simulium (mainly S. Dumnosum) (***The black fly***).

Life cycle: Nearly similar to Wuchereria bancrofti.

Clinical picture:

- Onchocercoma.
- ***Severe*** itching, skin rash, lesions and depigmentation (leopard skin).
- ***Ocular*** lesions leading to ***blindness*** (riverine blindness).
- There may be ***some*** degree of fibrosing obstructive lymphadenitis associated with ***elephantiasis*** and ***hanging groins*** (large redundant skin folds in the ***lower abdomen and groin***).

Diagnosis:

- Skin ***snips***.
- Aspiration of nodules can show ***microfilariae***, surgical removal and sectioning can reveal the adults.
- ***Mazzotti test:*** Administration of a ***small*** dose of ***DEC*** (50mg) is followed by a ***pruritic rash*** within 24 hours. This is a ***very dangerous test***, because it may ***lead to ocular.***

Treatment: DEC + General and surgical measures

6 - Visceral larva migrans

Inthe ***eyes***, they may cause chronic endophthalmitis and peripheral retinitis.

2) Protozoa

1 - Free living amoebae (Acanthamoeba spp.)

Amebic keratitis:

- Occurs in ***immunocompetent*** persons through contamination of ***water*** or ***contact lens*** solution.
- Corneal abrasions may ***enhance*** the infection.
- ***Keratitis*** leads to visual affection with ***severe pain*** and the lesion is irreversible.
- Sometimes the infection results in corneal perforation.

Normal eye

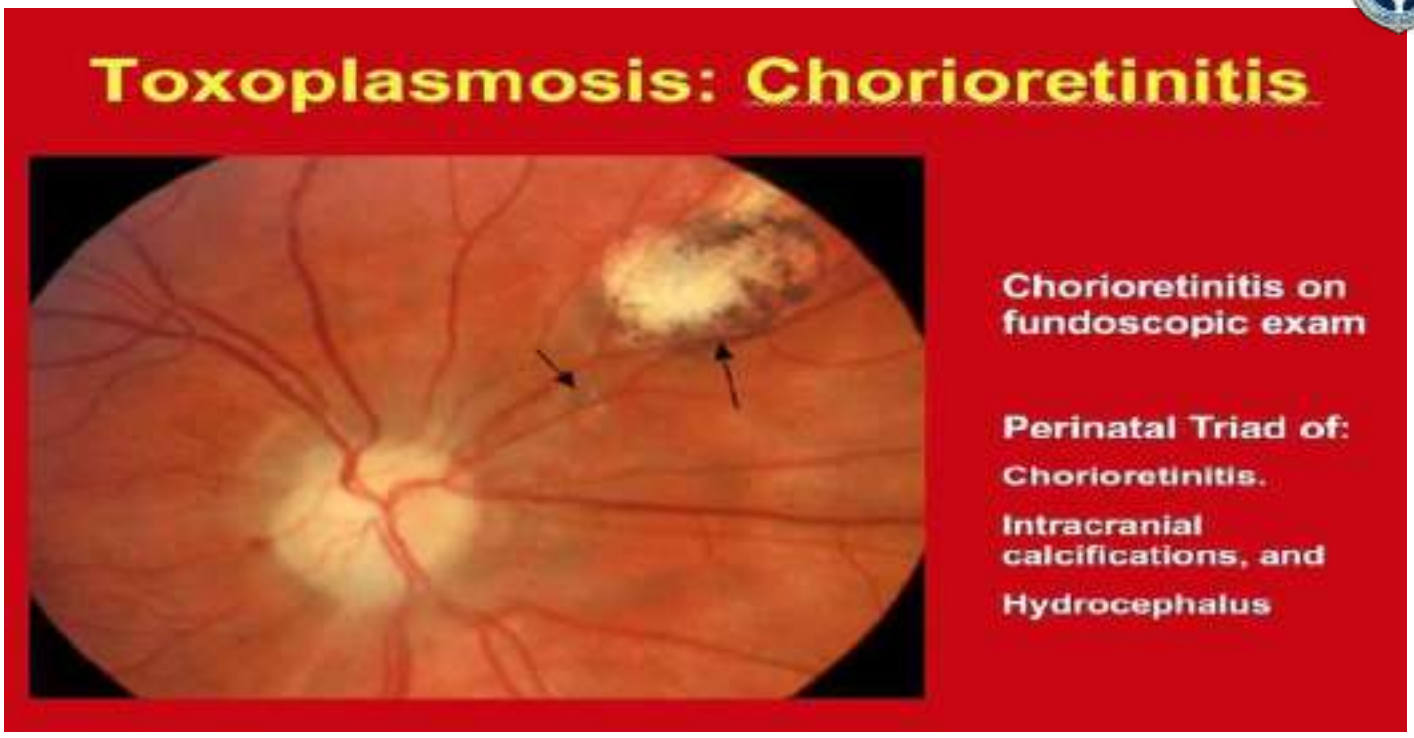
Acanthamoeba-infected eye

2-Trypanosoma cruzi (Romana 's sign)

- In the ***acute*** stage, ***trypanosomes*** enter the body through the conjunctiva of the ***eye***, they cause ***swelling*** of the eyelids and regional preauricular lymph node.
- This is known as ***Romana"s sign.*** It is common in ***children***. Parasites at this time can be found in ***tears*** and ***conjunctival swabs.***

3 - Toxoplasma gondii

- In ***chronic infection***, it may cause ***retinal "blind spots"*** or ***small infarcts***, which may ***coalesce*** in the macula causing ***blindness***.
- In ***congenital toxoplasmosis:*** ***chorioretinitis*** and ***blindness*** are the ***ocular manifestations***. Ocular manifestations may appear ***later*** in children.



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3) Arthropods

1-Ocular Myiasis

- ***Myiasis*** is a disease ***resulting*** from the infestation of ***human*** or ***animal tissues***
- or ***organs*** by larval stages of ***dipterous flies.***

Ocular myiasis (ophthalmomyiasis), this is either by:

1.the ***larvae feeding*** on the ***discharge*** of the inflamed eye (***semi specific***) e.g. ***Calliphoridae, Musca***

2.or by ***selective penetration*** of intact palpebral skin and conjunctiva, e.g. ***Oestrus ovis.***

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FACULITY OF
MEDICINE

Neuroscience

Pathology

NEU-316

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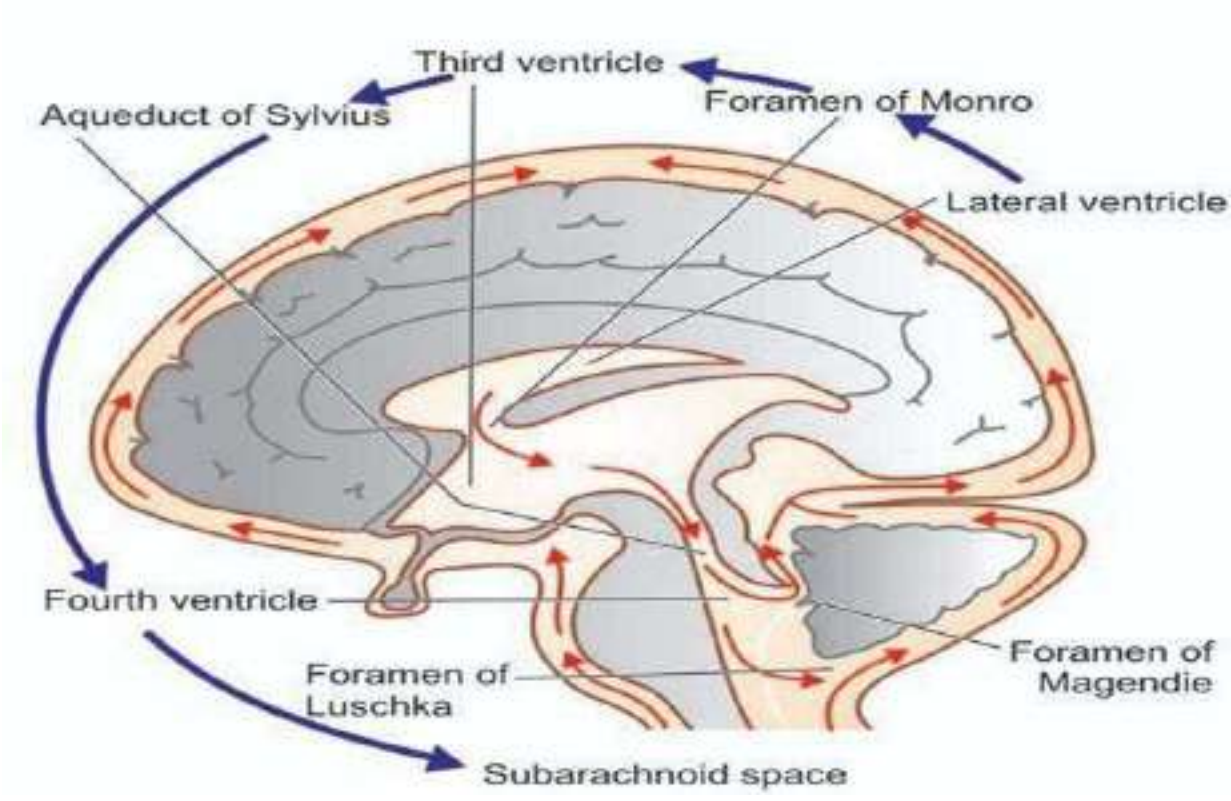
Basic pathology of the CNS

1 Cerebrospinal fluid (CSF)

- **Production and circulation:**
  - √ Produced by choroid plexus inthe lateral ventricles. Pass from the
  - √ lateral to the third ventricle through foramina of Monro (interventricular foramen). Pass from the third to the fourth ventricle through aqueduct of
  - √ Sylvius. Passes from the fourth ventricle to the sub- arachnoids space through foramina of Magendie and Luschka.
  - √
  - √ CSF then drains to the superior sagittal sinus through the arachnoid villi or granulations.



1) *hydrocephalus*

- **Definition:**

Abnormaldilatation of the ventricular systembyexcess CSF with atrophyofthebrain tissue.

- **Causes:**

I. IncreasedCSFproduction:

Choroidplexuses tumor as papilloma.

II. Obstructionofthe CSF flow:

A-Congenital causes:

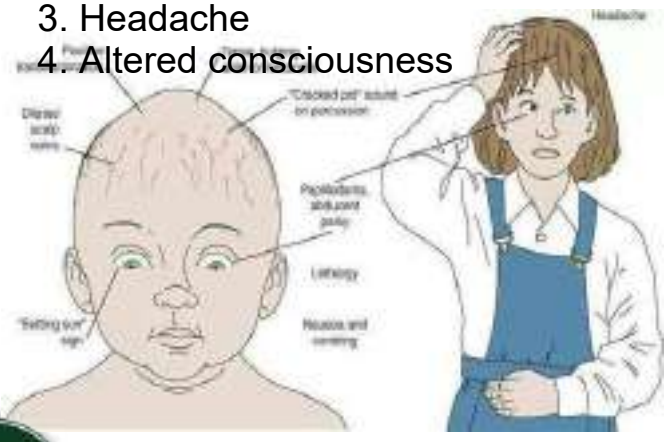
- Stenosisofthe aqueduct cerebri; foraminaofMonro and Sylvius.
- Stenosisofthe foramina of fourth ventricle;foramina of Magendie and Luschka.
- Arnold-Chiarimalformation.

B-Acquired causes:

- Meningitis → fibrous adhesions obstructingthe foramina of the fourth ventricleor the subarachnoid space.
- Brain abscessmay obstruct the pathwayofCSF.
- Brain tumorsmay also obstruct the pathwayofCSF

- **ClinicalFeatures of hydrocephalus** (Features of ↑ ICT)

1. Nausea
2. Vomiting
3. Headache
4. Altered consciousness



III-Decreased CSF drainage or absorption:

A. Congenital causes:

Aplasia of arachnoid villi

B. Acquired causes:

- Damage of arachnoid villi by inflammatory exudate or fibrosis.
- Obliteration of the subarachnoid space by subtentorial space occupying lesion → pushes the brain up against the skull bones.
- Thrombosis of the superior sagittal sinus.

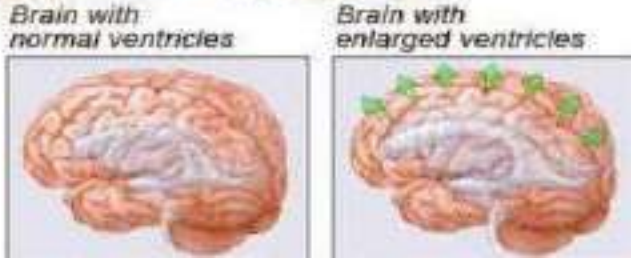
- **Types:**

1. Communicating hydrocephalus:

The obstruction is in the subarachnoid space around the mid brain. CSF can pass from the ventricular system to the subarachnoid space.

2. Non-communicating

hydrocephalus: The obstruction is within the ventricular system or in the roof of the fourth ventricle



2) *Increased intracranial tension (ICT)*

• Etiology:

1.Intracranial tumors

2.Intracranial inflammation:

A. Acute:

•Suppurative:septicmeningitis

•Non suppurative: viral meningitis

B. Chronic:

•Nonspecific: chronic abscess

•Specific:TB andGummaof syphilis

3.Intracranial vascular disorders

A. Aneurysms

B. Hemorrhages.

4.Hydrocephalus

• Symptoms:

Persistent headache
Vomiting
Blurred vision

• Effects:

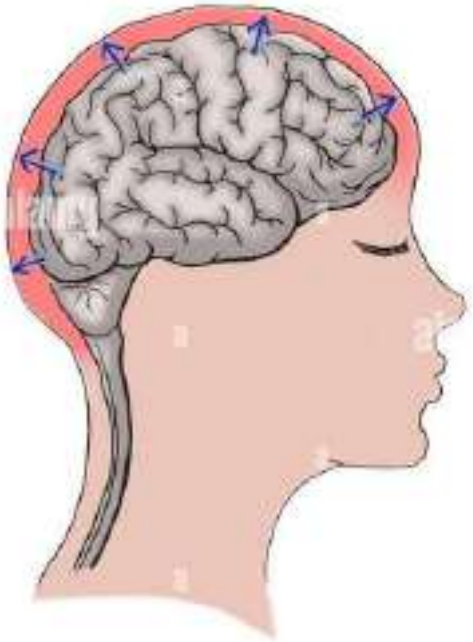
√ Flattened brain convolution on the same side of lesion.

√ Shift of midline structures to the opposite side.

√ Papillaedema: due to compression of retinal veins in the sub-arachenoid space.

√ Thinning of skull bone over the lesion in chronic cases.

√ Intracranial herniation: a)Tentorial b)Tonsillar



3) Cerebral edema

- **Definition:**
  - √ It isan excess accumulation of fluid in the intracellular or extracellular spaces of the brain, or
  - √ It is an accumulation of excess fluid within the brain parenchyma.
- **Clinically:**
  - √ The brain becomes soft and smooth
  - √ Gyri (ridges) become flattened
  - √ Sulci (grooves) become narrowed
  - √ Ventricular cavities become compressed

- **Symptoms:**
  - √ Nausea, vomiting, blurred vision, faintness
  - √ in severe cases, seizures and coma.
  - √ If brain herniation occurs, respiratory symptoms or respiratory arrest can also occur due to compression of the respiratory centers in the pons and medulla oblongata.

- **Etiology:**
  - √ Cerebral edema can result from trauma to the brain
  - √ or from non-traumatic causes such as ischemic stroke, tumors, or inflammation due to meningitis or encephalitis.
  - √ Vasogenic edema is caused by amyloid-modifying treatments.
  - √ The blood-brain barrier (BBB) or the blood-cerebrospinal fluid (CSF) barrier may break down, allowing fluid to accumulate in the brain's extracellular space.
  - √ Altered metabolism may cause brain cells to retain water, and dilution of the plasma may cause excess water to move into the brain cells.
  - √ Fast travel to high altitude without proper acclimatization can cause highaltitude cerebral edema

- **Gross picture:**
  - √ The edematous brain is softer and appears to overfill the cranial vault.
  - √ The gyri are flattened as a result of compression of the expanding brain by the dura matter and the inner surface of the skull. The intervening
  - √ sulci are narrowed. The ventricular cavities are compressed
  - √

- **Types:**

Fourtypes of cerebral edema have been identified:

a) Vasogenic edema
b) Cytotoxic edema
c) Osmotic edema
d) Interstitial edema

1) Vasogenic edema:	2) Cytotoxic edema:
● integrity of the normal blood-brain barrier; BBB is disturbed ● Subtypes of vasogenic edema include: **A.** Hydrostatic cerebral edema: This form of cerebral edema is seen in cases of malignant hypertension. **B.** Cerebral edema from brain tumors: Glioma can increase secretion of VEGF,which weakens the junctions of the BBB. **C.** High altitude cerebral edema (HACE): It is severe and sometimes fatal form of altitude sickness that results from capillary fluid leakage due to the effects of hypoxia on the mitochondria-rich endothelial cells of the BBB.	• It is an↑in the intra-cellular fluid secondary to neural, glial, or endothelial cell membrane injury e.g., generalized, hypoxic/ischemic insult, or exposure to some toxins.
3) Osmotic edema	4) Interstitial edema
● Normally, the osmolality of CSF and extracellular fluid in the brain is slightly lower than that of plasma. ● Plasma dilution decreases serum osmolality, resulting in a higher osmolality in the brain compared to the serum.	• Interstitial edema occurs in obstructive hydrocephalus due to a rupture of the CSF-brain barrier. • Interstitial cerebral edema differs from vasogenic edema as CSF contains almost no protein.

2 Pathology of cerebrovascular diseases

- **Introduction:**
  - √ Cerebrovascular diseases constitute a broad category of brain disorders; caused by pathologic processes that involve blood vessels.
  - √ They constitute a major cause of neurological mortality and morbidity.

- **Three main categories are:**
  1) Cerebral ischemia
  2) Cerebral infarction
  3) Intracranial haemorrhage
  4) Cerebral aneurysm

1) Cerebral ischemia

- The brain is a highly oxygen-dependent tissue. cerebral blood flow normally remains stable over a wide range of blood pressure and intracranial pressure
- because of vascular auto-regulation **Brain ischemia has two main**
- **patterns:**
  - ❖ Global cerebral ischemia
  - ❖ Focal cerebral ischemia

NOTE

Inglobal cerebralischemia:

Stroke: is abrupt onset of global neurological symptoms due to ischemia or hemorrhage with symptoms more than 24 hours

Transient ischemic attack (TIA): if symptoms resolve in less than 24 hrs, the term transient ischemic attack (TIA) is applied

1) Global cerebral ischemia	2) Focal cerebral ischemia
• **Definition:** ○ widespread ischemic / hypoxic injuryof brain.	• **Definition:** o Focal ischemia of a zone of braintissue due to occlusion of feeding artery. o Vascular collaterals of the brainare minimal or deficient; so branchesof cerebral arteries are termed "end arteries". o Arterial occlusion leads first to focal ischemia and then to infarction
• **Etiology:** due to severe systemic hypotension (systolic pressures < 50 mmHg) as in cases of cardiac arrest and different types of shock	• **Etiology:** Focal cerebral ischemia occurs due to either: ❖ **Cerebral embolism:** Sources of emboli are: √ Emboli from the heart: cardiac mural thrombi, myocardial dysfunction, √ valvular disease, and atrial fibrillation . Emboli from big arteries: atheromatous plaques of carotid arteries or aortic arch. ❖ **Cerebral thrombosis:** occurs on top atherosclerotic plaques of cerebral arteries commonly carotid bifurcation.
• **Fate:** depends on severity and duration of the insult. Mild ischemia >>>>> only a transient post-ischemic confusion state, with eventual complete recovery. ○ Severe ischemia >>>> wide neuronal death. Patients who survive usually have severe neurological impairment. Other patients meet the clinical criteria for brain death	• **Morphology:** ❖ **Grossly:** The macroscopic appearance changes over time. √ In early hours…. brain tissue looks normal √ After 48 hours…. the tissue becomes pale, soft and swollen. √ From 1-3 weeks…. the tissue liquefies, eventually leaving a fluid-filled cavity. ❖ **Microscopically** √ After 12 hours…. nuclear pyknosis and ischemic neuronal change with disintegration of myelin sheath. √ After 48 hours……infiltration by neutrophils followed by macrophages.

2) Cerebral infarction

- **Causes:**

√ Arterial occlusion is mainly due to atherosclerosis with thrombosis of internal carotid, cerebral, vertebral and basilar arteries.

√ Embolism from the left side of the heart.

√ Venous thrombosis associated with pregnancy and oral contraceptive pills.

- **Site:**

The commonest sites of thrombosis are basal ganglia, internal capsule and cerebral cortex.

- Cerebral infarction is an example of liquefactive necrosis
- It is changed into a fluid filled cavity surrounded by glial tissue (gliosis).

- **MORPHOLOGIC FEATURES:**

1 grossly
2 histologically

1) grossly	2) histologically
• Cerebral infarcts maybepale or red (hemorrhagic).	• Thesequentialchangesareasfollows:
• **Pale infarct** becomes evident 6-12 hours after its occurrence. 1. The affected area is soft and swollen 2. blurring of junction between grey and 3. white matter 4. Within 2-3 days, the infarct undergoes softening and disintegration. 5. Eventually, there is central liquefaction with peripheral firm glial reaction and 6. thickened leptomeninges, forming a cystic infarct	1. Initially,thereiseosinophilicneuronal necrosis 2. Theinfarctedareaisinfiltratedby neutrophils. 3. Afterthefirst2-3days,thereis progressiveinfiltrationbymacrophages 4. Thereisastrocyticandvascular proliferation. 5. In thefollowingweekstomonths,the macrophagesclearawaythenecrotic debris by phagocytosis followed by reactiveastrocytosis,oftenwithlittle fine fibrosis
• **Red (haemorrhagic) infarct** 1. It is red 2. superficially resembles a hematoma. 3. It is usually the result of fragmentation of occlusive arterial emboli or venous thrombosis.	6. Ultimately,after3-4monthsanoldcystic infarct is formed which shows a cyst traversedbysmallbloodvesselsandhas peripheral fibrillary gliosis. 7. Smallcavitaryinfarctsarecalledlacunar infarcts and are commonly found as a omplication of systemic hypertension.

3) *Intracranial hemorrhage*

- **Definition:**
  Hemorrhagewithin the cranial cavity.

- **Types:**

1-Meningeal Hemorrhage

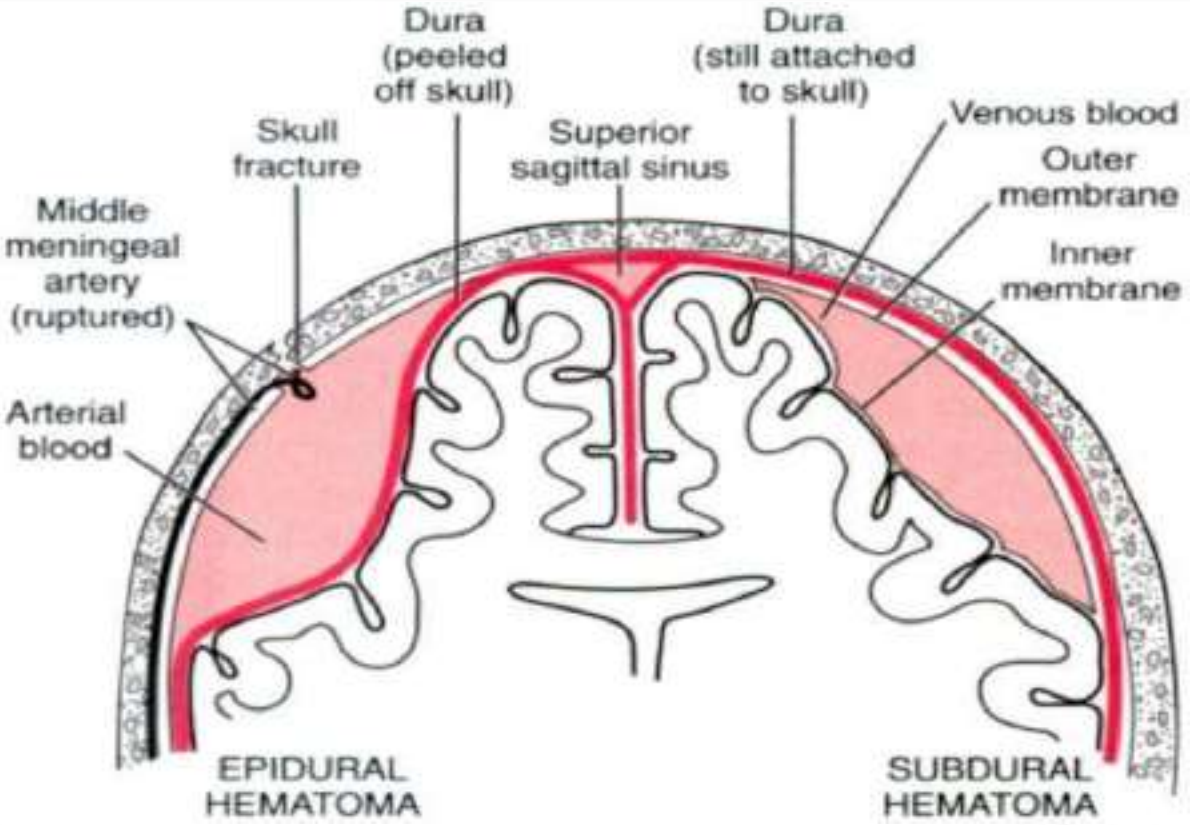
- Extradural hemorrhage
- Subdural hemorrhage
- Subarachnoid hemorrhage

2-Cerebral Hemorrhage

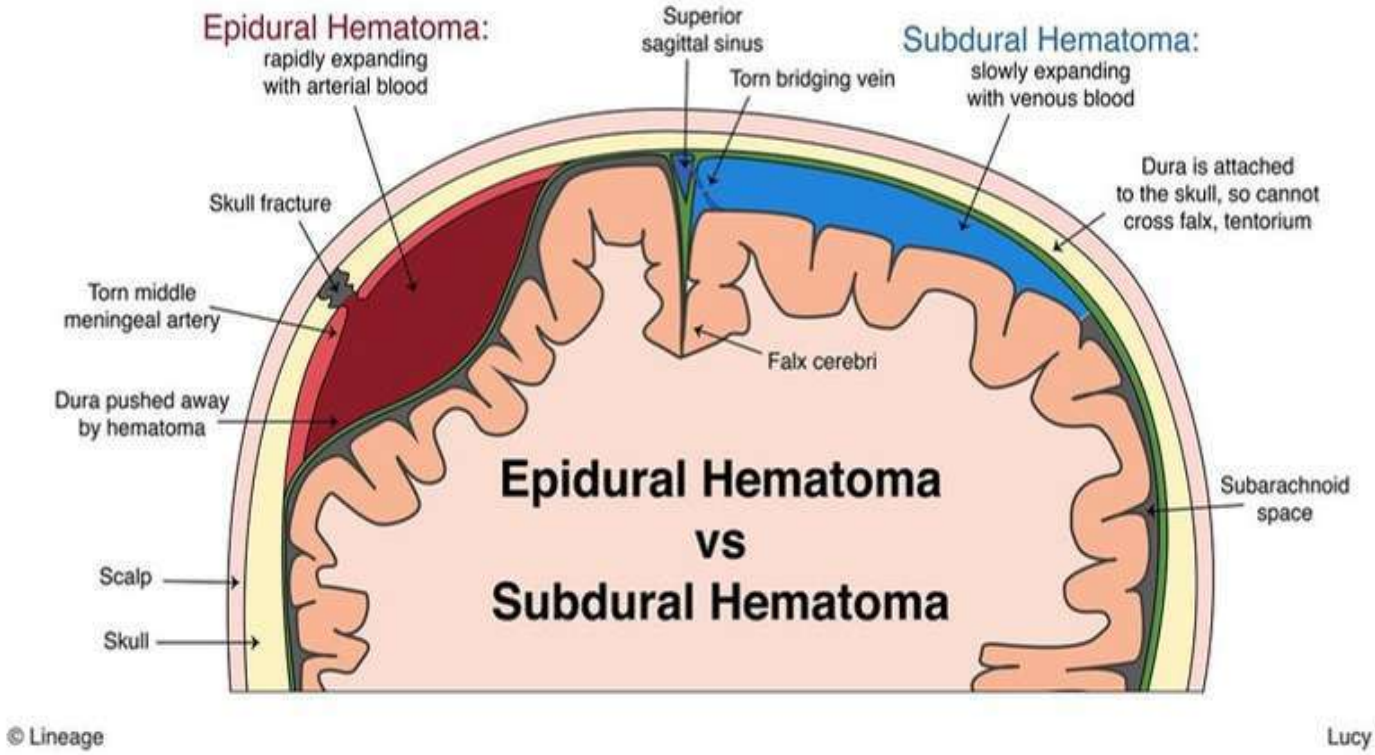
- Petechial hemorrhage
- Massive cerebral hemorrhage

Meningeal hemorrhage

1) Extradural (Epidural) hemorrhage	2) Subdural hemorrhage	3) subarachnoid Hemorrhage
• Blood accumulation between dura matter and skull	• Blood accumulation in subdural space.	• Blood accumulation in subarachnoid space
• Commonly by traumatic rupture of middle meningeal art	• Commonly by traumatic rupture of cerebral veins that cross the subdural space	• Can be caused by: √ Rupture of cerebral aneurysm (commonest, berry aneurysm) √ Traumatic laceration of brain √ Hemorrhagic blood diseases



Item	Epidural hemorrhage	Subdural hemorrhage
Location	Hematoma between skull and dura	Hematoma between dura and arachenoid
Involved vessels	**Arteries** - Middle meningeal artery (commonest) - Anterior ethmoid artery	**Veins** - Veins bridging sub-dural space
Clinically	Lucid interval followed by loss of consciousness	Gradually increasing headache and confusion



cerebral hemorrhage

- Mostly spontaneous (non-traumatic) hemorrhage.
- Accounts for 10-20% of all strokes.
- More in males.

❖ Location:

- Basal ganglion and thalamus (60%)
- Cerebral (20%)
- Cerebellum (13%)
- Pones (7%).

❖ Effects:

1.Hemiplegia.
2.Increased intracranial tension.
3.Compression of the brain stem and death due to tentorial herniation

❖ Causes:

- Hypertension: is the most common cause of non-traumatic hemorrhage due to:
  - Hyaline arteriolar sclerosis makes the arteriolar wall weaker and more vulnerable to rupture
  - The development of minute aneurysms in the cerebral vessels

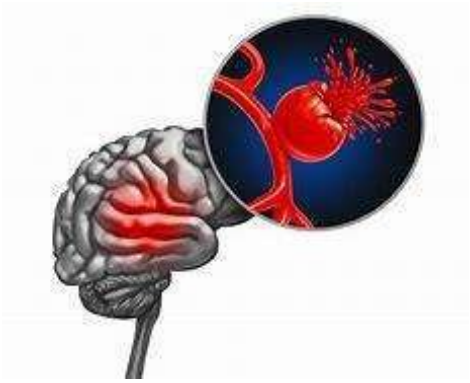
❖ Other causes:

- Rupture of cerebral aneurysms
- Neoplasms: bleeding into a degenerated tumor.
- Hemorrhagic blood diseases
- Vasculitis.
- Trauma.

1) Petechial cerebral hemorrhage:	2) Massive cerebral hemorrhage
• multiple small hemorrhages in the brain substance	• Extensive area of hemorrhages with destruction of the brain tissue.
• Capillary in origin	• Occurs mostly in one cerebral hemispheres due to hemorrhage from a branch of the middle cerebral artery.
• **caused by:** • meningitis • encephalitis • haemorrhagicblood diseases.	• **Common causes:** • Hypertension (commonest in old age). • Rupture of cerebral aneurysm (commonest in young age). • Haemorrhagic blood diseases. • Bleeding into a degenerated tumour. • Brain laceration by a trauma.

4) Cerebral aneurysm

- **Definition:**
  Localized dilatation of one (or more) branches of cerebral artery



- **Effect of cerebral aneurysms:**

1. Rupture leading to subarachnoid hemorrhage. Pressure effect
2. (increased ICT). Hydrocephaus
3. Thrombosis and calcification.
4. **Types:**

- 1. Congenital
  2. Mycotic
  3. Arteriosclerotic
  4. Arteriovenous

1) Congenital aneurysms (Berry aneurysm)	3) Arteriosclerotic aneurysm:
• Congenital weakness of the tunica media	• Fusiform in shape.
• Single or multiple small aneurysms (pin head)	• Develops in association with hypertension or cerebral atherosclerosis
• Commonly at the bifurcation of the cerebral arteries	• Sites: Internal carotid and basilar arteries
2) Mycotic aneurysms:	**4) Arteriovenous aneurysm:**
• Commonly due to subacute bacterial infection or Polyarteritis nodosa due to inflammatory weakness of the vascular wall	• Communication between the internal carotid artery and cavernous sinus.
• The commonest site is the branches of the middle cerebral artery	

3 Infections of the CNS

1) Meningitis

- **Definition:**
  inflammation of the meninges covering the brain and spinal cord.
  √ Rarely affects the dura
  √ (**Pachymeningitis**)
  More commonly affect the pia and arachnoid (**Leptomeningitis**)

Types:

❖ Septic (suppurative) meningitis
❖ Aseptic (non- suppurative) meningitis:

- √ Tuberculosis
- √ Viral
- √ Fungal
- √ syphilis

A) Septic meningitis

❖ Causative organisms

- **Meningococcal meningitis:**

• Meningococci is the most common causative organisms of meningococcal meningitis.
• Affects specially infants, children and young adults
• sporadic or epidemic forms.
• Transmitted by droplet infection
• The organism causes nasopharyngitis >>>>>> invade the blood stream >>>>> are carried to the meninges and choroid plexus

- **Otherpyogenic bacteria:**

- As streptococci, staphylococci, pneumococci and influenza bacilli:
- Rare causes for septic meningitis.
- They reach the meninges by the following routes:

√ **Blood spread:**
√ from a distant septic focus.
√ **Direct infection:** from sinusitis, otitis media and brain abscess.
Complicating compound fractures of the skull

❖ Gross Picture:

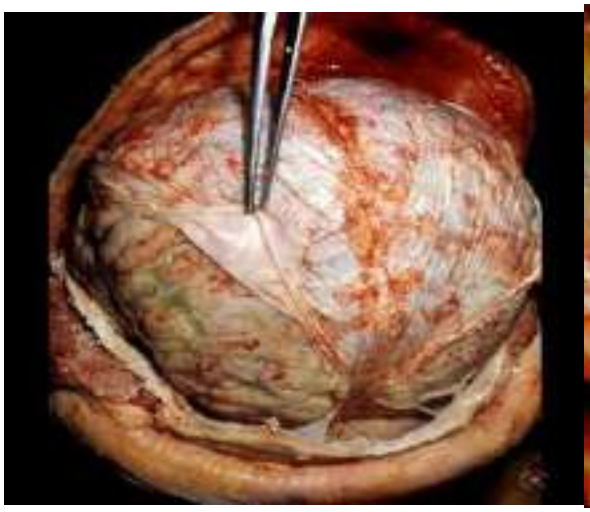
- The brainisswollen and edematous.
- Thickenedpia and arachnoid membranesand covered by purulent exudate

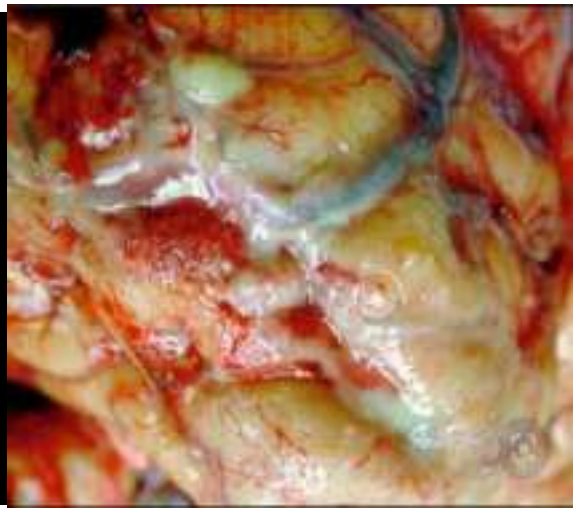
❖ Microscopic Picture:

•Theaffectedmeninges show dilated capillaries, infiltration by polymorphs, pus cellsand macrophages among fibrin threads.

- The meningeal and cortical vessels are dilated and congested. The
- subarachnoid space is filled with a yellowish purulent exudate. dilated
- ventricles and turbid CSF . The ependymal lining is necrotic rough,and the choroid plexus is congested.

•The subarachnoid space shows dilated congested vessels.





❖ CSF changes:

analysisofCSF shows:

- Increasedtension
- Presenceofmany PNLs, pus cells and bacteria.
- Elevatedprotein content.
- Decreasedsugar content.
- Culturewillshow the causative organism

❖ Clinicalpresentation

- Headache
- Irritability
- Neck stiffness
- Disturbed conscious level
- Photophobia

❖ Local effects and complications:	❖ General effects and complications:
• Thrombosis of the pial arteries causes infarctions in the brain and spinal cord.	• Septic thrombophlebitis causes septicemia and systemic pyemia
• Healing by fibrosis results in adhesions and obstruction of the foramina of Lushcka and Magendie which cause:	• Blood spread of bacteria causes acute bacterial endocarditis, pericarditis, empyema and arthritis.
• Compression of the cranial nerves; III, IV, VI and spinal nerve roots leading to diplopia, squint, ptosis.	• Adrenal cortical hemorrhage in severe cases causes acute adrenal cortical insufficiency due to septicemia
• Obstruction of the flow of CSF leading to hydrocephalus.	

B) *Viral (aseptic meningitis)*

- Less fulminant clinical course than septic meningitis.
- Self-limited and often treated symptomatically.
- Commonly due to entero-virus infection.
- There are no distinctive macroscopic findings except for brain swelling, which may be seen in only some instances.
- On microscopic examination, there is either no recognizable abnormality or a mild to moderate leptomeningeal lymphocytic infiltrate.
- **CSF shows:** Increased number of lymphocytes, moderate protein elevation and nearly normal glucose content

	Bacterial meningitis	Viral meningitis
Causative organism	Pyogenic bacteria mainly meningococci	Viruses mainly enterovirus
Course	progressive	Less fulminate clinical course , Self- limited
grossly	The brain is swollen and edematous. Thickened pia and arachnoid membranes and covered by purulent exudate The meningeal and cortical vessels are dilated and congested. The subarachnoid space is filled with a yellowish purulent exudate.	no distinctive macroscopic findings except for brain swelling
microscopic	The affected meninges show dilated capillaries, infiltration by polymorphs, pus cells and macrophages among fibrin threads. The subarachnoid space shows dilated congested vessels	no recognizable abnormality or a mild to moderate leptomeningeal lymphocytic infiltrate
CSF	Increased tension Presence of many PNLs, pus cells and bacteria. Elevated protein content. Decreased sugar content.	Increased number of lymphocytes, moderate protein elevation nearly normal glucose content
complicactions	More common	Less common

C) Tuberculous meningitis

- More common in children
- Affects the pia-arachnoid, its blood vessels and the superficial layer of the brain.
- It is due to the blood spread of a tuberculous focus in the body.

Pathological features:

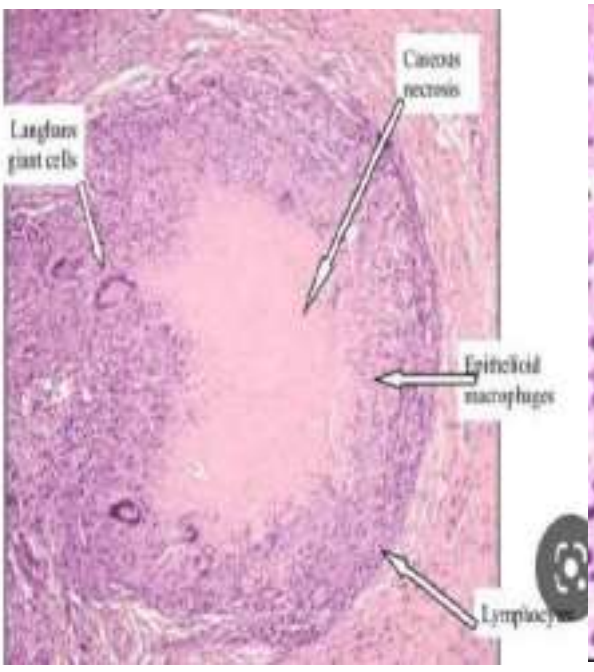
- The surface of the brain is covered by yellowish exudate that is marked over the base.
- Exudate is composed of epithelioid cells, lymphocytes, scanty giant cells, fibrin and shows areas of caseation.
- Healing by fibrosis may obstruct the roof of the fourth ventricle causing hydrocephalus.
- Tubercles on the small blood vessels of the brain, choroid plexus and lining of the ventricular cavities.

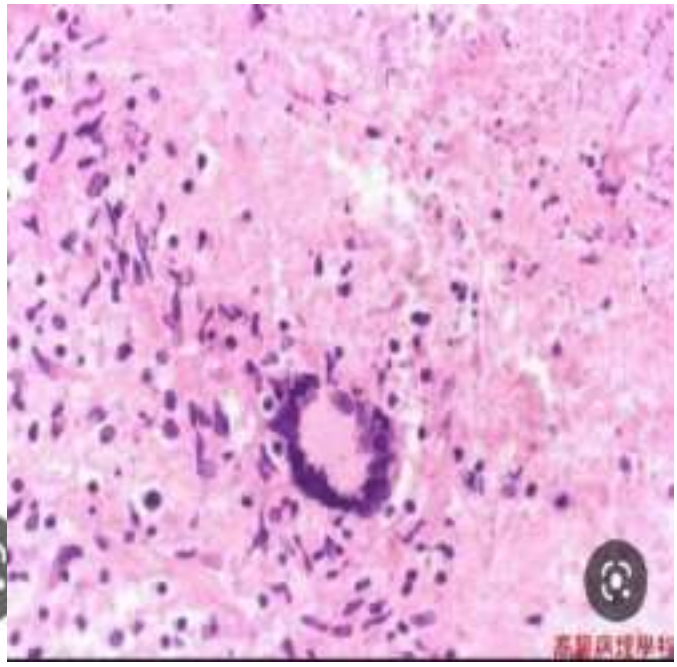
CSF changes:

- The fluid is clear and a fine network of fibrin forms on the surface on standing.
- Moderate increase in mononuclear inflammatory cells, mainly lymphocytes.
- The protein level is elevated.
- Glucose content is moderately reduced or
- normal.
- A small number of bacilli could be detected in smears.

Complications:

- Increased ICT
- Compression of the cranial nerves leading to paralysis.
- Hydrocephalus.
- May result in well circumscribed mass in
- the brain (Tuberculoma).
- Miliary tuberculosis.





Brain Abscess

Etiology:

Causative organisms:

- Pyogenicbacteria such as staphylococcus aureus, streptococcus, pneumococcusand E.coli

Routes of infections:

- **Direct spread:** from chronic suppurative focus as otitis media, sinusitis and mastoiditis.
- **Blood spread:** from acute bacterial endocarditis, bronchiectasis, lung abscess or empyema through the vertebral system of veins.
- **Infection from outside:** through compound fractures of the skull.

Sites:

- The site of abscess depends upon the route of infection.
- Otitis media → temporal or parietal lobe abscess.
- Mastoiditis → cerebellar abscess.
- Sinusitis, lung infection and fracture → frontal lobe abscess.
- Hematogenous abscesses from systemic pyemia are multiple the junction of the grey and white matter.

Grossly:

- Early brain tissue shows edema, softening and heavy neutrophilic infiltration.
- Active liquefactive necrosis is surrounded by zones of congestion.
- An abscess cavity filled with pus and lined by necrotic tissue develops.
- The affected part appears swollen, convolutions flattened, and meninges become thick and adherent.
- The abscess tends to spread inwards and rupture into the ventricles.
- If the pus is absorbed or evacuated spontaneous healing occurs by gliosis, but more commonly the abscess changes to chronic abscess.

Clinical and Complications:

- Progressive focal deficits.
- Increased ICT and brain herniation.
- Rupture of an abscess leads to ventriculitis, meningitis, and
- venous sinus thrombosis.

Septic meningitis, subdural abscess, extradural abscess andsinus thrombophlebitis.

CSF changes:

- Raised WBCs count
- Raised protein level.
- Normal glucose content

Encephalitis

- **Definition:** inflammation of the brain.
- **Itis caused by:**
  - Viralinfections such as rabies, poliomyelitis, herpes zoster and herpes simplex.
  - Pyogenic bacteria
  - Bacterial toxins as botulism.
  - Fungus infections as cryptococcosis.
  - Parasitic infections like malaria and toxoplasmosis.

Viral encephalitis

- That is an infection of the brain which is invariably associated with inflammation of the meninges (meningeo-encephalitis).
- The most characteristic features are perivascular and parenchymal mononuclear cell infiltrate.
- Certain viruses may form inclusion bodies.

DEMYELINATING DISEASES

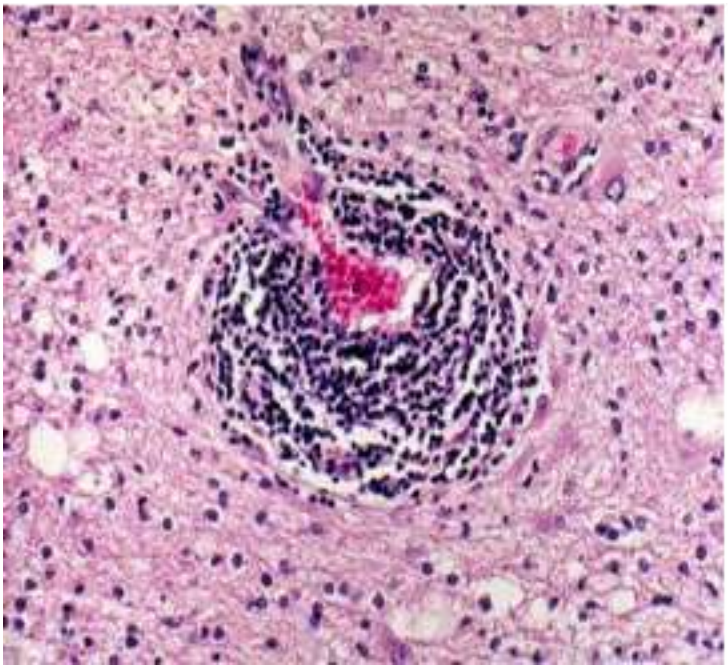
- Demyelinating diseases are an important group of neurological disorders which have the pathologic features of focal or patchy destruction of myelin sheaths in the CNS accompanied by an inflammatory response. Demyelination may affect the peripheral nervous system as well. The exact cause for demyelination is not known but currently viral infection and auto immunity are implicated in its pathogenesis. These conditions have known etiologies such as:

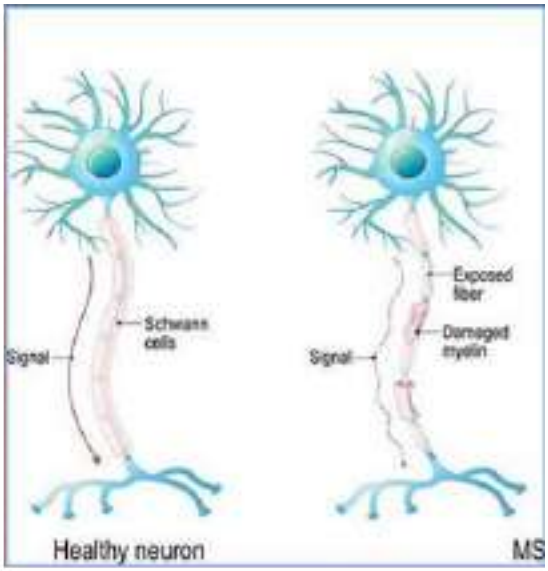
Multiple (Dissem i n ated) Sclerosis

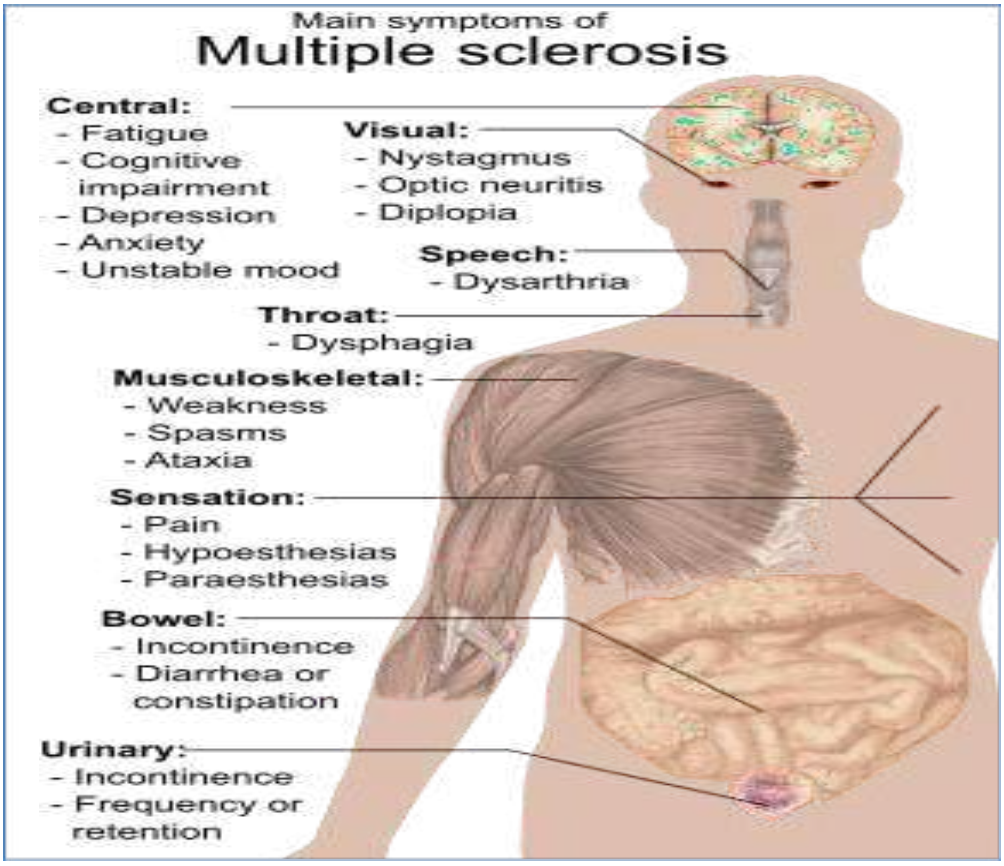
- Multiple or disseminated sclerosis is the most common of the CNS demyelinating diseases. The usual age at onset is 20 to 40 years. The disease presents recurrent attacks of focal neurologic disorder with predilection for involvement of the spinal cord, optic nerve and brain. The first attack usually begins with a single sign or symptom, most commonly optic neuritis, followed

- by recovery. As the disease becomes more progressive, remissions become infrequent and incomplete. The etiology of multiple sclerosis remains unknown but a role for genetic susceptibility, infectious agent and immunologic mechanism has been proposed. The pathologic hallmark is the presence of
- many discrete areas of demyelination termed plaques. **Grossly**, plaques appear as grey-pink, swollen, sharply defined, usually bilaterally symmetric areas in the white matter.

• **Microscopically:** In active enlarging plaques, the histologic features are accumulation of lymphocytes and macrophages around venules and at the plaque margin where demyelination is occurring. In addition, there is loss of oligodendrocytes and presence of lipid-laden macrophages in the plaque. The axons in the plaque are generally intact. In old inactive plaques, there is no perivascular inflammatory cell infiltrate and nearly total absence of oligodendrocytes. Demyelination in the plaque area is complete as there is only limited regeneration of myelin.







DEGENERATIVE DISEASES

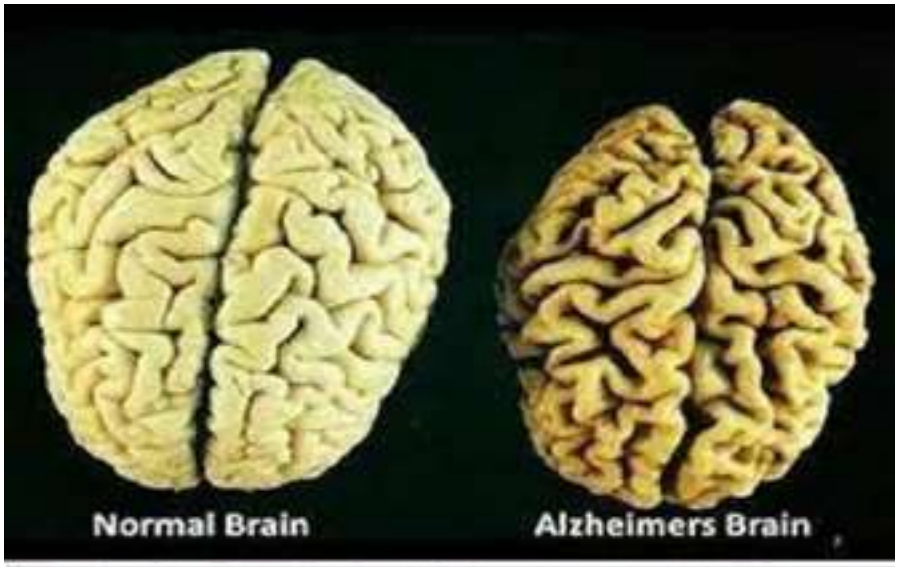
Degenerative diseases are disorders of unknown etiology, characterized by progressive loss of CNS neurons. The degenerative disorders usually begin insidiously and have a gradual progressive course over many years. The lesions have characteristic bilaterally symmetric distribution, and the degenerative disorders may be selectively affected, leaving others entirely intact.

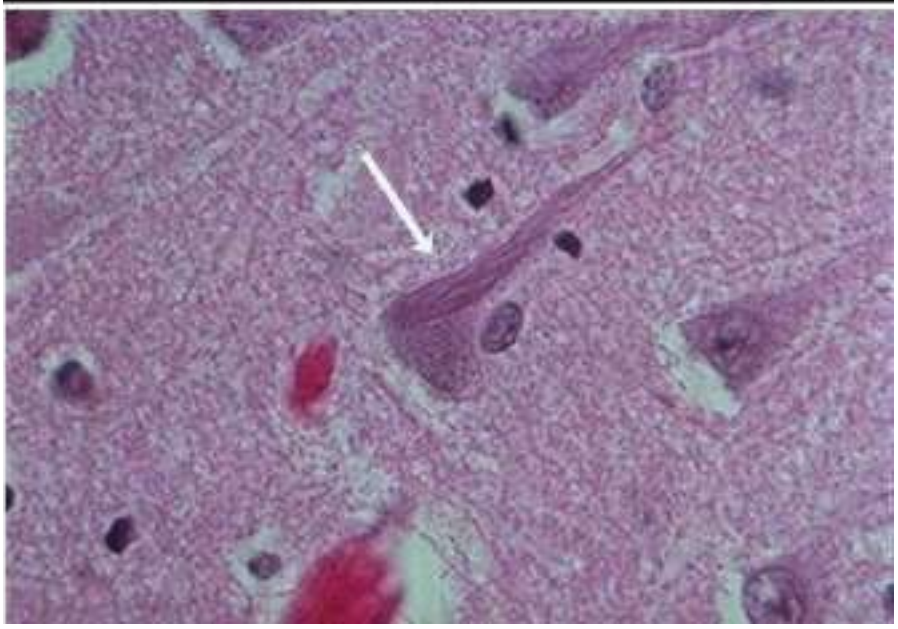
ALZHEIM ER'S DISEASE

Alzheimer's disease is the most common cause of dementia in the elderly. The condition occurs after 5th decade of life and its incidence progressively increases with advancing age. The exact cause is not known but a few factors are implicated in its etiology which includes positive family history and deposition of Ab amyloid derived from amyloid precursor protein (APP) forming neuritic senile plaques and neurofibrillary tangles. **Grossly,** the brain is often reduced in weight and bilaterally atrophic.

Microscopically: The main features are:

1. Senile neuritic plaque is the most conspicuous lesion and consists of focal area which has a central core containing Ab amyloid.
2. The neurofibrillary tangle is a filamentous collection of neurofilaments and neurotubules within the cytoplasm of neurons.





This is a neurofibrillary "tangle" of Alzheimer's disease. The tangle appears as long pink filaments in the cytoplasm. They are composed of cytoskeletal intermediate filaments.

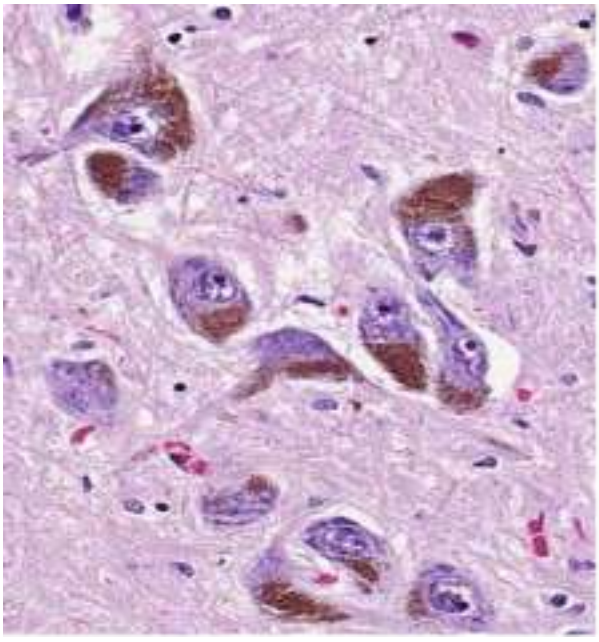
Parkinsonism

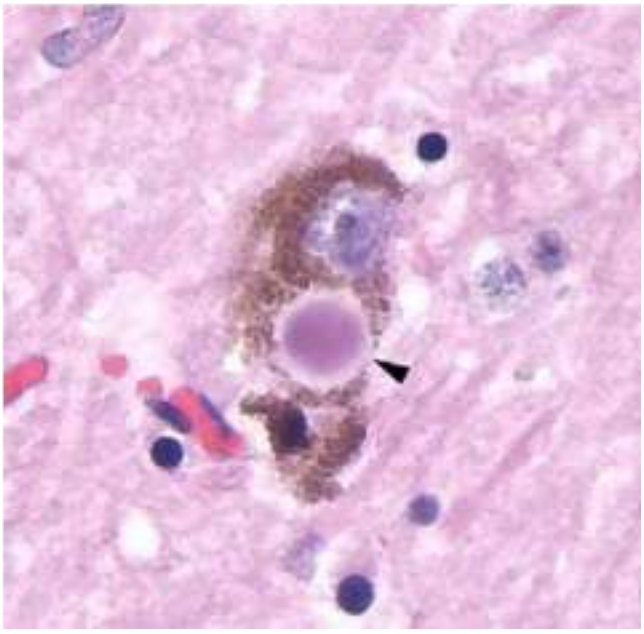
Parkinsonism is a syndrome of chronic progressive disorder of motor function and is clinically characterized by:

1. Tremors which are most conspicuous at rest and worsen with emotional stress.
2. Rigidity and disordered gait and posture.

Parkinsonism is caused by several degenerative diseases, trauma, toxic agents, and drugs (dopamine antagonists).**Grossly**, the brain is atrophic or may be normal.

- Microscopically: the hallmark is depigmentation of substantia nigra and locus ceruleus due to loss of neuromelanin pigment from neurons and accumulation of neuromelanin pigment in the glial cells. Some of the residual neurons in these areas contain intra cytoplasmic, eosinophilic, elongated inclusions called Lewy bodies.





TUMOURS OF THE CNS

-Tumours of the CNS may originate in the brain or spinal cord (primary tumours) or may spread to the brain from another primary site of cancer (metastatic tumours). More than one quarter of the CNS tumours are secondary metastases.

-A classification of intracranial tumours

I. 1.**TUMOURS OF NEUROGLIA (GLIOMAS)**

2. Astrocytoma

3. Oligodendroglioma

4. Ependymoma

II. Choroid plexus papilloma

1. **TUMOURS OF NEURONS**

2. Neuroblastoma.

3. Ganglioneuroblastoma

III. Ganglioneuroma

1. **POORLY-DIFFERENTIATED AND EMBRYONAL TUMOURS**

2. Medulloblastoma

3. Neuroblastoma

IV. PNET

1. **TUMOURS OF MENINGES**

2. Meningioma

V. Meningeal sarcoma

1. **NERVE SHEATH TUMOURS**

2. Schwannoma (neurilemmoma)

3. Neurofibroma

VI. Malignant nerve sheath tumour

METASTATIC TUMOURS

GLIOMAS

-The term glioma is used for all tumours arising from neuroglia, or from neuroectodermal epithelial tissue. Gliomas are disseminated to other parts of the CNS by CSF, but they rarely ever metastasize beyond the CNS They include tumours arising:

1. From astrocytes (astrocytoma and glioblastoma multiform);
2. From oligodendrocytes (oligodendroglioma);
3. From ependymal cells (ependymoma).
4. From choroid plexus (choroid plexus papilloma).

Astrocytomas

-Astrocytomas are the most common type of gliomas. They are found in 6th decade of life. They occur predominantly in the cerebral hemispheres and the spinal cord. In children and young adults, pilocytic astrocytoma arise in the optic nerves, cerebellum and brainstem. Astrocytomas have tendency to progress from low grade to higher grades of anaplasia. Low-grade astrocytomas grow slowly over several years than higher grades (anaplastic astrocytoma and glioblastoma multiform)

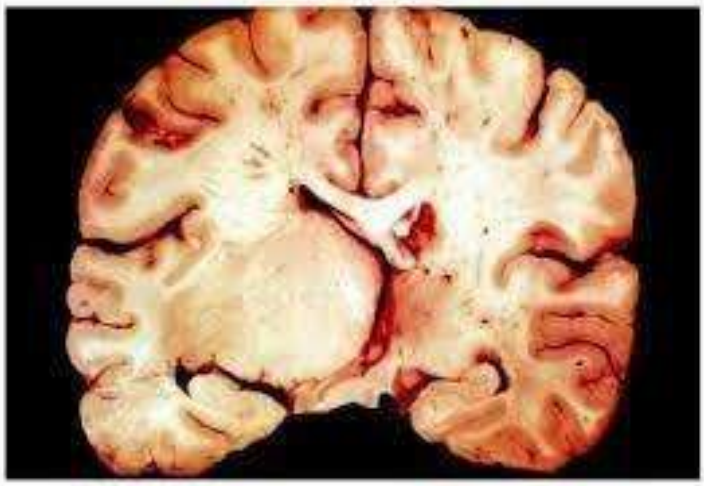
WHOclassificationofastrocytoma:into4gradesfromgrade I (lowgrade) to grade IV (glioblastoma multiform).

1. (WHO GRADE I) ASTROCYTOMA:

Also called as diffuse astrocytoma, it is a

low-grade tumour having good prognosis and includes Juvenile pilocytic astrocytoma. It occur in children and young adults in the cerebellum, third ventricle and optic nerve pathway. **Grossly**, it is usually cystic or solid and circumscribed.

-Microscopically: it is predominantly composed of fusiform pilocytic astrocytes having unusually long, wavy fibrillary processes.



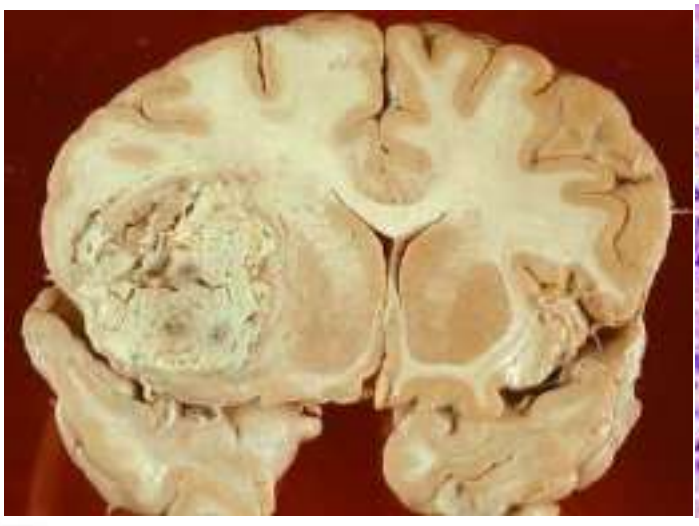
Pilocytic astrocytoma

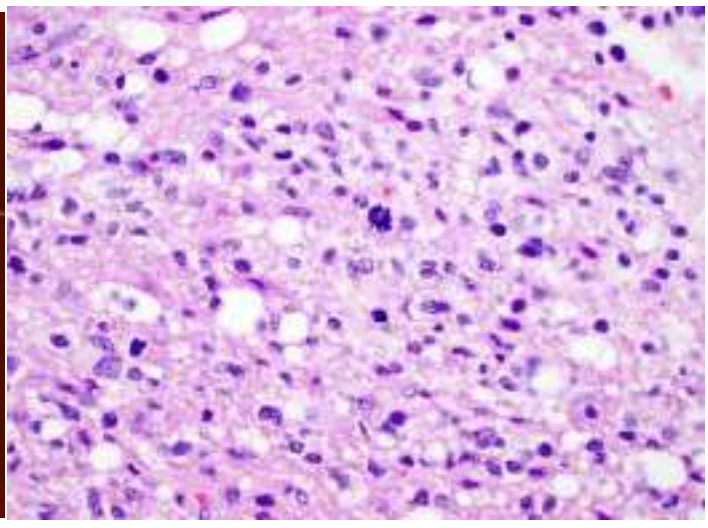
2. (WHO GRADE II) (WELL-DIFFERENTIATED) ASTROCYTOMA

It is also called fibrillary astro cytoma and is the most common form of glioma occurring in 3rd to 4th decades of life. **Grossly**, it is a poorly defined, grey-white tumour of variable size. The tumour distorts the underlying brain tissue. **Microscopically**, it is composed of well-differentiated astrocytes separated by variable amount of fibrillary background of astrocytic processes. Based on the type of astrocytes, three subtypes are distinguished: fibrillary, protoplasmic and gemistocytic astrocytoma.

3. WHO GRADE III (ANAPLASTIC) ASTROCYTOMA

It generally evolves from lower grade of astrocytoma. **Grossly** , it may not be distinguishable from the low-grade astrocytoma. **Microscopically** , it contains features of anaplasia (hypercellularity, pleomorphism, nuclear hyperchromatism and mitoses) and proliferation of vascular endothelium.





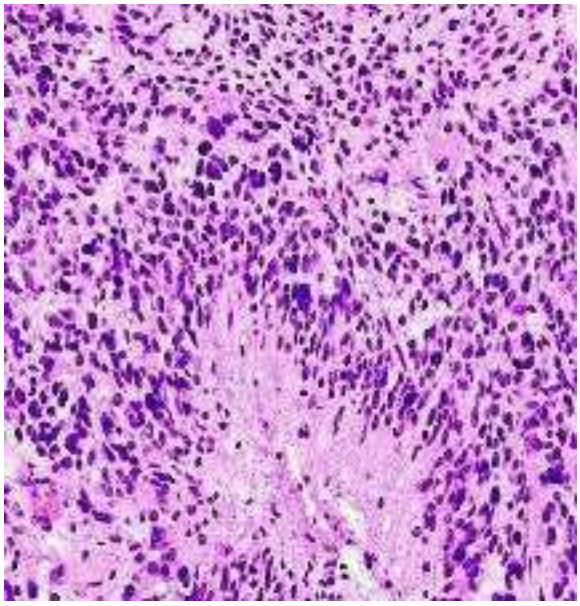
4. WHO GRADE IV ASTROCYTOMA (GLIOBLASTOMA MULTIFORME)

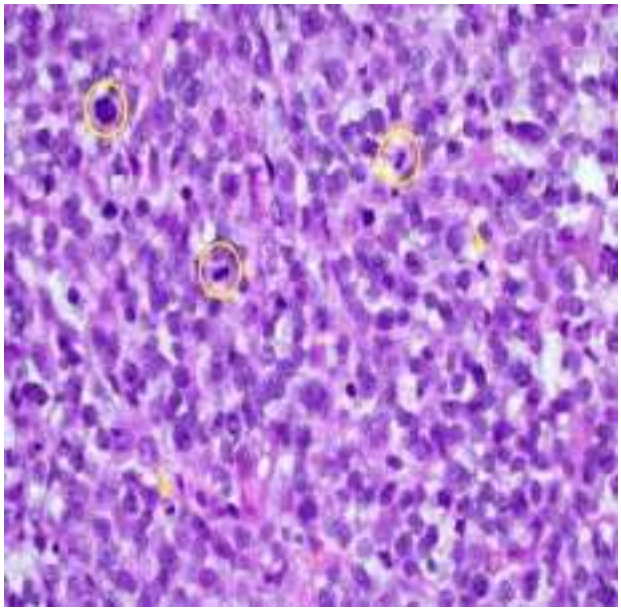
-Thistumour arisesby neoplastictransformation of mature astrocytes. It is the most aggressive of astrocytomas.

Glioblastoma multiforme Palisading necrosis Nuclear pleomrophism and mitosis

-Grossly some areas show grey-white appearance while others are yellow and soft with foci of haemorrhages and necrosis. The surrounding normal brain tissue is distorted and infiltrated by yellow tumour tissue.

-Histologically: It is highly anaplastic and hypercellular. The cell types show marked variation consisting of fusiform, pleomorphic and giant cells. Mitoses are frequent and glial fibrils are scanty. Some areas of tumour necrosis around which tumour cells may form pseudo palisading and microvascular endothelial proliferation.

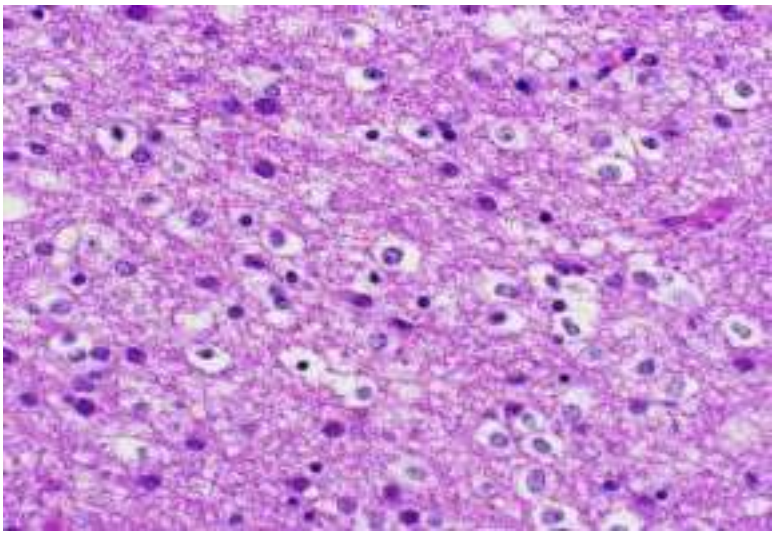




Oligodendroglioma

-Oligodendroglioma is a glioma of oligodendroglial origin. The tumour commonly presents in the 3rd to 4th decades of life. It occurs in the cerebral hemispheres, most commonly in the frontal lobes or within the ventricles. **Grossly**, oligodendroglioma is well-circumscribed, grey-white and gelatinous mass having cystic areas, foci of haemorrhages and calcification.

-Microscopically: the tumour is characterized by uniform cells with round to oval nuclei surrounded by a clear halo of cytoplasm and well-defined cell membranes. Tumour cells tend to cluster around the native neurons forming satellitosis. Typically, there are varying degrees of endothelial cell hyperplasia and foci of calcification. Anaplastic change may occur.



Oligodendroglioma

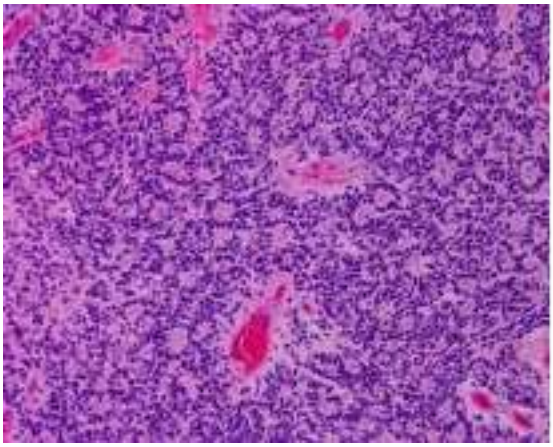
Ependymoma

-Ependymoma is a tumour derived from the layer of epithelium that lines the ventricles and the central canal of the spinal cord. It occurs chiefly in children and young adults (below 20 years of age). Typically, it is encountered in the fourth ventricle (posterior fossa tumour), the lateral ventricles, the third ventricle, and in adults, the spinal cord in the region of lumbar spine. Clinically, by virtue of their frequent location in the floor of the fourth ventricle, ependymomas are associated with obstructive hydrocephalus.

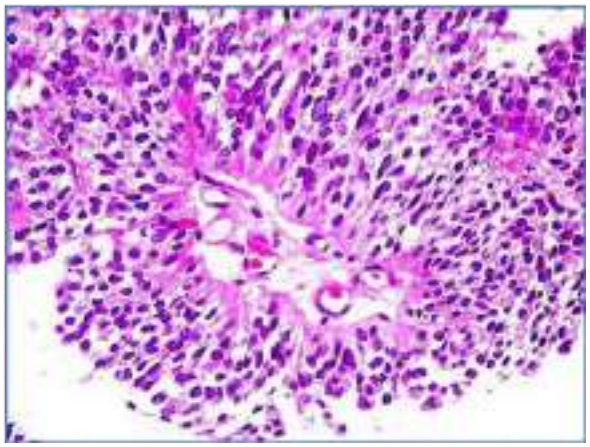
-The usual biologic behavior is of a slow growing tumour. -**Grossly**, ependymoma is a well-demarcated tumour.

-Microscopically: the tumour is composed of uniform epithelial (ependymal) cells forming rosettes and perivascular pseudorosettes.
Most tumours are well- differentiated but anaplastic variants are also recognized. Two variants of ependymoma deserve special mention: myxopapillary type and subependymoma. Myxopapillary ependymoma characteristically, it occurs in the region of cauda equina and originates from the filum terminale. It contains myxoid and papillary structures interspersed in the typical ependymal cells. It is a slow-growing tumour having a better prognosis.

Subependymoma It occurs as a small, asymptomatic, incidental solid nodule in the fourth and lateral ventricle of middle-aged or elderly patients. Areas of microcysts and calcification may be encountered. Histologically, it is composed of nests of uniform ependymal cells in a stroma of very dense, acellular, finely fibrillar background. Subependymoma is typically a very slow- growing tumour.



Ependymoma



Perivascular pseudorosettes

POORLY DIFFERENTIATED AND EMBRYONAL TUMOURS

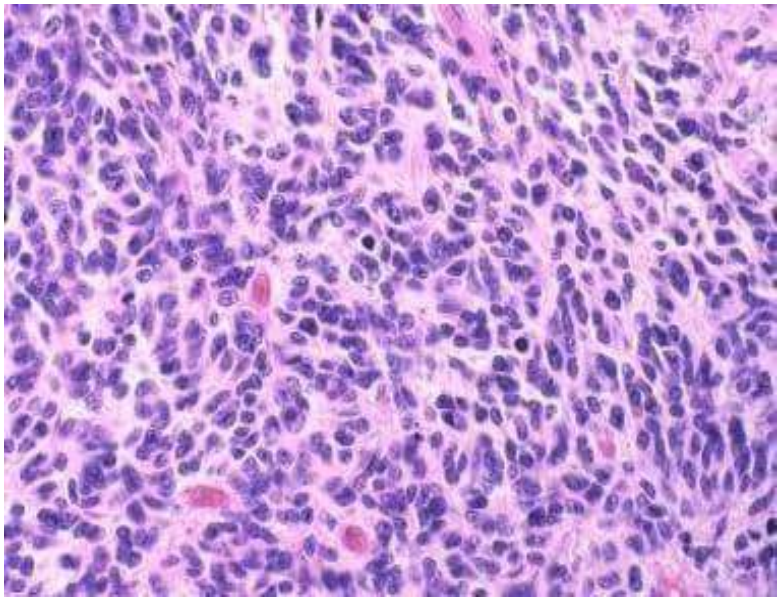
-CNS tumours composed of primitive undifferentiated cells include medulloblastoma, neuroblastoma and retinoblastoma.

Medulloblastom

-Medulloblastoma is the most common primitive neuroectodermal tumour, occur in patients over the age of 20 years. The most common location is the cerebellum in the region of roof of fourth ventricle, in the midline of cerebellum, in the vermis, and in the cerebellar hemispheres.

-Medulloblastoma is a highly malignant tumour. It invades locally and by the CSF to meninges, ventricles and subarachnoid space and has a tendency for widespread metastases to extra neural sites such as to lungs, liver, vertebrae and pelvis. **Grossly**, the tumour typically protrudes into the fourth ventricle as a soft, grey-white mass or invades the surface of the cerebellum. **-Microscopically:**

medulloblastoma is composed of small, poorly differentiated cells with ill-defined cytoplasmic processes and has a tendency to be arranged around blood vessels and occasionally forms pseudo rosettes (Homer Wright rosettes).



Medulloblastoma

TUMOURS OF MENINGES

Meningioma

-Meningiomas arise from the cap cell layer of the arachnoid. The most common sites are the lateral cerebral convexities, midline along the falx cerebri adjacent to the major venous sinuses parasagittally, and olfactory groove. Meningiomas are generally solitary. They are usually found in 2nd to 6th decades of life, with slight female preponderance.

-**Grossly**, meningioma is well-circumscribed, solid, spherical or hemispherical mass of varying size (1-10 cm). The tumour is generally firmly attached to the dura and indents the surface of the brain but rarely ever invades it.

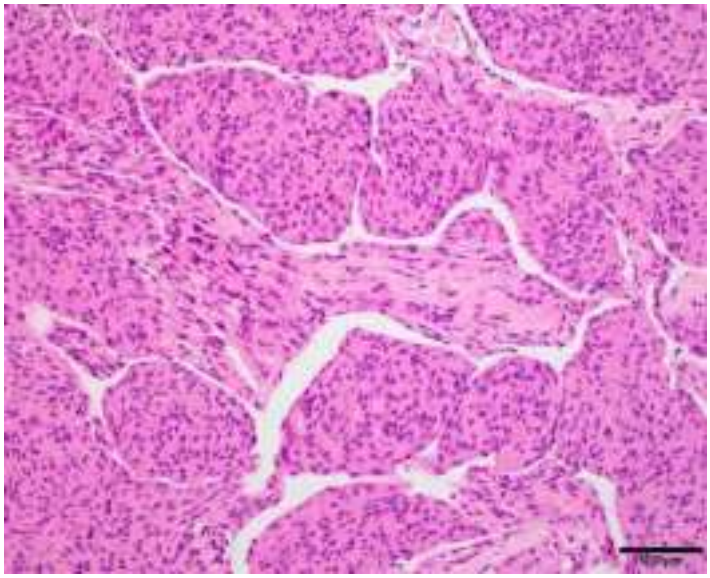
-Cut surface of the tumour is firm and fibrous, sometimes with foci of calcification.

Microscopically, meningiomas are divided into 5 sub types

1.**Meningotheliomatous(syncytial) meningioma:**

The tumour consists of solid masses of polygonal cells with poorly definedcell membranes (syncytial appearance). The cells have round to oval, central nuclei with abundant, finely granular cytoplasm. Some amount of collagenous stroma is present that divides the tumour into irregular lobules.

2 Fibrous (fibroblastic) meningioma
3 Transitional (mixed) meningioma
4 Angioblastic meningioma
5 Anaplastic meningioma



Meningeioma

NERVE SHEATH TUMOURS

-Tumours of the peripheral nerves are commonly benign and include schwannoma (neurilemmoma) and neurofibroma. Both of them arise from Schwann cells but neurofibroma contains large amount of collagen.

SCHWANNOMAS (NEURILEMMOMAS)

-Schwannomas or neurilemmomas arise from cranial and spinal nerve roots. An acoustic schwannoma or acoustic neuroma is an intracranial schwannoma located within the internal auditory canal originating from vestibular portion of the acoustic nerve. Schwannomas are tumours of adults except in von Recklinghausen's disease.

-Grossly, a schwannoma is an encapsulated, solid tumour that produces eccentric enlargement of the nerve root.

-Microscopically: the tumour is composed of fibrocellular bundles forming a whorled pattern. There are areas of dense and compact cellularity (Antoni A pattern) alternating with loose acellular areas (Antoni B pattern). Areas of Antoni A pattern show palisaded nuclei called Verocay bodies. Nerve fibres are usually found stretched over the capsule but not within the tumour.

NEUROFIBROMAS AND VON RECKLINGHAUSEN'S DISEASE

-Neurofibromas may occur as solitary, fusiform, cutaneous tumour of a single nerve, but more often are multiple associated with von Recklinghausen's disease. Solitary neurofibroma is a tumour of adults but multiple neurofibromas or neurofibromatosis is a hereditary disorder with autosomal dominant inheritance. Solitary neurofibroma is generally asymptomatic but patients with **-von Recklinghausen's disease have a triad of features**:

1. Multiple cutaneous neurofibromas.
2. Numerous pigmented skin lesions (café au lait' spots).
3. Pigmented iris.

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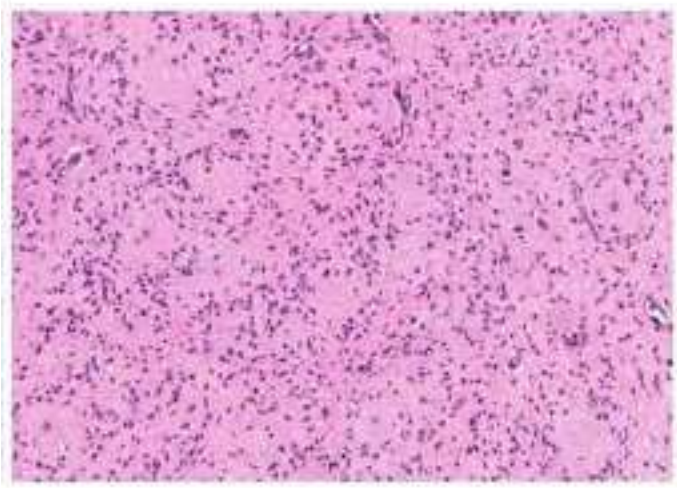
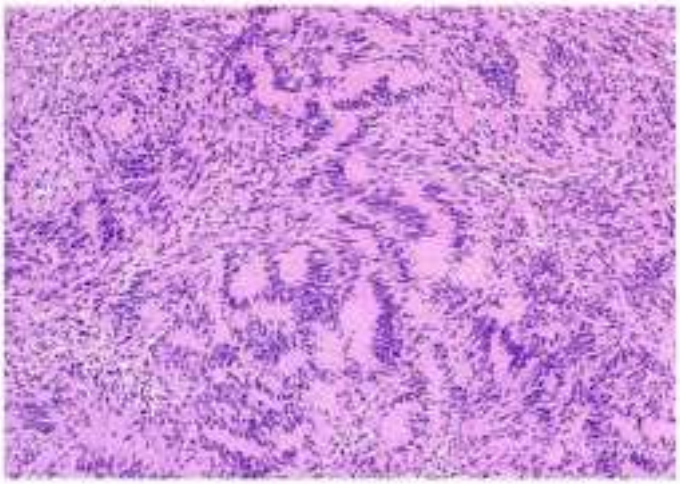
-Neurofibromatosis type 1: is a genetic disorder having mutation in chromosome 17 while **Neurofibromatosis type 2:** has mutation in chromosome 22. **Grossly**, neurofibroma is an unencapsulated tumour producing fusiform enlargement of the affected nerve. -Neurofibromatosis in von Recklinghausen's disease is characterized by numerous nodules of varying size, seen along the small cutaneous nerves. Neurofibromatosis may involve a group of nerves or may occur as multiple, oval and irregular swellings along the length of a nerve (plexiform neurofibroma).

-Microscopically: a neurofibroma is composed of bundles and interlacing fascicles of delicate and elongated spindle shaped cells having wavy nuclei. The cellular area is separated by loose collagen material. Neurofibromas have a tendency for local recurrences after excision.



Neurofibroma



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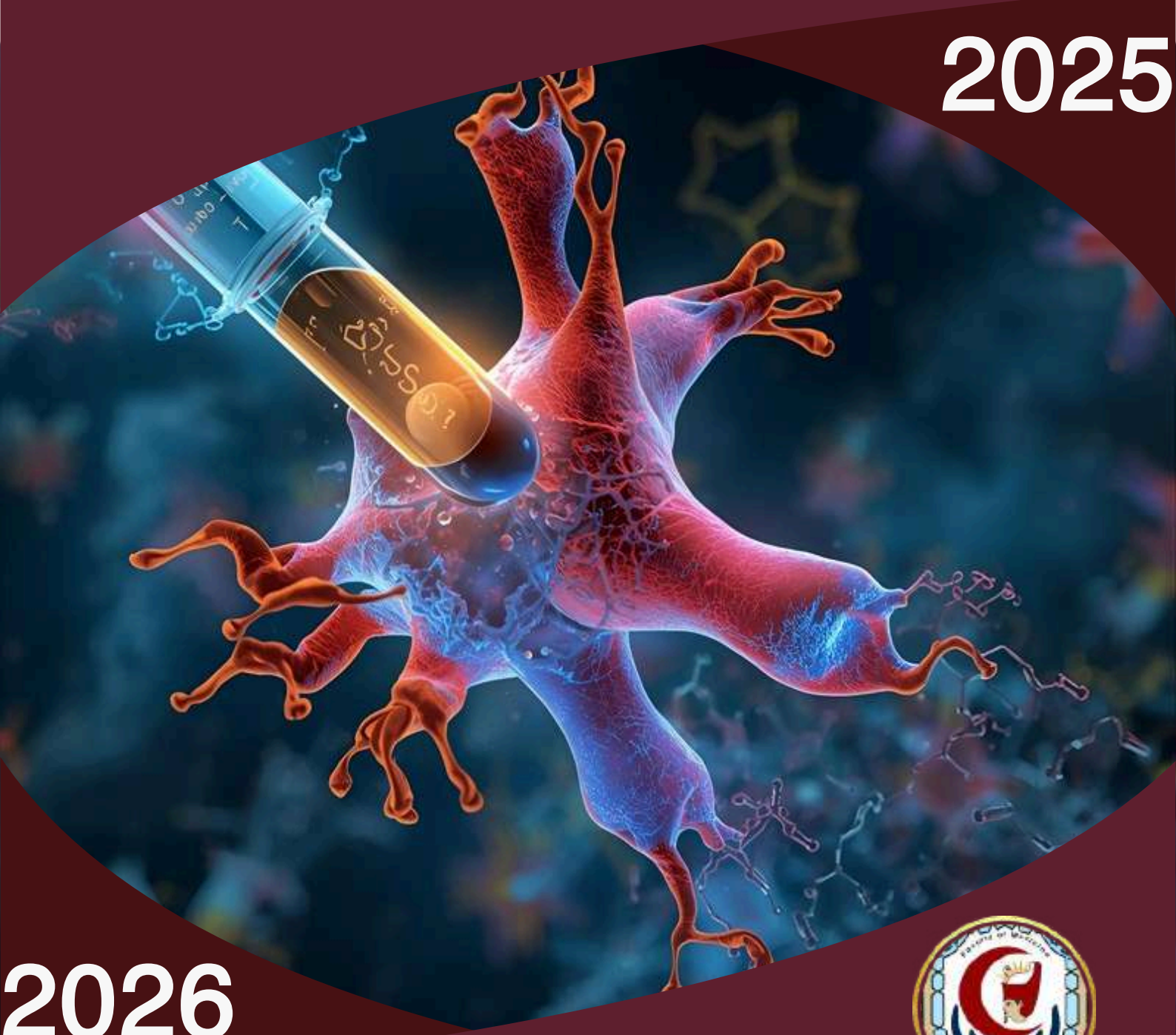

2026

SOHAG UNIVERSITY
FACULITY OF
MEDICINE

Neuroscience

Pharmacology

NEU-316

page	lecture
1	*General Anaesthetics*
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16	*SEDATIVE S-HYPNOTIC DR UGS*
72	*ATYPICAL SEDATIVE-HYPNOTICS*
54	*Pharmacotherapy of epilepsy*
72	*Parkinsonism*
83	*Opioid Analgesics*

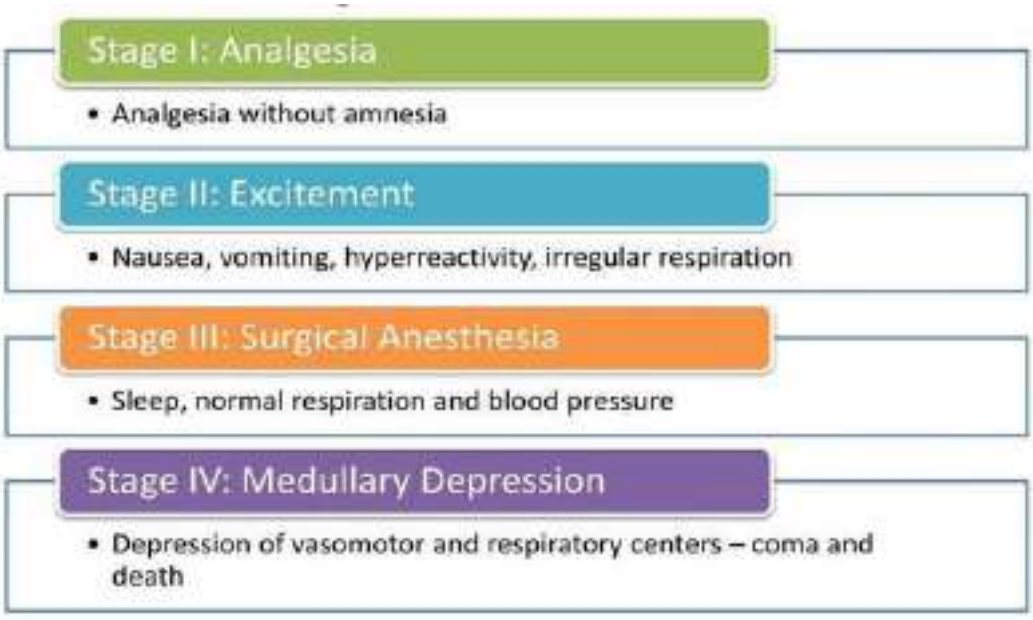
1 General Anaesthetics

ILOS

1. Define general anesthetics.
2. Classify general anesthetics.
3. Recognize adverse effects general anesthetics.
4. Enumerate pre-anesthetic medication and anesthetic adjuvant.
5. Explain mechanism of action of general anesthetics.

- **Anesthesia** means loss of sensation.
- **General Anesthetics:** A drug that brings about a reversible loss of consciousness and sensations to induce or maintain general anesthesia to facilitate surgery.
- **The neurophysiologic state** produced by general anesthetics is characterized by five primary effects: unconsciousness, amnesia, analgesia, inhibition of autonomic reflexes, and skeletal muscle relaxation. None of the currently available anesthetic agents when used alone can achieve allfiveof these desired effects.

Stages of General Anesthesia:



An ideal anaesthetic drug should

induce: Rapid

Smooth loss of consciousness

- Rapidly reversible discontinuation

Wide margin of safety.

Classification of general anaesthesia;

• according to channel of administration:

Inhalation Anesthetics:	Intravenous Anesthetics:
• Volatile liquids (Halogenated hydrocarbons): Halothane, Enflurane, Isoflurane, Desflurane, Sevoflurane. •Gasses (Nitrous oxide)	• Ultrashort-acting barbiturates (thiopental). • Non- barbiturates (Ketamine, propofol, etomidate).

Mechanism of Action of general anesthesia:

- There are many theories that explain mechanism of action; the most appropriate one is its actions on ligand-gated ion channels (GABAA and NMDA receptors).
- **Potentiation of inhibitory receptors (**GABAA and glycine)
- **Inhibition of excitatory receptors** (Ketamine & Nitrous Oxide).

Pharmacological Actions of Anesthetics:

- Irregular progressive descending depressionof CNS (Medullary centers are the last to be affected; Order of involvement is (cortex, basal ganglia and cerebellum, spinal cord and medullary centers).
- Respiratory depression and decrease ventilation.
- Hypotension due to direct vasodilatation, depression of myocardium and depression of the vasomotor Centre.

- All inhalational anesthetics produce dose-dependent reduction in renal and hepatic blood flow.
- All inhalational anesthetics produce varying degrees of skeletal muscle relaxation
- ***Hypothermia: due to :***
  1) Increased heat loss (peripheral vasodilatation).
  2) Decreased heat production (relaxation of skeletal muscles, depression of heat regulatory centre).

Toxicity of Anesthetic Agents:

1) Nephrotoxicity:

- Enflurane and sevoflurane: Generate nephrotoxic fluoride ions.
- Sevoflurane: degraded by CO2 absorbents in anaesthetic machine (produce nephrotic compound A which cause tubular necrosis.

2) Hematotoxicity:

- Megaloblastic anemia (nitrous oxide)
- All inhaled anesthetics: produce carbon monoxide which binds to hemoglobin, reducing oxygen delivery to tissues.

3) Hepatotoxicity:

- Hepatic dysfunction following surgery and general anaesthesia is most likely caused by hypovolemic shock, infection conferred by blood transfusion, or other surgical stresses rather than by volatile anesthetics toxicity (halothane hepatitis) due to immune response to hepatic proteins.

4) Malignant hyperthermia syndrome

- **A heritable genetic disorder** of skeletal muscle occurs in susceptible individuals exposed to volatile anesthetics while undergoing general anesthesia.
  (depolarizing muscle relaxant exaggerate it)
- **Manifestation:**
  a) Muscle rigidity
  b) Hyperthermia
  c) Tachycardia & hypercapnia
  d) Hyperkalaemia & metabolic acidosis

NOTE

The specific biochemical abnormality is an increase in free cytosolic calcium .concentration in skeletal muscle cells.

Treatment:

- Dantrolene (reduce calcium release from sarcoplasmic reticulum)

Appropriate Measures to reduce temperature and restore electrolyte a& acid base balance

Emergence & postoperative phenomenon:

Nausea and vomiting: action on the CTZ and vomiting center.

Hypertension and tachycardia: sympathetic nervous system regains its tone and is enhanced by pain.

Emergence excitement: restlessness, crying.

Postanaesthetic shivering: due to hypothermia (stopped by meperidine).

Pain (stopped by analgesic drugs "opioids –NSAID").

1) Inhalation Anesthetics

	Induction	Recovery	Side effects	Actions
Nitrous oxide	• Veryrapid	• Very rapid	• Megaloblasticanemia	• -Weakanesthetic (adjuvant) • -Analgesia (dental & Obstetric analgesia).
Halothan e	• Slow	• Slow	• -Hypotension (dose dependent) • -Fulminant hepatic necrosis • -Sensitize myocardium to arrhythmogenic effects of epinephrine	• -Well tolerated in children
Enflurane	• Slow	• Slow	• -Seizure activity (convulsion) • -Nephrotoxicity	• **Safer than halothane:** • (Regarding liver and heart)
Isoflurane	• Rapid	• Rapid	• -Hypotension (cardiac output is well maintained)	• **Safer than halothane:** • -No convulsion

				• Not sensitize the heart to arrhythmia of catcholamine • No hepatotoxicity
Desflurane	• Very rapid	• Very rapid	• -Irritates tracheobronchial tree • -Coughing, salivation & bronchospasm. -	• No nephrotoxicity • No Hepatotoxicity.
Sevoflurane	• Rapid	• Rapid	• Nephrotoxicity (degradation product produced by interaction of sevoflurane with CO2 absorbent soda lime in anesthesia machine.	• Not irritation to airway • Potent bronchodilator

2) Intravenous Anesthetics

These drugs can be used:

- As soleagents fordiagnostic and minor surgical procedures.
- As induction agents followed by maintenance with inhalational anesthetics for longer operations.
- Intravenous anesthetics (except ketamine) produce reduction in cerebral blood flow and cerebral oxygen consumption accompanied with decrease in intracranial pressure.

Thiopental	Induction	Recovery	Side effects	Actions
	• Rapid	• **Small dose:** rapid • **Large dose**: slow	• Respiratory depression, coughing, laryngospasm and bronchospasm.	• - Short duration (rapid tissue redistribution). • -Used for induction
Propofol	• Rapid	• Rapid	• - Respiratory depression	• Sedative agent in the intensive care unit • induction and maintenance

Etomidate	• Rapid	• Rapid	• Pain on injection • -Involuntary movements during induction • -Nausea and vomiting during recovery	• No cardiovascular or respiratory depressions
Ketamine	• Rapid (dissociative anesthesia).	• Slow (bad dreams, hallucinations).	• **Stimulates CVS:** • (↑ blood pressure, ↑ cardiac output, ↑ HR, ↑ intracranial pressure)	• useful in hypovolemic or cardiogenic shock

Anesthetic adjuvants & pre-anaesthetic medications

- A general anesthetic is rarely given as the sole agent.
- Anesthetic adjuncts usually are used to augment specific components of anesthesia, permitting lower doses of general anesthetics with fewer side effects.

1. Analgesics:

- **Nonsteroidal anti-inflammatory drugs** (for minor surgery).
- **Opioid analgesics:** morphine, meperidine (Meperidine reduce shivering).

2- Benzodiazepine (Midazolam, diazepam & lorazepam).

- Midazolam is preferred because its potency is greater, has rapid onset and shorter duration of action.
- Produce sedation and anterograde amnesia.

3- Dexmedetomidine (highly selective α2-adrenergic agonist that produces both sedation and analgesia with minimal respiratory depression.

4- Neuromuscular Blocking Agents (Depolarizing and non-depolarizing muscle relaxants.

5- Antihistaminic (HI-antagonists) eg, diphenhydraminc to prevent allergic reactions.

6- H2 blockers (ranitidine) (reduce gastric acidity and avoid acid reflux).

7- Atropine or scopolamine to prevent bradycardia and reduce respiratory secretion.

2 Local Anaesthetics

ILOs:

1-Define local anesthetics.

2-Recognize techniques of local anesthesia.

3-Classify local anesthetics.

4-Recognize adverse effects of local anesthetics.

5-Explain the mechanism of action of local anesthetics

Definition:

- Local anesthetics cause reversible loss of pain sensation without loss of consciousness or impairment of central control of vital functions.
- They may be used as the sole form of anesthesia, in combination with general anesthesia, and/or to provide postoperative analgesia.

Mechanism of action :

1) Local anesthetics act at cell membrane to prevent generation and conduction of nerve

2) impulses. The primary mechanism of action of local anesthetics is **blockade of voltage-**

3) **gated sodium channels.** It binds more readily to the Na+ channel during depolarization

4) (when it is open or activated). At higher concentrations, local anesthetics can block k+ channels. Also, high concentrations of local anesthetics can block Ca+2 channels.

Classification

1-Esters:

- Cocaine, Procaine
- Tetracaine, Benzocaine

2-Amides:

- Lidocaine, Mepivacaine, Prilocaine
- Bupivacaine, Ropivacaine

Undesirable effects

- Local anesthetics interfere with the function of all organs in which conduction or transmission of impulses occurs.
- Danger of such adverse reactions is proportional to concentration of local anesthetic achieved in the circulation.

1-CNS:

- Low concentrations: sleepiness, light headedness, visual and auditory disturbances and restlessness.
- An early symptom of toxicity: circumoral and tongue numbness and a metallic taste. At higher concentrations
- : nystagmus, muscle twitching followed by overt tonic clonic convulsions.

NOTE

Local anesthetics apparently cause depression of cortical inhibitory pathways thereby allowing unopposed activity of excitatory neuronal pathways. (Seizure activity) is then followed by generalized CNS depression.

2-Neurotoxicity:

- At high concentrations, Chloroprocaineand lidocaineare more neurotoxic than other local anesthetics when used for spinal anesthesia, producing transient neuropathic symptoms.

3-Cardiovascular symptoms:

- Local anesthetics blockcardiacsodiumchannelsandthus depress abnormal cardiac pacemaker activity, excitability and conduction.
- Except cocaine, they also depress strength of cardiac contraction and cause hypotension.
- Lidocaine is effective in arrhythmia associated with acute myocardial infarction by blocking activated and inactivated sodium channels.
- **Cocaine** differs from other LA in its cardiovascular effects (vasoconstriction, hypertension, cardiac arrhythmias).

4-Hematologic effects:

- Large doses (>8mg/kg) of prilocaineduringregionalanesthesia may lead to accumulation of its metabolite leading to methemoglobinemia.

5-Allergic reactions:

- Esters are metabolized to para-aminobenzoic acidderivatives.
- These metabolites are responsible for allergic reactions in a small percentage of patients.
- Amides are not metabolized to p-aminobenzoic acid; allergic reactions to amide local anesthetics are extremely rare.

6-Smooth muscles:

- They depress contractions initiallyingastrointestinal, vascular and bronchial smooth muscle (low concentrations may produce contraction).

Techniques for local anesthesia :

1-Surface anesthesia:

- Anesthesia of skin and mucous membranes of nose, mouth, throat, tracheobronchial tree, oesophagus and genitourinary tract.

2-Infiltration anesthesia:

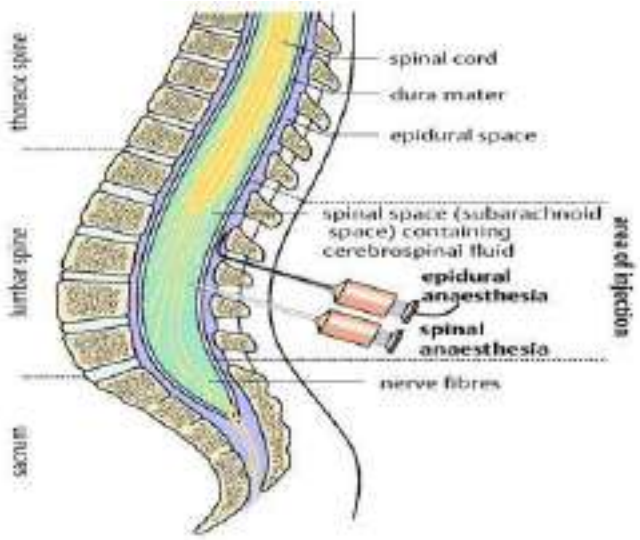
- Injection of local anesthetic directly into tissue without taking into consideration course of cutaneous nerves. Infiltration anesthesiacan be so superficial as to include only the skin.

3-Nerve block anesthesia:

- Injection of a local anesthetic into or around individual peripheral nerves or nerve plexuses, blockade of mixed peripheral nerves and nerve plexuses (brachial plexus block for procedures on the upper limb).

4-Spinal anesthesia:

- Injection of local anesthetic into subarachnoid space. It is performed through one of vertebral interspaces between second andfifth lumbar levels.
- Spinal anesthesia is a safe and effectivetechnique, Specially, during surgery involving lower abdomen, lower extremities and theperineum.

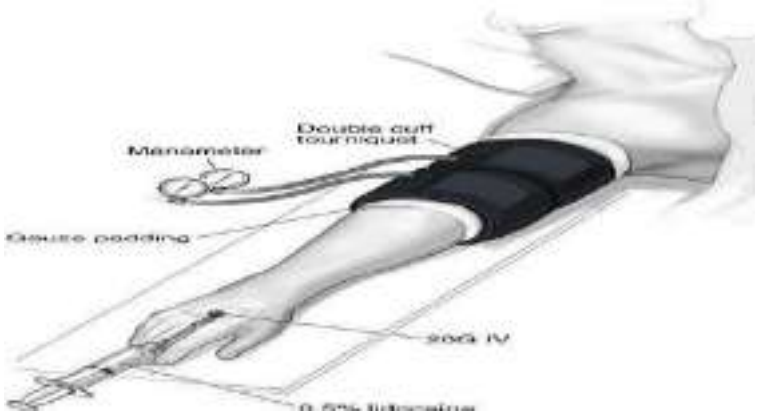


5-Epidural anesthesia:

- Injection of local anesthetic into epidural space.

6-Intravenous regional anesthesia:

- Used for short surgical procedures (<60 min) involving upper and lower extremities.
- It produced by IV injection of the anesthetic agent into a distal vein while circulation of limb is isolated with a proximally placed tourniquet.



NOTE

Small quantities of opioid injected intrathecally or epidurally produce segmental analgesia. Thus, spinal and epidural opioids are used in surgical procedures and for relief of postoperative and chronic pain.

3 SEDATIVE S-HYPNOTIC DR UGS

1) **Sedative (anxiolytic) agent** reduce anxiety and exert calming effect. **Hypnotic drug**
2) (encourage sleep) produces drowsiness and facilitates the onset and maintenance of a state of sleep that resembles natural sleep, and from which the recipient can be aroused easily. Hypnotic effects involve more pronounced depression of the central nervous system than sedation, and this can be achieved with many drugs in this class simply by increasing the dose. Individual drugs differ in the relationship between the dose and the degree of central nervous system depression. **Two examples of such dose-response relationships are shown:**

1) Barbiturates	2) Benzodiazepines and for certain newer hypnotics
● exhibit a linear dose response effect, an increase in dose higher than that needed for hypnosis may lead to a state of general anesthesia. ● At still higher doses, depression in the respiratory and vasomotor centers in the medulla, leading to coma and death.	● that have a similar mechanism of action, these drugs require proportionately greater dosage to achieve central nervous system depression

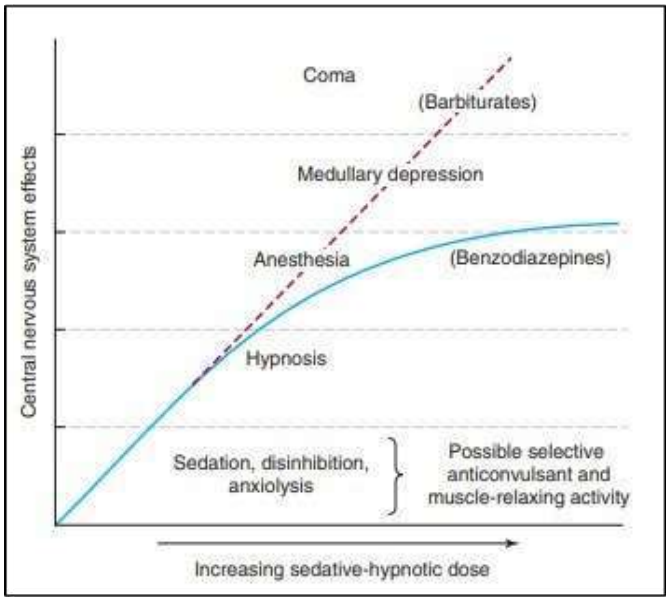


Fig.1. Relationships between dose of benzodiazepines and barbiturates and their CNS

Anxiolytic & hypnotics classification :

1- **Agonists of benzodiazepine receptors**:

- benzodiazepines, the "Z compounds"(Zolpidem & zaleplon)

2- **Atypical Sedative Hypnotic:**

- melatonin Receptor Agonists, 5HT1A Receptor Agonist, Orexin Antagonists.

3- **Barbiturates & chloral hydrate**

- rarely used now as sedative hypnotics because of their multiple side effects.

4- **Other classes of drugs that exert sedative effects:**

- **Antipsychotics and many antidepressant drugs:**

✓ The latter are currently used widely in the management of chronic anxiety disorders.

- **Antihistaminic agents:**

√ diphenhydramine, hydroxyzine, and promethazine cause sedation but commonly also exert marked effects on the peripheral autonomic nervous system.

√ Antihistaminic drugs with sedative effects are available as over-the-counter sleep aids.

- **β- Adrenergic blockers:**

√ used to treat some forms of anxiety when somatic effects (due to sympathetic over activity) as sweating, tremors, palpation and tachycardia are present.

√ *Currently,* the benzodiazepines are the safest and most commonly used sedative hypnotics.

√ Compared to barbiturates, benzodiazepines have the following pharmacological advantages:

A- Less depressant effect on respiration.

B- Less tendency for abuse and dependence.

C- Not liver enzyme inducers and thus have fewer drug interaction.

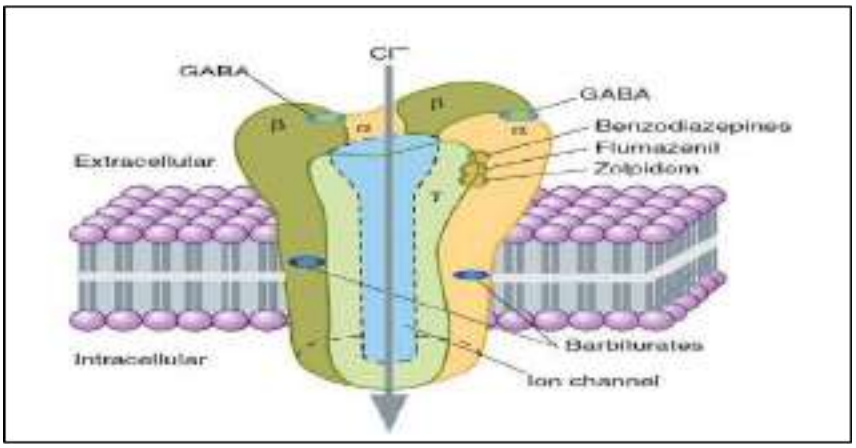
D- Little disturbance of rapid eye movement stage of sleep.

E- Better therapeutic index.

1) Benzodiazepines

Mechanism of action

- Receptors for benzodiazepines (BZ receptors) are located on the GABAA receptor molecule and thus are present in many brain regions.
- Binding of benzodiazepines facilitates the inhibitory actions of GABA, which are exerted through increased chloride ion conductance.
- Benzodiazepines increase the frequency of GABA-mediated chloride ion channel opening.



Benzodiazepine Binding Site Ligands:

- **Three types of ligand-benzodiazepine receptor interactions have been reported:**

(1) Agonists:

- agonists at the binding site increase the amount of chloride current generated by GABAA-receptor activation (facilitate GABA actions) clinically useful as antianxiety agent.

(2) Antagonists:

- *flumazenil*, which blocks the actions of benzodiazepines, zaleplon, and zolpidem but does not antagonize the actions of barbiturates (used in treatment of benzodiazepines overdose).

(3) Inverse agonists:

- (*β-carbolines*) act as inverse agonists of GABA-receptor.
- Its interaction with benzodiazepines sites on the GABAA receptor can produce anxiety and seizures.

Pharmacological Properties

1- Central NervousSystem:

- As the dose of a benzodiazepine is increased, sedation progresses to hypnosis and then to stupor, but the drugs do not cause a true general anesthesia because awareness usually persists, and immobility sufficient to allow surgery cannot be achieved. However, at
- "preanesthetic" doses, there is amnesia for events subsequent to administration of the drug. **Antianxiety:** In small doses, benzodiazepine reduces anxiety, tension and
- aggression. **Hypnotic:** 1- Benzodiazepines decrease sleep latency 2- Time in stage 1
- (descending drowsiness) of NREM sleep usually is **decreased.** 3- **Prominent decrease** in the time spent in slow-wave sleep (stages 3 and 4 of NREM sleep); reduces nightmares and night terrors.

4- The time spent in REM sleep usually is **shortened.**

5- Increases total sleep time, largely by increasing the time spent in stage 2 (which is the major fraction of NREM sleep).

Skeletal muscle relaxation.

Anticonvulsant.

NOTE

Benzodiazepines are thought by some gastroenterologists to improve a variety of "anxiety-related" gastrointestinal disorders such as stress ulcers.

Pharmacokinetics

1) Short-acting agents (t1/2 < 6 hours), including triazolam, oxazepam,

2) Intermediate-acting agents (t1/2 6-24 hours), including estazolam and temazepam.

3) Long-acting agents (t1/2 >24 hours), including flurazepam, diazepam, and quazepam.

- **Benzodiazepines have high lipid solubility**, so they cross the blood brain barrier easily.

  Metabolized in the liver by oxidation and conjugation.

- Some benzodiazepines give active metabolites like the parent drug, **eg: diazepam** Both metabolites are active as hypnotic and anxiolytic similar to diazepam.

- No correlation between the clinical duration of action and actual half-life of the parent drug

  e.g. **flurazepam** half-life is 3 hours but its active metabolite has a half-life of 50 hours.

- Not significantly induce the synthesis of hepatic CYPs.

Therapeutic uses

[a] Anxiety disorders:

- The longer acting benzodiazepineslike diazepam& lorazepam are often preferred as patients with anxiety may require treatment for prolonged period of time.
- **Alprazolam** has been used in the treatment of panic disorders.

[b] Sleep disorders:

- An ideal hypnotic agent would havea rapid onsetof action when taken at bedtime, a sufficiently sustained action to facilitate sleep throughout the night, and no residual action by the following morning.
- Among the benzodiazepines that are used commonly as hypnotic agents; **triazolam** short acting useful in initial insomnia i.e. difficult to enter into sleep, **temazepam** (intermediate acting) and **flurazepam** (long acting) which both are suitable in latent insomnia i.e early awakening.
- In practice, there appear to be some disadvantages to the use of agents that have a relatively rapid rate of disappearance, including the early-morning insomnia that is experienced by some patients.

[c] Seizure disorders:

- Benzodiazepines that are useful asanticonvulsantshavea long t1/2, and rapid entry into the brain.
- **Clonazepam** is useful in absence seizures and **diazepam** used in status epilepticus.

[d] Preanesthetic medication:

- Benzodiazepines are used for anxiolysis, amnesia, and sedation prior to induction of anesthesia or for sedation during procedures not requiring general anesthesia.
- The benzodiazepine most frequently used in the perioperative period is midazolam followed by diazepam and lorazepam.
- Midazolam is preferred because it's **rapid onset and shorter duration of action.**

[e] Skeletal muscle relaxants:

- Benzodiazepines may alleviate muscle spasticity in cerebral palsy or spinal cord lesions.

[f] Control withdrawal symptoms in alcoholics.

Untoward Effects

1. ***Residual effects***

√ may persist in the form of light-headedness, lassitude, increased reaction time, motor incoordination, impairment of mental and motor functions, confusion, and anterograde amnesia. All of these effects can greatly impair driving and other psychomotor skills,

√ especially if combined with ethanol.

2. ***Other relatively common side effects*** of benzodiazepines are ***weakness headache blurred vision vertigo nausea*** and ***vomiting epigastric distress*** , and ***diarrhea***.
3. ***Anticonvulsant benzodiazepines*** sometimes actually increase the frequency of seizures in patients with epilepsy.

4. Adverse Psychological Effects:

- ***Paradoxical effects*** nightmares especially during the first week of use and sometimes causes garrulousness, anxiety, irritability.
- ***Disinhibition*** **or** ***dyscontrol reactions***: they may produce bizarre behavior, hostility and rage.
- Such paradoxical or disinhibition reactions are rare and appear to be dose related.
- ***Dependence and abuse***: minimal.

5. ***Although overdosage with a benzodiazepine*** rarely causes severe cardiovascular or respiratory depression, ***therapeutic doses can further compromise respiration in patients with COPD or obstructive sleep apnea***.

Pharmacodynamic Interactions:

Ethanol increases both the rate of absorption of benzodiazepines and the associated CNS depression.

Novel Benzodiazepine Receptor Agonists:

- Hypnotics in this class are commonly referred to as "Z compounds".
- They include ***zolpidem, zaleplon.***
- Although the Z compounds are structurally unrelated to benzodiazepines and described as non-benzodiazepine hypnotics, their therapeutic efficacy as **hypnotics** is due to agonist effects on the benzodiazepine site of the GABAA receptor. Compared to benzodiazepines,
- Z compounds are less effective as anticonvulsants or muscle relaxants, which may be related to their relative selectivity for GABAA receptors containing the omega-1 subunit. The Z compounds have largely replaced benzodiazepines in the treatment of insomnia.
- Overdose with Z compounds are similar to that of benzodiazepine overdose and can be
- treated with the benzodiazepine antagonist flumazenil. Zaleplon and zolpidem are
- effective in relieving **sleep-onset insomnia.** Zaleplon and zolpidem have sustained
- hypnotic efficacy without occurrence of rebound insomnia on abrupt discontinuation.

Flumazenil: A Benzodiazepine Receptor Antagonist:

- Flumazenilbindswithhighaffinityto specificsites ontheGABAA receptor, where it competitively antagonizes the binding and effects of benzodiazepines and other ligands; it antagonizes both
- the effects of agonist and inverse-agonist. Flumazenil is available only for intravenous
- administration. Although absorbed rapidly after oral administration, <25% of the drug reaches
- the systemic circulation owing to extensive first-pass hepatic metabolism. Flumazenil used in the management of **benzodiazepine overdose**.

4 ATYPICAL SEDATIVE-HYPNOTICS

1) Melatonin Receptor Agonists

- **(Ramelteon and Tasimelteon).**
- Melatonin receptors are thought to be involved in maintaining circadian rhythms underlying the sleep-wake cycle.
- Ramelteon and tasimelteon activate melatonin receptors in the suprachiasmatic nuclei of the CNS and decrease the latency of sleep onset **with minimal rebound insomnia or withdrawal symptoms.**
- These agents have no direct effects on GABAergic neurotransmission in the central nervous system.
- Minimal potential for abuse.

2) 5HT1A Receptor Agonist

- **It has the following characteristics:**

1) It selectively binds to 5HT1A (serotonin) receptor acting as a partial agonist. It has no
2) interaction with benzodiazepines receptor or facilitates the action of the GABA. Its
3) **anxiolytic effect** does not appear before 2-4 weeks of its administration. So, it is suitable for chronic anxiety but not acute anxiety states. Also, it is not effective in

severe form of anxiety especially if associated with panic attacks.

4) It has **no hypnotic** or **anticonvulsant** or **muscle relaxant effects**.

5) **Side effects** may include restlessness and dysphoria.

Orexin Antagonists:

- Orexinisapeptidefound in the hypothalamus and is involved in wakefulness
- **Suvorexant**, an antagonist at OX1R and OX2R orexin receptors
- It has **hypnotic properties** and is approved for the treatment of insomnia.

3) Psychotropic Drugs

- **Psychotropic drugs:** are those which modifythe behavioral pattern of psychiatric patients.
- They do not eliminate the disease but prepare the patients to psychotherapy.

Psychotropic drugs are classified into:

[a] Psychotropic depressants (mood stabilizers):

1) Antipsychotics.

2) Antianxiety drugs.

[b] Psychotropic stimulants (mood elevators):

1) Antidepressants

Affective disorders: are psychic disturbances characterized primarily by changes of mood (depression or mania).

1. Depression: In endogenous depression (melancholia), mood is persistently low.

2. Mania*:* refers to the opposite condition of depression.

3. Bipolar affective disorder (manic-depressive illness): Characterized by episodes of mania alternative with depression.

NOTE

Although the patient may have suicidal thoughts, psychomotor retardation prevents suicidal impulses from being carried out.

4) Drugs Used In Affective Disorders :

1) ANTIDEPRESSANTS

2) MOODSTABILISERS

Psychotic disorders

- Schizophrenia.
- Manic phase of bipolar (manic-depressive) illness. Acute idiopathic psychotic illnesses
- (Psychoses). Other conditions marked by severe agitation. Psychosis is a severe mental
- disorder affecting about 0.5-1.0 % of populations, in which the patient behaves out of
- touch of reality with illogical and bizarre thinking, lives in his own world.

Symptoms of psychoses:

a) falsebeliefs(delusion)

b) Abnormal sensation (hallucination).

Psychoses have two components:

1) *Organiccomponent(dopaminehypothesis):* suggests that psychoses may be due to over-activity of dopaminergic action in mesolimbic system of the brain.

2) *Psychic component:* due to psychological and social stress.

- Schizophrenia has a strong genetic component.

The nature of schizophrenia:

1) Psychoticillnesscharacterisedby delusions, hallucinations and thought disorder (positive symptoms).

2) Flattening of emotional responses (negative symptoms).

- Its incidence is about 1% of population with a strong hereditary component

Dopamine theory:

- Pharmacological evidence is generalyconsistentwith dopamine overactivity theory.
- According to dopamine theory in schizophrenics hyperactivity of dopamine at D2 receptors in limbic system occurs.
- Antipsychotics do not cure psychoses. They often reduce the need for institutional care and enable patients to maintain relatively stable lives.

1) henothiazine derivatives

- Have similar pharmacological actions and therapeutic indications.
- **Include:**
  - √ Chlorpromazine
  - √ Thioridazine
  - √ Fluphenazine
- Chlorpromazine is the prototype drug.
- **All share in:**
  - √ Extrapyramidal side effects
  - √. Sedation and hypotensive effects.
  - √ Withmarked D-receptor blocking activity.

Antagonist Actions of Phenothiazine derivatives:

1) D_2receptors antagonist → antipsychotic effect. (Antidopaminergic)
2) M receptors antagonist → atropine-like effects. (Anticholinergic)
3) α-receptors antagonist → hypotension (Antiadrenergic)
4) D receptors antagonist in the nigrostriatal system → Extrapyramidal motor disturbances.
5) D receptors antagonist in area postrema → Antiemetic action.
6) D receptors antagonist in pituitary gland → Increased secretion of prolactin.
7) Histamine receptor *antagonist* in cerebral cortex→ Sedation (Antihistaminic), anti-serotonin and ganglion blocking activities.

Pharmacological actions:

1. Antidopaminergic effects: I-CNS:

a) Behavioral effects:

- Due to blocking D2 receptors (competitive antagonist) in mesolimbic system.
- ▯ aggressive and impulsive behavior and the patient become motionless, silent and he answers questions in a slow and monotonous voice.
- ▯ spontaneous motor activity.
- In high doses→ cataleptic effect (the body and the limbs are molded in various postures and remain immobile for long time).

b) Antiemetic effect:

- Blocking D2 receptors in chemoreceptor trigger zone (CTZ) →powerful antiemetic action. It is not effective in vomiting due to vestibular stimulation (motion sickness) or that caused by local gastrointestinal tract irritation.

c) Extrapyramidal effects:

- Blocking D2 receptors in basal ganglia → extrapyramidal side effects. L-Dopa (precursor of dopamine) thus exacerbates symptoms of schizophrenia.

d) Hyperprolactinemia effect:

- Blocking D2 receptors in hypothalamus –anterior pituitary axis → hyperprolactinemia (galactorrhea in female and decreased libido and impotence in male).

2. Effects on thermoregulatory centers:

→ disturbances → hypo-or hyperthermia.

3. Effects on analgesic or hypnotic actions:

→ they potentiate the analgesic and hypnotic actions of hypnotic and analgesic drugs.

II- ANS:

- Antiadrenergic (-blocker) → postural hypotension.
- Slight anticholinergics → antimuscarinic side effects.
- Slight antiserotonin, antihistaminic and slight ganglion blocking actions.

III- CVS:

Postural hypotension by I.V. chlorpromazinedueto:

- -blocking effect.
- -ve inotropic effect
- Direct vasodilation.
- Inhibition of centrally mediated reflexes.

Antiarrhythmic activity by chlorpromazine due to:

- Quinidine-like action
- Local anesthetic (membrane stabilizing effects

IV. Endocrinal effects:

- **~~Chlorpromazine block dopaminergicreceptorsinhypothalamic-adenohypophyseal axis:~~**

1) In female: galactorrhea, inhibit ovulation, infertility and suppress menses.

2) In male: decrease libido and decrease testicular weight.

V- Skeletal muscles:

- relaxant effect due to central mechanisminmedullaorbasal ganglia.

VI- Anti-hiccough:

- Chlorpromazine is highly effectiveinthetreatmentofintractable hiccough

Pharmacokinetics:

- Erratically absorbed from GIT.
- Highly lipophilic and concentrated in the brain.
- Highly bound to plasma protein.
- Elimination half-lives ranged from 12-24h
- Once daily dosing.
- Metabolized in the liver by the liver microsomal enzyme system through 2 steps “oxidation followed by conjugation with glucuronic acid”.
- Metabolites are excreted in the urine and persist in the body for long time and so can be detected in the urine even after several months after drug discontinuation.

Therapeutic uses:

1. Psychoses:

- Phenothiazines are most effectiveinexcitedpsychotic patients & in split mind (schizophrenia).
- Schizophrenia responds well to phenothiazine and electroconvulsive (ECT).
- Phenothiazine do not cure schizophrenic patients but just quieten and make him more co-operative and acceptable.
- The onset may be delayed to 2-5 weeks.

2. Behavioral disorders in children and senile:

- Children with aggressive destructivebehavior sometimesare improvedby phenothiazine.
- Senile patient's behavior can be controlled by phenothiazine.

3. Antiemetics and anti-hiccough:

- Chlorpromazine control vomiting ofcentraloriginbecauseitblocks D2 receptors in CTZ but does not control motion sickness.
- It is effective in the treatment of intractable hiccough (mechanism unknown).

Side effects and toxic manifestations:

1-Neurological or Extrapyramidal adverse effects

a) Parkinsoniansyndromewithmusclerigidity, tremors, maskface. Akathisia (motor
b) restlessness): compelling need of the patient to be in constant movement. Acute dystonic
c) reaction: spasm of muscles of face and neck leading to facial grimacing and torticollis. Neuroleptic malignant syndrome manifested by catatonia, extra- pyramidal signs, unstable

d) blood pressure, altered consciousness and hyperpyrexia. ,

e) Perioral tremor (rabbit syndrome).

f) Tardive dyskinesia that is stereotyped involuntary movements of tongue, lips, jaws and extremities.

The ***first four*** reactions usually appear soon after administration of the drug while the ***last two*** appear followingprolonged treatment.

2-Anticholinergic side effects including:

- dry mouth, blurredvision, urinaryretentionparticularlyinoldpatients with senile prostate, delayed gastricemptying.

3-Postural hypotension and sexual dysfunction ·

4-Allergic manifestations in the form of:

a- skin rash, contactdermatitis,photosensitivity,

b- b-intrahepatic obstructive jaundice.

c- blood dyscriasis in form of leucocytosis, leucopenia and agranulocytosis.

5-Abnormal pigmentation

- in regions exposed to the sunandwithhighdosespigmentaryretinopathy has been reported.

6-Gynecomastia, menstrual irregularities and lactation.

2) OTHER DRUG CLASSES

- Apart fromphenothiazinesother drug classes include butyrophenones, thioxanthenes and other heterocyclic compounds as loxapine, clozapine and risperidone.
- These resemble the phenothiazine derivatives pharmacologically with some differences between and within the groups as shown in table1.

Table 1, Some Representatives of Antipsychotic Drugs

Drug Classes	Daily Oral Dose Range* (mg)	Side-Effects		
		Extrapyramidal	*Sedative*	*Hypotensive*
Phenothiazines				
Chlorpromazine	200 – 800	++	+++	++
Thioridazine.	150 – 600	+	+++	+++
Fluphenazine.	2 – 20	++++	+	+
Butyrophenones				
Haloperidol	2 – 20	++++	+	+
Thioxanthenes				
Thiothixene	5 – 30	+++	++	++
Other heterocyclic compounds				
Loxapine	60 – 100	++	+	+
Clozapine	150 – 450	0	+++	+++
Risperidone	2 – 8	++	++	+++

**In psychiatric disorders it is usual to start treatment with low doses and are increased slowly to therapeutic levels.*

ANTIANXIETY DRUGS (Anxiolytic agents):

- Currently the most useful antianxiety drugsare the**benzodiazepines** and the 5HT $_{1A}$ receptor agonists. They have been discussed in previous section.

ANTIDEPRESSANT DRUGS:

- Drugsusedin the treatment of depression. The symptoms of depression are usually vague &
- unrecognized both by patients and by physicians. Patients with vague complaints that resist explanation as manifestations of somatic disorders should be suspected of being depressed.

Types of depression

Reactive depression:	Endogenous depression:
1) Precipitated by external factors (usually adverse life events) as: • Death of a loved one. • Failure • Major illness (MI or cancer) • Divorce. • Drugs (reserpine, alcohol). • Senility.	**1) No determined etiology** **2) Clinical picture include:** • Retardation in thoughts, movements or speech. • Feeling of guilty. • Occasional nightmares. • Early morning awakening. • Decreased appetite.

2) Clinical picture of marked sadness. More than 3) **60%** of all depressions. **The patient** blames 4) the situation rather than himself. **Responds** normally to moral, spiritual, social or friend 5) supports.	• Weight loss. 3) About **25%** of all depressions. 4) **The patient** blame himself rather than the situations. 5) **Responds** normally to medication

Symptoms of depression:

1) **Emotional symptoms:**

- Misery & apathy.
- Feelings of guilt & inadequacy.
- Loss of motivation.

2) **Biological symptoms:**

- Retardation of thought and movements.
- Sleep disturbances and loss of appetite.

Monoamine Theory of Depression

Depression iscausedby adeficitof monoamine transmitters (NEP and/or 5-HT) in the CNS. Pharmacological manipulation of monoamine transmission remains the most succesful therapeutic approach.

Most antidepressants exert important actions on the metabolism of monoamine neurotransmitters and their receptors, particularly norepinephrine and serotonin.

Antidepressant drugs

1) Tricyclic antidepressants (TCAs).
2) Monoamine oxidase inhibitors (MAOIs).
3) Selective Serotonin Re-uptake inhibitors (SSRIs)
4) Serotonin norepinephrine reuptake inhibitors (SNRIs) duloxetine – venlafaxine
5) Atypical antidepressants.
6) Agomelatine

1) Tricyclic Antidepressants (TCAs

Members:

- **TertiaryAmine:**Imipramine,Clomipramine, Amitriptyline, Doxepin and Trimipramine.
- **Secondary Amine**: Desipramine, nortriptyline, Maprotiline and Amoxapine.

NOTE

- All have a three-ring molecular core. So, named "tricyclic antidepressants ".
- Most share norepinephrine-reuptake inhibition.

Mechanism of action

- TCAsactby inhibitingreuptake of NEP and 5-HT by the nerve terminals → facilitating monoamine transmission in CNS.

- The tricyclic antidepressants do not block dopamine transport, thereby differing from CNS stimulants, including cocaine, methylphenidate and amphetamines.
- TCAs have antimuscarinic, antihistaminic, alpha receptor blocking and sodium channel blocking effects.

Pharmacological Actions of TCAs:

1) CNS:

- When given to depressed patients → mood elevation within 2-3 weeks.
- **The lag period in onset might be due to**:

√ The side effects which limit the rapid increase in dose and

- Possibility of other mechanism of action involving desensitization of autoinhibitory receptors (more acceptable mechanism).
- When given to normal subjects → anticholinergic side effects, sleepiness & light headedness.
- **Anticholinergic actions:** both central and peripheral → side effects. (atropine like action)

2) CVS:

- Orthostatic hypotension: due to □-blocking effect of TCAs.
- Antiarrhythmic properties: with therapeutic doses.
- Arrhythmogenic activities: with toxic doses.

Adverse effects:

1) **AnticholinergicSide Effects**:
   - dry mouth
   - blurred vision
   - constipation
   - urinary retention
2) **Postural hypotension**: due to -blocking effect
3) **Antihistamine:** drowsiness / sedation.
4) CNS effects as tremors, excitement, hallucination and deusions. Sedation & convulsions (with higher doses).
5) **Cardiac toxicity**: prolongation of QT interval, tachyarrhythmia "torsade de pointes "and precipitation of heart failure.
6) Allergic skin reaction, obstructive jaundice and agranulocytosis have been reported.

2))Monoamine oxidase inhibitors (MAOI)

- MAO is located intracellularly in nerve endings.
- Its main function is oxidative deamination of NEP, dopamine & 5-HT in the nerve endings.
- **MAO exists in two molecular forms:**

	MAO-A	MAO-B
Substrate preference	5-HT noradrenaline, dopamine and tyramine.	Dopamine and tyramine.
Inhibitors	**Selective:** Selegilin(B) Moclobemide(A).	**Nonselective:** Tranylcypromine Phenelzine.

- MAO inhibitors are not considered to be first-line antidepressants because of their adverse effects, dietary restriction, and drug interaction.

Mechanism of action:

- MAOIssharestructural similarities to endogenous amines which allows them to act as potential substrates for the enzyme.
- These compounds form irreversible covalent bonds with the enzyme, rendering them permanently inactive.
- **At doses used to treat depression, all MAO inhibitors behave as nonselective irreversible inhibitors of both MAO-A & MAO-B.**

- The effects on metabolism occur rapidly, action is long-lasting (weeks) because of irreversible inhibition of MAO (except for moclobemide).
- Due to irreversible inhibition of MAO, these drugs are called "**Hit and run**" drugs i.e., although drug become undetectable in the blood, its effect persists and gives outlasting actions.
- Termination of drug effects depends on the synthesis of fresh enzymes which takes several weeks.
- Adverse drug reactions may continue for 2-3 weeks after stopping drug administration.

Unwanted effects:

1) **PosturalHypotension:** **Clonidine-like effect** reduce sympathetic outflow. Evidence suggests that MAOIs elevate synaptic levels of norepinephrine, resulting in an increased stimulation of presynaptic alpha-2 receptors in the cardiovascular center in the brain stem.

2) This may also contribute to a reduction in sympathetic tone to the heart and vasculature.

3) **Tremors, excitement, insomnia** (even convulsions in overdose): due to excessive central stimulation. Antimuscarinic toxicity: dry mouth, blurred vision.

Interactions of MAO:

1.**~~Cheeseinteraction"or"Hypertensive crisis*~~**:

- Occurs when normally amines (mainly tyramine) are ingested (in aged cheese, yeast, chicken liver broad beans, chocolate, cream beer and wine).
- Tyramine causes NEP release from nerve ending → marked rise in BP.
- Tyramine is normally metabolized by MAO in the gut wall and liver and so little dietary tyramine reaches the systemic circulation.
- MAO inhibition allows all tyramine to be absorbed → hypertensive crisis, headache and intracranial hemorrhage.

Hypertensive crisis has also been reported in patients given TCA or sympathomimetic and MAOI simultaneously.

2. **Serotonin syndrome: MAOIs + SSRIs**:

- MAOI inhibits the metabolism of serotonin.
- SSRIs inhibit the reuptake of serotonin.
- Both effects accumulate serotonin in nerve endings and also in synapses → serotonin syndrome.
- This syndrome is manifested by hyperthermia, muscle rigidity and rapid changes in mental status.
- This syndrome is self-limiting after drug administration is discontinued.

3))Selective Serotonin Reuptake Inhibitors (SSRIs)

Examples:

- Fluoxetine
- Fluvoxamine
- Paroxetine
- Sertraline
- Citalopram

Advantages:

- Show selectivity with respect to 5-HT over NE uptake.
- Free from anticholinergic or cardiovascular adverse effects.
- Safe and tolerated by most patients.
- They do not cause “cheese reactions “.
- Wide margin of safety and so not useful to use as a means to commit suicide.

Disadvantages:

- Lesseffectivethan TCA in treating severe depression.
- Dangerous “serotonin reactions “if combined with MAOI manifested by hyperthermia, muscle rigidity, cardiovacular collapse.

Mild adverse reactions including:

1) **GIT:** nausea& vomiting.
2) **CNS:** headache, nervousness and insomnia.
3) **Sexual dysfunction:** impotence, ▯ libido and disturbance in orgasm and ejaculation.
4) They are not addicting but sudden discontinuation of their use results in both somatic and psychological withdrawal symptoms, mainly a flu-like state, dysphoric mood,
5) agitation (SSRIs discontinuation syndrome). QT prolongation by citalopram

4)SNRIs (Selective serotonin nor epinephrine re-uptake inhibitors

- Venlafaxine, desvenlafaxine and duloxetine.
- Inhibit the reuptake of both nor- epinephrine and serotonin.
- Effective in treating patients with depression and **neuropathic pain.**
- Differ from TCAs have no effect on adrenergic, muscarinic or histaminic receptors, thus less adverse effect.

5) Atypical antidepressants

Mechanisms of action are uncertain:

- weakmonoamineuptake blockers
- blocking α2 receptors on noradrenergic nerve terminals.

The main claims made for these newer agents are:

- fewer unwantedeffects(sedation,anticholinergicactivity)
- action with less delay
- efficacy in patients non-responsive to TCA or MAOI

Members:

- Bupropion (weak dopamine and NE reuptake inhibitor)
- Mirtazapine (α2 antagonist ,5HT antagonist and antihistaminic)
- Trazodone & nefazodone (weak 5HT & NE reuptake inhibitor and also antagonist at 5HT2a receptor in addition it has antihistaminic H1 blocking and sedating effect).

6) Agomelatine

- **Agomelatine** is a **melatonin receptor agonist (MT1 and MT2)and serotonin 5-HT2C and 5-HT2B receptor antagonist.**
- **Binding studies indicate that it has no effecton monoamine uptake andno affinity for adrenergic, histamine, cholinergic, dopamine, and benzodiazepine receptors, nor other serotonin receptors.**

Therapeutic Uses of Antidepressants

1) Depression:

- all types of antidepressant drugs take about 2 weeks to produce any beneficial effect.
- Effects of antidepressants are comparable to the effect of ECT.
- About 70-90 % of patients show satisfactory response with antidepressants.
- In acutely suicidal patients and in severe acute depression, ECT is the lifesaving treatment.

2) Nocturnal Enuresis:

- Pre-bedtime dose shows variable response in controlling enuresis in children and geriatrics (TCAs).

3) Attention Deficit Hyperactivity Disorders (ADHD):

- Antidepressants are considered the second choice to amphetamine in this disorder.

Advantages of antidepressants over CNS stimulants like amphetamine:

a) Provide more sustained improvement of the symptoms.

b) Do not induce abnormal movements associated with the use of amphetamine.

4) **Phobic anxiety syndrome, Obsessive compulsive disorders, Bulimia nervosa and posttraumatic stress disorders**:

- (uncontrolled episodes of eating followed by self- induced purging) and **post-traumatic stress disorder**.

5) **Severe psychosomatic disorders**:

- e.g., migraine, irritable bowel syndrome and sleep apnea.

6) **Smoking cessation (bupropion)and bupropion used as antidepressant in patient suffering from sexual dysfunction.**

7) **Neuropathic pain and fibromyalgia (SNRIs & TCAs).**

LITHIUM

- This agent is difficult to classify simply because it is used both to calm manic patients and prophylactically as a mood stabilizer in manic depressive illness.
- Lithium is a monovalent alkali metal ion of the same series as Na+ and K+.
- Unlike other psychotropic agents it does not have any psychotropic effects in man.
- It is not sedative, depressant or euphoriant.

Mechanism of Action

- The exact mechanism of action is unknown although speculations regarding altered distribution of ions in the CNS or effects on the turnover of biogenic monoamines have been implicated.
- However, recent evidence is in favor of the hypothesis that lithium reduces the release of inositol triphosphate and diacylglycerol which are important second messengers for both α-adrenergic and muscarinic transmission.

Therapeutic Uses

1) Prophylaxis or prevention of recurrence of manic depressive (bipolar affective) disorders.
   - It significantly decreases the frequency and severity of both manic and depressive attacks in about 70% of patients.

2) Treatment of acute mania. Because of its slow onset of action it is usually given along with a
3) neuroleptic drug for an immediate effect and then maintenance therapy can be carried out with lithium alone. Lithium may be useful in prophylaxis of recurrent unipolar depression.

Side effects and toxicity

- Lithium is a drug with low therapeutic index; therefore, concentrations in serum must be determined to facilitate the safe use of the drug.
- When the serum levels exceed 1.5 mEq/L, side effects are common and when exceed 2 mEq/L they start to be serious (CNS) and require urgent treatment.

1) **GI tract:** nausea, vomiting; diarrhea.
2) **CNS:** tremors, chorea, athetosis, convulsions, confusion, stupor.
3) **Kidney: nephrogenic diabetes insipidus** (thirst, polyuria) and renal cellular injury.
4) **Thyroid:** goiter and occasionally hypothyroidism.
5) Hypotension and cardiac dysrhythmias.
6) The use of lithium in early pregnancy can result in cardiovascular anomalies of the newborn.

- Lithium toxicity may be precipitated by water and sodium loss as by diuretics, diarrhea, vomiting and renal disease.

Administration:

- The carbonate salt of lithium is used as it is relatively less hygroscopic and less irritating to the gut. The drug should be given to maintain a serum concentration between 0.75-1.25 mEq/L. This can be attained by a dose of 900-1500 mg/day in outpatients and 1200-2400mg/day in hospitalized manic patients. The blood for lithium levels should be drawn12 hours after the last dose.

5 Pharmacotherapy of epilepsy

- Chronic neurological disease characterized by recurrent seizures (fits).
- **Seizures:** Episodes of brain dysfunction caused by ***excessive abnormal electrical discharge of cerebral neurons*** manifested by a transient alteration in behavior & muscle tone.
  Origin: Seizures almost arise often from cerebral cortex (not from other CNS structures).

Seizure Types

According to the characteristic seizure type & EEG changes:

1) :Partial seizures

1) Simple partial seizures	2) Complex partial seizures:
• It includes convulsions confined to a single limb or muscle group e.g., if a seizure involving motor cortex → clonic jerking of the body part controlled by this region of cortex. • *e.g.* , if motor cortex representing left thumb → clonic jerking of left thumb . • e.g., if somatosensory cortex representing left thumb → paresthesia of left thumb lasting approximating 20 to 60 seconds. • In general, in simple partial seizure there is no loss of consciousness.	**(psychomotor or temporal lobe epilepsy)** • It manifested by attacks of abnormal behavior with variable clinical manifestations and variable EEG changes with specific focal abnormalities in the anterior temporal region.

2) :Generalized seizures

a) Generalized tonic-clonic (*grand mal*) seizures:

- Aura → Loss of consciousness → Tonic convulsions for 1 min → Clonic jerking movement → flaccid relaxation → confusion & sleep.
- Aura is an abnormal sense that precedes loss of consciousness. It may be special sensory, autonomic, psychic or others.
- Characterized by major convulsions, tonic spasm of the whole body followed by synchronous clonic jerking. Convulsions are followed by depression of all central functions.
- If the seizures occur repeatedly without recovery of consciousness between the attacks, the disorder is called status epilepticus. It is a medical emergency and if left untreated may be fatal.

b) Absence (*petit mal*) seizures:

- Brief sudden attack of loss of consciousness associated with cessation of ongoing activities typically lasting for 10-15 sec (lasting less than 30 seconds).
- Mild or no motor activity e.g., eyelid blinking may occur.
- Begins in childhood (2-12 years) & may stop at 20 years or changed into tonic-clonic seizures.
- He will usually not fall. The EEG characteristically shows bilaterally symmetrical 3 per second spike and wave discharges.

Mechanism of Action of Antiepileptic Drugs

- In epilepsy the seizure activity is initiated by high frequency firing in an abnormal hyperexcitable focus that invades the normal brain tissue.
- The principal mechanisms of action of antiepileptic drugs involve:

1) Voltage operated ion channels

a- Na-channels:

- voltage dependent Na+ channels enter an inactive state following each action potential in which the channel does not allow the passage of Na+ into the cell; it is a refractory period.
- Stabilizing this inactive state with a concomitant prolongation of refractoriness leads to suppression of rapid repetitive firing in isolated neurons and protection against focal seizures.
- Phenytoin, carbamazepine, valproate and lamotrigine are thought to act by this mechanism.

b-Ca^{++} -channels:

- T-type Ca++ channels in thalamic neurons play a role in the 3 per second spike and wave discharge of absence seizures.
- Ethosuximide reduces the low threshold T-type Ca++ current in thalamic neurons.

2) Synaptic transmission

Enhancement of GABA-mediated inhibition can be produced by:

a) facilitating the GABA mediated opening of chloride channels as with benzodiazepines and phenobarbital.

b) promoting the synthesis or inhibiting the degradation of GABA as with valproate.

Classification Of Antiepileptic Drugs

According to seizure type (Therapeutic classification):

Seizure Type	1) Conventional Treatment	2) Recently developed drugs.
1) Partial seizures	• Carbamazepine • Phenytoin • Valproate	• Gabapentin • Lamotrigine • Topiramate • Levetiracetam • zonisamide
Generalized Tonic-Clonic	• Carbamazepine • Phenytoin • Valproate* • Phenobarbital • Primidone	• Lamotrigine • Topiramate
Generalized absence seizures:	• Ethosuximide • Valproate*	• Lamotrigine
Status epilepticus	• Diazepam • Phenytoin • Phenobarbitone	

NOTE

Thistable in dicate s the f ollowing:

- *Valproa te can be use d almost in e very typeof epilepsy.
- Onlyvalproate and ethosuximide are used effectively in absence seizure.

Phenytoin

- (200-400 mg/day)
- (Epanutin, Diphenyl hydantoin)

Mechanism of action

- Phenytoin produces stabilizing effect on neuronal (as well as cardiac) cell membrane.
- It slows the rate of recovery of voltage-activated Na+ channels from inactivation.
- This prolongs the refractory period and suppresses the episodes of repetitive neuronal firing.

Therapeutic Uses

1) Phenytoin is effective in generalized tonic-clonic and partial seizures.

   It is drug of choice because no sedation.It can also be given I.V. for control of status epilepticus.

2) It is used to treat ventricular dysrhythmias, paroxysmal atrial flutter or fibrillation and supraventricular dysrhythmias caused by digitalis (1st choice in digitalis dysrrhythmias).

3) Phenytoin is useful in some cases of trigeminal neuralgia if carbamazepine is ineffective or cannot be tolerated.

Side effects and toxicity

A) After prolonged use:

1) **Gingival (gum) hyperplasia(**inchildren).
2) **Cerebello-vestibular manifestations**: Ataxia, nystagmus, confusion & vertigo.
3) **GITirritation** (given after meals).
4) **Osteomalacia:** due to enzyme induction of Vit D metabolism → hypocalcemia.
5) **Cleft palate & hare lip** (fetal hydantoin syndrome):if given during first trimester.
6) **Hypoprothrombinemia** → bleeding (due to induction of vit K metabolism).
7) **Hirsutism**.
8) **Drug-drug interaction: :Phenytoin is a strong enzyme-inducer:**
   a) induction of its own metabolism (auto-induction).
   b) induction the metabolism of other AEDs (drug interaction)→ TDM is necessary.
9) **Hypersensitivity reactions*:*** skin rash, SLE, lymphadenopathy, and fatal hepatic necrosis.
10) **Hematological disturbances:blooddyscrasis**

B) After large I.V dose:

- **Cardiovascular collapse & CNSdepression.**

Carbamazepine

- (Tegretol; 600-1200 mg/day).
- Carbamazepine is related chemically to tricyclic antidepressants.
- Its anticonvulsant actions are similar to those of phenytoin; both have virtually identical actions in blocking Na+ channels in therapeutic concentrations.

Therapeutic Uses of carbamazepine

1) **Anti-Epileptic**: similar to phenytoin:
   - Drug of 1st choice → no sedation & effective.
   - Used in both generalized tonic clonic and partial seizures BUT worsen Petit mal epilepsy.
   - Hepatic microsomal enzyme-inducer.
   - If single drug therapy is unsuccessful, it can be given in combination with phenytoin.
2) **Antidepressant:** useful in manic-depressive illness (bipolar disorder).
3) **Treatment of Trigeminal neuralgia & peripheral neuropathy.**

Side effects

- **Cerebello-vestibular manifestations:** Ataxia, nystagmus, confusion, diplopia & vertigo (toxicity signs).
- Hyponatremia and water retention especially in elderly patients with cardiac disease.
- GI upset & anorexia.
- Bone marrow suppression → agranulocytosisand hepatic dysfunction.

VALPROIC ACID
)Depakene(

Mechanism of antiepileptic action:

1) **Na+ channel blocker:**
   - delay recovery & stabilizes neuronal cell membranes "similar to phenytoin" → effective in ***partial & grand mal epilepsy***.

2) **T-type Ca++ channel blocker:**
   - stabilize neuronal cell membrane "similar to ethosuximide"→ effective in ***absence seizure***.

3) **Gabamimetics:**

- Valproate can stimulate the activity of the GABA synthetic enzyme (glutamic acid decarboxylase) and inhibit GABA degradative enzymes (GABA transaminase and succinic semialdehyde dehydrogenase) → ↑ GABA levels.

4) **NMDA receptor antagonist.**

Toxicity

1) **Liver:**

- Fatal fulminant hepatitis "rare toxic complication".
- More frequent in children < 2 years & use multiple antiepileptic agents not preferred in children > 2 years.

2) **CNS:** sedation, ataxia & tremors.

3) **Hematological:** thrombocytopenia & ↓ platelet aggregation

4) **Teratogenic effect:** spina bifida (neural tube defect)

5) **Hair:** temporary loss of hair → thin curly hair.

6) LME inhibitor → ↑ metabolism of other AEDs.

7) **GIT:** Anorexia, nausea & vomiting.

8) Hypersensitivity reactions: skin rash, alopecia & acute pancreatitis.

Uses:

i.Broad-spectrum antiepileptic drug: in almost all types of epilepsy.

- Not used in status epilepticus because no parenteral therapy available.
- Not drug of choice because of hepatotoxicity & sedative effect.

ii.Drug of choice in mixed *petit mal + grand mal epilepsy*.

iiiother uses:

a) Mood stabilizer in bipolar affective disorder.

b) Migraine headache.

Phenobarbital

Phenobarbital is a long-acting barbiturate with a broad-spectrum anticonvulsant activity. It exerts a maximal anticonvulsant activity at doses below those required for hypnosis. The drug at therapeutic concentrations was found to reduce the excitatory effects of glutamate and to augment the inhibitory effects of GABA. Oral absorption of phenobarbital is complete but somewhat slow. It is bound to plasma proteins to the extent of 40 - 60%. About 25% of the dose is eliminated unchanged by the kidneys while the remainder is inactivated by hepatic microsomal enzymes and excreted in urine partly as glucuronide conjugate.

- Phenobarbital, like phenytoin and carbamazepine, induces drug metabolizing hepatic enzymes.

To xicity

a-Sedation (CNS depression):

- is most frequent, but tolerance develops during chronic medication.

b-Nystagmus and ataxia occur at excessive dosage.

c-Behavioral changes,

- skin rash, megaloblastic anemia and osteomalacia.
- Physical dependence occurs on prolonged use.
- Sudden withdrawal may result in grand mal seizures, severe tremors, vivid hallucinations and psychoses.

Therapeutic Uses

- Phenobarbital is an effective agent in generalized tonic - clonic and partial seizures. It has the advantages of low cost, efficacy and low toxicity. However, because of its sedative effects, its tendency to disturb behavior in children and its inability to completely control the seizures; it is considered to be alternative when therapy with carbamazepine or phenytoin is unsuccessful. Phenobarbital I.V. is also effective in statusepilepticus and other convulsive emergencies.

PRIMIDO NE

- **Primidone is deoxyphenobarbital;** it resembles phenobarbital in most of its effects.
- It is metabolized to two active compounds, phenobarbital (major metabolite) and phenylethylmalonamide.
- Thus, the anticonvulsant effects of primidone are attributed to both the drug and its active metabolites particularly phenobarbital.
- The drug is rapidly and almost completely absorbed from GIT, although individual variability can be great. Peak concentrations in plasma are usually observed 3 hours after ingestion.

Toxicity

- The most common complaints are sedation, vertigo, dizziness, nausea, vomiting, ataxia, diplopia and nystagmus. Serious side effects are relatively uncommon and include skin rash, blood dyscrasias, systemic lupus erythematosus, lymphadenopathy, acute psychotic reactions, megaloblastic anemia and osteomalacia.

Therapeutic uses

- Primidone is effective against generalizedtonic-clonic and partial seizures.
- While it may be effective alone in patientswho are refractory to other medications, it is generally used concurrently with phenytoinor carbamazepine.

ETHOSUXIMIDE

- Ethosuximide is a primary agent for the treatment of absence seizures.

Mechanism of action:

- probably involves Ca++ channels. The drug reduces the T-type calcium currents that provides a pacemaker current in thalamic neurons responsible for generating the rhythmic cortical discharge of an absence seizure attack. Absorption of ethosuximide from GIT appears to be complete. It is not significantly bound to plasma proteins. About 25% of the drug is excreted unchanged in urine and the remainder is metabolized by hepatic microsomal enzymes.

Toxicity

1) **GIT reactions:**

- nausea and vomiting.

2) **CNS effects:**

- drowsiness, dizziness euphoria, headache and parkinsonian like syndrome. Behavioral disturbances may occur particularly in patients with past history of psychosis.

3) **Skin reactions,**

- systemic lupus erythematosus and blood dyscrasias have been reported.

theraputic uses

- Ethosuximide is a drug of first choice in treatment of absence seizures.
- It is similar to valproate.

Benzodiazepines

- Besides the value of these drags as anxiolytics and hypnotics, some are also very effective in seizure control.
- Their beneficial effects reside largely in their ability to enhance GABA induced increases in chloride conductance.

CLONAZEPAM

- It is useful in the therapy of absence seizures.
- However, tolerance to its antiepileptic effects usually develops following 1-6 months of administration, after which some patients will not respond to clonazepam at any dosage.
- Clonazepam is well absorbed after oral administration and 85% is bound to plasma proteins. It is metabolized to inactive products.
- The principal side effects of long-term oral therapy with clonazepam include drowsiness, ataxia, dizziness, hypotonia, dysarthria, behavioral disturbances, anorexia and hyperphagia.
- Seizures are sometimes exacerbated.

DIAZEPAM

- Given I.V., it is the drag of choice for control of status epilepticus, but its short duration of action is a disadvantage. This short duration of action, as with other highly lipid soluble agents, is due to the phenomenon of redistribution. Diazepam is bound to plasma proteins to the extent of 99%. The main adverse effects that may result from I.V. administration of diazepam are cardiovascular and respiratory depression particularly if other anticonvulsants or CNS depressants have been administered previously. Lingual obstruction to respiration due to muscular relaxation is a serious complication.

NEW ANTIEPILEPTIC DRUGS

l-Gabapentin

- It is a simple analogue of GABA thatdoes notact on GABA receptors, although it may alter the release of GABA through unknown mechanism.
- It is used in conjunction with other drugs against partial and generalized tonic-clonic seizures. The side-effects of gabapentin appear to be less than with many antiepileptic drugs.

Lamotrigine

- Lamotrigine resembles phenytoin in the mechanismof action (blocking voltage-dependent Na+ channels) but additional actions on voltage activated Ca++ -channels or inhibition of synaptic release of glutamate might be involved.
- Lamotrigine is a broad-spectrum antiepileptic drug; it is used in the therapy of partial and generalized tonic-clonic seizures and is also effective against absence and myoclonic seizures.
- It proved to be useful in conjunction with other drugs in Lennox-Gastaut syndrome which is unusual disorder characterized by multiple seizure types, mental retardation and refractoriness to antiseizure medication.

Topiramate

- Topiramate, similar to phenytoin, blocks voltage-dependent Na+- channels.
- In addition, it potentiates the inhibitory effect of GABA and also depresses the excitatory action of AMPA-kainate subtypes of glutamate receptor.
- The drug is well tolerated and has broader spectrum, being effective against refractory partial seizures and generalized tonic-clonic seizures.
- It is also effective against Lennox- Gastaut syndrome and myoclonic seizures.

TREATMENT OF STATUS EPILEPTICUS

This is a medical emergency. The main lines of treatment are:

1) **Diazepam:**
   - 10 mg I.V. over the course of 2 minutes and the dose is repeated after 10 minutes if necessary.
   - This is usually effective in stopping the seizure for a brief period.
   - Alternatively, lorazepam 4 mg I.V. can replace diazepam and is repeated once after 10 minutes if necessary.

2) Regardless of the response to diazepam which is short-acting, **phenytoin 15-20 mg/kg**:
   - is given I.V. at rate of 50 mg/minute to provide initiation of long term seizure control and avoid relapse.
   - Fosphenytoin which is completely converted to phenytoin following I.V. administration can replace phenytoin.

3) **If seizures continue phenobarbital**:
   - is given in a dose of 10-20 mg /kg I.V. by slow or intermittent injection.

4) If these measures fail, **general anesthesia with ventilatory assistance and neuromuscular blockers** may be required.

An ideal antiepileptic drug should be:

1) effectiveinall types ofepilepsy.
2) rapidly acting with a long duration of action.
3) having minimal side effects and be non-toxic and non-addicting.
4) orally effective and inexpensive.

- Although no such drug exists, the present range of drugs can produce complete control of seizures in 50% of patients and reduce significantly their frequency in another 25%.
- In order to minimize toxicity, treatment with a single drug is usually sought. If one drug is not successful then a second drug may be substituted or added.
- Monitoring of plasma concentration is particularly useful to maintain therapeutic concentrations.
- However, for some drugs clinical effects do not correlate well with plasma concentrations and the ultimate therapeutic regimen must be determined by clinical assessment of effect and toxicity.
- Drug therapy is usually continued until there have been no seizures for at least 2 years.
- Once the decision to withdraw an antiepileptic drug is made, such withdrawal should be gradual over a period of months.
- The risk of status epilepticus is great with abrupt cessation of therapy

1 Parkinsonism

Itis a chronic and progressive disorder of basal ganglia in the CNS.

Other manifestations:

- Sweating.
- Salivation.
- Depression.

Causes:

I-Idiopathic:

Exposure to some neurotoxin (destruction of dopaminergic neurons).

- Occurrence of oxidation reactions with the generation of free radicals.
- Genetic factors.

III- Atherosclerosis.

IV-~~Wilson's disease~~.

V-~~post-traumatic~~

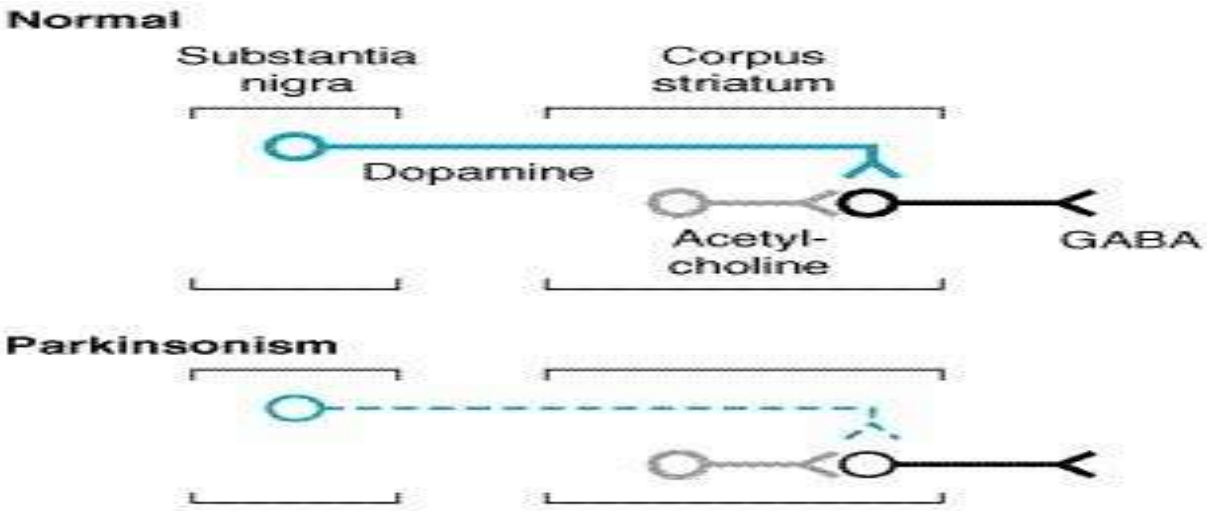
VI- Drug induced as:

- Dopamine receptor antagonists

  1- Anti-psychotic as: Chloropromazine Haloperidol
  2- Anti-emetics as: Metoclopramide.

- Adrenergic neuron blockers: Reserpine.
- Inhibition of dopamine synthesis: Alpha-methyl dopa
- Destruction of dopamine neurons:
  By toxin as MPTP (N-methyl-4-phenyl-1,2,3,6-tetrahydropyridine).

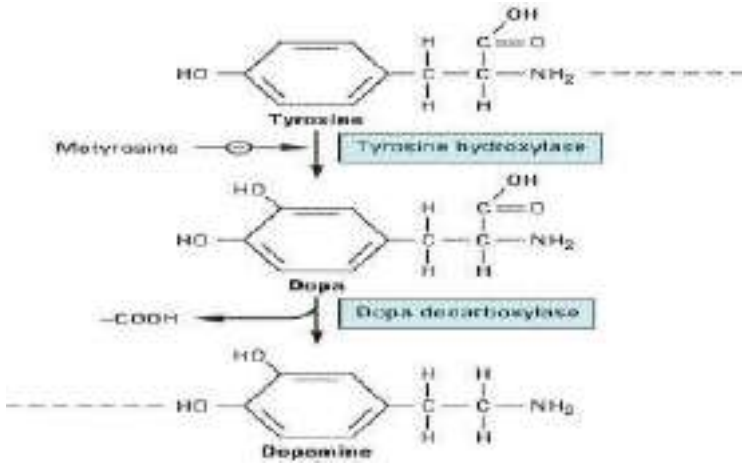
Pathophysiology



Drugs

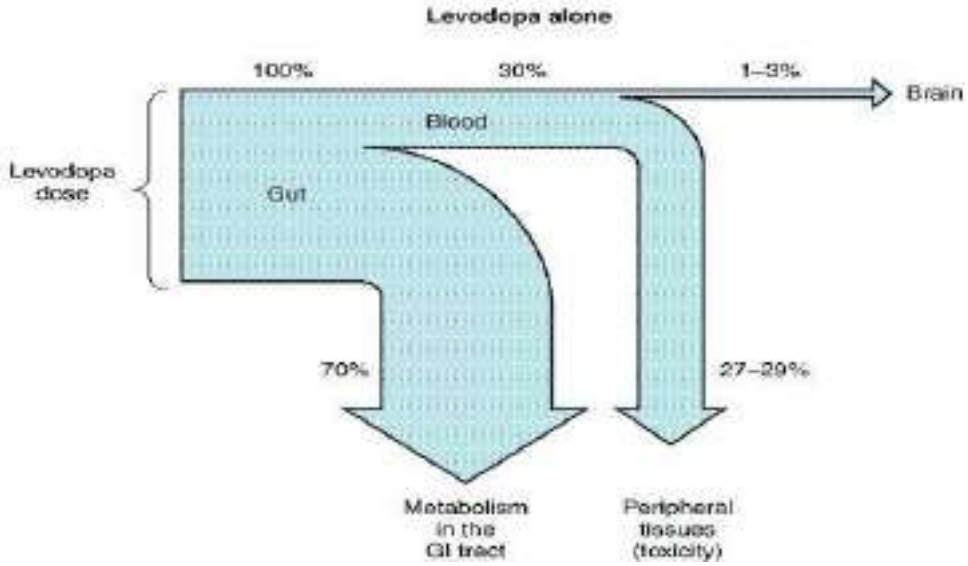
Drugs activating dopaminergic system	Drugs blocking cholinergic system
1- *Precursors ofdopamine:* • Levodopa.	**1- Atropine substitutes:** • Trihexyphenidyl. • Benztropine.
2- Dopamine receptor agonists: **a) Ergot derivatives:** • Bromocriptine. • Pergolide. **b) Nonergot derivatives:** • Pramipexole. • Ropinirole.	
3- Inhibitors of dopamine metabolizing enzymes: **a) MAO inhibitors:** • Selegiline. • Rasagiline. **b) COMT inhibitors:** • Tolcapone. • Entecapone.	
4- Drugs affecting synthesis, release, or reuptake of dopamine: • Amantadine.	

1) Levodopa



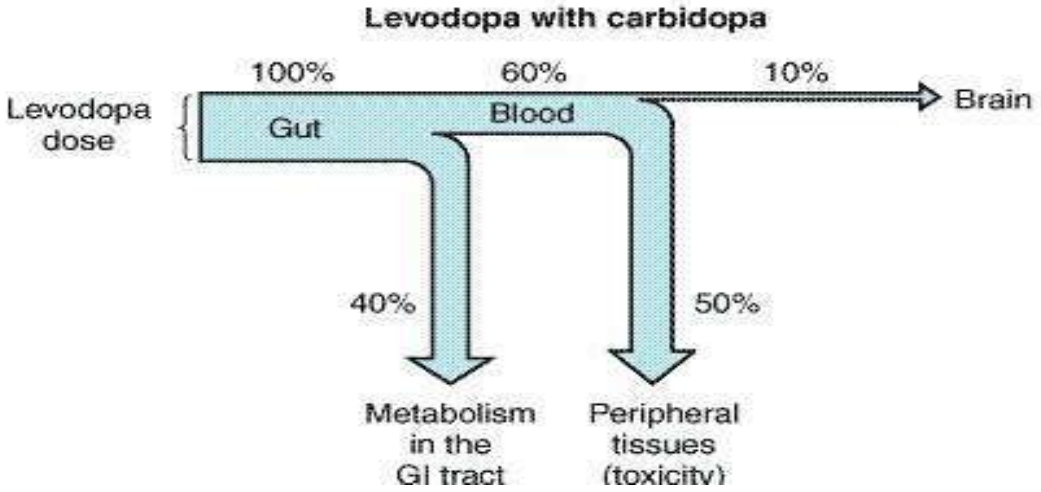
Mechanism of action

- Levodopa is decarboxylated to dopamine in C.N.S. Dopamine binds to dopamine receptors D1 and D2.
- The D2 receptors are located in basal ganglia.
- Only 1–3% of administered levodopa actually enters the brain, the remaining part ismetabolized in the blood by decarboxylation to dopamine, which does not penetrate the **B.B.B.**



When levodopa is given in combination with a **dopadecarboxylaseinhibitor Carbidopa** that does not penetrate the B.B.B, this leads to:

- The peripheral **metabolism** of levodopa is reduced.
- Plasma **levels** of levodopa are higher.
- Plasma **half-life** is longer.
- More dopa is available for **entry into the brain.**
- Reduce the **daily requirements** of levodopa by approximately 75%.



Clinical use

Levodopa therapy does **not stop the progression** of parkinsonism. The best results are obtained in the first 5 years of treatment

Fluctuations in response

a- End-of-dose akinesia:
related to the timing of doses i.e each dose of levodopa improves mobility for a period of time (1-2 h), but rigidity and akinesia return rapidly at the end of the dosing interval.

b- On-off phenomenon:
sudden and transient fluctuations in severity of

parkinsonism are unrelated to the timing of doses. In it, off-periods of marked akinesia alternate with on-periods of improved mobility but associated with marked dyskinesia. It occurs unpredictably but frequent during the day. It affects more than 50% of patients treated for 5 years or more with levodopa. The exact mechanism is unknown. The incidence of this phenomenon is decreased when plasma level of levodopa is maintained constant, through giving the drug by I.V infusion or by sustained release and transdermal formulations or giving the total daily dose in equal amounts every 2 hours rather than every 4-6 hours or by S.C injection of apomorphine or drug holidays.

Drug interactions

1- **Pyridoxine** (vitamin B6) enhances the extracerebral metabolism of levodopa so prevent its therapeutic effect.

2- Levodopa should not be given to patients taking **MAO- A inhibitors** because this leads to hypertensive crises.

Contraindications

- Psychotic illness.
- Closed angle glaucoma.
- Cardiac dysrhythmia.
- Active peptic ulcer.

2) *Dopamine agonists*

Drugs acting directly on dopamine receptors.

Bromocriptine

Itacts as agonist onD2 receptors in C.N.S, C.V.S, and G.I.T.

Therapeutic uses:

1- **Parkinson's disease** with levodopa.

2- **Hyperprolactinemia** which manifested by galactorrhea, amenorrhea and infertility.

3- **Acromegaly**.

Side effects:

- Nausea andVomiting.
- Digital vasospasm.
- Cardiac dysrhythmias.
- Mental disturbances.

Pergolide

- Itstimulatesboth **D1** and **D2** receptors.
- Itis more effective than bromocriptine.
- Itincreases"on-time" response so, allow the doses of levodopa to decrease.

Side effects:

- Nausea.
- Vomiting.
- **Valvular** diseases in 1/3 of patients.

Pramipexole

- Itis not an ergot derivative.
- Itis agonist on **D3** receptors.
- Itis used alone for **mild** parkinsonism.
- Itis also used in **advanced** disease, with levodopa.
- Ithas **antioxidant** activity by scavenge hydrogen peroxide.

Ropinirole

- Itis not an ergotderivative.
- Itis a relatively**pure D2** receptor agonist.
- Itis effectiveaspramipexole.

Side effects:

less than ergotderivatives.

- Rare side effectasan uncontrollable tendency to **fall asleep** at inappropriate times.

3) *Monoamine oxidase inhibitors*

Selegiline

- It is a selective inhibitor of MAO-B.
- It inhibits the breakdown of dopamine.
- It enhances and prolongs the antiparkinsonian effect of levodopa.
- It used for patients with fluctuating response with levodopa.
- It has **antioxidant** action.
- So, it reduces the disease **progression**.
- Its neuroprotective effect is due to its metabolite, **desmethylselegiline**.

Rasagiline

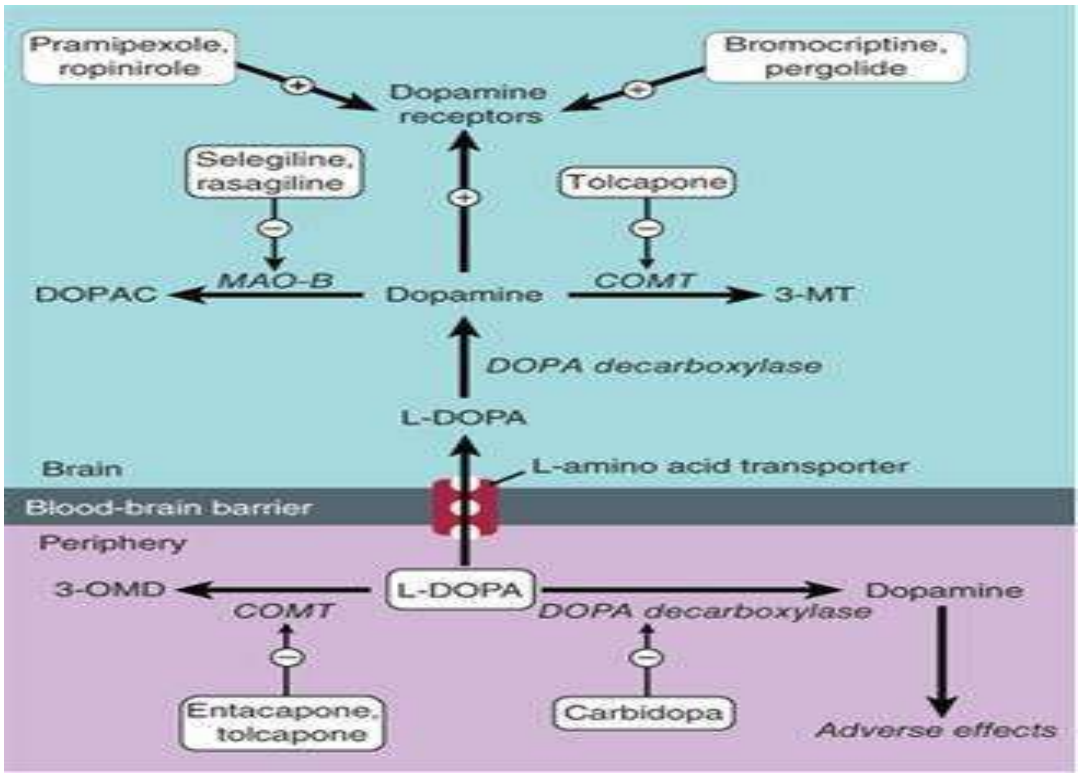
- It is more potent than selegiline in slowing disease progression neuroprotective effect.
- It is well effective in younger patients with early or mild symptoms.
- It is also used to **prolong the effects** of levodopa -carbidopa in patients with advanced disease.

Sideeffects

1- Selegiline andrasagiline may cause **insomnia** when taken later during the day.
2-It should notbetaken by patients receiving **TCAS**, or SRIS because of the risk of acute toxicinteractions.
3-The adverseeffects of **levodopa** may be increased by selegiline.

4) Catechol-O-methyl transferase inhibitors

- Tolcapone.
- Entecapone



1- Inhibition of dopa decarboxylase is associated with compensatory activation of other pathways of levodopa metabolism, as **COMT**.
2- Selective COMT inhibitors prolong the action of levodopa by diminishing its peripheral metabolism.

Tolcapone	Entacapone
• Central and peripheral effects.	• Peripheral effects.
• More potent.	• Less potent.
• Long duration of action.	• Short duration of action.
• Hepatotoxic.	• Not hepatotoxic.

Sideeffects

- Dyskinesias, and confusion.
- Nausea, diarrhea, and abdominal pain.
- Sleep disturbances.
- Tolcapone may cause an increase in liver enzyme levels.

5) *Amantadine*

- Itis an **antiviral** agent, was by chance found to have anti-parkinsonism properties.

Mechanism of action

- It is unclear.
- Itmaypotentiatedopaminergic function by:
- Affectthe**synthesis**,**release,** or **reuptake** of dopamine.
- Has anticholinergic effect
- Haseffectonglutamatereceptor

Clinical use

- Itislesspotentthanlevodopa.
- Itsbenefitsmaybe**short-lived**, and disappear afteronlyafewweeksoftreatment.

Sideeffects

1- CNSeffects:Restlessness,
2- depression, insomnia,
3- hallucinations. Livedoreticularisblushdiscoloration of the leg sometimes occurs
4- in patients takingamantadineandusually clears within a month after the drug is
5- withdrawn. Peripheral Headache,heartfailure,urinary retention. GIT disturbances **edema**

6) Anticholinergic drugs

Centrally acting antimuscarinic drugs.

- **Trihexyphenidyl.**
- **Benztropine.**

Clinical use

Antimuscarinic drugs improve the tremor and rigidity of parkinsonism.
Have little effect on bradykinesia.

Sideeffects

1- CNS effects, as drowsiness, hallucinations, and mood changes.
2- Dryness of the mouth.
3- Blurring of vision, increased intraocular pressure.
4- Urinary retention, constipation.

Contraindications

Itshould be **avoided** in patients with:

- Prostatic hyperplasia.
- Paralytic ileus.
- Glaucoma.

Concomitant administration of **TCAS** or **antihistamines** may precipitate some of the complicationsmentioned above.

2 Opioid Analgesics

Opioid analgesics are drugs which, while providing relief from pain, produce drowsiness, changes in mood and mental clouding.

Morphine and related opioids	Meperidine and congeners
• Morphine	• Meperidine
• Codeine	• Fentanyl
• Tramadol	• Diphenoxylate
• Heroin	• Loperamide
• Levorphanol	
Methadone and congeners	**Mixed agonist-antagonist and partial agonists**
• Methadone	• Pentazocine
• Propoxyphene	• Butorphanol
	• Nalbuphine
	• Buprenorphine

Opioid receptors

These receptors and the corresponding actions include:

1- μ (mu):

- **μ1:** supraspinal analgesia
- **μ2:** spinalanalgesia, respiratory depression, sedation, euphoria, miosis and decreasedGIT motility.

2- κ (Kappa): spinal analgesia, dysphoria, hallucinations, less intense miosis and respiratorydepression than do μ receptors.

3- δ (delta): theeffects in human beings are unclear although they may produce analgesiaatboth spinal and supraspinal levels.

Mechanism of action

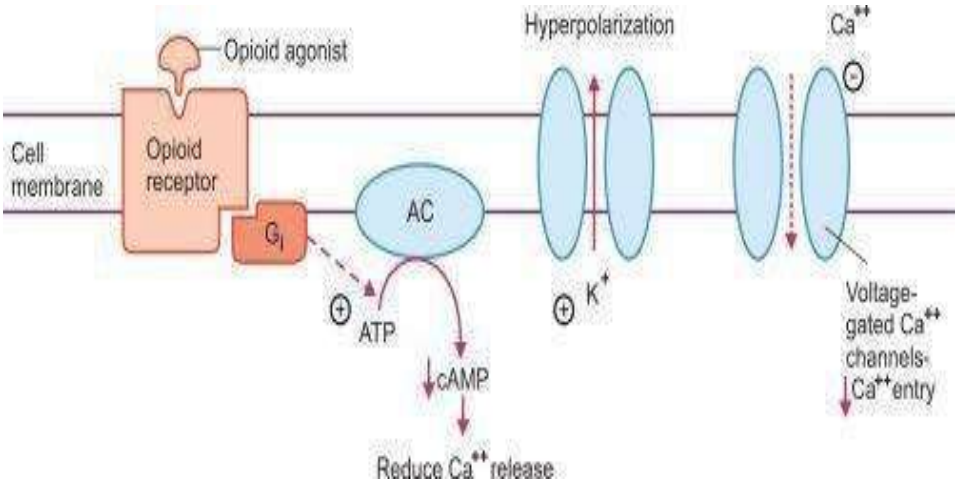
Opioid agonists produce analgesia by binding to specific G protein-coupled receptors that are located in CNS and other tissues.

A- The opioids have two well-established direct G protein-coupled actions on neurons:

(1) They close voltage-gated Ca2+ channels on presynaptic nerve terminals and thereby reduce transmitter release.

(2) They hyperpolarize and thus inhibit postsynaptic neurons by opening K+ channels.

B- Opioid receptors also regulate the activity of certain enzymes (inhibit adenyl cyclase and reduce the intracellular cAMP content).



Clinical use

1- Analgesia:

- Severe,constantoracute (lasting hours to days) pain is usually relieved with opioid analgesics,whereas sharp, intermittent pain does not appear to be as effectivelycontrolled.Also, chronic pain (lasting weeks to months) is not very satisfactorilymanagedwith opioids. For reliefofsuddenagonizing pain, morphine is usually given IV to produce immediatereliefandminimizes shock.
- Extensiveuseofmorphine in postoperative analgesia should be avoided as it may produceerespiratory depression, urinary retention and constipation, it may maskthesignsof recovery and prevent the early recognition of complications.

2- Cardiac asthma with Pulmonary Edema:

- The relief produced by intravenous morphine in dyspnea from pulmonary edema associated with left ventricular heart failure is remarkable. Proposed mechanisms include reduced anxiety (perception of shortness of breath), decrease respirato effort rcardiacpreload and reduced (reduced venous tone) and afterload (decreased peripheral resistance).

N.B.

If respiratory depression is a problem, furosemide may be preferred for the treatment of pulmonary edema. On the other hand, morphine can be particularly useful when treating painful myocardial ischemia with pulmonary edema. Nitro-glycerine is superior to morphine in this condition.

3- Cough:

- Suppression of cough (useful in dry non-productive cough) can be obtained at doses lower than those needed for analgesia eg, codeine. However, in recent years the use of opioid analgesics to suppress cough has diminished largely because a number of effective synthetic compounds have been developed that are neither analgesic nor addictive.

4- Diarrhea:

- Diarrheafrom almost any cause can be controlled with the opioid analgesics, but if diarrhea is associated with infection such use must not substitute for appropriate chemotherapy. Now synthetic surrogates with more selective gastrointestinal effects and few or noCNS effects, eg, diphenoxylate or loperamide, are used.

5- Shivering:

- Althoughallopioid agonists can reduce shivering, meperidine is reported to have themost pronounced anti-shivering properties.
- Meperidineapparently blocks shivering mainly through an action on subtypesofthe α2 adrenoceptor.

6- Applications in Anesthesia:

- The opioids are frequentlyusedas premedicant drugs before anesthesia and surgery becauseoftheirsedative, anxiolytic, and analgesic properties. They are also usedintraoperatively both as adjuvants to other anesthetic agents and, in highdoses(eg,fentanyl), as a primary component of the anesthetic regimen.
- Opioids are most commonlyusedin cardiovascular surgery and other types of high-risk surgeryinwhichaprimary goal is to minimize cardiovascular depression (eg. Fentanylorsufentanil). In such situations, mechanical respiratory assistancemustbeprovided.
- Opioids can also beusedasregional analgesics by administration into the epidural or subarachnoidspacesof the spinal column.

Sideeffects

1- Direct toxic effects of the opioid analgesics:

Whichareextensionsoftheir acutepharmacologicactions and include:

- Respiratory depression, nausea, vomiting,constipation (may be serious in old people and precipitate intestinal obstruction), increased intracranial pressure, postural hypotension accentuatedbyhypovolemia.
- Urinary retention, increased pressure in thebiliary tract, dizziness, mental clouding, itching around nose, urticaria (morefrequent with parenteral and spinal administration) and, behavioral restlessness, tremulousness, hyperactivity (in dysphoric reactions).

2- Tolerance:

- Althoughdevelopment of tolerance begins with the first dose of an opioid, tolerancegenerally does not become clinically manifest until after 2–3 weeks of frequentexposure to ordinary therapeutic doses. Tolerancedevelops most readily when large doses are given at short intervalsand is minimized by giving small amounts of drug with longer intervalsbetween doses.

Degrees ofTolerance:

High:Analgesia, euphoria, dysphoria, mental clouding, sedation, respiratory depression, antidiuresis, nausea, vomiting and Cough suppression.
Moderate:Bradycardia
Minimal or None: Miosis, constipation and convulsions.

Cross-tolerance: is an extremely important characteristic of the opioids, not only with respect to their analgesic actions but also to their euphoriant, sedative, and respiratory effects. Tolerance to morphine is attributed to adaptation of cells in the nervous system to the drug action (pharmacodynamics tolerance).

3- Dependence:

A- Physical dependence:

- The development of physicaldependenceisusually associated with tolerance to repeated administrationofanopioidofthe μ type. Failure to continue administeringthedrugresults in a characteristic withdrawal or abstinencesyndromethatreflectsanexaggerated rebound from the acute pharmacologiceffectsoftheopioid.
- The signs and symptomsofwithdrawalincluderhinorrhea, lacrimation, yawning, chills, piloerection,hyperventilation,hyperthermia, mydriasis, muscular aches, vomiting,diarrhea,anxiety,andaggression.
- With morphine, withdrawalsignsusuallystartwithin6–10 hours after the last dose. Peak effects areseenat36–48hours.By5days, most of the effects have disappeared, butsomemaypersistformonths.

❖ **Abstinence Syndrome**:

- Results not only fromdrugwithdrawalbutalso**antagonist-precipitated withdrawal** which canbeinducedinasubjectphysically dependent on opioids by administeringnaloxoneororanotherantagonist.
- Within 3 minutes afterinjectionoftheantagonist,signs and symptoms similar to those seen afterabruptdiscontinuanceappear, peaking in 10–20 minutes and continuesuntiltheantagonistiseliminated.

❖ **Treatment of morphine addiction:**

- Consists of its replacementbyanothernarcoticanalgesic that shows cross dependence with morphineanditswithdrawalsyndrome is mild.
  Methadone: orally is used.Oncethepatientisstabilized on methadone, its dose is gradually reducedby20%daily.
  Buprenorphine: (a partialagonist)isaseffectiveasmethadone.
  Levo-α-acetylmethadol: (μ full agonist) also used.

Another approach is to use: Clonidine: an α2 adrenergic agonist which is effective in suppressing autonomic hyperactivity manifestations observed in abstinence syndrome. Lofexidine: a similar drug with greater affinity for α2 receptors and less hypotension that limits usefulness of clonidine.

B- Psychologic Dependence:

- The euphoria, indifference tostimuli, and sedation usually caused by the opioid analgesics, especiallywhen injected intravenously, tend to promote their compulsive use.
- These factors constitute theprimary reasons for opioid abuse liability and are strongly reinforced by thedevelopment of physical dependence.

Contraindications

1- Use in asthmatic patients:

Is hazardous as it depressesthe respiratory center and cough reflex, releases histamine and tends to dry secretions, soif used, should be with extreme caution.

2- Use in Patients with Head Injuries:

- Carbon dioxide retention caused by respiratory depression results in cerebral
- vasodilation. In patients with elevated intracranial pressure, this may lead to lethal alterations in brain function. Morphine also produces miosis and mental clouding
- that may interfere with the diagnosis.

3- Use during Pregnancy and labor:

In pregnant women who arechronicallyusing opioids, the fetus may become physically dependent in uteroandmanifest withdrawal symptoms in the early postpartum period including irritability,shrillcrying, diarrhea, or even seizures. When

- morphine is administeredtothemother during labor, it crosses the placental barrier, and the fetus may exhibit(asphyxianeonatorum). So, morphine should be avoided if delivery is expectedwithin4-5hours.

4- Use in Patients with Impaired Pulmonary Function:

In patients with borderline respiratory reserve as in emphysema, kyphoscolioisis, severe obesity and chronic corpulmonale, the depressant properties of the opioid analgesics may lead to acuterespiratory failure.

5- Use in Acute Abdomen:

Pain is a chief diagnostic symptom, and its total abolition may interfere with the diagnosis.

6- Use in Patients with Impaired Hepatic or Renal Function:

Because morphine and its congeners are metabolized primarily in the liver, patients with hepatic dysfunction are not able to tolerate morphine. Half-life is prolonged in patients with impaired renal function, and morphine and its active glucuronide metabolite may accumulate; dosage can often be reduced in such patients.

7- Use in Patients with Endocrine Disease:

Patients with adrenal insufficiency (Addison's disease) and those with hypothyroidism (myxoedema) may have prolonged and exaggerated responses to opioids.

8- Use in Patients with Reduced Blood Volume: Are more susceptible to the hypotensive effect of morphine and therefore should be used with caution in any type of hemorrhage.

9- Use of Pure Agonists with Weak Partial Agonists:

When a weak partial agonist such as pentazocine isgiven to a patient also receiving a full agonist (eg, morphine), there is a risk ofdiminishing analgesia or even inducing astate of withdrawal; combining full agonist with partial agonist opioids should beavoided.

Acute opioid toxicity

Acute opioid toxicity may resultfromclinicaloverdosage,accidental overdosage in addicts, or attempts at suicide. It is difficult to define the exactamountofanyopioidthatistoxic or lethal to humans but in a normal pain free adult,adoseof60mgofmorphineis usually toxic and a dose of 200mg is usually fatal.

Symptoms and diagnosis:

- The **triad** of coma, pinpointpupils,anddepressedrespiration strongly suggests opioid poisoning. Urine formation is depressed,bodytemperaturefalls,andthe skin becomes cold and clammy. The skeletal musclesareflaccid,thejawisrelaxed, and the tongue may obstruct the airway. Frankconvulsionsoccasionallyoccur in infants and children. When death ensues, it is nearlyalwaysfromrespiratoryfailure. Even if respiration is restored,deathstillmayoccurasaresult of complications that develop during the periodofcoma,suchaspneumoniaor shock. Noncardiogenic pulmonary edemaisseencommonlywithopioid poisoning.
- **Treatment:**

Opioid antagonists can produce dramatic reversal of the severe respiratory depression; **naloxone** is the treatment of choice but should be used with care as it may producesevere withdrawal syndrome which is sometimes more life threatening thanthe respiratory depression itself.

Opioid antagonists

The antagonism is competitive in nature.

Naloxone

1- It has a pure antagonistic activity by binding to opioid receptors.
2-Usually given by injection and has a short duration of action.

Uses:

1- Treatment of acute opioid poisoning.
2- To counteract neonatal opioid-induced fetal asphyxia.
3- Diagnosis of opioid addiction.

Naltrexone

1- It is a pure antagonist with high oral efficacy (given orally).
2- It is more potent than naloxone and has longer duration of action (about 24 hours).

Uses:

1- It can be used in **ex-abusers** to opioids who are fully withdrawn as it will block the effects of self-administered opioids.
2- It is approved for treatment of alcoholism.